Fundamental Theorem of Matrix Product States

A Formalization Blueprint

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Chapter 1

Introduction

The central result is the Fundamental Theorem for translation-invariant matrix product vectors. The basic object is a family of matrices $A = \{A^i\}_{i=0}^{d-1}$, which assigns to each chain length N the vector $|V^{(N)}(A)\rangle$ with coefficients indexed by $\sigma = (i_1, \dots, i_N)$,

$$V^{(N)}(A)_\sigma = \mathrm{tr}(A^{i_1} \cdots A^{i_N}).$$

The main argument, for translation-invariant tensors, follows the canonical-form and basis-of-normal-tensors argument of [CPGSV16, CPGSV21], with the single-block injective case anchored in the earlier representation theorem of [PGVWC07].

Notation

$\mathbb{C}$, $\mathbb{N}$, $\mathbb{Z}$	complex numbers, natural numbers, integers
$M_D(\mathbb{C})$, $\mathrm{GL}_D(\mathbb{C})$	complex matrices and invertible complex matrices
$\mathrm{tr}(X)$, $\mathrm{span}_{\mathbb{C}}(S)$	trace and complex linear span
$A = \{A^i\}_{i=0}^{d-1}$	a translation-invariant MPS tensor
$w = (i_1, \dots, i_L)$, A^w	a word and the product $A^{i_1} \cdots A^{i_L}$
$\lvert V^{(N)}(A)\rangle$	the length-N matrix product vector
$V^{(N)}(A)_\sigma$	the coefficient of $\lvert V^{(N)}(A)\rangle$ at σ
$O_{AB}(N)$	the bilinear overlap $\sum_\sigma V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}$
$\mathcal{E}_A(X)$	the transfer map $\sum_i A^i X (A^i)^\dagger$
$A^{[L]}$	the length-L blocked tensor

In canonical form one writes

$$A^i = \bigoplus_{j=1}^{g} \mu_j A^i_j, \qquad B^i = \bigoplus_{k=1}^{h} \nu_k B^i_k, \tag{1.1}$$

where the displayed blocks are normal tensors. If $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every positive chain length N, then the bases of normal tensors have the same size. After relabelling the blocks of B, each matched pair satisfies

$$B^i_j = \zeta_j X_j A^i_j X_j^{-1}, \tag{1.2}$$

for some invertible matrix X_j and nonzero scalar ζ_j, and the equal case compares the coefficient power sums attached to the relabelled bases. The block gauges (1.2) and the weight comparison then combine into a single gauge conjugacy of the two weighted block-diagonal tensors (1.1). For a single injective block this reduces to the familiar conclusion $B^i = XA^iX^{-1}$ for all physical indices i, the single-block specialization of (1.2).

Chapters 2–11 carry out this argument. Chapter 2 fixes the matrix product vector notation, word evaluations, transfer maps, gauge equivalence, injectivity, blocking, and site-periodic tensor families. Chapter 3 proves the single-block theorem by trace pairings, linear extension, simplicity of matrix algebras, and Skolem–Noether. Chapters 4, 5, 6, and 7 develop the positive-map, multiplicative-domain, Perron–Frobenius, and overlap-decay tools used to control normal blocks. Chapters 8 and 9 supply the blocking, primitivity, irreducibility, and canonical-form reductions. Chapter 9 includes the invariant-projection splitting. Chapter 10 supplies bases of normal tensors and the power-sum comparison. The after-blocking reductions in Section 9.6.2 give the common primitive block decompositions used immediately in Chapter 11, which records the conditional theorem statements.

The chapters after Chapter 11 collect applications, alternative formulations, and later tensor-network material. Chapter 12 uses the Fundamental Theorem to obtain virtual symmetry gauges, projective bond representations, and string-order criteria; its Kraus-mixing step appeals only to the later channel-representation chapter, not to any additional input for the Fundamental Theorem. Parent Hamiltonians, correlations, examples, channel representations, normal forms, quantum dynamical semigroups, operator convexity, and entropy are then collected. A later draft treats the direct Fundamental Theorem for periodically repeated tensors in irreducible form, together with the compatibility theorems for physical-index rotation (Definition 12.1 in Chapter 12). The matrix-product-operator, periodic, algebraic, and PEPS draft chapters remain in the repository but are omitted from the generated blueprint while their statements are being stabilized. These later topics use the Fundamental Theorem, or develop later finite-dimensional channel theory, rather than enter its proof.

Chapter 2

Matrix Product Vectors

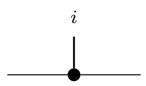
Fix a physical dimension d and a bond dimension D. This chapter studies the tensors that generate matrix product vector (MPV) families. All tensors here are translation-invariant with periodic boundary conditions (PBC).

2.1 Basic definitions

Definition 2.1 (MPS tensor). Lean ✓ Stmt A (translation-invariant, PBC) *MPS tensor* with physical dimension d and bond dimension D is a collection of matrices

$$\{A^i\}_{i=0}^{d-1}, \qquad A^i \in M_D(\mathbb{C}),$$

indexed by a physical index $i \in \{0, \dots, d-1\}$. Such a tensor defines an MPV family. Diagrammatically,

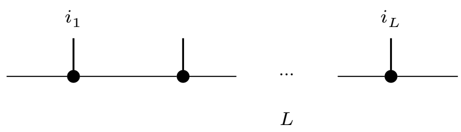


The black node denotes the tensor A, the horizontal legs are virtual, and the upper leg is the physical index i.

Definition 2.2 (Word evaluation). Lean ✓ Stmt Given a word $w = (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$, the *word evaluation* is the matrix product

$$A^w := A^{i_1} A^{i_2} \cdots A^{i_L} \in M_D(\mathbb{C}),$$

with the convention that the empty word evaluates to the identity: $A^\emptyset = \mathbb{1}_D$. In tensor-network notation,



in which each black node denotes the same local tensor A, the virtual legs remain open, and the physical legs are labelled by the word $(i_1, \dots, i_L)$.

Lemma 2.3 (Multiplicativity of word evaluation). Lean ✓ Stmt *For any words* w_1, w_2, $A^{w_1 \cdot w_2} = A^{w_1} A^{w_2}$.

Proof. ✓ Proof Immediate from the definition of A^w. □

Definition 2.4 (Matrix product vector). Lean ✓ Stmt The *matrix product vector* (MPV) of a tensor A at system size N is the vector

$$|V^{(N)}(A)\rangle = \sum_{i_1,\dots,i_N=0}^{d-1} \operatorname{tr}(A^{i_1} A^{i_2} \cdots A^{i_N}) |i_1, \dots, i_N\rangle \in (\mathbb{C}^d)^{\otimes N}.$$

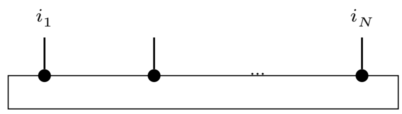
Equivalently, for a configuration $\sigma = (i_1, \dots, i_N) \in \{0, \dots, d-1\}^N$, we write

$$V^{(N)}(A)_\sigma := \operatorname{tr}(A^{i_1} A^{i_2} \cdots A^{i_N}).$$

The coefficient function $\sigma \mapsto V^{(N)}(A)_\sigma$ gives the components; the displayed ket is the corresponding vector in $(\mathbb{C}^d)^{\otimes N}$. For a general word w, we also write

$$c_w(A) := \operatorname{tr}(A^w).$$

The *MPV family* generated by A is the collection $\mathcal{V}(A) = \{|V^{(N)}(A)\rangle\}_{N \geq 1}$. The coefficient $V^{(N)}(A)_\sigma$ is the periodic contraction

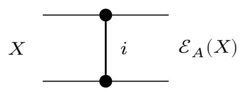


of N copies of the local tensor A, with the outer virtual legs closed by the trace.

Definition 2.5 (Transfer map). Lean ✓ Stmt The *transfer map* associated to a tensor A is the linear map $\mathcal{E}_A : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ defined by

$$\mathcal{E}_A(X) = \sum_{i=0}^{d-1} A^i X (A^i)^\dagger.$$

Diagrammatically, the transfer map is the double-layer contraction



in which the upper node denotes A, the lower node denotes $A^\dagger$, and the physical index is summed over between them.

Remark 2.6. ✓ Stmt The transfer map is automatically positive: if $X \geq 0$ then $\mathcal{E}_A(X) \geq 0$, because each summand $A^i X (A^i)^\dagger$ is positive semidefinite. The abstract positive-map framework appears in Chapter 4.

2.2 Site-periodic tensor families

Definition 2.7 (Site-periodic tensor family). Lean ✓ Stmt For a fixed period m, a *site-periodic tensor family* is a family of local tensors indexed by $\{0, \dots, m-1\}$, with local physical dimension d and bond dimension D.

Definition 2.8 (Periodic same-state and gauge-equivalence relations). Lean Lean ✓ Stmt For site-periodic tensors A, B at fixed period m:

- equality of the induced site-periodic states;
- cyclic gauge equivalence.

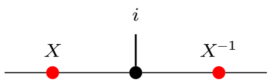
Lemma 2.9 (Equivalence structures for periodic relations). *Lean* *Lean* *✓ Stmt* *Both periodic relations are equivalence relations (reflexive, symmetric, transitive).*

2.3 Gauge equivalence and same MPV

Definition 2.10 (Gauge equivalence). Lean ✓ Stmt Two tensors A and B of the same bond dimension D are *gauge equivalent* if there exists an invertible matrix $X \in \mathrm{GL}_D(\mathbb{C})$ such that, for every $i \in \{0, \dots, d-1\}$,

$$B^i = XA^iX^{-1}.$$

In tensor-network notation,



the black node denotes the tensor named in the surrounding formula, the upper leg is the physical index i, and the two red side nodes denote the gauge matrices acting on the virtual legs.

Definition 2.11 (Same MPV). Lean ✓ Stmt Two tensors A and B (of the same bond dimension) *generate the same MPV family* if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every system size N.

Definition 2.12 (Same MPV - different bond dimensions). Lean ✓ Stmt For tensors A and B of possibly different bond dimensions D_1 and D_2, we write $\mathcal{V}(A) = \mathcal{V}(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every N.

Definition 2.13 (Positive-length same MPV - different bond dimensions). Lean ✓ Stmt For tensors A and B of possibly different bond dimensions, we write $\mathcal{V}^+(A) = \mathcal{V}^+(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every $N > 0$.

Lemma 2.14 (Length-zero MPV coefficient). *Lean* *✓ Stmt* *For any MPS tensor A with bond dimension D, the length-zero MPV coefficient is D, the trace of the identity on the bond space.*

Lemma 2.15 (Positive-length equality with equal bond dimensions). *Lean* *✓ Stmt* *If two tensors have the same positive-length MPV family and the same bond dimension, then they have the same MPV family at every length.*

Remark 2.16. ✓ Stmt Definition 2.11 is the same-bond-dimension specialization of Definition 2.12. We keep both because later results use both the same-dimension and different-dimension forms.

Definition 2.17 (Proportional MPV). Lean ✓ Stmt Two tensors A and B (of possibly different bond dimensions) generate weakly proportional MPV families if for every N there exists $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$.

Definition 2.18 (Nonzero proportional MPV). Lean ✓ Stmt Two tensors A and B (of possibly different bond dimensions) generate nonzero proportional MPV families if for every N there is a nonzero scalar $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$. This is the projective reading of the proportionality hypothesis in [CPGSV16, Theorem 2.10].

Definition 2.19 (Eventually nonzero proportional MPV). Lean ✓ Stmt Two tensors A and B are eventually nonzero proportional if, for all sufficiently large N, there is a nonzero scalar $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$. This auxiliary form is used with the eventual linear-independence statement Lemma 10.4.

Lemma 2.20 (Elementary nonzero proportionality rules). Lean Lean Lean Lean ✓ Stmt *Nonzero proportionality implies weak proportionality and eventual nonzero proportionality, is symmetric, and equality of MPV families is the special case with scalar* 1.

Proof. ✓ Proof Forgetting the nonvanishing condition gives weak proportionality. The eventual form follows by viewing an all-length statement as an eventual one. Symmetry follows by inverting the scalar c_N at each length, and equality is the case $c_N = 1$. □

Lemma 2.21 (Scaling of word evaluation). Lean ✓ Stmt *Let ζA denote the tensor with matrices $\{\zeta A^i\}_{i=0}^{d-1}$. Then for any word w,*

$$(\zeta A)^w = \zeta^{|w|} A^w,$$

where $|w|$ is the length of w.

Proof. ✓ Proof Induction on w: the base case gives $\zeta^0 = 1$, and each step picks up one factor of ζ. □

Lemma 2.22 (Scaling of MPV components). Lean ✓ Stmt *Let ζA denote the tensor with matrices $\{\zeta A^i\}_{i=0}^{d-1}$. Then for every N and every configuration $\sigma = (i_1, \dots, i_N)$,*

$$V^{(N)}(\zeta A)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

Proof. ✓ Proof Apply Lemma 2.21 to the word $w = (i_1, \dots, i_N)$ of length N, then take the trace; the scalar ζ^N pulls out by linearity. □

Definition 2.23 (Gauge-phase equivalence). Lean Lean ✓ Stmt Two tensors A and B are *gauge-phase equivalent* if there exist $X \in \mathrm{GL}_D(\mathbb{C})$ and $\zeta \in \mathbb{C} \setminus \{0\}$ such that, for every physical index i,

$$B^i = \zeta X A^i X^{-1}.$$

Proof. ✓ Proof Gauge equivalence is the special case $\zeta = 1$. □

Remark 2.24. ✓ Stmt After spectral-radius normalization one may restrict ζ to a unit-modulus phase $e^{i\phi}$. We allow an arbitrary nonzero scalar because later canonical-form statements keep the block-scaling factors explicit.

Lemma 2.25 (Gauge covariance of word evaluation). Lean ✓ Stmt *If $B^i = X A^i X^{-1}$ for all i, then for every word w,*

$$B^w = X A^w X^{-1}.$$

Proof. ✓ Proof Induction on the word length, using $XX^{-1} = \mathbb{1}$. □

Theorem 2.26 (Gauge invariance of MPVs). Lean ✓ Stmt *If A and B are gauge equivalent, then they generate the same MPV family.*

Proof. ✓ Proof Let $B^i = XA^iX^{-1}$. By Lemma 2.25, $B^w = XA^wX^{-1}$ for every word w. Then $\mathrm{tr}(B^w) = \mathrm{tr}(XA^wX^{-1}) = \mathrm{tr}(A^w)$ by cyclicity of the trace. □

Theorem 2.27 (Gauge-phase scaling of MPVs). Lean ✓ Stmt *If, for every physical index i,*

$$B^i = \zeta XA^iX^{-1},$$

then for every system size N and configuration σ one has

$$V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

Proof. ✓ Proof Combine gauge covariance of word evaluation with scalar scaling, then take traces. □

2.4 Injectivity and normality

Definition 2.28 (Injectivity). Lean ✓ Stmt A tensor A is *injective* if its matrices span the full matrix algebra: $\mathrm{span}_{\mathbb{C}}\{A^i : i = 0, \dots, d-1\} = M_D(\mathbb{C})$.

Definition 2.29 (L-block injectivity). Lean ✓ Stmt A tensor A is *L-block injective* if $\mathrm{span}_{\mathbb{C}}\{A^{i_1} \cdots A^{i_L} : (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L\} = M_D(\mathbb{C})$. An injective tensor is 1-block injective.

Lemma 2.30 (One-site injectivity gives one-block injectivity). Lean ✓ Stmt *If the one-site matrices $\{A^i\}_{i=0}^{d-1}$ span $M_D(\mathbb{C})$, then the length-one word products span $M_D(\mathbb{C})$, so A is 1-block injective.*

Proof. ✓ Proof The length-one words are exactly the letters $i \in \{0, \dots, d-1\}$, and evaluating a length-one word gives the matrix A^i. Thus the length-one word span is the same span as in one-site injectivity. □

Definition 2.31 (Normal tensor). Lean ✓ Stmt A tensor A is *normal* [CPGSV21] if there exists a finite L_0 such that A is L_0-block injective.

Remark 2.32. ✓ Stmt The algebraic characterization of normality in Definition 2.31 — eventual full matrix span, equivalently eventual block injectivity — is equivalent to the spectral characterization used in [CPGSV21, Definition IV.1]: after TP normalization, the transfer map $\mathcal{E}_A$ is a primitive channel, equivalently irreducible with trivial peripheral spectrum. This equivalence is proved in [SPGWC10, Proposition 3]. In what follows, the directions of the equivalence are supplied by Lemma 8.66, Theorem 8.75 for primitive normalized tensors with positive-definite fixed point, the Burnside route through Theorem 8.82 for the aperiodic algebraic criterion, and Theorem 9.31. In particular, every subsequent use of "normal" here refers to this single notion; Definition 9.26 employs the spectral formulation, but it describes the same class of tensors.

2.5 Canonical form

Definition 2.33 (Block-injective canonical form). Lean ✓ Stmt Here, a *block-injective canonical form* for a tensor consists of:

1. a number of blocks $r \geq 1$,
2. bond dimensions $D_1, \dots, D_r$,
3. injective tensors $\{A_k^i\}_{i=0}^{d-1}$ with $A_k^i \in M_{D_k}(\mathbb{C})$ for each block $k \in \{1, \dots, r\}$,
4. scaling factors $\mu_1, \dots, \mu_r \in \mathbb{C}$.

The associated tensor is the block-diagonal tensor

$$A^i = \bigoplus_{k=1}^{r} \mu_k A_k^i.$$

This block-injective formulation of the canonical form is used in [PGVWC07]; see also [CPGSV21] for the normal canonical form.

Remark 2.34. ✓ Stmt Here we use the block-injective canonical form: each block is injective in the sense of Definition 2.28. This is stronger than the NT-block canonical form of [CPGSV21], where the blocks are only normal in the spectral sense. Chapter 9 treats the normal canonical form needed for the blocking-based multi-block theory, while the periodic irreducible-form extension is kept separate. This section isolates the stronger block-injective form used in the injective and block-injective arguments.

Definition 2.35 (Block projection). Lean ✓ Stmt For a finite direct sum $\bigoplus_k \mathbb{C}^{D_k}$, the block projection P_j is the matrix that is the identity on the j-th summand and zero on all other summands.

Definition 2.36 (Block-diagonal matrix). Lean ✓ Stmt A matrix on $\bigoplus_k \mathbb{C}^{D_k}$ is block diagonal if it is a direct sum of square matrices, one on each summand.

Lemma 2.37 (Off-block characterization). *Lean ✓ Stmt A matrix on $\bigoplus_k \mathbb{C}^{D_k}$ is block diagonal if and only if every entry from one summand to a distinct summand is zero.*

Proof. ✓ Proof The forward direction is immediate from the definition of a direct sum of matrices. Conversely, if all off-block entries vanish, the diagonal blocks reconstruct the original matrix. □

Lemma 2.38 (Commutation with block projections). *Lean ✓ Stmt If a matrix on $\bigoplus_k \mathbb{C}^{D_k}$ commutes with every block projection P_j, then it is block diagonal.*

Proof. ✓ Proof Compare the (i, j) block of the identity $XP_j = P_jX$ for $i \neq j$. The right multiplication by P_j keeps that block, while the left multiplication by P_j kills it. Hence every off-block entry is zero, and Lemma 2.37 applies. □

Definition 2.39 (Block-diagonal tensor from blocks). Lean ✓ Stmt Given r blocks with bond dimensions $(D_k)_{k=1}^r$, per-block tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, the associated *block-diagonal tensor* is the tensor of total bond dimension $D = \sum_{k=1}^r D_k$ defined by

$$A^i = \bigoplus_{k=1}^{r} \mu_k A_k^i.$$

Lemma 2.40 (Word evaluation of a block-diagonal tensor). Lean ✓ Stmt *For a word w, the word evaluation of a block-diagonal tensor remains block diagonal:*

$$A^w = \bigoplus_k \mu_k^{|w|} A_k^w.$$

Proof. ✓ Proof Induct on the word and multiply block-diagonal matrices block by block; every letter contributes one scalar factor μ_k on the k-th block. □

Theorem 2.41 (MPV decomposition). Lean ✓ Stmt *The MPV of a block-diagonal tensor decomposes as*

$$|V^{(N)}(A)\rangle = \sum_{k=1}^{r} \mu_k^N |V^{(N)}(A_k)\rangle.$$

Equivalently, for each configuration σ one has

$$V^{(N)}(A)_\sigma = \sum_{k=1}^{r} \mu_k^N V^{(N)}(A_k)_\sigma.$$

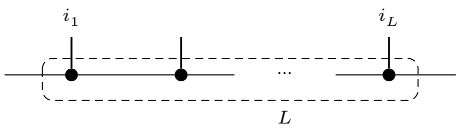
Proof. ✓ Proof By Lemma 2.40, the full word evaluation is block diagonal with blocks $\mu_k^N A_k^w$ for a word w of length N. Taking the trace of a block-diagonal matrix gives the sum of the block traces: $\operatorname{tr}(A^w) = \sum_k \mu_k^N \operatorname{tr}(A_k^w)$. □

2.6 Blocking

Definition 2.42 (Blocked tensor). Lean ✓ Stmt Given a tensor A and a blocking length L, the L-*blocked tensor* is the tensor with physical index set $\{0, \dots, d-1\}^L$ (hence physical dimension d^L) and the same bond dimension D, whose matrices are indexed by words $(i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$:

$$(A^{[L]})^{(i_1,\dots,i_L)} = A^{i_1} A^{i_2} \cdots A^{i_L}.$$

Blocking coarse-grains L neighbouring sites into one tensor: the inner virtual bonds are contracted, the L physical legs merge into a single composite index, and the two outer bonds remain as the bond of $A^{[L]}$.



Lemma 2.43 (Blocked word evaluation). Lean ✓ Stmt *If $w = (J_1, \dots, J_m)$ is a word in the blocked alphabet, where each J_k is an L-tuple in $\{0, \dots, d-1\}^L$, then concatenating the block words gives a word $\widetilde{w}$ of length mL in the original alphabet and one has*

$$(A^{[L]})^w = A^{\widetilde{w}}.$$

Proof. ✓ Proof Induct on the blocked word w and split off the first block word. Lemma 2.3 turns concatenation of blocks into multiplication of the corresponding evaluations. □

Lemma 2.44 (Blocking preserves MPV coefficients). Lean ✓ Stmt *For each basis state $\sigma = (\sigma_1, \dots, \sigma_N)$ of the blocked system, where each σ_k is an L-tuple in $\{0, \dots, d-1\}^L$, there exists a basis state $\tilde{\sigma} = (\tilde{\sigma}_1, \dots, \tilde{\sigma}_{NL})$ of the original system such that*

$$V^{(N)}(A^{[L]})_\sigma = V^{(NL)}(A)_{\tilde{\sigma}}.$$

Proof. ✓ Proof Concatenate the block words and apply the blocked word evaluation lemma (Lemma 2.43). □

Theorem 2.45 (Blocking preserves same MPV). Lean ✓ Stmt *If A and B generate the same MPV family, then for any blocking length L, the blocked tensors $A^{[L]}$ and $B^{[L]}$ also generate the same MPV family.*

Proof. ✓ Proof By Lemma 2.44, each MPV coefficient of the blocked tensor $A^{[L]}$ equals a coefficient of A at the corresponding concatenated configuration. The same holds for $B^{[L]}$ (with the same concatenated configuration). Since A and B agree on all configurations, so do $A^{[L]}$ and $B^{[L]}$. □

2.7 MPV overlap

Definition 2.46 (MPV as Hilbert-space vector). Lean ✓ Stmt We regard $|V^{(N)}(A)\rangle$ as an element of $\mathbb{C}^{d^N}$ indexed by configurations $\sigma \in \{0, \dots, d-1\}^N$, with σ-component $V^{(N)}(A)_\sigma$ as in Definition 2.4. This allows us to use the standard Euclidean inner product on $\mathbb{C}^{d^N}$.

Definition 2.47 (MPV inner product). Lean ✓ Stmt The *MPV inner product* between two tensors A and B at system size N is

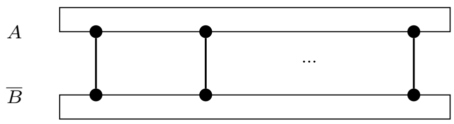
$$\langle V^{(N)}(A) | V^{(N)}(B) \rangle := \sum_{\sigma \in \{0,\dots,d-1\}^N} \overline{V^{(N)}(A)_\sigma} V^{(N)}(B)_\sigma,$$

i.e. the Euclidean inner product on $\mathbb{C}^{d^N}$ (conjugate-linear in the first argument).

Definition 2.48 (MPV overlap). Lean ✓ Stmt The *MPV overlap* between two tensors A (of bond dimension D_1) and B (of bond dimension D_2) at system size N is

$$O_{AB}(N) := \sum_{\sigma \in \{0,\dots,d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}.$$

Diagrammatically, the overlap contracts the MPV ring of A against the conjugate ring of B along their shared physical indices, both virtual rings closed by the trace:



This is the bilinear overlap (without complex conjugation) used later. By Lemma 2.49, it is the complex conjugate of the Hilbert-space inner product from Definition 2.47.

Lemma 2.49 (Overlap equals conjugate inner product). Lean ✓ Stmt $O_{AB}(N) = \overline{\langle V^{(N)}(A) | V^{(N)}(B) \rangle}$.

Proof. ✓ Proof The overlap sums $V_\sigma \overline{W_\sigma}$ while the inner product sums $\overline{V_\sigma} W_\sigma$. Complex conjugation swaps these. □

Lemma 2.50 (Overlap determined by positive-length MPV). Lean ✓ Stmt *If $V^{(N)}(A)_\sigma = V^{(N)}(A')_\sigma$ and $V^{(N)}(B)_\sigma = V^{(N)}(B')_\sigma$ for all $N > 0$ and all σ, then $O_{AB}(N) = O_{A'B'}(N)$ for all positive N.*

Proof. ✓ Proof Direct pointwise rewriting: if the MPV coefficients agree for all positive chain lengths, the overlap sums are identical. □

Chapter 3

The Single-Block Fundamental Theorem

An injective MPS tensor that generates the same matrix-product vectors as a second tensor of equal bond dimension is conjugate to it [CPGSV21, Corollary IV.5]: there is an invertible X with $B^i = XA^iX^{-1}$ for every physical index i. The argument is purely algebraic. Equality of the matrix-product vectors gives $\operatorname{tr}(A^w) = \operatorname{tr}(B^w)$ for every word w; the length-two and length-three traces produce a linear map T with $T(A^i) = B^i$ that is multiplicative and nonzero; a nonzero multiplicative endomorphism of the simple algebra $M_D(\mathbb{C})$ is a unital automorphism; and by Skolem–Noether every automorphism of $M_D(\mathbb{C})$ is conjugation by an invertible X. Evaluating at A^i closes the proof. The same conjugation, one normal block at a time, is the central step of the multi-block theorem in Chapter 11.

3.1 The multiplicative linear extension

Theorem 3.1 (Nondegeneracy of the trace pairing). Lean ✓ Stmt *For $M \in M_D(\mathbb{C})$, $\operatorname{tr}(MN) = 0$ for all $N \in M_D(\mathbb{C})$ if and only if $M = 0$.*

Proof. ✓ Proof Testing against the matrix unit E_{ji} gives $\operatorname{tr}(E_{ji}M) = M_{ij}$, so $\operatorname{tr}(MN) = 0$ for all N forces every entry of M to vanish. □

Lemma 3.2 (Same matrix-product vectors give trace agreement). Lean ✓ Stmt *If A and B generate the same matrix-product vectors, then $\operatorname{tr}(A^w) = \operatorname{tr}(B^w)$ for every word w.*

Proof. ✓ Proof The configuration $\sigma = w$ gives $V^{(|w|)}(A)_w = \operatorname{tr}(A^w)$. Equality of the matrix-product vectors gives $V^{(N)}(A) = V^{(N)}(B)$ for all N, so $\operatorname{tr}(A^w) = \operatorname{tr}(B^w)$. □

The trace conditions constrain every product of the matrices A^i. To bring the algebra structure of $M_D(\mathbb{C})$ to bear, record how a matrix pairs against the family $\{A^i\}$ under the trace.

Definition 3.3 (Trace pairing map). Lean ✓ Stmt For a tensor A, the trace pairing map $\Phi_A : M_D(\mathbb{C}) \to \mathbb{C}^d$ is

$$\Phi_A(M)_i = \operatorname{tr}(MA^i).$$

Theorem 3.4 (The trace pairing map is injective for injective tensors). Lean ✓ Stmt *If A is injective, then $\ker \Phi_A = \{0\}$.*

Proof. ✓ Proof If $\Phi_A(M) = 0$, then $\operatorname{tr}(MA^i) = 0$ for every i. The $\{A^i\}$ span $M_D(\mathbb{C})$, so $\operatorname{tr}(MN) = 0$ for all N, and $M = 0$ by Theorem 3.1. □

Lemma 3.5 (Injectivity transfers across a range inclusion). Lean ✓ Stmt *Let $\Phi, \Psi : V \to W$ be $\mathbb{C}$-linear maps with V finite-dimensional. If $\ker \Phi = \{0\}$ and $\operatorname{range} \Phi \subseteq \operatorname{range} \Psi$, then $\ker \Psi = \{0\}$.*

Proof. ✓ Proof By rank–nullity, $\ker \Phi = \{0\}$ gives $\dim \operatorname{range} \Phi = \dim V$, and the range inclusion gives

$$\dim V = \dim \operatorname{range} \Phi \le \dim \operatorname{range} \Psi \le \dim V,$$

so $\dim \operatorname{range} \Psi = \dim V$ and $\ker \Psi = \{0\}$. □

Theorem 3.6 (The linear extension exists and is unique). Lean ✓ Stmt *If A is injective and A, B generate the same matrix-product vectors, then there is a unique linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ with $T(A^i) = B^i$ for all i.*

Proof. ✓ Proof Length-two trace agreement gives $\Phi_A(A^i) = \Phi_B(B^i)$ for every i. The $\{A^i\}$ span $M_D(\mathbb{C})$, so $\operatorname{range} \Phi_A \subseteq \operatorname{range} \Phi_B$. With $\ker \Phi_A = \{0\}$ (Theorem 3.4), Lemma 3.5 gives $\ker \Phi_B = \{0\}$. Let g be a left inverse of Φ_B and set $T = g \circ \Phi_A$; then

$$T(A^i) = g(\Phi_A(A^i)) = g(\Phi_B(B^i)) = B^i.$$

Uniqueness holds because the $\{A^i\}$ span $M_D(\mathbb{C})$. □

Theorem 3.7 (The linear extension is multiplicative). Lean ✓ Stmt *Under the hypotheses of Theorem 3.6, the map T satisfies $T(MN) = T(M)T(N)$ for all $M, N \in M_D(\mathbb{C})$.*

Proof. ✓ Proof Length-two trace agreement gives $\Phi_B(T(A^i)) = \Phi_B(B^i) = \Phi_A(A^i)$; since the $\{A^i\}$ span $M_D(\mathbb{C})$ this extends to $\Phi_B \circ T = \Phi_A$, so $\operatorname{range} \Phi_A = \operatorname{range}(\Phi_B \circ T) \subseteq \operatorname{range} \Phi_B$. With $\ker \Phi_A = \{0\}$ (Theorem 3.4), Lemma 3.5 makes Φ_B injective. Length-three trace agreement gives, for all i, j, k,

$$\Phi_B(T(A^iA^j))_k = \Phi_A(A^iA^j)_k = \operatorname{tr}(A^iA^jA^k) = \operatorname{tr}(B^iB^jB^k) = \Phi_B(B^iB^j)_k,$$

so injectivity of Φ_B yields $T(A^iA^j) = B^iB^j$. Extend in two stages: for fixed j, the maps $M \mapsto T(MA^j)$ and $M \mapsto T(M)B^j$ agree on the spanning $\{A^i\}$, hence $T(MA^j) = T(M)B^j$ for all M; then for fixed M, the maps $N \mapsto T(MN)$ and $N \mapsto T(M)T(N)$ agree on the spanning $\{A^j\}$, hence $T(MN) = T(M)T(N)$ for all N. □

3.2 Inner automorphism and the single-block theorem

The rigidity of the full matrix algebra now finishes the proof. A nonzero multiplicative map on $M_D(\mathbb{C})$ is forced to be bijective (simplicity), unit-preserving (surjectivity), and conjugation by a single invertible matrix (Skolem–Noether). The next four results establish these steps and the nonvanishing of T, which the final theorem combines into the gauge X.

Theorem 3.8 (A nonzero multiplicative endomorphism of $M_D(\mathbb{C})$ is bijective). Lean ✓ Stmt *A nonzero multiplicative $\mathbb{C}$-linear endomorphism of $M_D(\mathbb{C})$ is bijective.*

Proof. ✓ Proof The kernel of a multiplicative linear map is a two-sided ideal. The algebra $M_D(\mathbb{C})$ is simple, so the kernel is $\{0\}$ or all of $M_D(\mathbb{C})$; the latter forces the map to be zero. Hence the map is injective, and surjective by finite-dimensionality. □

Lemma 3.9 (A surjective multiplicative map preserves the unit). Lean ✓ Stmt *If $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is a surjective multiplicative $\mathbb{C}$-linear map, then $T(\mathbb{1}) = \mathbb{1}$, so T is a $\mathbb{C}$-algebra homomorphism.*

Proof. ✓ Proof Choose X with $T(X) = \mathbb{1}$. Then $T(\mathbb{1}) = T(X)T(\mathbb{1}) = T(X\mathbb{1}) = T(X) = \mathbb{1}$. □

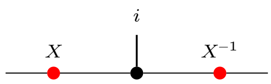
Lemma 3.10 (An injective tensor forces a nonzero partner). Lean ✓ Stmt *Let $D \geq 1$. If A is injective and A, B generate the same matrix-product vectors, then the B^i are not all zero.*

Proof. ✓ Proof If $B^i = 0$ for every i, then Lemma 3.2 on length-one words gives $\operatorname{tr}(A^i) = 0$ for all i. The $\{A^i\}$ span $M_D(\mathbb{C})$, so the trace functional vanishes identically, contradicting $\operatorname{tr}(\mathbb{1}) = D \neq 0$. □

Theorem 3.11 (Skolem–Noether). Lean ✓ Stmt *Every $\mathbb{C}$-algebra automorphism f of $M_D(\mathbb{C})$ is inner: there is $X \in \mathrm{GL}_D(\mathbb{C})$ with $f(M) = XMX^{-1}$ for all M.*

Proof. ✓ Proof Under the identification $M_D(\mathbb{C}) \cong \operatorname{End}_{\mathbb{C}}(\mathbb{C}^D)$, every algebra automorphism of $\operatorname{End}_{\mathbb{C}}(\mathbb{C}^D)$ is conjugation by an invertible linear map. Translating back gives the invertible X. □

Theorem 3.12 (Single-block Fundamental Theorem). Lean ✓ Stmt *Let A be an injective MPS tensor and B a tensor of the same bond dimension. If A and B generate the same matrix-product vectors, then there is $X \in \mathrm{GL}_D(\mathbb{C})$ with $B^i = XA^iX^{-1}$ for all i. In tensor-network notation the gauge conjugates the local tensor on its virtual legs, with the physical index i on the upper leg:*



Proof. ✓ Proof Theorem 3.6 gives the unique linear T with $T(A^i) = B^i$, and Theorem 3.7 makes it multiplicative. It is nonzero: otherwise $B^i = 0$ for all i, against Lemma 3.10. By Theorem 3.8, T is bijective, and by Lemma 3.9 it fixes $\mathbb{1}$, so T is a $\mathbb{C}$-algebra automorphism. Theorem 3.11 gives $X \in \mathrm{GL}_D(\mathbb{C})$ with $T(M) = XMX^{-1}$; evaluating at $M = A^i$ gives $B^i = XA^iX^{-1}$. □

Chapter 4

Quantum Channels and Positive Maps

This chapter develops the theory of positive maps and quantum channels on matrix algebras, following [Wol12].

4.1 Positive maps and channels

Definition 4.1 (Positive map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *positive* if $E(X) \geq 0$ whenever $X \geq 0$, where $X \geq 0$ means that X is positive semidefinite.

Remark 4.2. ✓ Stmt We use the Loewner order on matrices: $X \leq Y$ means $Y - X \geq 0$.

Definition 4.3 (Associated positive linear map). Lean ✓ Stmt A positive matrix map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ determines the same linear map regarded as a positive linear map: if $A \leq B$, then $E(A) \leq E(B)$.

Definition 4.4 (Trace-preserving map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *trace-preserving* if $\mathrm{tr}(E(X)) = \mathrm{tr}(X)$ for all X.

Definition 4.5 (Completely positive map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *completely positive* (CP) if it admits a *Kraus representation*: there exist operators $\{K_i\}_{i=0}^{r-1}$ with $K_i \in M_D(\mathbb{C})$ such that, for every $X \in M_D(\mathbb{C})$,

$$E(X) = \sum_{i=0}^{r-1} K_i X K_i^\dagger.$$

By Choi's theorem, this is equivalent to $E \otimes \mathrm{id}_n$ being positive for all n.

Theorem 4.6 (Entrywise complete positivity from a Kraus representation). *Lean* *✓ Stmt* *Let $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be completely positive in the Kraus sense. For every k and every positive block matrix $M \in M_k(M_D(\mathbb{C}))$, the entrywise image*

$$(E(M_{ab}))_{a,b}$$

is again positive in $M_k(M_D(\mathbb{C}))$.

Proof. ✓ Proof Write $E(X) = \sum_i K_i X K_i^\dagger$. For each i, let d_i be the block-diagonal matrix with K_i on every diagonal block. Then

$$(E(M_{ab}))_{a,b} = \sum_i d_i M d_i^\dagger.$$

Each term on the right is positive because conjugation preserves positivity, and a finite sum of positive matrices is positive. □

Theorem 4.7 (Kraus maps as abstract completely positive maps). Lean ✓ Stmt *Every Kraus-represented completely positive map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ determines a completely positive map between the matrix C^*-algebras in the entrywise positivity sense.*

Proof. ✓ Proof The defining entrywise positivity condition is exactly Theorem 4.6. □

Theorem 4.8 (CP maps are positive). Lean ✓ Stmt *Every completely positive map is positive.*

Proof. ✓ Proof If $X \geq 0$, then each summand $K_i X K_i^\dagger \geq 0$ (conjugation preserves positive semidefiniteness), so the sum is PSD. □

Theorem 4.9 (Channels are positive). Lean ✓ Stmt *Every quantum channel is positive.*

Proof. ✓ Proof A quantum channel is completely positive by definition, so Theorem 4.8 applies. □

Theorem 4.10 (Positive maps preserve Hermiticity). Lean ✓ Stmt *If E is positive and $X = X^\dagger$, then $E(X) = E(X)^\dagger$.*

Proof. ✓ Proof Regard E as the associated positive linear map (Definition 4.3). For positive linear maps between matrix C^*-algebras, $E(Y^*) = E(Y)^*$. Since $X = X^\dagger$ means $X = X^*$, one obtains $E(X)^\dagger = E(X)^* = E(X^*) = E(X)$. □

Definition 4.11 (Quantum channel). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is a *quantum channel* if it is both completely positive and trace-preserving (CPTP), following [Wol12].

Definition 4.12 (Density matrices). Lean Lean ✓ Stmt The set of *density matrices* in $M_D(\mathbb{C})$ is

$$\mathcal{D}_D = \{\rho \in M_D(\mathbb{C}) : \rho \geq 0, \operatorname{tr}(\rho) = 1\}.$$

Theorem 4.13 (Density matrices are compact). Lean ✓ Stmt *$\mathcal{D}_D$ is compact (in the entry-wise topology on $M_D(\mathbb{C})$).*

Proof. ✓ Proof The PSD cone is closed (preimage of the closed non-negative cone under continuous quadratic forms), and if $\rho \geq 0$ with $\operatorname{tr}(\rho) = 1$ then each entry satisfies $|\rho_{ij}|^2 \leq \rho_{ii}\rho_{jj} \leq 1$: the diagonal entries are non-negative and sum to 1, and the off-diagonal bound is the PSD Cauchy–Schwarz inequality. Hence the matrix entries are uniformly bounded, and Heine–Borel gives compactness. □

Theorem 4.14 (Density matrices are convex). Lean ✓ Stmt *$\mathcal{D}_D$ is convex.*

Proof. ✓ Proof Convex combination of PSD matrices is PSD, and trace is linear. □

Theorem 4.15 (Density matrices are nonempty). Lean ✓ Stmt *For $D \geq 1$, $\mathcal{D}_D \neq \emptyset$.*

Proof. ✓ Proof $\frac{1}{D}\mathbb{1}_D$ is a density matrix. □

Theorem 4.16 (Channels preserve density matrices). Lean ✓ Stmt *If E is a quantum channel, then $E(\mathcal{D}_D) \subseteq \mathcal{D}_D$.*

Proof. ✓ Proof Complete positivity implies positivity (Theorem 4.8), so $E(\rho) \geq 0$; trace preservation gives $\operatorname{tr}(E(\rho)) = \operatorname{tr}(\rho) = 1$. □

4.2 Irreducibility

Definition 4.17 (Orthogonal projection). Lean ✓ Stmt A matrix $P \in M_D(\mathbb{C})$ is an *orthogonal projection* if $P = P^\dagger$ and $P^2 = P$.

Theorem 4.18 (Orthogonal projections and involutive-algebra projections). Lean Lean ✓ Stmt *For a complex square matrix, the two descriptions $P = P^\dagger$, $P^2 = P$ and $P^* = P$, $P^2 = P$ are equivalent.*

Proof. ✓ Proof Since the involution on $M_D(\mathbb{C})$ is conjugate transpose, $P^* = P^\dagger$. Therefore

$$P^\dagger = P,\ P^2 = P \iff P^* = P,\ P^2 = P.$$

□

Theorem 4.19 (Complement of an orthogonal projection). Lean ✓ Stmt *If $P \in M_D(\mathbb{C})$ is an orthogonal projection, then $\mathbb{1} - P$ is an orthogonal projection.*

Proof. ✓ Proof In the involutive algebra $M_D(\mathbb{C})$, P satisfies $P^\dagger = P$ and $P^2 = P$. Hence

$$(\mathbb{1} - P)^\dagger = \mathbb{1} - P, \qquad (\mathbb{1} - P)^2 = \mathbb{1} - 2P + P^2 = \mathbb{1} - P.$$

Thus $\mathbb{1} - P$ is an orthogonal projection. □

Definition 4.20 (Irreducible map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *irreducible* if whenever P is an orthogonal projection satisfying $E(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$, then $P = 0$ or $P = \mathbb{1}$. This is [Wol12, Theorem 6.2(1)]. The definition applies to any linear map; complete positivity is not required.

Theorem 4.21 (Injectivity implies irreducibility). Lean ✓ Stmt *If A is an injective MPS tensor, then its transfer map $\mathcal{E}_A$ is irreducible.*

Proof. ✓ Proof If P is an invariant projection for $\mathcal{E}_A$, then $(\mathbb{1} - P)A^iP = 0$ for all i. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $(\mathbb{1} - P)MP = 0$ for all M, so $P = 0$ or $P = \mathbb{1}$. □

Theorem 4.22 (Transfer map is CP). Lean ✓ Stmt *The transfer map $\mathcal{E}_A(X) = \sum_i A^iX(A^i)^\dagger$ is completely positive.*

Proof. ✓ Proof It is already written in Kraus form, with Kraus operators $\{A^i\}$. □

Theorem 4.23 (Transfer map is a channel). Lean ✓ Stmt *If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, then the transfer map $\mathcal{E}_A(X) = \sum_i A^iX(A^i)^\dagger$ is a quantum channel (CPTP).*

Proof. ✓ Proof Complete positivity is the observation of Theorem 4.22. Trace preservation: $\mathrm{tr}(\mathcal{E}_A(X)) = \sum_i \mathrm{tr}(A^iX(A^i)^\dagger) = \mathrm{tr}((\sum_i (A^i)^\dagger A^i)X) = \mathrm{tr}(X)$ by cyclicity and the normalization hypothesis. □

4.3 Cesàro fixed-point theorem

Definition 4.24 (Cesàro mean). Lean ✓ Stmt The *Cesàro mean* of a linear map E at order N, applied to a starting matrix ρ_0, is

$$\sigma_N = \frac{1}{N}\sum_{n=0}^{N-1} E^n(\rho_0).$$

Theorem 4.25 (Cesàro telescope). Lean ✓ Stmt $E(\sigma_N) - \sigma_N = \frac{1}{N}(E^N(\rho_0) - \rho_0)$.

Proof. ✓ Proof Telescoping: $E(\sigma_N) = \frac{1}{N}\sum_{n=1}^{N} E^n(\rho_0)$ and subtracting σ_N cancels all but the first and last terms. □

Theorem 4.26 (Hermitian fixed-point decomposition). Lean ✓ Stmt *Let E be a quantum channel and $X = E(X)$ a Hermitian fixed point with decomposition $X = P_+ - P_-$ into orthogonal positive and negative parts ($P_\pm \geq 0$, $\mathrm{tr}(P_+P_-) = 0$). Then $E(P_+) = P_+$ and $E(P_-) = P_-$. This is [Wol12, Proposition 6.8].*

Proof. ✓ Proof Let Q be the support projection of P_+. Trace preservation and positivity give $\mathrm{tr}(QE(P_-)) = 0$ and $E(P_-) \geq 0$, so $QE(P_-) = 0$. The fixed-point equation $E(P_+) - E(P_-) = P_+ - P_-$ then identifies $E(P_\pm) = P_\pm$. □

Theorem 4.27 (Cesàro fixed-point theorem). Lean ✓ Stmt *Every quantum channel $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ (with $D \geq 1$) has a nonzero positive semidefinite fixed point: there exists $\rho \geq 0$, $\rho \neq 0$, with $E(\rho) = \rho$.*

Proof. ✓ Proof Start with any $\rho_0 \in \mathcal{D}_D$ (Theorem 4.15). Each Cesàro mean σ_N lies in $\mathcal{D}_D$ by applying channel preservation to every iterate, adding positive semidefinite matrices, rescaling by the positive average factor, and using trace linearity. By compactness (Theorem 4.13), extract a convergent subsequence $\sigma_{N_k} \to \rho$. The telescope identity (Theorem 4.25) gives $\|E(\sigma_N) - \sigma_N\| = \frac{1}{N}\|E^N(\rho_0) - \rho_0\| \to 0$, so $E(\rho) = \rho$ by continuity. In [Wol12, Theorem 6.11], existence is proved via Brouwer's fixed-point theorem; we use the Cesàro argument instead. □

4.4 Fixed-point projection

Definition 4.28 (Fixed-point projection). Lean ✓ Stmt Given a linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ and a fixed point ρ with $E(\rho) = \rho$ and $\mathrm{tr}(\rho) \neq 0$, the *fixed-point projection* is the rank-one map

$$P(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)}\rho.$$

This projects onto the span of ρ along the kernel of the trace functional. When the fixed point is unique up to scaling, this span is the full fixed-point space.

Theorem 4.29 (Power decomposition). Lean ✓ Stmt *If E is trace-preserving and $E(\rho) = \rho$, then for $n \geq 1$,*

$$E^n = P + (E - P)^n,$$

where P is the fixed-point projection.

Proof. ✓ Proof P is idempotent ($P^2 = P$), the fixed-point relation $E(\rho) = \rho$ gives $E \circ P = P$, and trace preservation gives $P \circ E = P$. These imply $(E - P) \circ P = 0$ and $P \circ (E - P) = 0$, so in the binomial expansion of $(P + (E - P))^n$ all cross terms vanish, leaving $E^n = P^n + (E - P)^n = P + (E - P)^n$. □

4.5 Peripheral spectrum and primitivity

Definition 4.30 (Peripheral spectrum). Lean ✓ Stmt The *peripheral spectrum* of a bounded linear operator T is the set of spectral values μ with $|\mu|$ equal to the spectral radius of T.

Definition 4.31 (Peripheral eigenvalues). Lean ✓ Stmt The *peripheral eigenvalues* of a linear map E on a finite-dimensional space are the eigenvalues λ on the unit circle: $|\lambda| = 1$. For a channel, the spectral radius is 1 [Wol12, Proposition 6.1], so these coincide with the eigenvalues of maximal modulus.

The condition $|\lambda| = 1$ is appropriate for channels, since the spectral radius of a channel is 1; for a general linear map with spectral radius $r \neq 1$ the condition would generalise to $|\lambda| = r$.

Remark 4.32 (Peripheral conventions for channels). ✓ Stmt The spectral-radius convention of Definition 4.30 is the usual bounded-operator notion. For channels the spectral radius is 1, so the channel arguments below use the unit-circle set of Definition 4.31.

Theorem 4.33 (1 is a peripheral eigenvalue). Lean ✓ Stmt *If E has a nonzero fixed point $\rho \neq 0$ with $E(\rho) = \rho$, then 1 is a peripheral eigenvalue of E.*

Proof. ✓ Proof ρ is an eigenvector with eigenvalue 1, and $|1| = 1$. □

Theorem 4.34 (Finiteness of peripheral eigenvalues). Lean ✓ Stmt *On a finite-dimensional space, the set of peripheral eigenvalues is finite.*

Proof. ✓ Proof Peripheral eigenvalues are a subset of all eigenvalues, which is a finite set in finite dimensions. □

Theorem 4.35 (Power-stable peripheral eigenvalues are roots of unity). Lean ✓ Stmt *Let E be a linear endomorphism on a finite-dimensional space. If μ is a peripheral eigenvalue of E and μ^n is an eigenvalue of E for every $n \geq 1$, then μ is a root of unity.*

Proof. ✓ Proof Since μ^n is an eigenvalue for every n, the pigeonhole principle on the finite eigenvalue set gives $\mu^a = \mu^b$ for some $a \neq b$, hence $\mu^{|a-b|} = 1$. □

Theorem 4.36 (Peripheral powers imply root of unity). Lean ✓ Stmt *If the set of peripheral eigenvalues of a linear endomorphism E on a finite-dimensional space is* closed under powers *($\mu \in \mathrm{peripheral}(E)$ and $n \geq 1$ imply $\mu^n \in \mathrm{peripheral}(E)$), then every peripheral eigenvalue is a root of unity.*

Proof. ✓ Proof The closure hypothesis provides $\mu^n \in \mathrm{peripheral}(E)$ for all n, which in particular means μ^n is an eigenvalue. Theorem 4.35 then gives $\mu^p = 1$ for some $p > 0$. □

4.5.1 Periodicity removal by powering

Remark 4.37. ✓ Stmt The results in this subsection are purely spectral: they use only finiteness of a set of complex numbers and the spectral mapping theorem for powers. In particular, they do not rely on complete positivity. For transfer maps, replacing a tensor by its p-blocked tensor replaces $\mathcal{E}_A$ by $\mathcal{E}_A^p$, so this powering step is the operator-theoretic form of blocking.

Lemma 4.38 (Common exponent for a finite set of roots of unity). Lean ✓ Stmt *Let $s \subseteq \mathbb{C}$ be a finite set. Assume that for every $\mu \in s$ there exists an exponent $p_\mu > 0$ with $\mu^{p_\mu} = 1$. Then there exists $p > 0$ such that $\mu^p = 1$ for all $\mu \in s$.*

Proof. ✓ Proof Take p to be the least common multiple of the exponents p_μ. □

Lemma 4.39 (Powering collapses peripheral eigenvalues). Lean ✓ Stmt *Let $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a linear map, and assume that E has a nonzero fixed point $\rho \neq 0$. Let $p > 0$. If every peripheral eigenvalue μ of E satisfies $\mu^p = 1$, then* $\mathrm{peripheral}(E^p) = \{1\}$.

Proof. ✓ Proof The fixed point ρ gives $1 \in \mathrm{peripheral}(E^p)$. For the reverse inclusion, let $\nu \in \mathrm{peripheral}(E^p)$. Spectral mapping gives $\nu = \mu^p$ for some $\mu \in \sigma(E)$. From $|\nu| = 1$ and $p > 0$ we obtain $|\mu| = 1$, hence $\mu \in \mathrm{peripheral}(E)$. The hypothesis $\mu^p = 1$ then forces $\nu = 1$. ☐

For a normalized MPS tensor, the transfer map is naturally trace-preserving but need not be unital. In general one should not expect a single gauge choice to make it both unital and trace-preserving.

Remark 4.40 (Bi-canonical special case). ✓ Stmt When a Kraus map happens to be both unital and trace-preserving (the "bi-canonical" case), the Kadison–Schwarz equality at peripheral eigenvectors (Theorem 4.46) shows directly that the peripheral eigenvalues are closed under powers, and hence are roots of unity by Theorem 4.36. It requires both normalizations simultaneously. The more general adjoint-fixed-point formulation below covers the case where only unitality and a positive definite adjoint fixed point are available.

4.5.2 The Kadison–Schwarz inequality and the multiplicative domain

Definition 4.41 (Kraus maps, adjoints, and normalizations). Lean Lean Lean Lean ✓ Stmt For operators $\{K_i\}_{i=0}^{d-1} \subseteq M_D(\mathbb{C})$, the associated *Kraus map* and *adjoint Kraus map* are

$$E(X) = \sum_i K_i X K_i^\dagger, \qquad E^*(X) = \sum_i K_i^\dagger X K_i.$$

The Kraus family is *unital* when $\sum_i K_i K_i^\dagger = \mathbb{1}$ and *trace-preserving* when $\sum_i K_i^\dagger K_i = \mathbb{1}$.

Theorem 4.42 (Kadison–Schwarz inequality). Lean ✓ Stmt *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a unital Kraus map. Then for every $X \in M_D(\mathbb{C})$,*

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [Wol12, Equation (5.2)].

Proof. ✓ Proof Apply the unital completely positive map entrywise to the positive 2×2 block matrix $\left(\begin{smallmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{smallmatrix}\right)$ and take the Schur complement of the lower-right block. ☐

Theorem 4.43 (Hilbert–Schmidt contraction). Lean ✓ Stmt *If a Kraus map is both unital and trace-preserving, then*

$$\mathrm{tr}(E(X)^\dagger E(X)) \leq \mathrm{tr}(X^\dagger X)$$

for all $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof The Kadison–Schwarz gap is positive semidefinite by Theorem 4.42. Trace preservation then identifies the traces of its two terms. ☐

Theorem 4.44 (KS gap decomposition). Lean ✓ Stmt *For a unital Kraus map,*

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_i R_i^\dagger R_i, \qquad R_i := X K_i^\dagger - K_i^\dagger E(X).$$

Proof. ✓ Proof Expand the right-hand side, distribute the products, and use $\sum_i K_i K_i^\dagger = \mathbb{1}$ to recover the Kadison–Schwarz gap. ☐

Theorem 4.45 (KS equality implies Kraus commutation). Lean ✓ Stmt *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then, for every i,*

$$X K_i^\dagger = K_i^\dagger E(X).$$

Proof. ✓ Proof The Kadison–Schwarz gap is a sum of positive semidefinite terms $\sum_i R_i^\dagger R_i$. If the gap vanishes, each R_i must vanish. □

Theorem 4.46 (KS equality for peripheral eigenvectors)**.** Lean ✓ Stmt *Let E be a Kraus map that is both unital and trace-preserving. If $E(X) = \mu X$ with $|\mu| = 1$, then*

$$E(X^\dagger X) = E(X)^\dagger E(X).$$

Proof. ✓ Proof The Kadison–Schwarz gap is positive semidefinite by Theorem 4.42. Trace preservation and the relation $E(X) = \mu X$ with $|\mu| = 1$ show that its trace is zero, hence the gap vanishes. □

Theorem 4.47 (Left multiplicative identity from KS equality)**.** Lean ✓ Stmt *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then for every $Y \in M_D(\mathbb{C})$,*

$$E(X^\dagger Y) = E(X)^\dagger E(Y).$$

Proof. ✓ Proof The commutation relation of Theorem 4.45 lets one move $X^\dagger$ past each Kraus operator inside the sum. □

Definition 4.48 (Multiplicative domain)**.** Lean ✓ Stmt The *multiplicative domain* of a Kraus map E is the set of matrices that are simultaneously left- and right-multiplicative for E.

Theorem 4.49 (Multiplicative domain is a $*$-subalgebra)**.** Lean ✓ Stmt *For a unital Kraus map, the multiplicative domain is a $*$-subalgebra of $M_D(\mathbb{C})$.*

Proof. ✓ Proof The multiplicative identities are stable under addition, multiplication, and adjoint, and they contain the identity matrix. □

4.5.3 Peripheral closure via adjoint fixed point

The following results replace the trace-preservation hypothesis by the existence of a positive definite fixed point for the *adjoint* Kraus map, matching the weighted-trace argument in [CPGSV16, Appendix A].

Lemma 4.50 (Peripheral closure via adjoint fixed point)**.** Lean ✓ Stmt *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital ($\sum_i K_i K_i^\dagger = \mathbb{1}$), and assume:*

1. *the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point $\rho > 0$, and*
2. *E is irreducible.*

Then $\mathrm{peripheral}(E)$ *is closed under powers: if $\mu \in$* $\mathrm{peripheral}(E)$ *then $\mu^n \in$* $\mathrm{peripheral}(E)$ *for every $n \in \mathbb{N}$.*

Proof. ✓ Proof Let $E(X) = \mu X$ with $|\mu| = 1$, and set $G := E(X^\dagger X) - E(X)^\dagger E(X)$. By Kadison–Schwarz, $G \geq 0$. Since $E^*(\rho) = \rho$,

$$\mathrm{tr}(\rho G) = \mathrm{tr}(\rho E(X^\dagger X)) - \mathrm{tr}(\rho E(X)^\dagger E(X)) = \mathrm{tr}(E^*(\rho) X^\dagger X) - \mathrm{tr}(\rho E(X)^\dagger E(X)) = \mathrm{tr}(\rho X^\dagger X) - |\mu|^2 \,\mathrm{tr}(\rho X^\dagger X) = 0.$$

Thus the trace of the Kadison–Schwarz gap against the adjoint fixed point vanishes. Because $\rho > 0$ and $G \geq 0$, this forces $G = 0$, so $E(X^\dagger X) = E(X)^\dagger E(X)$. Hence $X^\dagger X$ is a PSD fixed point, which is positive definite by irreducibility (Theorem 6.4). So X is invertible. By Theorem 4.45 (KS equality implies Kraus commutation), X commutes with each Kraus operator; iterating via the left multiplicative identity (Theorem 4.47) gives $E(X^n) = \mu^n X^n$, so μ^n is peripheral. □

Theorem 4.51 (Roots of unity via adjoint fixed point). Lean ✓ Stmt *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital, and assume that the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point and that E is irreducible. Then every peripheral eigenvalue of E is a root of unity.*

Proof. ✓ Proof Combine the closure-under-powers result (Lemma 4.50) with Theorem 4.36. □

Remark 4.52. ✓ Stmt Lemma 4.50 and Theorem 4.51 cover the bi-canonical (unital + TP) case as well: trace preservation gives $E^*(\mathbb{1}) = \mathbb{1}$, which is a positive definite fixed point of the adjoint. The adjoint-fixed-point formulation matches the argument in [CPGSV16, Appendix A], which uses a faithful invariant state rather than bi-canonicality. Theorem 4.51 establishes the root-of-unity conclusion in this adjoint-fixed-point formulation. In the same adjoint-fixed-point formulation, Chapter 7 proves the corresponding cyclic structure: the cyclic group of peripheral eigenvalues is Theorem 7.72, and the cyclic projection decomposition is Theorem 7.87.

Definition 4.53 (Channel period). Lean ✓ Stmt The *period* of a channel E is the number of peripheral eigenvalues (counted without multiplicity): $p(E) := |\mathrm{peripheral}(E)|$.

Definition 4.54 (Primitive map). Lean ✓ Stmt A linear map is *primitive* if its only peripheral eigenvalue is $\lambda = 1$, i.e. $\mathrm{peripheral}(E) = \{1\}$. For channels this corresponds to [Wol12, Theorem 6.7]. This definition applies to any linear endomorphism, not only to channels.

Theorem 4.55 (Primitivity equals period one). Lean ✓ Stmt *A linear map with a nonzero fixed point is primitive if and only if its period is* 1.

Proof. ✓ Proof If $\mathrm{peripheral}(E) = \{1\}$, then $|\mathrm{peripheral}(E)| = 1$. Conversely, if $|\mathrm{peripheral}(E)| = 1$, then since $1 \in \mathrm{peripheral}(E)$ (Theorem 4.33), the only element is 1. □

Remark 4.56 (Period-one characterization). ✓ Stmt The period $p(E)$ counts peripheral eigenvalues without multiplicity. Theorem 4.55 is therefore the period-one characterization of primitivity. The spectral estimates below are stated directly with Definition 4.54, since they use the absence of peripheral eigenvalues other than 1, not a separate counting hypothesis.

Theorem 4.57 (Complementary transfer-map gap for primitive maps). Lean ✓ Stmt *Let $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be trace-preserving, and let $\rho \neq 0$ satisfy $E(\rho) = \rho$ and $\mathrm{tr}(\rho) \neq 0$. Let*

$$P(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)}\rho$$

be the fixed-point projection associated to ρ. Assume that:

1. *E is primitive;*
2. *every eigenvalue μ of E satisfies $|\mu| \leq 1$;*
3. *whenever $E(X) = X$ and $\mathrm{tr}(X) = 0$, one has $X = 0$.*

Then every eigenvalue ν of $E - P$ satisfies $|\nu| < 1$.

Proof. ✓ Proof Let $(E - P)(X) = \nu X$ with $X \neq 0$. Then

$$E(X) = \nu X + P(X).$$

If $\mathrm{tr}(X) = 0$, then $P(X) = 0$, so $E(X) = \nu X$ and hence ν is an eigenvalue of E with $|\nu| \leq 1$. If $|\nu| = 1$, primitivity forces $\nu = 1$, so X is a trace-zero fixed point of E; by assumption this implies $X = 0$, a contradiction. Thus $|\nu| < 1$.

If instead $\operatorname{tr}(X) \neq 0$, trace preservation and $\operatorname{tr}(P(X)) = \operatorname{tr}(X)$ give

$$\operatorname{tr}(X) = \operatorname{tr}(E(X)) = \nu \operatorname{tr}(X) + \operatorname{tr}(P(X)) = \nu \operatorname{tr}(X) + \operatorname{tr}(X),$$

so $\nu \operatorname{tr}(X) = 0$ and therefore $\nu = 0$. In either case, $|\nu| < 1$. □

4.6 Fixed-point algebra

Definition 4.58 (Kraus fixed points). Lean ✓ Stmt The *fixed-point set* of a Kraus map $E(X) = \sum_i K_i X K_i^\dagger$ is $\operatorname{Fix}(E) = \{X \in M_D(\mathbb{C}) : E(X) = X\}$.

Theorem 4.59 (Fixed points lie in the multiplicative domain). Lean ✓ Stmt *Let E be a trace-preserving Kraus map whose Schrödinger-picture map has a positive definite fixed point $\rho > 0$. Then every adjoint fixed point belongs to the multiplicative domain of the adjoint Kraus family.*

Proof. ✓ Proof The Kadison–Schwarz equality for fixed points forces both one-sided multiplicative identities, so the fixed point lies in the full multiplicative domain. □

Theorem 4.60 (Fixed points form a $*$-subalgebra). Lean ✓ Stmt *Let E be a unital Kraus map whose adjoint map E^* has a positive definite fixed point $\rho > 0$. Then $\operatorname{Fix}(E)$ is a $*$-subalgebra of $M_D(\mathbb{C})$. This is [Wol12, Theorem 6.12].*

Proof. ✓ Proof Closure under addition and scalar multiplication is immediate from linearity. Closure under $\dagger$: if $E(X) = X$ then $E(X^\dagger) = X^\dagger$ because fixed points are stable under adjoint. Closure under multiplication: if $E(X) = X$ and $E(Y) = Y$, the weighted Kadison–Schwarz argument gives $E(X^\dagger X) = E(X)^\dagger E(X)$ and, applied to $X^\dagger$, also $E(XX^\dagger) = E(X)E(X)^\dagger$. Thus X lies in the multiplicative domain, and Theorem 4.47 yields $E(XY) = E(X)E(Y) = XY$. □

Remark 4.61. ✓ Stmt Theorem 4.59 is the trace-preserving / Heisenberg-picture counterpart of the same multiplicative-domain argument.

Theorem 4.62 (Projected elements lie in the multiplicative domain). Lean ✓ Stmt *Let E be a unital Kraus map whose adjoint map has a positive definite fixed point, and suppose $E^2 = E$. Then $E(X)$ lies in the multiplicative domain of E for every $X \in M_D(\mathbb{C})$.*

Proof. ✓ Proof Idempotence gives $E(E(X)) = E(X)$, so $E(X)$ is a fixed point. By the argument in Theorem 4.60, the Kadison–Schwarz gap for the fixed point $E(X)$ vanishes, hence

$$E(E(X)^\dagger E(X)) = E(X)^\dagger E(X).$$

Fixed points are closed under $\dagger$, so the same argument applied to $E(X)^\dagger$ gives

$$E(E(X)E(X)^\dagger) = E(X)E(X)^\dagger.$$

The vanishing of both Kadison–Schwarz gaps, again by the argument of Theorem 4.60, places $E(X)$ in the multiplicative domain. □

Theorem 4.63 (Choi–Effros left absorption). Lean ✓ Stmt *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(E(X)Y) = E(E(X)E(Y)).$$

Proof. ✓ Proof The previous theorem places $E(X)$ in the multiplicative domain, so $E(E(X)Y) = E(E(X))E(Y) = E(X)E(Y)$. Applying the same multiplicativity to $E(X)$ and $E(Y)$ gives $E(E(X)E(Y)) = E(E(X))E(E(Y))$, and idempotence rewrites this as $E(X)E(Y)$. □

Theorem 4.64 (Choi–Effros right absorption). Lean ✓ Stmt *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(XE(Y)) = E(E(X)E(Y)).$$

Proof. ✓ Proof The previous theorem places $E(Y)$ in the multiplicative domain, so $E(XE(Y)) = E(X)E(E(Y)) = E(X)E(Y)$. Applying the same multiplicativity to $E(X)$ and $E(Y)$ gives $E(E(X)E(Y)) = E(E(X))E(E(Y))$, and idempotence rewrites this as $E(X)E(Y)$. □

Theorem 4.65 (Choi–Effros symmetric absorption). Lean ✓ Stmt *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(E(X)Y) = E(XE(Y)).$$

Proof. ✓ Proof Both sides equal $E(E(X)E(Y))$ by the two preceding theorems. □

Theorem 4.66 (Choi–Effros projected product). Lean ✓ Stmt *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(E(X)E(Y)) = E(X)E(Y).$$

Proof. ✓ Proof The previous theorem places $E(X)$ and $E(Y)$ in the multiplicative domain. Therefore multiplicativity gives

$$E(E(X)E(Y)) = E(E(X))E(E(Y)).$$

Since E is idempotent under the standing hypotheses, $E(E(X)) = E(X)$ and $E(E(Y)) = E(Y)$, so

$$E(E(X)E(Y)) = E(X)E(Y).$$

□

Definition 4.67 (Adjoint fixed points). Lean ✓ Stmt The *adjoint fixed-point set* $\mathrm{Fix}(E^*) = \{X \in M_D(\mathbb{C}) : E^*(X) = X\}$, where $E^*(X) = \sum_i K_i^\dagger X K_i$.

Theorem 4.68 (Fixed points form a ∗-subalgebra in the Heisenberg picture). Lean Lean ✓ Stmt *If the Kraus map is trace-preserving and has a positive definite fixed point $\rho > 0$ with $E(\rho) = \rho$, then $\mathrm{Fix}(E^*)$ is a ∗-subalgebra of $M_D(\mathbb{C})$.*

Proof. ✓ Proof Apply Theorem 4.60 to the adjoint family $\{K_i^\dagger\}$. □

Definition 4.69 (Kraus commutant). Lean Lean ✓ Stmt The *Kraus commutant* is $\{K_i, K_i^\dagger\}' = \{X \in M_D(\mathbb{C}) : [X, K_i] = [X, K_i^\dagger] = 0 \forall i\}$. It is a ∗-subalgebra of $M_D(\mathbb{C})$.

Theorem 4.70 (Adjoint fixed points equal the Kraus commutant). Lean ✓ Stmt *Under the hypotheses of Theorem 4.68, the adjoint fixed-point ∗-subalgebra equals the Kraus commutant:*

$$\mathrm{Fix}(E^*) = \{K_i, K_i^\dagger\}'.$$

This is [Wol12, Theorem 6.13].

Proof. ✓ Proof (⊇): if X commutes with all K_i and $K_i^\dagger$, then $E^*(X) = \sum_i K_i^\dagger X K_i = X \sum_i K_i^\dagger K_i = X$ by trace preservation. (⊆): any ∗-subalgebra inside $\mathrm{Fix}(E^*)$ lies in the Kraus commutant, because for $X \in \mathrm{Fix}(E^*)$ with $X^\dagger X \in \mathrm{Fix}(E^*)$ the Kadison–Schwarz gap vanishes, which forces $XK_i = K_i X$ and $XK_i^\dagger = K_i^\dagger X$. □

Theorem 4.71 (Kraus commutant inside the adjoint fixed points). Lean ✓ Stmt *Under the hypotheses above, the Kraus commutant is the largest $*$-subalgebra contained in the adjoint fixed-point set.*

Proof. ✓ Proof By Theorem 4.70, every $*$-subalgebra inside the adjoint fixed points is automatically contained in the Kraus commutant, while the Kraus commutant itself is such a $*$-subalgebra. ☐

Chapter 5

Schwarz Inequalities and Multiplicative Domains

This chapter develops the Schwarz-inequality theory of [Wol12, Chapter 5]: the Kadison–Schwarz inequality for Kraus maps, the multiplicative domain as a $*$-subalgebra on which the map restricts to a $*$-homomorphism, the extension to normal, subnormal, and commuting-dominant inputs for positive subunital maps, order and spectral-interval contractivity, and the transpose-trace map as a Schwarz map that is not completely positive. The peripheral-eigenvector and normal-block consequences feed the Fundamental Theorem of Chapter 11. The operator-convexity and Jensen material from [Wol12, Chapter 5] is treated separately in Chapter 18.

5.1 Kadison–Schwarz inequality

A linear map is completely positive (Definition 4.5) if and only if it has a Kraus form. The Kadison–Schwarz inequality and the multiplicative-domain characterisation are proved first for Kraus maps. The transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is a Kraus map (Theorem 4.22), so these results apply directly to transfer maps.

Definition 5.1 (Kraus map). Lean ✓ Stmt Given operators $\{K_i\}_{i=0}^{d-1}$ with $K_i \in M_D(\mathbb{C})$, the *Kraus map* is $E(X) = \sum_{i=0}^{d-1} K_i X K_i^\dagger$. A linear map is completely positive (Definition 4.5) if and only if it can be written in this form.

Definition 5.2 (Adjoint Kraus map). Lean ✓ Stmt The *adjoint Kraus map* is $E^*(X) = \sum_{i=0}^{d-1} K_i^\dagger X K_i$. When the $\{K_i\}$ are the matrices of an MPS tensor A, this is the transfer map of the conjugate-transposed family $i \mapsto (A^i)^\dagger$.

Definition 5.3 (Unital Kraus map). Lean ✓ Stmt A Kraus map is *unital* if $\sum_{i=0}^{d-1} K_i K_i^\dagger = \mathbb{1}$. Equivalently, $E(\mathbb{1}) = \mathbb{1}$.

Definition 5.4 (Trace-preserving Kraus map). Lean ✓ Stmt A Kraus map is *trace-preserving* if $\sum_{i=0}^{d-1} K_i^\dagger K_i = \mathbb{1}$. Equivalently, the adjoint Kraus map is unital. This is the standard MPS normalization condition. In the later gauge language it is the left-canonical condition, so Kadison–Schwarz arguments are often applied to the adjoint map.

Theorem 5.5 (Kadison–Schwarz inequality). Lean ✓ Stmt *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a* unital *Kraus map (i.e. $\sum_i K_i K_i^\dagger = \mathbb{1}$). Then for all $X \in M_D(\mathbb{C})$,*

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [Wol12, Equation (5.2)].

Proof. ✓ Proof The 2×2 block matrix $\left[\begin{smallmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{smallmatrix}\right]$ is PSD ($= vv^\dagger$ with $v = [X^\dagger, \mathbb{1}]^T$). Applying the unital CP map blockwise preserves positive semidefiniteness (each block $K_i(\cdot)K_i^\dagger$ preserves PSD, and unitality ensures the $(2,2)$-block maps to $\mathbb{1}$). The Schur complement of the $(2,2)$-block $\mathbb{1}$ gives the result. □

Theorem 5.6 (Kadison–Schwarz for the adjoint channel). Lean ✓ Stmt *If $\sum_i K_i^\dagger K_i = \mathbb{1}$ (trace-preserving), then the adjoint Kraus map satisfies $E^*(X^\dagger X) \geq E^*(X)^\dagger E^*(X)$.*

Proof. ✓ Proof The adjoint operators $\{K_i^\dagger\}$ satisfy $\sum_i K_i^\dagger (K_i^\dagger)^\dagger = \sum_i K_i^\dagger K_i = \mathbb{1}$, so they form a unital family. Apply Theorem 5.5 to $\{K_i^\dagger\}$. □

5.1.1 Two-positive maps and the generalized Schwarz inequality

Definition 5.7 (k-positive map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is k-positive if $E \otimes \mathrm{id}_k$ is positive on $M_D(\mathbb{C}) \otimes M_k(\mathbb{C})$.

Definition 5.8 (Two-positive map). Lean ✓ Stmt A linear map is *two-positive* if it is 2-positive.

Theorem 5.9 (Completely positive maps are k-positive). Lean ✓ Stmt *Every completely positive map is k-positive for all $k \geq 1$.*

Theorem 5.10 ((k+1)-positive implies k-positive). Lean ✓ Stmt *Positivity under amplification is monotone in k.*

Theorem 5.11 (Completely positive implies two-positive). Lean ✓ Stmt *Every completely positive map is two-positive.*

Theorem 5.12 (Two-positive implies positive). Lean ✓ Stmt *Every two-positive map is positive.*

Definition 5.13 (Unital linear map). Lean ✓ Stmt A linear map E is unital if $E(\mathbb{1}) = \mathbb{1}$.

Theorem 5.14 (Kadison–Schwarz for unital two-positive maps). Lean ✓ Stmt *If E is unital and two-positive, then for all $X \in M_D(\mathbb{C})$,*

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

Proof. Form the 2×2 block matrix $\left(\begin{smallmatrix} \mathbb{1} & X \\ X^\dagger & X^\dagger X \end{smallmatrix}\right) \geq 0$. Apply $E \otimes \mathrm{id}_2$ (positive by 2-positivity) and read off the $(2,2)$-block Schur complement: $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$. □

Theorem 5.15 (Kraus-form Kadison–Schwarz as a special case). Lean ✓ Stmt *The Kraus-map Kadison–Schwarz inequality is recovered from the two-positive formulation.*

5.1.2 Douglas-type factorization lemmas

Theorem 5.16 (Range inclusion implies right factorization). Lean ✓ Stmt *If* $\operatorname{ran}(A) \subseteq \operatorname{ran}(B)$*, then there exists* C *such that* $A = BC$*.*

Proof. In finite dimensions, $\operatorname{ran}(A) \subseteq \operatorname{ran}(B)$ implies $A = B(B^\dagger B)^\dagger B^\dagger A$, where $(B^\dagger B)^\dagger$ denotes the Moore–Penrose pseudoinverse. Set $C = (B^\dagger B)^\dagger B^\dagger A$. □

Theorem 5.17 (Vectorwise range criterion implies factorization). Lean ✓ Stmt *If* $Av \in \operatorname{ran}(B)$ *for every vector* v*, then there exists* C *with* $A = BC$*.*

Theorem 5.18 (Hilbert–Schmidt contraction). Lean ✓ Stmt *If a Kraus map is both unital and trace-preserving, then* $\operatorname{tr}(E(X)^\dagger E(X)) \le \operatorname{tr}(X^\dagger X)$ *for all* $X \in M_D(\mathbb{C})$*.*

Proof. ✓ Proof By Theorem 5.5, $E(X^\dagger X) - E(X)^\dagger E(X) \ge 0$, so $\operatorname{tr}(E(X^\dagger X)) \ge \operatorname{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\operatorname{tr}(E(X^\dagger X)) = \operatorname{tr}(X^\dagger X)$. □

Lemma 5.19 (Trace pairing for Kraus map and adjoint map). Lean ✓ Stmt *For all* $X, Y \in M_D(\mathbb{C})$*,*

$$\operatorname{tr}(X\, E(Y)) = \operatorname{tr}(E^*(X)\, Y).$$

Lemma 5.20 (Positive-definite trace test). Lean ✓ Stmt *If* $A > 0$*,* $X \ge 0$*, and* $\operatorname{tr}(AX) = 0$*, then* $X = 0$*.*

Theorem 5.21 (Fixed-point peripheral eigenvectors satisfy KS equality). Lean ✓ Stmt *Let* E *be unital and let* E^* *be trace-preserving with a positive definite fixed point* $\rho > 0$*. If* $E(X) = \mu X$ *with* $|\mu| = 1$*, then*

$$E(X^\dagger X) = E(X)^\dagger E(X).$$

5.2 Multiplicative domain

Theorem 5.22 (KS gap decomposition). Lean ✓ Stmt *For a unital Kraus map* E*, the Kadison–Schwarz gap decomposes as*

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_{i=0}^{d-1} R_i^\dagger R_i,$$

where $R_i := XK_i^\dagger - K_i^\dagger E(X)$*.*

Proof. ✓ Proof Expand the right-hand side $\sum_i (XK_i^\dagger - K_i^\dagger E(X))^\dagger (XK_i^\dagger - K_i^\dagger E(X))$, distribute, and use unitality $\sum_i K_i K_i^\dagger = \mathbb{1}$ to identify the result with the KS gap. □

Theorem 5.23 (KS equality implies Kraus commutation). Lean ✓ Stmt *Let* E *be a unital Kraus map. If* $E(X^\dagger X) = E(X)^\dagger E(X)$ *for some* X*, then* $XK_i^\dagger = K_i^\dagger E(X)$ *for all* i*.*

Proof. ✓ Proof By Theorem 5.22, the KS gap is $\sum_i R_i^\dagger R_i$ with $R_i = XK_i^\dagger - K_i^\dagger E(X)$. If the gap vanishes, $\sum_i R_i^\dagger R_i = 0$. Each summand $R_i^\dagger R_i$ is PSD, so each must be zero, giving $R_i = 0$. □

Theorem 5.24 (KS equality for peripheral eigenvectors). Lean ✓ Stmt *Let* E *be a Kraus map that is both unital and trace-preserving. If* $E(X) = \mu X$ *with* $|\mu| = 1$*, then the KS gap vanishes:* $E(X^\dagger X) = E(X)^\dagger E(X)$*.*

Proof. ✓ Proof The gap $G := E(X^\dagger X) - E(X)^\dagger E(X)$ is PSD by Theorem 5.5. Its trace satisfies $\operatorname{tr}(G) = \operatorname{tr}(E(X^\dagger X)) - \operatorname{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\operatorname{tr}(E(X^\dagger X)) = \operatorname{tr}(X^\dagger X)$. Using $E(X) = \mu X$ and $|\mu| = 1$, $\operatorname{tr}(E(X)^\dagger E(X)) = |\mu|^2 \operatorname{tr}(X^\dagger X) = \operatorname{tr}(X^\dagger X)$. So $\operatorname{tr}(G) = 0$, and a PSD matrix with zero trace must be zero. □

Theorem 5.25 (Left multiplicative identity from KS equality). Lean ✓ Stmt *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then $E(X^\dagger Y) = E(X)^\dagger E(Y)$ for all $Y \in M_D(\mathbb{C})$.*

Proof. ✓ Proof From Theorem 5.23, $XK_i^\dagger = K_i^\dagger E(X)$ for all i. Taking conjugate transposes: $K_i X^\dagger = E(X)^\dagger K_i$. Then

$$\begin{aligned} E(X^\dagger Y) &= \sum_i K_i X^\dagger Y K_i^\dagger \\ &= \sum_i E(X)^\dagger K_i Y K_i^\dagger \\ &= E(X)^\dagger E(Y). \end{aligned}$$

□

Theorem 5.26 (Right multiplicative identity from KS equality). Lean ✓ Stmt *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then $E(YX) = E(Y)E(X)$ for all $Y \in M_D(\mathbb{C})$.*

Proof. ✓ Proof From Theorem 5.23, $XK_i^\dagger = K_i^\dagger E(X)$ for all i. Then

$$\begin{aligned} E(YX) &= \sum_i K_i Y X K_i^\dagger \\ &= \sum_i K_i Y K_i^\dagger E(X) \\ &= E(Y)E(X). \end{aligned}$$

□

5.2.1 Full multiplicative-domain structure

Definition 5.27 (Right and left multiplicative domains). Lean Lean ✓ Stmt For a Kraus map E, define

$$\mathcal{A}_R(E) := \{X \in M_D(\mathbb{C}) : \forall Y \in M_D(\mathbb{C}),\ E(XY) = E(X)E(Y)\},$$

and

$$\mathcal{A}_L(E) := \{X \in M_D(\mathbb{C}) : \forall Y \in M_D(\mathbb{C}),\ E(YX) = E(Y)E(X)\}.$$

We follow the convention that $\mathcal{A}_R(E)$ controls right multiplication and $\mathcal{A}_L(E)$ controls left multiplication. These are the two one-sided multiplicative domains from [Wol12, Equations (5.11)–(5.12)].

Definition 5.28 (Full multiplicative domain). Lean ✓ Stmt The *multiplicative domain* of E is

$$\mathcal{A}(E) := \mathcal{A}_R(E) \cap \mathcal{A}_L(E).$$

Theorem 5.29 (Multiplicative-domain characterization). Lean Lean ✓ Stmt *Let E be a unital Kraus map. Then*

$$X \in \mathcal{A}_R(E) \iff E(XX^\dagger) = E(X)E(X)^\dagger,$$

and

$$X \in \mathcal{A}_L(E) \iff E(X^\dagger X) = E(X)^\dagger E(X).$$

Together these are the two one-sided characterizations from [Wol12, Theorem 5.7].

Proof. ✓ Proof If X lies in a one-sided multiplicative domain, evaluate the defining identity at $Y = X^\dagger$. Conversely, the left-domain implication is exactly Theorem 5.25, and the right-domain implication follows by applying Theorem 5.25 to $X^\dagger$. □

Theorem 5.30 (One-sided multiplicative domains are subalgebras). Lean Lean ✓ Stmt *For a unital Kraus map E, both $\mathcal{A}_R(E)$ and $\mathcal{A}_L(E)$ are unital subalgebras of $M_D(\mathbb{C})$.*

Proof. ✓ Proof Closure under addition follows from linearity of E. Closure under multiplication: if $X, Y \in \mathcal{A}_R(E)$, then $E((XY)Z) = E(X)E(YZ) = E(X)E(Y)E(Z)$ for all Z, using Theorem 5.29 iteratively, and the $\mathcal{A}_L(E)$ case is analogous. Containment of $\mathbb{1}$ follows from unitality of E. □

Theorem 5.31 (Multiplicative domain is a $*$-subalgebra). Lean ✓ Stmt *For a unital Kraus map E, the full multiplicative domain $\mathcal{A}(E)$ is a $*$-subalgebra of $M_D(\mathbb{C})$. This is the algebraic content of [Wol12, Theorem 5.7].*

Proof. ✓ Proof If $X \in \mathcal{A}(E) = \mathcal{A}_R(E) \cap \mathcal{A}_L(E)$, then $E(X^\dagger X) = E(X)^\dagger E(X)$ and $E(XX^\dagger) = E(X)E(X)^\dagger$. The characterization of Theorem 5.29 shows $X^\dagger \in \mathcal{A}(E)$. Combined with Theorem 5.30, this gives a $*$-subalgebra. □

Theorem 5.32 (Restriction to multiplicative domain is a $*$-homomorphism). Lean ✓ Stmt *If E is unital, then the restricted map $E|_{\mathcal{A}(E)} : \mathcal{A}(E) \to M_D(\mathbb{C})$ is a $*$-algebra homomorphism. This is the homomorphism conclusion of [Wol12, Theorem 5.7].*

Proof. ✓ Proof Multiplicativity on $\mathcal{A}(E)$ is immediate from the definition. For $X \in \mathcal{A}(E)$, the Kraus commutation relation from Theorem 5.23 gives $E(X^\dagger) = E(X)^\dagger$, so the restriction preserves adjoints. □

5.2.2 Schwarz inequality for normal operators

Theorem 5.33 (CP Schwarz inequality for normal operators). Lean Lean ✓ Stmt *Let $E^*(X) = \sum_i K_i^\dagger X K_i$ be the adjoint Kraus map of a trace-preserving Kraus family, and let $A \in M_D(\mathbb{C})$ be normal. Then the Schwarz gap*

$$E^*(A^\dagger A) - E^*(A^\dagger)E^*(A) \tag{5.1}$$

is positive semidefinite. Equivalently,

$$E^*(A^\dagger)E^*(A) \leq E^*(A^\dagger A). \tag{5.2}$$

This is the completely positive case of [Wol12, Proposition 5.1].

Proof. ✓ Proof The positivity statement is exactly Theorem 5.6. The second formulation (5.2) is the Loewner-order reformulation of the first (5.1). □

Theorem 5.34 (Schwarz inequality for normal operators). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \leq \mathbb{1}$. If $A \in M_D(\mathbb{C})$ is normal, then*

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0.$$

This is [Wol12, Proposition 5.1].

Proof. ✓ Proof Since A is normal, diagonalize $A = UDU^\dagger$ with D diagonal and U unitary. Express $A^\dagger A = U|D|^2U^\dagger$ as a sum $\sum_{i,j} \bar{d}_i d_j P_i P_j$ where $P_i = Ue_ie_i^\dagger U^\dagger$. Each P_i is rank-one PSD, and the images $T(P_i)$ pairwise commute because they are images of mutually orthogonal projectors under a positive map. The Schwarz gap then reduces to a block quadratic form in the images, which is shown non-negative by a simultaneous-diagonalization argument: positivity of T ensures the scalar PSD matrix $(\langle v_k, T(P_iP_j)v_k\rangle)_{i,j}$ dominates the product $(\langle v_k, T(P_i)v_k\rangle\langle v_k, T(P_j)v_k\rangle)_{i,j}$, and combining over a common eigenbasis of the $T(P_i)$ yields the result. □

Remark 5.35. The proof follows the positivity-on-abelian argument of [Wol12, Proposition 1.6]: a positive map restricted to the commutative $*$-subalgebra generated by the spectral projectors of A satisfies the Schwarz inequality directly, without invoking the Arveson extension theorem.

5.2.3 Schwarz inequality for subnormal and commuting-dominant operators

Theorem 5.36 (Schwarz inequality for subnormal operators). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \le \mathbb{1}$. If $A \in M_D(\mathbb{C})$ is subnormal (i.e. there exists a normal operator on a larger space whose compression to $\mathbb{C}^D$ is A), then*

$$T(A^\dagger A) - T(A^\dagger)T(A) \ge 0.$$

This is [Wol12, Theorem 5.5].

Proof. ✓ Proof Embed A as the northwest corner of a normal operator N on $\mathbb{C}^D \oplus \mathbb{C}^E$. Extend T to a positive subunital map $\tilde{T}$ on the larger space (acting as T on the northwest block, zero elsewhere). Apply Theorem 5.34 to $\tilde{T}$ and N, then compress the resulting inequality back to $\mathbb{C}^D$. □

Theorem 5.37 (Schwarz inequality for commuting-dominant operators). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \le \mathbb{1}$. If $A \in M_D(\mathbb{C})$ admits a positive semidefinite* dominant *$B \ge 0$ with $A^\dagger A \le B$ and B commuting with A ($BA = AB$), then*

$$T(A^\dagger)T(A) \le T(B) \quad \text{and} \quad T(A)T(A^\dagger) \le T(B).$$

This is [Wol12, Theorem 5.6].

Proof. ✓ Proof First take $B > 0$. Since B commutes with A and $A^\dagger A \le B$, also $AA^\dagger \le B$, so $D_R = (B - A^\dagger A)^{1/2}$ and $D_L = (B - AA^\dagger)^{1/2}$ are positive semidefinite. The block matrix $N = \begin{pmatrix} A & D_L \\ D_R & -A^\dagger \end{pmatrix}$ is normal, with $N^\dagger N = NN^\dagger = B \oplus B$. As in the subnormal case, extend T to a positive subunital map on $M_{2D}(\mathbb{C})$, apply the normal-operator Schwarz inequality (Theorem 5.34) to that extension and N, and compress back to $\mathbb{C}^D$. Since N is normal, both $T(A^\dagger)T(A) \le T(B)$ and $T(A)T(A^\dagger) \le T(B)$ follow. For general $B \ge 0$, apply this to $B_\varepsilon = B + \varepsilon\mathbb{1} > 0$ and let $\varepsilon \to 0$. □

Theorem 5.38 (CP variant: Kadison–Schwarz for commuting-dominant inputs). Lean ✓ Stmt *Let E^* be the adjoint of a trace-preserving Kraus map. If $A \in M_D(\mathbb{C})$ admits a positive semidefinite dominant B commuting with A and satisfying $A^\dagger A \le B$, then $E^*(A^\dagger)E^*(A) \le E^*(B)$ and $E^*(A)E^*(A^\dagger) \le E^*(B)$.*

Proof. ✓ Proof Apply Theorem 5.37 to the adjoint Kraus map (which is unital under the trace-preservation hypothesis). □

5.2.4 Positive maps preserve order and spectral intervals

Theorem 5.39 (Positive maps are monotone). Lean ✓ Stmt *If $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is positive and $A \le B$, then $T(A) \le T(B)$.*

Proof. ✓ Proof Since $B - A \ge 0$, positivity gives $T(B-A) \ge 0$. By linearity, this is $T(B) - T(A) \ge 0$. $\square$

Theorem 5.40 (Positive maps preserve adjoints). Lean ✓ Stmt *If $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is positive, then $T(A^\dagger) = T(A)^\dagger$ for all $A \in M_D(\mathbb{C})$.*

Proof. ✓ Proof Regard T as the associated positive linear map (Definition 4.3). For positive linear maps between matrix C^*-algebras, $T(A^*) = T(A)^*$. Since the star on $M_D(\mathbb{C})$ is the conjugate transpose, $T(A^\dagger) = T(A^*) = T(A)^* = T(A)^\dagger$. $\square$

Theorem 5.41 (Order intervals are preserved). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \le \mathbb{1}$. If $a \le 0 \le b$ and $a\mathbb{1} \le A \le b\mathbb{1}$, then*

$$a\mathbb{1} \le T(A) \le b\mathbb{1}.$$

This is the matrix form of [Wol12, Equation (5.21)].

Proof. ✓ Proof By monotonicity, $T(a\mathbb{1}) \le T(A) \le T(b\mathbb{1})$. For the lower bound, $T(a\mathbb{1}) = aT(\mathbb{1})$. Since $a \le 0$ and $T(\mathbb{1}) \le \mathbb{1}$, we have $aT(\mathbb{1}) \ge a\mathbb{1}$. For the upper bound, $T(b\mathbb{1}) = bT(\mathbb{1}) \le b\mathbb{1}$ since $b \ge 0$. Therefore $a\mathbb{1} \le T(A) \le b\mathbb{1}$. $\square$

Theorem 5.42 (Spectral interval contractivity). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \le \mathbb{1}$, and let $A \in M_D(\mathbb{C})$ be Hermitian. If $\operatorname{spec}(A) \subseteq [a, b]$ with $a \le 0 \le b$, then*

$$\operatorname{spec}(T(A)) \subseteq [a, b].$$

This is [Wol12, Equation (5.21)].

Proof. ✓ Proof For Hermitian matrices, the inclusion $\operatorname{spec}(A) \subseteq [a, b]$ is equivalent to the order bounds $a\mathbb{1} \le A \le b\mathbb{1}$. Apply Theorem 5.41 and translate the resulting inequalities back into spectral bounds. $\square$

5.2.5 A positive Schwarz map that is not completely positive

Example 5.43 (Transpose-trace map on $M_2(\mathbb{C})$). Lean ✓ Stmt Consider the linear map $T : M_2(\mathbb{C}) \to M_2(\mathbb{C})$ defined by

$$T(A) = \frac{1}{2}A^T + \frac{1}{4}\operatorname{tr}(A)\mathbb{1}.$$

This is the map from [Wol12, Example 5.3].

Theorem 5.44 (The transpose-trace map is positive). Lean ✓ Stmt *The map from Example 5.43 is positive.*

Proof. ✓ Proof It is a non-negative linear combination of the transpose map and the map $A \mapsto \operatorname{tr}(A)\mathbb{1}$, and both of these maps send positive semidefinite matrices to positive semidefinite matrices. $\square$

Theorem 5.45 (The transpose-trace map satisfies the Schwarz inequality). Lean ✓ Stmt *For every $A \in M_2(\mathbb{C})$,*

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0,$$

where T is the map from Example 5.43.

Proof. ✓ Proof Write the Schwarz gap as $F(A)$. One checks that $F(A + \lambda\mathbb{1}) = F(A)$ for every scalar λ, so it is enough to treat the case $\mathrm{tr}(A) = 0$. In that case a direct 2×2 computation shows that $F(A)$ dominates $\frac{1}{2}(A^\dagger A)^T$, hence is positive semidefinite. □

Theorem 5.46 (The transpose-trace map is not completely positive). Lean ✓ Stmt *The map from Example 5.43 is not completely positive.*

Proof. ✓ Proof Compute the Choi matrix of T and test it on the antisymmetric vector $|01\rangle - |10\rangle$. The resulting expectation value is negative, so the Choi matrix is not positive semidefinite. □

Chapter 6

Perron–Frobenius Theory for Channels and Transfer Maps

This chapter establishes the Perron–Frobenius theory: existence, positive definiteness, and uniqueness of fixed points for injective and irreducible transfer maps, canonical gauge constructions, ergodicity of irreducible channels, the spectral-radius identification of Perron eigenvalues, and Wolf's spectral characterization of irreducibility. The conceptual source is Wolf's treatment of irreducible positive maps [Wol12, Chapter 6, especially Theorems 6.2–6.5 and Corollary 6.3] and [EHK78].

Theorem 6.1 (Brouwer fixed point on density matrices). Lean Lean Lean Lean Lean ✓ Stmt *Let $D > 0$. Every continuous map from the compact convex set of density matrices in $M_D(\mathbb{C})$ to itself has a fixed point.*

Proof. ✓ Proof The proof starts from Brouwer's theorem for products of simplices, transfers it to closed cubes, and then to compact retracts of finite-dimensional real normed spaces. The density-matrix set is such a compact retract, using the explicit density retraction. □

6.1 Positive definiteness

Remark 6.2. ✓ Stmt Wolf's general theory treats irreducible positive maps first ([Wol12, Theorem 6.2]), then derives non-degeneracy and positive definiteness ([Wol12, Theorem 6.3]). The injective-tensor version (Theorem 6.3) uses a shorter direct kernel argument; the irreducible CP-map version (Theorem 6.4) follows via the support-projection argument of [Wol12, Theorem 6.2].

Theorem 6.3 (PSD fixed point is positive definite). Lean ✓ Stmt *Let A be an injective MPS tensor and let $\rho \geq 0$, $\rho \neq 0$, be a fixed point of the transfer map $\mathcal{E}_A(\rho) = \rho$. Then ρ is positive definite.*

Proof. ✓ Proof Suppose $v^\dagger \rho v = 0$ for some $v \neq 0$. Since $\rho \geq 0$, this implies $\rho v = 0$. The fixed-point equation gives $v^\dagger \rho v = \sum_i ((A^i)^\dagger v)^\dagger \rho ((A^i)^\dagger v)$. Each term is non-negative; if the sum is zero, then $\rho (A^i)^\dagger v = 0$ for all i: the kernel of ρ is invariant under all $(A^i)^\dagger$. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, we have $\rho M v = 0$ for every matrix M. Given any vector w, choose M with $Mv = w$. Then $\rho w = 0$ for every w, so $\rho = 0$, contradicting $\rho \neq 0$. In [Wol12, Theorem 6.3(2)], the same conclusion is reached via the growth condition $(\mathbb{1} + \mathcal{E}_A)^{D-1}(\rho) > 0$ of [Wol12, Theorem 6.2, condition (2)]; here the direct kernel argument is shorter under injectivity. □

Theorem 6.4 (PSD fixed point is positive definite — irreducible version). Lean ✓ Stmt *If $\mathcal{E}_A$ is irreducible and $\rho \geq 0$, $\rho \neq 0$ is a fixed point, then ρ is positive definite.*

Proof. ✓ Proof Suppose ρ is not positive definite. Then ρ has a nontrivial kernel; let Q be the orthogonal projection onto the support of ρ (the complement of the kernel). A direct computation using the fixed-point equation $\mathcal{E}_A(\rho) = \rho$ shows that Q is an invariant projection for $\mathcal{E}_A$: it satisfies $Q\mathcal{E}_A(QXQ)Q = \mathcal{E}_A(QXQ)$ for all X. By irreducibility of $\mathcal{E}_A$ (Definition 4.20), every invariant projection is 0 or $\mathbb{1}$. If $Q = 0$ then $\rho = 0$, contradicting $\rho \neq 0$. If $Q = \mathbb{1}$ then ρ is positive definite, contradicting our assumption. Hence ρ must be positive definite. This is the support-projection direction of [Wol12, Theorem 6.2(1)]. □

Theorem 6.5 (Orthogonal trace condition). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, and let $A, B \geq 0$ be nonzero positive semidefinite matrices with* $\operatorname{tr}(BA) = 0$*. Then there exists t with $1 \leq t \leq D - 1$ such that*

$$\operatorname{tr}(B \cdot E^t(A)) > 0.$$

Proof. ✓ Proof The growth theorem for irreducible CP maps ([Wol12, Theorem 6.2(2)]) gives $(\mathbb{1} + E)^{D-1}(A) > 0$ (positive definite). Since $B \geq 0$ is nonzero, $\operatorname{tr}(B \cdot (\mathbb{1} + E)^{D-1}(A)) > 0$. Expanding by the binomial theorem:

$$\operatorname{tr}\left(B \cdot \sum_{k=0}^{D-1} \tbinom{D-1}{k} E^k(A)\right) > 0.$$

Each term $\operatorname{tr}(B \cdot E^k(A)) \geq 0$ since both factors are PSD. The $k = 0$ term vanishes by hypothesis ($\operatorname{tr}(BA) = 0$). Hence at least one $k \in \{1, \dots, D-1\}$ contributes a strictly positive term. This is [Wol12, Theorem 6.2(4)]: item (2) (growth condition) implies item (4) by a binomial expansion and PSD positivity argument. □

6.2 Uniqueness

Theorem 6.6 (Uniqueness of PSD fixed point). Lean ✓ Stmt *Let A be an injective MPS tensor. If $\rho, \sigma \geq 0$ are both nonzero fixed points of $\mathcal{E}_A$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.*

Proof. ✓ Proof Both ρ and σ are positive definite by Theorem 6.3. Write $\rho = SS^\dagger$ via a spectral square root and set $H = (S^\dagger)^{-1}\sigma S^{-1}$, which is Hermitian and positive definite. Let c_0 be the minimal eigenvalue of H. Then $\tau := \sigma - c_0\rho$ is a PSD fixed point which is *not* positive definite (it has a zero eigenvalue by construction of c_0). By Theorem 6.3, τ must be zero, giving $\sigma = c_0\rho$. In [Wol12, Theorem 6.3, item 2] uniqueness is proved via the growth condition of [Wol12, Theorem 6.2]; the same minimal-eigenvalue argument applies here, using the injective-tensor positive-definiteness (Theorem 6.3) in place of the growth condition. □

Remark 6.7. ✓ Stmt The proof produces a positive real scalar $c_0 > 0$: it is the minimal eigenvalue of the positive definite Hermitian matrix $H = (S^\dagger)^{-1}\sigma S^{-1}$.

Theorem 6.8 (Irreducible fixed-point uniqueness). Lean ✓ Stmt *Let A be an MPS tensor whose transfer map is irreducible. If $\rho, \sigma \geq 0$ are fixed points of $\mathcal{E}_A$, and if ρ is nonzero, then $\sigma = c\rho$ for some scalar c.*

Proof. ✓ Proof The irreducible positive-definiteness theorem (Theorem 6.4) upgrades every nonzero positive semidefinite fixed point to a positive definite one. The same minimal eigenvalue argument used in Theorem 6.6 then applies. □

Theorem 6.9 (Uniqueness of positive eigenvalue for irreducible CP maps). Lean ✓ Stmt *Let $D \geq 1$ and let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho, \sigma \geq 0$ are nonzero positive semidefinite matrices satisfying $E(\rho) = r_1\rho$ and $E(\sigma) = r_2\sigma$ for real scalars $r_1, r_2 > 0$. Then $r_1 = r_2$.*

Proof. ✓ Proof Extract Kraus operators K for E. Irreducibility of E passes to the adjoint transfer map $E^\dagger(\,\cdot\,) = \sum_i K_i^\dagger\,(\,\cdot\,)\,K_i$, which is a positive CP map. The Perron–Frobenius existence theorem (Theorem 6.22) together with the irreducible positive-definiteness upgrade (Theorem 6.4) gives a positive definite eigenvector $\tau > 0$ with eigenvalue $t > 0$: $E^\dagger(\tau) = t\,\tau$. For any nonzero PSD X with $E(X) = sX$, the trace pairing identity yields

$$s \cdot \mathrm{tr}(\tau X) = \mathrm{tr}(\tau \cdot E(X)) = \mathrm{tr}(E^\dagger(\tau) \cdot X) = t \cdot \mathrm{tr}(\tau X).$$

Since $\tau > 0$ and $X \geq 0$, $X \neq 0$, the scalar $\mathrm{tr}(\tau X) > 0$, so $s = t$. Applying this to both (ρ, r_1) and (σ, r_2) gives $r_1 = t = r_2$. This is [Wol12, Theorem 6.3, item 3], the non-degeneracy of the spectral-radius eigenvalue for irreducible CP maps; the proof follows Wolf's dual-map trace argument. □

6.3 Existence and the Perron–Frobenius theorem

Theorem 6.10 (Existence of PSD fixed point). Lean ✓ Stmt *Assume $D \geq 1$. Let A be an MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ (so that $\mathcal{E}_A$ is trace-preserving). Then there exists $\rho \geq 0$, $\rho \neq 0$, with $\mathcal{E}_A(\rho) = \rho$.*

Proof. ✓ Proof Under the normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, the transfer map is a quantum channel (Theorem 4.23). The Cesàro fixed-point theorem (Theorem 4.27) then provides the fixed point; injectivity is not needed for this step. □

Theorem 6.11 (Quantum Perron–Frobenius). Lean ✓ Stmt *Assume $D \geq 1$. Let A be an injective MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then the transfer map $\mathcal{E}_A$ has a unique PSD fixed point ρ (up to scaling), and ρ is positive definite.*

Proof. ✓ Proof Combine Theorems 6.10, 6.3, and 6.6. □

Theorem 6.12 (Quantum Perron–Frobenius for irreducible transfer maps). Lean ✓ Stmt *Assume $D \geq 1$. Let A be an MPS tensor such that its transfer map is irreducible and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ (so that $\mathcal{E}_A$ is trace-preserving). Then $\mathcal{E}_A$ has a unique positive definite fixed point, up to scalar multiple.*

Proof. ✓ Proof Existence is the channel fixed-point theorem (Theorem 6.10). Irreducibility upgrades the fixed point to positive definite, and Theorem 6.8 gives uniqueness up to scalar multiple. □

6.4 Right- and left-canonical gauges

Theorem 6.13 (Right-canonical (unital) gauge). Lean ✓ Stmt *Let $\mathcal{E}_A(\rho) = \rho$ and let S be an invertible matrix with $SS^\dagger = \rho$. Define the gauged operators $A'^i = S^{-1}A^iS$. Then $\sum_i A'^i(A'^i)^\dagger = \mathbb{1}$, i.e. the gauged Kraus map is unital. This is the right-canonical normalization.*

Proof. ✓ Proof Expand each summand: $(S^{-1}A^iS)(S^{-1}A^iS)^\dagger = S^{-1}A^i\,\underbrace{SS^\dagger}_{\rho}(A^i)^\dagger(S^\dagger)^{-1}$. Summing over i and using $\sum_i A^i\rho(A^i)^\dagger = \rho$ gives $S^{-1}\rho(S^\dagger)^{-1} = S^{-1}SS^\dagger(S^\dagger)^{-1} = \mathbb{1}$. □

Theorem 6.14 (Left-canonical (trace-preserving) gauge). Lean ✓ Stmt *Let σ be a fixed point of the adjoint transfer map $\sum_i (A^i)^\dagger \sigma A^i = \sigma$, and let S be invertible with $S^\dagger S = \sigma$. Define $A'^i = SA^iS^{-1}$. Then $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$, i.e. the gauged Kraus map is trace-preserving. This is the left-canonical normalization.*

Proof. ✓ Proof Expand each summand: $(SA^iS^{-1})^\dagger(SA^iS^{-1}) = (S^\dagger)^{-1}(A^i)^\dagger \underbrace{S^\dagger S}_{\sigma} A^iS^{-1}$. Summing over i and using $\sum_i (A^i)^\dagger \sigma A^i = \sigma$ gives $(S^\dagger)^{-1}\sigma S^{-1} = (S^\dagger)^{-1}S^\dagger SS^{-1} = \mathbb{1}$. □

Theorem 6.15 (Square-root trace-preserving gauge from an adjoint fixed point). Lean ✓ Stmt *Let σ be positive definite and satisfy $\sum_i (A^i)^\dagger \sigma A^i = \sigma$. Define*

$$B^i = \sigma^{1/2} A^i \sigma^{-1/2}.$$

Then B is trace-preserving: $\sum_i (B^i)^\dagger B^i = \mathbb{1}$.

Proof. ✓ Proof Since $\sigma^{1/2}$ is self-adjoint and $\sigma^{1/2}\sigma^{1/2} = \sigma$, each summand becomes $(\sigma^{1/2})^{-1}\,(A^i)^\dagger \sigma A^i\,\sigma^{-1/2}$. Summing over i and using the fixed-point equation leaves $(\sigma^{1/2})^{-1}\sigma\sigma^{-1/2} = \mathbb{1}$. □

Lemma 6.16 (Square-root trace-preserving gauge is a gauge equivalence). Lean ✓ Stmt *If σ is positive definite and $B^i = \sigma^{1/2}A^i\sigma^{-1/2}$, then A and B are gauge-equivalent.*

Proof. ✓ Proof The gauge matrix is $\sigma^{1/2}$, which is invertible because σ is positive definite. □

Remark 6.17. ✓ Stmt The right-canonical gauge of Theorem 6.13 and the left-canonical gauge of Theorem 6.14 are generally different similarity transforms: one is built from a fixed point of $\mathcal{E}_A$, the other from a fixed point of the adjoint transfer map. This section does not claim that a single gauge makes the transfer map simultaneously unital and trace-preserving.

6.5 Similarity preserves irreducibility

Lemma 6.18 (Similarity transform preserves irreducibility). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $C \in M_D(\mathbb{C})$ be invertible ($\det C \neq 0$), and let $c > 0$. Define the similarity-transformed map*

$$E'(X) = c \cdot C^{-1}\, E(C\, X\, C^\dagger)\, (C^\dagger)^{-1}.$$

Then E' is also irreducible.

Proof. ✓ Proof Suppose Q is an invariant projection for E': $Q \neq 0$, $Q \neq \mathbb{1}$, and $(\mathbb{1}-Q)E'(QXQ)Q = 0$ for all X. Setting $R = CQC^\dagger$, the support projection P of R is an invariant projection for E. Irreducibility of E forces $P = 0$ or $P = \mathbb{1}$. The invertibility of C then forces $Q = 0$ or $Q = \mathbb{1}$, a contradiction. This is the full similarity version of [Wol12, Proposition 6.6]; the scalar case ($C = \mathbb{1}$) gives the scaling result stated next. □

Lemma 6.19 (Similarity is an involution). Lean ✓ Stmt *For any invertible $C \in M_D(\mathbb{C})$ and any linear map E on $M_D(\mathbb{C})$, the similarity transforms by C and C^{-1} compose to the identity:*

$$X \mapsto C(C^{-1}\, E(C\, X\, C^\dagger)\, (C^\dagger)^{-1})\, C^\dagger = E(X).$$

Proof. ✓ Proof Direct computation: the inner conjugation $C \cdot (C^{-1}X(C^\dagger)^{-1}) \cdot C^\dagger$ collapses to X using $CC^{-1} = \mathbb{1}$ and $(C^\dagger)^{-1}C^\dagger = \mathbb{1}$, and the outer conjugation by $C(\cdot)C^\dagger$ then cancels the inner $C^{-1}(\cdot)(C^\dagger)^{-1}$ on $E(X)$ for the same reason. □

Lemma 6.20 (Irreducibility is invariant under similarity). Lean ✓ Stmt *For any invertible $C \in M_D(\mathbb{C})$ and any linear map E on $M_D(\mathbb{C})$, the similarity transform $X \mapsto C^{-1}\, E(C\, X\, C^\dagger)\, (C^\dagger)^{-1}$ is irreducible if and only if E is.*

Proof. ✓ Proof The forward direction follows from Lemma 6.18 applied with the inverse C^{-1} (whose determinant is also nonzero): if the similarity transform of E by C is irreducible, then so is its similarity transform by C^{-1}, which equals E by Lemma 6.19. The reverse direction is Lemma 6.18 itself (with $c = 1$). □

Theorem 6.21 (Scaling preserves irreducibility). Lean ✓ Stmt *If E is an irreducible map and $c \neq 0$, then cE is also irreducible. This is the scalar special case ($C = \mathbb{1}$) of [Wol12, Proposition 6.6].*

Proof. ✓ Proof An invariant projection for cE is an invariant projection for E (scale out the nonzero constant from the commutation relation). □

6.6 Perron–Frobenius eigenvector existence

This section establishes the existence of a positive definite eigenvector for the adjoint transfer map of an irreducible MPS tensor, and uses it to construct a TP-gauge normalization. The PSD-eigenvector step corresponds to [Wol12, Theorem 6.5] (general positive maps) and [EHK78]; the upgrade to positive definiteness uses the irreducible-map theory of [Wol12, Theorem 6.3]. The TP-gauge application follows [CPGSV16, Appendix A].

Theorem 6.22 (PSD eigenvector for positive maps). Lean ✓ Stmt *Let $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $D > 0$, and assume that $E(\sigma) \neq 0$ for every nonzero PSD matrix σ. Then there exist a nonzero PSD matrix ρ and a positive real $r > 0$ such that $E(\rho) = r\rho$.*

This is the finite-dimensional Perron–Frobenius theorem for positive maps on matrix algebras ([Wol12, Theorem 6.5]; see also [EHK78]). The proof applies Brouwer's fixed-point theorem to the normalization map $\rho \mapsto E(\rho)/\operatorname{tr}(E(\rho))$ on the compact convex set of density matrices.

Theorem 6.23 (Adjoint transfer map is nonzero). Lean ✓ Stmt *If some Kraus operator $A^i \neq 0$, then the adjoint transfer map $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ is nonzero.*

Proof. ✓ Proof If $\mathcal{E}_A^\dagger = 0$ then $\mathcal{E}_A^\dagger(\mathbb{1}) = 0$, so $\sum_i (A^i)^\dagger A^i = 0$. Since each summand $(A^i)^\dagger A^i$ is PSD, each $(A^i)^\dagger = 0$ and hence each $A^i = 0$. □

Theorem 6.24 (Positive-definite adjoint eigenvector for irreducible tensors). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive definite matrix σ and a positive real $r > 0$ such that $\mathcal{E}_A^\dagger(\sigma) = r\sigma$.*

This combines the general PF existence ([Wol12, Theorem 6.5]) with the irreducible upgrade to positive definiteness ([Wol12, Theorem 6.3(2)]); the application to the adjoint transfer map follows [CPGSV16, Appendix A].

Proof. ✓ Proof By Theorem 6.23, $\mathcal{E}_A^\dagger \neq 0$. Since $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ is a Kraus map, it is positive. Irreducibility of A transfers to the adjoint transfer map. More precisely: if P is invariant for $\mathcal{E}_A$ (i.e. $(\mathbb{1} - P)K_i P = 0$ for all Kraus operators K_i), then taking adjoints gives $PK_i^\dagger(\mathbb{1} - P) = 0$, which means $\mathbb{1} - P$ is invariant for $\mathcal{E}_A^\dagger$. Hence $\mathcal{E}_A$ is irreducible if and only if $\mathcal{E}_A^\dagger$ is irreducible, and together with $\mathcal{E}_A^\dagger \neq 0$ this implies $\mathcal{E}_A^\dagger(\tau) \neq 0$ for every nonzero PSD matrix τ. Hence Theorem 6.22 applies to $\mathcal{E}_A^\dagger$, giving $\sigma_0 \geq 0$ (PSD, nonzero) and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma_0) = r\sigma_0$. Rescale: set $T^i = r^{-1/2}(A^i)^\dagger$. Then $\mathcal{E}_T(\sigma_0) = \sigma_0$, and $\mathcal{E}_T$ is irreducible (by

Theorem 6.21, since irreducibility of $\mathcal{E}_A$ transfers to the adjoint and is preserved under scaling). Since $\mathcal{E}_T$ is irreducible with a PSD fixed point, the positive-definiteness upgrade (Theorem 6.4) gives σ_0 positive definite. □

Theorem 6.25 (TP-gauge existence for irreducible tensors). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive real r, a positive definite matrix σ, and a gauge-equivalent tensor B such that*

$$B^i = \sigma^{1/2} \cdot (r^{-1/2} A^i) \cdot \sigma^{-1/2}$$

is trace-preserving: $\sum_i (B^i)^\dagger B^i = \mathbb{1}$.

This is the "spectral rescaling + TP gauge" step of [CPGSV16, Appendix A]. The similarity is generally non-unitary.

Proof. ✓ Proof By Theorem 6.24, obtain σ positive definite and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma) = r\sigma$. Set $c = r^{-1/2}$ and $A'^i = cA^i$. Then $\mathcal{E}_{A'}^\dagger(\sigma) = \sigma$. By the square-root trace-preserving gauge construction (Theorem 6.15), $B^i = \sigma^{1/2} A'^i \sigma^{-1/2}$ satisfies $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. Lemma 6.16 gives the required gauge equivalence. □

Theorem 6.26 (Unital-gauge existence for irreducible tensors). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive real r, a positive definite matrix ρ, and a gauge-equivalent tensor B such that*

$$B^i = r^{-1/2} \rho^{-1/2} A^i \rho^{1/2}$$

is unital: $\sum_i B^i (B^i)^\dagger = \mathbb{1}$.

This is the Perron–Frobenius unital-gauge orientation used in [PGVWC07, Theorem 4, lines 765–770], stated for one irreducible nonzero block.

Proof. ✓ Proof Apply Theorem 6.24 to the conjugate-transposed Kraus family. Irreducibility transfers to that family, and the nonzero Kraus hypothesis is preserved by taking adjoints. Translating the resulting adjoint eigenvector equation back gives $\mathcal{E}_A(\rho) = r\rho$ with ρ positive definite and $r > 0$. The unital gauge theorem then gives the displayed unital representative, and the same square-root similarity gives gauge equivalence to $r^{-1/2}A$. □

6.7 Exponential positivity for irreducible CP maps

Theorem 6.27 (Exponential truncation is positive definite). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then the finite exponential truncation*

$$\sum_{k=0}^{D-1} \frac{t^k}{k!} E^k(A)$$

is positive definite. This is the finite-sum core of [Wol12, Theorem 6.2(3)].

Proof. ✓ Proof If a nonzero vector v annihilated this quadratic form, then every term $E^k(A)$ with $0 \leq k \leq D - 1$ would annihilate v. The growth theorem for irreducible CP maps (that is, $(\mathbb{1} + E)^{D-1}(A) > 0$ for irreducible E and nonzero PSD A, as established in the proof of Theorem 6.5) then forces $(\mathbb{1}+E)^{D-1}(A)$ to annihilate v, contradicting its positive definiteness. □

Theorem 6.28 (Exponential positivity for irreducible CP maps). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then*

$$\exp(tE)(A) = \sum_{k=0}^{\infty} \frac{t^k}{k!} E^k(A)$$

is positive definite. This is [Wol12, Theorem 6.2(3)].

Proof. ✓ Proof By Theorem 6.27, the first D terms already form a positive definite matrix. Every remaining term is positive semidefinite, so the full exponential series is the sum of a positive definite matrix and a positive semidefinite tail. □

6.8 Ergodicity of irreducible channels

Theorem 6.29 (Unique density-matrix fixed point for an irreducible channel). Lean ✓ Stmt *Let E be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$. Then there exists a unique density matrix σ such that $E(\sigma) = \sigma$. Moreover σ is positive definite. This is the qualitative form of [Wol12, Corollary 6.3].*

Proof. ✓ Proof Every Cesàro mean of a density matrix stays inside the compact convex set of density matrices, so a subsequence converges. Its limit is a fixed point by the telescoping argument, positive definiteness follows from irreducibility, and uniqueness comes from the uniqueness of positive semidefinite Perron eigenvectors at eigenvalue 1. □

Theorem 6.30 (Cesàro convergence for irreducible channels). Lean ✓ Stmt *Let E be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$, and let ρ be any density matrix. Then the Cesàro means*

$$\frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho)$$

converge to the unique positive definite density-matrix fixed point of E. This is [Wol12, Corollary 6.3].

Proof. ✓ Proof Every subsequential limit of the Cesàro means is again a density-matrix fixed point. By Theorem 6.29, there is only one such fixed point, so every convergent subsequence has the same limit. Compactness then forces the whole sequence to converge to that limit. □

6.9 Spectral radius at the Perron eigenvalue

Theorem 6.31 (Spectral radius is the Perron eigenvalue). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho > 0$ and $E(\rho) = r\rho$ with $r > 0$. Then the spectral radius of E is r. This is [Wol12, Theorem 6.3(4)].*

Proof. ✓ Proof Choose a positive definite eigenvector for the adjoint map with the same eigenvalue, gauge by its square root, and rescale by r^{-1}. The resulting map is trace-preserving and has a positive definite fixed point, so 1 is an eigenvalue. The spectral radius of a trace-preserving map F satisfies $\rho(F) \leq 1$, since $\operatorname{tr}(F^n(X)) = \operatorname{tr}(X)$ for all n bounds the growth of iterates. Hence its spectral radius is 1. Undoing the similarity and scalar rescaling recovers the spectral radius of E. □

Theorem 6.32 (Real spectral-radius identity). Lean ✓ Stmt *Under the hypotheses of Theorem 6.31, the real-valued spectral radius of E is also equal to r.*

Proof. ✓ Proof This is the real-valued reformulation of Theorem 6.31. □

6.10 Spectral characterization of irreducibility

Definition 6.33 (Spectral properties). Lean ✓ Stmt A completely positive map E on $M_D(\mathbb{C})$ has the *spectral properties* of [Wol12, Theorem 6.4] if there exist a positive real number r, a positive definite right eigenvector ρ, and a positive definite left eigenvector σ for the adjoint map such that $E(\rho) = r\rho$, $E^\dagger(\sigma) = r\sigma$, every positive semidefinite right eigenvector for eigenvalue r is a scalar multiple of ρ, and the spectral radius of E is equal to r.

Theorem 6.34 (Irreducibility gives the spectral properties). Lean ✓ Stmt *Every nonzero irreducible completely positive map on $M_D(\mathbb{C})$ has the spectral properties of Definition 6.33.*

Proof. ✓ Proof Combine Perron–Frobenius existence for the map and its adjoint with the uniqueness theorem for positive semidefinite Perron eigenvectors and the spectral-radius identity of Theorem 6.31. □

Theorem 6.35 (Spectral properties imply irreducibility). Lean ✓ Stmt *If a completely positive map on $M_D(\mathbb{C})$ has the spectral properties of Definition 6.33, then it is irreducible.*

Proof. ✓ Proof Gauge by the positive definite left eigenvector and rescale by the Perron eigenvalue to obtain a trace-preserving map with a positive definite fixed point. A nontrivial invariant projection would then produce, by Cesàro averaging in its corner algebra, a second positive semidefinite fixed point not proportional to the Perron vector, contradicting the uniqueness clause in Definition 6.33. □

Theorem 6.36 (Irreducibility iff the spectral properties hold). Lean ✓ Stmt *Let E be a nonzero completely positive map on $M_D(\mathbb{C})$. Then E is irreducible if and only if it has the spectral properties of Definition 6.33. This is [Wol12, Theorem 6.4].*

Proof. ✓ Proof Combine Theorems 6.34 and 6.35. □

Chapter 7

Transfer-Operator Gaps and Block Separation

This chapter proves that the MPV overlaps between distinct (non-gauge-phase-equivalent) MPS blocks decay to zero as system size grows. It also proves the corresponding self-overlap convergence to 1 under the complementary transfer-map gap hypothesis that encodes primitivity in the cases of interest. The argument is based on a spectral analysis of the *mixed transfer operator*, following [PGVWC07].

This chapter concerns gaps in the spectrum of transfer operators, writing $\rho_{\mathrm{spec}}(\cdot)$ for the spectral radius: $\rho_{\mathrm{spec}}(F_{AB}) < 1$ for mixed transfer operators, and $\rho_{\mathrm{spec}}(\mathcal{E}_A - P) < 1$ after removing the fixed-point projection. The unqualified phrase *spectral gap* is reserved for the energy gap of a many-body Hamiltonian in Chapter 13.

The algebraic mixed-transfer identities in the first two sections do not use normalization. Starting with the spectral-radius estimates, we assume the trace-preserving normalization

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}. \tag{7.1}$$

7.1 Mixed transfer operator

Definition 7.1 (Mixed transfer operator)**.** Lean ✓ Stmt The *mixed transfer operator* (or *cross transfer operator*) for two MPS tensors A and B of the same bond dimension D is the linear map $F_{AB} : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ defined by

$$F_{AB}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

Definition 7.2 (Rectangular mixed transfer operator)**.** Lean ✓ Stmt For MPS tensors A of bond dimension D_1 and B of bond dimension D_2 (possibly $D_1 \neq D_2$), the *rectangular mixed transfer operator* is $F_{AB}^{\mathrm{rect}} : M_{D_1 \times D_2}(\mathbb{C}) \to M_{D_1 \times D_2}(\mathbb{C})$ defined by

$$F_{AB}^{\mathrm{rect}}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

Theorem 7.3 (Self-transfer)**.** Lean ✓ Stmt *Setting $B = A$ recovers the ordinary transfer map: $F_{AA} = \mathcal{E}_A$.*

Proof. ✓ Proof Substituting $B = A$ into the defining sum $F_{AB}(X) = \sum_i A^i X (B^i)^\dagger$ gives $\sum_i A^i X (A^i)^\dagger = \mathcal{E}_A(X)$. $\square$

Lemma 7.4 (Iterated mixed transfer)**.** Lean ✓ Stmt *For all $X \in M_D(\mathbb{C})$ and $N \geq 0$,*

$$F_{AB}^N(X) = \sum_{\sigma \in \{0,\dots,d-1\}^N} A^\sigma X (B^\sigma)^\dagger,$$

where $A^\sigma = A^{\sigma_1} \cdots A^{\sigma_N}$ denotes the word evaluation (Definition 2.2).

Proof. ✓ Proof Induction on N: the base case is $F^0(X) = X$; the inductive step applies F_{AB} to each summand and re-indexes via $\{0,\dots,d-1\}^{N+1} \cong \{0,\dots,d-1\} \times \{0,\dots,d-1\}^N$. $\square$

7.2 MPV overlap as transfer trace

Recall from Definition 2.48 that

$$O_{AB}(N) = \sum_{\sigma \in \{0,\dots,d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}.$$

This section rewrites that scalar as the operator trace of the mixed transfer power.

Lemma 7.5 (Trace expansion over matrix units)**.** Lean ✓ Stmt *For any linear endomorphism T on $M_{D_1 \times D_2}(\mathbb{C})$, the operator trace can be expanded as*

$$\mathrm{Tr}(T) = \sum_{p=0}^{D_1-1} \sum_{q=0}^{D_2-1} (T(E_{pq}))_{pq},$$

where $E_{pq} \in M_{D_1 \times D_2}(\mathbb{C})$ is the matrix with 1 in position (p,q) and 0 elsewhere.

Proof. ✓ Proof The operator trace $\mathrm{Tr}(T)$ equals the sum of $\mathrm{tr}(T(E_{pq}) \cdot E_{qp})$ over all (p,q), and $\mathrm{tr}(M \cdot E_{qp}) = M_{pq}$. $\square$

Lemma 7.6 (Overlap equals transfer trace)**.** Lean ✓ Stmt *For tensors A, B of bond dimension $D \geq 1$,*

$$O_{AB}(N) = \mathrm{Tr}(F_{AB}^N).$$

Proof. ✓ Proof Expand the operator trace using Lemma 7.5 as a sum over matrix units E_{pq}. Apply Lemma 7.4 to evaluate $F_{AB}^N(E_{pq})$, extract the (p,q)-entry, and reassemble. The resulting double sum over (p,q) and σ equals $\sum_\sigma V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = O_{AB}(N)$. $\square$

Lemma 7.7 (Rectangular overlap as transfer trace)**.** Lean ✓ Stmt *For tensors A of bond dimension $D_1 \geq 1$ and B of bond dimension $D_2 \geq 1$,*

$$O_{AB}(N) = \mathrm{Tr}((F_{AB}^{\mathrm{rect}})^N).$$

Proof. ✓ Proof Expand the operator trace over the rectangular matrix-unit basis. Then apply the rectangular iterated-transfer formula and sum over the matrix-unit indices exactly as in Lemma 7.6. $\square$

7.3 Frobenius norm estimates

The transfer-operator gap proofs use the Frobenius (Hilbert–Schmidt) norm as the natural norm for the finite-dimensional matrix algebra. We write $\|\cdot\|_F$ for this norm; it coincides with, and we use it interchangeably with, the Hilbert–Schmidt norm $\|\cdot\|_{\mathrm{HS}}$ on finite-dimensional spaces. We use the squared version and an isometric embedding into Euclidean space.

Definition 7.8 (Frobenius norm squared). Lean ✓ Stmt For a (possibly rectangular) matrix $X \in M_{m\times n}(\mathbb{C})$, the *Frobenius norm squared* is

$$\|X\|_F^2 := \sum_{i=0}^{m-1}\sum_{j=0}^{n-1} |X_{ij}|^2.$$

Lemma 7.9 (Trace formula for Frobenius norm). *Lean* *✓ Stmt* $\|X\|_F^2 = \operatorname{Re}\operatorname{tr}(X^\dagger X)$.

Proof. ✓ Proof Expand the matrix product and trace entry by entry. □

Definition 7.10 (Euclidean-space embedding). Lean ✓ Stmt The map $\mathrm{vec} : M_{m\times n}(\mathbb{C}) \to \mathbb{C}^{mn}$ that flattens matrix entries is an isometric embedding with respect to the Frobenius norm: $\|\mathrm{vec}(X)\|^2 = \|X\|_F^2$.

Lemma 7.11 (Norm of embedded matrix). *Lean* *✓ Stmt* $\|\mathrm{vec}(X)\|^2 = \|X\|_F^2$.

Proof. ✓ Proof Expand both sides as sums over entries and use $|z|^2 = \bar{z}z$. □

7.4 Eigenvalue bound and transfer-operator gap

The eigenvalue bound comes from a uniform Frobenius-norm estimate obtained directly from the trace-preserving normalization (cf. [Wol12, Proposition 6.1] for the general operator-norm version via Russo–Dye). For the rigidity theorem, one gauges A and B separately using positive fixed points of their transfer maps to obtain unital Kraus families, and then applies the Kadison–Schwarz equality argument ([Wol12, Theorem 5.3]) to a block embedding of a mixed-transfer eigenvector.

Theorem 7.12 (Eigenvalue bound). *Lean* *✓ Stmt* *Let A, B be normalized MPS tensors. Every eigenvalue μ of F_{AB} satisfies $|\mu| \le 1$.*

Proof. ✓ Proof Expand $F_{AB}^n(X)$ as a sum over words and use Cauchy–Schwarz, together with $\sum_i (A^i)^\dagger A^i = \sum_i (B^i)^\dagger B^i = \mathbb{1}$ (Equation (7.1), applied to both tensors), to obtain a uniform Frobenius-norm bound $\|F_{AB}^n(X)\|_{\mathrm{HS}} \le C_D\|X\|_{\mathrm{HS}}$ for all n, where C_D depends only on D. If $F_{AB}(X) = \mu X$ and $X \neq 0$, then $|\mu|^n\|X\|_{\mathrm{HS}} = \|F_{AB}^n(X)\|_{\mathrm{HS}} \le C_D\|X\|_{\mathrm{HS}}$ for every n. Hence $|\mu| \le 1$.

The argument uses only the trace-preserving normalization (7.1) and does not appeal to Perron–Frobenius theory. □

Theorem 7.13 (Rectangular eigenvalue bound). *Lean* *✓ Stmt* *Let A and B be normalized MPS tensors of bond dimensions $D_1 \ge 1$ and $D_2 \ge 1$. Every eigenvalue μ of F_{AB}^{rect} satisfies $|\mu| \le 1$.*

Proof. ✓ Proof The word-expansion and Frobenius-norm estimate used in Theorem 7.12 apply equally to rectangular matrices: if $F_{AB}^{\mathrm{rect}}(X) = \mu X$ and $X \neq 0$, then $|\mu|^n\|X\|_{\mathrm{HS}} \le D_1\|X\|_{\mathrm{HS}}$ for every n. Hence $|\mu| \le 1$. □

Lemma 7.14 (Spectral radius bound). Lean ✓ Stmt *$\rho_{\text{spec}}(F_{AB}) \le 1$ for normalized tensors.*

Proof. ✓ Proof The spectral radius is the supremum of $|\mu|$ over all eigenvalues, and each satisfies $|\mu| \le 1$ by Theorem 7.12. □

Theorem 7.15 (Spectral radius ≥ 1 implies gauge-phase equivalence). Lean ✓ Stmt *Let A, B be injective normalized MPS tensors. If $\rho_{\text{spec}}(F_{AB}) \ge 1$, then A and B are gauge-phase equivalent.*

Proof. ✓ Proof Since $\rho_{\text{spec}}(F_{AB}) \le 1$ (Lemma 7.14) and $\rho_{\text{spec}}(F_{AB}) \ge 1$ by hypothesis, there exists an eigenvalue μ with $|\mu| = 1$ and eigenvector $X \ne 0$: $F_{AB}(X) = \mu X$. The proof proceeds in five steps.

Step 1 (Separate gauging). Choose positive definite fixed points ρ_A, ρ_B of the transfer maps $\mathcal{E}_A, \mathcal{E}_B$ (Theorem 6.11) and gauge each tensor separately to a unital Kraus family (Theorem 6.13): set $A'^i = \rho_A^{-1/2} A^i \rho_A^{1/2}$ and similarly for B. Let $X' = \rho_A^{-1/2} X \rho_B^{-1/2}$ denote the correspondingly gauged eigenvector.

Step 2 (Block Kraus family and off-diagonal embedding). Form the block Kraus operators $C^i = \left(\begin{smallmatrix} A'^i & 0 \\ 0 & B'^i \end{smallmatrix}\right)$. These constitute a unital Kraus family on $M_{2D}(\mathbb{C})$. The off-diagonal matrix $Y = \left(\begin{smallmatrix} 0 & X' \\ 0 & 0 \end{smallmatrix}\right)$ satisfies $E_C(Y) = \mu Y$.

Step 3 (KS equality and Kraus commutation). Because $|\mu| = 1$ and E_C is unital and trace-preserving, the Kadison–Schwarz equality for peripheral eigenvectors (Theorem 5.24) gives $E_C(Y^\dagger Y) = E_C(Y)^\dagger E_C(Y)$. By the Kraus commutation relation (Theorem 5.23), Y commutes with every block Kraus operator: $C^i Y = Y C^i$ for all i.

Step 4 (Intertwining identity and invertibility). Expanding the commutation relation block by block gives $A'^i X' = X' B'^i$ for every i. Since $\{A'^i\}$ span $M_D(\mathbb{C})$ (injectivity of A), X' is a nonzero intertwiner from $M_D(\mathbb{C})$ to $M_D(\mathbb{C})$. The left multiplicative identity (Theorem 5.25) gives $E_C(Y^\dagger Y) = E_C(Y^\dagger) E_C(Y)$, so $X'^\dagger X'$ is a positive semi-definite fixed point of $\mathcal{E}_{B'}$, positive definite by injectivity. Hence X' is invertible.

Step 5 (Skolem–Noether conclusion). The map $M \mapsto X' M (X')^{-1}$ is a $\mathbb{C}$-algebra automorphism of $M_D(\mathbb{C})$ sending B'^i to A'^i. By Skolem–Noether (Theorem 3.11) this is an inner automorphism. Undoing the separate gauges gives $B^i = \bar{\mu} X A^i X^{-1}$ for all i, establishing gauge-phase equivalence (Definition 2.23).

The gauging is applied to the individual transfer maps $\mathcal{E}_A, \mathcal{E}_B$ separately; the mixed transfer map F_{AB} need not be simultaneously unital and trace-preserving. □

Lemma 7.16 (Gauged peripheral eigenvector yields Kraus intertwining). Lean ✓ Stmt *Let A and B be normalized tensors, and let $X \ne 0$ satisfy $F_{AB}(X) = \mu X$ with $|\mu| = 1$. Choose positive-definite fixed points ρ_A, ρ_B of the transfer maps and gauge to unital families $A'^i = S_A^{-1} A^i S_A$ and $B'^i = S_B^{-1} B^i S_B$ with $S_A S_A^\dagger = \rho_A$ and $S_B S_B^\dagger = \rho_B$. Then the transported matrix $X' = S_A^{-1} X (S_B^\dagger)^{-1}$ is nonzero, the families A' and B' are unital, and for every i one has*

$$X'(B'^i)^\dagger = \mu (A'^i)^\dagger X', \qquad A'^i X' = \mu X' B'^i.$$

Proof. ✓ Proof Block-embed the gauged families and transported eigenvector into a single unital Kraus map, apply Kadison–Schwarz equality for the modulus-one eigenvalue, then extract the intertwining identities from the block structure. □

Lemma 7.17 (Intertwining yields gauge-phase equivalence). Lean ✓ Stmt *Let A and B be tensors of the same bond dimension. Suppose there are invertible gauges taking them to families A' and B', an invertible matrix X', and a scalar μ with $|\mu| = 1$ such that, for every physical index i,*

$$A'^i X' = \mu X' B'^i.$$

Then A and B are gauge-phase equivalent.

Proof. ✓ Proof Compose the two gauge matrices with the intertwiner to obtain a single invertible matrix conjugating A to a scalar multiple of B. □

Remark 7.18 (Two gauge conventions). ✓ Stmt The right-canonical gauge used here, $A'^i = \rho^{-1/2} A^i \rho^{1/2}$, makes $\{A'^i\}$ a *unital* Kraus family ($\sum_i A'^i (A'^i)^\dagger = \mathbb{1}$). A different convention, $\tilde{A}^i = \rho^{1/2} A^i \rho^{-1/2}$, makes the family *trace-preserving* ($\sum_i (\tilde{A}^i)^\dagger \tilde{A}^i = \mathbb{1}$). Both conventions appear in practice: the unital version via direct matrix square-root inversion (used in the transfer-operator gap proof above), and the trace-preserving version (used in the canonical-form reduction of Chapter 9).

Theorem 7.19 (Strict transfer-operator gap)**.** Lean ✓ Stmt *Let A, B be injective normalized MPS tensors that are* not *gauge-phase equivalent. Then $\rho_{\text{spec}}(F_{AB}) < 1$.*

Proof. ✓ Proof $\rho_{\text{spec}}(F_{AB}) \le 1$ by Lemma 7.14. If $\rho_{\text{spec}}(F_{AB}) = 1$, Theorem 7.15 gives gauge-phase equivalence, contradicting the hypothesis. □

7.5 Transfer-operator gap — rectangular case

Lemma 7.20 (Rectangular intertwining forces equal bond dimension)**.** Lean ✓ Stmt *Let A and B be injective tensors of bond dimensions D_1 and D_2. If there exist $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$ and $\mu \in \mathbb{C}$ such that, for every physical index i,*

$$X(B^i)^\dagger = \mu (A^i)^\dagger X, \qquad A^i X = \mu X B^i,$$

then $D_1 = D_2$.

Proof. ✓ Proof The intertwining relation shows that $\ker(X)$ is invariant under all generators of B. Since B is injective, its generators span the full matrix algebra, so $\ker(X)$ is trivial and $D_2 \le D_1$. Applying the same argument to $X^\dagger$ gives $D_1 \le D_2$. □

Theorem 7.21 (Rectangular transfer-operator gap)**.** Lean ✓ Stmt *Let A be an injective normalized tensor of bond dimension D_1 and B an injective normalized tensor of bond dimension D_2, with $D_1 \ne D_2$. Then $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$.*

Proof. ✓ Proof The same word-expansion argument as in Theorem 7.13 gives $|\mu| \le 1$ for every eigenvalue of F_{AB}^{rect}. If $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) = 1$, choose a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$.

Gauge A, B to unital families A', B' and transport X to a nonzero rectangular intertwiner X' with $A'^i X' = \mu X' B'^i$ for every i. This is the hypothesis of Lemma 7.20, so $D_1 = D_2$, contradicting the hypothesis. □

Theorem 7.22 (Rectangular overlap decay)**.** Lean ✓ Stmt *If $D_1 \ne D_2$ and both tensors are injective and normalized, then $O_{AB}(N) \to 0$ as $N \to \infty$.*

Proof. ✓ Proof The transfer-operator gap $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$ (Theorem 7.21) implies $(F_{AB}^{\text{rect}})^N \to 0$, so the operator trace converges to 0. By Lemma 7.7, $O_{AB}(N) = \operatorname{Tr}((F_{AB}^{\text{rect}})^N) \to 0$. □

7.6 MPV overlap decay

Theorem 7.23 (Transfer powers converge to zero)**.** Lean ✓ Stmt *Let A, B be injective normalized tensors that are not gauge-phase equivalent. Then $F_{AB}^n(X) \to 0$ for every $X \in M_D(\mathbb{C})$.*

Proof. ✓ Proof The transfer-operator gap $\rho_{\mathrm{spec}}(F_{AB}) < 1$ (Theorem 7.19) implies $\|F_{AB}^n\|_{\mathrm{op}} \to 0$ by the Gelfand formula, so $F_{AB}^n(X) \to 0$ for every X. □

Theorem 7.24 (Overlap decay). Lean ✓ Stmt *Let A, B be injective normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \to 0$ as $N \to \infty$.*

Proof. ✓ Proof By Lemma 7.6, $O_{AB}(N) = \mathrm{Tr}(F_{AB}^N)$. Expand the operator trace over matrix units (Lemma 7.5): $\mathrm{Tr}(F_{AB}^N) = \sum_{p,q}(F_{AB}^N(E_{pq}))_{pq}$. By Theorem 7.23, each $F_{AB}^N(E_{pq}) \to 0$ entry-wise. Since the sum is finite, $O_{AB}(N) \to 0$. □

7.7 Block separation

The block-separation results below record the overlap asymptotics used in the canonical-form separation of Chapter 9.

Theorem 7.25 (Cross-correlation decay). Lean ✓ Stmt *If A and B are injective normalized tensors that are not gauge-phase equivalent, then for every $X \in M_D(\mathbb{C})$ one has $\mathrm{tr}(F_{AB}^N(X)) \to 0$ as $N \to \infty$.*

Proof. ✓ Proof By Theorem 7.23, $F_{AB}^N(X) \to 0$ in operator norm, so $\mathrm{tr}(F_{AB}^N(X)) \to 0$. □

Theorem 7.26 (Self-correlation persists). Lean ✓ Stmt *If ρ is a fixed point of $\mathcal{E}_A$, then $\mathrm{tr}(\mathcal{E}_A^N(\rho)) = \mathrm{tr}(\rho)$ for all $N \geq 0$. In particular this applies to the distinguished QPF fixed point from Theorem 6.11.*

Proof. ✓ Proof Since ρ is a fixed point, $\mathcal{E}_A^N(\rho) = \rho$ for every N, so $\mathrm{tr}(\mathcal{E}_A^N(\rho)) = \mathrm{tr}(\rho)$. □

7.8 Transfer-operator gap under irreducible-TP hypotheses

The preceding transfer-operator gap results require *injectivity* of both tensors. The following variants weaken this to *irreducibility* combined with the trace-preserving normalization (7.1), following Cirac et al. [CPGSV17] and the irreducibility theory of [Wol12, Section 6.2]. The argument replaces the Kraus-commutation step with a Cauchy–Schwarz rigidity argument for the Perron–Frobenius fixed point.

Theorem 7.27 (Gauge-equivalence from irreducible-TP spectral radius). Lean ✓ Stmt *Let A, B be irreducible trace-preserving normalized MPS tensors of bond dimension D. If $\rho_{\mathrm{spec}}(F_{AB}) \geq 1$, then A and B are gauge-phase equivalent.*

Proof. ✓ Proof Choose a modulus-one eigenvector as in Theorem 7.15. The proof follows the five-step structure of Theorem 7.15, with one key adaptation. Steps 1–3 (separate gauging, block Kraus family, KS equality and Kraus commutation) proceed identically: the gauging uses the unique positive definite fixed points guaranteed by irreducibility (Theorem 6.11) and the right-canonical gauge (Theorem 6.13). The change is in Step 4: the Kadison–Schwarz equality gives $X'^{\dagger}X'$ as a nonzero positive semidefinite fixed point of $\mathcal{E}_{B'}$. Under injectivity this is immediately positive definite (the fixed point is unique up to scaling). Under the weaker hypothesis of irreducibility, Theorem 6.4 upgrades it to positive definite, giving invertibility of X'. Formally, Lemma 7.16 combines the separately gauged eigenvector into the required intertwining relations. Once X' is invertible, Lemma 7.17 upgrades the intertwining to gauge-phase equivalence. □

Theorem 7.28 (Strict transfer-operator gap (irreducible-TP)). Lean ✓ Stmt *Let A, B be irreducible trace-preserving normalized MPS tensors of the same bond dimension that are not gauge-phase equivalent. Then $\rho_{\text{spec}}(F_{AB}) < 1$.*

Proof. ✓ Proof Contrapositive of Theorem 7.27, combined with $\rho_{\text{spec}}(F_{AB}) \leq 1$ (Lemma 7.14). □

Theorem 7.29 (Overlap decay (irreducible-TP)). Lean ✓ Stmt *Let A, B be irreducible trace-preserving normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \to 0$ as $N \to \infty$.*

Proof. ✓ Proof Combine the transfer-operator gap (Theorem 7.28) with the overlap–trace identity (Lemma 7.6). □

Theorem 7.30 (Rectangular transfer-operator gap (irreducible-TP)). Lean ✓ Stmt *Let A (bond dimension D_1) and B (bond dimension D_2) be irreducible trace-preserving normalized tensors with $D_1 \neq D_2$. Then $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$.*

Proof. ✓ Proof Assume for contradiction that $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) = 1$. Choose a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C})$. Gauge A, B separately to unital families A', B' and transport X to $X' \neq 0$. The matrix $X'X'^{\dagger}$ ($D_1 \times D_1$) is a nonzero positive-semidefinite fixed point of $\mathcal{E}_{A'}$, so by Theorem 6.4 it is positive definite, hence invertible; thus X' has full row rank and $D_1 \leq D_2$. Symmetrically $X'^{\dagger}X'$ ($D_2 \times D_2$) is positive definite, giving full column rank and $D_2 \leq D_1$. Hence $D_1 = D_2$, contradicting the hypothesis. □

Theorem 7.31 (Rectangular overlap decay (irreducible-TP)). Lean ✓ Stmt *If $D_1 \neq D_2$ and both tensors are irreducible and trace-preserving normalized, then $O_{AB}(N) \to 0$ as $N \to \infty$.*

Proof. ✓ Proof Combine Theorem 7.30 with Lemma 7.7. □

7.9 Conditional expectation from a faithful fixed point

This section follows [Wol12, Section 6.4]; the conditional expectation is built from the fixed-point algebra of [Wol12, Theorem 6.14] via [Wol12, (1.40)]. The scalar conditional expectation E_σ is the special case where the fixed-point $*$-subalgebra is the scalar algebra $\mathbb{C} \cdot \mathbb{1}$ (the primitive-channel case). The general irreducible case with nontrivial period requires the Wedderburn block decomposition of [Wol12, Theorem 6.14].

Definition 7.32 (Conditional expectation). Lean ✓ Stmt Given a $*$-subalgebra $S \subseteq M_D(\mathbb{C})$, a linear map $P : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is a conditional expectation onto S if it is idempotent, unital, has range contained in S, and satisfies $P(X) = X$ for all $X \in S$.

Definition 7.33 (Scalar conditional expectation). Lean ✓ Stmt For $\sigma \geq 0$ with $\operatorname{tr}(\sigma) \neq 0$, define

$$E_\sigma(X) := \frac{\operatorname{tr}(\sigma X)}{\operatorname{tr}(\sigma)} \mathbb{1}.$$

Lemma 7.34 (Evaluation formula). Lean ✓ Stmt $E_\sigma(X) = \dfrac{\operatorname{tr}(\sigma X)}{\operatorname{tr}(\sigma)} \mathbb{1}$.

Lemma 7.35 (Unitality). Lean ✓ Stmt $E_\sigma(\mathbb{1}) = \mathbb{1}$.

Lemma 7.36 (Idempotence). Lean ✓ Stmt *$E_\sigma(E_\sigma(X)) = E_\sigma(X)$ for all X.*

Lemma 7.37 (Range is scalar). Lean ✓ Stmt *For every X, there exists $c \in \mathbb{C}$ with $E_\sigma(X) = c\,\mathbb{1}$.*

Lemma 7.38 (Fixes scalar operators). Lean ✓ Stmt *For $c \in \mathbb{C}$, $E_\sigma(c\,\mathbb{1}) = c\,\mathbb{1}$.*

Lemma 7.39 (Absorbs the adjoint map). Lean ✓ Stmt *If $T(\sigma) = \sigma$, then $E_\sigma \circ T^* = E_\sigma$.*

Lemma 7.40 (Adjoint map absorbs scalar projection). Lean ✓ Stmt *If T is trace-preserving, then $T^* \circ E_\sigma = E_\sigma$.*

Lemma 7.41 (Image lies in adjoint fixed points). Lean ✓ Stmt *For all X, $E_\sigma(X)$ is fixed by T^*.*

Theorem 7.42 (Scalar map is a conditional expectation). Lean ✓ Stmt *Let K be trace-preserving, let $\rho > 0$ be a positive-definite fixed point of the associated channel $T(\rho) = \sum_i K_i \rho K_i^\dagger$ (so $T(\rho) = \rho$), and assume every adjoint fixed point of T^* is a scalar multiple of $\mathbb{1}$. Then E_ρ is a conditional expectation onto the adjoint fixed-point $*$-subalgebra.*

Lemma 7.43 (Complete positivity of the scalar conditional expectation). Lean ✓ Stmt *For any positive semidefinite σ, the scalar conditional expectation E_σ is a completely positive map.*

Proof. ✓ Proof Write $S = \sqrt{\sigma}$ and define the Kraus family $K_{j,i} = |j\rangle\langle i|\,S$ indexed by $(j,i) \in \{1,\dots,D\}^2$. Using $|j\rangle\langle i|\,M\,|i\rangle\langle j| = M_{ii}|j\rangle\langle j|$ and summing first over i and then over j,

$$\sum_{j,i} K_{j,i}\,X\,K_{j,i}^\dagger = \sum_j |j\rangle\langle j|\,\mathrm{tr}(SXS) = \mathrm{tr}(\sigma X)\mathbb{1},$$

where we used cyclicity of the trace and $S^2 = \sigma$. Hence $E_\sigma = (\mathrm{tr}(\sigma))^{-1} \cdot (X \mapsto \sum_{j,i} K_{j,i} X K_{j,i}^\dagger)$ is the Kraus map scaled by the non-negative real number $(\mathrm{tr}(\sigma))^{-1}$, and therefore completely positive. □

Lemma 7.44 (σ-trace preservation of the scalar conditional expectation). Lean ✓ Stmt *If $\mathrm{tr}(\sigma) \neq 0$, then for every X*

$$\mathrm{tr}(\sigma\,E_\sigma(X)) = \mathrm{tr}(\sigma X).$$

Proof. ✓ Proof Direct computation: $\mathrm{tr}(\sigma \cdot \frac{\mathrm{tr}(\sigma X)}{\mathrm{tr}(\sigma)}\mathbb{1}) = \frac{\mathrm{tr}(\sigma X)}{\mathrm{tr}(\sigma)}\,\mathrm{tr}(\sigma) = \mathrm{tr}(\sigma X)$. □

7.10 Stationary support

This section develops the support-projection theory behind stationary states, following [Wol12, Section 6.4, Propositions 6.10–6.11].

Lemma 7.45 (Lower-triangular Kraus condition implies corner invariance). Lean ✓ Stmt *Let A be an MPS tensor and let P be an orthogonal projection. If, for every physical index i,*

$$(\mathbb{1} - P)A^i P = 0$$

then

$$P\mathcal{E}_A(PXP)P = \mathcal{E}_A(PXP)$$

for all $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof Expand the transfer map and use $(\mathbb{1} - P)A^i P = 0$ to reduce each Kraus term to the P-corner. □

Lemma 7.46 (Support projection invariance). Lean ✓ Stmt *Let E be a quantum channel and $\rho \geq 0$ a positive semidefinite fixed point ($E(\rho) = \rho$). Let P denote the support projection of ρ. Then*

$$PE(PXP)P = E(PXP)$$

for all $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof Write E in Kraus form $E(X) = \sum_i K_i X K_i^\dagger$. From $E(\rho) = \rho$ and $\rho \geq 0$ one deduces $(\mathbb{1} - P)K_i P = 0$ for all i. Apply Lemma 7.45. □

Definition 7.47 (Stationary state). Lean ✓ Stmt For an irreducible channel E on $M_D(\mathbb{C})$ (with $D \geq 1$), the *stationary state* is the unique density-matrix fixed point ρ_∞ satisfying $E(\rho_\infty) = \rho_\infty$, $\rho_\infty > 0$, and $\operatorname{tr}(\rho_\infty) = 1$. Uniqueness follows from the quantum Perron–Frobenius theorem (Theorem 6.11).

Definition 7.48 (Stationary support). Lean ✓ Stmt The *stationary support* of an irreducible channel E is the support projection of the stationary state (Definition 7.47).

Theorem 7.49 (Stationary support is full). Lean ✓ Stmt *For an irreducible channel E, the stationary support equals the identity:* $\operatorname{supp}(\rho_\infty) = \mathbb{1}$.

Proof. ✓ Proof The support projection P of ρ_∞ is an orthogonal projection satisfying $PE(PXP)P = E(PXP)$ for all X (Lemma 7.46). By irreducibility, $P = 0$ or $P = \mathbb{1}$. Since $\rho_\infty \neq 0$, we have $P \neq 0$, so $P = \mathbb{1}$. □

Remark 7.50 (Stationary-support formulations). The non-trivial formulations needed here are the following results from Wolf:

- the converse direction of [Wol12, Proposition 6.10]: if the support projection of every positive semidefinite fixed point is full, then the channel is irreducible;
- [Wol12, Proposition 6.11]: minimality of the stationary support among invariant projections, without assuming irreducibility.

These statements are not used elsewhere in this chapter.

7.10.1 Faithful compression onto the support sector

This subsection establishes the results on the corner algebra, $*$-structure, and compression of a PSD fixed point onto its support sector needed for [Wol12, Corollary 6.6]: the corner carries a $\mathbb{C}$-algebra and $*$-structure, the two standard presentations of the corner are identified, and the compression of a PSD fixed point onto its support sector is positive definite.

Lemma 7.51 (Support projection annihilates the kernel). Lean ✓ Stmt *Let $\rho \succeq 0$ on $\mathbb{C}^D$ with support projection $P = \operatorname{supp}(\rho)$. For every $v \in \mathbb{C}^D$, if $\rho\, v = 0$, then $P\, v = 0$.*

Proof. ✓ Proof Diagonalize $\rho = U \operatorname{diag}(\lambda)\, U^\dagger$ with $\lambda_j \geq 0$. Writing $w := U^\dagger v$, the hypothesis $\rho v = 0$ gives $\lambda_j\, w_j = 0$ for every j; hence $w_j = 0$ whenever $\lambda_j > 0$. The support projection acts as $\operatorname{diag}(\chi_{\lambda_j > 0})$ in the eigenbasis, so $P\, v = U \operatorname{diag}(\chi_{\lambda_j > 0})\, w = 0$. □

Definition 7.52 (Corner submodule / idempotent-corner identification). Lean ✓ Stmt For an idempotent $P \in M_D(\mathbb{C})$, the subspace $\{X \mid PXP = X\}$ and the corner $\{P\,X\,P \mid X \in M_D(\mathbb{C})\}$ are identified as $\mathbb{C}$-linear spaces by the identity on underlying matrices.

Remark 7.53 ($\mathbb{C}$-algebra and $*$-structure on the corner). Lean Lean Lean Lean ✓ Stmt The corner algebra associated to an idempotent already carries a ring structure with unit P. In the matrix case it also has $\mathbb{C}$-module and $\mathbb{C}$-algebra structures and, under the additional assumption $P^\dagger = P$, star, star ring, and star module structures over $\mathbb{C}$.

Theorem 7.54 (Faithful compression onto the support)**.** Lean ✓ Stmt *Let $\rho \succeq 0$ on $\mathbb{C}^D$ with support projection $P = \mathrm{supp}(\rho)$, and let $V : \mathbb{C}^D \to \mathbb{C}^k$ be a compression isometry satisfying $V V^\dagger = \mathbb{1}_k$ and $V^\dagger V = P$. Then the compression*

$$V \rho V^\dagger \in M_k(\mathbb{C})$$

is positive definite.

Proof. ✓ Proof Hermiticity follows from Hermiticity of ρ and the product-star identity. For strict positivity, let $w \neq 0$, set $u := V^\dagger w$, and note that $VV^\dagger = \mathbb{1}_k$ forces $u \neq 0$, while $V^\dagger V = P$ forces $Pu = u$. The quadratic form computes as $w^\dagger(V\rho V^\dagger)w = u^\dagger \rho u$, which is non-negative by positive semidefiniteness. If it vanishes, then $\rho u = 0$, and by Lemma 7.51 also $Pu = 0$, contradicting $Pu = u \neq 0$. □

7.11 Wedderburn decomposition of the fixed-point algebra

This section proves the structure theorem for the fixed-point algebra of a trace-preserving Kraus map, following [Wol12, Theorems 6.12–6.14]. The proof uses the Jacobson radical characterization of semisimplicity together with the Wedderburn–Artin structure theorem for finite-dimensional semisimple algebras over an algebraically closed field.

Theorem 7.55 (Fixed-point algebra is semisimple)**.** Lean ✓ Stmt *The adjoint-fixed-point $*$-subalgebra of a trace-preserving Kraus map on $M_D(\mathbb{C})$ is a semisimple ring.*

Proof. ✓ Proof The subalgebra is finite-dimensional (as a subalgebra of $M_D(\mathbb{C})$), hence Artinian. Following [Wol12, Theorem 6.14], it suffices to show the Jacobson radical J vanishes. If $x \in J$, then $x^*x \in J$ (left-ideal closure). The radical of an Artinian ring is nilpotent, so x^*x is nilpotent. A self-adjoint nilpotent matrix satisfies $\|x^*x\|^{2^k} = \|(x^*x)^{2^k}\| = 0$, hence $x^*x = 0$, and the C*-identity gives $x = 0$. □

Theorem 7.56 (Abstract Wedderburn–Artin decomposition)**.** Lean ✓ Stmt *There exist $n \in \mathbb{N}$ and dimensions $d_1, \dots, d_n \geq 1$ such that the adjoint-fixed-point $*$-subalgebra is $\mathbb{C}$-algebra isomorphic to*

$$\prod_{k=1}^{n} M_{d_k}(\mathbb{C}).$$

Proof. ✓ Proof Combine semisimplicity (Theorem 7.55) with the Wedderburn–Artin structure theorem for finite-dimensional semisimple algebras over an algebraically closed field. □

Theorem 7.57 (Fixed-point algebra decomposition with multiplicities)**.** Lean ✓ Stmt *There exist integers n, block sizes $d_1, \dots, d_n \geq 1$, multiplicities $m_1, \dots, m_n \geq 1$, and a $\mathbb{C}$-algebra isomorphism from the adjoint-fixed-point $*$-subalgebra to*

$$\prod_{k=1}^{n} M_{d_k}(\mathbb{C})$$

such that

$$\sum_{k=1}^{n} d_k m_k \leq D.$$

Proof. ✓ Proof Apply the abstract Wedderburn decomposition (Theorem 7.56). Since the adjoint-fixed-point algebra is realized as a subalgebra of the ambient matrix algebra $M_D(\mathbb{C})$, this realization also yields multiplicities m_k and the bound $\sum_k d_k m_k \leq D$. □

Remark 7.58 (Block form of the fixed-point algebra). Wolf's block form identifies the adjoint-fixed-point algebra, up to a unitary change of basis, with

$$0 \oplus \bigoplus_k M_{d_k}(\mathbb{C}) \otimes \mathbb{1}_{m_k}$$

together with the density-block form $M_{d_k}(\mathbb{C}) \otimes \rho_k$. The theorem above gives the product decomposition, multiplicities, and dimension bound needed from this block form.

Theorem 7.59 (Weighted fixed points form a $*$-subalgebra)**.** Lean ✓ Stmt *Let $T(X) = \sum_i K_i X K_i^\dagger$ be trace-preserving, and let $\rho > 0$ satisfy $T(\rho) = \rho$. If*

$$F_T := \{X \in M_D(\mathbb{C}) \mid T(X) = X\},$$

then $\rho^{-1/2} F_T \rho^{-1/2}$ is a $*$-subalgebra of $M_D(\mathbb{C})$.

Proof. ✓ Proof Set $B_i = \rho^{-1/2} K_i \rho^{1/2}$. The equation $T(\rho) = \rho$ implies that the family $\{B_i\}$ is unital, and trace preservation of T implies that ρ is a positive-definite fixed point of the adjoint map of the gauged family. Apply Theorem 4.60 to B. The identity

$$\sum_i B_i X B_i^\dagger = X \iff T(\rho^{1/2} X \rho^{1/2}) = \rho^{1/2} X \rho^{1/2}$$

identifies the fixed points of the gauged map with $\rho^{-1/2} F_T \rho^{-1/2}$. □

7.12 Further irreducibility and primitivity equivalences

This section collects several criteria and equivalences from [Wol12, Theorems 6.2, 6.7, 6.8] that are used later.

Theorem 7.60 (Exponential positivity condition)**.** Lean Lean ✓ Stmt *Let E be an irreducible completely positive map, $A \geq 0$ with $A \neq 0$, and $t > 0$. Then $\exp(tE)(A)$ is positive definite. Conversely, if $\exp(tE)(A)$ is positive definite for every such t, A, then E is irreducible (full equivalence* Lean *).*

Theorem 7.61 (Kraus-span equivalence)**.** Lean ✓ Stmt *Under normalization, primitivity in Wolf's sense is equivalent to eventual full Kraus rank.*

Theorem 7.62 (Combined primitivity equivalence)**.** Lean ✓ Stmt *Under normalization, primitivity in Wolf's sense is equivalent to the conjunction of eventual full Kraus rank, normality, and strong irreducibility.*

Remark 7.63 (Pairwise primitivity equivalences). Lean Lean Lean Lean ✓ Stmt These are the pairwise equivalences in [Wol12, Theorem 6.8].

Theorem 7.64 (Scalar fixed-point algebra case)**.** Lean ✓ Stmt *In the scalar fixed-point algebra setting, the weighted map $X \mapsto \frac{\mathrm{tr}(\rho X)}{\mathrm{tr}(\rho)} \mathbb{1}$ is a conditional expectation onto the adjoint fixed-point $*$-subalgebra.*

7.13 Multi-cycle block-permutation structure

This section develops the block-permutation structure underlying the cycle decomposition of [Wol12, Theorem 6.16]. The multi-cycle decomposition refines the single-cycle case to handle arbitrary peripheral periods.

Definition 7.65 (Block-permutation structure). Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$. A *block-permutation structure* for T consists of a finite index set ι, a family of orthogonal projections $P_k \in M_D(\mathbb{C})$ for $k \in \iota$, and a permutation $\sigma \in \mathrm{Sym}(\iota)$ (not necessarily a single cycle—in general σ may have multiple disjoint cycles), such that

$$T(P_{\sigma(k)}) = P_k, \quad T(P_k X) = T(P_k)\,T(X), \quad T(X P_k) = T(X)\,T(P_k)$$

for every $k \in \iota$ and $X \in M_D(\mathbb{C})$.

Definition 7.66 (Multi-cycle block-permutation structure). Lean ✓ Stmt A *multi-cycle decomposition* of $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ consists of a finite cycle index ι, a per-cycle period $m : \iota \to \mathbb{N}_{>0}$, a projection family $P_{c,k} \in M_D(\mathbb{C})$ for $c \in \iota$ and $k \in \{0, \dots, m(c)-1\}$ consisting of orthogonal projections. For every $c \in \iota$ and $k \in \{0, \dots, m(c)-1\}$, the per-cycle cyclic action is

$$T(P_{c,k+1}) = P_{c,k}.$$

The multiplicative-domain factorisations are $T(P_{c,k}\,X) = T(P_{c,k})\,T(X)$ and $T(X\,P_{c,k}) = T(X)\,T(P_{c,k})$.

Lemma 7.67 (Per-cycle corner preservation). Lean ✓ Stmt *For every cycle $c \in \iota$ and index $k \in \{0, \dots, m(c)-1\}$, the iterate $T^{m(c)}$ preserves the corner $P_{c,k} M_D(\mathbb{C})\,P_{c,k}$.*

Proof. ✓ Proof By induction on n, the n-th iterate T^n sends $P_{c,k+n}\,X\,P_{c,k+n}$ to $P_{c,k}\,T^n(X)\,P_{c,k}$, using the multiplicative-domain factorisations and the cyclic action $T(P_{c,j+1}) = P_{c,j}$ at each induction step. Specialising to $n = m(c)$ and using that the indices $k+m(c) = k$ in $\{0, \dots, m(c)-1\}$ yields corner preservation of $T^{m(c)}$. □

Lemma 7.68 (Common-period corner preservation). Lean ✓ Stmt *If $N \in \mathbb{N}$ is divisible by every per-cycle period $m(c)$, then T^N preserves every corner $P_{c,k} M_D(\mathbb{C})\,P_{c,k}$ of the decomposition.*

Proof. ✓ Proof Write $N = m(c)\,q$. Corner preservation is stable under taking powers, so $T^N = (T^{m(c)})^q$ preserves each corner by Lemma 7.67. □

Definition 7.69 (Flattening to a single-index block-permutation structure). Lean ✓ Stmt A multi-cycle decomposition gives a single-index block-permutation structure on the sigma index $\Sigma_{c\in\iota}\{0, \dots, m(c)-1\}$: the permutation is the sigma-product of per-cycle cyclic shifts $k \mapsto k+1$ (whose cycle decomposition has one cycle per $c \in \iota$), the projections are $(c, k) \mapsto P_{c,k}$, and the multiplicative-domain factorisations descend componentwise. The resulting map forgets the explicit cycle-indexing.

7.14 Peripheral eigenvalue group structure

This section proves the peripheral spectrum structure of [Wol12, Theorem 6.6] for irreducible Schwarz maps: the peripheral eigenvalues form a finite cyclic group. The trace-preserving refinement that the order divides the bond dimension is proved in Theorem 7.73.

Lemma 7.70 (Peripheral eigenvectors are invertible). Lean ✓ Stmt *Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. If $X \neq 0$ satisfies $E(X) = \mu X$ with $|\mu| = 1$, then X is invertible.*

Proof. ✓ Proof The KS equality gives $E(X^\dagger X) = X^\dagger X$, so $X^\dagger X$ is a nonzero PSD fixed point. Irreducibility upgrades it to positive definite, hence invertible, so $\det(X) \neq 0$. □

Lemma 7.71 (Peripheral eigenvalues: product closure). Lean ✓ Stmt *Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. If μ, ν are peripheral eigenvalues of E, then so is $\mu\nu$.*

Proof. ✓ Proof Take eigenvectors X, Y for μ, ν. The KS equality at X puts X in the multiplicative domain, so $E(YX) = E(Y)E(X) = (\nu\mu)(YX)$. Since X, Y are units (Lemma 7.70), $YX \neq 0$ and $|\mu\nu| = 1$. □

Theorem 7.72 (Peripheral eigenvalues: cyclic structure). Lean ✓ Stmt *Let E be an irreducible unital Schwarz map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point (trace-preservation is* not *required). Then there exist $m \geq 1$ and a primitive m-th root of unity γ such that the peripheral eigenvalues of E are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

Proof. ✓ Proof The peripheral eigenvalues form a finite subset of $\mathbb{C}$, contain 1, and are closed under multiplication and inversion, so they form a finite subgroup of $\mathbb{C}^\times$. Any finite subgroup of $\mathbb{C}^\times$ is cyclic. The generator γ has order $m = |\text{peripheral spectrum}|$. □

Theorem 7.73 (Peripheral eigenvalues form a cyclic group). Lean ✓ Stmt *Let E be an irreducible unital trace-preserving Schwarz map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. Then there exist $m \geq 1$ and a primitive m-th root of unity γ such that $m \mid D$ and the peripheral eigenvalues of E are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

Proof. ✓ Proof Apply Theorem 7.72 for the cyclic structure. Divisibility $m \mid D$ follows from the cyclic decomposition: m mutually orthogonal projections summing to $\mathbb{1}$ each have equal trace (by trace preservation), so $m \cdot \mathrm{tr}(P_0) = D$ with $\mathrm{tr}(P_0) \in \mathbb{N}$. □

Theorem 7.74 (Period divides bond dimension). Lean ✓ Stmt *Let E be an irreducible unital trace-preserving Schwarz map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. Let γ be a primitive m-th root of unity such that the peripheral eigenvalues are exactly $\{\gamma^k : k \in \{0, \dots, m-1\}\}$. Then $m \mid D$.*

Proof. ✓ Proof The cyclic decomposition gives orthogonal projections $P_0, \dots, P_{m-1}$ with $\sum_k P_k = \mathbb{1}$ and $E(P_{k+1}) = P_k$. Trace preservation gives $\mathrm{tr}(P_{k+1}) = \mathrm{tr}(P_k)$ for every k, hence

$$D = \mathrm{tr}(\mathbb{1}) = \sum_{k=0}^{m-1} \mathrm{tr}(P_k) = m\ \mathrm{tr}(P_0).$$

Since the trace of an orthogonal projection is an integer, $m \mid D$. □

7.14.1 Cyclic decomposition of irreducible Schwarz maps

Definition 7.75 (Corner preservation). Lean ✓ Stmt For an orthogonal projection P and linear map E, the corner $PM_D(\mathbb{C})P$ is preserved if $E(PXP) = PE(PXP)P$ for all X.

Definition 7.76 (Corner subspace). Lean ✓ Stmt The corner subspace associated with P is $\{PXP : X \in M_D(\mathbb{C})\}$.

Definition 7.77 (Restricted corner map). Lean ✓ Stmt The restriction of E to an invariant corner $PM_D(\mathbb{C})P$.

Definition 7.78 (Irreducible restriction on a corner). Lean ✓ Stmt Irreducibility of the corner-restricted map.

Definition 7.79 (Rank of an orthogonal projection). Lean ✓ Stmt The rank of an orthogonal projection $P \in M_D(\mathbb{C})$, i.e. the dimension of the corner algebra $PM_D(\mathbb{C})P$.

Lemma 7.80 (Compression isometry for an orthogonal projection). Lean ✓ Stmt *Let $P \in M_D(\mathbb{C})$ be an orthogonal projection of rank n. The corner algebra $PM_D(\mathbb{C})P$ is linearly isomorphic to the full matrix algebra $M_n(\mathbb{C})$. The isomorphism is built from the spectral diagonalisation of P by conjugating the top-left $n \times n$ block by the unitary diagonalising P.*

Lemma 7.81 (Rank equals trace for an orthogonal projection). Lean ✓ Stmt *The rank of an orthogonal projection P equals its trace: $n = \operatorname{tr} P$.*

Lemma 7.82 (Scalar fixed points for irreducible unital maps). Lean ✓ Stmt *If E is irreducible and unital, then every fixed point of E is a scalar multiple of $\mathbb{1}$.*

Proof. ✓ Proof If $E(X) = X$, then the Hermitian matrices

$$X + X^\dagger, \qquad i(X - X^\dagger)$$

are also fixed by E. The positive-semidefinite fixed-point uniqueness theorem gives $X + X^\dagger = a\mathbb{1}$ and $i(X - X^\dagger) = b\mathbb{1}$, hence

$$X = \frac{1}{2}(a - ib)\mathbb{1}.$$

□

Lemma 7.83 (Peripheral unitary eigenvector). Lean ✓ Stmt *For an irreducible unital Schwarz map with faithful adjoint fixed point, each peripheral eigenvalue admits a unitary eigenvector.*

Lemma 7.84 (Powers on a peripheral-unitary orbit). Lean ✓ Stmt *If $E(U) = \mu U$ with $|\mu| = 1$ and U unitary, then $E(U^n) = \mu^n U^n$ for all n.*

Lemma 7.85 (Normalized peripheral unitary). Lean ✓ Stmt *Peripheral unitary eigenvectors can be normalized compatibly with the faithful fixed-point state.*

Lemma 7.86 (Cyclic projections from a peripheral unitary). Lean ✓ Stmt *A primitive m-th peripheral root yields m orthogonal projections summing to $\mathbb{1}$ and cyclically permuted by the channel.*

Theorem 7.87 (Cyclic decomposition for irreducible Schwarz maps). Lean ✓ Stmt *Under irreducibility and faithful fixed point hypotheses, there is a full cyclic projection decomposition realizing the peripheral period.*

Lemma 7.88 (Power maps preserve each cyclic sector). Lean ✓ Stmt *Appropriate powers of the channel preserve each sector corner.*

Lemma 7.89 (Irreducibility of restricted sector dynamics). Lean ✓ Stmt *Each sector restriction is irreducible.*

Lemma 7.90 (Primitivity of restricted sector dynamics). Lean ✓ Stmt *Each sector restriction is primitive.*

Lemma 7.91 (Corner invariance at the period). Lean ✓ Stmt *The channel raised to the period preserves every cyclic corner.*

7.14.2 Group structure of the peripheral spectrum

Lemma 7.92 (Peripheral product closure). Lean ✓ Stmt *The product of two peripheral eigenvalues is peripheral.*

Lemma 7.93 (Peripheral inverse closure). Lean ✓ Stmt *The inverse of a peripheral eigenvalue is peripheral.*

Theorem 7.94 (Peripheral eigenvalue multiplicity one). Lean ✓ Stmt *Each peripheral eigenvalue has algebraic multiplicity one.*

Proof. ✓ Proof Given a peripheral unitary eigenvector U for γ, the multiplicative-domain identity gives $E(XU^\dagger) = XU^\dagger$ for any γ-eigenvector X. By the scalar fixed-point lemma for irreducible unital maps, $XU^\dagger = c \cdot \mathbb{1}$, hence $X = c \cdot U$ and the eigenspace is one-dimensional. ☐

7.15 Primitive overlap convergence (complementary transfer-map gap form)

The theorems in this section are stated directly in terms of the complementary map $E - P$. This is the analytic form needed for the later overlap argument, corresponding to the primitive-case specialization of [Wol12, Theorem 6.7] where T_ϕ is a rank-one projection. For primitive normalized transfer maps, Chapter 4 supplies this complementary transfer-map input under explicit hypotheses.

Theorem 7.95 (Complementary transfer-map gap implies trace convergence to 1). Lean ✓ Stmt *Let E be trace-preserving with fixed point ρ ($\mathrm{tr}(\rho) \neq 0$) and let P be the fixed-point projection (Definition 4.28). If $\rho_{\mathrm{spec}}(E - P) < 1$, then $\mathrm{Tr}(E^n) \to 1$ as $n \to \infty$.*

Proof. ✓ Proof By the power decomposition (Theorem 4.29), $E^n = P + (E - P)^n$ for $n \geq 1$. Since $\rho_{\mathrm{spec}}(E - P) < 1$, the Gelfand formula gives $(E - P)^n \to 0$, so $\mathrm{Tr}(E^n) \to \mathrm{Tr}(P)$. The operator trace of P equals 1: since $P(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)}\rho$, we have $\mathrm{Tr}(P) = \frac{\mathrm{tr}(\rho)}{\mathrm{tr}(\rho)} = 1$. ☐

Theorem 7.96 (Self-overlap convergence from a complementary transfer-map gap). Lean ✓ Stmt *Let A be a normalized MPS tensor with a nonzero PSD fixed point ρ of the transfer map. If $\rho_{\mathrm{spec}}(\mathcal{E}_A - P) < 1$ (where P is the fixed-point projection), then $O_{AA}(N) \to 1$ as $N \to \infty$.*

Proof. ✓ Proof By Lemma 7.6 (with $A = B$), $O_{AA}(N) = \mathrm{Tr}(\mathcal{E}_A^N)$. By Theorem 7.95, $\mathrm{Tr}(\mathcal{E}_A^N) \to 1$. ☐

Chapter 8

Wielandt Bound

This chapter proves a quantum analog of Wielandt's inequality: for a normal MPS tensor with bond dimension D, the cumulative span of matrix products stabilises after at most D^2 steps. The results follow [SPGWC10].

8.1 Cumulative span

Definition 8.1 (Word span). Lean ✓ Stmt The *word span* at length n is the subspace

$$S_n(A) = \mathrm{span}_{\mathbb{C}}\{A^w : |w| = n\} \subseteq M_D(\mathbb{C}).$$

This is [SPGWC10, Section II].

Definition 8.2 (Cumulative span). Lean ✓ Stmt The *cumulative span* at level n is

$$T_n(A) = S_0(A) + S_1(A) + \cdots + S_n(A).$$

Lemma 8.3 (Monotonicity of cumulative span). Lean ✓ Stmt *$T_n(A) \leq T_{n+1}(A)$ for all n.*

Proof. ✓ Proof T_{n+1} includes all words of length $\leq n+1$, which contains all words of length $\leq n$. ☐

Lemma 8.4 (Dimension bound). Lean ✓ Stmt *$\dim T_n(A) \leq D^2$ for all n.*

Proof. ✓ Proof $T_n(A) \subseteq M_D(\mathbb{C})$ and $\dim M_D(\mathbb{C}) = D^2$. ☐

Theorem 8.5 (Cumulative span stabilisation). Lean ✓ Stmt *If $T_n(A) = T_{n+1}(A)$, then $T_m(A) = T_n(A)$ for all $m \geq n$.*

Proof. ✓ Proof If $T_n = T_{n+1}$, then $S_{n+1} \subseteq T_n$. Left-multiplying any element of S_{n+1} by A^i gives an element of S_{n+2}, so $S_{n+2} \subseteq T_{n+1} = T_n$. By induction, $S_m \subseteq T_n$ for all $m > n$, hence $T_m = T_n$. ☐

Theorem 8.6 (Dimension growth). Lean ✓ Stmt *If $T_n(A) \neq T_{n+1}(A)$, then $\dim T_{n+1}(A) > \dim T_n(A)$.*

Proof. ✓ Proof $T_n \leq T_{n+1}$ (Lemma 8.3) and $T_n \neq T_{n+1}$ imply strict submodule inclusion, hence strict rank increase. ☐

Lemma 8.7 (Word span characterises block injectivity). Lean ✓ Stmt *$S_L(A) = M_D(\mathbb{C})$ if and only if A is L-block injective.*

Proof. ✓ Proof Both conditions assert that the span of all products of length L equals $M_D(\mathbb{C})$. □

Lemma 8.8 (Zero-length word span). Lean ✓ Stmt *If $D \geq 2$, then $S_0(A) \neq M_D(\mathbb{C})$.*

Proof. ✓ Proof The zero-length products consist only of the identity, so $S_0(A) = \mathrm{span}_{\mathbb{C}}\{\mathbb{1}\}$ has dimension 1. Since $\dim M_D(\mathbb{C}) = D^2 \geq 4$, this span cannot be all of $M_D(\mathbb{C})$. □

Corollary 8.9 (0-block injectivity). Lean ✓ Stmt *If $D \geq 1$ and A is 0-block injective, then $D = 1$.*

Proof. ✓ Proof By Lemma 8.7, 0-block injectivity gives $S_0(A) = M_D(\mathbb{C})$. Lemma 8.8 excludes $D \geq 2$, hence the nonzero bond dimension must be 1. □

Lemma 8.10 (Two-sided span from one nonzero matrix). Lean ✓ Stmt *If $C \in M_D(\mathbb{C})$ is nonzero, then*

$$\mathrm{span}_{\mathbb{C}}\{RCS : R, S \in M_D(\mathbb{C})\} = M_D(\mathbb{C}). \tag{8.1}$$

This is the matrix-factorization input used in [PGVWC07, Lemma 3].

Proof. ✓ Proof Choose a nonzero entry C_{pq}. Then every matrix unit E_{ij} is a scalar multiple of $E_{ip}CE_{qj}$, so the span in (8.1) contains the standard matrix basis. □

Lemma 8.11 (Exact words around a nonzero middle matrix). Lean ✓ Stmt *If A is L-block injective and $X \neq 0$, then*

$$\mathrm{span}_{\mathbb{C}}\{A^u X A^v : |u| = |v| = L\} = M_D(\mathbb{C}).$$

Proof. ✓ Proof By L-block injectivity, the length-L word products span $M_D(\mathbb{C})$. Bilinearity of $(R, S) \mapsto RXS$ reduces the preceding lemma to products with R and S chosen among those word products. □

8.2 Fitting decomposition

Definition 8.12 (Fitting decomposition). Lean ✓ Stmt A *Fitting decomposition* of a linear endomorphism $f : V \to V$ on a finite-dimensional vector space over an algebraically closed field consists of:

1. f is nilpotent on the generalized 0-eigenspace,
2. f is invertible on each generalized μ-eigenspace for $\mu \neq 0$,
3. the generalized eigenspaces span V,
4. the generalized eigenspaces are linearly independent.

Theorem 8.13 (Fitting decomposition exists). Lean ✓ Stmt *Every linear endomorphism on a finite-dimensional vector space over an algebraically closed field admits a Fitting decomposition.*

Proof. ✓ Proof Nilpotency on the zero generalized eigenspace follows from the definition of generalized eigenspaces. Invertibility on nonzero generalized eigenspaces follows because $f - \mu$ is nilpotent there, so $f = \mu(1 - (1 - f/\mu))$ is invertible. Spanning and independence are standard results for generalized eigenspaces over algebraically closed fields. In [SPGWC10] and [Wol12, Lemma 6.3(b)], the Jordan normal form is used directly. The argument is phrased in terms of generalized eigenspaces instead. □

Theorem 8.14 (Nilpotent power bound). Lean ✓ Stmt *If f is nilpotent on a space of dimension n, then $f^n = 0$.*

Proof. ✓ Proof A nilpotent endomorphism on an n-dimensional space has nilpotency index $\leq n$. ☐

8.3 Nonzero trace product

Theorem 8.15 (Cumulative span reaches $M_D(\mathbb{C})$ — explicit bound). Lean ✓ Stmt *If A is normal, then there exists $n \leq D^2$ with $T_n(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof By Theorem 8.6, if $T_n \neq T_{n+1}$ then $\dim T_{n+1} > \dim T_n$. Since $\dim T_n \leq D^2$ (Lemma 8.4), after at most D^2 steps either $T_n = M_D(\mathbb{C})$ or T_n has stabilised. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality (which requires $S_{L_0} = M_D(\mathbb{C})$ for some L_0, hence $T_{L_0} = M_D(\mathbb{C})$). ☐

Theorem 8.16 (Cumulative span reaches $M_D(\mathbb{C})$). Lean ✓ Stmt *If A is normal, then $T_{D^2}(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof Take n from Theorem 8.15; since $n \leq D^2$ and T is monotone, $T_{D^2} = M_D(\mathbb{C})$. ☐

Theorem 8.17 (Nonzero trace product). Lean ✓ Stmt *If A is normal, then there exists a word w of length $\leq D^2$ such that $\mathrm{tr}(A^w) \neq 0$. This is [SPGWC10, Lemma 1]. For the trace argument below, the nontrivial case is $D \geq 1$; when $D = 0$ the normality hypothesis is vacuous.*

Proof. ✓ Proof By Theorem 8.16, $\mathbb{1} \in T_{D^2}(A)$, so $\mathbb{1}$ is a linear combination of word products of length $\leq D^2$. In the nontrivial case $D \geq 1$, at least one has nonzero trace (since $\mathrm{tr}(\mathbb{1}) = D \neq 0$). ☐

Theorem 8.18 (Cumulative span reaches $M_D(\mathbb{C})$ — sharp bound). Lean ✓ Stmt *If A is normal, then $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$, where $d' = \dim S_1(A) = \mathrm{kr}(A)$. This is the sharp form of [SPGWC10, Lemma 1]: the cumulative span already reaches $M_D(\mathbb{C})$ by index $D^2 - d' + 1 \leq D^2$.*

Proof. ✓ Proof The key observation is that if $T_n(A) \neq T_{n+1}(A)$, the dimension strictly increases. Since $S_1(A) \subseteq T_1(A)$ and $\dim S_1(A) = d'$, we have $\dim T_1(A) \geq d'$. After at most $D^2 - d'$ additional strict-growth steps the dimension reaches $D^2 = \dim M_D(\mathbb{C})$. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality. Hence $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$. ☐

Theorem 8.19 (Sharp nonzero trace product). Lean ✓ Stmt *If A is normal, then there exists a word w of length $\leq D^2 - \mathrm{kr}(A) + 1$ such that $\mathrm{tr}(A^w) \neq 0$. This is [SPGWC10, Lemma 1] with the sharp length bound.*

Proof. ✓ Proof By Theorem 8.18, the identity $\mathbb{1}$ lies in $T_{D^2-d'+1}(A)$, hence is a linear combination of word products of length $\leq D^2 - d' + 1$. At least one such product has nonzero trace. ☐

Theorem 8.20 (Sharp nonzero trace product — positive length). Lean ✓ Stmt *If A is normal and $D \geq 2$, then there exists a word w of* positive *length $1 \leq |w| \leq D^2 - \mathrm{kr}(A) + 1$ such that $\mathrm{tr}(A^w) \neq 0$. This strengthens Theorem 8.19 by requiring $|w| \geq 1$, a feature needed for the blocking argument in the general Wielandt bound below.*

Proof. ✓ Proof Define the *positive-level* cumulative span $V_n = \mathrm{span}\{A^w : 1 \le |w| \le n\}$. For $D \ge 2$, the span $S_0(A) = \mathrm{span}\{\mathbb{1}\}$ has dimension 1, so $V_1 \supseteq S_1(A)$ has dimension $\ge d' \ge 1$. The same stabilization-or-strict-growth dichotomy shows $V_{D^2-d'+1} = M_D(\mathbb{C})$: if V_n stabilises and is not $M_D(\mathbb{C})$, then V_n is closed under left-multiplication by each A^i, so $S_N(A) \subseteq V_n \ne M_D(\mathbb{C})$ for all $N \ge 1$, contradicting normality. Since $V_{D^2-d'+1} = M_D(\mathbb{C})$, the identity $\mathbb{1}$ lies in $V_{D^2-d'+1}$ and is therefore a linear combination of word products of positive length. At least one such product has nonzero trace (since $\mathrm{tr}(\mathbb{1}) = D \ge 2 \ne 0$). ☐

8.4 Eigenvector extraction

Theorem 8.21 (Nonzero trace implies eigenvector). Lean ✓ Stmt *If $M \in M_D(\mathbb{C})$ has $\mathrm{tr}(M) \ne 0$, then M has a nonzero eigenvalue $\mu \ne 0$ and a corresponding eigenvector $\varphi \ne 0$ with $M\varphi = \mu\varphi$.*

Proof. ✓ Proof The trace is the sum of the eigenvalues of M, counted with multiplicity, so a nonzero trace gives a nonzero eigenvalue μ. Choosing a nonzero vector $\varphi \in \ker(M - \mu\mathbb{1})$ gives $M\varphi = \mu\varphi$. ☐

Theorem 8.22 (Eigenvalue extraction). Lean ✓ Stmt *If A is normal, then there exist a word w_0 with $|w_0| \le D^2$, a nonzero scalar $\mu \ne 0$, and a nonzero vector $\varphi \ne 0$ such that $A^{w_0}\varphi = \mu\varphi$.*

Proof. ✓ Proof By Theorem 8.17, there is a word w_0 with $\mathrm{tr}(A^{w_0}) \ne 0$ and $|w_0| \le D^2$. By Theorem 8.21, A^{w_0} has a nonzero eigenvector. ☐

8.5 Eigenvector spreading

Definition 8.23 (Cumulative vector span). Lean ✓ Stmt Given an MPS tensor A and a vector $\varphi \in \mathbb{C}^D$, the *cumulative vector span* is

$$K_n(A, \varphi) = \mathrm{span}_{\mathbb{C}}\{A^w\varphi : |w| \le n\} \subseteq \mathbb{C}^D.$$

This is [SPGWC10, proof of Lemma 2(a)].

Theorem 8.24 (Vector span stabilisation). Lean ✓ Stmt *If $K_n(A, \varphi) = K_{n+1}(A, \varphi)$, then $K_m(A, \varphi) = K_n(A, \varphi)$ for all $m \ge n$.*

Proof. ✓ Proof Same argument as Theorem 8.5: left-multiplication by A^i cannot escape the stabilised span. ☐

Lemma 8.25 (Vector span from matrix span). Lean ✓ Stmt *If $T_N(A) = M_D(\mathbb{C})$ and $\varphi \ne 0$, then $K_N(A, \varphi) = \mathbb{C}^D$.*

Proof. ✓ Proof Every matrix $M \in M_D(\mathbb{C})$ is in $T_N(A)$, so in particular every rank-one matrix sending φ to any target vector v is in T_N. Applying such matrices to φ recovers all of $\mathbb{C}^D$. ☐

Theorem 8.26 (Eigenvector spreading). Lean ✓ Stmt *Let A be a normal MPS tensor and let $A^{i_0}\varphi = \mu\varphi$ with $\mu \ne 0$ and $\varphi \ne 0$. Then $K_{D-1}(A, \varphi) = \mathbb{C}^D$. This is [SPGWC10, Lemma 2(a)].*

Proof. ✓ Proof The vector span K_n is monotone and $\dim K_n \le D$. If $K_n = K_{n+1}$ for some $n < D - 1$, the span stabilises (Theorem 8.24). But $T_{D^2}(A) = M_D(\mathbb{C})$ by Theorem 8.16, so Lemma 8.25 gives $K_{D^2}(A, \varphi) = \mathbb{C}^D$, contradicting early stabilisation. Hence $\dim K_n$ grows by ≥ 1 at each step, reaching D by step $D - 1$. ☐

8.6 Proof of the cumulative Wielandt bound

Theorem 8.27 (Wielandt bound). Lean ✓ Stmt *If A is a normal MPS tensor with bond dimension D, then $T_{D^2}(A) = M_D(\mathbb{C})$. In particular, the cumulative span stabilises at index at most D^2. This is the cumulative-span part of [SPGWC10, Theorem 1]. The cited theorem is stronger: via Lemma 2(b) it upgrades this to a fixed-length conclusion $S_N(A) = M_D(\mathbb{C})$, stated below as a separate statement.*

Proof. ✓ Proof Immediate from Theorem 8.16. □

Remark 8.28 (Explicit blocking-length estimates). ✓ Stmt Theorem 8.27 gives the D^2 cumulative-span bound. The full-rank index bound $\iota(A) \le (D^2-d+1)\cdot D^2$ ([SPGWC10, Theorem 1, case (1)]) is proved in Theorem 8.41 below. The left-canonical normal blocked-injectivity estimate D^4 is proved below in Theorem 8.48. The sharper bi-canonical blocking estimate $3D^5$ from [SPGWC10] is not proved here.

Definition 8.29 (One-step augmentation). Lean ✓ Stmt Let A be an MPS tensor with Kraus family $\{A^i\}_{i=0}^{d-1}$, and let $X \in M_D(\mathbb{C})$. The *one-step augmentation* of A by X is the tensor $\widetilde{A}$ with physical dimension $d+1$ whose first Kraus operator is X and whose remaining Kraus operators are those of A:

$$\widetilde{A}^0 = X, \qquad \widetilde{A}^{i+1} = A^i.$$

Lemma 8.30 (Kraus operators lie in the one-step span). Lean ✓ Stmt *Every Kraus operator A^i belongs to the one-step span $S_1(A)$.*

Proof. ✓ Proof This is the length-one word with letter i. □

Lemma 8.31 (Redundant one-step augmentation). Lean Lean Lean ✓ Stmt *Let $X \in S_1(A)$, and let $\widetilde{A}$ be the one-step augmentation of A by X. Then*

$$S_n(\widetilde{A}) = S_n(A)$$

for every $n \ge 0$.

Proof. ✓ Proof The new generator X is already in $S_1(A)$, and each original generator remains in the augmented family. Hence the one-step spans agree. Multiplying word spans and arguing by induction on the word length gives equality at every length. □

Lemma 8.32 (Normality and Kraus rank under one-step augmentation). Lean Lean ✓ Stmt *Let $X \in S_1(A)$, and let $\widetilde{A}$ be the one-step augmentation of A by X. If A is normal, then $\widetilde{A}$ is normal, and*

$$\mathrm{kr}(\widetilde{A}) = \mathrm{kr}(A).$$

Proof. ✓ Proof Since all exact word spans are unchanged by Lemma 8.31, eventual full word span for A gives eventual full word span for $\widetilde{A}$. The equality of Kraus ranks is the equality of the dimensions of the common one-step span. □

Theorem 8.33 (Wielandt case (2), invertible generator). Lean Lean ✓ Stmt *Let A be a normal MPS tensor with bond dimension D, and suppose that some Kraus operator A^{i_0} is invertible. Then*

$$S_{D^2-\mathrm{kr}(A)+1}(A) = M_D(\mathbb{C}), \qquad \iota(A) \le D^2 - \mathrm{kr}(A) + 1.$$

Proof. ✓ Proof The dimensions of the exact word spans cannot decrease, because left multiplication by the invertible Kraus operator is injective. If this dimension growth stopped before the full matrix algebra was reached, the same argument applied at all later lengths would contradict normality. Since $\dim S_1(A) = \mathrm{kr}(A)$ and $\dim M_D(\mathbb{C}) = D^2$, the full algebra is reached by length $D^2 - \mathrm{kr}(A) + 1$. The index bound follows from the definition of $\iota(A)$. □

Theorem 8.34 (Wielandt case (2), one-step invertible element). Lean ✓ Stmt *Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive. If $S_1(A)$ contains an invertible matrix X, then*

$$S_{D^2-\mathrm{kr}(A)+1}(A) = M_D(\mathbb{C}). \tag{8.2}$$

In the terminology of [SPGWC10], this is the case where the first application space $S_1(A)$ contains an invertible operator. This is the exact word-span form of [SPGWC10, Theorem 1, case (2)].

Proof. ✓ Proof Adjoin X as an extra first Kraus operator, obtaining $\widetilde{A}$. Primitivity and normalization give normality of A, hence normality of $\widetilde{A}$; the Kraus rank and all exact word spans are unchanged. The first Kraus operator of $\widetilde{A}$ is invertible, so Theorem 8.33 gives $S_{D^2-\mathrm{kr}(\widetilde{A})+1}(\widetilde{A}) = M_D(\mathbb{C})$. Transfer this equality back to A. □

Corollary 8.35 (Wielandt case (2), index bound). Lean ✓ Stmt *Under the hypotheses of Theorem 8.34,*

$$\iota(A) \leq D^2 - \mathrm{kr}(A) + 1,$$

where $\iota(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\}$.

Proof. ✓ Proof The word-span equality (8.2) gives an admissible value in the defining minimum for $\iota(A)$. □

Corollary 8.36 (Wielandt case (2), invertible Kraus operator). Lean Lean ✓ Stmt *Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive. If some Kraus operator A^{i_0} is invertible, then*

$$S_{D^2-\mathrm{kr}(A)+1}(A) = M_D(\mathbb{C}), \qquad \iota(A) \leq D^2 - \mathrm{kr}(A) + 1.$$

Proof. ✓ Proof Apply the one-step case to $X = A^{i_0}$, which belongs to $S_1(A)$ by Lemma 8.30. □

Lemma 8.37 (Rank-one extraction from a non-invertible eigenvector). Lean ✓ Stmt *Let A be a normal MPS tensor with bond dimension $D \geq 1$. Suppose that a Kraus operator A^{i_0} is non-invertible and that there exists a vector $\varphi \in \mathbb{C}^D$ and a nonzero scalar μ such that $A^{i_0}\varphi = \mu\varphi$. Then, for every $\psi \in \mathbb{C}^D$,*

$$|\varphi\rangle\langle\psi| \in S_{D^2-D+1}(A).$$

Proof. ✓ Proof Let r be the nilpotent index of left multiplication by A^{i_0}. The dimension-growth argument gives a level n_0 for which

$$(A^{i_0})^r S_{n_0}(A) = \{(A^{i_0})^r M : M \in M_D(\mathbb{C})\}.$$

Since $A^{i_0}\varphi = \mu\varphi$ and $\mu \neq 0$, iterating gives $(A^{i_0})^r\varphi = \mu^r\varphi$, hence

$$\varphi = \mu^{-r}(A^{i_0})^r\varphi \in \mathrm{range}((A^{i_0})^r).$$

Hence, for every $\psi \in \mathbb{C}^D$,

$$|\varphi\rangle\langle\psi| \in S_{r+n_0}(A).$$

The nilpotent-index estimate gives

$$r + n_0 \le D^2 - D + 1.$$

Since

$$A^{i_0}|\varphi\rangle\langle\psi| = |A^{i_0}\varphi\rangle\langle\psi| = \mu|\varphi\rangle\langle\psi|$$

and $\mu \neq 0$, the equality

$$|\varphi\rangle\langle\psi| = \mu^{-1}A^{i_0}|\varphi\rangle\langle\psi|$$

implies

$$|\varphi\rangle\langle\psi| \in S_k(A) \implies |\varphi\rangle\langle\psi| \in S_{k+1}(A)$$

because left multiplication by A^{i_0} maps $S_k(A)$ into $S_{k+1}(A)$. Iterating from $k = r + n_0$ and using $r + n_0 \le D^2 - D + 1$ gives

$$|\varphi\rangle\langle\psi| \in S_{D^2-D+1}(A).$$

□

Theorem 8.38 (Wielandt case (3), one-step non-invertible element)**.** Lean ✓ Stmt *Let A be an MPS tensor with bond dimension $D \ge 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive. If $S_1(A)$ contains a non-invertible matrix X with a nonzero eigenvalue, then*

$$S_{D^2}(A) = M_D(\mathbb{C}).$$

More explicitly, it is enough to have $X\varphi = \mu\varphi$ with $\varphi \neq 0$ and $\mu \neq 0$. In the terminology of [SPGWC10], this is the case where the first application space $S_1(A)$ contains a non-invertible operator with a nonzero eigenvalue. This is the exact word-span form of [SPGWC10, Theorem 1, case (3)].

Proof. ✓ Proof Adjoin X as an extra first Kraus operator. The augmented tensor is normal and has the same exact word spans as A. The eigenvector relation for X is now a one-generator eigenvector relation. The eigenvector-spreading argument gives full fixed-length vector span by length $D - 1$, and Lemma 8.37 gives the rank-one operators $|\varphi\rangle\langle\psi|$ in length $D^2 - D + 1$. Multiplying these two fixed-length spans gives S_{D^2} for the augmented tensor, hence for A. □

Corollary 8.39 (Wielandt case (3), index bound)**.** Lean ✓ Stmt *Under the hypotheses of Theorem 8.38,*

$$\iota(A) \le D^2.$$

Proof. ✓ Proof The equality $S_{D^2}(A) = M_D(\mathbb{C})$ gives an admissible value in the defining minimum for $\iota(A)$. □

Corollary 8.40 (Wielandt case (3), non-invertible Kraus operator)**.** Lean Lean ✓ Stmt *Let A be an MPS tensor with bond dimension $D \ge 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive. If some Kraus operator A^{i_0} is non-invertible and has an eigenvector $\varphi \neq 0$ with nonzero eigenvalue μ, then*

$$S_{D^2}(A) = M_D(\mathbb{C}), \qquad \iota(A) \le D^2.$$

Proof. ✓ Proof Apply the one-step case to $X = A^{i_0}$, which belongs to $S_1(A)$ by Lemma 8.30. □

Theorem 8.41 (Wielandt's inequality — general bound). Lean ✓ Stmt *Let A be a normalized MPS tensor ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$) that is primitive. Then*

$$\iota(A) \leq (D^2 - \mathrm{kr}(A) + 1) \cdot D^2,$$

where $\iota(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\}$ is the full-Kraus-rank index. This is [SPGWC10, Theorem 1, case (1)].

Proof. ✓ Proof

1. $D = 1$: In this case $\iota(A) = 0$, so the bound holds trivially.
2. $D \geq 2$: By Theorem 8.20, there exists a *positive-length* word w with $|w| \leq D^2 - d' + 1$ and $\mathrm{tr}(A^w) \neq 0$, where $d' = \mathrm{kr}(A)$. The matrix A^w has nonzero trace, hence a nonzero eigenvalue μ and eigenvector φ (Theorem 8.21). Set $n = |w|$ and let $B = A^{[n]}$ be the blocked tensor. Then B is normal (Theorem 8.52), and the encoding of w as a single block index gives $B^{i_0} = A^w$.
   (a) If A^w is invertible (invertible case), $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$.
   (b) If A^w is non-invertible (non-invertible case), the eigenvector φ spans a proper subspace, and the rank-one extraction argument gives $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$.

   In the invertible case, the invertible Kraus operator B^{i_0} allows direct dimension growth, giving $S_{D^2 - k_B + 1}(B) = M_D(\mathbb{C})$ where $k_B = \mathrm{kr}(B)$. In the non-invertible case, the bound $S_{D^2}(B) = M_D(\mathbb{C})$ follows from tracking indices through the rank-one extraction argument of [SPGWC10, Lemma 2(b)]; the separate qualitative fixed-length conclusion stated below does not supply this explicit bound. In both cases $\iota(A) \leq D^2 \cdot n \leq D^2 \cdot (D^2 - d' + 1)$.

□

Remark 8.42 (Quantitative and qualitative Wielandt bounds). ✓ Stmt The quantitative bound $\iota(A) \leq (D^2 - d' + 1) \cdot D^2$ is cited from [SPGWC10, Theorem 1, case (1)]. The blueprint's own lemmas prove the qualitative conclusion used on the fundamental-theorem route, namely Theorem 8.57: there exists N with $S_N(A) = M_D(\mathbb{C})$. The quantitative bound is not on the fundamental-theorem critical path.

8.7 Fixed-length matrix spanning

The preceding sections establish the cumulative-spanning part of the Lemma 2(a) argument from [SPGWC10]: eigenvector spreading gives a cumulative vector span $K_{D-1}(A, \varphi) = \mathbb{C}^D$. The next stage is to pass to a span at one fixed length, namely to find n_0 such that $S_{n_0}(A) = M_D(\mathbb{C})$ (i.e. word products of *exactly* one length span all matrices).

Definition 8.43 (Fixed-length vector span). Lean ✓ Stmt The *fixed-length vector span* at length n is

$$H_n(A, \varphi) = \mathrm{span}_{\mathbb{C}}\{A^w \varphi : |w| = n\} \subseteq \mathbb{C}^D.$$

This is $S_n(A)|\varphi\rangle$ in the notation of [SPGWC10].

Lemma 8.44 (Word span products). Lean ✓ Stmt $S_m(A) \cdot S_n(A) \subseteq S_{m+n}(A)$*: the product of word spans at lengths m and n is contained in the word span at length $m + n$.*

Proof. ✓ Proof Every product $A^{w_1}A^{w_2}$ with $|w_1| = m$ and $|w_2| = n$ equals $A^{w_1w_2}$ with $|w_1w_2| = m + n$. □

Lemma 8.45 (Eigenvector padding). Lean ✓ Stmt *Suppose $A^{i_0}\varphi = \mu\varphi$ with $\mu \neq 0$. Then for all n,*

$$K_n(A, \varphi) = H_n(A, \varphi).$$

In particular, if $K_n(A, \varphi) = \mathbb{C}^D$ then already $H_n(A, \varphi) = \mathbb{C}^D$.

Proof. ✓ Proof Any word w with $|w| \leq n$ can be padded to exact length n by appending $n - |w|$ copies of the index i_0. The eigenvector relation gives $A^{w \cdot i_0^k}\varphi = \mu^k \cdot A^w\varphi$, so the padded vector is a nonzero scalar multiple of the original. Hence every generator of K_n lies in H_n. The reverse inclusion $H_n \leq K_n$ is immediate. □

Theorem 8.46 (Fixed-length vector spanning implies matrix spanning). Lean ✓ Stmt *Assume:*

1. *$H_n(A, \varphi) = \mathbb{C}^D$ (length-n word products applied to φ span all of $\mathbb{C}^D$), and*
2. *for each standard basis vector e_j, the rank-one operator $|\varphi\rangle\langle e_j|$ lies in $S_m(A)$.*

Then $S_{n+m}(A) = M_D(\mathbb{C})$.

Proof. ✓ Proof Fix a matrix unit E_{ij}. By hypothesis (1), there exists $M_i \in S_n(A)$ with $M_i\varphi = e_i$. By hypothesis (2), $|\varphi\rangle\langle e_j| \in S_m(A)$. Their product satisfies

$$M_i \cdot |\varphi\rangle\langle e_j| = |M_i\varphi\rangle\langle e_j| = |e_i\rangle\langle e_j| = E_{ij},$$

and $E_{ij} \in S_{n+m}(A)$ by Lemma 8.44. Since the matrix units span $M_D(\mathbb{C})$, $S_{n+m}(A) = M_D(\mathbb{C})$. □

Lemma 8.47 (Blocking transfer). Lean ✓ Stmt *If the blocked tensor $A^{[L]}$ satisfies $S_n(A^{[L]}) = M_D(\mathbb{C})$, then $S_{nL}(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof Each length-n word in the blocked alphabet $\{0, \dots, d^L - 1\}$ decodes to a length-nL word in the original alphabet $\{0, \dots, d - 1\}$, so $S_n(A^{[L]}) \subseteq S_{nL}(A)$. □

Theorem 8.48 (Bounded injective blocking for left-canonical normal tensors). Lean ✓ Stmt *Let A be a left-canonical normal MPS tensor with bond dimension $D > 0$. Then there is a positive blocking length $L \leq D^4$ such that the blocked tensor $A^{[L]}$ is one-site injective.*

This is the blocked-injectivity form of the quantum Wielandt input used in the Fundamental Theorem for translation-invariant tensors. It corresponds to the statement in [CPGSV16, Section II, line 332] that every normal tensor becomes injective after at most D^4 blockings, using the index estimate of [SPGWC10, Theorem 1]. The Lean statement includes the left-canonical/trace-preserving hypothesis because that is the canonical-form context in which the blocked-injectivity input is consumed.

Proof. ✓ Proof The proof uses the full-Kraus-rank index bound from the quantum Wielandt inequality. Since A is normal, the index is attained; the one-step Kraus rank is nonzero in the left-canonical case, so the bound $(D^2 - \operatorname{rank} S_1(A) + 1)D^2$ is at most D^4. The passage from exact word-span fullness to injectivity of the blocked tensor is Lemma 8.47. □

Corollary 8.49 (Uniform injective blocking for separated normal-canonical families). Lean Lean ✓ Stmt *Let $(A_x)_{x \in X}$ be a finite separated normal-canonical family whose blocks have positive virtual bond dimension. Then there is a positive integer L such that every blocked tensor $A_x^{[L]}$*

is one-site injective. The same conclusion holds simultaneously for two finite separated normal-canonical families whose blocks have positive virtual bond dimension.

Here the family is a basis of normal-canonical representatives in the sense of Definition 10.9; the blocks are distinct and the weight moduli are non-increasing but need not be strictly ordered. It assumes that the representative family has already been selected; it does not reconstruct the full CPSV sector decomposition with repeated copies inside one sector. The restriction is recorded in `docs/paper-gaps/ft_one_copy_scope_restriction.tex`.

Proof. ✓ Proof Apply Theorem 8.48 to each block. Taking the product of the finitely many resulting positive blocking lengths gives a common positive multiple, and injectivity persists under positive further blocking. □

Remark 8.50 (Relation with the rank-one extraction lemma). ✓ Stmt In [SPGWC10], Lemma 2(b) combines a rank-one extraction for fixed-length spans with a product argument turning fixed-length vector spanning into fixed-length matrix spanning. Theorem 8.46 is the second step. Theorem 8.56 supplies the rank-one hypothesis in blocked form, and Theorem 8.57 then gives a blocked, fixed-length word-span saturation result.

Remark 8.51 (Use of the quantum Wielandt theorem in the MPS route). The source papers use the quantum Wielandt theorem as an input converting normality into injectivity after blocking. In the notation of this chapter, the input has two distinct conclusions. First, Theorem 8.27 gives cumulative saturation $T_{D^2}(A) = M_D(\mathbb{C})$. This alone is not the injectivity statement used in canonical form, because injectivity requires products of one fixed length. Second, Lemma 2(b) of [SPGWC10], formalized here through Theorem 8.57, supplies a length N for which $S_N(A) = M_D(\mathbb{C})$. This is the statement that matches the injectivity of Γ_N and can be transported through blocking by Lemma 8.47. Thus any later use of Wielandt in the Fundamental Theorem must specify whether it is using cumulative span, exact-length span, or the blocked version of the exact-length conclusion.

8.8 Blocked tensor and rectangular span

This section extends the Lemma 2(b) argument to the *blocked tensor* $A^{[L]}$ and develops the "rectangular span" used to produce rank-one elements.

Theorem 8.52 (Blocking preserves normality)**.** Lean ✓ Stmt *If A is normal and $L > 0$, then the blocked tensor $A^{[L]}$ is also normal.*

Proof. ✓ Proof Choose N with $S_N(A) = M_D(\mathbb{C})$. Then $\mathbb{1} \in S_N(A)$. For any $M \in S_N(A)$, write $\mathbb{1} = \sum_r c_r A^{w_r}$ with each $|w_r| = N$. Then

$$M = M \cdot \mathbb{1} = \sum_r c_r M A^{w_r} \in S_N(A) \cdot S_N(A) \subseteq S_{2N}(A) \tag{8.3}$$

by Lemma 8.44. Iterating the inclusion (8.3) gives $S_N(A) \subseteq S_{kN}(A)$ for every $k \geq 1$; taking $k = L$ yields $M_D(\mathbb{C}) = S_N(A) \subseteq S_{NL}(A)$. By Lemma 8.53, $S_{NL}(A) \subseteq S_N(A^{[L]})$, so $S_N(A^{[L]}) = M_D(\mathbb{C})$. □

Lemma 8.53 (Chunking: word span embeds into blocked word span)**.** Lean ✓ Stmt $S_{nL}(A) \subseteq S_n(A^{[L]})$ *for all n, L.*

Proof. ✓ Proof Every length-nL word in the original alphabet can be split into n consecutive chunks of length L, each of which is a single blocked Kraus operator. By induction on n, using the word span product lemma (Lemma 8.44). □

Theorem 8.54 (Word eigenvector gives blocked eigenvector). Lean ✓ Stmt *If $A^{w_0}\varphi = \mu\varphi$ for a word w_0 of length L, then $(A^{[L]})^{j_0}\varphi = \mu\varphi$, where j_0 is the encoded blocked index corresponding to w_0.*

Proof. ✓ Proof By definition, $(A^{[L]})^{j_0} = A^{w_0}$. □

Theorem 8.55 (Blocked fixed-length matrix spanning). Lean ✓ Stmt *Suppose A is normal and $L > 0$, and we have:*

1. *a word w_0 of length L with eigenvector $A^{w_0}\varphi = \mu\varphi$ ($\mu \neq 0$, $\varphi \neq 0$),*
2. *a word τ_0 of length L with transpose eigenvector $(A^{\tau_0})^T\psi = \nu\psi$ ($\nu \neq 0$, $\psi \neq 0$),*
3. *a rank-one element $\varphi\psi^T \in S_m(A^{[L]})$ for some m.*

Then there exists N such that $S_N(A) = M_D(\mathbb{C})$.
Remark. *The proof gives the explicit bound $N = ((D-1) + m + (D-1)) \cdot L$; the statement above gives only the existential conclusion.*

Proof. ✓ Proof Apply Theorem 8.46 to the blocked tensor $A^{[L]}$, which is normal by Theorem 8.52. The eigenvector and transpose eigenvector are transferred to single-index eigenvectors of $A^{[L]}$ by Theorem 8.54. Transfer back to the original tensor via Lemma 8.47. □

Theorem 8.56 (Rank-one extraction for blocked tensor). Lean ✓ Stmt *Suppose A is normal and $N_0 \geq 1$ with $S_{N_0}(A) = M_D(\mathbb{C})$. Then there exist words σ_0, τ_0 of length N_0, nonzero vectors φ, ψ, nonzero scalars μ, ν, and $m \in \mathbb{N}$ such that:*

1. *$A^{\sigma_0}\varphi = \mu\varphi$,*
2. *$(A^{\tau_0})^T\psi = \nu\psi$,*
3. *$\varphi\psi^T \in S_m(A^{[N_0]})$.*

Proof. ✓ Proof Set $B = A^{[N_0]}$. Since $S_{N_0}(A) = M_D(\mathbb{C})$, we have $S_1(B) = M_D(\mathbb{C})$ by definition of the blocked tensor. Hence $\mathbb{1} \in S_1(B)$, so in the nontrivial case $D \geq 1$ not all blocked generators can have trace zero. Choose j_0 with $\mathrm{tr}(B^{j_0}) \neq 0$. By Theorem 8.21, B^{j_0} has a nonzero eigenvector φ with eigenvalue $\mu \neq 0$. Writing j_0 as the blocked index corresponding to a word σ_0 of length N_0 gives $A^{\sigma_0}\varphi = \mu\varphi$. Applying the same argument to the transpose blocked tensor gives a word τ_0 of length N_0, a nonzero vector ψ, and $\nu \neq 0$ with $(A^{\tau_0})^T\psi = \nu\psi$. Since A is normal and $N_0 \geq 1$, the blocked tensor B is normal by Theorem 8.52. Therefore there exists M with $S_M(B) = M_D(\mathbb{C})$. In particular $\varphi\psi^T \in S_M(B)$. Taking $m = M$ proves (3). □

Theorem 8.57 (Fixed-length word-span saturation via blocking). Lean ✓ Stmt *If A is a normal MPS tensor with bond dimension D, then there exists N such that $S_N(A) = M_D(\mathbb{C})$. This gives the fixed-length spanning conclusion corresponding to [SPGWC10, Lemma 2(b)].*

Proof. ✓ Proof By normality, some N_0 has $S_{N_0}(A) = M_D(\mathbb{C})$. If $N_0 = 0$ we are done. Otherwise, Theorem 8.56 produces eigenvectors and a rank-one element $\varphi\psi^T \in S_m(A^{[N_0]})$. Theorem 8.55 then gives $S_{((D-1)+m+(D-1))\cdot N_0}(A) = M_D(\mathbb{C})$. □

Remark 8.58 (Blocking argument for fixed-length spanning). ✓ Stmt Theorem 8.57 is obtained by first extracting a rank-one operator in a blocked tensor (Theorem 8.56) and then applying the fixed-length matrix-spanning theorem (Theorem 8.55). Thus the fixed-length span of A is reached through a blocked auxiliary tensor rather than by a direct unblocked argument.

8.9 Primitive MPS tensors

Definition 8.59 (Primitive MPS tensor). Lean ✓ Stmt Assume $D > 0$. A tensor A together with a matrix $\rho \in M_D(\mathbb{C})$ is *primitive* if

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}, \qquad \rho \geq 0, \qquad \rho \neq 0, \qquad \mathcal{E}_A(\rho) = \rho,$$

and, if P is the fixed-point projection associated to ρ, then $\rho_{\mathrm{spec}}(\mathcal{E}_A - P) < 1$. When the choice of ρ is irrelevant, we simply say that A is a primitive MPS tensor. This is weaker than the primitive condition of [SPGWC10], which requires $\rho > 0$; see Remark 8.65. The stronger positive-definite hypothesis is imposed separately in Lemma 8.66 and Theorem 8.75.

Theorem 8.60 (Primitive overlap convergence). Lean ✓ Stmt *If A is a primitive MPS tensor, then $O_{AA}(N) \to 1$.*

Proof. ✓ Proof Immediate from the self-overlap convergence under a complementary transfer-map gap (Theorem 7.96). □

8.10 Primitivity and normality

Throughout this section, primitivity means the condition of Definition 8.59: the tensor is in TP gauge, it has a nonzero PSD fixed point, and the complementary map has spectral radius < 1.

Lemma 8.61 (Primitive MPS tensors have nonzero word products). Lean ✓ Stmt *If A is a primitive MPS tensor, then for every $n \geq 0$ there exists a word w of length* exactly *n with $A^w \neq 0$.*

Proof. ✓ Proof The PSD fixed point ρ of $\mathcal{E}_A$ satisfies $\mathcal{E}_A^n(\rho) = \rho \neq 0$. Since $\mathcal{E}_A^n(\rho) = \sum_{|w|=n} A^w \rho (A^w)^\dagger$, at least one summand is nonzero, hence some $A^w \neq 0$. □

Lemma 8.62 (Iterated transfer does not vanish). Lean ✓ Stmt *If A is a primitive MPS tensor, then $\mathcal{E}_A^n \neq 0$ for all $n \geq 0$.*

Proof. ✓ Proof By Lemma 8.61, there exists a word w of length n with $A^w \neq 0$, so $\mathcal{E}_A^n(\rho)$ has a nonzero summand. □

8.10.1 From primitivity to normality

The primitivity condition of Definition 8.59 combines TP normalization, a nonzero PSD fixed point, and a complementary transfer-map gap. It is connected to normality through the following results.

Lemma 8.63 (Fixed-point uniqueness from a complementary transfer-map gap). Lean ✓ Stmt *If A is a primitive MPS tensor with PSD fixed point ρ, then every fixed point σ of $\mathcal{E}_A$ is proportional to ρ: $\sigma = \frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)}\rho$.*

Proof. ✓ Proof Set $\sigma' = \sigma - \frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)}\rho$. Then $\mathrm{tr}(\sigma') = 0$ and $(\mathcal{E}_A - P_\rho)(\sigma') = \sigma'$. If $\sigma' \neq 0$, then 1 is an eigenvalue of $\mathcal{E}_A - P_\rho$, contradicting the complementary transfer-map gap $\rho_{\mathrm{spec}}(\mathcal{E}_A - P_\rho) < 1$. □

Lemma 8.64 (Complement powers tend to zero). Lean ✓ Stmt *If A is a primitive MPS tensor with PSD fixed point ρ, then $(\mathcal{E}_A - P_\rho)^n \to 0$ in operator norm.*

Proof. ✓ Proof The spectral radius of $\mathcal{E}_A - P_\rho$ is strictly less than 1 by the complementary transfer-map gap hypothesis. Apply the general result that powers of an operator with spectral radius < 1 tend to zero. □

Remark 8.65 (Primitivity: transfer-map gap vs. positive definiteness). ✓ Stmt The primitivity condition used in Section 8.10.1 has three parts: TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, a nonzero PSD fixed point ρ, and the complementary transfer-map gap $\rho_{\mathrm{spec}}(\mathcal{E}_A - P_\rho) < 1$. The notion of "primitive" in [SPGWC10, Proposition 3] additionally requires ρ to be positive definite (full rank). Without positive definiteness, the transfer-map gap condition alone does *not* imply irreducibility or normality. A counterexample is $D = 2$, $\rho = \mathrm{diag}(1, 0)$.

With the additional hypothesis $\rho > 0$, Lemma 8.66 gives irreducibility. Together with the Burnside argument below (Section 8.10.3) and Theorem 8.82, one obtains

$$\text{primitive} + \rho \text{ positive definite} \Rightarrow \text{irreducible} \Rightarrow \mathrm{alg}(A) = M_D(\mathbb{C}) \Rightarrow \text{normal}.$$

The last step uses the aperiodicity condition $\mathbb{1} \in S_1(A)$ from Remark 8.76. After restoring the positive-definite hypothesis, [SPGWC10, Proposition 3] identifies that primitive/strongly irreducible condition with eventual full Kraus rank, i.e. with our notion of normality.

8.10.2 Primitive implies irreducible

Lemma 8.66 (Primitive tensors with positive-definite fixed point are irreducible). *Lean* *✓ Stmt* *If A is a primitive MPS tensor with PSD fixed point ρ and ρ is positive definite, then A is an irreducible tensor. This is part of [SPGWC10, Proposition 3] (see also [CPGSV16, Section 2.3]).*

Proof. ✓ Proof By contrapositive. Assume A has an invariant projection P ($P \neq 0$, $P \neq \mathbb{1}$, $(\mathbb{1} - P)A^i P = 0$ for all i). Set $\sigma = P\rho P$.

1. *Invariance of σ under $\mathcal{E}_A^n$:* the projection relation $(\mathbb{1}-P)A^iP = 0$ implies $(\mathbb{1}-P)\cdot\mathcal{E}_A^n(\sigma) = 0$ for all n (by induction on n).
2. *Convergence:* By Lemma 8.64, $\mathcal{E}_A^n(\sigma) \to \frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)}\rho$. Hence $(\mathbb{1} - P) \cdot \frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)}\rho = 0$.
3. *Nonzero scalar:* Since $P \neq 0$ and ρ is positive definite, $\mathrm{tr}(P\rho P) > 0$, so $\frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)} \neq 0$. Therefore $(\mathbb{1} - P)\rho = 0$.
4. *Contradiction:* Since ρ is positive definite, it is a unit in $M_D(\mathbb{C})$. From $(\mathbb{1} - P)\rho = 0$ we get $\mathbb{1} - P = 0$, i.e. $P = \mathbb{1}$, contradicting $P \neq \mathbb{1}$.

□

8.10.3 From irreducibility to full spanning via Burnside's theorem

The irreducibility condition (Definition 9.16) is formulated in terms of invariant *projections*. For Burnside-style arguments it is more natural to work with invariant *subspaces* of $\mathbb{C}^D$ under the Kraus operators.

Definition 8.67 (Algebra span). Lean ✓ Stmt For an MPS tensor A, let $\mathrm{alg}(A)$ be the unital $\mathbb{C}$-subalgebra of $M_D(\mathbb{C})$ generated by the matrices $\{A^i\}_{i=0}^{d-1}$:

$$\mathrm{alg}(A) = \mathbb{C}\langle A^i : i = 0, \ldots, d-1 \rangle \subseteq M_D(\mathbb{C}).$$

Definition 8.68 (Invariant submodule). Lean ✓ Stmt A subspace $W \le \mathbb{C}^D$ is *A-invariant* if $A^i W \subseteq W$ for every i.

Definition 8.69 (Irreducible action). Lean ✓ Stmt The Kraus operators of A act *irreducibly* on $\mathbb{C}^D$ if every A-invariant subspace is trivial, i.e. equal to 0 or to $\mathbb{C}^D$.

Theorem 8.70 (Irreducible tensor implies irreducible action). Lean ✓ Stmt *If A is irreducible as a tensor (Definition 9.16), then the Kraus operators act irreducibly on $\mathbb{C}^D$ (Definition 8.69).*

Proof. ✓ Proof Let $W \le \mathbb{C}^D$ be an A-invariant subspace. In finite dimension the orthogonal projection P onto W exists. Invariance of W implies $(\mathbb{1} - P)A^i P = 0$ for all i. Thus P is a nontrivial invariant projection unless $W = 0$ or $W = \mathbb{C}^D$, contradicting irreducibility of the tensor. □

Theorem 8.71 (Burnside's theorem for Kraus algebras). Lean ✓ Stmt *Assume $D > 0$. If A acts irreducibly on $\mathbb{C}^D$, then the generated unital subalgebra is the full matrix algebra:*

$$\mathrm{alg}(A) = M_D(\mathbb{C}).$$

This is the complex finite-dimensional case of Burnside's theorem (equivalently, Jacobson's density theorem); see [Jac09].

Proof. ✓ Proof Transport $\mathrm{alg}(A)$ along the algebra equivalence $M_D(\mathbb{C}) \simeq_{\mathbb{C}} \mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$. The irreducible-action hypothesis makes $\mathbb{C}^D$ a simple module for this algebra. Over the algebraically closed field $\mathbb{C}$, Schur's lemma identifies the endomorphisms commuting with the action with scalars, and Jacobson density then forces the action map to be surjective onto all of $\mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$. □

Remark 8.72. ✓ Stmt This Burnside/Jacobson-density step is a substantial external algebraic ingredient in the chapter. We use it here as the finite-dimensional complex Burnside theorem, namely the statement that an irreducible matrix algebra action already generates the full endomorphism algebra.

8.10.4 From primitivity to strong irreducibility and normality

The results in this subsection remove the aperiodicity assumption from the normality conclusions. They connect primitivity in Definition 8.59, formulated via a complementary transfer-map gap, directly to normality (eventually full Kraus rank), using the convergence of the transfer-map iterates to the projection onto the fixed-point subspace.

Lemma 8.73 (Primitive tensors with positive-definite fixed point have irreducible transfer map). Lean ✓ Stmt *If A is a primitive MPS tensor with positive-definite fixed point ρ, then the transfer map $\mathcal{E}_A$ is irreducible.*

The proof uses fixed-point uniqueness (Lemma 8.63): every PSD fixed point is proportional to ρ, so Wolf's criterion for irreducibility (positive-definite fixed point implies irreducibility) applies directly.

Proof. ✓ Proof By Lemma 8.63, if $\sigma \ge 0$ and $\mathcal{E}_A(\sigma) = \sigma$, then $\sigma = \frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)}\rho$. Since $\rho > 0$, Wolf's uniqueness criterion for irreducibility applies. □

Lemma 8.74 (Primitive tensors with positive-definite fixed point are strongly irreducible). Lean ✓ Stmt *If A is a primitive MPS tensor with positive-definite fixed point ρ, then A is strongly irreducible in the sense of [SPGWC10, Proposition 3(c)]: $\mathcal{E}_A$ is irreducible, the fixed point ρ is positive definite, and the peripheral spectrum of $\mathcal{E}_A$ is $\{1\}$.*

Proof. ✓ Proof Combine Lemma 8.73 (irreducibility) with the peripheral-spectrum primitivity from the transfer-map gap hypothesis (Theorem 8.60). □

Theorem 8.75 (Primitive MPS with positive-definite fixed point is normal). Lean ✓ Stmt *If A is a primitive MPS tensor with positive-definite fixed point ρ, then A is normal (has eventually full Kraus rank). No aperiodicity assumption is needed.*

This is the key connection between primitivity in Definition 8.59, formulated via a complementary transfer-map gap, and normality; it removes the need for the aperiodicity-based argument.

Proof. ✓ Proof By Lemma 8.74, A is strongly irreducible, i.e. primitive in Wolf's sense. Under normalization, Theorem 7.62 identifies this primitivity condition directly with normality (equivalently, eventual full Kraus rank). □

8.11 Cumulative span to word span

Cumulative spanning ($T_N(A) = M_D(\mathbb{C})$, the union of word spans of all lengths $\leq N$) is weaker than normality ($S_L(A) = M_D(\mathbb{C})$ for a single length L). An aperiodicity hypothesis upgrades cumulative spanning to a single-length word span.

Remark 8.76 (Aperiodicity condition). ✓ Stmt We say A satisfies the *aperiodicity condition* if $\mathbb{1} \in S_1(A)$, i.e. the identity matrix lies in the span of the Kraus operators $\{A^i\}_{i=0}^{d-1}$. This is an additional hypothesis. The one-sided TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ does *not* imply $\mathbb{1} \in S_1(A)$; it only gives a quadratic identity among the Kraus operators. The aperiodicity condition is kept explicit in this section for generality. For the primitive condition with positive-definite fixed point (peripheral-spectrum primitivity together with a positive-definite fixed point), normality follows directly from Wolf's primitivity equivalence (Theorem 7.62) without passing through aperiodicity.

Remark 8.77 (Counterexample: cumulative spanning does not imply normality). ✓ Stmt In $M_2(\mathbb{C})$, let $A^0 = E_{12}$ and $A^1 = E_{21}$ (the off-diagonal matrix units). Then $\mathrm{alg}(A) = M_2(\mathbb{C})$ and $T_2(A) = M_2(\mathbb{C})$, but the word spans alternate: $S_n(A)$ contains only off-diagonal matrices for odd n and only diagonal matrices for even n. Hence $S_n(A) \neq M_2(\mathbb{C})$ for every n, and A is not normal. The aperiodicity condition $\mathbb{1} \in S_1(A)$ fails: $S_1(A) = \mathrm{span}_{\mathbb{C}}\{E_{12}, E_{21}\}$.

Theorem 8.78 (Word span monotonicity from aperiodicity). Lean ✓ Stmt *If $\mathbb{1} \in S_1(A)$, then $S_n(A) \leq S_{n+1}(A)$ for all n.*

Proof. ✓ Proof For any $M \in S_n(A)$, $M = M \cdot \mathbb{1} \in S_n(A) \cdot S_1(A) \subseteq S_{n+1}(A)$ by Lemma 8.44. □

Theorem 8.79 (Homogeneous word-span propagation under normalization). Lean Lean Lean ✓ Stmt *Suppose*

$$\sum_a A_a A_a^\dagger = \mathbb{1} \qquad \text{and} \qquad S_L(A) = M_D(\mathbb{C}).$$

Then, for every $m \geq L$,

$$S_m(A) = M_D(\mathbb{C}).$$

Equivalently, L-block injectivity propagates to every larger homogeneous length.

Proof. ✓ Proof If $S_n(A) = M_D(\mathbb{C})$, then for any $X \in M_D(\mathbb{C})$,

$$X = \left(\sum_a A_a A_a^\dagger\right) X = \sum_a A_a (A_a^\dagger X).$$

Since $A_a^\dagger X \in S_n(A)$, each summand lies in $S_1(A)S_n(A) \subseteq S_{n+1}(A)$. Thus $S_{n+1}(A) = M_D(\mathbb{C})$. Iterating this one-step implication gives the result for all $m \geq L$. $\square$

Theorem 8.80 (Cumulative span equals word span under aperiodicity). Lean ✓ Stmt *If $\mathbb{1} \in S_1(A)$, then $T_n(A) = S_n(A)$ for all n.*

Proof. ✓ Proof By Theorem 8.78, the word spans are monotone: $S_0 \leq S_1 \leq \cdots$. The supremum $T_n = S_0 + \cdots + S_n$ therefore equals the largest term S_n. $\square$

Theorem 8.81 (Cumulative spanning plus aperiodicity implies normality). Lean ✓ Stmt *If $T_N(A) = M_D(\mathbb{C})$ and $\mathbb{1} \in S_1(A)$, then A is normal.*

Proof. ✓ Proof By Theorem 8.80, $S_N(A) = T_N(A) = M_D(\mathbb{C})$. $\square$

Theorem 8.82 (Algebra span plus aperiodicity implies normality). Lean ✓ Stmt *If $\mathrm{alg}(A) = M_D(\mathbb{C})$ and $\mathbb{1} \in S_1(A)$, then A is normal.*

Proof. ✓ Proof Since $\mathrm{alg}(A) = M_D(\mathbb{C})$, there exists N with $T_N(A) = M_D(\mathbb{C})$ (by the ascending chain condition on finite-dimensional subspaces). Apply Theorem 8.81. $\square$

Remark 8.83 (Normal canonical form avoids the aperiodicity step). The aperiodicity results in Section 8.11 are needed only for the block-injective argument through Definition 2.31, where one upgrades cumulative spanning to eventual block injectivity. The normal-canonical theory used later in Chapters 9–10 does not pass through this step: it works directly with irreducible TP-normalized blocks whose blocked transfer maps are primitive and, once the corresponding weighted block data are available, places them in normal canonical form in the sense of [CPGSV16]. There are analogous cumulative-span lemmas for blocked irreducible tensors and eigenvector-spreading arguments, but they are auxiliary here and we do not list them separately.

8.12 Block injectivity from TP/primitive/irreducible blocks (relocated from Chapter 9)

Theorem 8.84 (TP-primitive irreducible tensor is injective after blocking). Lean ✓ Stmt *Let A be a left-canonical MPS tensor with $D > 0$, irreducible tensor, and primitive transfer map. Then some positive blocking $A^{[L]}$ is one-site injective.*

Proof. ✓ Proof If $D = 1$, trace preservation gives a nonzero one-site Kraus matrix, and every nonzero scalar matrix spans $M_1(\mathbb{C})$; hence A is one-site injective, so Lemma 2.30 gives the positive witness $L = 1$. Suppose now that $D \geq 2$. The normality theorem (Theorem 9.51) gives a block-injectivity witness L. If $L = 0$, Corollary 8.9 would force $D = 1$, a contradiction. Thus $L > 0$, and the blocked-chain equivalence identifies this L-block injectivity with one-site injectivity of the blocked tensor $A^{[L]}$. $\square$

Remark 8.85. See Theorem 8.52 for the formalized statement that blocking preserves normality.

Chapter 9

Canonical Form Reduction

This is the first of three chapters that together prove the Fundamental Theorem of matrix product states. Here an arbitrary translation-invariant MPS tensor is reduced to a weighted family of primitive blocks; Chapter 10 constructs the basis of normal tensors carried by those blocks and compares their coefficients, and Chapter 11 proves the theorem itself. The reduction follows [PGVWC07], [CPGSV16], [CPGSV21], and [Wol12, Chapter 6]. Its goal is to expose the primitive blocks inside the tensor. The transfer operator of an arbitrary tensor can be reducible and can have a period greater than one; the reduction splits it along its invariant subspaces and brings the period of each block down to one, so that each block is primitive, with peripheral spectrum $\{1\}$. The blocks are returned in a fixed gauge, and the sections below carry out the reduction. The periodic decomposition theorem of [PGVWC07, Theorem 5] also gives an explicit decomposition of the state vector into p-periodic summands; here we use only its blocked cyclic-sector consequence (Theorem 9.44), and that explicit state-vector decomposition lies outside this chapter.

Two normalizations appear. The unital orientation $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$ is used in [PGVWC07, Theorem 4]; the trace-preserving (left-canonical) orientation $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$ is used in the TP-gauge and after-blocking reductions below. They exchange under the conjugate-transposed Kraus family $K_i := (A^i)^\dagger$, whose transfer map is the Frobenius adjoint of $\mathcal{E}_A$.

Theorem 9.1 (Translation-invariant canonical form)**.** Lean ✓ Stmt *Let A be a translation-invariant MPS representation of bond dimension D. There is a positive scalar s and a finite family of blocks A_k, possibly empty, such that the globally rescaled tensor $s^{-1}A$ has the same MPV coefficients on every positive-length ring as*

$$\bigoplus_k \mu_k A_k^i.$$

The weights are positive and satisfy $1 \geq |\mu_k|$; if the family is nonempty, then some weight has norm 1. For each block,

$$\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}, \qquad \sum_i (A_k^i)^\dagger \Lambda_k A_k^i = \Lambda_k,$$

where Λ_k is diagonal, positive, and full-rank. Each block has positive bond dimension. Moreover the only fixed points of $\mathcal{E}_{A_k}(X) = \sum_i A_k^i X (A_k^i)^\dagger$ are the scalar multiples of $\mathbb{1}$: for every X,

$$\mathcal{E}_{A_k}(X) = X \Longrightarrow \exists\, c \in \mathbb{C}, X = c\,\mathbb{1}.$$

The total bond dimension of the decomposition is at most D.

This is the positive-length form of [PGVWC07, Theorem 4], with the global spectral-radius normalization step made explicit.

Remark 9.2 (Source proof order). The proof follows [PGVWC07, Theorem 4] in order: normalize the spectral radius and use a full-rank positive fixed point to obtain the unital gauge; derive the invariant support projection from a singular positive fixed point by the positivity contradiction; split the finite-ring trace into the supported and orthogonal summands; iterate, using a non-scalar fixed point to split further until each block has only scalar fixed points; and finally repeat the fixed-point argument for the dual map and diagonalize the resulting full-rank positive fixed point by a unitary gauge. The primitive-block results below are used only after their hypotheses have been obtained from this argument.

Remark 9.3 (CPSV blocked canonical form (source statement)). The blocked canonical-form reduction of [CPGSV16, Section 2.3] and [CPGSV21, Section IV.A] states that an arbitrary translation-invariant MPS tensor becomes, after blocking by some period $p \geq 1$, the direct sum of an all-zero summand and a weighted family of normal tensors $\bigoplus_k \mu_k B_k$ with $|\mu_k| \leq 1$ and at least one $|\mu_k| = 1$. The closest proved results here are Theorem 9.52 and Theorem 10.40; the source upper normalization $|\mu_k| \leq 1$ with a unit-modulus witness is the one remaining input, recorded in Remark 10.41.

9.1 Normal tensors and canonical forms

This section records the CPSV definitions of *normal tensor* and *canonical form* from [CPGSV16]. The basis of normal tensors is stated in Chapter 10. Write

$$\sigma_\partial(\mathcal{E}) = \{\lambda \in \sigma(\mathcal{E}) : |\lambda| = r(\mathcal{E})\}$$

for the peripheral spectrum of a transfer map $\mathcal{E}$, where $r(\mathcal{E})$ is its spectral radius. The stronger predicates used later (Definition 9.26 and the canonical-form predicate of Chapter 10) add left-canonical normalization and spectral primitivity. The weight moduli are taken non-increasing, not strictly decreasing; equal-modulus and repeated-copy sectors are retained and compared by the coefficient argument of Chapter 10.

Definition 9.4 (CPSV normal tensor (NT))**.** Lean ✓ Stmt A tensor A of physical dimension d and bond dimension D is a *normal tensor* if, after the spectral-radius normalization of [CPGSV16], (i) A admits no nontrivial invariant orthogonal projection and (ii) the associated CPM $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ has peripheral spectrum $\sigma_\partial(\mathcal{E}_A) = \{1\}$.

Remark 9.5. Condition (i) cannot be dropped. A reducible tensor may have peripheral spectrum $\{1\}$ while still admitting a nontrivial invariant projection, so condition (ii) alone does not characterize normality.

The CPSV *canonical form* (CF) of a tensor A is the normal-block decomposition

$$A^i \cong \bigoplus_{k=1}^{r} \mu_k A_k^i$$

with weights normalized by $|\mu_k| \leq 1$ and with at least one $|\mu_k| = 1$; see [CPGSV16, Section 2.3]. In the statements below the structural part is supplied by the stronger normal canonical form of Definition 9.26, and the global weight normalization is invoked as a separate source hypothesis where needed.

Remark 9.6 (Relation to the strong predicates). A tensor A which is both irreducible (Definition 9.16) and has primitive transfer map is normal in the sense of Definition 9.4. The reverse implications from the stronger predicates of Definition 9.26 and of Chapter 10 require the fundamental-theorem step or the algebraic-to-spectral normality equivalence, established later.

9.1.1 Normalization conventions

Remark 9.7 (Scalar normalization conventions). The reductions use four scalar facts: scaling preserves injectivity; the transfer map under a scalar ζ rescales by $|\zeta|^2$; unit-modulus phase rescaling preserves $\sum_i (A^i)^\dagger A^i = \mathbb{1}$; and the MPV of a weighted block-diagonal tensor factorizes as $\mu_k^N V^{(N)}(A_k)_\sigma = \eta_k^N |\mu_k|^N V^{(N)}(A_k)_\sigma$ with $\eta_k := \mu_k/|\mu_k|$.

Left-canonical normalization is required before the blocks can be compared. Without it the implication $\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k) \Rightarrow \forall k,\ \mathcal{V}(A_k) = \mathcal{V}(B_k)$ already fails for two scalar blocks: for $\mu = (1,2)$, $A = (1, 3/2)$, $B = (3, 1/2)$ the two direct sums generate identical MPV families but the first blocks differ at $N = 1$. Left-canonical normalization removes this particular ambiguity; the sectors are then matched, up to a permutation, by the exact fixed-length linear independence of Chapter 10, which requires no ordering of the weight moduli.

9.2 Invariant-subspace decomposition and irreducible blocks

The block decomposition rests on one principle: a positive semidefinite fixed point of $\mathcal{E}_A$ that is not full rank has a proper invariant support, which block-diagonalizes the Kraus operators without changing the MPV family. Iterating the resulting splitting decomposes any tensor into irreducible blocks.

Theorem 9.8 (Unitary conjugation preserves the MPV family). Lean ✓ Stmt *If U is a unitary matrix, then A and $U^\dagger A U$ generate the same MPV family.*

Proof. ✓ Proof By cyclicity of the trace, $\operatorname{tr}((U^\dagger A^{i_1} U) \cdots (U^\dagger A^{i_N} U)) = \operatorname{tr}(A^{i_1} \cdots A^{i_N})$. ☐

Theorem 9.9 (Invariant projection gives a two-block decomposition). Lean ✓ Stmt *Suppose P is an orthogonal projection satisfying, for every physical index i,*

$$(\mathbb{1} - P) A^i P = 0.$$

Then there exist n, m with $n + m = D$ and MPS tensors A_1, A_2 of bond dimensions n and m such that A and the two-block tensor $A_1 \oplus A_2$ generate the same MPV family in the sense of Definition 2.12.

Proof. ✓ Proof Diagonalize P by a unitary U. Since $P^2 = P$, the diagonal form of P has only 0- and 1-eigenvalues, so the basis splits as $n + m = D$. Conjugating A by U preserves the MPV family by Theorem 9.8. In this basis the relation $(\mathbb{1} - P)A^i P = 0$ makes each letter upper block triangular. Write

$$B^i = \begin{pmatrix} A_1^i & * \\ 0 & A_2^i \end{pmatrix}$$

for the conjugated letters in this split basis. Since $B^{i_1} \cdots B^{i_N}$ is again upper block triangular, its trace satisfies

$$\operatorname{tr}(B^{i_1} \cdots B^{i_N}) = \operatorname{tr}(A_1^{i_1} \cdots A_1^{i_N}) + \operatorname{tr}(A_2^{i_1} \cdots A_2^{i_N}).$$

Hence the block-diagonal tensor $A_1 \oplus A_2$ has the same MPV coefficients as the conjugated tensor. ☐

Theorem 9.10 (Nontrivial invariant projection gives strict two-block reduction). Lean ✓ Stmt *Under the hypotheses of Theorem 9.9, assume in addition that $P \neq 0$ and $P \neq \mathbb{1}$. Then there exist n, m with $n + m = D$, $n < D$, and $m < D$, together with block tensors A_1, A_2, such that A and $A_1 \oplus A_2$ generate the same MPV family in the sense of Definition 2.12.*

Proof. ✓ Proof Apply Theorem 9.9. In the diagonal form of P, the 1-eigenspace is nonempty because $P \neq 0$, and the 0-eigenspace is nonempty because $P \neq \mathbb{1}$. Hence both block sizes are positive. Since $n + m = D$, this implies $n < D$ and $m < D$. □

Definition 9.11 (Support projection). Lean ✓ Stmt For a positive semidefinite matrix $\rho \geq 0$, the *support projection* P is the orthogonal projection onto the range of ρ. Via the spectral decomposition $\rho = U \operatorname{diag}(\lambda_1, \dots, \lambda_D) U^\dagger$,

$$P = U \operatorname{diag}(\mathbf{1}_{\lambda_1 > 0}, \dots, \mathbf{1}_{\lambda_D > 0}) U^\dagger,$$

where $\mathbf{1}_{\lambda_j > 0}$ is 1 if $\lambda_j > 0$ and 0 otherwise.

Definition 9.12 (Invariant projection predicate). Lean ✓ Stmt An MPS tensor A *has an invariant projection* if there exists an orthogonal projection P with $P \neq 0$, $P \neq \mathbb{1}$, and $(\mathbb{1} - P) A^i P = 0$ for all i. A tensor is *irreducible* if it has no nontrivial invariant projection (Definition 9.16).

Theorem 9.13 (PSD fixed point gives invariant projection). Lean ✓ Stmt *Let $\rho \geq 0$ be a fixed point of $\mathcal{E}_A$, and let P be its support projection. Then $(\mathbb{1} - P) A^i P = 0$ for all i: the range of P is invariant under every A^i.*

Proof. ✓ Proof From $\mathcal{E}_A(\rho) = \rho$ and positivity of each summand $A^i \rho (A^i)^\dagger$, one gets $(\mathbb{1} - P) A^i \rho (A^i)^\dagger (\mathbb{1} - P) = 0$ for each i. Since ρ is supported on the range of P, this forces $(\mathbb{1} - P) A^i P = 0$.

This is the finite-dimensional case of [Wol12, Proposition 6.10]: the support projection of any stationary state is sub-harmonic. The equivalence between irreducibility and absence of nontrivial invariant projections is [Wol12, Theorem 6.2]; see also [CPGSV21, Section IV.A.1]. □

Theorem 9.14 (Two-block decomposition). Lean ✓ Stmt *If $\mathcal{E}_A$ has a nonzero PSD fixed point ρ that is not positive definite, then there exist MPS tensors A_1, A_2 of smaller bond dimensions with $\mathcal{V}(A) = \mathcal{V}(A_1 \oplus A_2)$.*

Proof. ✓ Proof The support projection P of ρ is neither 0 nor $\mathbb{1}$ (since $\rho \neq 0$ and ρ is not positive definite). Theorem 9.13 gives $(\mathbb{1} - P) A^i P = 0$, so a unitary change of basis block-diagonalizing P puts A^i in upper-triangular block form with diagonal blocks B^i_{11}, B^i_{22}. Theorem 9.8 preserves the MPV under the conjugation, and the trace of an upper-triangular product equals the sum of the diagonal-block traces, so deleting the off-diagonal blocks preserves all MPV coefficients. □

Theorem 9.15 (Non-scalar fixed point gives a further split). Lean Lean ✓ Stmt *Suppose A is unital, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If X is a Hermitian fixed point of $\mathcal{E}_A$ which is not a scalar multiple of $\mathbb{1}$, then there is a nonzero PSD fixed point ρ which is not positive definite, and consequently A admits a two-block MPV-equivalent decomposition with both block dimensions strictly smaller than D. This is the fixed-point shift and further decomposition step in [PGVWC07, Theorem 4].*

Proof. ✓ Proof Let $\lambda_{\max}$ be the largest eigenvalue of X. Then $\rho := \lambda_{\max} \mathbb{1} - X$ is positive semidefinite, singular, and nonzero (since X is not scalar), and unitality with the fixed-point equation for X gives $\mathcal{E}_A(\rho) = \rho$. Apply Theorem 9.14. The non-scalar hypothesis is needed because scalar multiples of $\mathbb{1}$ are fixed by every unital map and yield no nontrivial support projection. □

9.2.1 Iterated reduction to irreducible blocks

Iterating the two-block splitting on bond dimension decomposes any tensor into irreducible blocks; the remaining lemmas record the spectral facts about irreducible unital blocks and the algebra-spanning input used later.

Definition 9.16 (Irreducible tensor). Lean ✓ Stmt An MPS tensor A is *irreducible* if it has no nontrivial invariant projection.

Theorem 9.17 (Irreducible block decomposition). Lean ✓ Stmt *Every MPS tensor A admits a block-diagonal tensor $\bigoplus_{k=1}^{r} A_k$ with each A_k irreducible and $\mathcal{V}(A) = \mathcal{V}(\bigoplus_k A_k)$.*

Proof. ✓ Proof By strong induction on D. If A is irreducible, take $r = 1$. Otherwise a nontrivial invariant projection P gives the upper-triangular splitting of Theorem 9.14; deleting the off-diagonal part preserves the MPV family and yields two blocks of strictly smaller bond dimension. Apply the induction hypothesis to each block.

Both [PGVWC07, Theorem 4] and [CPGSV21, Section IV.A.1] obtain the block decomposition by the same invariant-projection splitting. ☐

Theorem 9.18 (Scalar fixed points in an irreducible unital block). Lean ✓ Stmt *If A is irreducible and unital, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and $\mathcal{E}_A(X) = X$, then X is a scalar multiple of $\mathbb{1}$. This is the final fixed-point assertion in the proof of [PGVWC07, Theorem 4].*

Proof. ✓ Proof Tensor irreducibility gives irreducibility of the completely positive transfer map. By Wolf's Theorem 6.6 for irreducible unital Kraus maps, every fixed point is scalar. ☐

Theorem 9.19 (Unital gauge from a full-rank Perron–Frobenius eigenvector). Lean Lean Lean ✓ Stmt *Let ρ be positive definite with $\mathcal{E}_A(\rho) = r\rho$ for some $r > 0$. For $B^i := r^{-1/2}\rho^{-1/2}A^i\rho^{1/2}$,*

$$\sum_i B^i(B^i)^\dagger = \mathbb{1}. \tag{9.1}$$

If $r = 1$, then the unscaled gauge $B^i := \rho^{-1/2}A^i\rho^{1/2}$ is unital and generates the same MPV family as A. This is the spectral-radius normalization and full-rank fixed-point gauge of [PGVWC07, Theorem 4].

Proof. ✓ Proof The positive square root of ρ is invertible. Substituting $B^i = r^{-1/2}\rho^{-1/2}A^i\rho^{1/2}$ into $\sum_i B^i(B^i)^\dagger$ and using $\mathcal{E}_A(\rho) = r\rho$ gives (9.1). For $r = 1$ the change from A to B is a similarity, so cyclicity of the trace gives equality of all MPV coefficients. ☐

Theorem 9.20 (Unitary diagonal positive fixed point in TP gauge). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $D > 0$. Then there exist a unitary U and a diagonal positive definite Λ such that, for $B^i := U^\dagger A^i U$,*

$$\sum_i (B^i)^\dagger B^i = \mathbb{1}, \qquad \mathcal{E}_B(\Lambda) = \Lambda.$$

Proof. ✓ Proof The TP hypothesis makes $\mathcal{E}_A$ a channel with a nonzero PSD fixed point. Irreducibility makes $\mathcal{E}_A$ an irreducible CP map, and Theorem 6.4 upgrades the fixed point to positive definite. The spectral theorem diagonalizes it by a unitary conjugation, which preserves the TP normalization. ☐

Remark 9.21 (Burnside input for cumulative spanning). ✓ Stmt The passage from irreducible tensors to full algebra spanning first passes through the irreducible action on $\mathbb{C}^D$. Burnside's theorem then says that the resulting irreducible subalgebra of $M_D(\mathbb{C})$ is all of $M_D(\mathbb{C})$. Once the algebra span is full, finite-dimensionality gives a single cumulative span that already equals $M_D(\mathbb{C})$.

Lemma 9.22 (The algebra span reaches a finite cumulative span)**.** Lean ✓ Stmt *If the generated unital algebra satisfies* $\mathrm{alg}(A) = M_D(\mathbb{C})$*, then there exists* N *such that* $T_N(A) = M_D(\mathbb{C})$*.*

Proof. ✓ Proof Every element of $\mathrm{alg}(A)$ lies in some cumulative span $T_n(A)$: the generators lie in $T_1(A)$, scalar multiples of the identity lie in $T_0(A)$, and the family $\{T_n(A)\}_{n\geq 0}$ is closed under addition and under multiplication with lengths adding. Since $M_D(\mathbb{C})$ is finite-dimensional, the ascending chain $T_0(A) \subseteq T_1(A) \subseteq \cdots$ stabilizes, so one can choose a single N with $T_N(A) = M_D(\mathbb{C})$. □

Lemma 9.23 (Irreducible action gives an irreducible tensor)**.** Lean ✓ Stmt *If the letters of* A *act irreducibly on* $\mathbb{C}^D$*, then* A *has no nontrivial invariant orthogonal projection.*

Proof. ✓ Proof If P were a nontrivial invariant orthogonal projection, then its range would be invariant under every letter of A. Irreducibility of the action forces this range to be either 0 or all of $\mathbb{C}^D$, so P must be 0 or $\mathbb{1}$, a contradiction. □

9.3 Normal canonical form

This section fixes the two canonical-form predicates used downstream: the block-injective form of Definition 9.24 and the spectral *normal* canonical form of Definition 9.26. It then records how the self-overlap clause and gauge-phase equivalence follow from primitivity, and closes with the periodicity-removal step that produces primitive transfer maps.

Recall from Definition 2.48 and Lemma 7.6 that the bilinear overlap is

$$O_{AB}(N) = \sum_{\sigma\in\{0,\dots,d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = \mathrm{Tr}(F_{AB}^N).$$

This notation enters the self-overlap clause of the canonical-form predicate below.

Definition 9.24 (Canonical form conditions)**.** Lean ✓ Stmt A collection of scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfies the *canonical form conditions* if:

1. each A_k is injective,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. the moduli are non-increasing: $|\mu_1| \geq |\mu_2| \geq \cdots \geq |\mu_r|$,
4. all $\mu_k \neq 0$,
5. the self-overlap converges: $O_{A_kA_k}(N) \to 1$ for each k.

Only this one-sided normalization is assumed; no unital condition is assumed here. Condition (5) is a primitivity hypothesis: for a primitive channel ([Wol12, Theorem 6.7(3)]), $T^n(\rho) \to \rho_\infty$ for every initial state, which forces the self-overlap to converge to 1. See also [Wol12, Theorem 6.8] for the completely-positive characterisation and Chapter 7 for the transfer-operator gap proof.

Remark 9.25 (Normalization convention). [PGVWC07, Theorem 4] uses the unital gauge $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$; here the blocks are in the dual TP gauge $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$. A positive-definite fixed point of the adjoint transfer map provides the change of gauge between the two (Theorem 6.14).

Definition 9.26 (Normal canonical form conditions). Lean ✓ Stmt Scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfy the *normal canonical form conditions* if:

1. each A_k is irreducible,
2. each A_k satisfies $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. each $\mathcal{E}_{A_k}$ is primitive (equivalently, $\sigma_\partial(\mathcal{E}_{A_k}) = \{1\}$),
4. $|\mu_1| \geq |\mu_2| \geq \cdots \geq |\mu_r|$,
5. all $\mu_k \neq 0$,
6. all bond dimensions are positive.

Here "normal" is the spectral sense of [CPGSV16], which by [SPGWC10, Proposition 3] is equivalent to the eventual block-injectivity formulation of Definition 2.31.

Theorem 9.27 (Self-overlap is derived in normal canonical form). Lean ✓ Stmt *If (μ_k, A_k) satisfies the normal canonical form conditions, then $O_{A_k A_k}(N) \to 1$ for every block k.*

Proof. ✓ Proof Fix a block k. Irreducibility and the TP normalization give a nonzero positive semidefinite fixed point ρ of $\mathcal{E}_{A_k}$ with fixed-point projector P; primitivity gives the complementary gap $\rho_{\mathrm{spec}}(\mathcal{E}_{A_k} - P) < 1$ (Theorem 4.57). Theorem 7.96 then yields $O_{A_k A_k}(N) \to 1$. ☐

Remark 9.28. See Theorem 7.27 for the modulus-one eigenvalue rigidity statement for irreducible trace-preserving blocks.

Lemma 9.29 (Mixed-transfer spectral radius from unit-modulus overlap). Lean ✓ Stmt *Let A and B have the same bond dimension. If $\|\langle V^{(N)}(A)|V^{(N)}(B)\rangle\| \to 1$, then the mixed transfer map has spectral radius at least 1.*

Proof. ✓ Proof If the mixed transfer spectral radius were < 1, the iterated mixed transfer map would tend to zero in operator norm, hence so would its trace; via $\mathrm{Tr}(F_{AB}^N) = \langle V^{(N)}(A)|V^{(N)}(B)\rangle$ the overlap would tend to 0, contradicting the hypothesis. ☐

Theorem 9.30 (Gauge-phase equivalence from unit-modulus overlap). Lean ✓ Stmt *Let A and B be irreducible MPS tensors of the same bond dimension with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. If $\|\langle V^{(N)}(A)|V^{(N)}(B)\rangle\| \to 1$, then A and B are gauge-phase equivalent.*

Proof. ✓ Proof By Lemma 9.29 the mixed transfer spectral radius is at least 1, and Theorem 7.27 then gives gauge-phase equivalence. ☐

9.3.1 Canonical form from primitivity

A normal block needs a primitive transfer map; an irreducible TP block may still have peripheral spectrum larger than $\{1\}$, namely the roots of unity coming from its period. Blocking by the least common multiple of those periods raises the transfer map to a power whose peripheral spectrum is $\{1\}$, removing the periodicity. In [CPGSV21] this is the passage from irreducibility to primitivity by blocking to period one.

Theorem 9.31 (Blocking yields a peripheral-spectrum primitive transfer map). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $D > 0$ and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then there exists a blocking length $p > 0$ such that $\sigma_\partial(\mathcal{E}_{A^{[p]}}) = \{1\}$.*

Proof. ✓ Proof Pass to the conjugate-transposed Kraus family $K_i := (A^i)^\dagger$, which is unital with irreducible Kraus map. The TP fixed-point theorem for $\mathcal{E}_A$ provides a positive definite fixed point for the adjoint of K, so Theorem 4.51 shows every peripheral eigenvalue is a root of unity. Since $\sigma_\partial(\mathcal{E}_A)$ is finite (Theorem 4.34), Lemma 4.38 gives a common exponent p with $\mu^p = 1$ for all $\mu \in \sigma_\partial(\mathcal{E}_A)$, and Lemma 4.39 gives $\sigma_\partial(\mathcal{E}_A^p) = \{1\}$. Blocking by p takes the pth power of the transfer map, so $\mathcal{E}_{A^{[p]}}$ is primitive. □

Remark 9.32. Theorem 9.31 is the periodicity-removal-by-blocking step of [CPGSV16, Appendix A] for an irreducible block already in TP gauge.

9.4 Left-canonical and dual diagonalization of irreducible blocks

Each irreducible block in trace-preserving gauge admits the PGVWC07 unital orientation together with a diagonal positive-definite dual fixed point (Theorem 9.33, the Perron–Frobenius gauge of Chapter 6, and Theorem 9.18); the full PGVWC07 canonical form is Theorem 9.1.

Theorem 9.33 (Left-canonical normalization for irreducible TP blocks). Lean ✓ Stmt *Let A be an irreducible MPS tensor in TP gauge ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$) with $D > 0$. Then there exist a unitary U and a diagonal positive definite Λ such that:*

1. *$B^i := U^\dagger A^i U$ is TP,*
2. *$\mathcal{E}_B(\Lambda) = \Lambda$,*
3. *Λ is diagonal and positive definite,*
4. *A and B generate the same MPV family.*

This is the left-canonical normalization step (Canonical Form II) of [CPGSV16, Appendix A]. The unitary conjugation is a second step inside TP gauge, performed only after TP normalization.

Proof. ✓ Proof Theorem 9.20 provides the unitary and the diagonal PD fixed point, and unitary conjugation preserves trace-preservation and the MPV family (Theorem 9.8). □

Theorem 9.34 (Dual diagonalization for irreducible unital blocks). Lean ✓ Stmt *Let A be an irreducible MPS tensor in unital gauge $\sum_i A^i (A^i)^\dagger = \mathbb{1}$ with $D > 0$. Then there exist a unitary U and a diagonal positive definite Λ such that, for $B^i := U^\dagger A^i U$,*

$$\sum_i B^i (B^i)^\dagger = \mathbb{1}, \qquad \sum_i (B^i)^\dagger \Lambda B^i = \Lambda, \tag{9.2}$$

and A, B generate the same MPV family. This is the dual-map fixed-point and unitary-diagonalization step of [PGVWC07, Theorem 4].

Proof. ✓ Proof Apply Theorem 9.33 to the conjugate-transposed family $i \mapsto (A^i)^\dagger$: unitality of A is trace preservation for this family, whose transfer map is the dual of $\mathcal{E}_A$. Translating back gives the equations (9.2). □

Lemma 9.35 (Common scalar rescaling of block weights). Lean ✓ Stmt *Multiplying every block weight in a block-diagonal MPS tensor by a scalar c multiplies every length-N MPV coefficient by c^N.*

Proof. ✓ Proof Expand the block-diagonal MPV coefficient as a finite sum over blocks and factor c^N out of the sum. □

Theorem 9.36 (Normal canonical form from a primitive block decomposition). Lean ✓ Stmt *Suppose A generates the same MPV family as $\bigoplus_k \mu_k A_k$, and that each A_k is irreducible, satisfies $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$, and has primitive transfer map, all $\mu_k \neq 0$, and all bond dimensions are positive. Then there exist a blocking length $p > 0$, weights (ν_k), and blocked tensors (B_k) such that $A^{[p]}$ generates the same MPV family as $\bigoplus_k \nu_k B_k$, and (ν_k, B_k) satisfies the normal canonical form conditions. The weight moduli need not be distinct; equal-modulus blocks are allowed.*

Proof. ✓ Proof The given blocks are already primitive, so take $p = 1$. Reorder the blocks by non-increasing weight modulus, with ties allowed, and set $\nu_k := \mu_k$ and $B_k := A_k$ in that order. Irreducibility, TP normalization, primitivity, nonzero weights, and positive bond dimensions are preserved, so the reordered family satisfies Definition 9.26 and generates the same MPV family as A. □

Remark 9.37 (Scope of Theorem 9.36). Theorem 9.36 assumes all six block hypotheses in advance and produces the normal canonical form by reordering. It is not the arbitrary-input existence theorem of [CPGSV16, Section 2.3]; that primitive-block reduction, including the zero-tail count, is Theorem 9.52 in Section 9.6.2.

9.5 Zero-block separation and the trace-preserving gauge

In an irreducible block decomposition some blocks may have all-zero Kraus operators. An all-zero block contributes $V^{(N)} = 0$ for every $N \geq 1$, so it adds nothing to the positive-length MPV family; at $N = 0$ the MPV coefficient is the bond dimension. The all-zero summand therefore enters the output not as a tensor but as a single count D_0 in the length-zero identity $D = D_0 + \sum_k D_k$, and all positive-length content is an equality of MPV families. This is the positive-length convention used throughout the reduction: comparisons of weighted block sums are stated for $N \geq 1$, with the zero block carried only by D_0. After separating the zero summand, each nonzero block is placed in the trace-preserving gauge, the furthest unconditional reduction available before periodicity removal.

Theorem 9.38 (Zero-block separation). Lean ✓ Stmt *Every MPS tensor A admits a block decomposition into a zero-block bond dimension D_0 and a finite family of nonzero blocks $(B_k)_{k=1}^r$, each irreducible, with at least one nonzero Kraus operator, and of positive bond dimension, recorded by two facts. The positive-length equality of MPV families: for every $N \geq 1$ and every σ,*

$$V^{(N)}(A)_\sigma = V^{(N)}(\textstyle\bigoplus_{k=1}^r B_k)_\sigma.$$

The length-zero dimension identity

$$D = D_0 + \sum_{k=1}^{r} D_k,$$

where D_0 is the total bond dimension of the all-zero blocks.

Proof. ✓ Proof Apply the irreducible block decomposition (Theorem 9.17) and partition the blocks by whether each has a nonzero Kraus operator; D_0 accumulates the bond dimensions of the all-zero blocks. For every all-zero irreducible block C_j and every $N \geq 1$, $V^{(N)}(C_j)_\sigma = 0$, so

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r B_k)_\sigma.$$

At $N = 0$ the MPV coefficient is the bond dimension, giving $D = D_0 + \sum_{k=1}^r D_k$. □

Theorem 9.39 (Blockwise TP-gauge normalization of irreducible blocks). Lean ✓ Stmt *Suppose A generates the same MPV family as $\bigoplus_{k=1}^r B_k$, where each B_k is irreducible with at least one nonzero Kraus operator. Then there are nonzero weights $(\mu_k)_{k=1}^r$ and blocks $(C_k)_{k=1}^r$ such that, for every N and every σ,*

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r \mu_k C_k)_\sigma = \sum_{k=1}^r \mu_k^N V^{(N)}(C_k)_\sigma,$$

each C_k irreducible and left-canonical, $\sum_i (C_k^i)^\dagger C_k^i = \mathbb{1}$, with positive bond dimension. This is the blockwise form used in the arbitrary-tensor trace-preserving gauge below.

Proof. ✓ Proof The Perron–Frobenius TP-gauge construction is applied to each irreducible block. Gauge equivalence preserves irreducibility and the MPV family, and the positive scaling factors are absorbed into the weights μ_k. □

9.5.1 Trace-preserving gauge for arbitrary tensors

Theorem 9.40 (TP-gauge reduction for arbitrary tensors with zero-block separation). Lean ✓ Stmt *For any MPS tensor A, there exist a zero-block bond dimension D_0 and nontrivial blocks $(B_k)_{k=1}^r$ with nonzero weights $(\mu_k)_{k=1}^r$, each block irreducible, left-canonical ($\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$), and of positive bond dimension, such that for every $N \geq 1$ and every σ,*

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r \mu_k B_k)_\sigma,$$

and the zero-block contribution is recorded by the length-zero identity

$$D = D_0 + \sum_{k=1}^r D_k.$$

Proof. ✓ Proof Theorem 9.38 supplies D_0 and the nonzero irreducible blocks $(C_k)_{k=1}^r$. For every $N \geq 1$ and every σ,

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r C_k)_\sigma.$$

The blockwise TP-gauge theorem then replaces each C_k by a left-canonical block B_k with weight μ_k. Again for every $N \geq 1$ and every σ,

$$V^{(N)}(\bigoplus_{k=1}^r C_k)_\sigma = V^{(N)}(\bigoplus_{k=1}^r \mu_k B_k)_\sigma.$$

The length-zero identity $D = D_0 + \sum_{k=1}^r D_k$ is preserved. □

9.6 Blocking, period removal, and primitive blocks

Blocking by an integer p carries MPV equalities, weights, transfer maps, and identifications of the blocked physical alphabet. The first consequence is the existence of a common blocking period under which every transfer map in a finite family becomes primitive; the period-removal and primitive-block reductions then proceed from that common period.

Theorem 9.41 (Common blocking period for a block family). Lean ✓ Stmt *Let $(B_k)_{k=1}^r$ be a finite family of positive-bond-dimension MPS tensors, each admitting a blocking period p_k making its transfer map primitive. Then $p = \mathrm{lcm}(p_1, \dots, p_r)$ is a common period: $\mathcal{E}_{B_k^{[p]}}$ is primitive for every k.*

Proof. ✓ Proof Each p_k divides p, so primitivity of $\mathcal{E}_{B_k^{[p_k]}}$ carries over to $\mathcal{E}_{B_k^{[p]}}$ via the divisibility identity for transfer-map powers. □

The following stability statements are used when a primitive irreducible block is blocked again for a common-refinement or injectivity length. This later blocking is distinct from the period-removal length that produced the sector block.

Theorem 9.42 (Blocking preserves irreducibility for primitive irreducible blocks). Lean ✓ Stmt *Let A be left-canonical, irreducible, primitive, of positive bond dimension. For every $P > 0$, the blocked tensor $A^{[P]}$ is irreducible: there is no orthogonal projection Q with $Q \neq 0$, $Q \neq \mathbb{1}$ such that, for every P-blocked index α,*

$$(\mathbb{1} - Q)(A^{[P]})^\alpha Q = 0.$$

Proof. ✓ Proof The blocked transfer map is the corresponding positive power of the original one:

$$\mathcal{E}_{A^{[P]}}(X) = \mathcal{E}_A^P(X).$$

Hence a positive-definite fixed point of $\mathcal{E}_A$ remains fixed by $\mathcal{E}_{A^{[P]}}$. Primitivity gives uniqueness of positive semidefinite fixed points for the blocked map, hence its transfer map is irreducible, which is the irreducibility criterion for $A^{[P]}$. □

Theorem 9.43 (Extra blocking of primitive irreducible sector blocks). Lean ✓ Stmt *Let A be trace-preserving, primitive, and tensor-irreducible. For every $k > 0$,*

$$\sum_\alpha ((A^{[k]})^\alpha)^\dagger (A^{[k]})^\alpha = \mathbb{1}, \qquad \sigma_\partial(\mathcal{E}_{A^{[k]}}) = \{1\},$$

and $A^{[k]}$ is tensor-irreducible. This k is a later common-refinement or injectivity length, distinct from the period-removal length that produced the sector block.

Proof. ✓ Proof Trace preservation follows from blocking a trace-preserving tensor, primitivity from a positive power of a primitive transfer map, and irreducibility from Theorem 9.42. □

9.6.1 Period removal by cyclic sectors

The step needed below is the period-removing blocking described in [CPGSV16, Section 2.3]: after blocking each irreducible TP block by the least common multiple of its peripheral periods, the result has no nontrivial p-periodic vectors. The cyclic-sector decomposition comes from the peripheral spectrum of the adjoint transfer map [Wol12, Theorem 6.6].

Theorem 9.44 (Period removal by blocking). Lean ✓ Stmt *Let A be a left-canonical irreducible tensor. Then there is a period-removing length $m \geq 1$ such that $A^{[m]}$ admits:*

1. *left-canonical sector tensors $(C_k)_{k=1}^m$ of nonzero bond dimension,*
2. *a unit-weight direct-sum MPV decomposition $A^{[m]} \sim \bigoplus_{k=1}^m C_k$,*
3. *orthogonal projections $(P_k)_{k=1}^m$ summing to $\mathbb{1}$ with $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$,*

4. *commutation relations* $P_k A_i^{[m]} = A_i^{[m]} P_k$ *for every blocked letter* i,

5. *the sector trace formula* $f_{C_k}(\sigma) = \operatorname{tr}(P_k A_{\sigma_1}^{[m]} \cdots A_{\sigma_N}^{[m]})$,

6. *linear isomorphisms onto the compression corners* $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k M_D(\mathbb{C}) P_k$,

7. *transfer-intertwining identities* $\varphi_k(\mathcal{E}_{C_k^\dagger}(X)) = \mathcal{E}_{(P_k A^{[m]})^\dagger}(\varphi_k(X))$,

8. *multiplicativity and adjoint compatibility* $\varphi_k(XY) = \varphi_k(X)\varphi_k(Y)$ *and* $\varphi_k(X^\dagger) = \varphi_k(X)^\dagger$.

This is the cyclic-sector part of the period-removal step of [CPGSV16, Section 2.3]; the projections P_k *come from the peripheral spectrum of the adjoint transfer map ([Wol12, Theorem 6.6]).*

Proof. ✓ Proof The cyclic decomposition of the adjoint transfer map gives projections (P_k) with $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$. Iterating the cyclic relation m times gives $(\mathcal{E}_{A^\dagger})^m(P_k) = P_k$, and the blocked-transfer identity $\mathcal{E}_{(A^{[m]})^\dagger} = (\mathcal{E}_{A^\dagger})^m$ identifies these with the adjoint transfer map of $A^{[m]}$. Apply the block decomposition from adjoint-fixed projections. □

Theorem 9.45 (Unconditional primitive blocked sectors). Lean ✓ Stmt *Let* A *be a left-canonical irreducible tensor whose adjoint transfer map has peripheral spectrum* $\{\gamma^j : 0 \le j < m\}$ *for a primitive* m*th root of unity* γ, *with blocked cyclic-sector decomposition* $(C_u)_{u=1}^m$, $(P_u)_{u=1}^m$, *and linear isomorphisms* $\varphi_u : M_{\dim C_u}(\mathbb{C}) \xrightarrow{\sim} P_u M_D(\mathbb{C}) P_u$. *Assume explicitly that the* P_u *are orthogonal projections summing to* $\mathbb{1}$ *and satisfying* $\mathcal{E}_{A^\dagger}(P_{u+1}) = P_u$, *that each sector has nonzero bond dimension, and that the compression maps satisfy*

$$\varphi_u(\mathcal{E}_{C_u^\dagger}(X)) = \mathcal{E}_{(P_u A^{[m]})^\dagger}(\varphi_u(X)), \qquad \varphi_u(XY) = \varphi_u(X)\varphi_u(Y), \qquad \varphi_u(X^\dagger) = \varphi_u(X)^\dagger.$$

Here $P_u A^{[m]}$ *denotes the blocked tensor with letters* $P_u (A^{[m]})^\alpha$. *Then every sector tensor satisfies* $\sigma_\partial(\mathcal{E}_{C_u}) = \{1\}$ *and is tensor-irreducible.*

Proof. ✓ Proof Theorem 9.67 gives irreducibility of $(\mathcal{E}_A^\dagger)^m$ on each corner P_u. The compression equivalences φ_u intertwine the adjoint transfer map of C_u with the corresponding corner restriction of $(\mathcal{E}_A^\dagger)^m$. The peripheral spectrum of the blocked adjoint map is

$$\sigma_\partial((\mathcal{E}_A^\dagger)^m) = \{(\gamma^j)^m : 0 \le j < m\} = \{1\},$$

because γ is a primitive mth root of unity. Hence each nonzero irreducible corner restriction is primitive. Primitivity and irreducibility are preserved across φ_u; then use Theorem 9.59 to pass from the adjoint blocked map to the primal transfer map of C_u. □

Theorem 9.46 (Period removal with primitive irreducible sectors). Lean ✓ Stmt *Let* A *be a left-canonical irreducible tensor. Then there is a period* $m \ge 1$ *and a unit-weight decomposition of* $A^{[m]}$ *into sector blocks* $(C_k)_{k=1}^m$, *each of which is left-canonical, has primitive transfer map, is tensor-irreducible, and has nonzero bond dimension. Equivalently, the cyclic-sector decomposition of* $A^{[m]}$ *already satisfies the trace-preservation, primitivity, and irreducibility hypotheses needed for the common-blocking step (Lemma 9.66).*

Proof. ✓ Proof Left-canonicality makes $K_i := (A^i)^\dagger$ unital, irreducibility passes to the adjoint map, and the quantum Perron–Frobenius theorem provides a positive-definite fixed point; the cyclic peripheral-spectrum theorem and Theorem 9.44 then produce the cyclic-sector decomposition. Keep the projections and compression equivalences, and apply Theorem 9.45 to the sector blocks. □

Theorem 9.47 (Supported corner compression with a support isometry). Lean ✓ Stmt *Let A be a tensor normalized on the support of an orthogonal projection P, so that $PA^iP = A^i$ and $\sum_i (A^i)^\dagger A^i = P$. Then there are a corner tensor C, a linear isomorphism*

$$\varphi : M_{\dim C}(\mathbb{C}) \xrightarrow{\sim} PM_D(\mathbb{C})P,$$

and a rectangular isometry $V : \mathbb{C}^{\dim C} \to \mathbb{C}^D$ such that

$$V^\dagger V = \mathbb{1}, \qquad VV^\dagger = P, \qquad \varphi(X) = VXV^\dagger.$$

Moreover $\varphi(C^i) = A^i$ for every physical letter, and for every word $\sigma = (\sigma_1, \dots, \sigma_N)$,

$$V^{(N)}(C)_\sigma = \operatorname{tr}\big(PA^{\sigma_1} \cdots A^{\sigma_N}\big).$$

Proof. ✓ Proof Diagonalize the projection P by a unitary change of basis. The inclusion of the support coordinates gives a rectangular isometry V with $V^\dagger V = \mathbb{1}$ and $VV^\dagger = P$. Compress the letters of A to this support and let φ be the corresponding corner isomorphism. The support relation $PA^iP = A^i$ gives the corner-letter identity. For each word $\sigma = (\sigma_1, \dots, \sigma_N)$ the MPV component is obtained from the trace computation

$$V^{(N)}(C)_\sigma = \operatorname{tr}(C^{\sigma_1} \cdots C^{\sigma_N}) = \operatorname{tr}(V^\dagger A^{\sigma_1} \cdots A^{\sigma_N} V) = \operatorname{tr}(PA^{\sigma_1} \cdots A^{\sigma_N}).$$

The final equality uses cyclicity of the trace and $VV^\dagger = P$. □

Theorem 9.48 (Commuting projections give sector isometries). Lean ✓ Stmt *Let a left-canonical tensor A commute with a family of orthogonal projections (P_k) summing to $\mathbb{1}$. Then A decomposes into compressed sector tensors (C_k), with a unit-weight direct-sum MPV decomposition, corner isomorphisms $\varphi_k : M_{\dim C_k}(\mathbb{C}) \to P_k M_D(\mathbb{C}) P_k$, and rectangular isometries V_k satisfying*

$$V_k^\dagger V_k = \mathbb{1}, \qquad V_k V_k^\dagger = P_k, \qquad \varphi_k(X) = V_k X V_k^\dagger.$$

The sector letters obey $\varphi_k(C_k^i) = P_k A^i P_k$.

Proof. ✓ Proof Apply Theorem 9.47 to the compression of A on each sector P_k. Orthogonality and $\sum_k P_k = \mathbb{1}$ give the direct-sum decomposition. For every word $\sigma = (\sigma_1, \dots, \sigma_N)$ the partition of unity gives

$$V^{(N)}(A)_\sigma = \operatorname{tr}\Big(\big(\sum_k P_k\big) A^{\sigma_1} \cdots A^{\sigma_N}\Big) = \sum_k \operatorname{tr}(P_k A^{\sigma_1} \cdots A^{\sigma_N}) = \sum_k V^{(N)}(C_k)_\sigma,$$

which is the unit-weight matrix-product-vector identity. □

Theorem 9.49 (Adjoint-fixed projections give sector isometries). Lean ✓ Stmt *Let A be left-canonical and let (P_k) be orthogonal projections summing to $\mathbb{1}$ and fixed by the adjoint transfer map. Then the same sector decomposition and support isometries of Theorem 9.48 exist. In particular, the adjoint fixed-point condition $\mathcal{E}_A^\dagger(P_k) = P_k$ gives*

$$P_k A^i = A^i P_k,$$

which gives the corner-letter identity

$$\varphi_k(C_k^i) = P_k A^i P_k.$$

Proof. ✓ Proof The fixed-point equation $\mathcal{E}_A^\dagger(P_k) = P_k$, together with left-canonicality $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, places P_k in the multiplicative domain of $\mathcal{E}_A^\dagger$, forcing

$$P_k A^i = A^i P_k$$

for every letter A^i. The result then follows from Theorem 9.48. □

Theorem 9.50 (Spectral cyclic sectors with support isometries)**.** Lean ✓ Stmt *Let A be a left-canonical irreducible tensor, and suppose the adjoint transfer map $\mathcal{E}_{A^\dagger}$ has a positive-definite fixed point, is irreducible as a positive map, and has peripheral spectrum*

$$\sigma_\partial(\mathcal{E}_{A^\dagger}) = \{\gamma^j : j \in \mathbb{Z}/m\mathbb{Z}\}$$

for a primitive mth root of unity γ, with $m \geq 1$. Then the spectral cyclic-sector decomposition of $A^{[m]}$ has compression maps which may be chosen with rectangular support isometries V_k satisfying

$$V_k^\dagger V_k = \mathbb{1}, \qquad V_k V_k^\dagger = P_k, \qquad \varphi_k(X) = V_k X V_k^\dagger.$$

These isometries also realize the blocked corner letters $\varphi_k(C_k^\alpha) = P_k(A^{[m]})^\alpha P_k$.

Proof. ✓ Proof The cyclic peripheral-spectrum decomposition applied to $\mathcal{E}_{A^\dagger}$ provides projections P_k satisfying $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$. Iterating m times gives $(\mathcal{E}_{A^\dagger})^m(P_k) = P_k$, and the identity $(\mathcal{E}_{A^\dagger})^m = \mathcal{E}_{(A^{[m]})^\dagger}$ shows that each P_k is fixed by the adjoint transfer map of $A^{[m]}$. Applying Theorem 9.49 to that blocked tensor gives the compressed sector tensors and the support isometries. □

9.6.2 Reduction to primitive blocks

This subsection completes the reduction after period removal. A TP-primitive-irreducible block is normal, and an arbitrary tensor becomes, after a positive blocking length, a weighted family of left-canonical primitive blocks with the zero-block dimension recorded separately. For two tensors with the same MPV family, the cyclic sectors can be expressed over one common blocked alphabet, which is the form used by the Fundamental Theorem.

Theorem 9.51 (TP-primitive irreducible tensor is normal)**.** Lean ✓ Stmt *Let A be a left-canonical MPS tensor with $D > 0$, irreducible transfer map, and peripheral spectrum $\{1\}$. Then A is normal.*

Proof. ✓ Proof Irreducibility upgrades the PSD fixed point to positive definite (Theorem 6.4), and Theorem 8.75 then gives normality. □

An arbitrary translation-invariant tensor reduces, after blocking, to a weighted direct sum of primitive irreducible blocks.

Theorem 9.52 (Arbitrary tensor to primitive block decomposition)**.** Lean ✓ Stmt *For any MPS tensor A, there exist $p \geq 1$, a zero-tail bond dimension D_0, nonzero weights $(\mu_k)_{k=1}^r$, and blocks $(B_k)_{k=1}^r$ such that, for every blocked σ of positive length N,*

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\textstyle\bigoplus_{k=1}^r \mu_k B_k)_\sigma.$$

The zero-block contribution is recorded separately by the length-zero identity $D_0 + \sum_{k=1}^r D_k = D$. Each block is left-canonical and primitive,

$$\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}, \qquad \sigma_\partial(\mathcal{E}_{B_k}) = \{1\},$$

and each bond dimension is positive.

Proof. ✓ Proof Theorem 9.40 supplies the zero-block dimension, the left-canonical nontrivial blocks, the positive-length equality, and the identity $D = D_0 + \sum_k D_k$. Theorem 9.41 provides a common blocking period p making all transfer maps primitive, Lemma 9.65 keeps the blocked weights nonzero, and Lemma 9.63 carries the positive-length equality through blocking. □

Theorem 9.53 (Cyclic sectors after the zero-block split). Lean ✓ Stmt *If A and B generate the same MPV family, then after the zero-block split and the trace-preserving gauge there are weighted nonzero-block tensors $\bigoplus_j \mu_j^A A_j$ and $\bigoplus_k \mu_k^B B_k$ with, for every $N \geq 1$ and σ,*

$$V^{(N)}(A)_\sigma = V^{(N)}(\textstyle\bigoplus_j \mu_j^A A_j)_\sigma, \qquad V^{(N)}(B)_\sigma = V^{(N)}(\textstyle\bigoplus_k \mu_k^B B_k)_\sigma,$$

and the two nonzero parts agree:

$$V^{(N)}(\textstyle\bigoplus_j \mu_j^A A_j)_\sigma = V^{(N)}(\textstyle\bigoplus_k \mu_k^B B_k)_\sigma.$$

Each nonzero block has primitive irreducible cyclic sectors: for each j, there is $m_j \geq 1$ with a unit-weight sector decomposition $A_j^{[m_j]} \sim \bigoplus_{u=1}^{m_j} C_{j,u}^A$, and similarly for B_k.

Proof. ✓ Proof Apply Theorem 9.40 separately to A and B. Its positive-length conclusion gives, for $N \geq 1$,

$$V^{(N)}(A)_\sigma = V^{(N)}(\textstyle\bigoplus_j \mu_j^A A_j)_\sigma, \qquad V^{(N)}(B)_\sigma = V^{(N)}(\textstyle\bigoplus_k \mu_k^B B_k)_\sigma.$$

Chaining these through $V(A) = V(B)$ identifies the two nonzero parts. Finally apply Theorem 9.46 to every trace-preserving irreducible nonzero-weight block. □

Theorem 9.54 (Common blocking length for cyclic sectors). Lean ✓ Stmt *If two tensors generate the same MPV family, then the cyclic-sector decompositions on the two sides can be expressed at one common positive blocking length p. For every $N \geq 1$ and every blocked σ, each blocked tensor equals its powered nonzero part,*

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\textstyle\bigoplus_j (\mu_j^A)^p (A_j)^{[p]})_\sigma, \qquad V^{(N)}(B^{[p]})_\sigma = V^{(N)}(\textstyle\bigoplus_k (\mu_k^B)^p (B_k)^{[p]})_\sigma,$$

and the two powered nonzero parts agree,

$$V^{(N)}\big(\textstyle\bigoplus_j (\mu_j^A)^p (A_j)^{[p]}\big)_\sigma = V^{(N)}\big(\textstyle\bigoplus_k (\mu_k^B)^p (B_k)^{[p]}\big)_\sigma.$$

Proof. ✓ Proof Start from Theorem 9.53. With p_A, p_B the least common multiples of the period-removal lengths on the two sides, use $p = \mathrm{lcm}(p_A, p_B)$. The prescribed-common-multiple lemma constructs both one-sided sector families at this length, and the positive-length blocking lemmas carry the one-sided tensor/nonzero-part equalities over to the common alphabet; for every $N \geq 1$,

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}\big(\textstyle\bigoplus_j (\mu_j^A)^p (A_j)^{[p]}\big)_\sigma.$$

The same applies on the B side, and the two-sided blocking lemma gives equality of the two powered nonzero parts. □

Theorem 9.55 (Unconditional common primitive block decompositions). Lean ✓ Stmt *Let A and B have the same MPV family. Then there is a positive blocking length at which both blocked tensors have the same positive-length MPV family as weighted direct sums of trace-preserving, primitive, tensor-irreducible blocks with positive bond dimensions and nonzero weights. The two weighted nonzero-sector families have the same positive-length MPV family.*

Proof. ✓ Proof Apply Theorem 9.54 to choose the common blocking length and the two common cyclic-sector families. For either one of these families, write G_k for its cyclic-sector blocks, and write (ν_x, C_x) for the weights and blocks of the flattened common-sector family from Lemma 9.73. The common-alphabet nonzero-part identity gives, for every length N and blocked word σ,

$$V^{(N)}(\bigoplus_k \mu_k^p G_k^{[p]})_\sigma = V^{(N)}(\bigoplus_x \nu_x C_x)_\sigma,$$

Applying this on the two sides of the comparison yields a positive length p and weighted primitive block families such that, for every $N \geq 1$ and blocked word σ,

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\bigoplus_x \mu_x^A A_x)_\sigma, \qquad V^{(N)}(B^{[p]})_\sigma = V^{(N)}(\bigoplus_y \mu_y^B B_y)_\sigma,$$

with the two weighted nonzero parts equal at every positive length. The structural properties and the nonvanishing of the weights come from the common-sector families. □

A weighted block sum with distinct sectors is the simplest sector decomposition: one block per sector, each of multiplicity one. Equal-modulus and repeated-copy sectors are not single scalar powers, already for $A = C \oplus (-C)$ with $V^{(N)}(A)_\sigma = (1 + (-1)^N)\, V^{(N)}(C)_\sigma$; they are compared by the basis-of-normal-tensors construction of Chapter 10.

Definition 9.56 (Single-copy sector decomposition). Lean ✓ Stmt Given nonzero weights $(\mu_k)_{k=1}^r$ and blocks $(B_k)_{k=1}^r$, the *single-copy sector decomposition* has r sectors, one basis block B_k per sector, multiplicity 1, and sector weight μ_k. As a finite sum,

$$\sum_{k=1}^{r} \mu_k^N V^{(N)}(B_k)_\sigma = V^{(N)}(\bigoplus_k \mu_k B_k)_\sigma.$$

Lemma 9.57 (Single-copy sector decomposition preserves the represented family). Lean ✓ Stmt *The single-copy sector decomposition of $(\mu_k, B_k)_k$ represents the same MPS family as $\bigoplus_k \mu_k B_k$.*

Proof. ✓ Proof Expanding the sector decomposition gives $\sum_k \mu_k^N V^{(N)}(B_k)_\sigma$, the finite block-sum expression for $\bigoplus_k \mu_k B_k$. □

9.7 Blocking and weighted direct sums

Blocking by a length p commutes with weighted direct sums up to the pth power of each weight: for nonzero weights (μ_k) and blocks (B_k),

$$\left(\bigoplus_k \mu_k B_k\right)^{[p]} \sim \bigoplus_k \mu_k^p\, B_k^{[p]},$$

where $\sim$ is equality of MPV families. When two block families are reblocked to a common length $p = m_k e_k$, the p-blocked alphabet is identified with the e_k-fold tensor power of the m_k-blocked alphabet, so each sector reblocked through its own period agrees with the directly p-blocked block.

Lemma 9.58 (Canonical form from peripheral primitivity). Lean ✓ Stmt *Let $(\mu_k, A_k)_{k=1}^r$ be nonzero weights and injective left-canonical blocks with positive bond dimensions, non-increasing moduli $|\mu_1| \geq \cdots \geq |\mu_r| > 0$, and $\sigma_\partial(\mathcal{E}_{A_k}) = \{1\}$. Then $(\mu_k, A_k)_{k=1}^r$ satisfies the canonical-form conditions of Definition 9.24, and $O_{A_k A_k}(N) \to 1$ for every k.*

Proof. ✓ Proof The injectivity, left-canonical equation, weight ordering, and nonzero-weight clauses are exactly the hypotheses. For each block, peripheral primitivity gives the complementary spectral gap $\rho_{\mathrm{spec}}(\mathcal{E}_{A_k} - P) < 1$ by Theorem 4.57. Theorem 7.96 then yields $O_{A_k A_k}(N) \to 1$. □

Lemma 9.59 (Adjoint primitivity equivalence). Lean ✓ Stmt *Let $\mathcal{E}$ be a finite-dimensional completely positive map. Then $\mathcal{E}$ is primitive if and only if its Frobenius adjoint $\mathcal{E}^\dagger$ is primitive; equivalently, $\sigma_\partial(\mathcal{E}) = \{1\}$ iff $\sigma_\partial(\mathcal{E}^\dagger) = \{1\}$, since the peripheral spectrum of $\mathcal{E}^\dagger$ is the complex conjugate of that of $\mathcal{E}$.*

Lemma 9.60 (Primitivity under blocking by a multiple). Lean ✓ Stmt *Let A have positive bond dimension and let $p_0, p \geq 1$ with $p_0 \mid p$. If $\mathcal{E}_{A^{[p_0]}}$ is primitive, then $\mathcal{E}_{A^{[p]}}$ is primitive.*

Lemma 9.61 (Blocking the weighted block sum). Lean ✓ Stmt *For nonzero weights $(\mu_k)_{k=1}^r$, blocks $(B_k)_{k=1}^r$, and any $p \geq 1$,*

$$(\bigoplus_k \mu_k B_k)^{[p]} \sim \bigoplus_k \mu_k^p \, B_k^{[p]},$$

where $\sim$ denotes generation of the same MPV family at every length.

Lemma 9.62 (Positive-length equality survives blocking). Lean ✓ Stmt *If two tensors have the same MPV coefficients at every positive length, then so do their blocked tensors after any positive blocking length.*

Lemma 9.63 (Positive-length blocking of a weighted block sum). Lean ✓ Stmt *If A and $\bigoplus_k \mu_k B_k$ have the same MPV coefficients at every positive length, then after any positive blocking length p, $A^{[p]}$ and $\bigoplus_k \mu_k^p B_k^{[p]}$ have the same MPV coefficients at every positive length.*

Lemma 9.64 (Positive-length equality of powered block sums). Lean ✓ Stmt *If $\bigoplus_j \mu_j^A A_j$ and $\bigoplus_k \mu_k^B B_k$ have the same MPV coefficients at every positive length, then after any positive common blocking length p, $\bigoplus_j (\mu_j^A)^p A_j^{[p]}$ and $\bigoplus_k (\mu_k^B)^p B_k^{[p]}$ have the same MPV coefficients at every positive length.*

Lemma 9.65 (Nonzero weights survive blocking). Lean ✓ Stmt *If $\mu \in \mathbb{C}$ is nonzero and $p \geq 1$, then $\mu^p \neq 0$; blocking preserves the nonzero-weight hypothesis of a weighted block decomposition.*

Lemma 9.66 (Common reblocking of cyclic-sector families). Lean ✓ Stmt *Given a finite family $(B_k)_{k=1}^r$, each with primitive irreducible cyclic sectors, there is a common blocking length p such that each $B_k^{[p]}$ admits a unit-weight direct-sum decomposition over a single common blocked alphabet $d^{[p]}$ into left-canonical blocks with primitive transfer maps, tensor irreducibility, and positive bond dimensions.*

Lemma 9.67 (Sector irreducibility for the adjoint blocked map). Lean ✓ Stmt *Let A be a left-canonical irreducible MPS tensor with $D > 0$, let m be the period of its cyclic-sector decomposition, and let $(P_u)_{u=1}^m$ be the cyclic projections of the adjoint transfer map with $\mathcal{E}_{A^\dagger}(P_{u+1}) = P_u$ and $\sum_u P_u = \mathbb{1}$. Then the restriction of $(\mathcal{E}_{A^\dagger})^m$ to each corner $P_u M_D(\mathbb{C}) P_u$ is irreducible.*

Lemma 9.68 (Iterated blocking at the transfer-map level). Lean ✓ Stmt *For any MPS tensor A and $m, n \geq 0$,*

$$\mathcal{E}_{(A^{[m]})^{[n]}} = \mathcal{E}_{A^{[mn]}}.$$

Definition 9.69 (Primitive irreducible cyclic-sector decomposition). Lean ✓ Stmt *A has primitive irreducible cyclic sectors* of period $m \geq 1$ if there is a family $(C_k)_{k=1}^m$ with $A^{[m]} \sim \bigoplus_{k=1}^m C_k$ (unit weights), each C_k left-canonical, with primitive transfer map, tensor-irreducible, and of positive bond dimension.

Lemma 9.70 (Common reblocking at a prescribed multiple). Lean ✓ Stmt *With hypotheses as in Lemma 9.66, and given any common multiple p of the per-block period-removal lengths (m_k), the common reblocking can be performed at the prescribed length p.*

Lemma 9.71 (Structural properties of the common reblocked sectors). Lean ✓ Stmt *Each sector tensor in the common reblocked cyclic-sector family of Lemma 9.66 is trace-preserving ($\sum_i (B^i)^\dagger B^i = \mathbb{1}$), has primitive transfer map ($\sigma_\partial(\mathcal{E}_B) = \{1\}$), is tensor-irreducible, and has positive bond dimension.*

Lemma 9.72 (Common-alphabet flattening). Lean ✓ Stmt *With $p = m_k e_k$ for each k, the iterated blocked tensor $(B_k^{[m_k]})^{[e_k]}$ agrees with $B_k^{[p]}$ in MPV family, after the canonical identification of the p-blocked alphabet $d^{[p]}$ with the e_k-fold tensor power of the m_k-blocked alphabet $d^{[m_k]}$.*

Lemma 9.73 (Reindexed nonzero-part identity). Lean ✓ Stmt *The powered nonzero-part decomposition $\bigoplus_k \mu_k^p B_k^{[p]}$ carries over under the common-alphabet identification of Lemma 9.72: the common-alphabet sectors have the same nonzero-part decomposition as the direct p-blocked tensors.*

Lemma 9.74 (Direct blocking in the common alphabet). Lean ✓ Stmt *For every original block B_k, direct blocking at the common length $p = m_k e_k$ agrees, as an MPV family, with the same block transported to the common blocked alphabet. Write $\widetilde{B}_k$ for this common-alphabet image. Then*

$$V^{(N)}(B_k^{[p]})_\sigma = V^{(N)}(\widetilde{B}_k)_\sigma.$$

Chapter 10

Basis of Normal Tensors

A *basis of normal tensors* (BNT) for an MPS tensor A is a finite family of normal blocks $\{A_j\}$ whose MPV states $\{|V^{(N)}(A_j)\rangle\}$ span $|V^{(N)}(A)\rangle$ at every length and are eventually linearly independent. Two payoffs follow. First, equal MPV families force a bijective gauge-phase matching of the basis sectors $B_k = e^{i\phi_k} X_k A_{j_k} X_k^{-1}$. Second, the sector coefficients $c_N^{(j)} = \sum_q \mu_{j,q}^N$ recover the per-copy weight multisets by Newton–Girard. The references are [PGVWC07], [CPGSV16], and [CPGSV21].

10.1 Bases of normal tensors

Definition 10.1 (Basis of Normal Tensors (BNT)). Lean ✓ Stmt A *basis of normal tensors* (BNT) for a total tensor A_{tot} consists of g block tensors $(A_1, \dots, A_g)$ with bond dimensions $(D_1, \dots, D_g)$ such that each A_j is normal (Definition 2.31); at each system size N the MPV of A_{tot} lies in the span of the block MPVs $\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g}$; and for sufficiently large N those block MPVs are linearly independent. This is [CPGSV21, Definition IV.2]. In the canonical-form characterization of [CPGSV16, Section 2.3], a BNT is a minimal family of representatives for $\widetilde{A}_k^i = e^{i\phi_k} X_k A_j^i X_k^{-1}$ among the normal blocks.

Definition 10.2 (Basis of normal tensors, coefficient form). Lean ✓ Stmt The coefficient form of a basis of normal tensors: each block A_j is a CPSV normal tensor (Definition 9.4), the MPV of A is given at every length by explicit coefficients $c_j^{(N)} \in \mathbb{C}$ with $|V^{(N)}(A)\rangle = \sum_{j=1}^{g} c_j^{(N)} |V^{(N)}(A_j)\rangle$, and the block MPVs $\{|V^{(N)}(A_j)\rangle\}$ are eventually linearly independent.

The eventual linear-independence clause is produced from the asymptotic overlaps: diagonal overlaps tending to one and off-diagonal overlaps tending to zero make the Gram matrix converge to the identity.

Lemma 10.3 (Eventual linear independence from finite-index overlap orthonormality). Lean ✓ Stmt *Let I be a finite index set and let $(A_i)_{i\in I}$ be a finite family of tensors such that $O_{A_iA_i}(N) \to 1$ for each i, and $O_{A_iA_j}(N) \to 0$ whenever $i \neq j$. Then the MPV states $\{|V^{(N)}(A_i)\rangle\}_{i\in I}$ are linearly independent for all sufficiently large N.*

Proof. ✓ Proof The Gram matrix of the family $\{|V^{(N)}(A_i)\rangle\}_{i\in I}$ converges entrywise to the identity matrix. Hence for all sufficiently large N it is invertible, so the MPV states are linearly independent. □

Lemma 10.4 (Eventual linear independence from overlap orthonormality). Lean ✓ Stmt *Let $(A_j)_{j=1}^g$ be a finite family of tensors such that $O_{A_jA_j}(N) \to 1$ for each j and $O_{A_iA_j}(N) \to 0$ for $i \neq j$. Then the MPV states $\{|V^{(N)}(A_j)\rangle\}_{j=1}^g$ are linearly independent for all sufficiently large N.*

Proof. ✓ Proof Apply Lemma 10.3 with $I = \{1, \dots, g\}$. □

The same conclusion is often more convenient with an explicit threshold rather than the eventual quantifier.

Theorem 10.5 (Existential BNT independence from overlap limits). Lean ✓ Stmt *If the self-overlaps of a finite block family tend to 1 and the cross-overlaps of distinct blocks tend to 0, then the MPV states are linearly independent for all sufficiently large system sizes.*

Proof. ✓ Proof This is Lemma 10.4, restated with an explicit threshold N_0. □

Eventual linear independence is also what converts an equality of two finite linear combinations into equality of their coefficients; this is the mechanism behind the coefficient identities of Chapter 11.

Lemma 10.6 (Eventual coefficient extraction from linear independence). Lean ✓ Stmt *If a finite family of vectors is linearly independent for all sufficiently large N, and two finite linear combinations of that family are equal for all sufficiently large N, then the corresponding coefficients are equal for all sufficiently large N.*

Proof. ✓ Proof Move the two linear combinations to one side. Linear independence forces each coefficient difference to vanish at every sufficiently large length. □

When two BNT families must be compared against each other, the same Gram-matrix criterion applies to their disjoint union, provided the mixed overlaps also decay. This is the form used in the exact sector matching of Section 10.6.

Lemma 10.7 (Eventual linear independence for two families). Lean ✓ Stmt *Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be finite families of tensors. Suppose the overlaps inside each family are asymptotically orthonormal, and suppose every mixed overlap $O_{A_jB_k}(N)$ tends to 0. Then the combined MPV family*

$$\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g_A} \cup \{|V^{(N)}(B_k)\rangle\}_{k=1}^{g_B}$$

is linearly independent for all sufficiently large N.

Proof. ✓ Proof Apply the finite-index Gram-matrix criterion to the disjoint union of the two index sets. The within-family hypotheses give the diagonal and off-diagonal limits inside each block, and the mixed overlap hypothesis gives the off-diagonal limits between the two blocks. □

We now isolate the separation condition that supplies the off-diagonal decay and combine it with the canonical-form hypotheses.

Definition 10.8 (No gauge-phase-equivalent distinct blocks). Lean ✓ Stmt A finite block family has no gauge-phase-equivalent distinct blocks when, for any two distinct indices with the same bond dimension, the corresponding blocks are not gauge-phase equivalent.

Definition 10.9 (Normal canonical form with BNT separation). Lean ✓ Stmt A family is in *normal canonical form with BNT separation* if it satisfies Definition 9.26 and distinct blocks of the same bond dimension are not gauge-phase equivalent. The weight moduli are non-increasing but need not be strictly decreasing, so equal-modulus blocks are allowed; the grouping into a

BNT is governed by gauge-phase equivalence, not by distinctness of moduli. Repeated equal-modulus copies of one basis tensor are not separate blocks here; they appear as the weights $\mu_{j,q}$ in $c_N^{(j)} = \sum_q \mu_{j,q}^N$ in the sector decomposition below.

The off-diagonal decay needed for the BNT criterion follows from the separation condition together with the overlap-decay theorems of the preceding chapter.

Theorem 10.10 (Cross-overlap decay from explicit BNT hypotheses). Lean ✓ Stmt *Let $(A_j)_{j=1}^r$ be a family in which each A_j is injective and left-canonical, $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$, and distinct equal-dimension blocks are not gauge-phase equivalent. Then $O_{A_j A_k}(N) \to 0$ for every pair $j \neq k$.*

Proof. ✓ Proof If the bond dimensions differ, Theorem 7.22 gives the decay. If the bond dimensions agree, the non-equivalence hypothesis excludes gauge-phase equivalence, so Theorem 7.24 gives the decay. □

Lemma 10.11 (BNT from explicit blockwise hypotheses). Lean ✓ Stmt *Let $(\mu_k, A_k)_{k=1}^r$ be a weighted block family. Assume each A_k is injective and left-canonical, $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$, that $O_{A_k A_k}(N) \to 1$ for each k, and that distinct equal-dimension blocks are not gauge-phase equivalent. Then the blocks form a basis of normal tensors for the block-diagonal tensor $\bigoplus_k \mu_k A_k$.*

Proof. ✓ Proof Injective blocks are normal. The block-diagonal decomposition theorem gives the MPV expansion of $\bigoplus_k \mu_k A_k$ in terms of the block MPVs. The self-overlap limits are part of the hypotheses, and Theorem 10.10 supplies the off-diagonal decay. Therefore Lemma 10.4 gives the eventual linear independence required in Definition 10.1. □

The normal-canonical formulation encodes normality spectrally (primitive transfer map), the BNT definition algebraically (eventual block injectivity).

Remark 10.12 (Comparing the two normality conditions). The spectral normality of Definition 9.26 and the algebraic normality of Definition 2.31 agree: the Wielandt theory of Chapter 8 supplies the implication from primitivity to eventual block injectivity used here.

Lemma 10.13 (Restricted normal-CF-BNT hypotheses give a BNT). Lean ✓ Stmt *If a family is in normal canonical form with BNT separation and each block is also normal in the algebraic sense, then the blocks form a basis of normal tensors for the associated block-diagonal tensor. The blocks here are distinct representatives.*

Proof. ✓ Proof The normal-canonical hypotheses already give the block-diagonal MPV expansion, the self-overlap normalization, and the eventual linear independence coming from irreducible trace-preserving overlap decay. Supplying algebraic normality for each block adds the remaining clause in Definition 10.1. □

10.2 Permutation rigidity for bases of normal tensors

When two bases of normal tensors generate the same MPV subspace at every length, the blocks must agree up to a permutation and per-block gauge phases. The argument turns the equality of spans into an invertible change-of-basis matrix, then analyses its asymptotics through the overlaps.

Lemma 10.14 (Invertible change of basis). Lean ✓ Stmt *Let $v_1, \dots, v_g$ and $w_1, \dots, w_g$ be two linearly independent families in a complex vector space with the same span. Then there exists an invertible matrix $U \in \mathrm{GL}_g(\mathbb{C})$ such that*

$$w_j = \sum_{i=1}^{g} U_{ij} v_i$$

for every j.

Proof. ✓ Proof Because the two families span the same subspace, each w_j has a unique expansion in the basis $\{v_i\}_{i=1}^g$; these coefficients form the columns of U. If $Ua = 0$, then the corresponding linear relation among the w_j is trivial because the w_j are linearly independent. Thus U is injective, hence invertible. ☐

Lemma 10.15 (Eventual change of basis from equal spans). Lean ✓ Stmt *If two finite families of blocks have asymptotically orthonormal overlaps within each family and their MPV spans agree for every system size N, then for all sufficiently large N there exists an invertible matrix U_N expressing the MPV states of one family as linear combinations of the MPV states of the other.*

Proof. ✓ Proof By Lemma 10.4, both families are linearly independent for all sufficiently large N. At such an N, the span equality lets us apply Lemma 10.14. ☐

The change-of-basis matrix is now used to match the blocks. We first treat equal block counts, then remove that restriction.

Theorem 10.16 (Permutation rigidity for equal block counts). Lean ✓ Stmt *Let $(A_j)_{j=1}^g$ and $(B_j)_{j=1}^g$ be injective families of positive bond dimension satisfying trace-preserving normalization, with self-overlaps tending to 1 and off-diagonal overlaps tending to 0 within each family. If the spans of the MPV states agree for every system size N, then there is a permutation π of $\{1, \dots, g\}$ such that A_j and $B_{\pi(j)}$ have the same bond dimension and are gauge-phase equivalent.*

Proof. ✓ Proof For large N, Lemma 10.15 gives an invertible matrix U_N relating the two MPV families. Fix j and write $|V^{(N)}(B_j)\rangle = \sum_i u_{ij}^{(N)} |V^{(N)}(A_i)\rangle$. With $G_A(N) = (O_{A_i A_\ell}(N))_{i,\ell}$ and $b_j(N) = (O_{A_i B_j}(N))_i$, this gives

$$G_A(N)\overline{u_j(N)} = b_j(N), \qquad O_{B_j B_j}(N) = u_j(N)^\top G_A(N)\overline{u_j(N)}.$$

If all mixed overlaps $O_{A_i B_j}(N)$ tended to 0, then $b_j(N) \to 0$; since $G_A(N) \to I$, one obtains $\overline{u_j(N)} \to 0$ and hence $u_j(N) \to 0$. Thus $O_{B_j B_j}(N) \to 0$, contradicting $O_{B_j B_j}(N) \to 1$. Thus each B_j has a matched block $A_{f(j)}$ with non-decaying mixed overlap. Theorems 7.22 and 7.24 then force equal bond dimension and gauge-phase equivalence for each match. Off-diagonal decay within the B-family makes f injective, hence bijective, and its inverse is the required permutation. ☐

Lemma 10.17 (Permutation rigidity with asymptotically orthonormal overlaps). Lean ✓ Stmt *Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be two families of injective tensors of positive bond dimension satisfying the TP normalization $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$ and $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$, whose self-overlaps converge to 1 and whose cross-overlaps within each family converge to 0. If for every system size N,*

$$\mathrm{span}_{\mathbb{C}}\{|V^{(N)}(A_j)\rangle : 1 \le j \le g_A\} = \mathrm{span}_{\mathbb{C}}\{|V^{(N)}(B_k)\rangle : 1 \le k \le g_B\},$$

then $g_A = g_B$, and there is a permutation π of the common block index set such that A_j and $B_{\pi(j)}$ have the same bond dimension and are gauge-phase equivalent for each j.

Proof. ✓ Proof By Lemma 10.4, both families are linearly independent for all sufficiently large N. At such an N, the common span has the same dimension when computed from the A-family or the B-family, so $g_A = g_B$. After identifying the two index sets, Theorem 10.16 yields the permutation conclusion. □

The two preceding results obtain the block matching from the asymptotic orthonormality of the overlaps. The same matching is obtained by the exact sector comparison of Section 10.6, which pairs the sectors of two canonical forms through a bijection fixed by their bond dimensions and gauge-phase equivalences. The proof of the Fundamental Theorem in Chapter 11 proceeds through the sector comparison.

The same overlap dichotomy also shows that the MPV states of distinct normal blocks are non-proportional at arbitrarily large lengths; this is an input to the exact sector matching later in the chapter.

Theorem 10.18 (Non-equivalent irreducible TP blocks are eventually non-proportional). Lean Lean ✓ Stmt *Let A and B be irreducible trace-preserving blocks whose self-overlaps converge to 1. If they are not gauge-phase equivalent, then for every lower bound N_0 there is $N \geq N_0$ such that $|V^{(N)}(A)\rangle$ is not proportional to $|V^{(N)}(B)\rangle$. Consequently, in an already separated same-dimensional BNT block family, any two distinct blocks have such non-proportional MPV states at arbitrarily large lengths.*

Proof. ✓ Proof Suppose that for all sufficiently large N one had $|V^{(N)}(A)\rangle = c_N|V^{(N)}(B)\rangle$. Then

$$O_{AA}(N) = |c_N|^2 O_{BB}(N), \qquad O_{AB}(N) = c_N O_{BB}(N). \tag{10.1}$$

Since $O_{AA}(N) \to 1$ and $O_{BB}(N) \to 1$, the first identity in (10.1) gives $|c_N| \to 1$, and hence $|O_{AB}(N)| \to 1$. The irreducible trace-preserving overlap decay theorem gives $O_{AB}(N) \to 0$ when $A \nsim_{\mathrm{gp}} B$, a contradiction. □

10.3 Newton–Girard identities and power-sum recovery

The coefficients $c_N^{(j)} = \sum_q \mu_{j,q}^N$ that appear in the two-layer BNT expansion are power sums of the copy weights. Recovering the individual weights from these sums is a Newton–Girard problem: equal power sums determine equal characteristic polynomials, hence equal multisets of roots.

Theorem 10.19 (Newton–Girard trace recursion). Lean Lean ✓ Stmt *If $A, B \in M_n(\mathbb{C})$ satisfy $\mathrm{tr}(A^k) = \mathrm{tr}(B^k)$ for all $k \geq 1$, then A and B have the same characteristic polynomial. It is enough to assume this equality for $1 \leq k \leq n$.*

Proof. ✓ Proof Let e_m denote the mth elementary symmetric polynomial in the eigenvalues, with $e_0 = 1$. The Newton–Girard recursion

$$me_m = \sum_{i=1}^{m} (-1)^{i-1} e_{m-i}\, \mathrm{tr}(A^i)$$

expresses the coefficients of the characteristic polynomial in terms of the power-sum traces $\mathrm{tr}(A^i)$. Equal power sums for A and B therefore give equal coefficients, hence equal characteristic polynomials. □

The matched-sector comparison below applies the power-sum identities to a single matched pair, after the BNT permutation and phase gauges have identified which basis tensors are compared: the coefficient $\sum_q \mu_{j,q}^N$ then equals $\sum_q (\nu_{j,q} e^{i\phi_j})^N$, and Newton–Girard recovers the weight multiset with multiplicity.

Theorem 10.20 (Finite-range equal power sums imply equal multisets). Lean Lean Lean Lean ✓ Stmt *Let $\alpha, \beta : \{0, \dots, n-1\} \to \mathbb{C}$. If $\sum_i \alpha_i^k = \sum_i \beta_i^k$ for $1 \le k \le n$, then the multisets $\{\alpha_i\}$ and $\{\beta_i\}$ are equal.*

Two variants relax the equal-cardinality assumption. If $\alpha : \{0, \dots, m-1\} \to \mathbb{C}$ and $\beta : \{0, \dots, n-1\} \to \mathbb{C}$ have no zero entries and their power sums agree for $1 \le k \le \max(m, n)$, then $m = n$ and the multisets are equal. Without a nonzero hypothesis, the list form $\sum_{k=1}^{x_a} \lambda_{a,k}^N = \sum_{k=1}^{x_b} \lambda_{b,k}^N$ for $1 \le N \le \max(x_a, x_b)$ gives equality of the nonzero multisets; if it also holds at $N = 0$ then $x_a = x_b$ and the full lists agree.

Proof. ✓ Proof Apply the finite-range Newton–Girard theorem to $\operatorname{diag}(\alpha)$ and $\operatorname{diag}(\beta)$ and extract the roots of $\prod_i (X - \alpha_i)$ and $\prod_i (X - \beta_i)$. For unequal cardinalities, pad both lists with zeros to length $\max(m, n)$; the nonzero hypothesis makes the padded-zero counts $\max(m, n) - m$ and $\max(m, n) - n$, forcing $m = n$. Without that hypothesis, delete the zero entries first (unchanged for positive N); the equality at $N = 0$ gives $x_a = x_b$. □

Theorem 10.21 (Equal power sums imply equal multisets). Lean ✓ Stmt *Let $\alpha, \beta : \{0, \dots, n-1\} \to \mathbb{C}$. If $\sum_i \alpha_i^k = \sum_i \beta_i^k$ for all $k \ge 1$, then the multisets $\{\alpha_i\}$ and $\{\beta_i\}$ are equal.*

Proof. ✓ Proof Construct diagonal matrices $\operatorname{diag}(\alpha)$ and $\operatorname{diag}(\beta)$. The power-sum hypothesis gives $\operatorname{tr}(\operatorname{diag}(\alpha)^k) = \operatorname{tr}(\operatorname{diag}(\beta)^k)$. By Theorem 10.19, the characteristic polynomials agree: $\prod_i (X - \alpha_i) = \prod_i (X - \beta_i)$. Extracting roots gives multiset equality. □

10.4 The BNT canonical form on a sector decomposition

This section introduces the BNT canonical-form predicate of [CPGSV16, Section 2], defined on a sector decomposition with raw sector weights $\mu_{j,q}$ and coefficients $c_N^{(j)} = \sum_q \mu_{j,q}^N$; no equal-modulus factorization is assumed.

10.4.1 Sector decompositions and phase matching

The sector decomposition and the MPV phase-equivalence predicates introduced here feed the BNT canonical-form predicate below.

Definition 10.22 (Sector weights and sector decomposition). Lean Lean ✓ Stmt A sector-weight tuple over g basis blocks consists of multiplicities $r_j > 0$, nonzero weights $\mu_{j,q}$ for $q \in \{1, \dots, r_j\}$, and the sector coefficients

$$c_{S,j}^{(N)} := \sum_{q=1}^{r_j} \mu_{j,q}^N.$$

A *sector decomposition* consists of the basis blocks $(A_j)_{j=1}^g$ together with these multiplicities and weights.

Lemma 10.23 (Sector decomposition formula). Lean ✓ Stmt *For a sector decomposition S with associated tensor A_{tot}, the MPV satisfies*

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = \sum_{j=1}^{g} c_{S,j}^{(N)} V^{(N)}(A_j)_\sigma.$$

Two blocks are identified for canonical-form purposes when their MPV families differ by a scalar power. The next three definitions formalize this relation, the common cover it generates between two families, and the phase classes of a single family.

Definition 10.24 (MPV phase equivalence for nonzero-weight blocks). Lean ✓ Stmt Two blocks A and B are MPV-phase equivalent if there is a nonzero scalar ζ such that, for every system size N and every physical word,

$$V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

The two bond dimensions are allowed to differ.

Definition 10.25 (Common MPV phase cover). Lean ✓ Stmt For two finite nonzero-weight block families, a common MPV phase cover consists of a third finite family of blocks, a map from each nonzero-weight block family onto that common family, and MPV-phase equivalences between each block and its common representative. Both maps are required to be surjective, so the common family contains no unused representative.

Definition 10.26 (MPV phase classes). Lean Lean Lean ✓ Stmt For a finite family of blocks, put $j \sim k$ when there is a nonzero scalar factor ζ such that

$$|V^{(N)}(B_k)\rangle = \zeta^N |V^{(N)}(B_j)\rangle$$

for every length N. The associated tuple enumerates the finite quotient, chooses one representative per class, records the scalar relating the representative to each member, and proves that distinct representatives are not MPV-phase equivalent.

Definition 10.27 (BNT canonical form). Lean ✓ Stmt Given a sector decomposition P, the BNT canonical-form predicate records the minimal BNT hypotheses of [CPGSV16, CPGSV21] without any equal-modulus or strict-ordering condition on the raw sector weights. It asserts that each basis block has positive bond dimension, is irreducible, and left-canonical; the normalized self-overlap of every basis block converges to 1; the basis MPV family is eventually linearly independent; distinct same-dimension basis blocks are not gauge-phase equivalent after identifying equal bond dimensions; and the [CPGSV16, Section 2.3] canonical-form normalization holds,

$$\forall (j,q),\ |\mu_{j,q}| \le 1, \qquad \exists\, (j_*, q_*) : |\mu_{j_*,q_*}| = 1.$$

The MPV expansion takes the raw two-layer form

$$V^{(N)}(P)_\sigma = \sum_{j=1}^{g} \left(\sum_{q=1}^{r_j} \mu_{j,q}^N \right) V^{(N)}(P_j)_\sigma,$$

matching [CPGSV16, Section 2.3] and [CPGSV21, Definition IV.2]. The canonical-form normalization admits every example of the source paper (in particular $C \oplus D$, $C \oplus (-C)$, $C \oplus (1/2)C$, and $C \oplus (-C) \oplus (1/2)C$) and is not derivable from the per-block normality hypotheses alone.

The matching arguments below read off three facts from a BNT canonical form: the raw coefficient identity, the cross-overlap decay between distinct basis blocks, and the combined-family linear independence.

Lemma 10.28 (Raw two-layer sector coefficient identity). Lean ✓ Stmt *For every sector decomposition P, length N, and basis index j,*

$$c_N^{(j)} = \sum_{q=1}^{r_j} \mu_{j,q}^N.$$

This is [CPGSV16, Section 2.3].

Proof. ✓ Proof Immediate from the definition of the sector coefficient. □

Lemma 10.29 (Cross-overlap decay between distinct basis blocks). Lean ✓ Stmt *If P is a BNT canonical form and $j \neq k$, then $O_{P_j P_k}(N) \to 0$.*

Proof. ✓ Proof Case on the bond dimensions. If they differ, the dimension-mismatch overlap-decay theorem applies. If they agree, the basis-distinctness clause rules out gauge-phase equivalence, so the equal-dimension overlap-decay theorem applies. □

Lemma 10.30 (Combined-family eventual linear independence). Lean ✓ Stmt *Let P and Q each be a BNT canonical form, and suppose every mixed overlap $O_{P_j Q_k}(N)$ tends to 0. Then the union family*

$$\{|V^{(N)}(P_j)\rangle\}_{j=1}^{g_P} \cup \{|V^{(N)}(Q_k)\rangle\}_{k=1}^{g_Q}$$

is linearly independent for all sufficiently large N. This is [CPGSV16, Appendix A].

Proof. ✓ Proof Feed the per-family normalized self-overlaps, the within-family cross-overlap decay (Lemma 10.29), and the inter-family decay hypothesis into the two-family overlap-orthonormality criterion (Lemma 10.7). □

Lemma 10.31 (Unit-modulus weight witness from the canonical-form normalization). Lean ✓ Stmt *Under the BNT canonical form, the [CPGSV16, Section 2.3] unit-modulus existential provides indices (j_*, q_*) with $|\mu_{j_*,q_*}| = 1$.*

Proof. ✓ Proof The canonical-form normalization includes a unit-modulus weight $|\mu_{j_*,q_*}| = 1$. □

Example 10.32 ($C \oplus (1/2)C$, unequal-modulus admissibility). Lean ✓ Stmt The single-sector two-copy decomposition with raw weights $(1, 1/2)$ is a BNT canonical form, so unequal-modulus copies are admissible: $|1/2| \leq 1$ and the unit witness is the first copy.

10.4.2 Prepared-block construction of BNT canonical forms

The construction below separates the after-blocking preparation of suitable blocks from the normalized-weight BNT construction: the prepared blocks are trace-preserving, primitive, irreducible, and injective; normalized weights then quotient gauge-phase-equivalent blocks into sectors to produce the canonical form of Definition 10.27.

Lemma 10.33 (Common injective blocking for primitive irreducible families). Lean ✓ Stmt *Let $(B_k)_{k=1}^r$ be a finite family of positive-bond-dimension MPS tensors, each trace-preserving, primitive, and tensor-irreducible. Then there is a single $L \geq 1$ such that each $B_k^{[L]}$ is one-site injective, and this blocking preserves trace preservation, primitivity, and irreducibility for every block.*

Proof. ✓ Proof For each block, Theorem 8.84 gives a positive length at which it becomes one-site injective; the product of these lengths is a common length. Injectivity persists under positive further blocking, and Theorem 9.43 preserves the other hypotheses. □

Theorem 10.34 (Prepared BNT blocks after blocking). Lean ✓ Stmt *Let A be an MPS tensor. Then there is a blocking length $p \geq 1$, nonzero weights μ_k, and blocks B_k over the p-blocked alphabet such that:*

1. *each block has positive bond dimension;*
2. *each block is trace-preserving, primitive, tensor-irreducible, and one-site injective;*

3. *$A^{[p]}$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family at every positive length.*

No weight-modulus normalization is asserted at this stage.

Proof. ✓ Proof Apply the common primitive irreducible block decomposition to (A, A), giving a positive blocking length and a nonzero weighted family of trace-preserving primitive irreducible blocks with the positive-length MPV identity. Lemma 10.33 supplies one common further blocking making every block one-site injective while preserving the other hypotheses; finally identify the iterated blocked alphabet with the direct one. □

With the prepared blocks in hand, normalized weights produce a BNT canonical form by quotienting gauge-phase-equivalent blocks into sectors.

Theorem 10.35 (BNT from normal blocks with normalized weights)**.** Lean ✓ Stmt *Let $r \in \mathbb{N}$, and for each $k = 1, \ldots, r$ let $D_k \in \mathbb{N}$, $\mu_k \in \mathbb{C}$, and let B_k be an MPS tensor of physical dimension d and bond dimension D_k. Assume:*

1. ***positive bond dimensions:*** *$D_k > 0$ for all k,*
2. ***TP normalization:*** *$\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$,*
3. ***primitivity:*** *$\mathcal{E}_{B_k}$ is primitive for each k,*
4. ***irreducibility:*** *each B_k is irreducible,*
5. ***nonzero weights:*** *$\mu_k \neq 0$ for all k,*
6. ***bounded weights:*** *$|\mu_k| \leq 1$ for all k,*
7. ***global unit witness:*** *$|\mu_k| = 1$ for some k.*

Then there exists a sector decomposition P such that P and $\bigoplus_{k=1}^r \mu_k B_k$ generate the same MPV family at every length, and P satisfies the basis-of-normal-tensors conditions (Definition 10.2) together with the multiplicity, span, and weight-bound conditions of Definition 10.27.

Proof. ✓ Proof The phase-class quotient groups gauge-phase-equivalent blocks into sectors with unit-modulus per-copy scalars. Under the TP, primitive, and irreducible hypotheses the phase-class construction applies, and for distinct representatives B_j, $B_{j'}$ $(j \neq j')$ the cross-overlap decays,

$$\lim_{N\to\infty} O_{B_j B_{j'}}(N) = 0, \qquad \lim_{N\to\infty} O_{B_j B_j}(N) = 1,$$

which gives the eventual linear-independence condition with no injectivity required. The resulting sector decomposition inherits the span and multiplicity conditions, and the bounded-weight and global-unit hypotheses give the weight-norm conditions. □

The BNT construction preserves total bond dimension because CF summands assigned to the same BNT representative have equal bond dimension. Indeed, suppose trace-preserving primitive irreducible NTs X and Y of positive bond dimension have MPV families related by $|V^{(N)}(Y)\rangle = \zeta^N |V^{(N)}(X)\rangle$ with $\zeta \neq 0$. The normalized self-overlaps give $|O_{XX}(N)| \to 1$. The phase relation gives the cross-overlap identity

$$O_{YX}(N) = \zeta^N O_{XX}(N), \qquad |O_{YX}(N)| = |\zeta|^N |O_{XX}(N)|.$$

Since the same normalization forces $|\zeta| = 1$, this gives $|O_{YX}(N)| \to 1$. If their bond dimensions differed, the rectangular overlap-decay theorem would give

$$O_{YX}(N) \to 0,$$

a contradiction.

Lemma 10.36 (Total dimension after grouping by BNT representatives). Lean ✓ Stmt *Suppose CF summands are grouped by BNT representative, every original copy weight is nonzero, and every summand assigned to a representative has that representative's bond dimension. Then the total bond dimension of the resulting BNT sector decomposition is the sum of the original block dimensions.*

Proof. ✓ Proof The collapsed decomposition counts one representative dimension for each assigned copy. Let $e(j,q)$ denote the original CF summand assigned to the qth copy of representative P_j. Since $(j,q) \mapsto e(j,q)$ reindexes the original summands and each assigned summand has the representative's bond dimension,

$$\sum_{j=1}^{g} \sum_{q=1}^{r_j} \dim(P_j) = \sum_{j=1}^{g} \sum_{q=1}^{r_j} D_{e(j,q)} = \sum_{k=1}^{r} D_k.$$

This is the equality between the collapsed total dimension and the original direct-sum dimension. □

Lemma 10.37 (Prepared phase-equivalent NTs have the same bond dimension). Lean ✓ Stmt *Let X and Y be trace-preserving, primitive, irreducible NTs of positive bond dimension. If their MPV families differ by a nonzero scalar power, then X and Y have the same bond dimension.*

Proof. ✓ Proof Choose $\zeta \neq 0$ with $|V^{(N)}(Y)\rangle = \zeta^N |V^{(N)}(X)\rangle$ for every length N. The normalized self-overlaps give $|O_{XX}(N)| \to 1$, and the phase relation gives

$$O_{YX}(N) = \zeta^N O_{XX}(N), \qquad |O_{YX}(N)| = |\zeta|^N |O_{XX}(N)|.$$

Since the same normalization forces $|\zeta| = 1$, this gives $|O_{YX}(N)| \to 1$. If the two bond dimensions differed, the rectangular overlap-decay theorem would force

$$O_{YX}(N) \to 0,$$

contradicting the preceding limit. □

Lemma 10.38 (Prepared BNT representative classes preserve bond dimension). Lean ✓ Stmt *In a prepared family of positive-bond-dimension trace-preserving primitive irreducible NTs, every CF summand assigned to a BNT representative has that representative's bond dimension.*

Proof. ✓ Proof Apply Lemma 10.37 to each representative and each assigned summand. □

Lemma 10.39 (Prepared collapsed BNT total dimension). Lean ✓ Stmt *For prepared trace-preserving primitive irreducible NTs of positive bond dimension with all copy weights nonzero, grouping CF summands by BNT representative preserves the total bond dimension: the collapsed sector decomposition has total bond dimension equal to the sum of the original block dimensions.*

Proof. ✓ Proof Every assigned summand shares the bond dimension of its representative, by the equal-dimension fact above. The displayed regrouping identity in Lemma 10.36 then identifies the collapsed sum over pairs (j,q) with $\sum_{k=1}^{r} D_k$, the original direct-sum dimension. □

Combining the after-blocking preparation with the normalized-weight construction gives the arbitrary-input statement, conditional on the weight normalization.

Theorem 10.40 (Conditional BNT form after blocking for an arbitrary tensor). Lean Lean ✓ Stmt *Let A be an MPS tensor. Then there exists a blocking length $p \geq 1$, nonzero weights μ_k, and blocks B_k over the blocked alphabet, such that:*

1. *every block has positive bond dimension;*

2. *every block is TP-normalized, primitive, irreducible, and injective;*

3. $A^{[p]}$ *and* $\bigoplus_k \mu_k B_k$ *generate the same MPV family at every positive length.*

Moreover, under $|\mu_k| \le 1$ *for every* k *and* $|\mu_k| = 1$ *for at least one* k*, there exists a sector decomposition* P *over the blocked alphabet in the BNT canonical form of Definition 10.27 such that* $A^{[p]}$ *and* P *generate the same MPV family at every positive length; the identity upgrades to all lengths once the total bond dimension of* P *equals that of* $A^{[p]}$*.*

Proof. ✓ Proof The after-blocking reduction supplies the TP-normalized primitive irreducible injective blocks and the positive-length MPV identity. The normalized-weight hypotheses are the choice that [CPGSV16, Section 2.3] makes; Theorem 10.35 then produces the sector decomposition, following the BNT selection of [CPGSV16, Appendix A]. At length zero the MPV coefficient is the bond dimension, so equal total bond dimensions supply the missing length-zero coefficient. The remaining rescaling step is recorded in Remark 10.41. □

Remark 10.41 (Arbitrary-input reduction). The reduction is conditional on the two weight-normalization hypotheses of Theorem 10.40.

10.5 Coefficient comparison and copy-weight recovery

The exact matching of Section 10.6 produces, on each matched pair, a coefficient identity. This section turns that identity into equality of the per-copy weight multisets [CPGSV16, Appendix A]. No analytic non-decay estimate enters: the fixed-length linear independence isolates each sector coefficient exactly.

10.5.1 Matched-sector weight multiset equality

Given a matched pair $(j_0, \tilde{k}_0)$ and a nonzero scalar ζ with $c_N^{(j_0)}(P) = \zeta^N c_N^{(\tilde{k}_0)}(Q)$ past a threshold N_0, this section records two conclusions: the multiset equality of the per-copy weights from power-sum recovery (Theorem 10.21), and its upgrade to an explicit per-copy permutation. Together they give the $\mu_{j,q} = \nu_{j,q}\, e^{i\varphi_j}$ recovery of [CPGSV16, Appendix A]. The coefficient identity is supplied, in the equal-MPV case, by Lemma 11.3 of Chapter 11; in the proportional case it is the remaining hypothesis of the conditional proportional global gauge (Theorem 11.10).

Lemma 10.42 (Matched-sector weight multiset equality)**.** Lean ✓ Stmt *Let* P, Q *be sector decompositions with weights* $\mu_{j,q}$ *and* $\nu_{k,q'}$*, and fix indices* j_0*,* $\tilde{k}_0$*, and a nonzero scalar* $\zeta \in \mathbb{C} \setminus \{0\}$*. Suppose that for every* $N > N_0$*,*

$$c_N^{(j_0)}(P) = \zeta^N \cdot c_N^{(\tilde{k}_0)}(Q). \tag{10.2}$$

Then the per-sector multiplicities agree, $r_{j_0}(P) = r_{\tilde{k}_0}(Q)$*, and the rescaled per-copy weight multisets coincide. This is [CPGSV16, Appendix A] read on a single matched sector.*

Proof. ✓ Proof Expand both sides of the coefficient identity (10.2) into per-copy power sums $\sum_q \mu_{j_0,q}^N$ and $\sum_{q'} \nu_{\tilde{k}_0,q'}^N$, and distribute ζ^N through the right-hand sum to obtain, for every $N > N_0$,

$$\sum_q \mu_{j_0,q}^N = \sum_{q'} (\zeta \cdot \nu_{\tilde{k}_0,q'})^N.$$

Extend to all positive exponents and apply the unequal-cardinality multiset-recovery lemma (Theorem 10.20), using $\zeta \neq 0$ together with the weight non-vanishing assumption to ensure the input weights are nonzero. This yields the multiplicity match and the multiset equality. $\square$

Lemma 10.43 (Matched-sector weight-permutation extractor). Lean ✓ Stmt *Under the hypotheses of Lemma 10.42, the per-sector multiplicities agree and there is a permutation τ of the per-copy index set identifying the individual weights up to the gauge phase: for every q,*

$$\nu_{\tilde{k}_0,\tau(q)} = \zeta^{-1} \cdot \mu_{j_0,q}.$$

This is the per-copy realisation of [CPGSV16, Appendix A]: the multiset equality from Lemma 10.42 is upgraded to a concrete indexing between the matched P- and Q-copies.

Proof. ✓ Proof Invoke Lemma 10.42 to obtain the cardinality match and the multiset equality. From this multiset-map equality extract a permutation matching the elements pointwise (proved by induction on the cardinality). Compose with the cardinality identification and cancel ζ via $\zeta \neq 0$ to convert the gauge factor on the right of the pointwise identity to ζ^{-1} on the left. $\square$

10.6 Strong existential matching by exact linear independence

The exact matching reads [CPGSV16, Appendix A] as a statement at fixed length: equal MPV families force each basis sector of one form to pair with a basis sector of the other. The non-vanishing of the sector coefficients isolates the matched sector exactly, with no limit of a single-sector projection.

Lemma 10.44 (Sector coefficient is not eventually zero). Lean ✓ Stmt *For a BNT canonical form P and any sector j, the sector coefficient $c^{(N)}_{P,j} = \sum_q \mu^N_{j,q}$ is not eventually zero in N. Only nonvanishing of the copy weights is used, not any modulus condition.*

Proof. ✓ Proof If $c^{(N)}_{P,j} = 0$ for all $N \geq M$, the signed geometric-sequence vanishing lemma forces $\sum_q \mu^k_{j,q} = 0$ at every exponent k, in particular $\sum_q 1 = r_j = 0$ at $k = 0$, contradicting $r_j > 0$. Repeated ratios are handled by the same extrapolation, so distinct weights are not needed. $\square$

Theorem 10.45 (Exact single-sector matching). Lean ✓ Stmt *Let P, Q be two BNT canonical forms with the same MPV family (at all positive lengths). For every sector j of P there exist a sector k of Q with matched bond dimension $\dim_P(j) = \dim_Q(k)$, a gauge-phase equivalence after identifying the matched bond dimensions, and a non-decaying cross-overlap. No per-sector unit-modulus hypothesis is required.*

Proof. ✓ Proof Suppose j decays against every sector of Q, and let T be the set of P-sectors with this property. For $j' \notin T$ the overlap dichotomy turns a non-decaying overlap with some $Q_{k(j')}$ into a gauge-phase equivalence, hence a scalar $\omega_{j'} \in \mathbb{C}^\times$ with $V^{(N)}(P_{j'}) = \omega^N_{j'} V^{(N)}(Q_{k(j')})$, so $V^{(N)}(P_{j'})$ lies in the span of $\{V^{(N)}(Q_k)\}_k$. The family $\{P_{j'} : j' \in T\} \cup \{Q_k\}_k$ then has all pairwise cross-overlaps decaying, hence is eventually linearly independent (Lemma 10.30). Rewriting the common-MPV identity over this family,

$$\sum_{j' \in T} c^{(j')}_N V^{(N)}(P_{j'}) = \sum_k \Big(c^{(k)}_N - \sum_{j' \notin T,\, k(j')=k} c^{(j')}_N \omega^N_{j'} \Big) V^{(N)}(Q_k),$$

exact coefficient extraction forces $c_N^{(j)} = 0$ at each fixed large N, contradicting Lemma 10.44. Hence a matched sector k exists; the gauge-phase equivalence is recovered from the non-decaying overlap by the irreducible trace-preserving overlap dichotomy. $\square$

Theorem 10.46 (Strong existential matching, full-basis form)**.** Lean ✓ Stmt *Let P, Q be two BNT canonical forms with the same MPV family. Then for every sector k of Q there exist a sector j of P with matched bond dimension $\dim_P(j) = \dim_Q(k)$, a gauge-phase equivalence after identifying the matched bond dimensions, and a non-decaying cross-overlap.*

Proof. ✓ Proof Apply Theorem 10.45 with the roles of P and Q swapped and the same-MPV hypothesis flipped by symmetry of gauge-phase equivalence and overlap convergence. $\square$

10.6.1 Bijective matching by symmetry

Applying the exact matching once more with P and Q swapped gives the reverse direction; the basis-distinctness clause forces injectivity both ways, so $g_a = g_b$ and the source relabelling $B_k = e^{i\phi_k} X_k A_{j_k} X_k^{-1}$ holds on the full basis [CPGSV16, Appendix A].

Theorem 10.47 (Bijective matching by symmetry)**.** Lean ✓ Stmt *Let P, Q be two BNT canonical forms with the same MPV family. Then there is a bijection*

$$\beta : J(Q) \simeq J(P)$$

such that for every $k \in J(Q)$ the bond dimensions match, the basis tensors $P_{\beta(k)}$ and Q_k are gauge-phase equivalent after identifying the matched bond dimensions, and the basis cross-overlap does not tend to zero.

Proof. ✓ Proof Theorem 10.45 applied to (Q, P) gives a matching map $m : J(Q) \to J(P)$, and applied to (P, Q) a matching map $J(P) \to J(Q)$.

Injectivity of m. If $m(k_1) = m(k_2) = j$, the two witnesses give a common centre P_j gauge-phase equivalent, after identifying bond dimensions, to both Q_{k_1} and Q_{k_2}; transitivity yields a gauge-phase equivalence $Q_{k_1} \sim Q_{k_2}$ under $\dim_Q(k_1) = \dim_Q(k_2)$. The basis-distinctness clause on Q then forces $k_1 = k_2$; the reverse map is injective by symmetry.

Bijection. The two injections give equal finite sector counts, so m is a bijection $\beta : J(Q) \simeq J(P)$. $\square$

Chapter 11

Proof of the Fundamental Theorem

For BNT canonical forms P and Q, write the weighted direct sums over normal-tensor sectors $P^i = \bigoplus_j \mu_j P_j^i$ and $Q^i = \bigoplus_k \nu_k Q_k^i$. The proportional theorem gives a bijection β of the normal-tensor sectors with unit-modulus phases ζ_k, invertible matrices X_k, and $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$; the equal-MPV corollary collapses the per-block phases into a single X with $Q_{\mathrm{tot}}^i = X P_{\mathrm{tot}}^i X^{-1}$. Given the sector matching of Chapter 10, extract the per-sector coefficient identities $c_N^{(\beta(k))}(P) = \zeta_k^N\, c_N^{(k)}(Q)$ from eventual linear independence, recover the copy weights by finite power-sum comparison, and combine the block gauges into $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$.

Throughout, "eventually" abbreviates "for all sufficiently large N": there is N_0 such that the stated relation holds for every $N > N_0$. The comparison is made at a fixed such N, where BNT linear independence detects a vanishing coefficient even when $|\mu| < 1$ lets it decay.

11.1 Coefficient identity, copy weights, and the block-diagonal gauge

The bijective matching theorem of Chapter 10 (Theorem 10.47) gives the sector phases ζ_k and block gauges X_k. The coefficient comparison of [CPGSV16, Appendix MPV proof, lines 1187–1188] then turns the equality of total MPV families into the per-sector coefficient identity $c_N^{(\beta(k))}(P) = \zeta_k^N\, c_N^{(k)}(Q)$. Neither step uses a per-sector unit-modulus hypothesis.

Lemma 11.1 (Unit phase from matched normalized BNT blocks)**.** Lean ✓ Stmt *Let P and Q be BNT canonical forms. If, for every length N and word σ, a sector P_j and a sector Q_k have MPV families related by*

$$V^{(N)}(Q_k)_\sigma = \zeta^N V^{(N)}(P_j)_\sigma \tag{11.1}$$

then $|\zeta| = 1$.

Proof. ✓ Proof The BNT normalization gives self-overlap limits of modulus 1 for both sectors. The MPV identity (11.1) gives, for every N,

$$O_{Q_kQ_k}(N) = |\zeta|^{2N}\, O_{P_jP_j}(N).$$

Since $O_{Q_kQ_k}(N) \to 1$ and $O_{P_jP_j}(N) \to 1$, taking limits forces $|\zeta|^{2N} \to 1$ and hence $|\zeta| = 1$. □

Theorem 11.2 (Proportional sector matching with unit phases). Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Then there are a bijection $\beta : J(Q) \simeq J(P)$, bond-dimension equalities $\dim P_{\beta(k)} = \dim Q_k$, scalars $\zeta_k \in \mathbb{C}$, and block gauges X_k such that $|\zeta_k| = 1$ and, for every sector k and physical index i,*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1} \tag{11.2}$$

Equivalently, for every length N and word σ,

$$V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma.$$

This is [CPGSV16, Appendix MPV theorem statement, lines 1167–1170, and proof line 1182].

Proof. ✓ Proof Apply the proportional sector-matching theorem to obtain the bijection and dimension-compatible gauge-phase equivalences. Choosing the gauges and scalars gives $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$, and therefore

$$O_{Q_k Q_k}(N) = |\zeta_k|^{2N} O_{P_{\beta(k)} P_{\beta(k)}}(N).$$

The normalized self-overlap limits give $O_{Q_k Q_k}(N) \to 1$ and $O_{P_{\beta(k)} P_{\beta(k)}}(N) \to 1$, so $|\zeta_k|^{2N} \to 1$ and hence $|\zeta_k| = 1$. The MPV scalar-power identity follows from the same gauge equation (11.2): $V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma$. □

Lemma 11.3 (Full-basis coefficient identity from matched sectors). Lean ✓ Stmt *Let P and Q be BNT canonical forms with the same MPV family, matched by a bijection $\beta : J(Q) \simeq J(P)$ with bond-dimension equalities and a gauge-phase equivalence between Q_k and $P_{\beta(k)}$ for every k. Then each sector carries a unit-modulus phase ζ_k, and eventually*

$$c_N^{(\beta(k))}(P) = \zeta_k^N \, c_N^{(k)}(Q). \tag{11.3}$$

This is the equal-MPV coefficient comparison of [CPGSV16, Appendix MPV proof, lines 1187–1188].

Proof. ✓ Proof For each matched sector, the gauge-phase equivalence gives $V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma$. Lemma 11.1 gives $|\zeta_k| = 1$. Applying Lemma 11.4 to the same phases gives $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ for all sufficiently large N. □

Lemma 11.4 (Full-basis coefficient identity from matched phases). Lean ✓ Stmt *Let P be a BNT canonical form, let Q be a sector decomposition, and assume P_{tot} and Q_{tot} have the same MPV family. Suppose that there is a bijection*

$$\beta : J(Q) \simeq J(P)$$

and scalars $\zeta_k \in \mathbb{C}$ such that, for every $k \in J(Q)$, every length N, and every spin configuration σ of length N,

$$V^{(N)}(Q_k)_\sigma = \zeta_k^N \, V^{(N)}(P_{\beta(k)})_\sigma. \tag{11.4}$$

Then for every $k \in J(Q)$ there is N_0 such that, for all $N > N_0$,

$$c_N^{(\beta(k))}(P) = \zeta_k^N \, c_N^{(k)}(Q).$$

This is the coefficient-comparison step of [CPGSV16, Appendix MPV proof, lines 1187–1188] in full-basis form.

Proof. ✓ Proof For every N, expand the equality of total MPVs as

$$\sum_{j\in J(P)} c_N^{(j)}(P)|V^{(N)}(P_j)\rangle = \sum_{k\in J(Q)} c_N^{(k)}(Q)|V^{(N)}(Q_k)\rangle.$$

The phase relation (11.4) gives $|V^{(N)}(Q_k)\rangle = \zeta_k^N|V^{(N)}(P_{\beta(k)})\rangle$, so reindexing the Q-sum by β onto the P-basis yields

$$\sum_{j\in J(P)} c_N^{(j)}(P)|V^{(N)}(P_j)\rangle = \sum_{j\in J(P)} \zeta_{\beta^{-1}(j)}^N c_N^{(\beta^{-1}(j))}(Q)|V^{(N)}(P_j)\rangle.$$

For all sufficiently large N the P-basis MPV family $\{|V^{(N)}(P_j)\rangle\}_{j\in J(P)}$ is linearly independent, by the BNT eventual linear independence of P. Lemma 10.6 then extracts the coefficient equalities $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ from this eventual linear independence, for all sufficiently large N. □

The coefficient identity (11.3) is now converted, sector by sector, into a matching of the individual copy weights.

Definition 11.5 (Copy-weight matching for matched BNT sectors). Lean ✓ Stmt Let the sectors of two decompositions P and Q be matched by $\beta : J(Q) \simeq J(P)$, with sector phases ζ_k. A copy-weight matching consists of copy permutations τ_k such that, for all sectors k and copies q,

$$\nu_{k,q} = \zeta_k^{-1}\mu_{\beta(k),\tau_k(q)}.$$

This is the copy-level conclusion of [CPGSV16, Appendix MPV proof, line 1188].

Lemma 11.6 (Copy-weight matching from matched-sector coefficient identities). Lean ✓ Stmt *Let the sectors of P and Q be matched by $\beta : J(Q) \simeq J(P)$, with nonzero sector phases ζ_k. If the matched-sector coefficient identities*

$$c_N^{(\beta(k))}(P) = \zeta_k^N\, c_N^{(k)}(Q) \tag{11.5}$$

hold eventually, then the phases ζ_k determine a copy-weight matching $\nu_{k,q} = \zeta_k^{-1}\mu_{\beta(k),\tau_k(q)}$. This is, sector by sector, [CPGSV16, Appendix MPV proof, line 1188].

Proof. ✓ Proof Eventually the coefficient identity (11.5) in sector k reads

$$\sum_r \mu_{\beta(k),r}^N = \zeta_k^N \sum_q \nu_{k,q}^N = \sum_q (\zeta_k\nu_{k,q})^N.$$

Newton–Girard finite power-sum comparison (Lemma 10.43) applies sector by sector, producing a copy permutation τ_k with $\mu_{\beta(k),\tau_k(q)} = \zeta_k\nu_{k,q}$; since $\zeta_k \neq 0$ this is $\nu_{k,q} = \zeta_k^{-1}\mu_{\beta(k),\tau_k(q)}$. □

The matched block gauges and copy weights now combine into a single block-diagonal gauge.

Theorem 11.7 (Global gauge from matched sectors and copy weights). Lean ✓ Stmt *Suppose that the sectors of two sector decompositions P and Q are matched by a bijection $\beta : J(Q) \simeq J(P)$, with bond-dimension equalities, copy permutations τ_k, nonzero phases ζ_k, and block gauges X_k. If, for every sector k, copy q, and physical index i,*

$$Q_k^i = \zeta_k\, X_k P_{\beta(k)}^i X_k^{-1}, \qquad \nu_{k,q} = \zeta_k^{-1}\mu_{\beta(k),\tau_k(q)}, \tag{11.6}$$

then, in the flattened Q-coordinates,

$$Q_{\text{flat}}^i = X\, P_{\text{matched}}^i X^{-1}, \qquad X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

This is the direct-sum gauge construction of [CPGSV16, Appendix MPV proof, lines 1189–1192], separated from the coefficient-comparison step that supplies the weight identities.

Proof. ✓ Proof For each matched copy,

$$\nu_{k,q} Q_k^i = (\zeta_k^{-1} \mu_{\beta(k),\tau_k(q)})(\zeta_k X_k P_{\beta(k)}^i X_k^{-1}) = \mu_{\beta(k),\tau_k(q)} X_k P_{\beta(k)}^i X_k^{-1}.$$

Thus every flattened weighted block is conjugated by its sector gauge X_k. Lemma **??** combines these block conjugacies into the direct-sum gauge $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$. □

Theorem 11.8 (Global gauge from a copy-weight matching)**.** Lean ✓ Stmt *Suppose that the sectors of P and Q are matched, with bond-dimension equalities, nonzero phases, and block gauges satisfying*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}.$$

If the same phases also give a copy-weight matching, then the flattened Q-tensor is obtained from the matched-coordinate P-tensor by the direct-sum gauge

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

This is only a reformulation of Theorem 11.7.

Proof. ✓ Proof The copy-weight matching supplies $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k),\tau_k(q)}$. Together with $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$, these are exactly the two displayed hypotheses (11.6) of Theorem 11.7. □

We can now state the proportional global gauge, first from a copy-weight matching and then from the coefficient identities that produce one.

Theorem 11.9 (Conditional proportional global gauge from a copy-weight matching)**.** Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Then the proportional sector-matching theorem gives β, the bond-dimension equalities, the unit phases ζ_k, and the block gauges X_k. If the remaining proportional coefficient comparison supplies a copy-weight matching for these same phases, then the flattened Q-tensor is obtained from the matched-coordinate P-tensor by the direct-sum gauge*

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

Proof. ✓ Proof The proportional sector-matching theorem supplies β, X_k, and unit phases ζ_k satisfying $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$. Since $|\zeta_k| = 1$, each ζ_k is nonzero. A copy-weight matching gives $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k),\tau_k(q)}$, so Theorem 11.8 gives the direct-sum gauge. □

Theorem 11.10 (Conditional proportional global gauge from coefficient identities)**.** Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Then there are a bijection $\beta : J(Q) \simeq J(P)$, bond-dimension equalities, unit-modulus phases ζ_k, and block gauges X_k such that*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$$

for every sector k and physical index i. If these same phases satisfy the matched-sector coefficient identities

$$c_N^{(\beta(k))}(P) = \zeta_k^N \, c_N^{(k)}(Q)$$

for all sufficiently large N, then they determine a copy-weight matching and hence the flattened Q-tensor is obtained from the matched-coordinate P-tensor by the direct-sum gauge

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

Proof. ✓ Proof The sector-matching theorem gives β, X_k, and $|\zeta_k| = 1$ with $Q^i_k = \zeta_k X_k P^i_{\beta(k)} X_k^{-1}$. Since $\zeta_k \neq 0$, the coefficient identities $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ produce $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k),\tau_k(q)}$ by Lemma 11.6. Theorem 11.9 then applies to this copy-weight matching. □

In the equal-MPV case all ingredients are unconditional.

Lemma 11.11 (Equal-MPV matched copy-weight witnesses). Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors have the same MPV family. Then there are a bijection $\beta : J(Q) \simeq J(P)$, bond-dimension equalities, unit-modulus phases ζ_k, and block gauges X_k such that*

$$Q^i_k = \zeta_k X_k P^i_{\beta(k)} X_k^{-1}.$$

For the same phases there is a copy-weight matching: after a permutation of the copies inside each matched sector,

$$\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k),\tau_k(q)}.$$

Proof. ✓ Proof The full-basis matching theorem gives the bijection and block gauges. The self-overlap comparison

$$\langle V_N(Q_k), V_N(Q_k)\rangle = |\zeta_k|^{2N} \langle V_N(P_{\beta(k)}), V_N(P_{\beta(k)})\rangle$$

together with the BNT normalization limits on both sides forces $|\zeta_k| = 1$. The matched MPV phase identities also give the eventual coefficient identities

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q).$$

Lemma 11.6 applies the finite power-sum comparison in each sector and gives the copy-weight identities. □

Theorem 11.12 (Full-basis global gauge in matched coordinates). Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors have the same MPV family. Then there exist:*

- *a bijection $\beta : J(Q) \simeq J(P)$,*
- *bond-dimension equalities $\dim P_{\beta(k)} = \dim Q_k$,*
- *copy-number equalities $r_{\beta(k)}(P) = r_k(Q)$ and copy permutations τ_k,*
- *unit-modulus phases ζ_k with $|\zeta_k| = 1$ and block gauges X_k,*

such that, for every sector k, copy q, and physical index i,

$$Q^i_k = \zeta_k\, X_k P^i_{\beta(k)} X_k^{-1}, \qquad \nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k),\tau_k(q)}.$$

Moreover, if

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$$

is the flattened block-diagonal gauge, then, for every physical index i, the flattened Q-coordinates satisfy

$$Q^i_{\text{flat}} = X\, P^i_{\text{matched}}\, X^{-1}.$$

This is the explicit direct-sum gauge construction of [CPGSV16, Appendix MPV proof, lines 1189–1192].

Proof. ✓ Proof Lemma 11.11 gives β, the bond-dimension identifications, the phases ζ_k, the block gauges X_k, and copy permutations τ_k satisfying

$$\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k),\tau_k(q)}.$$

Theorem 11.8 combines the block conjugacies into the flattened direct-sum conjugation by $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$. □

Theorem 11.13 (Literal equal-MPV gauge after sector-coordinate permutation)**.** Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors have the same MPV family. Then the total bond dimensions agree; write their common value as D. There is*

$$Y \in \mathrm{GL}_D(\mathbb{C}) \tag{11.7}$$

such that, for every physical index i,

$$Q^i_{\mathrm{tot}} = Y \, P^i_{\mathrm{tot}} \, Y^{-1},$$

where P^i_{tot} and Q^i_{tot} are both regarded as matrices in $M_D(\mathbb{C})$ using the total bond-dimension equality. This is the BNT form of [CPGSV16, Corollary II.2 and Appendix MPV proof, lines 1189–1192].

Proof. ✓ Proof Apply Theorem 11.12 to obtain the flattened gauge after the sectors of P have been reindexed by the bijection between basis blocks of P and Q. The sector-coordinate permutation ρ, induced by this bijection, the copy permutations, and the bond-dimension equalities, identifies the reindexed P-side direct sum with the total tensor P_{tot}, transported along the total bond-dimension equality. Composing the permutation gauge for ρ with the sector-permuted gauge gives the single matrix Y of (11.7). □

11.2 The equal and proportional cases

Theorem 11.14 (Multi-block fundamental theorem for normal tensors)**.** Lean ✓ Stmt *Let P and Q be BNT canonical forms (Definition 10.27), with bases of normal tensors $(P_j)_{j=1}^{g_a}$ and $(Q_k)_{k=1}^{g_b}$. If the total tensors P_{tot}, Q_{tot} generate eventually nonzero proportional MPV families — for all sufficiently large N there is a nonzero scalar c_N with $V^{(N)}(P_{\mathrm{tot}}) = c_N V^{(N)}(Q_{\mathrm{tot}})$ — then $g_a = g_b$. After relabelling the basis of normal tensors of P by a bijection $\beta : J(Q) \simeq J(P)$, there are scalars $\zeta_k \in \mathbb{C}$ with $|\zeta_k| = 1$ and invertible matrices Y_k such that, for every k and every physical index i,*

$$Q^i_k = \zeta_k \, Y_k \, P^i_{\beta(k)} \, Y_k^{-1}.$$

Thus the paired normal tensors have the same bond dimension ($\dim P_{\beta(k)} = \dim Q_k$) and the same gauge-phase class.

Proof. ✓ Proof The proportional sector-matching Theorem 11.2, applied to the BNT canonical forms P and Q, supplies the bijection β, the bond-dimension equalities, the unit phases ζ_k, and the block gauges Y_k with $Q^i_k = \zeta_k Y_k P^i_{\beta(k)} Y_k^{-1}$; the matching needs no per-sector unit-modulus hypothesis. □

This is [CPGSV16, Theorem 2.10]; see also [CPGSV21, Theorem IV.4 and Corollary IV.5]. The statement is given on the BNT canonical-form surface (Definition 10.27), which is the paper's "A and B in canonical form" with chosen bases of normal tensors; the reduction of an *arbitrary* proportional-MPV pair to a common primitive block decomposition is Theorem 9.54 of Chapter 9.

Theorem 11.15 (Equal BNT canonical forms are globally conjugate). Lean ✓ Stmt *Let P and Q be BNT canonical forms (Definition 10.27) whose total tensors P_{tot}, Q_{tot} generate the same MPV family for every system size N. Then their total bond dimensions coincide; writing the common value as D, there is one invertible matrix $X \in \mathrm{GL}_D(\mathbb{C})$ such that, for every physical index i,*

$$Q^i_{\mathrm{tot}} = X\, P^i_{\mathrm{tot}}\, X^{-1}.$$

This is [CPGSV16, Corollary 2.11] on the BNT canonical-form surface; the canonical form of [CPGSV16, Section 2.3], a direct sum of normal tensors with scalar weights, is a BNT canonical form. The reduction of an arbitrary *pair of same-MPV tensors to a common BNT canonical form is Chapter 9.*

Proof. ✓ Proof This is Theorem 11.13. The BNT comparison of Chapter 10 provides the basis relabelling, the per-block gauge-phase equivalences, and the copy-weight multiset matching, which combine into the block-diagonal global gauge. The exact linear-independence matching needs no per-sector unit-modulus hypothesis, and the BNT canonical form needs no one-site injectivity. ∎

This is also [CPGSV21, Corollary IV.5].

Chapter 12

Symmetries and String Order

This chapter collects the symmetry-theoretic consequences of the MPS Fundamental Theorem [PGVWC07], following the symmetry analysis of [PGWS+08]. For injective tensors, physical on-site symmetries determine virtual gauges, those gauges define projective representations on the bond space, and virtual intertwiners enter the string-order criteria.

12.1 Physical Symmetries Give Virtual Gauges

Definition 12.1 (Physical-index rotation). Lean ✓ Stmt Given a matrix $M \in M_d(\mathbb{C})$ and an MPS tensor A, the *physical-index rotation* is the tensor $(A_M)^i := \sum_j M_{ij} A^j$. When the matrix is a unitary u, the rotation is written A_u.

Corollary 12.2 (Periodic symmetry yields gauge equivalence up to a root of unity). Lean ✓ Stmt *Suppose that for any two tensors in irreducible form II with the same MPV family there is $m > 0$ making them gauge equivalent up to an m-th root of unity. Let A be in irreducible form II and let M be a matrix such that $B := A_M$ is also in irreducible form II and $\mathcal{V}(A) = \mathcal{V}(B)$. Then there is $m > 0$ such that A and B are gauge equivalent up to an m-th root of unity.*

Proof. ✓ Proof Apply the assumed equivalence to A and B, which are in irreducible form II with $\mathcal{V}(A) = \mathcal{V}(B)$. ☐

12.2 Virtual Representation Theorem

With a full group of on-site symmetries (rather than just a single unitary), the resulting gauges determine a projective representation on the bond space.

Definition 12.3 (Twisted tensor). Lean ✓ Stmt Given a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$ on the physical index, the *g-twisted tensor* is defined, for each $i \in \{0, \dots, d-1\}$, by

$$\widetilde{A}^i_g := \sum_{j=0}^{d-1} U(g)_{ij} A^j.$$

Definition 12.4 (On-site symmetry). Lean ✓ Stmt A tensor A is *on-site symmetric* under a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$ if $\mathcal{V}(A) = \mathcal{V}(\widetilde{A}_g)$ for every $g \in G$.

Lemma 12.5 (Functoriality of twisting). Lean ✓ Stmt *The twisted tensor satisfies the composition law*

$$\widetilde{A}_{gh} = \widetilde{(\widetilde{A}_h)}_g,$$

i.e. twisting by gh is the same as first twisting by h and then by g.

Proof. ✓ Proof Expanding both sides using $U(gh) = U(g)\,U(h)$, then swapping the order of summation. □

Lemma 12.6 (Identity twist). Lean ✓ Stmt *Twisting by the identity group element is trivial: $\widetilde{A}_1 = A$.*

Proof. ✓ Proof Since $U(1) = \mathbb{1}_d$, each component reduces to $\sum_j \delta_{ij}\, A^j = A^i$. □

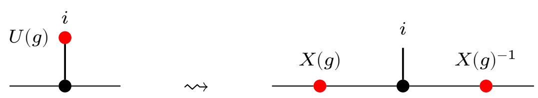
Lemma 12.7 (Symmetric twists are gauge equivalent). Lean ✓ Stmt *If A is injective and on-site symmetric under U, then for every $g \in G$ the twisted tensor $\widetilde{A}_g$ is gauge equivalent to A.*

Proof. ✓ Proof On-site symmetry gives $\mathcal{V}(A) = \mathcal{V}(\widetilde{A}_g)$, and the single-block Fundamental Theorem turns this equality of MPV families into gauge equivalence. □

Remark 12.8 (Fundamental-Theorem route for injective symmetry). The injective virtual-representation theorem is not proved from string-order rigidity. The on-site symmetry condition first identifies A and $\widetilde{A}_g$ as the same MPV family; the single-block Fundamental Theorem gives the gauge in Lemma 12.7. Gauge uniqueness and the twist composition law then give Theorem 12.15. The string-order results later in this chapter use this virtual representation as an input.

Theorem 12.9 (Virtual symmetry equation). Lean ✓ Stmt *If A is injective and on-site symmetric under U, then for every $g \in G$ there exist an invertible matrix $X(g) \in \mathrm{GL}_D(\mathbb{C})$ and a nonzero scalar $\varphi(g) \in \mathbb{C}^\times$ (in fact $\varphi = 1$ in the single-block case) such that $\widetilde{A}^i_g = \varphi(g)\, X(g)\, A^i\, X(g)^{-1}$ for all i.*

Proof. ✓ Proof The physical action is transferred to the virtual level by the same local move that replaces the twisted tensor by a gauge-conjugated one:



Apply Lemma 12.7 to obtain $X(g)$, then set $\varphi(g) = 1$. □

Lemma 12.10 (Gauge uniqueness up to scalar). Lean ✓ Stmt *Let A be an injective tensor. If $X, Y \in \mathrm{GL}_D(\mathbb{C})$ both satisfy $B^i = X\, A^i\, X^{-1} = Y\, A^i\, Y^{-1}$ for all i, then there exists a nonzero scalar $u \in \mathbb{C}^\times$ such that $Y = u \cdot X$.*

Proof. ✓ Proof The matrix $Z = Y^{-1}X$ commutes with every A^i. Since A is injective, $\{A^i\}$ spans $M_D(\mathbb{C})$, so Z lies in the centre of $M_D(\mathbb{C})$. By Lemma **??**, Z is a scalar matrix. Since Z is invertible, that scalar is nonzero, and $Y = u \cdot X$. □

Theorem 12.11 (Gauge uniqueness from equality of MPV families). Lean ✓ Stmt *Let A be an injective tensor and assume that A and B generate the same MPV family. If $X, Y \in \mathrm{GL}_D(\mathbb{C})$ both satisfy $B^i = X\, A^i\, X^{-1} = Y\, A^i\, Y^{-1}$ for all i, then there is a nonzero scalar $u \in \mathbb{C}^\times$ such that $Y = u \cdot X$.*

Proof. ✓ Proof This is exactly Lemma 12.10; the extra MPV hypothesis specifies the setting in which the two gauges are usually produced. □

Definition 12.12 (Scalar 2-cochain). Lean ✓ Stmt A *scalar 2-cochain* on a group G is a function $\omega : G \times G \to \mathbb{C}^\times$.

Definition 12.13 (Projective representation). Lean ✓ Stmt Given a scalar 2-cochain ω, a *projective representation* of G on $\mathbb{C}^D$ is a map $\rho : G \to \mathrm{GL}_D(\mathbb{C})$ satisfying, for all $g, h \in G$,

$$\rho(g)\,\rho(h) = \omega(g,h)\,\rho(gh).$$

Lemma 12.14 (Associativity forces the cocycle condition). Lean ✓ Stmt *If ρ is a projective representation with factor system ω and $D > 0$, then ω satisfies the 2-cocycle identity:*

$$\omega(g,h)\,\omega(gh,k) = \omega(g,hk)\,\omega(h,k).$$

Proof. ✓ Proof Expanding $\rho(g)\,(\rho(h)\rho(k))$ and $(\rho(g)\rho(h))\rho(k)$ using the projective multiplication law and equating via matrix associativity. The hypothesis $D > 0$ allows cancellation of the invertible matrix factor. □

Theorem 12.15 (Virtual representation theorem for injective MPS). Lean ✓ Stmt *Let A be an injective MPS tensor that is on-site symmetric under a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$. Then there exist:*

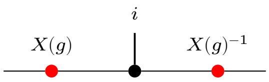
1. *a scalar 2-cochain $\omega : G \times G \to \mathbb{C}^\times$, and*
2. *a map $\rho : G \to \mathrm{GL}_D(\mathbb{C})$ satisfying, for all $g, h \in G$,*

$$\rho(g)\,\rho(h) = \omega(g,h)\,\rho(gh).$$

such that, defining the gauge matrices $X(g) := \rho(g^{-1})$, one has, for every $g \in G$ and $i \in \{0, \dots, d-1\}$,

$$\widetilde{A}^i_g = X(g)\,A^i\,X(g)^{-1}.$$

In tensor-network notation, for each fixed g this is the local gauge relation



with the black node denoting the tensor named in the surrounding formula and the two red side nodes acting on the virtual legs. In particular, ρ is a projective representation of G on the bond space $\mathbb{C}^D$.

Proof. ✓ Proof Since A is injective and on-site symmetric, each twisted tensor $\widetilde{A}_g$ is gauge equivalent to A (Lemma 12.7). Choose $X(g) \in \mathrm{GL}_D(\mathbb{C})$ with $\widetilde{A}^i_g = X(g)\,A^i\,X(g)^{-1}$ for all i.

By functoriality (Lemma 12.5), the product $X(h)\,X(g)$ also intertwines A with $\widetilde{A}_{gh}$. Since $X(gh)$ does the same, gauge uniqueness (Lemma 12.10) gives a nonzero scalar $\omega'(h,g)$ with $X(h)\,X(g) = \omega'(h,g)\,X(gh)$.

Defining $\rho(g) := X(g^{-1})$ converts this to the standard law: $\rho(g)\,\rho(h) = \omega(g,h)\,\rho(gh)$ where $\omega(g,h) := \omega'(h^{-1}, g^{-1})$. □

Corollary 12.16 (2-cocycle identity for the factor system). Lean ✓ Stmt *If $D > 0$, the factor system ω from Theorem 12.15 satisfies the 2-cocycle identity for all $g, h, k \in G$:*

$$\omega(g,h)\,\omega(gh,k) = \omega(g,hk)\,\omega(h,k).$$

Proof. ✓ Proof Theorem 12.15 gives ρ and ω. By Lemma 12.14, the cocycle identity follows from associativity of matrix multiplication and invertibility of the representation matrices. □

12.3 Cohomology Classes of Virtual Symmetry Cocycles

Two cocycles may differ by a coboundary yet represent the same projective-representation class. The following definitions and results make this precise and establish the quotient $H^2(G, \mathbb{C}^\times)$.

Definition 12.17 (Coboundary). Lean ✓ Stmt A scalar 2-cocycle ω is a *coboundary* if there exists $\varphi\colon G \to \mathbb{C}^\times$ such that, for all $g, h \in G$,

$$\omega(g, h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}.$$

Definition 12.18 (Cohomologous cocycles). Lean ✓ Stmt Two cocycles ω_1, ω_2 are *cohomologous* (written $\omega_1 \sim \omega_2$) if there exists $\varphi\colon G \to \mathbb{C}^\times$ such that, for all $g, h \in G$,

$$\omega_1(g, h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g, h).$$

Theorem 12.19 (Cohomology relation is an equivalence relation). Lean ✓ Stmt *The relation "cohomologous to" is reflexive, symmetric, and transitive on the set of scalar 2-cocycles.*

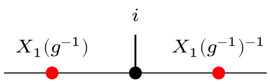
Proof. ✓ Proof Reflexivity: take $\varphi = 1$. Symmetry: replace φ by φ^{-1}. Transitivity: compose the witnesses φ and ψ pointwise. □

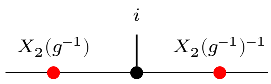
Lemma 12.20 (Coboundary iff cohomologous to the trivial cocycle). Lean ✓ Stmt *A cocycle ω is a coboundary if and only if $\omega \sim \mathbb{1}$, where $\mathbb{1}(g, h) = 1$ for all g, h.*

Proof. ✓ Proof Both directions follow by absorbing or introducing the factor $\mathbb{1}(g, h) = 1$ via $x \cdot 1 = x$. □

Theorem 12.21 (Gauge independence of the cocycle class). Lean ✓ Stmt *If the bond dimension satisfies $D \geq 1$ and two projective representations ρ_1, ρ_2 (with cocycles ω_1, ω_2) both arise as virtual representations of the same injective MPS tensor A under an on-site symmetry U, then $\omega_1 \sim \omega_2$.*

Proof. ✓ Proof For each g, gauge uniqueness gives a nonzero scalar $f(g)$ with $X_2(g^{-1}) = f(g)\, X_1(g^{-1})$. Equivalently, for each fixed g the two local gauges



and


describe the same twisted tensor, so they differ by a scalar. Substituting into the projective law and cancelling $X_1(g \cdot h)$ yields $\omega_2(g, h) = f(g)\, f(h)\, f(gh)^{-1}\omega_1(g, h)$. □

Definition 12.22 (Multiplicative 2-cocycle condition). Lean ✓ Stmt A scalar 2-cochain $\omega\colon G \times G \to \mathbb{C}^\times$ is a *2-cocycle* if it satisfies, for all $g, h, k \in G$,

$$\omega(g, h)\omega(gh, k) = \omega(g, hk)\omega(h, k).$$

Definition 12.23 (Second cohomology quotient). Lean ✓ Stmt The second cohomology set is the quotient

$$H^2(G, \mathbb{C}^\times) := Z^2(G, \mathbb{C}^\times)/\sim,$$

where $\sim$ is the coboundary-equivalence relation.

Definition 12.24 (Projective equivalence). Lean ✓ Stmt Two projective representations are *projectively equivalent* when their factor systems are cohomologous.

Theorem 12.25 (Projective equivalence and cohomologous cocycles). Lean ✓ Stmt *Let ρ_1, ρ_2 be projective representations with factor systems ω_1, ω_2. Then*

$$\rho_1 \sim_{\text{proj}} \rho_2 \iff \omega_1 \sim \omega_2.$$

Proof. ✓ Proof This is immediate from the definition of projective equivalence as cohomology-class equality for factor systems. □

12.3.1 Non-triviality via the commutator phase

For an abelian symmetry group the cohomology class of a factor system is detected by an anti-symmetric pairing, which gives a computable test for non-triviality.

Definition 12.26 (Commutator phase). Lean ✓ Stmt The *commutator phase* of a 2-cochain ω at a pair g, h is $\beta_\omega(g, h) := \omega(g, h)\,\omega(h, g)^{-1}$. It is defined for any $\mathbb{C}^\times$-valued 2-cochain, the cocycle condition is not needed.

Theorem 12.27 (The commutator phase is a class invariant). Lean ✓ Stmt *If $\omega_1 \sim \omega_2$ and $gh = hg$, then $\beta_{\omega_1}(g, h) = \beta_{\omega_2}(g, h)$.*

Proof. ✓ Proof Write $\omega_1(g, h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g, h)$ for a coboundary witness φ. Substituting into the ratio and using $gh = hg$ and the commutativity of $\mathbb{C}^\times$,

$$\beta_{\omega_1}(g, h) = \frac{\varphi(g)\varphi(h)\varphi(gh)^{-1}\,\omega_2(g, h)}{\varphi(h)\varphi(g)\varphi(hg)^{-1}\,\omega_2(h, g)} = \frac{\omega_2(g, h)}{\omega_2(h, g)} = \beta_{\omega_2}(g, h),$$

the φ-factors cancelling since $\varphi(gh) = \varphi(hg)$. □

Definition 12.28 (Non-trivial cohomology class). Lean ✓ Stmt A factor system ω has a *non-trivial class* when it is not cohomologous to the constant cocycle 1.

Theorem 12.29 (Commutator-phase non-triviality test). Lean ✓ Stmt *If a commuting pair g, h has $\beta_\omega(g, h) \neq 1$, then ω has a non-trivial class.*

Proof. ✓ Proof The trivial cocycle has commutator phase 1 at every pair, so by Theorem 12.27 any cocycle cohomologous to it has $\beta = 1$ at every commuting pair. Contrapositively, $\beta_\omega(g, h) \neq 1$ for one commuting pair forces ω to be non-cohomologous to the trivial cocycle. □

12.4 Permutation-Based Physical Symmetry

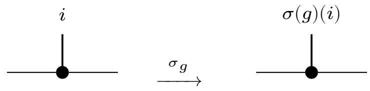
The linear-combination twist $\widetilde{A}^i_g = \sum_j U(g)_{ij}\,A^j$ is the standard form used in the virtual representation theorem. The following alternative formulation replaces the matrix action by a permutation of the physical index; it is useful for symmetries that act by reordering basis states rather than by a general linear map.

Definition 12.30 (Permutation symmetry datum). Lean ✓ Stmt An *on-site symmetry datum* for a group G on a physical space of dimension d consists of a matrix-valued map $u : G \to M_d(\mathbb{C})$ and a physical-index action $\sigma : G \to \mathrm{End}(\{0, \ldots, d{-}1\})$.

Definition 12.31 (Permutation-twisted tensor). Lean ✓ Stmt Given an on-site symmetry datum (u, σ), the *permutation-twisted tensor* at group element g is

$$(A^{\sigma_g})^i := A^{\sigma(g)(i)}.$$

In tensor-network notation the local tensor is unchanged and only the physical leg is relabelled:



Lemma 12.32 (Identity permutation twist). Lean ✓ Stmt *If $\sigma(1) = \mathrm{id}$, then $(A^{\sigma_1})^i = A^i$ for all i.*

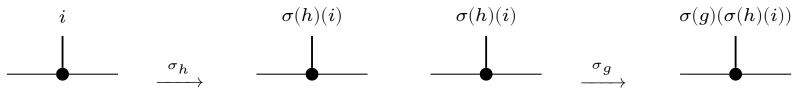
Proof. ✓ Proof Immediate from $\sigma(1)(i) = i$ for all i. □

Composition order. The linear-combination twist (Lemma 12.5) composes as $\widetilde{(\widetilde{A}_h)}_g = \widetilde{A}_{gh}$, applying h first then g. The permutation twist below uses the same left-action convention $\sigma(gh) = \sigma(g) \circ \sigma(h)$, so $(A^{\sigma_g})^{\sigma_h} = A^{\sigma_{gh}}$; the inner permutation $\sigma(h)$ acts first on the index.

Lemma 12.33 (Composition of permutation twists). Lean ✓ Stmt *If $\sigma(gh) = \sigma(g) \circ \sigma(h)$ for all $g, h \in G$, then*

$$A^{\sigma_{gh}} = (A^{\sigma_g})^{\sigma_h}.$$

Proof. ✓ Proof Expanding both sides: $(A^{\sigma_{gh}})^i = A^{\sigma(g)(\sigma(h)(i))} = ((A^{\sigma_g})^{\sigma_h})^i$. Diagrammatically this is just two consecutive relabellings of the same local tensor:



□

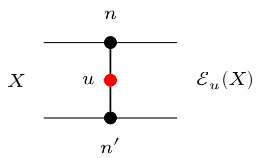
12.5 String Order and Local Symmetry Equivalence

The string order parameter [PGWS+08] detects whether an on-site unitary u acts as a local symmetry on the state generated by an injective MPS tensor.

Definition 12.34 (Twisted transfer map). Lean ✓ Stmt Given an MPS tensor A and a physical-index matrix u, the *twisted transfer map* is

$$\mathcal{E}_u(X) := \sum_{n,n'} \langle n'|u|n\rangle A_n\, X\, A^\dagger_{n'}.$$

Diagrammatically one inserts the physical operator u on the doubled local transfer cell:

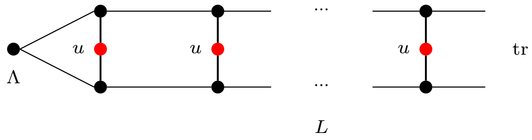


The upper and lower black nodes denote A_n and $A^\dagger_{n'}$, and the red node labelled u couples the physical legs.

Definition 12.35 (String order parameter). Lean ✓ Stmt The *string order parameter* for twist u and stationary state Λ at length L is

$$R_L(u) := \mathrm{tr}(\Lambda \cdot \mathcal{E}_u^L(\mathbb{1})).$$

In tensor-network notation this is the length-L doubled chain with a copy of u inserted on each physical rung:



The boundary matrix Λ is contracted into the virtual boundary, the red nodes labelled u sit on the physical rungs, and the right boundary is traced.

Definition 12.36 (Physical string correlator)**.** Lean ✓ Stmt For physical operators $x, y \in M_d(\mathbb{C})$, a twist $u \in M_d(\mathbb{C})$, and boundary state Λ, the transfer-form string correlator of [PGWS+08, display SOPMP, lines 176–181] is

$$S_N(x, y, u) := \operatorname{tr}\!\big(\Lambda\, \mathcal{E}_x\, \mathcal{E}_u^N(\mathcal{E}_y(\mathbb{1}))\big).$$

This is the physical-endpoint expression corresponding to $\langle\Psi|\, x \otimes u^{\otimes N} \otimes y\, |\Psi\rangle$.

Definition 12.37 (Physical string order)**.** Lean Lean ✓ Stmt A tensor has physical string order, in the sense of [PGWS+08, display SOP, lines 112–122], if there exist a unitary $u \neq \mathbb{1}$ and physical endpoint operators x, y for which the absolute value of the correlator has a positive limit: for some $s > 0$,

$$|S_N(x, y, u)| \longrightarrow s.$$

The comparison with virtual boundary matrices is a separate endpoint-realizability statement.

Definition 12.38 (Boundary string order parameter)**.** Lean ✓ Stmt For a boundary state Λ and boundary matrices $X, Y \in M_D(\mathbb{C})$, the *boundary string order parameter* is

$$R_L(u; X, Y) := \operatorname{tr}\!\big(\Lambda X \mathcal{E}_u^L(Y)\big).$$

The ordinary string order parameter is the special case $X = Y = \mathbb{1}$. The matrices X, Y act on the virtual level. In the string correlator of [PGWS+08] the string of twists is instead flanked by local operators x, y acting on physical sites; expanding such operators in the transfer picture produces particular virtual matrices, and [PGWS+08] argues that for injective tensors, physical operators on sufficiently many sites produce every virtual matrix. That comparison is not formalized here, and the results below are stated for the virtual form.

Definition 12.39 (Local symmetry)**.** Lean ✓ Stmt An MPS tensor A has *local symmetry* under a unitary u with respect to a boundary state Λ if there exist a unitary matrix V and a phase μ with $|\mu| = 1$ such that $V^\dagger \Lambda V = \Lambda$ and, for every physical index i,

$$\sum_j u_{ij}\, A^j = \mu\, V\, A^i\, V^\dagger.$$

The unitary V intertwines the physical action of u with a virtual conjugation, and the boundary state Λ is invariant under V.

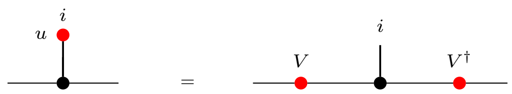
Definition 12.40 (String order)**.** Lean ✓ Stmt String order *exists* for u with respect to a boundary state Λ if there exist boundary matrices X, Y and a positive constant $c > 0$ such that $c \le \|R_L(u; X, Y)\|$ for all L, where $R_L(u; X, Y)$ is the boundary-refined string-order parameter. Thus some boundary-refined string-order sequence does not decay to zero.

String order is here an existence statement over the virtual matrices X, Y, for a fixed twist u; no claim is made for any particular boundary choice. Definition 12.37 records the physical-endpoint source definition, which quantifies existentially over u, x, y and requires a positive limit rather than a lower bound uniform in the length; that paper also notes that the correlator for a particular endpoint choice can vanish even when the symmetry is present.

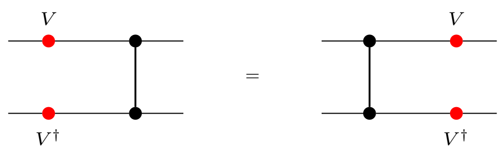
Lemma 12.41 (String order prevents decay to zero). Lean ✓ Stmt *If A has string order for u with respect to a boundary state Λ, then for some boundary matrices X, Y the sequence $R_L(u; X, Y)$ does not tend to 0 as $L \to \infty$.*

Proof. ✓ Proof A uniform lower bound $c \leq \|R_L(u; X, Y)\|$ is incompatible with convergence to 0. □

Definition 12.42 (Local physical intertwining). Lean ✓ Stmt The local physical intertwining relation is $\sum_j u_{ij}\, A^j = V\, A^i\, V^\dagger$. Locally, the physical action of u can therefore be pushed to the virtual legs:



Definition 12.43 (Transfer-map covariance). Lean ✓ Stmt Transfer-map covariance is the identity that the transfer map commutes with virtual conjugation: $\mathcal{E}(VXV^\dagger) = V\,\mathcal{E}(X)\,V^\dagger$. In doubled-layer notation, the virtual operators slide through the local transfer cell:

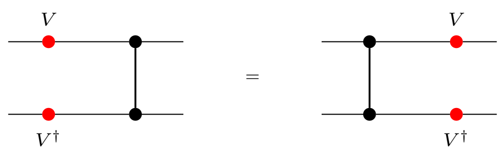


Definition 12.44 (Commutation in the doubled transfer picture). Lean ✓ Stmt In the transfer-map language, the commutation of the doubled transfer matrix $E = \sum_i A_i \otimes \bar{A}_i$ with $V \otimes \bar{V}$ is

$$V\,\mathcal{E}_1(X)\,V^\dagger = \mathcal{E}_1(VXV^\dagger).$$

Theorem 12.45 (Transfer-map covariance and doubled commutation are equivalent). Lean ✓ Stmt *Transfer-map covariance and commutation in the doubled transfer picture are equivalent for any matrix V.*

Proof. ✓ Proof Both sides express the same local picture with opposite orientation:



The two formulas are the same identity read from opposite sides. □

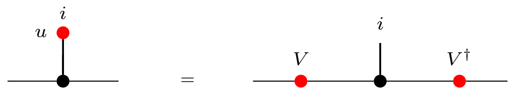
Lemma 12.46 (Unitary Kraus mixing). Lean ✓ Stmt *If u is unitary, then replacing Kraus operators $\{A_i\}$ by their unitary mixtures $\{\sum_j u_{ij} A_j\}$ preserves the channel:*

$$\sum_i \left(\sum_j u_{ij} A_j\right) Y \left(\sum_j u_{ij} A_j\right)^\dagger = \sum_i A_i\, Y\, A_i^\dagger.$$

Proof. ✓ Proof This is exactly the usual invariance of a Kraus decomposition under a unitary change of Kraus operators; see Theorem 16.20. □

Theorem 12.47 (Local physical intertwining implies transfer-map covariance). Lean ✓ Stmt *If V and u are unitary and the local physical intertwining relation holds, then the transfer map is covariant.*

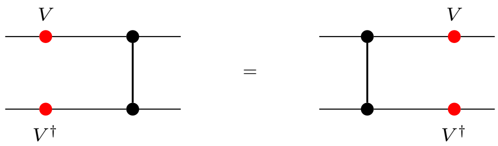
Proof. ✓ Proof Starting from



one inserts $V^\dagger V = \mathbb{1}$ so that each local transfer cell is written in terms of conjugated Kraus operators, then replaces those operators using the intertwining relation. After that, the physical u-boxes disappear by the unitary Kraus-mixing identity. □

Theorem 12.48 (Transfer-map covariance implies local physical intertwining). Lean ✓ Stmt *For an injective MPS, if V is unitary and the transfer map is covariant, then there exists a unitary u satisfying the local physical intertwining relation.*

Proof. ✓ Proof Set $B^i := V\,A^i\,V^\dagger$. The transfer-map covariance relation



implies that the two Kraus families B and A define the same transfer map. Kraus freedom for equal channels converts $\mathcal{E}_B = \mathcal{E}_A$ into a unitary Kraus mixing matrix u with $B^i = \sum_j u_{ij} A^j$. □

Definition 12.49 (Twisted companion tensor). Lean ✓ Stmt For an MPS tensor A and a physical-index matrix u, define the companion tensor B_u by

$$B_u^n := \sum_{n'=0}^{d-1} \overline{u_{n'n}}\, A^{n'}.$$

Lemma 12.50 (Twisted transfer as a mixed transfer operator). Lean ✓ Stmt *The twisted transfer map of A is the mixed transfer operator of the pair (A, B_u):*

$$\mathcal{E}_u = F_{A,B_u}.$$

Proof. ✓ Proof Expand both definitions and rearrange the double sum. □

Theorem 12.51 (Twisted-transfer eigenvalues have modulus at most one). Lean ✓ Stmt *Let A be an injective MPS tensor with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and let u satisfy $uu^\dagger = \mathbb{1}$. If*

$$\mathcal{E}_u(X) = \lambda X$$

for some nonzero $X \in M_D(\mathbb{C})$, then $|\lambda| \le 1$.

Proof. ✓ Proof By Lemma 12.50, the twisted transfer map is the mixed transfer operator F_{A,B_u}. Because B_u is obtained from A by unitary mixing of the Kraus operators, the transfer maps of A and B_u coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common trace-preserving gauge. Theorem 7.12 then applies to the mixed transfer operator in that gauge, and similarity preserves the eigenvalue λ. □

Theorem 12.52 (Modulus-one twisted eigenvalue implies gauge-phase equivalence). Lean ✓ Stmt *Let A be an injective MPS tensor with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and let u satisfy $uu^\dagger = \mathbb{1}$. If*

$$\mathcal{E}_u(X) = \lambda X \tag{12.1}$$

for some nonzero $X \in M_D(\mathbb{C})$ with $|\lambda| = 1$, then A is gauge-phase equivalent to the companion tensor B_u.

Proof. ✓ Proof By Lemma 12.50, the twisted transfer map is the mixed transfer operator F_{A,B_u}. Because B_u is obtained from A by unitary mixing of the Kraus operators, the transfer maps of A and B_u coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common trace-preserving gauge. In that gauge the eigenvalue equation (12.1) becomes a modulus-one peripheral eigenvalue for an irreducible mixed transfer operator, so Theorem 7.27 yields gauge-phase equivalence. Undoing the common trace-preserving gauge gives the claim for A and B_u. □

Lemma 12.53 (Spectral radius < 1 forces boundary-string decay). Lean ✓ Stmt *If $\rho(\mathcal{E}_u) < 1$, then, as $L \to \infty$, for all boundary matrices $X, Y, \Lambda \in M_D(\mathbb{C})$ one has*

$$R_L(u; X, Y) \longrightarrow 0.$$

Proof. ✓ Proof In finite dimension, $\rho(\mathcal{E}_u) < 1$ implies $\mathcal{E}_u^L \to 0$ in operator norm. Composing with the fixed trace pairing $Z \mapsto \mathrm{tr}(\Lambda X Z)$ gives the stated scalar convergence. □

Theorem 12.54 (String order gives gauge-phase equivalence with the companion tensor). Lean ✓ Stmt *Let A be injective, let $\Lambda \in M_D(\mathbb{C})$, and assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If string order exists for u with boundary parameter Λ, then A is gauge-phase equivalent to the companion tensor B_u.*

Proof. ✓ Proof If $\rho(\mathcal{E}_u) < 1$, Lemma 12.53 forces every boundary string-order sequence to decay to zero, contradicting Definition 12.40. Hence $\rho(\mathcal{E}_u) \geq 1$. Theorem 12.51 shows that every eigenvalue has modulus at most 1, so some peripheral eigenvalue has modulus exactly 1. Applying Theorem 12.52 then gives the gauge-phase equivalence. □

Theorem 12.55 (Gauge-phase equivalence with the companion tensor yields a virtual unitary). Lean ✓ Stmt *Let A be injective, assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If A is gauge-phase equivalent to the companion tensor B_u, then there exist a unitary V and a phase μ with $|\mu| = 1$ such that, for every physical index i,*

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger. \tag{12.2}$$

Proof. ✓ Proof Start from an invertible intertwiner X between A and B_u. The positive matrix $Q := XX^\dagger$ is a positive fixed point of $\mathcal{E}_A$. By uniqueness of positive fixed points for an injective normalized tensor, Q is a positive scalar multiple of the identity. Rescaling X by the square root of that scalar produces a unitary V, and the original intertwining relation becomes the phased covariance equation (12.2), with $|\mu| = 1$. □

Theorem 12.56 (Virtual unitary gives a twisted-transfer eigenmatrix). Lean ✓ Stmt *Assume $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ and that, for every physical index i,*

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger$$

with V unitary. Then V is an eigenmatrix of the twisted transfer map:

$$\mathcal{E}_u(V) = \mu V.$$

Proof. ✓ Proof Expand $\mathcal{E}_u(V)$, insert the intertwining relation into each local term, and use $V^\dagger V = \mathbb{1}$ together with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. ☐

Theorem 12.57 (Boundary-state invariance of the virtual unitary). Lean ✓ Stmt *Let A be injective, assume $uu^\dagger = \mathbb{1}$, and let Λ be a positive definite boundary state with $\operatorname{tr}(\Lambda) = 1$ and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. If V is a unitary and μ a phase satisfying, for every physical index i,*

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger, \tag{12.3}$$

then

$$V^\dagger \Lambda V = \Lambda.$$

Proof. ✓ Proof Relation (12.3) shows that $V^\dagger \Lambda V$ is another positive fixed point of the adjoint transfer map. Since Λ is the normalized positive fixed point, uniqueness forces $V^\dagger \Lambda V = \Lambda$. ☐

Lemma 12.58 (Local symmetry implies string order). Lean ✓ Stmt *Assume $\operatorname{tr}(\Lambda) = 1$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If u is a local symmetry of A with respect to Λ, then string order exists for u.*

Proof. ✓ Proof Choose the boundary matrices $X = V^\dagger$ and $Y = V$, where V is the unitary from the local-symmetry datum. By Theorem 12.56, one has $\mathcal{E}_u^L(V) = \mu^L V$. Therefore

$$R_L(u; V^\dagger, V) = \mu^L \operatorname{tr}(\Lambda) = \mu^L,$$

whose norm is constantly 1. ☐

Theorem 12.59 (Local symmetry iff a unitary peripheral eigenmatrix exists). Lean ✓ Stmt *Let A be injective, and assume $uu^\dagger = \mathbb{1}$, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\operatorname{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then u is a local symmetry if and only if there exist a unitary $V \in M_D(\mathbb{C})$ and a phase μ with $|\mu| = 1$ such that*

$$\mathcal{E}_u(V) = \mu V.$$

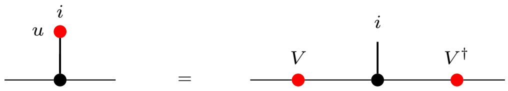
By Theorem 12.51, this witness form is equivalent to the peripheral-spectrum condition $\rho(\mathcal{E}_u) = 1$.

Proof. ✓ Proof The forward direction is Theorem 12.56. For the reverse direction, start from a unitary peripheral eigenmatrix. Theorem 12.52 gives gauge-phase equivalence with the companion tensor; then Theorem 12.55 yields a virtual unitary and phase satisfying the local covariance relation, and Theorem 12.57 gives the boundary-state invariance needed for local symmetry. ☐

Theorem 12.60 (String order $\Leftrightarrow$ local symmetry). Lean ✓ Stmt *For an injective MPS, assume $uu^\dagger = \mathbb{1}$, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\operatorname{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then string order for u exists iff u is a local symmetry.*

Proof. ✓ Proof If string order exists, Theorem 12.54 gives a gauge-phase equivalence with the companion tensor. Theorem 12.55 extracts a virtual unitary and phase from that equivalence, and Theorem 12.57 shows that the same unitary preserves the boundary state, so one obtains local symmetry. Conversely, Lemma 12.58 turns local symmetry into a non-decaying boundary string-order sequence. ☐

Theorem 12.61 (Virtual unitary from string order). Lean ✓ Stmt *For an injective MPS, let $\Lambda \in M_D(\mathbb{C})$, and assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If string order exists for u with boundary parameter Λ, then there exists a virtual unitary V and a unit-modulus scalar μ such that $\sum_j u_{ij} A^j = \mu V A^i V^\dagger$ for all i. Diagrammatically this is the same local intertwining relation, but only up to an overall unit scalar on the virtual side:*



Proof. ✓ Proof String order first gives gauge-phase equivalence with the companion tensor by Theorem 12.54. Theorem 12.55 then normalizes the resulting intertwiner to a unitary and produces the phase μ. □

12.6 SPT Phase Labels and String-Order Universality

Two injective symmetric tensors are defined to be in the same symmetry-protected topological (SPT) phase when their virtual representation cocycles are cohomologous. The identification of this cocycle class with phase equivalence under symmetric gapped paths is the classification theorem of Chen, Gu, and Wen [CGW11] and Schuch, Pérez-García, and Cirac [SPGC11, Section II.F], which is not formalized here (Remark 12.63). The theorems of this section show that string order holds for every injective symmetric tensor with canonical normalisations, so existence of string order is a symmetry diagnostic and does not separate SPT phases (Remark 12.66).

Definition 12.62 (Same SPT phase)**.** Lean ✓ Stmt Injective tensors A and B, both symmetric under an on-site representation $U : G \to \mathrm{GL}_d(\mathbb{C})$, are in the *same SPT phase* if there exist virtual cocycles ω_A, ω_B with corresponding projective representations that intertwine the respective twisted tensors and satisfy $\omega_A \sim \omega_B$ (cohomologous).

Remark 12.63 (The classification theorem is not formalized here)*.* For injective tensors with a fixed on-site symmetry and no symmetry breaking, two tensors can be connected by a symmetric gapped path if and only if their virtual cocycle classes in $H^2(G, \mathrm{U}(1))$ coincide ([CGW11]; [SPGC11, Section II.F]). Neither direction of that equivalence is established here: this chapter constructs the cocycle label and proves its gauge independence (Theorem 12.21), and Definition 12.62 takes cohomologous cocycles as the definition of phase equality.

Theorem 12.64 (String order holds for injective symmetric MPS)**.** *Lean ✓ Stmt Let A be an injective tensor, symmetric under a unitary representation U. Assume $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite,* $\mathrm{tr}(\Lambda) = 1$*, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then string order exists for every group element g.*

Proof. ✓ Proof The virtual representation theorem produces $\rho(g^{-1}) \in \mathrm{GL}_D(\mathbb{C})$ satisfying $\sum_j U(g)_{ij}\, A^j = \rho(g^{-1})\, A^i\, \rho(g^{-1})^{-1}$. A direct computation using $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ shows that $\mathcal{E}_{U(g)}(\rho(g^{-1})) = \rho(g^{-1})$, so $\rho(g^{-1})$ is a nonzero eigenvector with eigenvalue 1. Theorem 12.52 gives gauge-phase equivalence with the companion tensor, and Theorem 12.55 extracts a virtual unitary and phase. By Theorem 12.57, this unitary preserves the boundary state, so Lemma 12.58 gives string order. □

Theorem 12.65 (String order existence agrees for any two injective symmetric tensors)**.** *Lean ✓ Stmt If A and B are injective, both on-site symmetric under a unitary representation U, and if $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, $\mathcal{E}_B(\mathbb{1}) = \mathbb{1}$, Λ_A and Λ_B are positive definite,* $\mathrm{tr}(\Lambda_A) = 1$, $\mathrm{tr}(\Lambda_B) = 1$, *$\mathcal{E}_A^\dagger(\Lambda_A) = \Lambda_A$, and $\mathcal{E}_B^\dagger(\Lambda_B) = \Lambda_B$, then for every $g \in G$, string order exists for A with twist $U(g)$ if and only if it exists for B. In particular, since Definition 12.62 implies on-site symmetry for both tensors, this theorem applies to tensors in the same SPT phase; it applies equally to symmetric tensors in different SPT phases.*

Proof. ✓ Proof Both directions follow from Theorem 12.64, which shows that string order holds universally for injective symmetric tensors with canonical normalisations. □

Remark 12.66 (String order detects symmetry, not the SPT phase). Theorem 12.64 applies to every injective symmetric tensor with canonical normalisations, irrespective of its cocycle class, so tensors in different SPT phases all have string order for every group element. Existence of string order is equivalent to the local symmetry itself (Theorem 12.60); the invariant that separates SPT phases is the cohomology class of the virtual representation cocycle (Definition 12.62).

Chapter 13

Parent Hamiltonians

For a translation-invariant matrix product state with tensor A, the *parent Hamiltonian* is the translation-invariant sum of local terms, each the orthogonal projector onto the complement of the length-L local space $G_L(A) = \mathrm{im}(\Gamma_L)$. This is the space $\mathcal{G}_L^A$ of [PGVWC07]; we write $\Gamma_L(X)(\sigma) = \mathrm{tr}(A^\sigma X)$, which is the same as the source convention $\mathrm{tr}(XA^\sigma)$ by cyclicity of the trace. We follow the construction of [PGVWC07] and [CPGSV21], identifying the N-site space with functions on the computational basis,

$$\{0, \dots, d-1\}^N \to \mathbb{C},$$

rather than as an explicit tensor product.

The chapter establishes the ground-space structure in increasing generality. For an injective tensor, when $N \geq 2$ and $1 < L \leq N$, the parent Hamiltonian is frustration-free and the periodic MPS vector $V^{(N)}(A)$ is its unique ground state, obtained from the intersection property of the local ground spaces. For a normal tensor the corresponding conclusion follows, under the stated closure-property and length hypotheses, from the inverting-and-growing-back argument at the injectivity length. The nearest-neighbor section separates the translated parent-term commutation and zero-energy equations for $V^{(N)}(A)$ from the source ground-space spanning clause in the pure-state theorem relating renormalization fixed points, zero correlation length, and nearest-neighbor commuting parent Hamiltonians in [CPGSV16]. The spectral-gap section records the source anticommutator martingale condition for overlapping local terms, together with cyclic norm-compression estimates as sufficient stronger hypotheses. For a tensor in block-injective canonical form $A^i = \bigoplus_j \mu_j A_j^i$ with a basis of normal tensors $\{A_j\}$, [CPGSV21, Theorem IV.16] and [PGVWC07, Theorem 12] state that the periodic parent-Hamiltonian ground space is spanned by the vectors $\{V^{(N)}(A_j)\}$. The sections below record the open-segment recursion and the conditional reductions for the boundary-condition comparison in the periodic ring.

13.1 The N-site Hilbert space

Definition 13.1 (N-site Hilbert space). Lean ✓ Stmt For local dimension d and chain length N, the N-site Hilbert space is

$$\mathcal{H}_N^{(d)} := \{0, \dots, d-1\}^N \to \mathbb{C}.$$

This is the computational-basis identification used throughout the chapter for periodic-chain states.

13.2 Local ground space $G_L(A)$

Fix an MPS tensor $A = \{A^i\}_{i=0}^{d-1}$ with $A^i \in M_D(\mathbb{C})$ and a block length $L \geq 1$. For a word $w = (i_1, \dots, i_L)$, write $A^w := A^{i_1} \cdots A^{i_L}$. The boundary-condition parametrization is

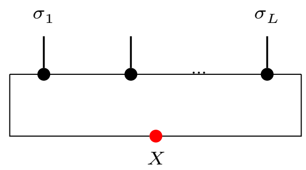
$$\Gamma_L : M_D(\mathbb{C}) \to (\{0, \dots, d-1\}^L \to \mathbb{C}), \qquad \Gamma_L(X)(\sigma) := \operatorname{tr}(A^\sigma X),$$

where A^σ abbreviates $A^{\sigma(0)} \cdots A^{\sigma(L-1)}$. By cyclicity of the trace this is the same as the convention of [PGVWC07], namely $\operatorname{tr}(XA^\sigma)$ for the local space $\mathcal{G}_L^A$.

Definition 13.2 (Ground-space map). Lean ✓ Stmt The map Γ_L above is a $\mathbb{C}$-linear map

$$\Gamma_L : M_D(\mathbb{C}) \to (\{0, \dots, d-1\}^L \to \mathbb{C}).$$

In tensor-network notation,



the upper chain denotes the word tensor A^σ with open physical indices $\sigma_1, \dots, \sigma_L$, and the lower red node denotes the boundary matrix X inserted on the closing virtual bond before taking the trace. The chain length L is fixed by the surrounding formula, not by an extra label inside the diagram.

Lemma 13.3 (Evaluation formula). Lean ✓ Stmt *For every boundary matrix $X \in M_D(\mathbb{C})$ and every length-L word σ,*

$$\Gamma_L(X)(\sigma) = \operatorname{tr}(A^\sigma X).$$

Definition 13.4 (Local ground space). Lean ✓ Stmt The *local ground space* is the image

$$G_L(A) := \operatorname{im}(\Gamma_L) \subseteq (\{0, \dots, d-1\}^L \to \mathbb{C}).$$

Lemma 13.5 (Dimension bound). Lean ✓ Stmt *For every L,*

$$\dim G_L(A) \leq D^2.$$

Proof. ✓ Proof Since $G_L(A)$ is the range of a linear map from $M_D(\mathbb{C})$,

$$\dim G_L(A) \leq \dim M_D(\mathbb{C}) = D^2.$$

□

Lemma 13.6 (Ambient-space dimension). Lean ✓ Stmt *The above coefficient space satisfies*

$$\dim(\{0, \dots, d-1\}^L \to \mathbb{C}) = d^L.$$

Lemma 13.7 (Nontriviality criterion). Lean ✓ Stmt *If $d^L > D^2$, then $G_L(A)$ is a proper subspace of the local Hilbert space:*

$$G_L(A) \neq (\{0, \dots, d-1\}^L \to \mathbb{C}).$$

Proof. ✓ Proof If $G_L(A)$ were the whole space, its dimension would be d^L by Lemma 13.6; Lemma 13.5 gives the contradiction $d^L \leq D^2$. □

13.3 Parent interaction and chain Hamiltonian

The canonical parent interaction at range L is

$$h_L(A) = 1 - \Pi_{G_L(A)} = \Pi_{G_L(A)^\perp}, \qquad \ker h_L(A) = G_L(A).$$

The sources also allow any positive local interaction with this kernel; the orthogonal projector is the canonical representative used here.

Definition 13.8 (Local ground space under the canonical ℓ^2 identification). Lean ✓ Stmt Let

$$\iota_L : (\{0, \dots, d-1\}^L \to \mathbb{C}) \simeq \ell^2(\{0, \dots, d-1\}^L)$$

be the canonical computational-basis identification. The local ground space under this ℓ^2 identification is

$$G_L^{\mathrm{ES}}(A) := \iota_L(G_L(A)) \subseteq \ell^2(\{0, \dots, d-1\}^L).$$

Lemma 13.9 (Membership under the canonical ℓ^2 identification). *Lean* *✓ Stmt* *For $v \in \ell^2(\{0, \dots, d-1\}^L)$,*

$$v \in G_L^{\mathrm{ES}}(A) \iff \iota_L^{-1} v \in G_L(A).$$

Proof. ✓ Proof This is the defining equation $G_L^{\mathrm{ES}}(A) = \iota_L(G_L(A))$. □

Definition 13.10 (Canonical parent interaction). Lean ✓ Stmt The parent interaction at range L is the canonical orthogonal projector

$$h_L(A) := \Pi_{G_L(A)^\perp} = 1 - \Pi_{G_L(A)}, \qquad \ker h_L(A) = G_L(A),$$

on the local L-site space.

Lemma 13.11 (Parent interaction annihilates ground-space elements). *Lean* *✓ Stmt* *For any $v \in G_L(A)$, we have $h_L(A)\, v = 0$.*

Proof. ✓ Proof Since $h_L(A)$ is the orthogonal projector onto $G_L(A)^\perp$, it annihilates every element of $G_L(A)$. □

Definition 13.12 (Restriction to a cyclic interval). Lean ✓ Stmt For $\sigma \in \{0, \dots, d-1\}^N$ and a starting site i, write

$$\sigma|_{[i,i+L-1]} \in \{0, \dots, d-1\}^L$$

for the word defined by

$$(\sigma|_{[i,i+L-1]})(r) = \sigma(i + r \bmod N), \qquad 0 \le r < L.$$

Definition 13.13 (Prescribing a cyclic interval). Lean ✓ Stmt If $L \le N$ and $\tau \in \{0, \dots, d-1\}^L$, write

$$\sigma^{[i,i+L-1] \leftarrow \tau}$$

for the configuration defined by

$$\sigma^{[i,i+L-1]\leftarrow\tau}(k) = \begin{cases} \tau(r), & k \equiv i + r \pmod N \text{ for some } 0 \le r < L, \\ \sigma(k), & \text{otherwise.} \end{cases}$$

The index r in the first case is unique because $L \le N$.

Lemma 13.14 (Extracting a replaced window). Lean ✓ Stmt *If $L \le N$, then restricting to the same cyclic interval after prescribing its values recovers the inserted word:*

$$\left(\sigma^{[i,i+L-1]\leftarrow\tau}\right)|_{[i,i+L-1]} = \tau.$$

Lemma 13.15 (Replacing an extracted window). Lean ✓ Stmt *If $L \le N$, then prescribing on a cyclic interval the values already carried by σ leaves σ unchanged:*

$$\sigma^{[i,i+L-1]\leftarrow\sigma|_{[i,i+L-1]}} = \sigma.$$

Proof. ✓ Proof Case-split on whether the offset $(k-i+N) \bmod N$ lies in $[0, L)$: inside the cyclic interval the prescribed value is the original value of σ; outside the interval both sides already agree with σ. □

For a periodic chain of length N, the parent Hamiltonian is the sum of translated copies of the local interaction.

Definition 13.16 (Translated local term). Lean ✓ Stmt For each site index $i \in \{0, \dots, N-1\}$, the translated local term is determined by

$$(h_i\psi)(\sigma) = \begin{cases} \left[h_L(A)(\tau \mapsto \psi(\sigma^{[i,i+L-1]\leftarrow\tau}))\right](\sigma|_{[i,i+L-1]}), & L \le N, \\ 0, & L > N. \end{cases}$$

Definition 13.17 (Parent Hamiltonian). Lean ✓ Stmt The chain Hamiltonian is

$$H_N(A, L) := \sum_{i=0}^{N-1} h_i.$$

Definition 13.18 (Frustration-free condition). Lean ✓ Stmt A vector ψ is a frustration-free ground state of the parent Hamiltonian when each local interaction annihilates it:

$$\forall\, i \in \{0, \dots, N-1\}, \quad h_i\psi = 0.$$

Lemma 13.19 (Periodic MPS vector window membership). Lean ✓ Stmt *For any window of L consecutive sites starting at position i in an N-site periodic chain ($L \le N$), the restriction of the periodic MPS vector $V^{(N)}(A)$ to that window lies in $G_L(A)$.*

Proof. ✓ Proof The boundary matrix is the product of A-matrices on the complement sites. Trace cyclicity rotates the full N-site product so that window indices come first, matching the ground-space map definition. □

Lemma 13.20 (Local term annihilates the periodic MPS vector). Lean ✓ Stmt *For each site i, $h_i\, V^{(N)}(A) = 0$.*

Proof. ✓ Proof Combines periodic-MPS-vector window membership with the fact that the parent interaction annihilates ground-space elements. □

Lemma 13.21 (Parent Hamiltonian annihilates the periodic MPS vector). Lean ✓ Stmt *For every N with $L \le N$, the periodic MPS vector satisfies*

$$H_N(A, L)\, V^{(N)}(A) = 0.$$

Proof. ✓ Proof Sum of zeros, since each local term annihilates the periodic MPS vector. □

Lemma 13.22 (Frustration-freeness of the periodic MPS vector). Lean ✓ Stmt *The periodic MPS vector is a frustration-free ground state: for every site i,*

$$h_i V^{(N)}(A) = 0.$$

Proof. ✓ Proof Immediate from Lemma 13.20. □

13.4 Intersection property and unique ground state

This section develops the intersection property of local ground spaces for injective MPS and the resulting uniqueness of the ground state on the periodic chain.

13.4.1 Restriction maps

Given a state ψ on $L+1$ sites, we define two restriction maps.

Remark 13.23 (Word and restriction identities). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Appending a last letter to a word corresponds to right multiplication by the corresponding tensor letter, while prepending a first letter corresponds to left multiplication. The restriction maps are the associated linear maps on the state spaces under the computational-basis identification, together with their pointwise formulas.

Definition 13.24 (Left restriction)**.** Lean ✓ Stmt Fix the last physical index j:

$$\sigma \longmapsto \psi(\sigma_1, \dots, \sigma_L, j).$$

Definition 13.25 (Right restriction)**.** Lean ✓ Stmt Fix the first physical index i:

$$\sigma \longmapsto \psi(i, \sigma_1, \dots, \sigma_L).$$

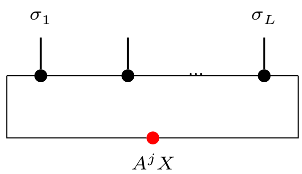
Definition 13.26 (Left ground membership)**.** Lean ✓ Stmt A state ψ on $L+1$ sites satisfies the left ground condition if for every j, the state obtained by fixing the last index to j lies in $G_L(A)$.

Definition 13.27 (Right ground membership)**.** Lean ✓ Stmt A state ψ on $L+1$ sites satisfies the right ground condition if for every i, the state obtained by fixing the first index to i lies in $G_L(A)$.

13.4.2 Forward direction: ground space restricts

Lemma 13.28 (Left restriction)**.** Lean ✓ Stmt *If $\psi \in G_{L+1}(A)$, i.e. $\psi(\sigma) = \operatorname{tr}(A^\sigma X)$ for some X, then for each j, the state obtained by fixing the last index to j lies in $G_L(A)$ with witness $A^j X$.*

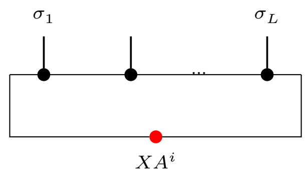
Proof. ✓ Proof The restriction simply absorbs the last tensor into the boundary matrix:



Direct computation: $\psi(\sigma_1, \dots, \sigma_L, j) = \operatorname{tr}(A^{\sigma_1} \cdots A^{\sigma_L} \cdot A^j \cdot X) = \Gamma_L(A^j X)(\sigma)$. $\square$

Lemma 13.29 (Right restriction)**.** Lean ✓ Stmt *If $\psi \in G_{L+1}(A)$ with $\psi(\sigma) = \operatorname{tr}(A^\sigma X)$, then for each i, the state obtained by fixing the first index to i lies in $G_L(A)$ with witness XA^i.*

Proof. ✓ Proof On the right restriction one first rotates the trace and then reads the resulting boundary matrix on the shortened window:



By trace cyclicity: $\psi(i, \sigma_1, \dots, \sigma_L) = \operatorname{tr}(A^i A^\sigma X) = \operatorname{tr}(A^\sigma \cdot XA^i) = \Gamma_L(XA^i)(\sigma)$. □

13.4.3 Injectivity of the ground-space map

Theorem 13.30 (Ground-space map injectivity). Lean ✓ Stmt *If A is injective and $L \geq 1$, then Γ_L is injective.*

Proof. ✓ Proof If $\Gamma_L(X) = 0$ then $\operatorname{tr}(A^\sigma X) = 0$ for every length-L word σ. Since A is injective, the products $\{A^\sigma\}_\sigma$ span $M_D(\mathbb{C})$, so the trace pairing gives $X = 0$. □

Theorem 13.31 (Ground-space map injectivity from word span). Lean ✓ Stmt *If the length-L products $\{A^\sigma : |\sigma| = L\}$ span $M_D(\mathbb{C})$, then Γ_L is injective.*

Proof. ✓ Proof If $\Gamma_L(X) = 0$, then $\operatorname{tr}(A^\sigma X) = 0$ for every word σ of length L. The spanning hypothesis turns this into $\operatorname{tr}(MX) = 0$ for every $M \in M_D(\mathbb{C})$, and the trace pairing gives $X = 0$. □

Lemma 13.32 (Ground-space map injectivity after blocking). Lean ✓ Stmt *If the length-L_0 products span the full matrix algebra,*

$$\operatorname{span}\{A^\omega : |\omega| = L_0\} = M_D(\mathbb{C}),$$

then Γ_{L_0} is injective.

Proof. ✓ Proof The hypothesis is precisely the fixed-length spanning condition needed by Theorem 13.31. □

Theorem 13.33 (Ground-space dimension). Lean ✓ Stmt *If A is injective and $L \geq 1$, then* $\dim G_L(A) = D^2$.

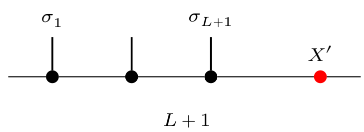
Proof. ✓ Proof The upper bound $\dim G_L(A) \leq D^2$ is Lemma 13.5. Injectivity of Γ_L gives $\dim G_L(A) = \dim M_D(\mathbb{C}) = D^2$. □

13.4.4 The intersection property

Theorem 13.34 (One inclusion of the intersection property). Lean ✓ Stmt *If A is injective and $L \geq 2$, then the nontrivial inclusion in the intersection property holds:*

$$G_L^{\text{left}} \cap G_L^{\text{right}} \subseteq G_{L+1}(A).$$

Proof. ✓ Proof The "inverting and growing back" argument compares the left and right restrictions on their common $(L-1)$-site overlap and then reconstructs the full $(L+1)$-site word from a single boundary matrix:



1. From the right ground condition, for each physical index i there exists a unique $Y_i \in M_D(\mathbb{C})$ (by injectivity of Γ_L) such that $\psi(i, \sigma) = \operatorname{tr}(A^\sigma Y_i)$.
2. Similarly, from the left ground condition, for each j there exists a unique Z_j with $\psi(\sigma, j) = \operatorname{tr}(A^\sigma Z_j)$.
3. Matching on the $(L-1)$-site overlap and using nondegeneracy of the trace pairing: $A^j Y_i = Z_j A^i$ for all i, j.
4. Since A is injective, the matrices $\{A^j\}$ span $M_D(\mathbb{C})$ (Definition 2.28), so $\mathbb{1} = \sum_j c_j A^j$ for some coefficients $c_j \in \mathbb{C}$. Then $Y_i = \mathbb{1} \cdot Y_i = \sum_j c_j A^j Y_i = \sum_j c_j Z_j A^i = X' A^i$, where $X' := \sum_j c_j Z_j$.
5. By trace cyclicity, $\psi(\sigma) = \operatorname{tr}(A^\sigma X')$, so $\psi \in G_{L+1}(A)$.

□

Theorem 13.35 (Intersection property). Lean ✓ Stmt *For injective A and $L \geq 2$:*

$$G_L^{\text{left}} \cap G_L^{\text{right}} = G_{L+1}(A),$$

equivalently $\psi \in G_{L+1}(A) \iff \psi \in G_L^{\text{left}} \cap G_L^{\text{right}}$.

Proof. ✓ Proof Immediate from the two forward-direction lemmas and Theorem 13.34. □

13.4.5 Unique ground state

Contiguous and cyclic interval restrictions

Remark 13.36 (Contiguous interval restrictions). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt For a non-wrapping interval $[s, s+M-1] \subseteq \{0, \dots, N-1\}$ and a full configuration ρ supplying the complementary values, the restriction map is

$$(\operatorname{Res}^\rho_{s,M} \psi)(\omega) = \psi\big(\rho^{[s,s+M-1]\leftarrow\omega}\big)\,.$$

The listed identities are the equations for $s = 0$, $M = N$, for deleting the first or last site, and for first fixing K sites and then restricting to $[s+K, s+K+L-1]$.

Remark 13.37 (Cyclic interval restrictions). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt On the periodic chain, for a full configuration ρ supplying the complementary values, put

$$\delta_i(k) := (k + N - i) \bmod N, \qquad \rho^{\text{cyc}}_{i,L,\omega}(k) := \begin{cases} \omega(\delta_i(k)), & \delta_i(k) < L, \\ \rho(k), & \delta_i(k) \geq L. \end{cases}$$

The restriction to the cyclic interval beginning at i is

$$(\operatorname{Res}^\rho_{i,L} \psi)(\omega) = \psi\Big(\rho^{\text{cyc}}_{i,L,\omega}\Big)\,,$$

with all site labels read modulo N. If $L \leq N$, this is the same configuration as $\rho^{[i,i+L-1]\leftarrow\omega}$. If two boundary conditions agree whenever $\delta_i(k) \geq L$, their restrictions are equal. In particular, filling the interval inside the outside configuration leaves the restriction unchanged:

$$\operatorname{Res}^{\rho^{\text{cyc}}_{i,L,\omega}}_{i,L} \psi = \operatorname{Res}^\rho_{i,L} \psi.$$

For every $r < L$, the inserted word is recovered:

$$\rho^{\mathrm{cyc}}_{i,L,\omega}(i+r) = \omega(r).$$

Here site labels are taken modulo N. The remaining listed identities say that deleting the final site of an $(L+1)$-interval gives the L-interval beginning at i; if $L+1 \le N$, deleting the first site gives the L-interval beginning at $i+1$; and the cyclic formula agrees with the contiguous formula when the interval does not wrap around the ring.

Lemma 13.38 (One-site restrictions of cyclic-window boundary conditions)**.** Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *If a cyclic $(L+1)$-window is represented by a boundary matrix Y, then deleting one endpoint gives*

$$\Gamma_{L+1}(Y)\big|_{\mathit{last}=b} = \Gamma_L(A^bY), \qquad \Gamma_{L+1}(Y)\big|_{\mathit{first}=a} = \Gamma_L(YA^a).$$

Hence two adjacent cyclic windows whose restrictions agree on the common length-L overlap have boundary matrices satisfying

$$Y_1A^a = A^bY_2.$$

Consequently, for a finite chain of adjacent overlaps indexed by $0 \le r < n$,

$$Y_rA^{a_r} = A^{b_r}Y_{r+1},$$

one has

$$Y_0A^{a_0}\cdots A^{a_{n-1}} = A^{b_0}\cdots A^{b_{n-1}}Y_n.$$

In particular, if the two endpoint words are named by the equations

$$a_r = \rho(i_0+r), \qquad b_r = \rho(i_0+r+L+1),$$

with site labels read modulo N, then the same transport identity is written as

$$Y_0A^a = A^bY_n.$$

Proof. ✓ Proof The first two identities are the cyclic first- and last-site deletion identities, followed by the corresponding ground-space restriction formulas. For adjacent windows, equality of the completed outside configurations gives the common-overlap equality

$$\Gamma_L(Y_0A^a) = \Gamma_L(A^bY_n).$$

Injectivity of Γ_L gives $Y_0A^a = A^bY_n$. Iterating the displayed adjacent identity gives the word-product identity. $\square$

Lemma 13.39 (Iterated contiguous-window intersection)**.** Lean ✓ Stmt *If an N-site state satisfies the L-site ground condition on every non-wrapping contiguous window (with $L \ge 2$), then it lies in $G_N(A)$.*

Proof. ✓ Proof The induction repeatedly merges two neighbouring admissible windows $[i, i+L-1]$ and $[i+1, i+L]$ into the larger window $[i, i+L]$ by applying Theorem 13.34 to their common overlap $[i+1, i+L-1]$. $\square$

Definition 13.40 (Suffix restriction)**.** Lean ✓ Stmt Fix a prefix $u \in \{0, \ldots, d-1\}^K$. The suffix restriction of an $(K+L)$-site state ψ is the L-site state given by

$$(R^{\mathrm{tail}}_u\psi)(\sigma) = \psi(u, \sigma).$$

Definition 13.41 (Suffix ground membership). Lean ✓ Stmt A state ψ on $K + L$ sites lies in the suffix ground condition if every fixed prefix u gives a suffix state in $G_L(A)$.

Lemma 13.42 (Ground-space suffix restriction). Lean Lean Lean Lean ✓ Stmt *If $\psi \in G_{K+L}(A)$, then every fixed-prefix suffix restriction of ψ lies in $G_L(A)$. For a vector in $G_{L+1}(A)$, fixing either endpoint gives the expected left or right multiplication of its boundary matrix.*

Proof. ✓ Proof Write $\psi(\rho) = \mathrm{tr}(A^\rho X)$ for a boundary matrix X. Fixing the final site gives the boundary matrix $A^j X$, while fixing the first site gives XA^i by cyclicity of the trace. Fixing a prefix u gives

$$\psi(u, \sigma) = \mathrm{tr}(A^u A^\sigma X) = \mathrm{tr}(A^\sigma X A^u),$$

so the suffix restriction again has the form of a ground-space vector. □

Lemma 13.43 (Left-tail compatibility). Lean ✓ Stmt *Assume ψ on $K + L_0 + 1$ sites satisfies the left ground condition on the first $K + L_0$ sites and the suffix ground condition on every fixed prefix of length K. Then there exist matrices $(Z_j)_j$ and $(Y_u)_u$ such that*

$$\psi(\sigma, j) = \mathrm{tr}(A^\sigma Z_j), \qquad \psi(u, \tau) = \mathrm{tr}(A^\tau Y_u),$$

and for all j and u,

$$Z_j A^u = A^j Y_u.$$

Proof. ✓ Proof Choose Z_j from the long left-window condition and Y_u from the suffix ground condition. The two restrictions give the same vector in $G_{L_0}(A)$:

$$\Gamma_{L_0}(Z_j A^u) = \Gamma_{L_0}(A^j Y_u).$$

Injectivity of Γ_{L_0}, obtained from L_0-block injectivity, gives $Z_j A^u = A^j Y_u$. □

Theorem 13.44 (Inverting step for word compatibility). Lean ✓ Stmt *Assume A is L_0-block-injective with $L_0 > 0$. If matrices $(Z_j)_j$ satisfy*

$$Z_j A^\sigma = A^j Y_\sigma$$

for every word σ of length K, then there exists a matrix X such that $Z_j = A^j X$ for all j.

Proof. ✓ Proof For $K = 1$, the hypothesis gives $Z_j A^i = A^j Y_i$ for every letter i. Every blocked word of length L_0 is nonempty, so this identity extends to the coefficients of the blocked tensor $A^{[L_0]}$. Since $A^{[L_0]}$ is injective, the identity matrix is a linear combination of those blocked coefficients, and substituting this decomposition yields $Z_j = A^j X$ for a common matrix X. For larger K, strip the first letter: for each i, the matrices $Z_j A^i$ satisfy the same compatibility hypothesis with words of length $K - 1$, so the induction hypothesis reduces the problem to the case $K = 1$. □

Theorem 13.45 (Growing-back step at the injectivity length). Lean ✓ Stmt *Assume A is L_0-block-injective with $L_0 > 0$. If an element ψ on $K + L_0 + 1$ sites satisfies the left ground condition on the first $K + L_0$ sites and the suffix ground condition on every prefix of length K, then $\psi \in G_{K+L_0+1}(A)$.*

Proof. ✓ Proof Lemma 13.43 produces matrices $(Z_j)_j$ and $(Y_u)_u$ with $Z_j A^u = A^j Y_u$ for every prefix word u of length K. Theorem 13.44 gives a common right factor X such that $Z_j = A^j X$ for all j. Therefore each last-site restriction of ψ agrees with the corresponding last-site restriction of the ground- space vector determined by X. Since every configuration is obtained by adjoining its final letter to its initial $(K + L_0)$-tuple, the two states coincide, so ψ lies in $G_{K+L_0+1}(A)$. □

Theorem 13.46 (Open-chain iteration of the intersection property). Lean ✓ Stmt *Let A be L_0-block-injective with $D \geq 1$ and $L_0 > 0$. If an N-site state satisfies the ground-space constraint on every non-wrapping contiguous interval of length $L_0 + 1$, for every fixed choice of the complementary sites, with $N \geq L_0 + 1$, then it lies in $G_N(A)$.*

Proof. ✓ Proof Induct on the extra length beyond $L_0 + 1$. The induction hypothesis gives the long left-window ground condition after deleting the last site, while Definition 13.40 and the contiguous-interval restriction identities above identify every fixed-prefix suffix with one of the assumed length-$L_0 + 1$ intervals. Theorem 13.45 then grows back the rightmost site. □

Lemma 13.47 (Open-chain iteration for a sequence of subspaces). Lean ✓ Stmt *Let $S_m \subseteq (\mathbb{C}^d)^{\otimes m}$ be subspaces, and let $0 < L \leq N$. Suppose that every non-wrapping length-L interval of ψ, for every fixed choice of the complementary sites, lies in S_L. Suppose also that, for every $m \geq L$,*

$$\mathbb{C}^d \otimes S_m \cap S_m \otimes \mathbb{C}^d = S_{m+1}.$$

Then $\psi \in S_N$.

Proof. ✓ Proof Induct on the length m of a non-wrapping interval. The case $m = L$ is the assumed local condition. If all length-m intervals lie in S_m, then the two one-site restrictions of a length-$m + 1$ interval are adjacent length-m intervals. Hence the length-$m + 1$ interval lies in $\mathbb{C}^d \otimes S_m \cap S_m \otimes \mathbb{C}^d$, and the displayed identity places it in S_{m+1}. Taking $m = N$ and the interval starting at the first site gives $\psi \in S_N$. □

Lemma 13.48 (Periodic constraints and propagated subspaces). Lean ✓ Stmt *Let $S_m \subseteq (\mathbb{C}^d)^{\otimes m}$ be subspaces, and let $0 < L \leq N$. Suppose that $G_L(A) \subseteq S_L$. Suppose also that, for every $m \geq L$,*

$$\mathbb{C}^d \otimes S_m \cap S_m \otimes \mathbb{C}^d = S_{m+1}.$$

Then the periodic local constraints imply

$$\mathcal{G}_{N,L}(A) \subseteq S_N.$$

Proof. ✓ Proof If $\psi \in \mathcal{G}_{N,L}(A)$, then every length-$L$ cyclic interval of ψ lies in $G_L(A)$, and hence in S_L. A non-wrapping interval is the cyclic interval starting at the same site. Therefore all non-wrapping length-L intervals lie in S_L, and Lemma 13.47 gives $\psi \in S_N$. □

Lemma 13.49 (Cyclic-window range antitonicity). Lean Lean ✓ Stmt *On a nonempty periodic chain, longer cyclic-window constraints imply all shorter cyclic-window constraints: if $L' \leq L \leq N$, then $\mathcal{G}_{N,L}(A) \subseteq \mathcal{G}_{N,L'}(A)$.*

Proof. ✓ Proof For every cyclic window vector $v \in G_{L+1}(A)$ and every last letter b,

$$v|_{\text{last}=b} \in G_L(A).$$

Thus $\mathcal{G}_{N,L+1}(A) \subseteq \mathcal{G}_{N,L}(A)$, and iteration gives the general antitonicity. □

Theorem 13.50 (Open-chain containment from cyclic local constraints). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$ and $D \geq 1$. If an N-site periodic-chain state satisfies all cyclic ground-space constraints at some range $L_0 + 1 \leq L \leq N$, then it lies in the open-chain ground space $G_N(A)$.*

Proof. ✓ Proof Lemma 13.49 reduces the cyclic hypotheses to range $L_0 + 1$. On every non-wrapping interval, the cyclic and contiguous restriction maps agree, so Theorem 13.46 applies. □

13.5 Periodic-boundary comparison for injective tensors

On the periodic ring the two cyclic supports that cross the periodic-boundary cut expose the same complementary word, and matching their boundary matrices forces X to commute with every A^j. This section derives the boundary-crossing compatibilities $C_\tau^+ A^j X = Y_\tau A^j$ and $X A^j C_\tau^- = A^j Y_\tau$ and combines them into the commutation $XA^j = A^j X$.

Lemma 13.51 (Boundary-crossing compatibility at the last site). Lean ✓ Stmt *Assume A is L_0-block-injective with $L_0 > 0$ and $M \geq L_0$. If the reduced cyclic interval crossing the last site has boundary matrices Y_τ, then for every physical letter j one has*

$$C_\tau^+ A^j X = Y_\tau A^j, \tag{13.1}$$

where C_τ^+ is the complementary word seen by that cyclic interval.

Proof. ✓ Proof The trace identity for this cyclic interval is tested against all words of length L_0. Injectivity of Γ_{L_0}, which follows from L_0-block injectivity, turns these test-word identities into the displayed matrix identity (13.1). ☐

Lemma 13.52 (Boundary-crossing compatibility at the second boundary position). Lean ✓ Stmt *Assume A is L_0-block-injective with $L_0 > 0$ and $M \geq L_0$. If the second boundary-crossing reduced cyclic interval has boundary matrices Y_τ, then*

$$X A^j C_\tau^- = A^j Y_\tau$$

for every physical letter j, where C_τ^- is the complementary word seen from the second cyclic position.

Proof. ✓ Proof Factor the cyclic word at the second cyclic position, rotate the trace, and use injectivity of Γ_{L_0} to remove the length-L_0 test word. ☐

Theorem 13.53 (The two boundary-crossing compatibilities). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$ and $D \geq 1$. Suppose $\psi \in \mathcal{G}_{N,L}(A)$, $N \geq 2$, $L_0 < L \leq N$, and $\psi = \Gamma_N(X)$ for a boundary matrix X. Then there are two families of boundary matrices Y_τ^+ and Y_τ^-, indexed by fixed complement values τ, such that for every physical letter j,*

$$C_\tau^+ A^j X = Y_\tau^+ A^j, \qquad X A^j C_\tau^- = A^j Y_\tau^-, \tag{13.2}$$

where C_τ^+ and C_τ^- are the complementary words exposed by the two boundary-crossing cyclic intervals, both of reduced length $L_0 + 1$.

Proof. ✓ Proof First use Lemma 13.49 to reduce the cyclic constraints from range L to range $L_0 + 1$. Then apply the two cyclic-interval compatibility lemmas at the two boundary positions to obtain the displayed one-sided identities (13.2). ☐

Lemma 13.54 (Boundary-crossing equations for $\Gamma_{M+1}(X)$). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, $D \geq 1$, and $L_0 \leq M$. Suppose $\psi = \Gamma_{M+1}(X)$, and suppose matrices $Y_i(\tau)$ represent the length-$(L_0 + 1)$ cyclic restrictions*

$$\mathrm{Res}^\tau_{i,L_0+1}(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Then, for every physical letter j and boundary condition τ,

$$A^{\tau_{L_0}} \cdots A^{\tau_{M-1}} A^j X = Y_M(\tau) A^j, \qquad X A^j A^{\tau_1} \cdots A^{\tau_{M-L_0}} = A^j Y_{M+1-L_0}(\tau). \tag{13.3}$$

Proof. ✓ Proof Substitute $\psi = \Gamma_{M+1}(X)$ in the two cyclic restrictions at the last site and at the second boundary-crossing support. Lemmas 13.51 and 13.52 then give the two equations in (13.3). □

Lemma 13.55 (One-sided products for a common complement word)**.** Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, $D \geq 1$, and $L_0 \leq M$. Suppose $\psi = \Gamma_{M+1}(X)$, and suppose matrices $Y_i(\tau)$ represent the length-(L_0+1) cyclic restrictions*

$$\mathrm{Res}^{\tau}_{i,L_0+1}(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

For every boundary letter η, every physical letter j, and every complementary word μ,

$$Y_M(\tau^+_\eta(\mu))A^j = A^\mu A^j X, \qquad XA^jA^\mu = A^j Y_{M+1-L_0}(\tau^-_\eta(\mu)).$$

Proof. ✓ Proof Substitute $\tau^+_\eta(\mu)$ and $\tau^-_\eta(\mu)$ into Lemma 13.54. The exposed-complement identities of Lemma 13.56 identify both complementary products with A^μ. □

Lemma 13.56 (Common complement word for the two boundary supports)**.** Lean Lean Lean Lean ✓ Stmt *On a chain of length $N = M+1$, with $L_0 \leq M$, fix a physical letter η used outside the complementary sites and a word μ of length $M+1-(L_0+1)$. Write $\tau^+_\eta(\mu)$ for the boundary condition at the support crossing the last site and $\tau^-_\eta(\mu)$ for the boundary condition at the second boundary-crossing support. Their exposed complements are both exactly μ.*

Proof. ✓ Proof Put the same letter η on the remaining sites. Fill the physical sites $L_0, \dots, M-1$ with μ for the support crossing the last site, and fill the sites $1, \dots, M-L_0$ with μ for the second boundary-crossing support. The two complement-extraction identities are then immediate from the index arithmetic. The same outside-site calculation gives, for any ρ with $\rho_{k+L_0} = \mu_k$,

$$\mathrm{Res}^{\rho}_{M,L_0+1}\,\psi = \mathrm{Res}^{\tau^+_\eta(\mu)}_{M,L_0+1}\,\psi,$$

and, for any ρ with $\rho_{k+1} = \mu_k$,

$$\mathrm{Res}^{\rho}_{M+1-L_0,L_0+1}\,\psi = \mathrm{Res}^{\tau^-_\eta(\mu)}_{M+1-L_0,L_0+1}\,\psi.$$

□

Lemma 13.57 (Boundary equations from a common complement word)**.** Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Let A be L_0-block-injective, with $L_0 > 0$, and let $L_0 \leq M$. Suppose two length-(L_0+1) restrictions of a state ψ at the same cyclic support are represented by*

$$\mathrm{Res}^{\rho}_{i,L_0+1}\,\psi = \Gamma_{L_0+1}(Y_\rho), \qquad \mathrm{Res}^{\tau}_{i,L_0+1}\,\psi = \Gamma_{L_0+1}(Y_\tau).$$

If these two restrictions are equal, then for every physical letter j,

$$Y_\rho A^j = Y_\tau A^j.$$

In particular, the last-site boundary-crossing condition satisfies

$$\rho_{k+L_0} = \mu_k \Longrightarrow Y_\rho A^j = Y_{\tau^+_\eta(\mu)} A^j,$$

and the second boundary-crossing condition satisfies

$$\rho_{k+1} = \mu_k \Longrightarrow Y_\rho A^j = Y_{\tau^-_\eta(\mu)} A^j.$$

Consequently, for every word σ of length L_0, the two displayed implications remain true after right multiplication by A^σ.

Proof. ✓ Proof Apply the first-letter restriction to both equal length-(L_0+1) restrictions. The two resulting length-L_0 vectors are represented by $\Gamma_{L_0}(Y_\rho A^j)$ and $\Gamma_{L_0}(Y_\tau A^j)$, and block injectivity makes Γ_{L_0} injective. The two displayed special cases follow from Lemma 13.56. The length-L_0 word forms are obtained by multiplying these equations on the right by A^σ. □

Lemma 13.58 (Reduction to compatible boundary conditions)**.** Lean ✓ Stmt *Let A be L_0-block-injective, with $L_0 > 0$, and let $L_0 \leq M$. Let ψ be a state on the chain of length $M+1$, and let the matrices $Y_i(\tau)$ represent its length-(L_0+1) cyclic restrictions:*

$$\mathrm{Res}^{\tau}_{i,L_0+1}(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Fix a boundary letter η and a complementary word μ. Let ρ^+ and ρ^- be two boundary conditions satisfying

$$\rho^+_{k+L_0} = \mu_k, \qquad \rho^-_{k+1} = \mu_k.$$

If the adjacent-window comparison gives, for every physical letter j and every word σ of length L_0,

$$Y_M(\rho^+)A^jA^\sigma = Y_{M+1-L_0}(\rho^-)A^jA^\sigma,$$

then the same product equation holds for $\tau^+_\eta(\mu)$ and $\tau^-_\eta(\mu)$:

$$Y_M(\tau^+_\eta(\mu))A^jA^\sigma = Y_{M+1-L_0}(\tau^-_\eta(\mu))A^jA^\sigma.$$

Proof. ✓ Proof By Lemma 13.57,

$$Y_M(\rho^+)A^j = Y_M(\tau^+_\eta(\mu))A^j, \qquad Y_{M+1-L_0}(\rho^-)A^j = Y_{M+1-L_0}(\tau^-_\eta(\mu))A^j.$$

Multiplying on the right by A^σ and applying the hypothesis gives the displayed product equation. □

Theorem 13.59 (Pointwise reduction to compatible boundary conditions)**.** Lean ✓ Stmt *Let A be L_0-block-injective, with $L_0 > 0$, and let $L_0 \leq M$. Let ψ be a state on the chain of length $M+1$, and let the matrices $Y_i(\tau)$ represent its length-(L_0+1) cyclic restrictions:*

$$\mathrm{Res}^{\tau}_{i,L_0+1}(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Fix a boundary letter η and a complementary word μ. Suppose that, for each physical letter j and word σ of length L_0, boundary conditions $\rho^+_{j,\sigma}$ and $\rho^-_{j,\sigma}$ satisfy, for every complementary position k,

$$\rho^+_{j,\sigma}(k+L_0) = \mu_k, \qquad \rho^-_{j,\sigma}(k+1) = \mu_k.$$

If, for every j and σ,

$$Y_M(\rho^+_{j,\sigma})A^jA^\sigma = Y_{M+1-L_0}(\rho^-_{j,\sigma})A^jA^\sigma,$$

then, for every j and σ,

$$Y_M(\tau^+_\eta(\mu))A^jA^\sigma = Y_{M+1-L_0}(\tau^-_\eta(\mu))A^jA^\sigma.$$

Proof. ✓ Proof Fix j and σ. Lemma 13.57 gives

$$Y_M(\rho^+_{j,\sigma})A^j = Y_M(\tau^+_\eta(\mu))A^j, \qquad Y_{M+1-L_0}(\rho^-_{j,\sigma})A^j = Y_{M+1-L_0}(\tau^-_\eta(\mu))A^j.$$

Multiply the two identities on the right by A^σ and use the assumed product equation for the chosen pair j, σ. □

Lemma 13.60 (Endpoint words for the periodic-boundary windows). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, and let $L_0 \le M$. Let ψ be a state on the chain of length $M+1$. Suppose that matrices $Y_r(\rho)$, for $0 \le r \le L_0 - 1$, represent the length-(L_0+1) cyclic restrictions beginning at the sites $M+1-L_0+r$:*

$$\mathrm{Res}^{\rho}_{M+1-L_0+r,L_0+1}\,\psi = \Gamma_{L_0+1}(Y_r(\rho)).$$

Then

$$Y_0(\rho)A^{\rho_{M+1-L_0}} \cdots A^{\rho_{M-1}} = A^{\rho_1} \cdots A^{\rho_{L_0-1}} Y_{L_0-1}(\rho).$$

For $L_0 = 1$, both products are empty.

Proof. ✓ Proof Use Lemma 13.38 for the L_0-1 adjacent restrictions starting at $i_0 = M+1-L_0$. For $0 \le r < L_0 - 1$, the two endpoint words are determined by

$$i_0 + r = M + 1 - L_0 + r, \qquad i_0 + r + L_0 + 1 \equiv r + 1 \pmod{M+1},$$

Substituting these two words in the iterated transport identity gives the displayed product equation. □

Lemma 13.61 (Boundary-condition product for the periodic-boundary windows). Lean Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, and let $L_0 \le M$. Let ψ be a state on the chain of length $M+1$, and suppose matrices $Y_i(\rho)$ represent the length-(L_0+1) cyclic restrictions*

$$\mathrm{Res}^{\rho}_{i,L_0+1}(\psi) = \Gamma_{L_0+1}(Y_i(\rho)).$$

Then, for every boundary condition ρ,

$$Y_{M+1-L_0}(\rho)A^{\rho_{M+1-L_0}} \cdots A^{\rho_{M-1}} = A^{\rho_1} \cdots A^{\rho_{L_0-1}} Y_M(\rho).$$

For $L_0 = 1$, both products are empty. The same equation remains true after multiplying both sides on the right by any matrix R.

Proof. ✓ Proof Apply Lemma 13.38 to the $L_0 - 1$ restrictions beginning at $M+1-L_0+r$, with

$$Y_r(\rho) = Y_{M+1-L_0+r}(\rho).$$

With $i_0 = M+1-L_0$, the site labels satisfy

$$i_0 + r = M + 1 - L_0 + r, \qquad i_0 + r + L_0 + 1 \equiv r + 1 \pmod{M+1}$$

hence, for $r = 0, \dots, L_0 - 2$,

$$Y_{M+1-L_0+r}(\rho)A^{\rho_{M+1-L_0+r}} = A^{\rho_{r+1}} Y_{M+2-L_0+r}(\rho).$$

Multiplying these $L_0 - 1$ equations from left to right gives the displayed product equation. The right-multiplied form follows by multiplying this equality by R. □

Lemma 13.62 (Transport followed by the one-sided boundary equation). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$ and $D \ge 1$, and let $L_0 \le M$. Let $\psi = \Gamma_{M+1}(X)$, and suppose that matrices $Y_i(\rho)$ represent all length-(L_0+1) cyclic restrictions of ψ. Then, for every boundary condition ρ and every physical letter j,*

$$Y_{M+1-L_0}(\rho)A^{\rho_{M+1-L_0}} \cdots A^{\rho_{M-1}} A^j = A^{\rho_1} \cdots A^{\rho_{L_0-1}} A^{\rho_{L_0}} \cdots A^{\rho_{M-1}} A^j X.$$

For $\rho = \tau^-_\eta(\mu)$ this is the transport identity supplied by the adjacent-window argument, with the single factor A^j following $Y_{M+1-L_0}(\tau^-_\eta(\mu))$. The padded identity, whose corresponding factor is $A^j A^\sigma$, is supplied separately.

Proof. ✓ Proof Lemma 13.61 gives

$$Y_{M+1-L_0}(\rho)A^{\rho_{M+1-L_0}} \cdots A^{\rho_{M-1}} A^j = A^{\rho_1} \cdots A^{\rho_{L_0-1}} Y_M(\rho) A^j.$$

The one-sided equation for the window beginning at M gives

$$Y_M(\rho)A^j = A^{\rho_{L_0}} \cdots A^{\rho_{M-1}} A^j X.$$

Substitution gives the displayed formula. □

Lemma 13.63 (Long boundary-condition product for the periodic-boundary windows). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, and let $L_0 \leq M$. Let ψ be a state on the chain of length $M+1$, and suppose matrices $Y_i(\rho)$ represent the length-(L_0+1) cyclic restrictions*

$$\mathrm{Res}^{\rho}_{i,L_0+1}(\psi) = \Gamma_{L_0+1}(Y_i(\rho)).$$

Then, for every boundary condition ρ,

$$Y_M(\rho)A^{\rho_M}A^{\rho_0} \cdots A^{\rho_{M-L_0}} = A^{\rho_{L_0}} \cdots A^{\rho_M}A^{\rho_0}Y_{M+1-L_0}(\rho).$$

Each side has $M+2-L_0$ one-site factors, with site labels read modulo $M+1$.

Proof. ✓ Proof Apply Lemma 13.38 to the $M+2-L_0$ restrictions beginning at $M+r$ modulo $M+1$. With $i_0 = M$, the site labels satisfy

$$i_0 + r \equiv M + r, \qquad i_0 + r + L_0 + 1 \equiv L_0 + r \pmod{M+1}.$$

The final window is

$$i_0 + M + 2 - L_0 \equiv M + 1 - L_0 \pmod{M+1}.$$

Substituting these site labels in the iterated product identity gives the displayed equation. □

Lemma 13.64 (One matrix Y_μ from compared boundary matrices). Lean ✓ Stmt *Fix a physical letter η used outside the complementary sites. Suppose the two one-sided boundary matrix families Y^+ and Y^- have been extracted from the cyclic windows. If, for every word on the complementary sites μ, these boundary matrices agree on the boundary conditions built from η,*

$$Y^+_{\tau^+_\eta(\mu)} = Y^-_{\tau^-_\eta(\mu)},$$

then there is a single family Y_μ satisfying

$$A^\mu A^b X = Y_\mu A^b, \qquad X A^a A^\mu = A^a Y_\mu$$

for all letters a, b.

Proof. ✓ Proof Define Y_μ to be the first boundary matrix evaluated on the boundary condition $\tau^+_\eta(\mu)$. Lemma 13.56 rewrites the first cyclic-window complement as μ, while the assumed comparison at $\tau^-_\eta(\mu)$ rewrites the second boundary matrix to the same Y_μ. □

Lemma 13.65 (Commutation from two one-sided equations). Lean ✓ Stmt *Suppose a single family of boundary matrices Y_μ, indexed by words μ on the complementary sites, satisfies the two one-sided identities*

$$A^\mu A^b X = Y_\mu A^b, \qquad X A^a A^\mu = A^a Y_\mu \tag{13.4}$$

for every pair of physical letters a, b. Then X commutes with every word product $A^a A^\mu A^b$ obtained by adjoining one letter on each side of the word A^μ.

Proof. ✓ Proof Multiply the second identity in (13.4) on the right by A^b and the first identity on the left by A^a:

$$XA^aA^\mu A^b = A^aY_\mu A^b = A^aA^\mu A^bX.$$

□

Theorem 13.66 (Boundary matrix commutation from periodic closure). Lean ✓ Stmt *Let A be injective with $D \geq 1$, and let $\psi = \Gamma_N(X)$ for some boundary matrix $X \in M_D(\mathbb{C})$. If every cyclic restriction of length L of ψ lies in $G_L(A)$, with $N \geq 2$ and $1 < L \leq N$, then, for every $j = 0, \dots, d-1$,*

$$XA^j = A^jX$$

Proof. ✓ Proof The periodic boundary condition produces matrix identities of the form $XMW = MY$ for arbitrary test matrices M and complementary words W. Spanning first over the length-$(L-1)$ tails and then over the complementary words, injectivity forces $(XM - MX)W = 0$ for every W, hence $XM = MX$ for every matrix M. In particular, X commutes with each letter A^j. □

13.6 Periodic unique ground state

This section gives three periodic ground-space facts: the span of the periodic MPS vector $V^{(N)}(A) = \Gamma_N(\mathbb{1})$, its membership in every cyclic window constraint $\mathcal{G}_{N,L}(A)$, and its nonvanishing for block-injective tensors. Together with the boundary-matrix commutation $XA^j = A^jX$ of the previous section, these establish that the periodic ground space is spanned by $V^{(N)}(A)$; the one-dimensionality conclusion is completed downstream.

Definition 13.67 (Span of the periodic MPS vector). Lean ✓ Stmt The subspace spanned by the MPS vector $V^{(N)}(A)(\sigma) = \mathrm{tr}(A^{\sigma_0} \cdots A^{\sigma_{N-1}})$.

Lemma 13.68 (The periodic MPS vector is the ground-space map at the identity). Lean ✓ Stmt *For every chain length N,*

$$V^{(N)}(A) = \Gamma_N(\mathbb{1}).$$

Lemma 13.69 (The periodic MPS vector lies in the local ground space). Lean ✓ Stmt $V^{(N)}(A) \in G_N(A)$ *for every N, with boundary matrix $X = I$.*

Proof. ✓ Proof $V^{(N)}(A)(\sigma) = \mathrm{tr}(A^\sigma) = \mathrm{tr}(A^\sigma \cdot I) = \Gamma_N(I)(\sigma)$. □

Definition 13.70 (One-dimensional ground space). Lean ✓ Stmt A finite-dimensional ground space S is one-dimensional if $\dim S = 1$.

Theorem 13.71 (One-dimensionality criterion). Lean ✓ Stmt *For finite-dimensional S, $\dim S = 1$ iff there exists a nonzero $\psi_0 \in S$ such that every $\psi \in S$ is of the form $c\,\psi_0$.*

Proof. ✓ Proof Apply the standard finite-dimensional criterion that a nonzero space has dimension one exactly when all of its vectors are proportional to a single nonzero vector. □

Definition 13.72 (Periodic parent-Hamiltonian ground space). Lean ✓ Stmt For $N > 0$ and $L \leq N$, define the *periodic parent-Hamiltonian ground space*

$$\mathcal{G}_{N,L}(A) := \{\psi \in (\{0, \dots, d-1\}^N \to \mathbb{C}) : \forall i \in \{0, \dots, N-1\},\ \psi|_{[i,i+L-1]} \in G_L(A)\},$$

where the condition $\psi|_{[i,i+L-1]} \in G_L(A)$ means that for every choice of physical indices outside the window, the restriction of ψ to that window lies in $G_L(A)$. The one-index notation $G_L(A)$

denotes the local ground space $\mathcal{G}_L$ of [CPGSV21, Section IV.C]. The two-index notation $\mathcal{G}_{N,L}(A)$ used here denotes the periodic intersection of those local constraints over all translated length-L windows on the N-site ring. When $N = 0$ or $L > N$, set $\mathcal{G}_{N,L}(A) := \top$ by convention.

Lemma 13.73 (Cyclic-window representation matrices inside the periodic ground space). Lean ✓ Stmt *If $0 < N$, $L \leq N$, and $\psi \in \mathcal{G}_{N,L}(A)$, then for every cyclic window and every outside configuration there is a boundary matrix Y such that the corresponding restricted L-site state is $\Gamma_L(Y)$.*

Proof. ✓ Proof This is the defining cyclic-window condition for membership in $\mathcal{G}_{N,L}(A)$, together with the definition of the local ground space as the range of Γ_L. □

Lemma 13.74 (The periodic MPS vector satisfies every cyclic window constraint). Lean ✓ Stmt *If $0 < N$ and $L \leq N$, then*

$$V^{(N)}(A) \in \mathcal{G}_{N,L}(A).$$

Theorem 13.75 (Periodic MPS vector nonvanishing for block-injective tensors). Lean ✓ Stmt *If A is L_0-block-injective with $L_0 > 0$, $D \geq 1$, and $N \geq L_0 + 1$, then $V^{(N)}(A) \neq 0$ in the N-site space.*

Proof. ✓ Proof Suppose for contradiction that $V^{(N)}(A) = 0$, i.e. $\mathrm{tr}(A^\sigma) = 0$ for every word σ of length N. For any words w and u with $|w| = N - L_0$ and $|u| = L_0$, we have $\mathrm{tr}(A^w A^u) = 0$. Since A is L_0-block-injective, $\{A^u : |u| = L_0\}$ spans $M_D(\mathbb{C})$. By nondegeneracy of the trace pairing, $A^w = 0$ for every $|w| = N - L_0$. Repeating the argument with words of length $N - 2L_0$ (since $|w'| + L_0 = N - L_0$, the product $A^{w'} A^{u_1}$ has length $N - L_0$, so $A^{w'} A^{u_1} = 0$ by the previous step) and iterating, we eventually reach $\mathrm{tr}(A^u) = 0$ for $|u| = L_0$, which gives $I = 0$, a contradiction. □

13.7 Injectivity-length reduction for the closure property

This section implements the inverting-and-growing-back reduction from [CPGSV21, Section IV.C]: a commutation or annihilation relation for length-m word products is reduced to the injectivity length L_0, then used in the periodic-boundary closure-property step. Throughout, $A^w := A^{i_1} \cdots A^{i_L}$ denotes the product along a word $w = (i_1, \ldots, i_L)$, Γ_N sends a boundary matrix to its open-chain vector, and A being L_0-*block-injective* means $\mathrm{span}\{A^u : |u| = L_0\} = M_D(\mathbb{C})$, so a relation valid on every length-L_0 product extends by linearity to all of $M_D(\mathbb{C})$.

Theorem 13.76 (Common right factor from suffix equations). Lean ✓ Stmt *Assume A is L_0-block-injective with $L_0 > 0$. Let $\{Z_u\}_{|u|=L_0}$ be a family indexed by words of length L_0. If for some $K \geq 0$ and every suffix word w of length K there exists $Y_w \in M_D(\mathbb{C})$ such that, for every word u of length L_0,*

$$Z_u A^w = A^u Y_w,$$

then there exists $X \in M_D(\mathbb{C})$ with $Z_u = A^u X$ for every word u of length L_0.

Proof. ✓ Proof Induct on K. For $K = 0$ the empty suffix gives $Z_u = A^u Y_\emptyset$. For the successor step, fix the first physical index i and apply the induction hypothesis to the shortened suffix. This gives a matrix X_i satisfying

$$Z_u A^i = A^u X_i$$

for every word u of length L_0. Extending this letter equation to all length-L_0 words and decomposing $\mathbb{1}$ in the spanning family $\{A^u : |u| = L_0\}$ yields a common factor X. □

Theorem 13.77 (Common boundary matrix from spanning word identities). Lean ✓ Stmt *Let A be a tensor whose length-K word products span $M_D(\mathbb{C})$. Let $\{F_b\}_{b\in B}$ and $\{Z_b\}_{b\in B}$ be two families of matrices. If, for every length-K word w, there is a matrix Y_w such that, for every $b \in B$,*

$$Z_b A^w = F_b Y_w,$$

then there is a single matrix Y such that, for every $b \in B$,

$$Z_b = F_b Y.$$

Proof. ✓ Proof Suppose $Z_b M_1 = F_b Y_1$ and $Z_b M_2 = F_b Y_2$ for every $b \in B$. Then

$$Z_b(M_1 + M_2) = Z_b M_1 + Z_b M_2 = F_b(Y_1 + Y_2).$$

For a scalar c,

$$Z_b(cM_1) = cZ_b M_1 = F_b(cY_1).$$

Thus the displayed identity holds for every matrix in the span of the length-K word products. Since this span is all of $M_D(\mathbb{C})$, apply it to the identity matrix. □

Theorem 13.78 (Injectivity-length commutation from length-m commutation). Lean ✓ Stmt *If A is L_0-block-injective with $L_0 > 0$, $m \geq L_0$, and $X \in M_D(\mathbb{C})$ commutes with every length-m product A^ω, then X already commutes with every length-L_0 product A^u.*

Proof. ✓ Proof Write each length-m word as a concatenation uw with $|u| = L_0$ and $|w| = m - L_0$. The hypothesis gives

$$XA^u A^w = A^u A^w X = A^u(A^w X).$$

Thus the family $Z_u := XA^u$ satisfies the hypothesis of Theorem 13.76. One obtains $XA^u = A^u R$ for all $|u| = L_0$. Decomposing $\mathbb{1}$ in the span of the length-L_0 words shows $R = X$, hence $XA^u = A^u X$. □

Theorem 13.79 (Centrality from length-m commutation). Lean ✓ Stmt *If A is L_0-block-injective with $L_0 > 0$, $m \geq L_0$, and $X \in M_D(\mathbb{C})$ commutes with every length-m word product, then X commutes with every matrix in $M_D(\mathbb{C})$.*

Proof. ✓ Proof By Theorem 13.78, X commutes with every length-L_0 word product. These products span $M_D(\mathbb{C})$ by block injectivity, so linearity gives $XM = MX$ for every $M \in M_D(\mathbb{C})$. □

Lemma 13.80 (Amplifying fixed-length word commutation). Lean ✓ Stmt *If a boundary matrix X commutes with every word product of length m, then it commutes with every word product whose length is a multiple $q\,m$.*

Proof. ✓ Proof Proceed by induction on q. For $q = 0$, the claim is trivial. For the inductive step, let a word of length $(q+1)m$ be given and write it as a length-m prefix followed by a length-$q\,m$ suffix. The prefix commutes with X by hypothesis, and the suffix commutes by the induction hypothesis. □

Lemma 13.81 (Padded complement annihilation). Lean Lean ✓ Stmt *If A is L_0-block-injective, $q \geq 1$, and $k \leq qL_0$, then the two one-sided annihilation statements are:*

$$ZA^w = 0 \quad (|w| = k) \qquad \Longrightarrow \qquad Z = 0.$$

The left-handed form is

$$A^w Z = 0 \quad (|w| = k) \qquad \Longrightarrow \qquad Z = 0.$$

Proof. ✓ Proof The hypothesis $\mathrm{span}\{A^u : |u| = L_0\} = M_D(\mathbb{C})$ implies the same spanning statement at every positive multiple qL_0. Each length-qL_0 word factors as a length-k prefix followed by padding, so Z annihilates all generators of the full span and hence annihilates the identity. For the left-handed form, factor each length-qL_0 word as a prefix followed by a length-k suffix and use the same spanning argument. □

Lemma 13.82 (One-sided boundary matrices are unique). Lean Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$. If two boundary matrices have the same product with every one-site tensor on one fixed side, then the two boundary matrices are equal.*

Proof. ✓ Proof If $Y_1 A^j = Y_2 A^j$ for every physical index j, then for every nonempty word w, induction on the word gives

$$Y_1 A^w = Y_2 A^w.$$

Since A is L_0-block-injective,

$$\mathrm{span}\{A^w : |w| = L_0\} = M_D(\mathbb{C}).$$

Thus the left multiplication maps $M \mapsto Y_1 M$ and $M \mapsto Y_2 M$ are equal on $M_D(\mathbb{C})$, and evaluating on the identity gives $Y_1 = Y_2$. The other one-sided statement is the same argument, with right multiplication in place of left multiplication. □

Lemma 13.83 (Generator commutation from length-m commutation). Lean ✓ Stmt *Under the hypotheses of Theorem 13.79, one has $XA^i = A^i X$ for every physical index i.*

Proof. ✓ Proof Apply Theorem 13.79 with $M = A^i$. □

Theorem 13.84 (MPS-line containment from length-m commutation). Lean ✓ Stmt *Let A be injective after blocking $L_0 > 0$ sites, and let $m \geq L_0$. If a boundary matrix X commutes with every length-m word product, then*

$$\Gamma_N(X) \in \mathrm{span}\{V^{(N)}(A)\}.$$

Proof. ✓ Proof Theorem 13.79 makes X commute with the whole matrix algebra. The center of $M_D(\mathbb{C})$ consists of scalar matrices, so $X = c\mathbb{1}$, and therefore $\Gamma_N(X) = cV^{(N)}(A)$. □

Theorem 13.85 (MPS-line containment from positive-length commutation). Lean ✓ Stmt *Let A be injective after blocking $L_0 > 0$ sites. If X commutes with every word product of some positive length m, then*

$$\Gamma_N(X) \in \mathrm{span}\{V^{(N)}(A)\}.$$

Proof. ✓ Proof Lemma 13.80 amplifies the length-m commutation hypothesis to length $L_0 m \geq L_0$. Then Theorem 13.84 applies. □

Theorem 13.86 (MPS-line containment from two one-sided equations). Lean ✓ Stmt *Let A be injective after blocking $L_0 > 0$ sites. Suppose there are matrices Y_μ such that*

$$A^\mu A^b X = Y_\mu A^b, \qquad X A^a A^\mu = A^a Y_\mu,$$

for every pair of physical letters a, b. Then

$$\Gamma_N(X) \in \mathrm{span}\{V^{(N)}(A)\}.$$

Proof. ✓ Proof The identities in (13.4) give commutation with all words of length $|\mu| + 2$, a positive length. The positive-length containment theorem then applies. □

Theorem 13.87 (Boundary equality implies containment in the MPS line). Lean ✓ Stmt *Let A be injective after blocking $L_0 > 0$ sites, and fix a physical letter η used outside the complementary sites. Suppose that, for every boundary condition τ and every physical letter j,*

$$C_\tau^+ A^j X = Y_\tau^+ A^j, \qquad X A^j C_\tau^- = A^j Y_\tau^-,$$

where C_τ^+ and C_τ^- are the complementary words exposed by the two boundary-crossing cyclic windows. Suppose also that the two boundary matrices satisfy

$$Y^+_{\tau_\eta^+(\mu)} = Y^-_{\tau_\eta^-(\mu)},$$

for every word μ on the complementary sites, then

$$\Gamma_N(X) \in \operatorname{span}\{V^{(N)}(A)\}.$$

Proof. ✓ Proof Lemma 13.64 gives a family Y_μ satisfying

$$A^\mu A^b X = Y_\mu A^b, \qquad X A^a A^\mu = A^a Y_\mu.$$

Theorem 13.86 applies. □

Theorem 13.88 (Closure-property step for periodic chains). Lean ✓ Stmt *Let A be injective after blocking $L_0 > 0$ sites and let $D \geq 1$. Assume $N \geq 2$ and $L_0 < L \leq N$. Suppose that, for every physical letter η used in the two boundary conditions and every boundary matrix obtained from an N-site chain ground state, the identities*

$$A^\mu A^j X = Y^+_{\tau_\eta^+(\mu)} A^j, \qquad X A^j A^\mu = A^j Y^-_{\tau_\eta^-(\mu)},$$

hold for every physical letter j and every complementary word μ, and are supplemented, for every μ, by

$$Y^+_{\tau_\eta^+(\mu)} = Y^-_{\tau_\eta^-(\mu)}.$$

Then

$$\mathcal{G}_{N,L}(A) \subseteq \operatorname{span}\{V^{(N)}(A)\}.$$

Proof. ✓ Proof The cyclic-to-open-chain reduction writes any chain ground state as $\Gamma_N(X)$. The two boundary-crossing cyclic windows give

$$A^\mu A^j X = Y^+_{\tau_\eta^+(\mu)} A^j, \qquad X A^j A^\mu = A^j Y^-_{\tau_\eta^-(\mu)}.$$

Together with

$$Y^+_{\tau_\eta^+(\mu)} = Y^-_{\tau_\eta^-(\mu)},$$

Theorem 13.87 finishes. □

13.8 Periodic-boundary comparison for normal tensors

This section isolates the closure-property part of the periodic-boundary argument for normal tensors. In the source proof, after the intersection property has grown the open interval, the same inverting-and-growing-back argument is applied when closing the boundaries [CPGSV21, Section IV.C]. For a vector $\psi = \Gamma_{M+1}(X)$, the closure property asks that the two boundary-crossing restrictions $\mathrm{Res}^{\tau_\eta^+(\mu)}_{M,L_0+1}(\psi)$ and $\mathrm{Res}^{\tau_\eta^-(\mu)}_{M+1-L_0,L_0+1}(\psi)$ agree. The symbols $\tau_\eta^\pm(\mu)$ index these restrictions, and the matrices $Y_i(\tau)$ below satisfy

$$\mathrm{Res}^{\tau}_{i,L_0+1}(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Once the one-sided boundary products are known, equality of the restrictions follows from the boundary-matrix commutation relation $XA^j = A^jX$ for every physical letter j. The block-window identity

$$XA^\alpha A^\nu = A^\alpha Y_\nu$$

is derived from the cyclic-window constraints crossing the periodic cut.

Lemma 13.89 (Block-to-letter translation of the boundary block matrix equation). Lean ✓ Stmt
Let A be an MPS tensor and let $L_0, K \geq 0$. Let X be a matrix, and let Y_{c_b} be a matrix for each iterated block index c_b of the complement. Suppose that, for every block letter b of the alphabet $\{0, \dots, d-1\}^{L_0}$ and every iterated block index c_b of the complement, with $w(b)$ the length-L_0 word of b and $\widetilde{w(c_b)}$ the length-L_0K complement word obtained by concatenating the blocks of c_b,

$$XA^{w(b)}A^{\widetilde{w(c_b)}} = A^{w(b)}Y_{c_b}.$$

Then there is a family Y'_c, indexed by the length-L_0K words c, with

$$XA^sA^c = A^sY'_c$$

for every length-L_0 word s and every length-L_0K word c.

Proof. ✓ Proof Each length-L_0 word s is the word $w(b)$ of its block letter b, and each length-L_0K word c regroups into K blocks, recovering it as the concatenated complement word $\widetilde{w(c_b)}$ of an iterated block index c_b; both identifications are bijective. Substituting these representations into the hypothesis and setting $Y'_c = Y_{c_b}$ gives the asserted equation for every s and c. □

Lemma 13.90 (Boundary matrix commutation from block-window equations). Lean ✓ Stmt
Let A be L_0-block-injective with $L_0 > 0$. Let X be a boundary matrix, and let Y_v be a matrix for each complementary word v of length K. Suppose that, for every word u of length L_0 and every word v of length K,

$$XA^uA^v = A^uY_v.$$

Then, for every physical letter j,

$$XA^j = A^jX.$$

This is the boundary-matrix commutation step obtained from the equations $XA^uA^v = A^uY_v$, for all length-L_0 words u and all length-K words v, as in [CPGSV21, Section IV.C, lines 2049–2090].

Proof. ✓ Proof By Lemma 8.7,

$$\mathrm{span}\{A^u : |u| = L_0\} = M_D(\mathbb{C}).$$

Hence, for every $M \in M_D(\mathbb{C})$ and every complementary word v of length K, the displayed hypothesis extends by linearity in the length-L_0 block to

$$XMA^v = MY_v$$

Taking $M = \mathbb{1}$ gives

$$Y_v = XA^v.$$

Taking $M = A^j$ and substituting the preceding identity gives, for every word v of length K,

$$(XA^j - A^jX)A^v = 0$$

Since $K \leq (K+1)L_0$ and $K + 1 \geq 1$, Lemma 13.81 gives $XA^j - A^jX = 0$. □

Lemma 13.91 (Trace identity from the boundary-crossing cyclic window). Lean ✓ Stmt *Let A be a tensor with virtual dimension at least one, and let $L_0 > 0$ and $L_0 < M$. Let*

$$\psi = \Gamma_{M+1}(X)$$

and assume that every cyclic restriction of length $L_0 + 1$ belongs to $G_{L_0+1}(A)$. Then there are boundary matrices Y_ν, indexed by nonempty complementary words ν of length $M + 1 - (L_0 + 1)$, such that, for every physical letter j and every word α of length L_0,

$$\operatorname{tr}\left(A^j X A^\alpha A^\nu\right) = \operatorname{tr}\left(A^j A^\alpha Y_\nu\right).$$

Proof. ✓ Proof Let Y_ν be the matrix representing the cyclic restriction whose window begins at the last site. The cyclic word is

$$\alpha\, \nu\, j$$

after deleting the first letter from the window and reading the complement, so the ground-space representation gives

$$\operatorname{tr}\left(A^\alpha A^\nu A^j X\right) = \operatorname{tr}\left(A^j A^\alpha Y_\nu\right).$$

Cyclicity of the trace gives

$$\operatorname{tr}\left(A^j X A^\alpha A^\nu\right) = \operatorname{tr}\left(A^\alpha A^\nu A^j X\right).$$

Therefore

$$\operatorname{tr}\left(A^j X A^\alpha A^\nu\right) = \operatorname{tr}\left(A^j A^\alpha Y_\nu\right).$$

□

Lemma 13.92 (One-site-injective boundary block-window identity). Lean ✓ Stmt *Let A be a tensor with virtual dimension at least one whose single-site matrices span the full matrix algebra. Let $L_0 > 0$ and $L_0 < M$, and let*

$$\psi = \Gamma_{M+1}(X).$$

If every cyclic restriction of length L_0+1 belongs to $G_{L_0+1}(A)$, then there are boundary matrices Y_ν, indexed by nonempty complementary words ν of length $M+1-(L_0+1)$, such that, for every word α of length L_0,

$$XA^\alpha A^\nu = A^\alpha Y_\nu.$$

Proof. ✓ Proof For every physical letter j, Lemma 13.91 gives

$$\operatorname{tr}\left(A^j X A^\alpha A^\nu\right) = \operatorname{tr}\left(A^j A^\alpha Y_\nu\right)$$

Put

$$Z = XA^\alpha A^\nu - A^\alpha Y_\nu.$$

For every physical letter j, the preceding identity gives

$$\operatorname{tr}(A^j Z) = 0.$$

Since the matrices A^j span the full matrix algebra, the trace pairing gives

$$(\forall j,\ \mathrm{tr}(A^j Z) = 0) \quad \Longrightarrow \quad Z = 0.$$

Hence $XA^\alpha A^\nu = A^\alpha Y_\nu$. ☐

Theorem 13.93 (Trace identities with length-L_0 probes from cyclic-window constraints). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, $L_0 < M$, and $D \geq 1$. Let*

$$\psi = \Gamma_{M+1}(X)$$

and assume that every cyclic restriction of length $L_0 + 1$ belongs to $G_{L_0+1}(A)$. Then there are boundary matrices Y_ν, indexed by nonempty complementary words ν of length $M+1-(L_0+1)$, such that, for every word α of length L_0 and every word β of length L_0,

$$\mathrm{tr}\left(A^\beta X A^\alpha A^\nu\right) = \mathrm{tr}\left(A^\beta A^\alpha Y_\nu\right).$$

Proof. ✓ Proof Choose representation matrices for every cyclic restriction of length L_0+1. Fix α and ν. The boundary-crossing equations move the matrix X through the first boundary letter. In local boundary notation this has the form

$$XA^{\rho_0}A^{\rho_1\cdots\rho_{M-L_0}} = A^{\rho_0}Y_{M+1-L_0}(\rho).$$

The adjacent boundary-window product then transports this comparison through the remaining $L_0 - 1$ boundary letters:

$$Y_{M+1-L_0}(\rho)A^{\rho_{M+1-L_0}\cdots\rho_{M-1}} = A^{\rho_1\cdots\rho_{L_0-1}}Y_M(\rho).$$

Finally, the outside-label uniqueness lemma identifies the final boundary matrix with the matrix Y_ν attached to the complementary word ν. Thus

$$XA^\alpha A^\nu = A^\alpha Y_\nu.$$

Multiplying on the left by A^β and taking the trace gives the asserted identity for every length-L_0 word β. ☐

Theorem 13.94 (Length-word trace separation for the boundary block window). Lean ✓ Stmt *Let A be L_0-block-injective. Let X be a boundary matrix, and let Y_ν be a matrix for each complementary word ν of length K. Suppose that, for every word α of length L_0, every word β of length L_0, and every complementary word ν,*

$$\mathrm{tr}\left(A^\beta X A^\alpha A^\nu\right) = \mathrm{tr}\left(A^\beta A^\alpha Y_\nu\right).$$

Then, for every word α of length L_0 and every complementary word ν,

$$XA^\alpha A^\nu = A^\alpha Y_\nu.$$

Proof. ✓ Proof The trace identities say exactly that

$$\Gamma_{L_0}(XA^\alpha A^\nu) = \Gamma_{L_0}(A^\alpha Y_\nu)$$

for each fixed pair (α, ν). Since A is L_0-block-injective, Γ_{L_0} is injective by Lemma 13.32. Hence

$$XA^\alpha A^\nu = A^\alpha Y_\nu.$$

☐

Lemma 13.95 (Matrix identity $XA^\alpha A^\nu = A^\alpha Y_\nu$ from cyclic-window constraints). Lean ✓ Stmt
Let A be L_0-block-injective with $L_0 > 0$, $L_0 < M$, and $D \geq 1$. Let

$$\psi = \Gamma_{M+1}(X)$$

and assume that every cyclic restriction of length $L_0 + 1$ belongs to $G_{L_0+1}(A)$. Then there are boundary matrices Y_ν, indexed by nonempty complementary words ν of length $M + 1 - (L_0 + 1)$, such that for every word α of length L_0,

$$XA^\alpha A^\nu = A^\alpha Y_\nu.$$

Proof. ✓ Proof Theorem 13.93 gives, for every word β of length L_0,

$$\operatorname{tr}\left(A^\beta X A^\alpha A^\nu\right) = \operatorname{tr}\left(A^\beta A^\alpha Y_\nu\right).$$

Theorem 13.94 then gives the displayed matrix equation. □

Lemma 13.96 (Boundary restrictions from commutation and one-sided products). Lean ✓ Stmt
Let A be L_0-block-injective with $L_0 > 0$. Let X be a boundary matrix, let W_η and V_η be matrix families indexed by a physical letter η, and let μ be a word. Suppose that, for every physical letter j,

$$XA^j = A^j X$$

Suppose also that, for every η and every j,

$$W_\eta A^j = A^\mu A^j X, \qquad XA^j A^\mu = A^j V_\eta.$$

Then, for every η and every j,

$$W_\eta A^j = V_\eta A^j.$$

Proof. ✓ Proof The commutation relation $XA^j = A^j X$ extends by induction: for every word w,

$$XA^w = A^w X$$

Hence the second one-sided equation gives, for every k,

$$A^k V_\eta = XA^k A^\mu = A^k X A^\mu = A^k A^\mu X.$$

Lemma 13.82 gives $V_\eta = A^\mu X$. The first one-sided equation then gives

$$W_\eta A^j = A^\mu A^j X = A^\mu X A^j = V_\eta A^j.$$

□

Lemma 13.97 (Products with one-site tensors determine the boundary matrix). Lean ✓ Stmt
Let A be L_0-block-injective with $L_0 > 0$. Fix two matrix families Y^+ and Y^- and the two reindexed boundary conditions $\tau_\eta^+(\mu)$ and $\tau_\eta^-(\mu)$. If, for every physical letter j,

$$Y^+_{\tau^+_\eta(\mu)} A^j = Y^-_{\tau^-_\eta(\mu)} A^j,$$

then

$$Y^+_{\tau^+_\eta(\mu)} = Y^-_{\tau^-_\eta(\mu)}.$$

Proof. ✓ Proof Apply Lemma 13.82 to the two matrices $Y^+_{\tau^+_\eta(\mu)}$ and $Y^-_{\tau^-_\eta(\mu)}$. □

Lemma 13.98 (Outside labels determine boundary-restriction matrices). Lean Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$ and $L_0 \le M$. If an outside configuration ρ has the same word μ as $\tau_\eta^+(\mu)$ on the sites outside the last-site cyclic window, then the two matrices representing that restriction agree:*

$$Y_\rho = Y_{\tau_\eta^+(\mu)}.$$

The same assertion holds for the second boundary-crossing window: if ρ has the same outside word as $\tau_\eta^-(\mu)$, then

$$Y_\rho = Y_{\tau_\eta^-(\mu)}.$$

This is the boundary-matrix independence step used to choose one representative outside configuration for each outside word in the closure-property matrix comparison.

Proof. ✓ Proof Equality of the cyclic restrictions gives, for every physical letter j,

$$Y_\rho A^j = Y_{\tau_\eta^\pm(\mu)} A^j.$$

Lemma 13.82 identifies the two matrices from these one-site products. □

Lemma 13.99 (First-letter restrictions determine a vector). Lean ✓ Stmt *Let R_j be restriction to first letter j:*

$$(R_j\phi)(\sigma_1, \dots, \sigma_L) = \phi(j, \sigma_1, \dots, \sigma_L).$$

For $\phi, \psi \in (\mathbb{C}^d)^{\otimes(L+1)}$,

$$(\forall j,\ R_j\phi = R_j\psi) \Longrightarrow \phi = \psi.$$

Proof. ✓ Proof At $(\sigma_0, \sigma_1, \dots, \sigma_L)$ the required equality is

$$\phi(\sigma_0, \sigma_1, \dots, \sigma_L) = (R_{\sigma_0}\phi)(\sigma_1, \dots, \sigma_L) = (R_{\sigma_0}\psi)(\sigma_1, \dots, \sigma_L) = \psi(\sigma_0, \sigma_1, \dots, \sigma_L).$$

□

Lemma 13.100 (Auxiliary restriction equality for the closure property). Lean ✓ Stmt *Let $L_0 > 0$ and $L_0 \le M$, and set*

$$B^+_{\eta,\mu}(\psi) = \mathrm{Res}^{\tau_\eta^+(\mu)}_{M, L_0+1}(\psi), \qquad B^-_{\eta,\mu}(\psi) = \mathrm{Res}^{\tau_\eta^-(\mu)}_{M+1-L_0, L_0+1}(\psi).$$

With R_j as in Lemma 13.99,

$$(\forall j,\ R_j B^+_{\eta,\mu}(\psi) = R_j B^-_{\eta,\mu}(\psi)) \Longrightarrow B^+_{\eta,\mu}(\psi) = B^-_{\eta,\mu}(\psi).$$

Proof. ✓ Proof Apply Lemma 13.99 to the two length-(L_0+1) restrictions, using the displayed family of first-letter equalities. □

Lemma 13.101 (Boundary-crossing restriction from right products). Lean ✓ Stmt *Let A be an MPS tensor, let $L_0 > 0$, and let $L_0 \le M$. Suppose the two boundary-crossing restrictions of length $L_0 + 1$ are represented by boundary matrices $Y^+_{\eta,\mu}$ and $Y^-_{\eta,\mu}$:*

$$B^+_{\eta,\mu}(\psi) = \Gamma_{L_0+1}(Y^+_{\eta,\mu}), \qquad B^-_{\eta,\mu}(\psi) = \Gamma_{L_0+1}(Y^-_{\eta,\mu}).$$

If, for every physical letter j,

$$Y^+_{\eta,\mu} A^j = Y^-_{\eta,\mu} A^j,$$

then

$$B^+_{\eta,\mu}(\psi) = B^-_{\eta,\mu}(\psi).$$

Proof. ✓ Proof Fix j and take the first-letter restriction of the two displayed length-(L_0+1) restrictions. These are

$$\Gamma_{L_0}(Y^{+}_{\eta,\mu}A^j), \qquad \Gamma_{L_0}(Y^{-}_{\eta,\mu}A^j).$$

The one-site product equality gives, for every j,

$$\Gamma_{L_0}(Y^{+}_{\eta,\mu}A^j) = \Gamma_{L_0}(Y^{-}_{\eta,\mu}A^j).$$

Lemma 13.99 identifies the two original restrictions. □

Lemma 13.102 (Boundary restrictions from first-letter products)**.** Lean ✓ Stmt *Let $L_0 > 0$ and $L_0 \le M$. Let $Y_i(\tau)$ represent the length-(L_0+1) cyclic restrictions of ψ. Fix a complementary word μ. If, for every boundary letter η and physical letter j,*

$$Y_M(\tau^+_\eta(\mu))A^j = Y_{M+1-L_0}(\tau^-_\eta(\mu))A^j,$$

then for every η,

$$\mathrm{Res}^{\tau^+_\eta(\mu)}_{M,L_0+1}(\psi) = \mathrm{Res}^{\tau^-_\eta(\mu)}_{M+1-L_0,L_0+1}(\psi).$$

Proof. ✓ Proof Apply Lemma 13.101 for each boundary letter η, using the corresponding first-letter product equations. □

Lemma 13.103 (Equality of the two restrictions crossing the periodic cut)**.** Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$ and $D \ge 1$, let $L_0 < M$, and let*

$$\psi \in \mathcal{G}_{M+1,L_0+1}(A), \qquad \psi = \Gamma_{M+1}(X).$$

For every boundary letter η and complementary word μ, the two restrictions crossing the periodic cut satisfy

$$\mathrm{Res}^{\tau^+_\eta(\mu)}_{M,L_0+1}(\psi) = \mathrm{Res}^{\tau^-_\eta(\mu)}_{M+1-L_0,L_0+1}(\psi).$$

Proof. ✓ Proof The cyclic-window representation lemma gives matrices $Y_M(\tau^+_\eta(\mu))$ and $Y_{M+1-L_0}(\tau^-_\eta(\mu))$ such that

$$\mathrm{Res}^{\tau^+_\eta(\mu)}_{M,L_0+1}(\psi) = \Gamma_{L_0+1}\big(Y_M(\tau^+_\eta(\mu))\big), \qquad \mathrm{Res}^{\tau^-_\eta(\mu)}_{M+1-L_0,L_0+1}(\psi) = \Gamma_{L_0+1}\big(Y_{M+1-L_0}(\tau^-_\eta(\mu))\big).$$

Theorem 13.120 gives, for every j,

$$Y_M(\tau^+_\eta(\mu))A^j = Y_{M+1-L_0}(\tau^-_\eta(\mu))A^j.$$

Lemma 13.101 identifies the two length-(L_0+1) restrictions. □

Lemma 13.104 (Boundary-matrix product from boundary-crossing restrictions)**.** Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, and let $L_0 \le M$. Let ψ be a state on the chain of length $M+1$, and let $Y_i(\tau)$ be matrices satisfying*

$$\mathrm{Res}^{\tau}_{i,L_0+1}(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Suppose that, for every outside letter η,

$$\mathrm{Res}^{\tau^+_\eta(\mu)}_{M,L_0+1}(\psi) = \mathrm{Res}^{\tau^-_\eta(\mu)}_{M+1-L_0,L_0+1}(\psi).$$

Then there are boundary conditions $\rho^+_{j,\sigma}$ and $\rho^-_{j,\sigma}$ such that, for every complementary position k,

$$\rho^+_{j,\sigma}(k+L_0) = \mu_k, \qquad \rho^-_{j,\sigma}(k+1) = \mu_k,$$

and, for every physical letter j and every word σ of length L_0,

$$Y_M(\rho^+_{j,\sigma})A^jA^\sigma = Y_{M+1-L_0}(\rho^-_{j,\sigma})A^jA^\sigma.$$

Proof. ✓ Proof For the product indexed by j and σ, specialize the preceding equality at the outside letter $\eta = j$. Take $\rho^+_{j,\sigma} = \tau^+_j(\mu)$ and $\rho^-_{j,\sigma} = \tau^-_j(\mu)$. The complement equations are the two identities of Lemma 13.56. Fix j. Applying the first-letter restriction to the displayed equality gives

$$\Gamma_{L_0}\left(Y_M(\tau^+_j(\mu))A^j\right) = \Gamma_{L_0}\left(Y_{M+1-L_0}(\tau^-_j(\mu))A^j\right),$$

where the two sides are identified by Lemma 13.38. By Lemma 13.32,

$$Y_M(\tau^+_j(\mu))A^j = Y_{M+1-L_0}(\tau^-_j(\mu))A^j.$$

Right multiplication by A^σ gives the asserted product equation. $\square$

Lemma 13.105 (Boundary-matrix product from right products)**.** Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, and let $L_0 \le M$. Let ψ be a state on the chain of length $M+1$, and let $Y_i(\tau)$ be matrices satisfying*

$$\mathrm{Res}^\tau_{i,L_0+1}(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Fix a complementary word μ of length $M+1-(L_0+1)$. Suppose that, for every boundary letter η and every physical letter j,

$$Y_M(\tau^+_\eta(\mu))A^j = Y_{M+1-L_0}(\tau^-_\eta(\mu))A^j.$$

Then there are boundary conditions $\rho^+_{j,\sigma}$ and $\rho^-_{j,\sigma}$ such that, for every complementary position k,

$$\rho^+_{j,\sigma}(k+L_0) = \mu_k, \qquad \rho^-_{j,\sigma}(k+1) = \mu_k,$$

and, for every physical letter j and every word σ of length L_0,

$$Y_M(\rho^+_{j,\sigma})A^jA^\sigma = Y_{M+1-L_0}(\rho^-_{j,\sigma})A^jA^\sigma.$$

Proof. ✓ Proof For each boundary letter η, Lemma 13.101 gives

$$\mathrm{Res}^{\tau^+_\eta(\mu)}_{M,L_0+1}(\psi) = \mathrm{Res}^{\tau^-_\eta(\mu)}_{M+1-L_0,L_0+1}(\psi).$$

Applying these equalities, Lemma 13.104 gives the displayed boundary conditions and product equation. $\square$

Lemma 13.106 (Boundary-crossing comparison after multiplication by words)**.** Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, and let $L_0 \le M$. Fix a complementary word μ, a matrix X, and a family of matrices $Y_i(\tau)$. Suppose that, for every boundary letter η, every physical letter j, and every word σ of length L_0,*

$$Y_{M+1-L_0}(\tau^-_\eta(\mu))A^jA^\sigma = A^\mu A^j X A^\sigma.$$

Then, for every η and j,

$$Y_{M+1-L_0}(\tau^-_\eta(\mu))A^j = A^\mu A^j X.$$

Proof. ✓ Proof Fix η and j, and set

$$Z_{\eta,j} = Y_{M+1-L_0}(\tau^-_\eta(\mu))A^j - A^\mu A^j X.$$

The hypothesis gives, for every σ with $|\sigma| = L_0$,

$$Z_{\eta,j}A^\sigma = 0.$$

Since A is L_0-block-injective, the products A^σ with $|\sigma| = L_0$ span the full matrix algebra. Applying Lemma 13.81 gives $Z_{\eta,j} = 0$. $\square$

Lemma 13.107 (Boundary-crossing comparison after left multiplication by words). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, and let $L_0 \le M$. Fix a complementary word μ, a matrix X, and a family of matrices $Y_i(\tau)$. Suppose that, for every boundary letter η, every physical letter j, and every pair of words σ, α of length L_0,*

$$A^\alpha \left(Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma\right) = A^\alpha \left(A^\mu A^j X A^\sigma\right).$$

Then, for every η, j, and σ,

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma = A^\mu A^j X A^\sigma.$$

Proof. ✓ Proof Fix η, j, and σ, and set

$$Z_{\eta,j,\sigma} = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma - A^\mu A^j X A^\sigma.$$

The hypothesis says that $A^\alpha Z_{\eta,j,\sigma} = 0$ for every word α of length L_0. The left-handed form of Lemma 13.81 gives $Z_{\eta,j,\sigma} = 0$. □

Lemma 13.108 (Boundary-matrix product from the boundary-crossing comparison). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, and let $L_0 \le M$. Let ψ be a state on the chain of length $M+1$, and let $Y_i(\tau)$ be matrices satisfying*

$$\mathrm{Res}^\tau_{i,L_0+1}(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Fix a complementary word μ and a matrix X. Suppose that, for every boundary letter η and every physical letter j,

$$Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X.$$

Suppose also that, for every η, j, and every word σ of length L_0,

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma = A^\mu A^j X A^\sigma.$$

Then there are boundary conditions $\rho^+_{j,\sigma}$ and $\rho^-_{j,\sigma}$ such that, for every complementary position k,

$$\rho^+_{j,\sigma}(k+L_0) = \mu_k, \qquad \rho^-_{j,\sigma}(k+1) = \mu_k,$$

and, for every physical letter j and every word σ of length L_0,

$$Y_M(\rho^+_{j,\sigma})A^j A^\sigma = Y_{M+1-L_0}(\rho^-_{j,\sigma})A^j A^\sigma.$$

Proof. ✓ Proof Lemma 13.106 gives, for every η and j,

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j = A^\mu A^j X.$$

Combining this equation with

$$Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X$$

gives

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Lemma 13.105 then gives the displayed boundary conditions and product equation. □

Lemma 13.109 (Boundary-matrix product from the left-multiplied boundary-crossing comparison). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, and let $L_0 \le M$. Let ψ be a state on the chain of length $M+1$, and let $Y_i(\tau)$ be matrices satisfying*

$$\mathrm{Res}^{\tau}_{i,L_0+1}(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Fix a complementary word μ and a matrix X. Suppose that, for every boundary letter η and every physical letter j,

$$Y_M(\tau^+_\eta(\mu))A^j = A^\mu A^j X.$$

Suppose also that, for every η, j, and every pair of words σ, α of length L_0,

$$A^\alpha \left(Y_{M+1-L_0}(\tau^-_\eta(\mu))A^j A^\sigma\right) = A^\alpha \left(A^\mu A^j X A^\sigma\right).$$

Then there are boundary conditions $\rho^+_{j,\sigma}$ and $\rho^-_{j,\sigma}$ satisfying the complementary-word equations

$$\rho^+_{j,\sigma}(k+L_0) = \mu_k, \qquad \rho^-_{j,\sigma}(k+1) = \mu_k,$$

and the product equation

$$Y_M(\rho^+_{j,\sigma})A^j A^\sigma = Y_{M+1-L_0}(\rho^-_{j,\sigma})A^j A^\sigma.$$

Proof. ✓ Proof Lemma 13.107 gives

$$Y_{M+1-L_0}(\tau^-_\eta(\mu))A^j A^\sigma = A^\mu A^j X A^\sigma.$$

Lemma 13.108 then gives the displayed boundary conditions and product equation. □

Theorem 13.110 (Boundary-matrix product from left-multiplied equations). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, $L_0 < M$, and $D \ge 1$. Let*

$$\psi = \Gamma_{M+1}(X)$$

be an open-chain representation, and let $Y_i(\tau)$ represent the length-(L_0+1) cyclic restrictions of ψ. Fix a complementary word μ. Suppose that, for every boundary letter η, physical letter j, and length-L_0 words σ, α,

$$A^\alpha \left(Y_{M+1-L_0}(\tau^-_\eta(\mu))A^j A^\sigma\right) = A^\alpha \left(A^\mu A^j X A^\sigma\right).$$

Then there are boundary conditions $\rho^+_{j,\sigma}$ and $\rho^-_{j,\sigma}$ satisfying the complementary-word equations

$$\rho^+_{j,\sigma}(k+L_0) = \mu_k, \qquad \rho^-_{j,\sigma}(k+1) = \mu_k,$$

and the product equation

$$Y_M(\rho^+_{j,\sigma})A^j A^\sigma = Y_{M+1-L_0}(\rho^-_{j,\sigma})A^j A^\sigma.$$

Proof. ✓ Proof Lemma 13.55 gives the equation for the boundary-crossing window beginning at M,

$$Y_M(\tau^+_\eta(\mu))A^j = A^\mu A^j X.$$

The assumed left-multiplied coordinate equation and Lemma 13.109 then give the displayed boundary conditions and product equation. □

Theorem 13.111 (Boundary restrictions from the left-multiplied comparison). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, $L_0 \le M$, and $D \ge 1$. Let*

$$\psi = \Gamma_{M+1}(X)$$

be an open-chain representation, and let $Y_i(\tau)$ represent the length-(L_0+1) cyclic restrictions of ψ. Fix a complementary word μ. Suppose that, for every boundary letter η, physical letter j, and length-L_0 words α, σ,

$$A^\alpha \left(Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j A^\sigma \right) = A^\alpha \left(A^\mu A^j X A^\sigma \right).$$

Then, for every boundary letter η,

$$\mathrm{Res}^{\tau_\eta^+(\mu)}_{M,L_0+1}(\psi) = \mathrm{Res}^{\tau_\eta^-(\mu)}_{M+1-L_0,L_0+1}(\psi).$$

Proof. ✓ Proof The assumed equations and Lemma 13.107 give, for every η, j, and σ,

$$Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j A^\sigma = A^\mu A^j X A^\sigma.$$

Lemma 13.106 removes the right word:

$$Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j = A^\mu A^j X.$$

The equation for the boundary-crossing window beginning at M, from Lemma 13.55, gives

$$Y_M(\tau_\eta^+(\mu)) A^j = A^\mu A^j X.$$

Hence the first-letter products agree, and Lemma 13.102 identifies the two restrictions. □

Theorem 13.112 (Equality of the boundary-crossing restrictions). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, $L_0 < M$, and $D \ge 1$. Let*

$$\psi = \Gamma_{M+1}(X)$$

be an open-chain representation. Assume that for all i and τ,

$$\mathrm{Res}^{\tau}_{i,L_0+1}(\psi) \in G_{L_0+1}(A).$$

In local coordinate notation, fix a nonempty word μ on the sites outside the two cyclic windows. For every outside letter η, choose the two outside configurations so that, for each index k of this outside word,

$$\tau_\eta^+(\mu)_{k+L_0} = \mu_k, \qquad \tau_\eta^-(\mu)_{k+1} = \mu_k,$$

and put the letter η on the remaining sites. The notation $\tau_\eta^\pm(\mu)$ is a coordinate parametrization for the comparison used at the periodic boundary in [CPGSV21, Section IV.C, lines 2078–2079]; it is not notation from the source. The corresponding boundary-crossing restriction equality is

$$\mathrm{Res}^{\tau_\eta^+(\mu)}_{M,L_0+1}(\psi) = \mathrm{Res}^{\tau_\eta^-(\mu)}_{M+1-L_0,L_0+1}(\psi).$$

Proof. ✓ Proof Choose matrices $Y_i(\tau)$ representing the length-(L_0+1) restrictions. The one-sided boundary-product lemma gives, for every outside letter η and physical letter j,

$$Y_M(\tau_\eta^+(\mu)) A^j = A^\mu A^j X, \qquad X A^j A^\mu = A^j Y_{M+1-L_0}(\tau_\eta^-(\mu)).$$

Lemma 13.95 gives matrices Y_ν satisfying, for every word α of length L_0 and every complementary word ν,

$$XA^\alpha A^\nu = A^\alpha Y_\nu.$$

Lemma 13.90 then gives, for every physical letter k, the commutation identity

$$XA^k = A^k X.$$

Lemma 13.96 gives

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Lemma 13.102 then identifies the two boundary-crossing restrictions, giving the periodic-boundary comparison of [CPGSV21, lines 2078–2079]. □

13.8.1 Reverse comparison for boundary-crossing restrictions

This subsection runs the comparison in the reverse direction: from equality of the two cyclic restrictions it recovers the first-letter product equation $Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j$ and the left-multiplied boundary comparison.

Lemma 13.113 (First-letter products from equal boundary restrictions). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$ and $L_0 \leq M$. Let ψ be a vector on $M+1$ sites, and suppose the two length-(L_0+1) boundary restrictions are represented by*

$$\Gamma_{L_0+1}(Y_M(\tau_\eta^+(\mu))), \qquad \Gamma_{L_0+1}(Y_{M+1-L_0}(\tau_\eta^-(\mu))).$$

If these two restrictions are equal, then for every physical letter j,

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Thus the first-letter restriction of the preceding equality is precisely the displayed product equation.

Proof. ✓ Proof Restricting both sides to the first physical letter j gives

$$\Gamma_{L_0}\left(Y_M(\tau_\eta^+(\mu))A^j\right) = \Gamma_{L_0}\left(Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j\right).$$

Since A is L_0-block-injective, Γ_{L_0} is injective, and the displayed product equality follows. □

Lemma 13.114 (Left-multiplied comparison from equal boundary restrictions). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$ and $L_0 \leq M$. Let ψ be a vector on $M+1$ sites, and let $Y_i(\tau)$ represent the length-(L_0+1) cyclic restrictions of ψ. Assume that for every boundary letter η,*

$$\mathrm{Res}^{\tau_\eta^+(\mu)}_{M,L_0+1}(\psi) = \mathrm{Res}^{\tau_\eta^-(\mu)}_{M+1-L_0,L_0+1}(\psi).$$

Then for every boundary letter η, physical letter j, and length-L_0 words α, σ,

$$A^\alpha\left(Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma\right) = A^\alpha\left(Y_M(\tau_\eta^+(\mu))A^jA^\sigma\right).$$

Proof. ✓ Proof The preceding lemma gives

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Multiplying on the left by A^α and on the right by A^σ gives the desired comparison. □

Theorem 13.115 (Left-multiplied comparison of boundary matrices)**.** Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, $L_0 < M$, and $D \geq 1$. Let*

$$\psi = \Gamma_{M+1}(X)$$

be an open-chain representation, and let $Y_i(\tau)$ represent the length-(L_0+1) cyclic restrictions of ψ. Fix a nonempty complementary word μ. Then for every boundary letter η, physical letter j, and length-L_0 words α, σ,

$$A^{\alpha}\left(Y_{M+1-L_0}(\tau_{\eta}^{-}(\mu))A^{j}A^{\sigma}\right) = A^{\alpha}\left(Y_M(\tau_{\eta}^{+}(\mu))A^{j}A^{\sigma}\right).$$

Proof. ✓ Proof The matrices $Y_i(\tau)$ give the local ground-space representations needed in the preceding lemma. Once the preceding boundary-restriction equality is proved, it gives

$$\mathrm{Res}^{\tau_{\eta}^{+}(\mu)}_{M,L_0+1}(\psi) = \mathrm{Res}^{\tau_{\eta}^{-}(\mu)}_{M+1-L_0,L_0+1}(\psi).$$

Applying the left-multiplied comparison for equal boundary restrictions gives

$$A^{\alpha}\left(Y_{M+1-L_0}(\tau_{\eta}^{-}(\mu))A^{j}A^{\sigma}\right) = A^{\alpha}\left(Y_M(\tau_{\eta}^{+}(\mu))A^{j}A^{\sigma}\right).$$

□

Theorem 13.116 (Left-multiplied coordinate comparison at the periodic boundary)**.** Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, $L_0 < M$, and $D \geq 1$. Let*

$$\psi = \Gamma_{M+1}(X)$$

be an open-chain representation, and let $Y_i(\tau)$ represent the length-(L_0+1) cyclic restrictions of ψ. Fix a nonempty complementary word μ. Then for every boundary letter η, physical letter j, and length-L_0 words α, σ,

$$A^{\alpha}\left(Y_{M+1-L_0}(\tau_{\eta}^{-}(\mu))A^{j}A^{\sigma}\right) = A^{\alpha}\left(A^{\mu}A^{j}XA^{\sigma}\right).$$

Proof. ✓ Proof Once the comparison of the two cyclic restrictions is available, it gives

$$A^{\alpha}\left(Y_{M+1-L_0}(\tau_{\eta}^{-}(\mu))A^{j}A^{\sigma}\right) = A^{\alpha}\left(Y_M(\tau_{\eta}^{+}(\mu))A^{j}A^{\sigma}\right).$$

The one-sided equation for the boundary-crossing window beginning at M gives

$$Y_M(\tau_{\eta}^{+}(\mu))A^{j} = A^{\mu}A^{j}X.$$

Substitution gives the displayed identity. □

Lemma 13.117 (Letterwise comparison of boundary-crossing restrictions)**.** Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$ and $D \geq 1$, let $L_0 < M$, and let*

$$\psi \in \mathcal{G}_{M+1,L_0+1}(A), \qquad \psi = \Gamma_{M+1}(X).$$

Let R_j denote restriction to first physical letter j. For every outside letter η, complementary word μ, and physical letter j, one has

$$R_j\,\mathrm{Res}^{\tau_{\eta}^{+}(\mu)}_{M,L_0+1}(\psi) = R_j\,\mathrm{Res}^{\tau_{\eta}^{-}(\mu)}_{M+1-L_0,L_0+1}(\psi).$$

This is the one-letter product form of the local comparison corresponding to [CPGSV21, lines 2078–2079].

Proof. ✓ Proof Let $Y_i(\tau)$ be the representation matrices supplied by Lemma 13.73. By Theorem 13.120,

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Fixing the first physical letter in the two length-(L_0+1) windows gives

$$R_j \operatorname{Res}_{M,L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \Gamma_{L_0}\left(Y_M(\tau_\eta^+(\mu))A^j\right),$$
$$R_j \operatorname{Res}_{M+1-L_0,L_0+1}^{\tau_\eta^-(\mu)}(\psi) = \Gamma_{L_0}\left(Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j\right).$$

The two restrictions are therefore equal. □

Theorem 13.118 (Boundary-matrix product for the two boundary-crossing restrictions)**.** Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$ and $D \geq 1$. Let ψ be a state on the chain of length $M+1$ with open-chain representation $\psi = \Gamma_{M+1}(X)$. Let $Y_i(\tau)$ be matrices satisfying*

$$\operatorname{Res}_{i,L_0+1}^{\tau}(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

If $L_0 < M$, then for every nonempty complementary word μ there are boundary conditions $\rho_{j,\sigma}^+$ and $\rho_{j,\sigma}^-$, depending on the physical letter j and the word σ of length L_0, such that, for every complementary position k,

$$\rho_{j,\sigma}^+(k+L_0) = \mu_k, \qquad \rho_{j,\sigma}^-(k+1) = \mu_k,$$

and, for every j and σ,

$$Y_M(\rho_{j,\sigma}^+)A^jA^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^jA^\sigma.$$

Proof. ✓ Proof Theorem 13.116 gives, for every boundary letter η, physical letter j, and words α, σ of length L_0,

$$A^\alpha\left(Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma\right) = A^\alpha\left(A^\mu A^j X A^\sigma\right).$$

Theorem 13.110 applies this left-multiplied comparison together with the one-sided boundary products and gives the displayed boundary conditions $\rho_{j,\sigma}^+$ and $\rho_{j,\sigma}^-$, including

$$Y_M(\rho_{j,\sigma}^+)A^jA^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^jA^\sigma.$$

□

Theorem 13.119 (Product equality at the periodic boundary)**.** Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$ and $D \geq 1$. Let $\psi \in \mathcal{G}_{M+1,L_0+1}(A)$ have an open-chain representation $\psi = \Gamma_{M+1}(X)$. Let $Y_i(\tau)$ be matrices satisfying*

$$\operatorname{Res}_{i,L_0+1}^{\tau}(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

If $L_0 < M$, then for every boundary letter η, complementary word μ, physical letter j, and word σ of length L_0,

$$Y_M(\tau_\eta^+(\mu))A^jA^\sigma = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma.$$

Proof. ✓ Proof Theorem 13.118 gives boundary conditions $\rho_{j,\sigma}^+$ and $\rho_{j,\sigma}^-$ satisfying

$$\rho_{j,\sigma}^+(k+L_0) = \mu_k, \qquad \rho_{j,\sigma}^-(k+1) = \mu_k,$$

and

$$Y_M(\rho_{j,\sigma}^+)A^jA^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^jA^\sigma.$$

Theorem 13.59 replaces these auxiliary conditions by the displayed boundary conditions:

$$Y_M(\tau_\eta^+(\mu))A^jA^\sigma = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma,$$

which is the product equality needed for the periodic-boundary comparison. □

Theorem 13.120 (Right-product equality at the periodic boundary)**.** Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$ and $D \geq 1$. Let $\psi \in \mathcal{G}_{M+1,L_0+1}(A)$ have an open-chain representation $\psi = \Gamma_{M+1}(X)$. Let $Y_i(\tau)$ be matrices satisfying*

$$\mathrm{Res}^{\tau}_{i,L_0+1}(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

If $L_0 < M$, then for every letter η used in the two boundary conditions, every word μ on the complementary sites, and every physical letter j,

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

This is the one-site product equality used to compare the two boundary matrices in the periodic-boundary comparison.

Proof. ✓ Proof Fix j and set

$$Z_j = Y_M(\tau_\eta^+(\mu))A^j - Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Theorem 13.119 gives, after subtracting the two sides, for every word σ of length L_0,

$$Z_jA^\sigma = 0.$$

Since A is L_0-block-injective,

$$\mathrm{span}\{A^\sigma : |\sigma| = L_0\} = M_D(\mathbb{C}).$$

Applying Lemma 13.81 gives $Z_j = 0$, hence

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

□

Theorem 13.121 (Equality of the two boundary-matrix families)**.** Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$ and $D \geq 1$, let $N \geq 2$, $L_0 + 1 < N$, and $L_0 < L \leq N$, and let $\psi \in \mathcal{G}_{N,L}(A)$ have an open-chain representation $\psi = \Gamma_N(X)$. Suppose Y^+ and Y^- are two families of matrices obtained from the two boundary-crossing cyclic-window constraints, and that they satisfy, for every physical letter j and every boundary condition τ,*

$$A^{\tau_{L_0}\cdots\tau_{N-2}}A^jX = Y^+_\tau A^j, \qquad XA^jA^{\tau_1\cdots\tau_{N-L_0-1}} = A^jY^-_\tau.$$

For every physical letter $\eta \in \{0,\dots,d-1\}$ and every word μ on the complementary sites, define two boundary conditions by

$$\tau_\eta^+(\mu)_i = \begin{cases} \mu_{i-L_0}, & L_0 \leq i < N-1, \\ \eta, & \textit{otherwise}, \end{cases} \qquad \tau_\eta^-(\mu)_i = \begin{cases} \mu_{i-1}, & 1 \leq i < N-L_0, \\ \eta, & \textit{otherwise}. \end{cases}$$

Then the two families agree on these two boundary conditions:

$$Y^+_{\tau_\eta^+(\mu)} = Y^-_{\tau_\eta^-(\mu)}.$$

This is the boundary-matrix comparison required for equality of the two restrictions crossing the periodic cut.

Proof. ✓ Proof The reduced cyclic-window compatibilities at the boundary give, for every physical letter j,

$$A^\mu A^j X = Y^+_{\tau^+_\eta(\mu)} A^j, \qquad X A^j A^\mu = A^j Y^-_{\tau^-_\eta(\mu)}.$$

By Lemma 13.97, it is enough to prove, for every j,

$$Y^+_{\tau^+_\eta(\mu)} A^j = Y^-_{\tau^-_\eta(\mu)} A^j.$$

Write $N = M + 1$. The two boundary-crossing supports start at M and $M + 1 - L_0$. Let $Y_i(\tau)$ denote the cyclic-window representation matrix given by Lemma 13.73 at the window beginning at i. The first step identifies the two abstract boundary-condition families with the two boundary-restriction matrices:

$$Y^+_{\tau^+_\eta(\mu)} = Y_M(\tau^+_\eta(\mu)), \qquad Y^-_{\tau^-_\eta(\mu)} = Y_{M+1-L_0}(\tau^-_\eta(\mu)).$$

Together with the preceding identifications, Theorem 13.120 gives, for every physical letter j,

$$Y^+_{\tau^+_\eta(\mu)} A^j = Y^-_{\tau^-_\eta(\mu)} A^j.$$

Lemma 13.97 then gives the desired equality of boundary matrices. □

13.8.2 Normal unique-ground-state consequences

In the injective case, for $N \geq 2$ and $1 < L \leq N$, the chain ground space equals $\mathrm{span}\{V^{(N)}(A)\}$, so the parent Hamiltonian has a unique ground state in that length range. In the normal case the same conclusion is the payoff of the section, given the boundary-crossing comparison and the length hypotheses $N \geq 2$, $L_0 + 1 < N$, and $L_0 < L \leq N$.

Theorem 13.122 (Chain ground space is spanned by the MPS vector in the injective case)**.** Lean ✓ Stmt *If A is injective, $D \geq 1$, $N \geq 2$, and $1 < L \leq N$, then*

$$\mathcal{G}_{N,L}(A) = \mathrm{span}\{V^{(N)}(A)\}.$$

Proof. ✓ Proof Let $\psi \in \mathcal{G}_{N,L}(A)$. The non-wrapping window conditions and Lemma 13.39 imply $\psi \in G_N(A)$, so $\psi = \Gamma_N(X)$ for some boundary matrix X. The periodic boundary condition now applies Theorem 13.66 and forces X to commute with every letter A^j. Since A is injective, the letters generate the full matrix algebra, hence X is scalar. Therefore ψ is proportional to $V^{(N)}(A)$, while the reverse inclusion is Lemma 13.74. □

Theorem 13.123 (Normal tensor: $\mathcal{G}_{N,L}(A) \subseteq \mathrm{span}\{V^{(N)}(A)\}$)**.** Lean ✓ Stmt *If A is normal, $D \geq 1$, and injective after blocking $L_0 > 0$ sites, with $N \geq 2$, $L_0 + 1 < N$, and $L_0 < L \leq N$, then*

$$\mathcal{G}_{N,L}(A) \subseteq \mathrm{span}\{V^{(N)}(A)\}.$$

Proof. ✓ Proof For a chain ground state, the open-chain containment gives $\psi = \Gamma_N(X)$. The two boundary-crossing cyclic-window constraints give

$$A^\mu A^j X = Y^+_{\tau^+_\eta(\mu)} A^j, \qquad X A^j A^\mu = A^j Y^-_{\tau^-_\eta(\mu)}.$$

The boundary-crossing comparison is the equality

$$Y^+_{\tau^+_\eta(\mu)} = Y^-_{\tau^-_\eta(\mu)}.$$

These equations imply

$$\psi \in \mathrm{span}\{V^{(N)}(A)\}.$$

□

Theorem 13.124 (Chain ground space is spanned by the MPS vector in the normal case). Lean ✓ Stmt *If A is normal, $D \geq 1$, and injective after blocking $L_0 > 0$ sites, with $N \geq 2$, $L_0 + 1 < N$, and $L_0 < L \leq N$, then*

$$\mathcal{G}_{N,L}(A) = \operatorname{span}\{V^{(N)}(A)\}.$$

Theorem 13.125 (Unique ground state on the periodic chain). Lean ✓ Stmt *For an injective tensor A with $D \geq 1$, if $N \geq 2$ and $1 < L \leq N$, then the periodic chain ground space is one-dimensional.*

Proof. ✓ Proof By Theorem 13.122, the ground space is the line $\operatorname{span}\{V^{(N)}(A)\}$. It remains to show that the MPV is nonzero. If $V^{(N)}(A) = 0$, then Lemma 13.68 gives

$$\Gamma_N(\mathbb{1}) = \Gamma_N(0).$$

Since A is injective and $N > 0$, Theorem 13.30 forces $\mathbb{1} = 0$, a contradiction. Therefore the spanning line is one-dimensional. □

Theorem 13.126 (Unique ground state after blocking). Lean ✓ Stmt *If A becomes injective after blocking $L_0 > 0$ sites and $D \geq 1$, the parent Hamiltonian with interaction range $2L_0$ has a unique ground state on every periodic chain with $N \geq 2L_0$ and $L_0 + 1 < N$.*

Theorem 13.127 (Unique ground state for normal tensors at range $L_0 + 1$). Lean ✓ Stmt *If A is normal, becomes injective after blocking $L_0 > 0$ sites, and $D \geq 1$, then the parent Hamiltonian with interaction range $L_0 + 1$ has a unique ground state on every periodic chain with $L_0 + 1 < N$.*

13.9 Translated parent-term commutation

This section concerns the commutation equation for translated parent terms in a periodic chain. For normal tensors it relates renormalization fixed points to the tensors whose length-two parent terms commute, $h_i(A,2)h_j(A,2) = h_j(A,2)h_i(A,2)$; the implication from commuting terms back to a renormalization fixed point uses a left-canonical normalization.

The nearest-neighbor commuting parent Hamiltonian condition in [CPGSV16, Definition 3.9 and Theorem 3.10] is distinct from the parent commuting Hamiltonian condition of Appendix D.2, which is stated for local projectors Q_{AX} and Q_{XB}. The latter appears in the decorrelation section that follows. In Definition 3.9 the parent-Hamiltonian condition also says that, for $N > L$, the ground space is spanned by the periodic MPS vectors associated to the BNT components; nearest-neighbor means the specialization $L = 2$.

Definition 13.128 (Translated parent-term commutation). Lean ✓ Stmt The parent Hamiltonian with block length L on N sites has *commuting* local terms if $h_ih_j = h_jh_i$ for all $i, j \in \{0, \ldots, N-1\}$. This records the translated commutation equation $[\tau_j(P_L), P_L] = 0$ from [CPGSV16, Definition 3.9]; the same source definition also requires the ground space to be spanned, for $N > L$, by the periodic MPS vectors of the BNT components.

Definition 13.129 (Nearest-neighbor translated parent-term commutation). Lean ✓ Stmt The nearest-neighbor commuting condition is the length-2 specialization of the preceding commutation equation. For every $i, j \in \{0, \ldots, N-1\}$,

$$h_ih_j = h_jh_i.$$

The source definition [CPGSV16, Definition 3.9] also requires the corresponding parent Hamiltonian to have $L = 2$ and to have the prescribed ground-space spanning property for $N > 2$.

Definition 13.130 (Commutation and zero-energy equations for nearest-neighbor parent terms). Lean ✓ Stmt This condition records the commutation equation and the zero-energy equation for the nearest-neighbor parent terms:

$$h_i h_j = h_j h_i, \qquad h_i V^{(N)}(A) = 0.$$

This is the ground-vector part of the condition appearing in [CPGSV16, Theorem 3.10(iii)]: the MPS vector has zero energy for the translated two-site parent terms, and those terms commute. The missing source assertion is that the whole parent-Hamiltonian ground space is spanned by the corresponding periodic MPS vectors from the BNT components, as recorded in Definition 13.131.

Definition 13.131 (Parent-Hamiltonian ground-space spanning condition). Lean ✓ Stmt Let B be a canonical-form tensor with BNT components $A_1, \dots, A_g$. The source parent-Hamiltonian spanning condition is that, for every $N > L$,

$$\ker H_L^{(N)}(B) = \operatorname{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

This is the ground-space clause in [CPGSV16, Definition 3.9]. It is separate from the commutation equation $[\tau_j(P_L), P_L] = 0$ and from the zero-energy equation for a single periodic MPS vector.

Definition 13.132 (Nearest-neighbor commuting parent-Hamiltonian ground-space condition). Lean ✓ Stmt For a fixed chain length N, this condition is the conjunction of the nearest-neighbor commutation equation, the zero-energy equation for the MPS vector, and the ground-space spanning equation:

$$h_i h_j = h_j h_i, \qquad h_i V^{(N)}(B) = 0, \qquad \ker H_2^{(N)}(B) = \operatorname{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

The source statement in [CPGSV16, Theorem 3.10(iii)] quantifies this condition over all $N > 2$.

Definition 13.133 (All-chain nearest-neighbor commuting parent-Hamiltonian ground-space condition). Lean ✓ Stmt This is the source quantification in [CPGSV16, Theorem 3.10(iii)]. For every $N > 2$, the fixed-chain condition of Definition 13.132 holds:

$$h_i h_j = h_j h_i, \qquad h_i V^{(N)}(B) = 0, \qquad \ker H_2^{(N)}(B) = \operatorname{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

This is a named condition. It does not prove the ground-space spanning equation for a concrete canonical-form tensor.

Remark 13.134 (Elementary consequences of the commuting definition). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt The nearest-neighbor condition is the length-2 specialization of the general commuting condition. The ground-vector condition above contains this commuting equation together with the frustration-free equation for $V^{(N)}(A)$; it does not assert the source ground-space spanning property. For $N \geq 2$, Lemma 13.22 with $L = 2$ gives

$$h_i\, V^{(N)}(A) = 0$$

for every site i. The fixed-chain ground-space condition also contains

$$\ker H_2^{(N)}(B) = \operatorname{span}\{V^{(N)}(A_j) : j = 1, \dots, g\},$$

and quantifying this condition over all $N > 2$ gives the nearest-neighbor case of the spanning condition above. Conversely, the all-chain condition is exactly the conjunction of the all-chain ground-vector equations and the nearest-neighbor instance of the spanning condition above. This is precisely the condition named in Definition 13.133.

Lemma 13.135 (Zero-energy BNT vectors lie in the parent-Hamiltonian kernel). Lean Lean ✓ Stmt *Let B be a canonical-form tensor with BNT components $A_1, \dots, A_g$. If every component vector has zero energy for the parent Hamiltonian,*

$$H_L^{(N)}(B)V^{(N)}(A_j) = 0 \qquad (1 \leq j \leq g),$$

then

$$\operatorname{span}\{V^{(N)}(A_j) : j = 1, \dots, g\} \subseteq \ker H_L^{(N)}(B).$$

It is enough to assume the local frustration-free equations $h_i V^{(N)}(A_j) = 0$ for every i and j. This is the easy inclusion in the source ground-space spanning equation of [CPGSV16, Definition 3.9]; the reverse inclusion is the remaining spanning direction.

Proof. ✓ Proof The kernel is a linear subspace. Hence it is enough to check the generators $V^{(N)}(A_j)$. The first hypothesis gives this directly. Under the local frustration-free equations,

$$H_L^{(N)}(B)V^{(N)}(A_j) = \sum_i h_i V^{(N)}(A_j) = 0.$$

□

Theorem 13.136 (Renormalization fixed points have commuting nearest-neighbor parent terms). Lean Lean Not ready *If A has nonzero virtual dimension and is a normal renormalization fixed point, then for every chain length $N \geq 2$ the two-site parent terms commute. This theorem records the commutation part of the nearest-neighbor source condition, not the ground-space spanning part. This implication is cited here from the structural characterization theorem for renormalization fixed points in [CPGSV16]; the source proof says that RFP ⇒ NNCPH is immediate from that theorem. The present formalization still supplies this implication through the axiom tagged above; the source derivation from the Appendix B product-of-pairs form remains to be proved.*

Theorem 13.137 (Renormalization fixed points satisfy the NNCPH commutation and zero-energy equations). Lean Lean Not ready *If A has nonzero virtual dimension and is a normal renormalization fixed point, then for every chain length $N \geq 2$ the MPS vector $V^{(N)}(A)$ satisfies the commutation and zero-energy equations for nearest-neighbor parent terms:*

$$h_i h_j = h_j h_i, \qquad h_i V^{(N)}(A) = 0.$$

The zero-energy equation is the usual parent-Hamiltonian frustration-freeness statement. The commuting part is still inherited from the axiom tagged above, so this entry is not yet the full source implication.

Proof. Not ready Conditional on Theorem 13.136, the first equation is immediate. The second is Lemma 13.22 with $L = 2$. □

Remark 13.138 (Product-of-pairs equations for nearest-neighbor parent-term commutation). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt The Appendix B structural form supplies

$$A^i = X \Lambda U^i X^{-1}.$$

The change of basis does not alter periodic MPS coefficients, so the corresponding two-site amplitude is

$$\phi_2(a, b) = V^{(2)}(A)(a, b) = V^{(2)}(\Lambda U)(a, b).$$

The product-of-pairs input is the even-chain factorization

$$V^{(2n)}(A)(\sigma) = \prod_{p=0}^{n-1} \phi_2(\sigma_{2p}, \sigma_{2p+1}).$$

The two-site parent projectors must also be identified with idempotents p_i satisfying

$$h_i(A, 2) = p_i, \qquad p_i^2 = p_i, \qquad p_i p_j = p_j p_i.$$

These equations give

$$h_i(A, 2)h_j(A, 2) = h_j(A, 2)h_i(A, 2),$$

which is the nearest-neighbor commuting conclusion for every finite chain. The even-chain factorization and the two-site projector identification are the inputs supplied by Appendix B of [CPGSV16]; given them, the local terms commute. Together with Lemma 13.22, they also give the zero-energy equation for $V^{(N)}(A)$, still without asserting the source ground-space spanning condition.

Theorem 13.139 (All-chain nearest-neighbor commuting parent-Hamiltonian ground spaces imply a renormalization fixed point). Lean Lean Not ready *Let B be normal and in left-canonical form. Suppose that the family A_j is a BNT for B, and that the nearest-neighbor parent Hamiltonian of B has the all-chain ground-space property of Definition 13.133: for every $N > 2$ the length-two translated parent interactions commute,*

$$h_i(B, 2)h_j(B, 2) = h_j(B, 2)h_i(B, 2),$$

the vector $V^{(N)}(B)$ has zero energy, and

$$\ker H_2^{(N)}(B) = \operatorname{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

Then B is a renormalization fixed point. The present formalization still supplies this reverse implication through the axiom tagged above, which stands for Beigi's one-dimensional nearest-neighbor commuting-Hamiltonian ground-space theorem and the identification of the resulting locally orthogonal states with the BNT of B.

13.10 Decorrelation and the idempotent-product parent-commuting condition

This section develops the idempotent-product formulation used for the decorrelation implication from Appendix D. It is not the full source condition. In Appendix D.2 of [CPGSV16], the parent commuting Hamiltonian condition is stated for local projectors Q_{AX} and Q_{XB} with

$$[Q_{AX}, Q_{XB}] = 0, \qquad K_{AXB} = (K_{AX} \otimes H_B) \cap (H_A \otimes K_{XB}).$$

Writing $P_{AX} = 1 - Q_{AX}$ and $P_{XB} = 1 - Q_{XB}$ gives, for the idempotent P_K used below, the identity $P_{AX}P_{XB} = P_K$. The definition below records this idempotent-product consequence, not the tensor-product locality, orthogonal-projector, or ground-space-intersection hypotheses of Appendix D.2.

Definition 13.140 (Decorrelation). Lean ✓ Stmt Given an endomorphism P_K and sets of observables $\mathcal{O}_A$, $\mathcal{O}_B$, regions A and B are *decorrelated* w.r.t. K when $P_K \circ O_A \circ (1 - P_K) \circ O_B \circ P_K = 0$ for all $O_A \in \mathcal{O}_A$, $O_B \in \mathcal{O}_B$.

Definition 13.141 (Idempotent-product parent-commuting structure). Lean ✓ Stmt The structure used below records only the idempotent-product consequence of the parent commuting Hamiltonian condition of [CPGSV16, Appendix D.2]. It consists of commuting idempotent endomorphisms P_{AX} and P_{XB} such that, for the idempotent P_K associated to K,

$$P_{AX} \circ P_{XB} = P_K.$$

In the source, this is the equation obtained from local projectors Q_{AX} and Q_{XB} satisfying $[Q_{AX}, Q_{XB}] = 0$ and $K_{AXB} = (K_{AX} \otimes H_B) \cap (H_A \otimes K_{XB})$.

Theorem 13.142 (Idempotence gives tautological idempotent-product data). *Lean* *✓ Stmt* *If*

$$P_K \circ P_K = P_K,$$

then P_K has idempotent-product data.

Proof. ✓ Proof Take $P_{AX} = P_{XB} = P_K$. Their commutativity is immediate, and the product identity is exactly the assumed idempotence of P_K. □

Lemma 13.143 (Cancellation for commuting idempotents). *Lean* *✓ Stmt* *If P and Q are commuting idempotents, then*

$$Q \circ (1 - P \circ Q) \circ P = 0.$$

Proof. ✓ Proof The cancellation is the computation

$$Q(1 - PQ)P = QP - QPQP = QP - QPPQ = QP - QP = 0,$$

using commutativity and idempotence. □

Theorem 13.144 (Idempotence of the product projector). *Lean* *✓ Stmt* *If $P_K = P_{AX} \circ P_{XB}$ is idempotent-product data, then*

$$P_K \circ P_K = P_K.$$

Proof. ✓ Proof Since P_{AX} and P_{XB} are idempotent and commute, their product $P_{AX} \circ P_{XB}$ is idempotent. Substituting $P_K = P_{AX} \circ P_{XB}$ gives the claim. □

Theorem 13.145 (Absorption by the left projector). *Lean* *✓ Stmt* *If $P_K = P_{AX} \circ P_{XB}$ is idempotent-product data, then*

$$P_{AX} \circ P_K = P_K.$$

Proof. ✓ Proof Using $P_K = P_{AX} \circ P_{XB}$ and the idempotence of P_{AX},

$$P_{AX} \circ P_K = P_{AX} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

□

Theorem 13.146 (Absorption by the right projector). *Lean* *✓ Stmt* *If $P_K = P_{AX} \circ P_{XB}$ is idempotent-product data, then*

$$P_K \circ P_{XB} = P_K.$$

Proof. ✓ Proof Using $P_K = P_{AX} \circ P_{XB}$ and the idempotence of P_{XB},

$$P_K \circ P_{XB} = P_{AX} \circ P_{XB} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

□

Theorem 13.147 (Cross-absorption by the right projector). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is idempotent-product data, then*

$$P_{XB} \circ P_K = P_K.$$

Proof. ✓ Proof By commutativity, $P_{XB} \circ P_K = P_{XB} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} \circ P_{XB}$, and idempotence of P_{XB} yields $P_{XB} \circ P_K = P_K$. □

Theorem 13.148 (Cross-absorption by the left projector). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is idempotent-product data, then*

$$P_K \circ P_{AX} = P_K.$$

Proof. ✓ Proof By commutativity, $P_K \circ P_{AX} = P_{AX} \circ P_{XB} \circ P_{AX} = P_{AX} \circ P_{AX} \circ P_{XB}$, and idempotence of P_{AX} yields $P_K \circ P_{AX} = P_K$. □

Theorem 13.149 (Reverse product identity). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is idempotent-product data, then*

$$P_{XB} \circ P_{AX} = P_K.$$

Proof. ✓ Proof This is immediate from commutativity of P_{AX} and P_{XB} together with the identity $P_{AX} \circ P_{XB} = P_K$. □

Theorem 13.150 (Commuting complementary projectors). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is idempotent-product data and*

$$Q_{AX} = 1 - P_{AX}, \qquad Q_{XB} = 1 - P_{XB},$$

then

$$Q_{AX} \circ Q_{XB} = Q_{XB} \circ Q_{AX}.$$

Proof. ✓ Proof Expanding both products and using $P_{AX} \circ P_{XB} = P_{XB} \circ P_{AX}$ shows that the complementary projectors commute. □

Theorem 13.151 (Idempotent-product data imply decorrelation). Lean ✓ Stmt *If P_{AX} and P_{XB} are idempotent endomorphisms satisfying $P_{AX} \circ P_{XB} = P_K$ and $P_{AX} \circ P_{XB} = P_{XB} \circ P_{AX}$, and A-observables commute with P_{XB} while B-observables commute with P_{AX}, then regions A and B are decorrelated w.r.t. K. This is the idempotent-product form of the backward direction of [CPGSV16, Appendix D, Section D.2, Proposition D.3].*

Proof. ✓ Proof The key identity is $P_{XB} \circ (1 - P_{AX} \circ P_{XB}) \circ P_{AX} = 0$. Substituting $P_K = P_{AX} \circ P_{XB}$ and using $O_A \circ P_{XB} = P_{XB} \circ O_A$ and $O_B \circ P_{AX} = P_{AX} \circ O_B$ gives

$$P_K \circ O_A \circ (1 - P_K) \circ O_B \circ P_K = P_{AX} \circ O_A \circ [P_{XB} \circ (1 - P_{AX} \circ P_{XB}) \circ P_{AX}] \circ O_B \circ P_{XB} = 0.$$

□

Theorem 13.152 (Decorrelating idempotent-product data). Lean ✓ Stmt *Corollary of Theorem 13.151: if P_K has idempotent-product data, the A-observables commute with P_{XB}, and the B-observables commute with P_{AX}, then regions A and B are decorrelated w.r.t. K.*

Proof. ✓ Proof Apply Theorem 13.151 to the witnessing endomorphisms P_{AX} and P_{XB} supplied by the idempotent-product data. □

Lemma 13.153 (Frustration-free identity). Lean ✓ Stmt *For any endomorphisms P and Q,*

$$(1 - P) + (1 - Q) - (1 - P) \circ (1 - Q) = 1 - P \circ Q.$$

Proof. ✓ Proof Expanding the left-hand side and cancelling gives the result. □

Theorem 13.154 (Idempotent-product Hamiltonian identity). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is idempotent-product data, then*

$$Q_{AX} + Q_{XB} - Q_{AX} \circ Q_{XB} = 1 - P_K,$$

where $Q_{AX} = 1 - P_{AX}$ and $Q_{XB} = 1 - P_{XB}$ are the complementary projectors. This is the algebraic identity obtained from the complementary projectors in [CPGSV16, Appendix D, Section D.2]; it is not the full source parent commuting Hamiltonian condition.

Proof. ✓ Proof Applying Lemma 13.153 to P_{AX} and P_{XB}, and substituting $P_{AX} \circ P_{XB} = P_K$. □

Theorem 13.155 (Ground-space membership). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is idempotent-product data, then*

$$P_K v = v \quad \Longleftrightarrow \quad P_{AX} v = v \text{ and } P_{XB} v = v.$$

This is the algebraic consequence corresponding to equation (D.2) from [CPGSV16]: $K_{AXB} = (K_{AX} \otimes H_B) \cap (H_A \otimes K_{XB})$. The source equation itself also carries the tensor-product local-subspace hypotheses.

Proof. ✓ Proof ($\Rightarrow$) If $P_K v = v$, then $P_{AX}(P_K v) = (P_{AX} \circ P_K)v = P_K v = v$ by left absorption, and similarly for P_{XB}.

($\Leftarrow$) If $P_{AX} v = v$ and $P_{XB} v = v$, then $P_K v = (P_{AX} \circ P_{XB})v = P_{AX}(P_{XB} v) = P_{AX} v = v$. □

13.11 Spectral gap

A frustration-free Hamiltonian is *gapped* if the smallest nonzero eigenvalue, equivalently the gap above the ground space, is bounded away from zero uniformly in the chain length N. For MPS parent Hamiltonians, the martingale method of Fannes–Nachtergaele–Werner and Nachtergaele [FNW92, Nac96], as recalled by Cirac–Perez-Garcia–Schuch–Verstraete [CPGSV21, Section IV.C] and in the later formulation of Kastoryano and Lucia [KL18], reduces a uniform gap to an anticommutator estimate for overlapping local terms. Principal angles between the corresponding local ground spaces give the geometric interpretation of this estimate. The results below state the anticommutator condition for the cyclic-window local terms and also record cyclic norm-compression estimates as sufficient stronger hypotheses.

Definition 13.156 (Parent ground space under the canonical ℓ^2 identification). Lean ✓ Stmt Denote by $\mathcal{G}^{\mathrm{ES}}_{N,L}(A)$ the parent ground space after identifying the coefficient space with the Hilbert space $\ell^2(\{0, \dots, d-1\}^N)$.

Definition 13.157 (Parent Hamiltonian under the canonical ℓ^2 identification). Lean ✓ Stmt Denote by $H^{\mathrm{ES}}_{N,L}(A)$ the conjugate of $H_N(A, L)$ by the canonical identification between the coefficient space and $\ell^2(\{0, \dots, d-1\}^N)$.

Theorem 13.158 (Kernel under the canonical ℓ^2 identification). Lean ✓ Stmt *Under the canonical ℓ^2 identification of the N-site state space, the parent ground space is exactly the kernel of the corresponding parent Hamiltonian.*

Proof. ✓ Proof This is an immediate unraveling of the definitions: both constructions are obtained from the same parent Hamiltonian by conjugating with the canonical identification between the site space and the ℓ^2 space, so the ground space is precisely the kernel after this identification. □

Lemma 13.159 (Membership criterion under the canonical ℓ^2 identification). Lean ✓ Stmt *A vector belongs to $\mathcal{G}^{\mathrm{ES}}_{N,L}(A)$ exactly when the parent Hamiltonian annihilates it.*

Remark 13.160 (Local projectors under the canonical ℓ^2 identification). Lean Lean Lean Lean ✓ Stmt Let $P^{\mathrm{ES}}_L(A)$ denote the orthogonal projector onto $(G^{\mathrm{ES}}_L(A))^\perp$ on the canonical ℓ^2 L-site Hilbert space. For a cyclic window starting at i and an outside configuration τ, the map $R_{i,\tau}$ restricts an N-site vector to that length-L cyclic window with the outside configuration fixed by τ. The local term $h_{i,\mathrm{ES}}$ is the corresponding N-site local projector, and $R^\dagger_{i,\tau}P^{\mathrm{ES}}_L(A)R_{i,\tau}$ is the basic positive summand in its cyclic-averaging formula.

Lemma 13.161 (Positivity of the Hilbert-space local summands). Lean Lean Lean ✓ Stmt *The Hilbert-space local projector $P^{\mathrm{ES}}_L(A)$ is positive. Consequently, every summand $R^\dagger_{i,\tau}P^{\mathrm{ES}}_L(A)R_{i,\tau}$ is positive, and so is the finite sum over τ.*

Proof. ✓ Proof Orthogonal projectors are positive. Conjugation by $R_{i,\tau}$ preserves positivity, and finite sums of positive operators remain positive. □

Lemma 13.162 (Cyclic averaging for the Hilbert-space local terms). Lean Lean ✓ Stmt *The Hilbert-space parent Hamiltonian is the sum of the Hilbert-space local terms $H_{N,\mathrm{ES}} = \sum_i h_{i,\mathrm{ES}}$, and each local term admits the exact cyclic-averaging formula*

$$h_{i,\mathrm{ES}} = d^{-L}\sum_\tau R^\dagger_{i,\tau}P^{\mathrm{ES}}_L(A)R_{i,\tau}. \tag{13.5}$$

Proof. ✓ Proof Unfold the Hilbert-space local term and compare it configuration-by-configuration with the cyclic restrictions. The d^{-L} factor compensates for the d^L choices of the window coordinates in the ambient configuration parameter τ. □

Lemma 13.163 (Positivity of the Hilbert-space Hamiltonian). Lean Lean ✓ Stmt *Each Hilbert-space local term $h_{i,\mathrm{ES}}$ is positive, hence the full Hilbert-space parent Hamiltonian $H_{N,\mathrm{ES}}$ is positive.*

Proof. ✓ Proof Apply the averaging formula (13.5): $h_{i,\mathrm{ES}}$ is a non-negative scalar multiple of a finite sum of positive conjugates of $P^{\mathrm{ES}}_L(A)$, hence is itself positive. Summing over i yields positivity of $H_{N,\mathrm{ES}}$. □

Lemma 13.164 (Hilbert-space local terms are symmetric projections). Lean Lean ✓ Stmt *The Hilbert-space local projector $P^{\mathrm{ES}}_L(A)$ is a symmetric projection, and every Hilbert-space local term $h_{i,\mathrm{ES}}$ on the N-site chain is a symmetric projection.*

Proof. ✓ Proof The local operator $P^{\mathrm{ES}}_L(A)$ is an orthogonal projection onto $(G^{\mathrm{ES}}_L(A))^\perp$. For $L \le N$, restricting $h_{i,\mathrm{ES}}v$ to a cyclic window gives $P^{\mathrm{ES}}_L(A)$ applied to the corresponding restriction of v; applying $h_{i,\mathrm{ES}}$ a second time therefore applies $(P^{\mathrm{ES}}_L(A))^2 = P^{\mathrm{ES}}_L(A)$ on that window. The positivity result above supplies symmetry, and the case $L > N$ is the zero projection by definition. □

Lemma 13.165 (Kernel of the Hilbert-space parent interaction). Lean ✓ Stmt *The Hilbert-space local interaction $P^{\mathrm{ES}}_L(A)$ annihilates exactly the Hilbert-space local ground space:*

$$P^{\mathrm{ES}}_L(A)v = 0 \iff v \in G^{\mathrm{ES}}_L(A).$$

Proof. ✓ Proof Since $P^{\mathrm{ES}}_L(A)$ is the orthogonal projector onto $(G^{\mathrm{ES}}_L(A))^\perp$, its kernel is the original subspace $G^{\mathrm{ES}}_L(A)$. □

Lemma 13.166 (Local-term kernel via cyclic restrictions). Lean Lean ✓ Stmt *For $L \le N$, a Hilbert-space local term $h_{i,\mathrm{ES}}$ annihilates a vector v iff every restriction of v to the cyclic length-L window starting at i, with the outside configuration fixed, lies in $G_L^{\mathrm{ES}}(A)$.*

Proof. ✓ Proof Restricting $h_{i,\mathrm{ES}}v$ to the same cyclic window gives $P_L^{\mathrm{ES}}(A)$ applied to the corresponding restriction of v. Lemma 13.165 identifies the vanishing of this local projector with membership in $G_L^{\mathrm{ES}}(A)$. □

Theorem 13.167 (Intersection property for two adjacent local kernels). Lean Lean Lean ✓ Stmt *If A is injective and $L \ge 2$, then on an $(L+1)$-site chain the kernels of the two adjacent Hilbert-space L-site local terms have common kernel equal to the Hilbert-space $(L+1)$-site MPS ground space:*

$$h_{0,\mathrm{ES}}v = 0 \quad \text{and} \quad h_{1,\mathrm{ES}}v = 0 \quad \iff \quad v \in G_{L+1}^{\mathrm{ES}}(A).$$

Proof. ✓ Proof Lemma 13.166 translates the two local kernel assumptions into the left and right L-site ground conditions. The open-chain intersection property, Theorem 13.34, then grows back the shared $(L-1)$-site overlap to the $(L+1)$-site ground space. The converse uses the forward restriction lemmas for ground-space vectors and the same kernel–restriction characterization. □

Lemma 13.168 (Non-overlap positivity for Hilbert-space local terms). Lean Lean Lean Lean Lean ✓ Stmt *Let $L \le N$ and let $h_{i,\mathrm{ES}}, h_{j,\mathrm{ES}}$ be Hilbert-space local terms whose cyclic L-windows are site-disjoint: no site of the periodic chain has cyclic offset $< L$ from both i and j. Then the terms commute pointwise: for every vector v,*

$$h_{i,\mathrm{ES}}(h_{j,\mathrm{ES}}v) = h_{j,\mathrm{ES}}(h_{i,\mathrm{ES}}v),$$

and their ordered cross term is non-negative:

$$0 \le \mathrm{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle.$$

Proof. ✓ Proof In coordinates, $h_{i,\mathrm{ES}}$ applies $P_L^{\mathrm{ES}}(A)$ only to the i-window variables and leaves the complementary variables fixed. Site disjointness gives two replacement identities: replacing the i-window does not change the extracted j-window, and the i- and j-window replacements commute. Therefore the two embedded local operators commute. The preceding projection-geometry identity for commuting symmetric projections gives the non-negative ordered cross term. □

Lemma 13.169 (Quadratic form to norm bound). Lean ✓ Stmt *Let H be a positive semidefinite self-adjoint operator on a finite-dimensional complex Hilbert space, and suppose that, for all v,*

$$\gamma\,\langle v, Hv\rangle \le \langle Hv, Hv\rangle$$

i.e. $H^2 \ge \gamma H$ as a quadratic form, for some $\gamma > 0$. Then for every v in the orthogonal complement of $\ker H$,

$$\gamma\,\|v\| \le \|Hv\|.$$

In particular, every positive eigenvalue of H is at least γ. The positive-semidefiniteness hypothesis is essential: without it, $H = -\mathrm{Id}$ with $\gamma = 2$ satisfies the quadratic-form inequality vacuously but violates the norm bound.

Proof. ✓ Proof By the finite-dimensional spectral theorem, H diagonalises in an orthonormal eigenbasis with real eigenvalues λ_k. For a unit eigenvector ψ_k with eigenvalue λ_k the hypothesis gives $\gamma\,\lambda_k \le \lambda_k^2$, whence $\lambda_k \le 0$ or $\lambda_k \ge \gamma$. Since H is positive semidefinite (its quadratic form is ≥ 0), the negative case is excluded and every nonzero eigenvalue satisfies $\lambda_k \ge \gamma$. Writing $v = \sum_k c_k\psi_k$ and projecting onto $(\ker H)^\perp$ keeps only terms with $\lambda_k \ge \gamma$, so $\|Hv\|^2 = \sum_k |c_k|^2\lambda_k^2 \ge \gamma^2 \sum_k |c_k|^2 = \gamma^2\|v\|^2$. □

13.11.1 Martingale condition for a finite family of projections

Lemma 13.170 (Parent-Hamiltonian quadratic-form reduction). Lean Lean ✓ Stmt *Let $H^{\mathrm{ES}}_{N,L}(A)$ be the Hilbert-space representative of the parent Hamiltonian. If a constant $\gamma > 0$ satisfies the uniform quadratic-form estimate*

$$\gamma \operatorname{Re}\langle H^{\mathrm{ES}}_{N,L}(A)v, v\rangle \leq \operatorname{Re}\langle H^{\mathrm{ES}}_{N,L}(A)v, H^{\mathrm{ES}}_{N,L}(A)v\rangle \tag{13.6}$$

for every $N \geq 2L$ and every vector v, then

$$\gamma \|v\| \leq \|H^{\mathrm{ES}}_{N,L}(A)v\|$$

for every $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$. In particular, with $L > 1$ and $\gamma = 1/(4L)$, the explicit gap-bound conclusion follows once this same quadratic-form estimate (13.6) is supplied.

Proof. ✓ Proof The Hilbert-space Hamiltonian is positive by Lemma 13.163, and its Hilbert-space ground space is its kernel by Theorem 13.158. Therefore the abstract spectral-theorem conversion of Lemma 13.169 applies directly. The explicit constant is positive because $L > 1$ implies $4L > 0$. □

Remark 13.171 (Projection-geometry identities for row-sum arguments). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt The projection-geometry reduction uses the following elementary Hilbert-space identities: the diagonal identity $\operatorname{Re}\langle Pv, Pv\rangle = \operatorname{Re}\langle Pv, v\rangle$ for symmetric projections; nonnegativity of this quadratic form; nonnegativity of $\operatorname{Re}\langle Pv, Qv\rangle$ for commuting symmetric projections P, Q; the Cauchy–Schwarz conversion from $\|PQv\| \leq c\|Pv\|$ to $\operatorname{Re}\langle Pv, Qv\rangle \geq -c \operatorname{Re}\langle Pv, v\rangle$; finite-sum expansions of $\operatorname{Re}\langle(\sum_i P_i)v, v\rangle$ and $\operatorname{Re}\langle(\sum_i P_i)v, (\sum_i P_i)v\rangle$; the diagonal/off-diagonal split of the ordered double sum; the aggregate off-diagonal estimate obtained from row sums $\sum_{j\neq i} c_{ij} \leq 1$; and the corresponding aggregate estimate for the anticommutator martingale condition.

Lemma 13.172 (Ordered projection row-sum quadratic-form reduction). Lean ✓ Stmt *Let P_i be a finite family of symmetric projections and let $H = \sum_i P_i$. If the ordered off-diagonal forms obey*

$$\operatorname{Re}\langle P_i v, P_j v\rangle \geq -(1-\gamma)c_{ij} \operatorname{Re}\langle P_i v, v\rangle \qquad (i \neq j),$$

with row sums $\sum_{j\neq i} c_{ij} \leq 1$ and $\gamma \leq 1$, then $H^2 \geq \gamma H$ as a quadratic form.

Proof. ✓ Proof Expand $\operatorname{Re}\langle Hv, Hv\rangle$ as an ordered double sum. Symmetric idempotence identifies each diagonal term with $\operatorname{Re}\langle P_i v, v\rangle$, while the row-summable ordered cross-term bounds control the off-diagonal contribution by $-(1-\gamma)\sum_i \operatorname{Re}\langle P_i v, v\rangle$. □

Lemma 13.173 (Projection anticommutator row-sum reduction). Lean ✓ Stmt *Let P_i be a finite family of symmetric projections and let $H = \sum_i P_i$. Suppose the anticommutator forms obey*

$$\operatorname{Re}\langle P_i v, P_j v\rangle + \operatorname{Re}\langle P_j v, P_i v\rangle \geq -(1-\gamma)c_{ij}(\operatorname{Re}\langle P_i v, v\rangle + \operatorname{Re}\langle P_j v, v\rangle) \qquad (i \neq j),$$

with row sums $\sum_{j\neq i} c_{ij} \leq 1$, column sums $\sum_{i\neq j} c_{ij} \leq 1$, and $\gamma \leq 1$. Then $H^2 \geq \gamma H$ as a quadratic form.

Proof. ✓ Proof Expand $\operatorname{Re}\langle Hv, Hv\rangle$ as an ordered double sum. The anticommutator bound controls the sum of each ordered term with its reversed term. The row and column estimates bound the coefficient-weighted diagonal contribution by twice $\sum_i \operatorname{Re}\langle P_i v, v\rangle$; after dividing by two, the same quadratic-form estimate follows. □

Lemma 13.174 (Finite-overlap projection anticommutator reduction). Lean ✓ Stmt *Let P_i be a finite family of symmetric projections and let $\gamma \le 1$. Suppose an interaction predicate $i \sim j$ has at most m off-diagonal neighbours in each row and in each column, where $m > 0$. If noninteracting pairs have non-negative anticommutator form and interacting pairs satisfy*

$$\mathrm{Re}\langle P_i v, P_j v\rangle + \mathrm{Re}\langle P_j v, P_i v\rangle \ge -(1-\gamma)\frac{1}{m}(\mathrm{Re}\langle P_i v, v\rangle + \mathrm{Re}\langle P_j v, v\rangle),$$

then $H = \sum_i P_i$ satisfies $H^2 \ge \gamma H$ as a quadratic form.

Proof. ✓ Proof Choose $c_{ij} = 1/m$ on interacting pairs and $c_{ij} = 0$ otherwise. The row and column cardinality bounds give the corresponding coefficient sums, and the noninteracting anticommutator non-negativity supplies the zero-coefficient estimates. Apply Lemma 13.173. □

Lemma 13.175 (Finite-overlap projection row-sum reduction). Lean ✓ Stmt *Let P_i be a finite family of symmetric projections and let $\gamma \le 1$. Suppose an interaction predicate $i \sim j$ has at most m off-diagonal neighbours in each row for some positive integer m. If noninteracting ordered cross terms are non-negative and, on each interacting pair,*

$$\mathrm{Re}\langle P_i v, P_j v\rangle \ge -(1-\gamma)\frac{1}{m}\,\mathrm{Re}\langle P_i v, v\rangle,$$

then $H = \sum_i P_i$ satisfies $H^2 \ge \gamma H$ as a quadratic form.

Proof. ✓ Proof Choose $c_{ij} = 1/m$ on interacting pairs and $c_{ij} = 0$ otherwise. The row cardinality bound gives $\sum_{j\ne i} c_{ij} \le 1$; noninteracting nonnegativity supplies the zero-coefficient cross-term estimates. Apply Lemma 13.172. □

Lemma 13.176 (Finite-overlap projection norm-compression reduction). Lean Lean Lean ✓ Stmt *Let P_i be a finite family of symmetric projections and let $\gamma \le 1$. Suppose an interaction predicate has at most $m > 0$ off-diagonal neighbours in each row, noninteracting ordered cross terms are non-negative, and interacting pairs satisfy the norm-compression estimate*

$$\|P_i P_j v\| \le (1-\gamma)\frac{1}{m}\|P_i v\|. \tag{13.7}$$

Then $H = \sum_i P_i$ satisfies $H^2 \ge \gamma H$ as a quadratic form. The same conclusion holds from a separate compression coefficient η whenever $\eta \le (1-\gamma)/m$. In particular, if $0 \le \eta$ and $\eta m < 1$, then

$$H^2 \ge (1-\eta m)H$$

and $1-\eta m > 0$.

Proof. ✓ Proof Cauchy–Schwarz and symmetric idempotence turn the norm-compression estimate into $\mathrm{Re}\langle P_i v, P_j v\rangle \ge -(1-\gamma)m^{-1}\,\mathrm{Re}\langle P_i v, v\rangle$. If a separate coefficient η is used, the assumption $\eta \le (1-\gamma)/m$ first weakens that estimate to (13.7). The finite-overlap row-sum reduction then applies. For the strict form take $\gamma = 1-\eta m$. □

Lemma 13.177 (Quadratic-form estimate from the martingale condition). Lean ✓ Stmt *For a fixed chain length N, suppose the Hilbert-space local terms $h_{i,\mathrm{ES}}$ satisfy the ordered row-sum estimate*

$$\mathrm{Re}\langle h_{i,\mathrm{ES}} v, h_{j,\mathrm{ES}} v\rangle \ge -(1-\gamma)c_{ij}\,\mathrm{Re}\langle h_{i,\mathrm{ES}} v, v\rangle \qquad (i \ne j), \tag{13.8}$$

with $\sum_{j\ne i} c_{ij} \le 1$ and $\gamma \le 1$. Then the Hilbert-space parent Hamiltonian satisfies

$$\gamma\,\mathrm{Re}\langle H^{\mathrm{ES}}_{N,L} v, v\rangle \le \mathrm{Re}\langle H^{\mathrm{ES}}_{N,L} v, H^{\mathrm{ES}}_{N,L} v\rangle$$

for every vector v.

Proof. ✓ Proof Lemma 13.164 supplies the symmetric-projection hypothesis for the family $P_i = h_{i,\mathrm{ES}}$. Apply Lemma 13.172 to this family and use $H^{\mathrm{ES}}_{N,L} = \sum_i h_{i,\mathrm{ES}}$. □

Lemma 13.178 (Parent-Hamiltonian gap from ordered local row bounds)**.** Lean ✓ Stmt *Assume uniformly for every $N \geq 2L$ that the Hilbert-space local terms satisfy the ordered row-sum estimate (13.8) with $\gamma = 1/(4L)$. If $L > 1$, then $1/(4L) > 0$ and, for every $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$,*

$$\frac{1}{4L}\|v\| \leq \|H^{\mathrm{ES}}_{N,L}(A)v\|.$$

Proof. ✓ Proof The fixed-chain row-sum reduction supplies the quadratic-form estimate with $\gamma = 1/(4L)$ for every admissible N. The positivity and kernel identification of the Hilbert-space Hamiltonian then allow Lemma 13.170 to convert this quadratic-form estimate into the norm lower bound on the orthogonal complement of the ground space. □

Lemma 13.179 (Parent-Hamiltonian gap from the anticommutator martingale condition)**.** Lean Lean ✓ Stmt *Assume uniformly for every $N \geq 2L$ that the Hilbert-space local terms satisfy the anticommutator martingale estimate*

$$\mathrm{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle + \mathrm{Re}\langle h_{j,\mathrm{ES}}v, h_{i,\mathrm{ES}}v\rangle \geq -c_{ij}\Big(1 - \frac{1}{4L}\Big)(\mathrm{Re}\langle h_{i,\mathrm{ES}}v, v\rangle + \mathrm{Re}\langle h_{j,\mathrm{ES}}v, v\rangle)$$

for $i \neq j$, with row and column coefficient sums bounded by one. If $L > 1$, then $1/(4L) > 0$ and, for every $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$,

$$\frac{1}{4L}\|v\| \leq \|H^{\mathrm{ES}}_{N,L}(A)v\|.$$

Proof. ✓ Proof The fixed-chain anticommutator row-sum reduction supplies the quadratic-form estimate with $\gamma = 1/(4L)$ for every admissible N. The positivity and kernel identification of the Hilbert-space Hamiltonian then convert this quadratic-form estimate into the norm lower bound on the orthogonal complement of the ground space. □

Lemma 13.180 (Parent-Hamiltonian finite-overlap anticommutator reduction)**.** Lean Lean ✓ Stmt *Fix $L > 1$ and set $m = 2(L-1)$. Suppose uniformly for every $N \geq 2L$ that the overlap relation on local windows has at most m off-diagonal neighbours in each row and in each column. If non-overlapping local terms have non-negative anticommutator form, and overlapping local terms satisfy*

$$\mathrm{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle + \mathrm{Re}\langle h_{j,\mathrm{ES}}v, h_{i,\mathrm{ES}}v\rangle \geq -\Big(1 - \frac{1}{4L}\Big)\frac{1}{m}(\mathrm{Re}\langle h_{i,\mathrm{ES}}v, v\rangle + \mathrm{Re}\langle h_{j,\mathrm{ES}}v, v\rangle),$$

then the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$.

Proof. ✓ Proof Apply the finite-overlap anticommutator projection reduction to the Hilbert-space local terms. The resulting quadratic-form estimate has $\gamma = 1/(4L)$, and Lemma 13.170 converts it to the norm lower bound. □

Lemma 13.181 (Parent-Hamiltonian finite-overlap martingale quadratic reduction)**.** Lean ✓ Stmt *For a fixed chain length N, let m be a positive integer and let $\gamma \leq 1$. Suppose an overlap predicate on local windows has at most m overlapping off-diagonal windows in each row. If non-overlapping local terms have non-negative ordered cross terms, and overlapping local terms satisfy the finite-overlap principal-angle estimate*

$$\mathrm{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle \geq -(1-\gamma)\frac{1}{m}\,\mathrm{Re}\langle h_{i,\mathrm{ES}}v, v\rangle,$$

and the Hilbert-space local terms are symmetric projections, then the Hilbert-space Hamiltonian satisfies ${H^{\mathrm{ES}}_{N,L}}^2 \geq \gamma H^{\mathrm{ES}}_{N,L}$ *as a quadratic form.*

Proof. ✓ Proof Apply the finite-overlap projection reduction with $P_i = h_{i,\mathrm{ES}}$. □

Lemma 13.182 (Finite-overlap norm-compression reduction). Lean Lean ✓ Stmt *For a fixed chain length* N*, let* $m > 0$ *and* $\gamma \leq 1$*. Suppose an overlap predicate on local windows has at most* m *overlapping off-diagonal windows in each row, non-overlapping local terms have non-negative ordered cross terms, and overlapping local terms satisfy*

$$\|h_{i,\mathrm{ES}}(h_{j,\mathrm{ES}}v)\| \leq (1-\gamma)\frac{1}{m}\|h_{i,\mathrm{ES}}v\|.$$

Then ${H^{\mathrm{ES}}_{N,L}}^2 \geq \gamma H^{\mathrm{ES}}_{N,L}$ *as a quadratic form. The same conclusion holds from a separate coefficient* η *satisfying* $\eta \leq (1-\gamma)/m$*.*

Proof. ✓ Proof Apply the finite-overlap norm-compression projection reduction to the family $P_i = h_{i,\mathrm{ES}}$. If a separate coefficient η is used, first weaken the compression estimate using $\eta \leq (1-\gamma)/m$. □

13.11.2 Cyclic-window overlap estimates

Definition 13.183 (Cyclic-window support). Lean ✓ Stmt For a length-L cyclic window on $\{0, \dots, N-1\}$ starting at i, its support is

$$S_i = \{i, i+1, \dots, i+L-1\} \pmod N,$$

with repeated visits counted once.

Lemma 13.184 (Cyclic-window support offsets). Lean ✓ Stmt *If* $L \leq N$*, then a site* k *belongs to the cyclic support* S_i *exactly when its cyclic offset from* i *is below* L*:*

$$k \in S_i \iff (k-i) \bmod N < L.$$

Proof. ✓ Proof Write $k = i + r \pmod N$ for a representative offset r. Membership in S_i gives an offset $r < L$, and conversely an offset below L gives the corresponding point of the support. □

Definition 13.185 (Cyclic-window overlap). Lean ✓ Stmt Two length-L cyclic windows overlap, written $i \sim_L j$, when their supports meet: $S_i \cap S_j \neq \emptyset$.

Lemma 13.186 (Cyclic-window overlap symmetry). Lean ✓ Stmt *For length-*L *cyclic windows,* $i \sim_L j$ *if and only if* $j \sim_L i$*.*

Proof. ✓ Proof Both assertions say that the two cyclic supports have a common site. □

Lemma 13.187 (Concrete non-overlap positivity for cyclic supports). Lean Lean ✓ Stmt *If* $L \leq N$ *and the cyclic supports* S_i *and* S_j *do not meet, then the Hilbert-space local terms have non-negative ordered cross term:*

$$0 \leq \mathrm{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle.$$

Proof. ✓ Proof The support-offset characterization converts failure of $S_i \cap S_j \neq \emptyset$ into site-disjointness of the two cyclic windows. The disjoint-window positivity lemma then applies. □

Lemma 13.188 (Cyclic-window overlap row cardinality). Lean ✓ Stmt *If $2L \le N$ and $L > 1$, then for every $i \in \{0, \dots, N-1\}$,*

$$\#\{j \ne i : i \sim_L j\} \le 2(L-1).$$

Proof. ✓ Proof If S_i and S_j meet, choose offsets $a, b < L$ with $i + a \equiv j + b \pmod N$. Since $2L \le N$, the offset from i to j is either the clockwise distance $a - b \in \{1, \dots, L-1\}$ or the counterclockwise distance $b-a \in \{1, \dots, L-1\}$; the zero distance is excluded by $j \ne i$. Thus every overlapping off-diagonal start lies among the $L-1$ clockwise starts or the $L-1$ counterclockwise starts. □

Lemma 13.189 (Parent-Hamiltonian gap from a finite-overlap martingale estimate). Lean ✓ Stmt *Fix $L > 1$ and set $m = 2(L-1)$. Suppose uniformly for every $N \ge 2L$ that the hypotheses of the fixed-chain finite-overlap martingale reduction hold with $\gamma = 1/(4L)$ and this value of m. Then $1/(4L) > 0$, and the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^\perp$.*

Proof. ✓ Proof The fixed-chain finite-overlap reduction gives the required quadratic-form estimate for every admissible N. Lemma 13.170 converts that estimate to the norm lower bound. □

Lemma 13.190 (Parent-Hamiltonian gap from a cyclic-window martingale estimate). Lean ✓ Stmt *Fix $L > 1$ and let $i \sim_L j$ mean that the cyclic supports of the length-L windows at i and j meet. Suppose uniformly for every $N \ge 2L$ that overlapping off-diagonal terms satisfy*

$$\mathrm{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle \ge -(1 - \tfrac{1}{4L})\frac{1}{2(L-1)}\,\mathrm{Re}\langle h_{i,\mathrm{ES}}v, v\rangle. \tag{13.9}$$

Then $1/(4L) > 0$, and the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^\perp$.

Proof. ✓ Proof Lemma 13.188 gives the row-cardinality bound $m = 2(L-1)$ for the overlap relation among length-L local windows, and Lemma 13.164 supplies the symmetric projection structure of each Hilbert-space local term. If two cyclic supports do not meet, the corresponding windows are site-disjoint, so Lemma 13.187 gives the required non-negative ordered cross term. Apply Lemma 13.189 with these facts and the overlapping-window estimate (13.9). □

Lemma 13.191 (Parent-Hamiltonian gap from an overlapping cyclic martingale estimate). Lean ✓ Stmt *Fix $L > 1$ and let $i \sim_L j$ mean that the cyclic supports of the length-L windows at i and j meet. Suppose uniformly for every $N \ge 2L$ that overlapping off-diagonal terms satisfy*

$$\mathrm{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle \ge -(1 - \tfrac{1}{4L})\frac{1}{2(L-1)}\,\mathrm{Re}\langle h_{i,\mathrm{ES}}v, v\rangle. \tag{13.10}$$

Then the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^\perp$.

Proof. ✓ Proof Apply Lemma 13.190. Its non-overlap positivity and finite-family hypotheses are already internal to the preceding reduction; the overlapping estimate (13.10) supplies its remaining ordered cross-term condition. □

Lemma 13.192 (Parent-Hamiltonian gap from a cyclic anticommutator estimate). Lean ✓ Stmt *Fix $L > 1$ and let $i \sim_L j$ mean that the cyclic supports of the length-L windows at i and j meet. Suppose uniformly for every $N \ge 2L$ that overlapping off-diagonal terms satisfy*

$$\mathrm{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle + \mathrm{Re}\langle h_{j,\mathrm{ES}}v, h_{i,\mathrm{ES}}v\rangle \ge -\Big(1 - \frac{1}{4L}\Big)\frac{1}{2(L-1)}\big(\mathrm{Re}\langle h_{i,\mathrm{ES}}v, v\rangle + \mathrm{Re}\langle h_{j,\mathrm{ES}}v, v\rangle\big).$$

Then the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^\perp$.

Proof. ✓ Proof The cyclic overlap relation is symmetric, and Lemma 13.188 gives at most $2(L-1)$ overlapping off-diagonal windows in each row and column. If two cyclic supports do not meet, Lemma 13.187 gives non-negative ordered cross terms in both orders. Apply the finite-overlap anticommutator reduction. □

Lemma 13.193 (Parent-Hamiltonian gap from sufficient cyclic norm compression)**.** Lean ✓ Stmt *Fix $L > 1$. Suppose uniformly for every $N \geq 2L$ and every overlapping off-diagonal pair that*

$$\|h_{i,\mathrm{ES}}(h_{j,\mathrm{ES}}v)\| \leq (1 - \tfrac{1}{4L})\frac{1}{2(L-1)}\|h_{i,\mathrm{ES}}v\|.$$

Then the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$.

Proof. ✓ Proof For an overlapping pair, Cauchy–Schwarz and $h^2_{i,\mathrm{ES}} = h_{i,\mathrm{ES}}$ give

$$\begin{aligned}
\operatorname{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle &= \operatorname{Re}\langle h_{i,\mathrm{ES}}v, h_{i,\mathrm{ES}}h_{j,\mathrm{ES}}v\rangle \\
&\geq -\|h_{i,\mathrm{ES}}v\|\,\|h_{i,\mathrm{ES}}h_{j,\mathrm{ES}}v\| \\
&\geq -\Big(1 - \frac{1}{4L}\Big)\frac{1}{2(L-1)}\|h_{i,\mathrm{ES}}v\|^2 \\
&= -\Big(1 - \frac{1}{4L}\Big)\frac{1}{2(L-1)}\operatorname{Re}\langle h_{i,\mathrm{ES}}v, v\rangle.
\end{aligned}$$

This is the ordered lower bound used by the preceding overlapping-cyclic martingale reduction. □

Lemma 13.194 (Parent-Hamiltonian gap from a cyclic overlap compression constant)**.** Lean Lean Lean ✓ Stmt *Fix $L > 1$, and let $m = 2(L-1)$. Suppose uniformly for every $N \geq 2L$ and every overlapping off-diagonal pair that*

$$\|h_{i,\mathrm{ES}}(h_{j,\mathrm{ES}}v)\| \leq \eta\,\|h_{i,\mathrm{ES}}v\|.$$

If $\gamma > 0$, $\gamma \leq 1$, and $\eta \leq (1-\gamma)/m$, then $0 < \gamma$ and, for every admissible N and every $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$,

$$\gamma\|v\| \leq \|H^{\mathrm{ES}}_{N,L}(A)v\|.$$

In particular, if $\eta \geq 0$ and $\eta m < 1$, then $0 < 1 - \eta m$ and, for every admissible N and every $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$,

$$(1-\eta m)\|v\| \leq \|H^{\mathrm{ES}}_{N,L}(A)v\|.$$

This strict compression form is a sufficient strengthening of the anticommutator martingale estimate in Cirac–Perez-Garcia–Schuch–Verstraete [CPGSV21, Section IV.C]. It does not prove the remaining MPS-specific estimate; it isolates the finite-row martingale argument once such a strengthening is supplied.

Proof. ✓ Proof The overlap-cardinality lemma gives at most $m = 2(L-1)$ overlapping off-diagonal neighbours in each row, while the non-overlap positivity lemma handles disjoint supports. The finite-overlap norm-compression reduction gives the quadratic-form estimate with the chosen γ, and the positive quadratic-form criterion converts it to the norm lower bound. The strict form is the same statement with $\gamma = 1 - \eta m$. □

13.11.3 Martingale gap consequences

Theorem 13.195 (Martingale criterion for spectral gap). Not ready *Consider a frustration-free Hamiltonian*

$$H = \sum_{i=1}^{n} h_i$$

with projector interaction terms ($h_i^2 = h_i$). Suppose that for every unordered pair $\{i,j\}$ of overlapping terms there exist constants $c_{\{i,j\}} \geq 0$ and $\gamma > 0$ satisfying

$$h_i h_j + h_j h_i \geq -c_{\{i,j\}}(1-\gamma)(h_i + h_j), \tag{13.11}$$

and suppose further that the coefficients attached to the terms overlapping h_i satisfy

$$\sum_{\substack{j \neq i \\ \{i,j\} \text{ overlapping}}} c_{\{i,j\}} \leq 1$$

for every i. Then the smallest nonzero eigenvalue of H satisfies $\lambda^+_{\min}(H) \geq \gamma$.

Proof. Not ready The relevant geometry is a family of translated length-L projector windows on the ring; two terms h_i and h_j interact only when their supports overlap, equivalently when the cyclic distance $d_N(i,j)$ is smaller than L. From $h_i^2 = h_i$,

$$H^2 = \sum_i h_i + \sum_{\substack{i<j \\ \text{overlapping}}} (h_i h_j + h_j h_i) + \sum_{\substack{i<j \\ \text{non-overlapping}}} (h_i h_j + h_j h_i).$$

Non-overlapping projectors commute ($h_i h_j = h_j h_i$), so their product $h_i h_j$ is itself a projector and hence PSD; thus the last sum is ≥ 0. For the overlapping sum, the martingale condition (13.11) gives $h_i h_j + h_j h_i \geq -c_{\{i,j\}}(1-\gamma)(h_i + h_j)$. Summing over unordered pairs $\{i,j\}$:

$$\sum_{\{i,j\},\text{overlap}} (h_i h_j + h_j h_i) \geq -(1-\gamma) \sum_{\{i,j\},\text{overlap}} c_{\{i,j\}}(h_i + h_j).$$

Collecting terms for each h_i:

$$\sum_{\{i,j\},\text{overlap}} c_{\{i,j\}}(h_i + h_j) = \sum_i \Big(\sum_{\substack{j \neq i \\ \text{overlap}}} c_{\{i,j\}} \Big) h_i \leq H,$$

where the last inequality uses the row-sum hypothesis $\sum_{\substack{j \neq i \\ \{i,j\} \text{ overlapping}}} c_{\{i,j\}} \leq 1$. Combining, $H^2 \geq H - (1-\gamma)H = \gamma H$. For any eigenvector with $H\psi = \lambda\psi$ and $\lambda > 0$, this gives $\lambda^2 \geq \gamma\lambda$, hence $\lambda \geq \gamma$ [KL18]. □

Lemma 13.196 (MPS anticommutator input for the martingale condition). Not ready *Let A be injective and fix an interaction range $L \geq 2$ as in Definition 13.10. Let $h_L(A) := \Pi_{G_L(A)^\perp}$ be the L-site parent interaction, and for each chain length and site i let h_i denote its translate as in Definition 13.16. For every pair of overlapping terms h_i and h_j with $0 < d_N(i,j) < L$, where $d_N(i,j) := \min(|i-j|, N - |i-j|)$ is the cyclic distance on the ring (i.e. whose L-site supports share at least one site under periodic boundary conditions), the source martingale argument asks for constants $c_{\{i,j\}}$ and $\gamma > 0$, independent of N, for which the martingale condition of*

Theorem 13.195 holds. Cirac–Perez-Garcia–Schuch–Verstraete [CPGSV21, Section IV.C] use the martingale estimate

$$h_i h_j + h_j h_i \geq -c_{ij}(1-\gamma)(h_i + h_j)$$

and interpret it as a lower bound on the smallest non-zero principal angle between the local ground spaces $\ker h_i$ *and* $\ker h_j$*. In the Hilbert-space formulation above, the source condition is the corresponding quadratic-form inequality*

$$\mathrm{Re}\langle h_i v, h_j v\rangle + \mathrm{Re}\langle h_j v, h_i v\rangle \geq -c_{ij}(1-\gamma)\big(\mathrm{Re}\langle h_i v, v\rangle + \mathrm{Re}\langle h_j v, v\rangle\big),$$

with row-summable coefficients. Lemma 13.179 records the formal reduction from this estimate to the spectral gap. A directed norm-compression estimate, independent of N, such as

$$\|h_i h_j v\| \leq \eta \|h_i v\|, \qquad \eta\, 2(L-1) < 1,$$

for every overlapping off-diagonal pair is a sufficient stronger hypothesis; Theorem 13.199 records that consequence. (When $L = 1$ the interaction terms have disjoint supports, so there are no overlapping pairs and the spectral gap follows directly without the martingale argument.)

Proof. Not ready The cited martingale paragraph compares the anticommutator estimate with principal angles between the local ground spaces. If $q_i := I - h_i$ and $q_j := I - h_j$ denote the corresponding ground-space projectors, the Kastoryano–Lucia estimate cited there concerns

$$\delta(\ell) = \|q_i q_j - P\|,$$

where P projects onto $\ker(h_i + h_j) = \ker h_i \cap \ker h_j$ and ℓ is the overlap length. What remains is to derive the anticommutator estimate above, with its blocking convention and constants, for the cyclic-window local terms. The intersection property identifies $\ker h_i \cap \ker h_j$ with the parent ground space on the union of the two windows, but it does not by itself supply this numerical inequality. The directed excitation-projection estimate

$$\|h_i h_j v\| \leq \eta \|h_i v\|, \qquad \eta\, 2(L-1) < 1.$$

remains useful as a sufficient strengthening of the source condition.

Row-sum bound. Since $L \geq 2$, each site i overlaps with at most $2(L-1)$ neighbours (those j with $0 < d_N(i,j) < L$). Set $c_{\{i,j\}} := \frac{1}{2(L-1)}$ for every unordered overlapping pair and $c_{\{i,j\}} := 0$ otherwise. These constants depend only on the cyclic distance $d_N(i,j)$. Then $\sum_{\substack{j \neq i \\ \{i,j\} \text{ overlapping}}} c_{\{i,j\}} \leq 2(L-1) \cdot \frac{1}{2(L-1)} = 1$, satisfying the row-sum hypothesis of Theorem 13.195.

Anticommutator constant. It remains to compare the cited principal-angle estimates with the cyclic-window anticommutator statement

$$h_i h_j + h_j h_i \geq -c_{\{i,j\}}(1-\gamma)(h_i + h_j).$$

If this estimate is obtained with the displayed row-summable coefficients, Lemma 13.179 gives the gap. A sufficient stronger route is the norm-compression statement

$$\|h_i h_j v\| \leq \eta \|h_i v\|, \qquad \eta\, 2(L-1) < 1.$$

If such an η is obtained, Theorem 13.199 gives the gap constant $1 - \eta\, 2(L-1)$. The frequently displayed fixed-coefficient compression form is the specialization with

$$\eta = \left(1 - \frac{1}{4L}\right)\frac{1}{2(L-1)}, \qquad 1 - \eta\, 2(L-1) = \frac{1}{4L}.$$

In the corresponding anticommutator form the martingale condition is

$$h_i h_j + h_j h_i \geq -\left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)} (h_i + h_j)$$

for every overlapping off-diagonal pair, and a sufficient one-sided strengthening is

$$\|h_i h_j v\| \leq \left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)} \|h_i v\|.$$

A qualitative statement that the relevant smallest non-zero principal angles are positive is not by itself enough: the martingale reduction needs a numerical anticommutator bound with row-summable coefficients, or a stronger norm-compression bound with $\eta\, 2(L-1) < 1$. Thus the remaining comparison is whether the principal-angle estimates cited in [FNW92, Nac96, KL18] supply such a cyclic-window estimate, with constants independent of N, under the convention used here. ☐

Theorem 13.197 (Gap bound from cyclic overlap compression). Lean ✓ Stmt *Let A be a tensor and let $L > 1$. Suppose that, for every $N \geq 2L$ and every overlapping off-diagonal pair of cyclic length-L windows,*

$$\|h_{i,\mathrm{ES}}(h_{j,\mathrm{ES}} v)\| \leq \left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)} \|h_{i,\mathrm{ES}} v\|.$$

Then the parent Hamiltonian has the explicit gap constant

$$\gamma_L := \frac{1}{4L} > 0.$$

Concretely, if $N \geq 2L$ and $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$, then

$$\gamma_L \|v\| \leq \|H^{\mathrm{ES}}_{N,L}(A) v\|.$$

Proof. ✓ Proof The local averaging, positivity, and symmetric-projection statements are now established in Lemmas 13.162, 13.163, and 13.164. Lemma 13.170 reduces the present theorem to the remaining MPS-specific task: deriving the uniform quadratic-form estimate. Lemma 13.178 establishes the finite-sum algebra from ordered cross-term row bounds, while Lemma 13.179 records the source anticommutator martingale condition. Lemmas 13.181 and 13.189 state the common finite-overlap ordered-cross-term specialization with $m = 2(L-1)$. Lemma 13.168 supplies the non-overlap positivity statement for site-disjoint windows, and Lemma 13.187 translates this to the concrete support-disjoint cyclic windows. Lemma 13.188 gives the finite-overlap row-cardinality bound. Lemma 13.190 combines these results with the finite-overlap reduction, so the only additional input is the ordered anticommutator or cross-term estimate for overlapping cyclic windows. Lemma 13.191 states the same overlap-only formulation, and Lemma 13.193 states a sufficient norm-compression formulation. Equivalently, this is the constant-compression theorem below with

$$\eta = \left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)}.$$

Since

$$1 - \eta\, 2(L-1) = \frac{1}{4L},$$

the displayed hypothesis gives the stated lower bound with $\gamma_L = 1/(4L)$. ☐

Theorem 13.198 (Gap bound from a cyclic anticommutator estimate). Lean ✓ Stmt *Let $L > 1$. Suppose that for every $N \geq 2L$ and every overlapping off-diagonal pair of cyclic length-L windows,*

$$h_{i,\mathrm{ES}}h_{j,\mathrm{ES}} + h_{j,\mathrm{ES}}h_{i,\mathrm{ES}} \geq -\left(1 - \frac{1}{4L}\right)\frac{1}{2(L-1)}(h_{i,\mathrm{ES}} + h_{j,\mathrm{ES}})$$

in quadratic form. Then the parent Hamiltonian has the explicit gap constant

$$\gamma_L := \frac{1}{4L} > 0.$$

Concretely, if $N \geq 2L$ and $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$, then

$$\gamma_L\|v\| \leq \|H^{\mathrm{ES}}_{N,L}(A)v\|.$$

Proof. ✓ Proof This is Lemma 13.192 stated as the public conditional gap theorem, with the anticommutator inequality retained as an explicit hypothesis. □

Theorem 13.199 (Gap bound from a strict cyclic compression constant). Lean ✓ Stmt *Let $L > 1$. Put $m = 2(L-1)$. Suppose that there is a constant $\eta \geq 0$ such that $\eta m < 1$, and suppose that, for every $N \geq 2L$ and every overlapping off-diagonal pair of cyclic length-L windows,*

$$\|h_{i,\mathrm{ES}}(h_{j,\mathrm{ES}}v)\| \leq \eta\,\|h_{i,\mathrm{ES}}v\|.$$

Then the parent Hamiltonian has the gap constant $1 - \eta m > 0$. Concretely, if $N \geq 2L$ and $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$, then

$$(1 - \eta m)\|v\| \leq \|H^{\mathrm{ES}}_{N,L}(A)v\|.$$

Proof. ✓ Proof Lemma 13.194, with $\gamma = 1 - \eta m$, applies to the displayed hypothesis

$$\|h_{i,\mathrm{ES}}(h_{j,\mathrm{ES}}v)\| \leq \eta\,\|h_{i,\mathrm{ES}}v\|.$$

It gives $0 < 1 - \eta m$ and the stated norm lower bound. The preceding fixed-coefficient theorem is the specialization $\eta = (1 - 1/(4L))/(2(L-1))$. □

Theorem 13.200 (Conditional spectral gap for MPS parent Hamiltonians). Lean Lean ✓ Stmt *For a tensor A and a range $L > 1$, assume either the cyclic-window anticommutator estimate of Theorem 13.198 or the stronger overlapping cyclic-window norm-compression estimate of Theorem 13.197. Then there exists $\gamma > 0$ such that for every $N \geq 2L$ and every vector $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$ one has*

$$\gamma\|v\| \leq \|H^{\mathrm{ES}}_{N,L}(A)v\|.$$

By Lemma 13.169, this implies the usual uniform lower bound on every nonzero eigenvalue of the parent Hamiltonian.

Proof. ✓ Proof This follows immediately from the corresponding explicit theorem by taking γ to be the constant produced there. □

Theorem 13.201 (Conditional spectral gap from a strict cyclic compression constant). Lean ✓ Stmt *For a tensor A and a range $L > 1$, suppose there is a constant $\eta \geq 0$ such that $\eta\,2(L-1) < 1$ and, for every $N \geq 2L$ and every overlapping off-diagonal pair of cyclic length-L windows,*

$$\|h_{i,\mathrm{ES}}(h_{j,\mathrm{ES}}v)\| \leq \eta\,\|h_{i,\mathrm{ES}}v\|.$$

Then there exists $\gamma > 0$ such that, for every $N \geq 2L$ and every vector $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$,

$$\gamma\|v\| \leq \|H^{\mathrm{ES}}_{N,L}(A)v\|.$$

Proof. ✓ Proof This follows from Theorem 13.199 by taking $\gamma = 1 - \eta\,2(L-1)$. □

13.12 Block injectivity and degenerate ground spaces

For an MPS with more than one block in canonical form, after blocking to block-injective canonical form, the physical–virtual correspondence is restricted to block-diagonal boundary conditions. In the notation of [CPGSV21, lines 2113–2117], set

$$\widetilde{A}^{\omega} := (A_1^{\omega}, \dots, A_g^{\omega}) \in \prod_{j=1}^{g} M_{D_j}(\mathbb{C}).$$

Then block injectivity is the statement

$$\forall X = (X_j)_{j=1}^{g} \in \bigoplus_{j=1}^{g} M_{D_j}(\mathbb{C}), \quad \exists (c_\omega(X))_\omega, \quad X = \sum_\omega c_\omega(X)\widetilde{A}^{\omega}.$$

Equivalently, with the same coefficient family,

$$\forall j, \qquad X_j = \sum_\omega c_\omega(X) A_j^{\omega}.$$

The strengthened parent-Hamiltonian theorem in [CPGSV21, Theorem IV.16, lines 2126–2128] is

$$N \geq L_0 + 1 \quad \Longrightarrow \quad \ker H_N = \operatorname{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

The statements below record the corresponding periodic-boundary ground-space formulation; the passage to $\ker H_N$ is kept separate.

Definition 13.202 (Periodic-boundary ground space for the block-diagonal tensor). Not ready For a basis of normal tensors $A = (A_j)_{j=1}^{g}$ and nonzero weights μ_j, let

$$B := \bigoplus_{j=1}^{g} \mu_j A_j.$$

The ground space on the N-site ring considered in this section is

$$\mathcal{G}_{N,L}(B).$$

Lemma 13.203 (Block-diagonal periodic-boundary ground space). Not ready *By definition, for $B = \bigoplus_{j=1}^{g} \mu_j A_j$ one has*

$$\mathcal{G}_{N,L}(B) = \mathcal{G}_{N,L}\left(\bigoplus_{j=1}^{g} \mu_j A_j\right).$$

Definition 13.204 (Vector space generated by a basis of normal tensors). Lean ✓ Stmt For a basis of normal tensors $A_1, \dots, A_g$, the vector space generated by their length-N matrix product vectors is

$$\operatorname{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

Definition 13.205 (Fixed-length block-injective canonical-form condition). Lean ✓ Stmt For a fixed blocked length m, the fixed-length form of the block-injective canonical-form condition is

$$\operatorname{span}\{(A_1^{\omega}, \dots, A_g^{\omega}) : |\omega| = m\} = \prod_{j=1}^{g} M_{D_j}(\mathbb{C}).$$

Equivalently,

$$\forall X = (X_j)_j \in \bigoplus_{j=1}^{g} M_{D_j}(\mathbb{C}), \quad \exists (c_\omega(X))_{|\omega|=m}, \quad (X_j)_{j=1}^{g} = \sum_{|\omega|=m} c_\omega(X)(A_1^\omega, \dots, A_g^\omega).$$

This is the fixed-length form of the block-injective canonical-form condition of [CPGSV21, lines 2113–2117].

Lemma 13.206 (Each normal-tensor vector lies in the periodic-boundary ground space)**.** Not ready *If $1 < L$ and $N \geq L + 1$, then for each normal tensor A_j in the basis the vector $V^{(N)}(A_j)$ lies in $\mathcal{G}_{N,L}(\bigoplus_k \mu_k A_k)$.*

Proof. Not ready Write Γ_N^B for the length-N boundary-to-state map of the tensor B, and let $\iota_j(Z)$ be the block-diagonal matrix with Z in the j-th block and 0 elsewhere. The direct-sum evaluation gives

$$\Gamma_N^{\oplus_k \mu_k A_k}\big(\iota_j(\mu_j^{-N} X)\big) = \Gamma_N^{A_j}(X).$$

Taking $X = I$ gives the vector $V^{(N)}(A_j)$, and Lemma 13.74 applied to the individual block gives its membership in $\mathcal{G}_{N,L}(\bigoplus_k \mu_k A_k)$. □

Lemma 13.207 (Inclusion of a block ground space)**.** Lean ✓ Stmt *For each block index j with $\mu_j \neq 0$, and whenever $0 < N$ and $L \leq N$, the periodic-boundary ground space of the block A_j embeds into the corresponding ground space of the direct-sum tensor:*

$$\mathcal{G}_{N,L}(A_j) \subseteq \mathcal{G}_{N,L}\left(\bigoplus_k \mu_k A_k\right).$$

Proof. ✓ Proof The definition by length-L windows on the ring reduces the inclusion to the local block-embedding statement that each ground vector of A_j on the window lifts to the direct sum by the identity

$$\Gamma_L^{\oplus_k \mu_k A_k}\big(\iota_j(\mu_j^{-L} X)\big) = \Gamma_L^{A_j}(X).$$

Here ι_j inserts a matrix in the j-th diagonal block and has zero off that block. Thus $\iota_j(\mu_j^{-L} X)$ has exactly the local vector of the j-th block on that window. □

Lemma 13.208 (Forward inclusion from the block ground spaces)**.** Lean ✓ Stmt *Assume $\mu_j \neq 0$ for every block index j. Whenever $0 < N$ and $L \leq N$, the linear sum of the blockwise ground spaces with periodic boundary conditions is contained in the direct-sum tensor's ground space with periodic boundary conditions:*

$$\sum_j \mathcal{G}_{N,L}(A_j) \subseteq \mathcal{G}_{N,L}\left(\bigoplus_k \mu_k A_k\right),$$

where j ranges over all block indices.

Proof. ✓ Proof Take the linear sum of the inclusions

$$\mathcal{G}_{N,L}(A_j) \subseteq \mathcal{G}_{N,L}\left(\bigoplus_k \mu_k A_k\right)$$

over all block indices j. □

Lemma 13.209 (Forward inclusion for the direct-sum tensor). Not ready *Let $A^i = \bigoplus_j \mu_j A^i_j$ be in block-injective canonical form, where $A_1, \dots, A_g$ is a basis of normal tensors. The linear sum of the blockwise ground spaces with periodic boundary conditions is contained in the ground space of the block-diagonal tensor on the N-site ring:*

$$\sum_j \mathcal{G}_{N,L}(A_j) \subseteq \mathcal{G}_{N,L}\left(\bigoplus_k \mu_k A_k\right).$$

Proof. Not ready The block-injective canonical-form hypothesis supplies $\mu_j \neq 0$ for every j, and the block-diagonal periodic-boundary ground space is $\mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$ by definition. □

13.13 Block separation and intersection

This section establishes that block-separating words make the simultaneous block-word tuples $\omega \mapsto (A_1^\omega, \dots, A_g^\omega)$ span the full product algebra $\prod_j M_{D_j}(\mathbb{C})$ at a single length, and that this single-length span, together with the normalization $\sum_a A^j_a A^{j\dagger}_a = I$, yields the one-step intersection identity $\bigvee_j G_{n+2}(A^j)$ used for the degenerate parent-Hamiltonian ground space.

Lemma 13.210 (Block-diagonal commutant reduction from span extension). Not ready *Let S be a family of virtual matrices. If each sector projection satisfies*

$$P_j \in \operatorname{span} S$$

and a boundary matrix X commutes with every M in S,

$$XM = MX,$$

then, for every sector j,

$$XP_j = P_j X,$$

and hence, for distinct sectors j and k,

$$P_j X P_k = 0.$$

Thus X is block diagonal with respect to the sector decomposition.

Proof. Not ready The equations $XM = MX$ are linear in M, so they hold for every $M \in \operatorname{span} S$. Substituting $M = P_j$ gives $XP_j = P_j X$. Multiplying this identity on the left by P_j and on the right by P_k with $j \neq k$ gives $P_j X P_k = 0$. □

Lemma 13.211 (Block projections from product-algebra span). Not ready *Let $T_a = (T_a(i))_i$ be a family of tuples spanning the full product algebra $\prod_i M_{n_i}(\mathbb{C})$, and let $c_i \neq 0$ be nonzero scalars. Then, for every sector k, the sector projection P_k lies in the span of the scaled dependent block-diagonal matrices*

$$\bigoplus_i c_i T_a(i).$$

Proof. Not ready Apply the linear map $(M_i)_i \mapsto \bigoplus_i c_i M_i$ to the full-span product-algebra element with $(c_k)^{-1} I$ in the k-th component and zero in every other component. Its image is exactly P_k. □

Lemma 13.212 (Sector projections from blockwise word span). Not ready *If the simultaneous length-m block-word tuples*

$$\omega \longmapsto (A_j^\omega)_j$$

span the full product algebra $\prod_j M_{D_j}(\mathbb{C})$ and every μ_j is nonzero, then the pulled-back length-m word products of $\bigoplus_j \mu_j A_j$ span every virtual sector projection P_j. If a virtual matrix X commutes with the pulled-back word product for every word ω of length m,

$$X\left(\bigoplus_j \mu_j^m A_j^\omega\right) = \left(\bigoplus_j \mu_j^m A_j^\omega\right) X,$$

then, for distinct sectors j and k,

$$P_j X P_k = 0.$$

Proof. Not ready First embed the product algebra linearly as dependent block-diagonal matrices, scaling the j-th block by μ_j^m. Since $\mu_j^m \neq 0$, the product-algebra span contains the tuple with $(\mu_j^m)^{-1} I$ in one component and zero elsewhere; its scaled diagonal image is exactly P_j. The word-evaluation formula for the direct-sum tensor identifies these scaled diagonal matrices with the pulled-back direct-sum word products, and the commutant conclusion follows from Lemma 13.210. ☐

Definition 13.213 (Pairwise block-separating equations). Lean Lean ✓ Stmt Fix a finite target set T of block indices. A length-S block-separating equation for k against T is a choice of coefficients c_ω such that

$$\sum_{|\omega|=S} c_\omega A_\ell^\omega = \begin{cases} I, & \ell = k, \\ 0, & \ell \in T. \end{cases}$$

The pairwise version requires this equation for each ordered pair of distinct blocks, with $T = \{j\}$.

Definition 13.214 (Trace-dual pair product-span condition). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt For two blocks A and B, the length-S pair word tuple is $w \mapsto (A^w, B^w)$. We also use the subalgebra of $M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C})$ generated by the one-letter pairs (A_i, B_i). The pair product-span condition says that the length-S tuples span $M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C})$. Equivalently, in the trace-dual formulation of the same product-span condition, the only pair of test matrices (Δ_A, Δ_B) whose trace pairing with every length-S tuple vanishes is $(0, 0)$. The cumulative variant replaces one fixed length by all word lengths at most S, and the all-length variant allows every finite word.

Lemma 13.215 (Three-block trace reduction from a nonzero middle matrix). Lean Lean ✓ Stmt *Let A and B be tensors with virtual bond dimensions D_A and D_B. Assume that A is L-block injective. Let $\Delta_A \in M_{D_A}(\mathbb{C})$ with $\Delta_A \neq 0$, and let $\Delta_B \in M_{D_B}(\mathbb{C})$. Suppose that, for all words u, v, t of length L,*

$$\operatorname{tr}(\Delta_A A_v A_t A_u) + \operatorname{tr}(\Delta_B B_v B_t B_u) = 0.$$

Then for every $Z \in M_{D_A}(\mathbb{C})$ there is a matrix $W \in M_{D_B}(\mathbb{C})$ such that, for all words t of length L,

$$\operatorname{tr}(Z A_t) + \operatorname{tr}(W B_t) = 0.$$

This is the trace-dual algebraic step in the two-block case of the direct-sum lemma of [PGVWC07].

Proof. ✓ Proof Since $\Delta_A \neq 0$, Lemma 8.11 writes Z as a linear combination of matrices $A_u \Delta_A A_v$. For one such term, cyclicity of the trace rewrites the three-block relation as

$$\operatorname{tr}(A_u \Delta_A A_v A_t) + \operatorname{tr}(B_u \Delta_B B_v B_t) = 0.$$

Taking the same linear combination of the matrices $B_u \Delta_B B_v$ gives the required W, and linearity gives the general case. ☐

Lemma 13.216 (Finite-chain image inclusion from a three-block relation)**.** Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Under the hypotheses of Lemma 13.215, the length-L spaces $\mathcal{G}_L^A$ and $\mathcal{G}_L^B$ satisfy*

$$\mathcal{G}_L^A \subseteq \mathcal{G}_L^B.$$

If the corresponding nonzero middle matrix exists on both sides, then $\mathcal{G}_L^A = \mathcal{G}_L^B$.

Proof. ✓ Proof Write a vector in $\mathcal{G}_L^A$ as $t \mapsto \operatorname{tr}(A_t Z)$. The preceding lemma supplies W with $\operatorname{tr}(ZA_t) + \operatorname{tr}(WB_t) = 0$ for every word t. Cyclicity of the trace gives

$$\operatorname{tr}(A_t Z) = \operatorname{tr}(ZA_t) = -\operatorname{tr}(WB_t) = \operatorname{tr}(B_t(-W)),$$

so the vector lies in $\mathcal{G}_L^B$. The equality statement follows by applying the same argument with the two blocks exchanged. ☐

Lemma 13.217 (Dimension step for two-block local spaces)**.** Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *If A and B are length-L block-injective, the map $Z \mapsto (t \mapsto \operatorname{tr}(A_t Z))$ has image dimension D_A^2, and similarly for B. Therefore an inclusion $\mathcal{G}_L^A \subseteq \mathcal{G}_L^B$, together with $D_B \leq D_A$, forces $D_A = D_B$ and $\mathcal{G}_L^A = \mathcal{G}_L^B$. In particular, the three-block trace relation from Lemma 13.216 gives this equality whenever the ordering $D_B \leq D_A$ is available. If the inclusion comes from the strictly larger block, $D_B < D_A$, then the same dimension step already contradicts the existence of the three-block trace relation. More generally, let H be a source hypothesis which excludes the simultaneous conclusion*

$$D_A = D_B, \qquad \mathcal{G}_L^A = \mathcal{G}_L^B.$$

Then the three-block intersection vanishes:

$$H \Longrightarrow \mathcal{G}_{3L}^A \cap \mathcal{G}_{3L}^B = \{0\}.$$

With injectivity at the three-block length, the strict-size branch gives

$$D_B < D_A, \quad \mathcal{G}_{3L}^A \cap \mathcal{G}_{3L}^B = \{0\} \Longrightarrow \operatorname{span}\{(A^\omega, B^\omega) : |\omega| = 3L\} = M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C}).$$

One such source hypothesis is non-proportionality of the periodic MPS vectors at a sufficiently long chain length:

$$\mathbb{C}\, V^{(N)}(A) \neq \mathbb{C}\, V^{(N)}(B).$$

Indeed,

$$\mathcal{G}_L^A = \mathcal{G}_L^B \Longrightarrow \mathcal{G}_{N,L}(A) = \mathcal{G}_{N,L}(B) \Longrightarrow \mathbb{C}\, V^{(N)}(A) = \mathbb{C}\, V^{(N)}(B),$$

by equality of the periodic-boundary constraint spaces, followed by injective uniqueness. For same-dimensional normalized BNT representatives that are not gauge-phase equivalent, the non-equivalence condition supplies this non-proportionality at arbitrarily large chain lengths; the *direct-sum injectivity hypotheses remain explicit. Combining the equal-dimension and unequal-dimension branches over all ordered block pairs gives one common positive length S_0 with*

$$\operatorname{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = S_0\} = \prod_{j=1}^{g} M_{D_j}(\mathbb{C}).$$

For an ordered pair of blocks, the trace-separation conclusion has the form

$$[\forall|\omega| = S,\ \operatorname{tr}(\Delta_A A^\omega) + \operatorname{tr}(\Delta_B B^\omega) = 0] \Longrightarrow \Delta_A = \Delta_B = 0.$$

Equivalently, the homogeneous pair span is full:

$$\operatorname{span}\{(A^\omega, B^\omega) : |\omega| = S\} = M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C}).$$

Concatenating words propagates this equality to positive multiples of S, and hence to a finite period window for every ordered pair. Combining that window with the conditional period-window homogenization theorem gives a single length S_0 such that

$$\operatorname{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = S_0\} = \prod_{j=1}^{g} M_{D_j}(\mathbb{C}).$$

The same conclusion applies to the corresponding normal canonical-form hypotheses with a basis of normal tensors once one-site injectivity is supplied explicitly, since those hypotheses carry the irreducibility and normalization assumptions used in the direct-sum comparison.

Proof. ✓ Proof Block injectivity makes the boundary-to-chain map injective, so $\dim \mathcal{G}_L^A = D_A^2$ and $\dim \mathcal{G}_L^B = D_B^2$. Inclusion gives $D_A^2 \le D_B^2$, while $D_B \le D_A$ gives the reverse inequality. Hence $D_A = D_B$, and equality of dimensions inside the inclusion gives $\mathcal{G}_L^A = \mathcal{G}_L^B$. In the strict-size branch this contradicts $D_B < D_A$. For the conditional directness statement, suppose $\psi \in \mathcal{G}_{3L}^A \cap \mathcal{G}_{3L}^B$ has boundary representations

$$\psi = \Gamma_{3L}^A(X) = \Gamma_{3L}^B(Y).$$

If $X = 0$, injectivity of the A boundary-to-chain map gives $\psi = 0$. Otherwise equality of the coefficient formulas gives

$$\forall|t| = 3L, \qquad \operatorname{tr}(XA^t) - \operatorname{tr}(YB^t) = 0,$$

equivalently

$$\forall|t| = 3L, \qquad \operatorname{tr}(XA^t) + \operatorname{tr}((-Y)B^t) = 0.$$

The dimension step applied to this homogeneous three-block trace relation gives $D_A^2 \le D_B^2$, contradicting $D_B < D_A$ in the strict-size branch. In the equal-size branch, equal local spaces impose the same length-L constraints on the N-site ring:

$$\mathcal{G}_L^A = \mathcal{G}_L^B \Longrightarrow \mathcal{G}_{N,L}(A) = \mathcal{G}_{N,L}(B).$$

The injective uniqueness theorem identifies these periodic ground spaces with the one-dimensional MPS spans,

$$\mathcal{G}_{N,L}(A) = \mathbb{C}\, V^{(N)}(A), \qquad \mathcal{G}_{N,L}(B) = \mathbb{C}\, V^{(N)}(B),$$

and hence

$$\mathbb{C}\, V^{(N)}(A) = \mathbb{C}\, V^{(N)}(B),$$

contradicting non-proportionality of the two MPS vectors. □

Lemma 13.218 (All-length pair trace separation has finite cumulative length). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *If no nonzero trace functional vanishes on all finite pair words, then there is a finite S such that vanishing on every word of length at most S already forces the test pair to be zero. For a finite family of ordered block pairs, these bounds can be replaced by one common bound.*

Proof. ✓ Proof Write

$$W_S(A,B) = \operatorname{span}\{(A^w, B^w) : |w| \le S\}, \qquad W_\infty(A,B) = \operatorname{span}\{(A^w, B^w) : w \text{ finite}\}.$$

The trace-dual criterion is the equivalence

$$W_\infty(A,B) = M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C}) \iff \left[\ \forall w,\ \operatorname{tr}(\Delta_A A^w) + \operatorname{tr}(\Delta_B B^w) = 0\ \right] \Longrightarrow (\Delta_A, \Delta_B) = (0,0).$$

Since the product algebra is finite-dimensional and

$$W_0(A,B) \subseteq W_1(A,B) \subseteq \cdots, \qquad \bigvee_{S \ge 0} W_S(A,B) = W_\infty(A,B),$$

there is an index S_0 with

$$W_{S_0}(A,B) = W_\infty(A,B) = M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C}).$$

Applying the finite trace-duality criterion at S_0 gives

$$\left[\ \forall w,\ |w| \le S_0 \Longrightarrow \operatorname{tr}(\Delta_A A^w) + \operatorname{tr}(\Delta_B B^w) = 0\ \right] \Longrightarrow (\Delta_A, \Delta_B) = (0,0).$$

For a finite family of ordered block pairs, choose such an S_{kj} for each pair and replace them by

$$S = \max_{k \ne j} S_{kj}.$$

Monotonicity of W_S gives the same finite cumulative trace separation for every pair. This is the cumulative span at all lengths at most S, not a single homogeneous length, since the length-zero word already contributes the identity to the cumulative span. □

Lemma 13.219 (Pair algebra projections are full). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *If one of the two blocks is injective, then the corresponding coordinate projection of the pair algebra generated by the simultaneous one-letter pairs is the full matrix algebra. Every finite pair word lies in this pair algebra, and the all-word linear span is the submodule generated by the pair algebra. If the pair algebra is the graph of an algebra automorphism of the matrix algebra, Skolem–Noether turns this graph case into gauge-phase equivalence, so under non-gauge-equivalence the graph branch is impossible. For two blocks of unequal bond dimensions, the graph branch is also impossible because it would identify matrix algebras with different finite dimensions over $\mathbb{C}$.*

Proof. ✓ Proof Projecting the pair generators to one coordinate gives exactly the one-block generator family. Injectivity says that the linear span of this family is the full matrix algebra, and the linear span is contained in the unital algebra it generates. The pair-word membership follows by induction on the word, using closure of the generated algebra under multiplication. Conversely, the all-word linear span is closed under componentwise multiplication, so it contains the algebra generated by the pair generators. In the graph case, each generator satisfies $B_i = \varphi(A_i)$; applying Skolem–Noether to φ gives an ordinary gauge, hence a gauge-phase equivalence with scalar 1. □

Lemma 13.220 (Conditional padding from exact homogeneous identity witnesses). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Homogeneous identity-padding witnesses multiply by concatenation of words. Therefore, if one has a positive period and a full residue window of homogeneous pair spans, then the simultaneous identity pair lies in all sufficiently large homogeneous pair spans. Combined with all-length pair trace separation, this period-window input gives one homogeneous pair trace-separation length. This does not prove the block-separation span condition of [PGVWC07] by itself: fullness of spans up to a bounded length is weaker than the exact homogeneous input used here.*

Proof. ✓ Proof The length-zero word gives the identity only at length zero, so it is not a positive-length padding theorem. The useful padding facts are homogeneous: if the identity pair is in the span at lengths m and n, componentwise multiplication puts it in the span at length $m+n$. A period witness and a complete residue window therefore give homogeneous padding at every sufficiently large length, which is the input needed by the all-length-to-fixed-length implication. □

Lemma 13.221 (Pair trace separation gives pairwise block separation). Lean Lean Lean Lean Lean Lean ✓ Stmt *At a fixed length S, the pair trace-separation criterion is dual to full pair product-span. Therefore a pair trace-separation witness supplies a linear combination of word evaluations that is identity on the first block and zero on the second. A common witness for all ordered distinct pairs gives pairwise block-separating words at the same length.*

Proof. ✓ Proof If the pair span were proper, a nonzero linear functional would vanish on it. Nondegeneracy of the matrix trace pairing represents that functional as (Δ_A, Δ_B), contradicting pair trace separation. Once the pair span is the whole product algebra, the tuple $(I, 0)$ is in the span. Using the same coefficients on the ambient block family gives the required pairwise block-separating equation. □

Lemma 13.222 (Block-separating equations from pairwise separation). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *If every ordered pair of distinct blocks admits a common-length pairwise block-separating equation, then for each k there are coefficients $c_\omega^{(k)}$ with*

$$\sum_{|\omega|=(g-1)S} c_\omega^{(k)} A_j^\omega = \begin{cases} I, & j = k, \\ 0, & j \neq k. \end{cases}$$

Hence common block injectivity plus pairwise separation implies the finite-length blockwise physical–virtual correspondence of the block-injectivity proposition in [CPGSV16, Section 2.3] and the block-injective canonical-form trace-separation condition.

Proof. ✓ Proof The empty word gives the identity tuple, corresponding to an empty target set. The span of word tuples is closed under pointwise multiplication because the product of a length-L word and a length-S word is the concatenated length-$(L+S)$ word. Inducting over the finite set of blocks different from k and multiplying the corresponding pairwise equations keeps the value I on k and introduces one additional zero block at each step. Extracting coefficients from the final tuple-span membership gives the coefficient form of the displayed block-separating equation. □

Lemma 13.223 (Homogeneous padding for blockwise product spans). Lean Lean ✓ Stmt *Suppose the length-L simultaneous block-word tuples span $\prod_j M_{D_j}(\mathbb{C})$, and suppose the simultaneous identity tuple $(I, \ldots, I)$ lies in the span of the length-S simultaneous block-word tuples. Then the length-$L+S$ simultaneous block-word tuples span $\prod_j M_{D_j}(\mathbb{C})$. In particular, two full homogeneous spans at lengths L and S give a full homogeneous span at length $L+S$.*

Proof. ✓ Proof Let $M = (M_j)_j$ be a target tuple. Write

$$M_j = \sum_{|u|=L} a_u A_u^j, \qquad I = \sum_{|v|=S} b_v A_v^j$$

simultaneously for every block j. Multiplying the two expansions gives

$$M_j = M_j I = \sum_{|u|=L} \sum_{|v|=S} a_u b_v A_u^j A_v^j = \sum_{|u|=L} \sum_{|v|=S} a_u b_v A_{uv}^j.$$

The words uv have length $L + S$, so the target tuple lies in the length-$L + S$ span. □

Lemma 13.224 (Period-window propagation for blockwise product spans). Lean Lean ✓ Stmt *Let $p > 0$. Suppose the length-p simultaneous block-word tuples span $\prod_j M_{D_j}(\mathbb{C})$, and suppose that for every $0 \le r < p$, the length-$s+r$ simultaneous block-word tuples span $\prod_j M_{D_j}(\mathbb{C})$. Then there is a length N such that, for every $n \ge N$, the length-n simultaneous block-word tuples span $\prod_j M_{D_j}(\mathbb{C})$. More concretely, one may take $N = s$: writing*

$$n = s + r + qp, \qquad 0 \le r < p,$$

the span at length $s + r$ propagates to the span at length n by multiplying q times by the period-p span.

Proof. ✓ Proof The Euclidean division of $n - s$ by p gives $n = s + r + qp$ with $0 \le r < p$. The assumed residue-window span gives the full product algebra at length $s + r$. Applying homogeneous padding with the length-p span once gives the full product algebra at length $s+r+p$; iterating gives the full product algebra at length $s + r + qp$. □

Lemma 13.225 (Blockwise word span from block-separating equations). Lean Lean ✓ Stmt *Let J be a finite set of r blocks, and let $(A_j)_{j\in J}$ be a finite family of tensors with common block injectivity at length L. Suppose there are length-S block-separating equations: for each block k,*

$$\sum_{|\omega|=S} c_\omega^{(k)} A_j^\omega = \begin{cases} I, & j = k, \\ 0, & j \ne k. \end{cases}$$

Then the simultaneous length-$(L + S)$ block-word evaluations span $\prod_j M_{D_j}(\mathbb{C})$. If the block-separating equations are first obtained from pairwise block-separating equations at length S, the same conclusion holds at length $L + (r - 1)S$.

Proof. ✓ Proof Block injectivity writes each target block matrix as a length-L linear combination of word evaluations. For a target tuple $(M_j)_j$, write M_k in this form in block k, multiply by the block-separating equation for k, and sum over k. Thus, if $M_k = \sum_{|u|=L} a_u^{(k)} A_k^u$ and $\sum_{|v|=S} b_v^{(k)} A_j^v = \delta_{j,k} I$, then for every block j,

$$M_j = \sum_{k\in J} \sum_{|u|=L} \sum_{|v|=S} a_u^{(k)} b_v^{(k)} A_j^u A_j^v.$$

The right side is a linear combination of length-$(L + S)$ word tuples, so these tuples span the full product algebra. If one starts with pairwise separation instead, first use Lemma 13.222 to multiply the pairwise equations into the full equations. □

13.13.1 Trace identities for block intersections

Once the simultaneous block-word tuples span the full product algebra at a single length, nondegeneracy of the product trace pairing turns blockwise trace equalities into blockwise matrix identities $A_b^j C_a^j = A_b^j E^j A_a^j$. The lemmas here use those identities to place left-boundary trace decompositions in the block sum $\bigvee_j G_{n+2}(A^j)$ and to obtain the one-step block-intersection identity.

Lemma 13.226 (BNT block separation gives a finite blockwise product span)**.** Lean Lean Lean Lean ✓ Stmt *Let $A^1, \dots, A^r$ be normalized representatives of a basis of normal tensors: each representative is irreducible, the family is left-canonical, the self-overlaps are normalized, and no two distinct same-dimensional representatives are gauge-phase equivalent. Fix $L_0 > 0$. Assume that, for every block j, the map*

$$\Gamma_{L_0}^{A^j} : M_{D_j}(\mathbb{C}) \to (\mathbb{C}^d)^{\otimes L_0}, \qquad X \longmapsto \sum_{|u|=L_0} \operatorname{tr}(X A_u^j)|u\rangle$$

is injective, and that the homogeneous word spans at the two source lengths are full:

$$\operatorname{span}\{A_u^j : |u| = L_0 + 1\} = M_{D_j}(\mathbb{C}), \qquad \operatorname{span}\{A_v^j : |v| = 3(L_0 + 1)\} = M_{D_j}(\mathbb{C}).$$

Then for every ordered pair $k \neq j$ there are coefficients $c_\omega^{k,j}$ such that

$$\sum_{|\omega|=3(L_0+1)} c_\omega^{k,j} A_\omega^k = I, \qquad \sum_{|\omega|=3(L_0+1)} c_\omega^{k,j} A_\omega^j = 0.$$

Consequently the simultaneous block-word tuples at length

$$(L_0 + 1) + (r - 1)3(L_0 + 1)$$

span the full product algebra

$$\prod_{j=1}^{r} M_{D_j}(\mathbb{C}).$$

Proof. ✓ Proof Lemma 13.217 gives homogeneous trace separation at length $3(L_0 + 1)$ for every ordered pair of distinct normalized BNT representatives. Lemma 13.221 converts these trace-separation equations into the displayed pairwise block-separating equations. Multiplying the equations for the $r - 1$ blocks different from k gives a block-separating equation for k, and Lemma 13.225 then supplies the displayed product span. □

Lemma 13.227 (Blockwise matrix equality from trace pairings)**.** Lean ✓ Stmt *Let $A^1, \dots, A^g$ be block tensors whose length-m simultaneous word tuples span the product algebra. If, for every word w of length m, two blockwise matrix families X_j and Y_j satisfy*

$$\sum_j \operatorname{tr}(X_j A_w^j) = \sum_j \operatorname{tr}(Y_j A_w^j)$$

then $X_j = Y_j$ for every block j.

Proof. ✓ Proof For every word w of length m,

$$\sum_j \operatorname{tr}\big((X_j - Y_j)A_w^j\big) = \sum_j \operatorname{tr}(X_j A_w^j) - \sum_j \operatorname{tr}(Y_j A_w^j) = 0.$$

Nondegeneracy of the product trace pairing, together with the spanning hypothesis, gives $X_j - Y_j = 0$ for every block j. □

Lemma 13.228 (Boundary identities from block-sum membership). Lean ✓ Stmt *Let $A^1, \dots, A^g$ be block tensors whose length-m simultaneous word tuples span the product algebra. Let $C_a^j \in M_{D_j}(\mathbb{C})$ be matrices indexed by blocks j and physical letters a. Suppose that, for every pair of boundary letters a, b and every word w of length m, a vector ψ has the left-boundary trace form*

$$\psi(a, w, b) = \sum_j \operatorname{tr}\Big(A_b^j C_a^j A_w^j\Big)$$

and if

$$\psi \in \bigvee_j G_{m+2}(A^j),$$

then there are matrices E_j such that, for every block j and all a, b,

$$A_b^j C_a^j = A_b^j E_j A_a^j$$

Proof. ✓ Proof Write $\psi = \sum_j \phi_j$ with $\phi_j \in G_{m+2}(A^j)$. For each j, choose E_j whose image under the length-$m+2$ ground-space map for A^j is ϕ_j. Comparing the coefficient of the word awb in the two expressions for ψ gives

$$\sum_j \operatorname{tr}\Big(A_b^j C_a^j A_w^j\Big) = \sum_j \operatorname{tr}\Big(A_b^j E_j A_a^j A_w^j\Big).$$

Lemma 13.227 gives the displayed blockwise identities. $\square$

Lemma 13.229 (Blockwise boundary compatibility from traces). Lean ✓ Stmt *Let $A^1, \dots, A^g$ be block tensors with the following common word-span property: for a fixed m, the simultaneous tuples*

$$w \longmapsto \left(A_w^1, \dots, A_w^g\right), \qquad |w| = m,$$

span the full product algebra $\prod_j M_{D_j}(\mathbb{C})$. Suppose that matrices $C_a^j, D_b^j \in M_{D_j}(\mathbb{C})$ satisfy, for all physical indices a, b and all words w of length m,

$$\sum_j \operatorname{tr}\Big(A_b^j C_a^j A_w^j\Big) = \sum_j \operatorname{tr}\Big(D_b^j A_a^j A_w^j\Big).$$

Then

$$A_b^j C_a^j = D_b^j A_a^j$$

for every block j and all a, b.

Proof. ✓ Proof For fixed a, b, set

$$\Delta_j = A_b^j C_a^j - D_b^j A_a^j.$$

The assumed trace equality says that

$$\sum_j \operatorname{tr}\Big(\Delta_j A_w^j\Big) = 0$$

for every word w of length m. Since the tuples $(A_w^1, \dots, A_w^g)$ span the product algebra, the product trace pairing is nondegenerate and hence every Δ_j is zero. $\square$

Lemma 13.230 (Word-valued boundary compatibility from traces). Lean Lean ✓ Stmt *Let $A^1, \dots, A^g$ be block tensors with the common word-span property at length m. Suppose that $C^j_\rho \in M_{D_j}(\mathbb{C})$ is indexed by a block j and a word ρ of some fixed length M, and that $D^j_\beta \in M_{D_j}(\mathbb{C})$ is indexed by a block j and a word β of some fixed length K. If, for every word ρ of length M, every word β of length K, and every middle word w of length m,*

$$\sum_j \operatorname{tr}\left(A^j_\beta C^j_\rho A^j_w\right) = \sum_j \operatorname{tr}\left(D^j_\beta A^j_\rho A^j_w\right),$$

then

$$A^j_\beta C^j_\rho = D^j_\beta A^j_\rho$$

for every block j and all words β, ρ. In particular, taking $D^j_\beta = X_j A^j_\beta$ gives

$$A^j_\beta C^j_\rho = (X_j A^j_\beta) A^j_\rho.$$

Proof. ✓ Proof Fix β and ρ, and set

$$\Delta_j = A^j_\beta C^j_\rho - D^j_\beta A^j_\rho.$$

The trace equality says that

$$\sum_j \operatorname{tr}(\Delta_j A^j_w) = 0$$

for every word w of length m. The same nondegenerate product trace-pairing argument as in Lemma 13.229 gives $\Delta_j = 0$ for every j. The specialization $D^j_\beta = X_j A^j_\beta$ is immediate. □

Lemma 13.231 (Fixed complementary-word compatibility from traces). Lean ✓ Stmt *Let $A^1, \dots, A^g$ be block tensors with the common word-span property at length m. Fix a complementary word ρ of length M and matrices $C^j_\rho, X_j \in M_{D_j}(\mathbb{C})$. If, for every wrapped word β of length K and every middle word w of length m,*

$$\sum_j \operatorname{tr}\left(A^j_\beta C^j_\rho A^j_w\right) = \sum_j \operatorname{tr}\left((X_j A^j_\beta) A^j_\rho A^j_w\right),$$

then

$$A^j_\beta C^j_\rho = (X_j A^j_\beta) A^j_\rho$$

for every block j and every wrapped word β.

Proof. ✓ Proof Fix β and set

$$\Delta_j = A^j_\beta C^j_\rho - (X_j A^j_\beta) A^j_\rho.$$

The trace equality gives

$$\sum_j \operatorname{tr}(\Delta_j A^j_w) = 0$$

for every word w of length m. The common word-span property and the product trace pairing imply $\Delta_j = 0$ for every j. □

Lemma 13.232 (Normalized boundary matrix identities). Lean Lean Lean Lean Lean ✓ Stmt *Let $(A_a)_{a\in I}$, $(C_a)_{a\in I}$, and $(D_b)_{b\in I}$ be matrices in $M_D(\mathbb{C})$, indexed by a finite alphabet. If*

$$\sum_a A_a A_a^\dagger = \mathbb{1}, \qquad A_b C_a = D_b A_a$$

for all $a, b \in I$, and if

$$E = \sum_a C_a A_a^\dagger,$$

then

$$D_b = A_b E, \qquad A_b C_a = A_b E A_a$$

for all $a, b \in I$.

In particular, the same calculation applies when the alphabet is the finite set of words of a fixed length and A_β denotes $A_{\beta_1} \cdots A_{\beta_K}$.

The same argument also applies when the normalized family is indexed by a finite set J, while $(A_b)_{b \in I}$ and $(D_b)_{b \in I}$ are indexed by a set I. If $(B_a)_{a \in J}$ and $(C_a)_{a \in J}$ are indexed by J, and if for all $a \in J$ and $b \in I$,

$$\sum_a B_a B_a^\dagger = \mathbb{1}, \qquad A_b C_a = D_b B_a$$

then with

$$E = \sum_a C_a B_a^\dagger$$

one has

$$D_b = A_b E, \qquad A_b C_a = A_b E B_a.$$

This includes the case where I and J are word sets of different lengths.

Proof. ✓ Proof Multiply D_b by the normalization:

$$D_b = D_b \sum_a A_a A_a^\dagger = \sum_a D_b A_a A_a^\dagger = \sum_a A_b C_a A_a^\dagger = A_b E.$$

Substituting this in $A_b C_a = D_b A_a$ gives $A_b C_a = A_b E A_a$. For two alphabets, the corresponding step is

$$D_b = D_b \sum_a B_a B_a^\dagger = \sum_a D_b B_a B_a^\dagger = \sum_a A_b C_a B_a^\dagger = A_b E,$$

where $E = \sum_a C_a B_a^\dagger$. Substituting this in $A_b C_a = D_b B_a$ gives $A_b C_a = A_b E B_a$. □

Lemma 13.233 (Explicit C^j, D^j, E^j identities from trace decompositions)**.** Lean ✓ Stmt *Let $A^1, \dots, A^g$ be block tensors with the common word-span property at length m. Fix matrices X_j, and put*

$$D_\beta^j = X_j A_\beta^j.$$

Suppose that C_ρ^j is indexed by a block j and a complementary word ρ of length M, and assume that, for every wrapped word β of length K, every complementary word ρ, and every middle word w of length m,

$$\sum_j \operatorname{tr}\!\left(A_\beta^j C_\rho^j A_w^j\right) = \sum_j \operatorname{tr}\!\left(D_\beta^j A_\rho^j A_w^j\right).$$

If each block satisfies

$$\sum_a A_a^j A_a^{j\,\dagger} = I,$$

then for every block j there is a matrix

$$E^j = \sum_\rho C_\rho^j \left(A_\rho^j\right)^\dagger$$

such that, for every wrapped word β and complementary word ρ,

$$D^j_\beta = A^j_\beta E^j, \qquad A^j_\beta C^j_\rho = A^j_\beta E^j A^j_\rho.$$

This is the calculation in [PGVWC07, Theorem 12, proof lines 1446–1451], with the adjoint on A^j_ρ required by the displayed normalization.

Proof. ✓ Proof Lemma 13.230 gives

$$A^j_\beta C^j_\rho = D^j_\beta A^j_\rho.$$

The single-letter normalization gives the word normalization

$$\sum_\rho A^j_\rho \left(A^j_\rho\right)^\dagger = I.$$

Lemma 13.232, applied to the two word alphabets, gives

$$E^j = \sum_\rho C^j_\rho \left(A^j_\rho\right)^\dagger, \qquad D^j_\beta = A^j_\beta E^j, \qquad A^j_\beta C^j_\rho = A^j_\beta E^j A^j_\rho.$$

□

Lemma 13.234 (Block-diagonal boundary C^j, D^j, E^j identities from trace decompositions)**.** Lean ✓ Stmt *Let $A^1, \ldots, A^g$ be block tensors with the common word-span property at length m. Fix $N \in \mathbb{N}$, complex numbers μ_j, and block-diagonal boundary matrices X_j, and put*

$$D^j_\beta = (\mu_j^N X_j) A^j_\beta.$$

Suppose that, for every wrapped word β of length K, every complementary word ρ of length M, and every middle word w of length m,

$$\sum_j \operatorname{tr}\left(A^j_\beta C^j_\rho A^j_w\right) = \sum_j \operatorname{tr}\left(((\mu_j^N X_j) A^j_\beta) A^j_\rho A^j_w\right).$$

If each block satisfies

$$\sum_a A^j_a A^{j\,\dagger}_a = I,$$

then for every block j there is a matrix

$$E^j = \sum_\rho C^j_\rho \left(A^j_\rho\right)^\dagger$$

such that, for every wrapped word β and complementary word ρ,

$$(\mu_j^N X_j) A^j_\beta = A^j_\beta E^j, \qquad A^j_\beta C^j_\rho = A^j_\beta E^j A^j_\rho.$$

This is the boundary-matrix calculation in [PGVWC07, Theorem 12, proof lines 1446–1451], after substituting the block-diagonal boundary matrix $D^j_\beta = (\mu_j^N X_j) A^j_\beta$.

Proof. ✓ Proof Apply Lemma 13.233 with $D^j_\beta = (\mu_j^N X_j) A^j_\beta$. □

Lemma 13.235 (Complementary-word boundary identities from a compatibility hypothesis). Lean ✓ Stmt *Let A_β be the products over wrapped words of length K, let A_ρ be the products over complementary words of length M, and let $X \in M_D(\mathbb{C})$ be a matrix. Suppose that the complementary-word products satisfy*

$$\sum_\rho A_\rho A_\rho^\dagger = \mathbb{1}$$

and that there are matrices C_ρ, indexed by complementary words, such that for every wrapped word β and complementary word ρ,

$$A_\beta C_\rho = (XA_\beta)A_\rho.$$

Then, for every complementary word ρ, with

$$E_\rho = \left(\sum_\gamma C_\gamma A_\gamma^\dagger\right) A_\rho,$$

one has, for every wrapped word β,

$$(XA_\beta)A_\rho = A_\beta E_\rho.$$

Proof. ✓ Proof Apply Lemma 13.232 with $D_\beta = XA_\beta$. It gives

$$A_\beta C_\rho = A_\beta E A_\rho$$

for $E = \sum_\gamma C_\gamma A_\gamma^\dagger$. With $E_\rho = EA_\rho$, the compatibility hypothesis gives

$$(XA_\beta)A_\rho = A_\beta C_\rho = A_\beta E A_\rho = A_\beta E_\rho.$$

□

Lemma 13.236 (Complementary-word boundary identities from trace decompositions). Lean ✓ Stmt *Let $A^1, \ldots, A^g$ be block tensors with the common word-span property at length m. Fix matrices X_j. Suppose that C_ρ^j is indexed by a block j and a complementary word ρ of length M. Assume that, for every wrapped word β of length K, every complementary word ρ, and every middle word w of length m,*

$$\sum_j \operatorname{tr}\left(A_\beta^j C_\rho^j A_w^j\right) = \sum_j \operatorname{tr}\left((X_j A_\beta^j) A_\rho^j A_w^j\right).$$

If each block satisfies

$$\sum_a A_a^j A_a^{j\dagger} = I,$$

then, for every block j and every complementary word ρ, there is a matrix $E_{j,\rho}$ such that, for every wrapped word β,

$$(X_j A_\beta^j) A_\rho^j = A_\beta^j E_{j,\rho}.$$

Proof. ✓ Proof Lemma 13.230 applied with $D_\beta^j = X_j A_\beta^j$ gives

$$A_\beta^j C_\rho^j = (X_j A_\beta^j) A_\rho^j.$$

The hypothesis $\sum_a A_a^j A_a^{j\dagger} = I$ extends to words of length M:

$$\sum_\rho A_\rho^j A_\rho^{j\dagger} = I$$

. Lemma 13.235 gives the matrices $E_{j,\rho}$. □

Lemma 13.237 (A left-boundary summand is a ground-space vector). Lean ✓ Stmt *Let A be a block tensor, and let $C_a, E \in M_D(\mathbb{C})$ satisfy*

$$A_b C_a = A_b E A_a$$

for all physical indices a, b. For a word $\sigma = (\sigma_1, \dots, \sigma_{n+2})$, define

$$\alpha(\sigma) = \operatorname{tr}\left(A_{\sigma_{n+2}} C_{\sigma_1} A_{\sigma_2} \cdots A_{\sigma_{n+1}}\right).$$

Then $\alpha \in G_{n+2}(A)$.

Proof. ✓ Proof The boundary identity gives

$$A_{\sigma_{n+2}} C_{\sigma_1} = A_{\sigma_{n+2}} E A_{\sigma_1}.$$

Hence, by cyclicity of the trace,

$$\alpha(\sigma) = \operatorname{tr}\left(A_{\sigma_1} A_{\sigma_2} \cdots A_{\sigma_{n+2}} E\right),$$

which is the ground-space parametrization of E. □

Lemma 13.238 (Left-boundary summands are ground-space vectors). Lean ✓ Stmt *Let $A^1, \dots, A^g$ be block tensors, and let $C^j_a, E^j \in M_{D_j}(\mathbb{C})$ satisfy*

$$A^j_b C^j_a = A^j_b E^j A^j_a$$

for every block j and all physical indices a, b. For a word $\sigma = (\sigma_1, \dots, \sigma_{n+2})$, define

$$\alpha_j(\sigma) = \operatorname{tr}\left(A^j_{\sigma_{n+2}} C^j_{\sigma_1} A^j_{\sigma_2} \cdots A^j_{\sigma_{n+1}}\right).$$

Then

$$\sum_j \alpha_j \in \bigvee_j G_{n+2}(A^j).$$

Proof. ✓ Proof Fix j. By the assumed boundary identity and cyclicity of the trace,

$$\operatorname{tr}\left(A^j_{\sigma_{n+2}} C^j_{\sigma_1} A^j_{\sigma_2} \cdots A^j_{\sigma_{n+1}}\right) = \operatorname{tr}\left(A^j_{\sigma_1} A^j_{\sigma_2} \cdots A^j_{\sigma_{n+2}} E^j\right).$$

Thus α_j is the image of E^j under the parametrization of $G_{n+2}(A^j)$. Hence each α_j lies in $G_{n+2}(A^j)$, and the finite sum lies in the supremum of these subspaces. □

Lemma 13.239 (Left-boundary trace decompositions lie in the block ground spaces). Lean ✓ Stmt *Let $A^1, \dots, A^g$ be block tensors with common word-span separation at length n, and suppose that*

$$\sum_a A^j_a A^{j\dagger}_a = \mathbb{1}$$

for every block j. Suppose a vector $\psi \in (\mathbb{C}^d)^{\otimes(n+2)}$ has the left-boundary trace decomposition

$$\psi = \sum_j \alpha_j, \qquad \alpha_j(i_1, \dots, i_{n+2}) = \operatorname{tr}\left(A^j_{i_{n+2}} C^j_{i_1} A^j_{i_2} \cdots A^j_{i_{n+1}}\right),$$

and suppose that the two trace decompositions agree for every middle word:

$$\sum_j \operatorname{tr}\left(A^j_b C^j_a A^j_w\right) = \sum_j \operatorname{tr}\left(D^j_b A^j_a A^j_w\right).$$

Then

$$\psi \in \bigvee_j G_{n+2}(A^j).$$

Proof. ✓ Proof The trace-decomposition equality and the common word-span hypothesis give

$$A_b^j C_a^j = D_b^j A_a^j$$

for every j, a, b. By the normalization,

$$D_b^j = D_b^j \sum_a A_a^j A_a^{j\dagger} = \sum_a A_b^j C_a^j A_a^{j\dagger} = A_b^j E^j, \qquad E^j := \sum_a C_a^j A_a^{j\dagger}.$$

Hence $A_b^j C_a^j = A_b^j E^j A_a^j$. Lemma 13.238 then gives

$$\psi = \sum_j \alpha_j \in \bigvee_j G_{n+2}(A^j).$$

□

Lemma 13.240 (One-step block intersection from block restrictions)**.** Lean ✓ Stmt *Let $A^1, \dots, A^g$ be block tensors with common word-span separation at length n, and suppose*

$$\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$$

for every block j. If $\psi \in (\mathbb{C}^d)^{\otimes(n+2)}$ satisfies

$$\psi(a, -) \in \bigvee_j G_{n+1}(A^j), \qquad \psi(-, b) \in \bigvee_j G_{n+1}(A^j)$$

for every physical index a, b, then

$$\psi \in \bigvee_j G_{n+2}(A^j).$$

Proof. ✓ Proof Choose finite decompositions of the fixed-boundary vectors:

$$\psi(a, i_2, \dots, i_{n+2}) = \sum_j \operatorname{tr}\left(A_{i_2}^j \cdots A_{i_{n+2}}^j C_a^j\right),$$

and

$$\psi(i_1, \dots, i_{n+1}, b) = \sum_j \operatorname{tr}\left(A_{i_1}^j \cdots A_{i_{n+1}}^j D_b^j\right).$$

Evaluating these two decompositions on (a, w, b), where w has length n, gives

$$\sum_j \operatorname{tr}\left(A_b^j C_a^j A_w^j\right) = \sum_j \operatorname{tr}\left(D_b^j A_a^j A_w^j\right).$$

The first decomposition also gives

$$\psi = \sum_j \alpha_j, \qquad \alpha_j(i_1, \dots, i_{n+2}) = \operatorname{tr}\left(A_{i_{n+2}}^j C_{i_1}^j A_{i_2}^j \cdots A_{i_{n+1}}^j\right).$$

Lemma 13.239 therefore gives

$$\psi \in \bigvee_j G_{n+2}(A^j).$$

□

Lemma 13.241 (One-step block intersection characterization). Lean ✓ Stmt *Under the hypotheses of Lemma 13.240, an $(n+2)$-site vector satisfies*

$$\psi \in \bigvee_j G_{n+2}(A^j)$$

if and only if

$$\psi(a,-) \in \bigvee_j G_{n+1}(A^j), \qquad \psi(-,b) \in \bigvee_j G_{n+1}(A^j)$$

for every physical index a, b.

Proof. ✓ Proof The reverse implication is Lemma 13.240. For the forward implication, write

$$\psi = \sum_j \gamma_j, \qquad \gamma_j \in G_{n+2}(A^j).$$

Each γ_j has fixed-boundary restrictions in $G_{n+1}(A^j)$:

$$\gamma_j(a,-) \in G_{n+1}(A^j), \qquad \gamma_j(-,b) \in G_{n+1}(A^j).$$

Taking the finite sum over j gives the two restrictions of ψ in $\bigvee_j G_{n+1}(A^j)$. □

Lemma 13.242 (Subspace form of the one-step block intersection). Lean ✓ Stmt *Under the hypotheses of Lemma 13.240, set*

$$S_n := \bigvee_j G_{n+1}(A^j).$$

Then

$$\{\psi : \forall b,\ \psi(-,b) \in S_n\} \cap \{\psi : \forall a,\ \psi(a,-) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

Proof. ✓ Proof Lemma 13.241 gives the displayed equivalence for each vector ψ. Interpreting the two sides of that equivalence as subspaces gives the stated equality. □

13.13.2 Eventual block-intersection consequences

Propagating the single-length product span through a period window gives the block-intersection identity for all sufficiently large n, both from a general period window and from the normalized BNT representative hypotheses. The full product span at length n also makes the sum of the local spaces $G_n(A^j)$ direct.

Lemma 13.243 (Period-window form of the block intersection identity). Lean ✓ Stmt *Let $p > 0$. Suppose the length-p simultaneous block-word tuples span $\prod_j M_{D_j}(\mathbb{C})$, and suppose that for every $0 \le r < p$, the length-$s + r$ simultaneous block-word tuples span $\prod_j M_{D_j}(\mathbb{C})$. Assume also the normalization*

$$\sum_a A_a^j A_a^{j\,\dagger} = I$$

for every block j. Then there is a length N such that, for every $n \ge N$, if

$$S_n := \bigvee_j G_{n+1}(A^j),$$

then

$$\{\psi : \forall b,\ \psi(-,b) \in S_n\} \cap \{\psi : \forall a,\ \psi(a,-) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

Proof. ✓ Proof Lemma 13.224 gives a length N such that the simultaneous block-word tuples span $\prod_j M_{D_j}(\mathbb{C})$ at every length $n \ge N$. For each such n, Lemma 13.242 applies and gives the displayed subspace equality. $\square$

Lemma 13.244 (BNT block separation gives one block-intersection step)**.** Lean ✓ Stmt *Under the hypotheses of Lemma 13.226, assume in addition the normalization*

$$\sum_a A_a^j A_a^{j\,\dagger} = I$$

for every block j. Let

$$n = (L_0 + 1) + (r-1)3(L_0+1), \qquad S_n = \bigvee_j G_{n+1}(A^j).$$

Then

$$\{\psi : \forall b,\ \psi(-,b) \in S_n\} \cap \{\psi : \forall a,\ \psi(a,-) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

Proof. ✓ Proof Lemma 13.226 gives

$$\mathrm{span}\{(A_w^1, \dots, A_w^r) : |w| = n\} = \prod_j M_{D_j}(\mathbb{C}).$$

Substituting this span into Lemma 13.242, together with the normalization, gives the stated subspace equality. $\square$

Lemma 13.245 (BNT block separation with a block-injectivity window)**.** Lean ✓ Stmt *Let $A^1, \dots, A^r$ be normalized representatives of a basis of normal tensors: each representative is irreducible, the family is left-canonical, the self-overlaps are normalized, and no two distinct same-dimensional representatives are gauge-phase equivalent. Assume $L > 1$, and assume that, for every block j,*

$$\mathrm{span}\{A_a^j : a = 0, \dots, d{-}1\} = M_{D_j}(\mathbb{C}), \qquad \mathrm{span}\{A_u^j : |u| = L\} = M_{D_j}(\mathbb{C}), \qquad \mathrm{span}\{A_v^j : |v| = 3L\} = M_{D_j}(\mathbb{C}).$$

Set

$$S = 3L, \qquad q = (r-1)S.$$

Suppose that $p > 0$ and let s_0 be a starting length. Assume that, for every block j,

$$\mathrm{span}\{A_u^j : |u| = p\} = M_{D_j}(\mathbb{C}),$$

and that, for every $0 \le s < p + q$ and every block j,

$$\mathrm{span}\{A_v^j : |v| = s_0 + s\} = M_{D_j}(\mathbb{C}).$$

Assume also the normalization

$$\sum_a A_a^j A_a^{j\,\dagger} = I$$

for every block j. Then there is N such that, for every $n \ge N$, if

$$S_n := \bigvee_j G_{n+1}(A^j),$$

then

$$\{\psi : \forall b,\ \psi(-,b) \in S_n\} \cap \{\psi : \forall a,\ \psi(a,-) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

Proof. ✓ Proof The direct-sum dimension step gives, for each ordered pair of distinct blocks $k \neq j$, coefficients $c_\omega^{k,j}$ of length S such that

$$\sum_{|\omega|=S} c_\omega^{k,j} A_\omega^k = I, \qquad \sum_{|\omega|=S} c_\omega^{k,j} A_\omega^j = 0.$$

Multiplying these equations over the $r-1$ blocks $j \neq k$ gives coefficients $c_\omega^{(k)}$ of total length q satisfying

$$\sum_{|\omega|=q} c_\omega^{(k)} A_\omega^j = \begin{cases} I, & j = k, \\ 0, & j \neq k. \end{cases}$$

Combining this block-separating equation with block injectivity at length p gives

$$\operatorname{span}\{(A_w^1, \dots, A_w^r) : |w| = p+q\} = \prod_j M_{D_j}(\mathbb{C}).$$

Combining the same separating equations with the assumed block-injectivity window gives the full product span at lengths s_0+q+s, for $0 \le s < p+q$. The period-window form of Lemma 13.243 then gives the displayed equality for all sufficiently large n. □

Lemma 13.246 (Normalized block-separating equations give all large product spans)**.** Lean Lean ✓ Stmt *Let $A^1, \dots, A^r$ be blocks satisfying the normalization*

$$\sum_a A_a^j A_a^{j\dagger} = I$$

and assume that every block is injective at length L:

$$\operatorname{span}\{A_u^j : |u| = L\} = M_{D_j}(\mathbb{C}).$$

If there are block-separating equations of length S, then, for every $n \ge L + (r-1)S$,

$$\operatorname{span}\{(A_w^1, \dots, A_w^r) : |w| = n\} = \prod_j M_{D_j}(\mathbb{C}).$$

In particular, for normalized BNT representatives satisfying the displayed unital normalization, if $L_0 > 0$ and $\Gamma_{L_0}^{A^j}$ is injective for every block j, the same conclusion holds with

$$S = 3(L_0+1), \qquad n \ge (L_0+1) + (r-1)3(L_0+1).$$

Proof. ✓ Proof By Theorem 8.79, the normalization propagates the length-L span equality to every larger homogeneous length:

$$m \ge L \implies \operatorname{span}\{A_u^j : |u| = m\} = M_{D_j}(\mathbb{C}).$$

For $n \ge L + (r-1)S$, take $m = n - (r-1)S$. Combining the length-m block span with the $r-1$ block-separating equations gives the displayed product span at length n. In the BNT case, the injectivity of $\Gamma_{L_0}^{A^j}$ gives the full span at length L_0, and the normalization propagates it to the lengths L_0+1 and $3(L_0+1)$. Lemma 13.226 then supplies the block-separating equations with $S = 3(L_0+1)$. □

Lemma 13.247 (Product span makes the sum of local spaces direct). Lean Lean ✓ Stmt *Suppose*

$$\mathrm{span}\{(A^1_w, \dots, A^r_w) : |w| = n\} = \prod_j M_{D_j}(\mathbb{C}).$$

Then the sum of the local spaces $G_n(A^j)$ *is direct: if*

$$\phi_j \in G_n(A^j), \qquad \sum_j \phi_j = 0,$$

then $\phi_j = 0$ *for every* j. *In particular, for the normalized BNT hypotheses above, this directness holds for every*

$$n \ge (L_0 + 1) + (r - 1)3(L_0 + 1).$$

Proof. ✓ Proof Write

$$\phi_j(\sigma) = \mathrm{tr}(A^j_\sigma X_j).$$

Evaluating $\sum_j \phi_j = 0$ on each word w of length n gives

$$\sum_j \mathrm{tr}(A^j_w X_j) = 0.$$

Since $\mathrm{tr}(A^j_w X_j) = \mathrm{tr}(X_j A^j_w)$, the product-span equation gives, for every $(M_1, \dots, M_r) \in \prod_j M_{D_j}(\mathbb{C})$,

$$\sum_j \mathrm{tr}(X_j M_j) = 0.$$

Taking only the j-th component nonzero and using nondegeneracy of the trace pairing yields $X_j = 0$, hence $\phi_j = 0$. The BNT statement follows by substituting Lemma 13.246. ☐

13.14 Block-diagonal boundary conditions for periodic ground spaces

For a block-diagonal tensor $B = \bigoplus_{j=1}^g \mu_j A_j$ with $\mu_j \neq 0$, this section proves implications from explicit block-diagonal boundary identities to the component membership

$$\Gamma^{A_j}_N(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

These implications supply the boundary-condition step in the periodic ring used by [CPGSV21, Theorem IV.16].

Lemma 13.248 (Normalized BNT representatives give the large-length block intersection). Lean Lean Lean ✓ Stmt *Let* $A^1, \dots, A^r$ *be normalized representatives of a basis of normal tensors: each representative is irreducible, the family is left-canonical, the self-overlaps are normalized, and no two distinct same-dimensional representatives are gauge-phase equivalent. Suppose* $L_0 > 0$, *and for every block* j *the map*

$$\Gamma^{A^j}_{L_0} : M_{D_j}(\mathbb{C}) \to (\mathbb{C}^d)^{\otimes L_0}$$

is injective. Assume also the normalization

$$\sum_a A^j_a A^{j\,\dagger}_a = I.$$

Set

$$N = (L_0 + 1) + (r - 1)3(L_0 + 1).$$

For every $n \geq N$, set

$$S_n := \bigvee_j G_{n+1}(A^j),$$

and the two consecutive sums are internal direct sums:

$$\bigvee_j G_{n+1}(A^j) = \bigoplus_j G_{n+1}(A^j), \qquad \bigvee_j G_{n+2}(A^j) = \bigoplus_j G_{n+2}(A^j).$$

With this notation,

$$\{\psi : \forall b,\ \psi(-,b) \in S_n\} \cap \{\psi : \forall a,\ \psi(a,-) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

Proof. ✓ Proof Lemma 13.246 gives

$$\operatorname{span}\{(A^1_w, \dots, A^r_w) : |w| = n\} = \prod_j M_{D_j}(\mathbb{C})$$

for every $n \geq N$. Substituting this product-span equation into Lemma 13.242, together with the normalization equation, gives the displayed intersection identity. Lemma 13.247 at lengths $n+1$ and $n+2$ gives the two directness statements. □

Lemma 13.249 (Normalized BNT representatives give an eventual block intersection)**.** Lean ✓ Stmt *Let $A^1, \dots, A^r$ be normalized representatives of a basis of normal tensors: each representative is irreducible, the family is left-canonical, the self-overlaps are normalized, and no two distinct same-dimensional representatives are gauge-phase equivalent. Suppose $L_0 > 0$, and for every block j the map*

$$\Gamma^{A^j}_{L_0} : M_{D_j}(\mathbb{C}) \to (\mathbb{C}^d)^{\otimes L_0}$$

is injective. Assume also the normalization

$$\sum_a A^j_a A^{j\dagger}_a = I.$$

Then there is a length N such that, for every $n \geq N$, with

$$S_n := \bigvee_j G_{n+1}(A^j),$$

one has

$$\{\psi : \forall b,\ \psi(-,b) \in S_n\} \cap \{\psi : \forall a,\ \psi(a,-) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

Proof. ✓ Proof Apply Lemma 13.248 and take

$$N = (L_0 + 1) + (r - 1)3(L_0 + 1).$$

□

Lemma 13.250 (Block-diagonal boundary-condition formula). Lean ✓ Stmt *Let*

$$B = \bigoplus_{j=1}^{g} \mu_j A_j.$$

For block-diagonal boundary conditions X_j*,*

$$\Gamma_L^B\left(\bigoplus_{j=1}^{g} X_j\right) = \sum_{j=1}^{g} \Gamma_L^{A_j}(\mu_j^L X_j).$$

Proof. ✓ Proof This is the diagonal-block trace formula specialized to a block-diagonal boundary matrix. The j-th diagonal block of $\bigoplus_k X_k$ is X_j, giving the displayed sum. □

Lemma 13.251 (One block in a block-diagonal local ground space). Lean ✓ Stmt *Let*

$$B = \bigoplus_{k=1}^{g} \mu_k A_k.$$

If $\mu_j \neq 0$*, then, for every length* L*,*

$$G_L(A_j) \subseteq G_L(B).$$

Proof. ✓ Proof A vector in $G_L(A_j)$ has the form $\Gamma_L^{A_j}(X)$. Put $\mu_j^{-L}X$ in the j-th diagonal block of a boundary matrix for B and put zero in every other block. If $\iota_j(Z)$ denotes this block insertion, Lemma 13.250 gives

$$\Gamma_L^B\big(\iota_j(\mu_j^{-L}X)\big) = \sum_{k=1}^{g} \Gamma_L^{A_k}\big(\mu_k^L(\iota_j(\mu_j^{-L}X))_{kk}\big) = \Gamma_L^{A_j}(X).$$

Hence $\Gamma_L^{A_j}(X) \in G_L(B)$. □

Lemma 13.252 (Local ground space of a block-diagonal tensor). Lean ✓ Stmt *Let*

$$B = \bigoplus_{j=1}^{g} \mu_j A_j, \qquad \mu_j \neq 0 \quad (1 \leq j \leq g).$$

Then, for every length L*,*

$$G_L(B) = \bigvee_{j=1}^{g} G_L(A_j).$$

Proof. ✓ Proof A boundary matrix X for B has diagonal blocks X_j, and block-diagonality gives

$$\Gamma_L^B(X) = \sum_{j=1}^{g} \Gamma_L^{A_j}(\mu_j^L X_j).$$

This proves $G_L(B) \subseteq \bigvee_j G_L(A_j)$. The reverse inclusion is Lemma 13.251 for each block j. □

Lemma 13.253 (Periodic constraints for a block-diagonal tensor). Lean ✓ Stmt *Let*

$$B = \bigoplus_{j=1}^{g} \mu_j A_j, \qquad \mu_j \neq 0 \quad (1 \leq j \leq g).$$

Set

$$S_M := \bigvee_{j=1}^{g} G_M(A_j).$$

If, for every $M \geq L$,

$$\mathbb{C}^d \otimes S_M \cap S_M \otimes \mathbb{C}^d = S_{M+1},$$

then

$$\mathcal{G}_{N,L}(B) \subseteq S_N.$$

Proof. ✓ Proof Lemma 13.252 gives $G_L(B) = S_L$. Apply Lemma 13.48 to the sequence S_M. □

Lemma 13.254 (Periodic constraints lie in the block local-space sum). Lean ✓ Stmt *Let $A_1, \dots, A_g$ be normalized representatives of a basis of normal tensors: each representative is irreducible, the family is left-canonical, the self-overlaps are normalized, and no two distinct same-dimensional representatives are gauge-phase equivalent. Suppose $L_0 > 0$, and for every block j the map*

$$\Gamma_{L_0}^{A^j} : M_{D_j}(\mathbb{C}) \to (\mathbb{C}^d)^{\otimes L_0}$$

is injective. Assume also the normalization

$$\sum_a A_a^j A_a^{j\dagger} = I.$$

For nonzero scalars μ_j, set

$$B = \bigoplus_{j=1}^{g} \mu_j A_j, \qquad S_M := \bigvee_{j=1}^{g} G_M(A_j).$$

If

$$L \geq (L_0 + 1) + (g-1)3(L_0+1) + 1,$$

then, for every $N \geq L$,

$$\mathcal{G}_{N,L}(B) \subseteq S_N.$$

Proof. ✓ Proof Put

$$T = (L_0 + 1) + (g-1)3(L_0+1).$$

For $M \geq L$, one has $M - 1 \geq T$. Applying Lemma 13.248 at $n = M - 1$ gives

$$\mathbb{C}^d \otimes S_M \cap S_M \otimes \mathbb{C}^d = S_{M+1}$$

for every $M \geq L$. Lemma 13.253 then propagates the local constraint from L to N. □

Lemma 13.255 (Periodic constraints and internality of the block local-space sum). Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.254, set*

$$S_N := \bigvee_{j=1}^{g} G_N(A_j).$$

Then, for every $N \geq L$,

$$\mathcal{G}_{N,L}(B) \subseteq S_N,$$

and the sum defining S_N is internal: if

$$\phi_j \in G_N(A_j), \qquad \sum_{j=1}^{g} \phi_j = 0,$$

then $\phi_j = 0$ for every j.

Proof. ✓ Proof The inclusion is Lemma 13.254. Since $N \geq L$ and

$$L \geq (L_0 + 1) + (g - 1)3(L_0 + 1) + 1,$$

the product-span directness range of Lemma 13.247 applies at length N:

$$\operatorname{span}\{(A^1_w, \dots, A^g_w) : |w| = N\} = \prod_j M_{D_j}(\mathbb{C}) \quad \Longrightarrow \quad \ker\left(\bigoplus_{j=1}^{g} G_N(A_j) \longrightarrow (\mathbb{C}^d)^{\otimes N}, \quad (\phi_j)_j \longmapsto \sum_j \phi_j\right) = 0.$$

$\square$

Lemma 13.256 (Periodic chain vectors decompose in the local block spaces)**.** Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.254, every vector*

$$\psi \in \mathcal{G}_{N,L}\left(\bigoplus_{j=1}^{g} \mu_j A_j\right)$$

has a unique family $(\psi_j)_{j=1}^{g}$ such that

$$\psi = \sum_{j=1}^{g} \psi_j, \qquad \psi_j \in G_N(A_j).$$

Proof. ✓ Proof Lemma 13.255 gives

$$\mathcal{G}_{N,L}\left(\bigoplus_{j=1}^{g} \mu_j A_j\right) \subseteq \bigvee_{j=1}^{g} G_N(A_j),$$

and the spaces $G_N(A_j)$ are independent: for every j and every $x_j \in G_N(A_j)$,

$$\sum_j x_j = 0 \quad \Longrightarrow \quad x_j = 0.$$

Thus every $\psi \in \mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$ has a unique decomposition $\psi = \sum_j \psi_j$ with $\psi_j \in G_N(A_j)$. $\square$

Lemma 13.257 (Block-diagonal boundary conditions for periodic chain vectors)**.** Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.254, every vector*

$$\psi \in \mathcal{G}_{N,L}\left(\bigoplus_{j=1}^{g} \mu_j A_j\right)$$

admits block-diagonal boundary conditions X_j such that

$$\psi = \Gamma^B_N\left(\bigoplus_{j=1}^{g} X_j\right), \qquad \Gamma^{A_j}_N(\mu_j^N X_j) \in G_N(A_j)$$

for every j.

Proof. ✓ Proof Lemma 13.256 gives

$$\psi = \sum_{j=1}^{g} \phi_j, \qquad \phi_j \in G_N(A_j).$$

Choose Y_j with $\phi_j = \Gamma_N^{A_j}(Y_j)$, and set $X_j = \mu_j^{-N} Y_j$. Since $\mu_j \neq 0$,

$$\Gamma_N^{A_j}(\mu_j^N X_j) = \phi_j.$$

The block-diagonal boundary formula then gives

$$\Gamma_N^B\left(\bigoplus_{j=1}^{g} X_j\right) = \sum_{j=1}^{g} \Gamma_N^{A_j}(\mu_j^N X_j) = \psi.$$

□

The lemma gives

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in G_N(A_j).$$

This is not enough for periodic-boundary membership when a local window crosses the chosen cut. The boundary-crossing comparison of Lemma 13.278 concerns the separate assertion

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

13.14.1 Boundary-crossing component constraints

A cyclic length-L window either stays inside the chosen linear interval ($i + L \leq N$) or crosses the boundary cut ($N < i + L$). The non-wrapping case is immediate from trace cyclicity, while the crossing case requires a boundary-matrix identity $\mu_j^N X_j A_\beta^j = A_\beta^j Y$. The lemmas below establish the single-block periodic membership $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$ in both cases.

Lemma 13.258 (Cyclic restrictions of block-diagonal boundary vectors)**.** Lean ✓ Stmt *Let $B = \bigoplus_{j=1}^{g} \mu_j A_j$, and fix block-diagonal boundary conditions X_j. For every cyclic restriction $R_{i,\tau}$ from N sites to L sites,*

$$R_{i,\tau}\left(\Gamma_N^B\left(\bigoplus_{j=1}^{g} X_j\right)\right) = \sum_{j=1}^{g} R_{i,\tau}\left(\Gamma_N^{A_j}(\mu_j^N X_j)\right).$$

Proof. ✓ Proof Apply Lemma 13.250 at length N, then use linearity of the cyclic restriction $R_{i,\tau}$. □

Lemma 13.259 (Local direct-sum constraint from block-diagonal boundary conditions)**.** Lean ✓ Stmt *Let $B = \bigoplus_{j=1}^{g} \mu_j A_j$, with $\mu_j \neq 0$, and suppose*

$$\psi = \Gamma_N^B\left(\bigoplus_{j=1}^{g} X_j\right), \qquad \psi \in \mathcal{G}_{N,L}(B).$$

Then every local constraint of ψ gives

$$\sum_{j=1}^{g} R_{i,\tau}\left(\Gamma_N^{A_j}(\mu_j^N X_j)\right) \in \bigvee_{j=1}^{g} G_L(A_j).$$

Proof. ✓ Proof The hypothesis $\psi \in \mathcal{G}_{N,L}(B)$ says that the left-hand side before decomposition lies in $G_L(B)$. Lemma 13.258 rewrites this local vector as the displayed sum, and Lemma 13.252 identifies $G_L(B)$ with $\bigvee_j G_L(A_j)$.

□

Lemma 13.260 (Cyclic windows not crossing the cut)**.** Lean ✓ Stmt *Fix block-diagonal boundary conditions X_j. If the cyclic window beginning at i stays inside the chosen linear interval, $i+L \leq N$, then for every block j,*

$$R_{i,\tau}\Big(\Gamma_N^{A_j}(\mu_j^N X_j)\Big) \in G_L(A_j).$$

Proof. ✓ Proof When the window stays inside the chosen linear interval, the boundary matrix remains outside the window. Let A_{left} be the product on the sites before the window and A_{right} the product on the sites after the window. Trace cyclicity gives

$$R_{i,\tau}\Big(\Gamma_N^{A_j}(\mu_j^N X_j)\Big) = \Gamma_L^{A_j}\big(A_{\text{right}}\, \mu_j^N X_j\, A_{\text{left}}\big)\,.$$

Hence the restricted vector lies in $G_L(A_j)$. □

Lemma 13.261 (Boundary-crossing coefficient formula)**.** Lean ✓ Stmt *Assume $0 < N$, $L \leq N$, and let the cyclic interval beginning at i cross the boundary cut, so $N < i+L$. Write $a = i+L-N$. For a local word σ of length L, set*

$$\beta = \sigma_{N-i} \cdots \sigma_{L-1}, \qquad \rho = \tau_a \cdots \tau_{i-1}, \qquad \alpha = \sigma_0 \cdots \sigma_{N-i-1}.$$

Then the corresponding coefficient of the j-th block component is

$$R_{i,\tau}\Big(\Gamma_N^{A_j}(\mu_j^N X_j)\Big)(\sigma) = \operatorname{tr}\Big(A^j_\beta A^j_\rho A^j_\alpha\, \mu_j^N X_j\Big)\,.$$

Proof. ✓ Proof Let ω be the N-site word obtained by inserting σ in the cyclic interval beginning at i and by using τ outside that interval. Since the interval crosses the cut,

$$\omega = \beta\rho\alpha.$$

Hence

$$\Gamma_N^{A_j}(\mu_j^N X_j)(\omega) = \operatorname{tr}\Big(A^j_\beta A^j_\rho A^j_\alpha\, \mu_j^N X_j\Big)\,.$$

This is exactly $R_{i,\tau}\Big(\Gamma_N^{A_j}(\mu_j^N X_j)\Big)(\sigma)$. □

Lemma 13.262 (Boundary-crossing trace decomposition from the local constraint)**.** Lean ✓ Stmt *Assume $0 < N$, $L \leq N$, and let the cyclic interval beginning at i cross the boundary cut, so $N < i+L$. Put $a = i+L-N$. For a local word σ of length L, write*

$$\beta = \sigma_{N-i} \cdots \sigma_{L-1}, \qquad \rho = \tau_a \cdots \tau_{i-1}, \qquad \alpha = \sigma_0 \cdots \sigma_{N-i-1}.$$

These are the formal cut-adapted word coordinates used to compare with the source decompositions involving $C^j_{i_1}$ and $D^j_{i_{m+1}}$. If

$$\sum_j R_{i,\tau}\Big(\Gamma_N^{A_j}(\mu_j^N X_j)\Big) \in \bigvee_j G_L(A_j),$$

then there are matrices $C^j_{i,\tau}$ such that, for every σ,

$$\sum_j \operatorname{tr}\Big(A^j_\beta C^j_{i,\tau} A^j_\alpha\Big) = \sum_j \operatorname{tr}\Big(((\mu_j^N X_j) A^j_\beta) A^j_\rho A^j_\alpha\Big)\,.$$

Proof. ✓ Proof Decompose the local vector in $\bigvee_j G_L(A_j)$ as a sum of vectors $\Gamma_L^{A_j}(C^j_{i,\tau})$. Evaluating at σ gives

$$\sum_j \operatorname{tr}\left(A^j_{\alpha\beta} C^j_{i,\tau}\right).$$

Trace cyclicity rewrites this as $\sum_j \operatorname{tr}(A^j_\beta C^j_{i,\tau} A^j_\alpha)$. On the other hand, Lemma 13.261 gives the second trace expression for the same local coefficient sum. ☐

Theorem 13.263 (Fixed boundary-crossing C^j, D^j comparison from the local block-sum constraint)**.** Lean ✓ Stmt *Assume $0 < N$, $L \leq N$, and let the cyclic interval beginning at i cross the boundary cut, so $N < i + L$. Put $a = i + L - N$ and $\rho = \tau_a \cdots \tau_{i-1}$. Suppose also that the block tensors have the common word-span property at length $N - i$. If*

$$\sum_j R_{i,\tau}\left(\Gamma_N^{A_j}(\mu_j^N X_j)\right) \in \bigvee_j G_L(A_j),$$

then there are matrices $C^j_{i,\tau}$ such that, for every block j and every word β of length a,

$$A^j_\beta C^j_{i,\tau} = ((\mu_j^N X_j) A^j_\beta) A^j_\rho.$$

This theorem is the fixed cyclic-interval coordinate form of the Perez-Garcia–Verstraete–Wolf–Cirac comparison $A^j_{i_{m+1}} C^j_{i_1} = D^j_{i_{m+1}} A^j_{i_1}$; it does not assert the full periodic-boundary ground-space statement.

Proof. ✓ Proof Lemma 13.262 gives matrices $C^j_{i,\tau}$ with

$$\sum_j \operatorname{tr}\left(A^j_\beta C^j_{i,\tau} A^j_\alpha\right) = \sum_j \operatorname{tr}\left(((\mu_j^N X_j) A^j_\beta) A^j_\rho A^j_\alpha\right)$$

for every word α of length $N - i$ and every word β of length a. The fixed complementary-word trace comparison, applied with X_j replaced by $\mu_j^N X_j$, gives the displayed matrix identity. ☐

Theorem 13.264 (Fixed boundary-crossing component in the local block space)**.** Lean ✓ Stmt *Assume $0 < N$, $L \leq N$, and let the cyclic interval beginning at i cross the boundary cut, so $N < i + L$. Suppose that the block tensors have the common word-span property at length $N - i$ and that*

$$\sum_j R_{i,\tau}\left(\Gamma_N^{A_j}(\mu_j^N X_j)\right) \in \bigvee_j G_L(A_j).$$

Then each block component of the fixed boundary-crossing restriction lies in its local block space:

$$R_{i,\tau}\left(\Gamma_N^{A_j}(\mu_j^N X_j)\right) \in G_L(A_j).$$

This is the corresponding fixed-interval block-diagonal boundary-condition consequence.

Proof. ✓ Proof Theorem 13.263 gives matrices $C^j_{i,\tau}$ with

$$A^j_\beta C^j_{i,\tau} = ((\mu_j^N X_j) A^j_\beta) A^j_\rho.$$

Thus $E = C^j_{i,\tau}$ satisfies the boundary matrix identity required in Lemma 13.265, and the displayed membership follows for every block j. ☐

Lemma 13.265 (Boundary-crossing cyclic intervals from the boundary matrix identity). Lean ✓ Stmt *Assume $0 < N$, $L \le N$, and let the cyclic interval beginning at i cross the boundary cut, so $N < i + L$. Put $a = i + L - N$. Suppose that there is a matrix E such that for every word β of length a,*

$$\mu_j^N X_j A^j_\beta A^j_{\tau_a} \cdots A^j_{\tau_{i-1}} = A^j_\beta E.$$

Then

$$R_{i,\tau}\Big(\Gamma_N^{A_j}(\mu_j^N X_j)\Big) \in G_L(A_j).$$

Proof. ✓ Proof Write a length-L word in the boundary-crossing interval as $\alpha\beta$, where α is the segment inside $\{i, \dots, N-1\}$ and β is the segment after the interval wraps past the cut, inside $\{0, \dots, a-1\}$. The full N-site word appearing in the restriction is then

$$\beta\, \tau_a \cdots \tau_{i-1}\, \alpha.$$

Trace cyclicity gives

$$\operatorname{tr}\Big(A^j_\beta A^j_{\tau_a} \cdots A^j_{\tau_{i-1}} A^j_\alpha \mu_j^N X_j\Big) = \operatorname{tr}\Big(A^j_\alpha \mu_j^N X_j A^j_\beta A^j_{\tau_a} \cdots A^j_{\tau_{i-1}}\Big).$$

The assumed boundary matrix identity changes the last expression to

$$\operatorname{tr}\Big(A^j_\alpha A^j_\beta E\Big),$$

which is the coefficient of $\Gamma_L^{A_j}(E)$ at the word $\alpha\beta$. Hence the restricted vector lies in $G_L(A_j)$. ☐

Lemma 13.266 (Last crossing-window component trace form). Lean ✓ Stmt *Write $N = M + 1$ and let the local length be $m + 2 \le M + 1$. Consider the cyclic interval beginning at the last site M. For an outside configuration τ, put*

$$\rho = \tau_{m+1} \cdots \tau_{M-1}, \qquad C^j_a = A^j_\rho A^j_a (\mu_j^{M+1} X_j).$$

Then the j-th component of the last crossing-window restriction has the left-boundary trace form

$$R_{M,\tau}\Big(\Gamma_{M+1}^{A_j}(\mu_j^{M+1} X_j)\Big) = \Phi^j, \qquad \Phi^j(a, w, b) = \operatorname{tr}\Big(A^j_b C^j_a A^j_w\Big).$$

Proof. ✓ Proof The cyclic interval beginning at M reads the last site first and then wraps to the sites $0, \dots, m$. Thus, for the local word awb, the full word in the trace is

$$w\, b\, \rho\, a.$$

Hence the j-th coefficient is

$$\operatorname{tr}\Big(A^j_w A^j_b A^j_\rho A^j_a (\mu_j^{M+1} X_j)\Big).$$

Trace cyclicity rotates this to

$$\operatorname{tr}\Big(A^j_b A^j_\rho A^j_a (\mu_j^{M+1} X_j) A^j_w\Big),$$

which is the displayed left-boundary trace form. ☐

Lemma 13.267 (Last crossing-window trace form). Lean ✓ Stmt *Write $N = M + 1$ and let the local length be $m + 2 \le M + 1$. Consider the cyclic interval beginning at the last site M. For an outside configuration τ, put*

$$\rho = \tau_{m+1} \cdots \tau_{M-1}, \qquad C^j_a = A^j_\rho A^j_a (\mu_j^{M+1} X_j).$$

Then the block sum of the last crossing-window restrictions has the left-boundary trace form

$$\sum_j R_{M,\tau}\Big(\Gamma_{M+1}^{A_j}(\mu_j^{M+1} X_j)\Big) = \sum_j \Phi^j, \qquad \Phi^j(a, w, b) = \operatorname{tr}\Big(A^j_b C^j_a A^j_w\Big).$$

Proof. ✓ Proof Sum Lemma 13.266 over the block index. $\square$

Lemma 13.268 (Last crossing-window coefficient comparison). Lean ✓ Stmt *In the setting of Lemma 13.267, assume that the length-m simultaneous word tuples span the product algebra and that*

$$\sum_j R_{M,\tau}\left(\Gamma^{A_j}_{M+1}(\mu_j^{M+1}X_j)\right) \in \bigvee_j G_{m+2}(A^j).$$

Then there are matrices E_j such that, for every block j and all boundary letters a, b,

$$A^j_b(A^j_\rho A^j_a(\mu_j^{M+1}X_j)) = A^j_b E_j A^j_a.$$

Proof. ✓ Proof Lemma 13.267 writes the local block sum in the form

$$\Phi(a,w,b) = \sum_j \operatorname{tr}\left(A^j_b C^j_a A^j_w\right), \qquad C^j_a = A^j_\rho A^j_a(\mu_j^{M+1}X_j).$$

Applying Lemma 13.228 gives the displayed blockwise equations. $\square$

Lemma 13.269 (Last crossing-window component membership). Lean ✓ Stmt *In the setting of Lemma 13.267, assume that the length-m simultaneous word tuples span the product algebra and that*

$$\sum_j R_{M,\tau}\left(\Gamma^{A_j}_{M+1}(\mu_j^{M+1}X_j)\right) \in \bigvee_j G_{m+2}(A^j).$$

Then, for every block j,

$$R_{M,\tau}\left(\Gamma^{A_j}_{M+1}(\mu_j^{M+1}X_j)\right) \in G_{m+2}(A^j).$$

Proof. ✓ Proof Lemma 13.268 gives matrices E_j with

$$A^j_b C^j_a = A^j_b E_j A^j_a.$$

By Lemma 13.266, the j-th cyclic restriction is the left-boundary component determined by C^j_a. The displayed identity makes that component equal to $\Gamma^{A_j}_{m+2}(E_j)$, hence it lies in $G_{m+2}(A^j)$. $\square$

Lemma 13.270 (Last crossing-window coefficients from a block-diagonal chain vector). Lean ✓ Stmt *Let $B = \bigoplus_j \mu_j A^j$, and assume $\mu_j \neq 0$ for every block j. Suppose*

$$\psi = \Gamma^B_{M+1}\left(\bigoplus_j X_j\right), \qquad \psi \in \mathcal{G}_{M+1,m+2}(B).$$

Assume also that the length-m simultaneous word tuples span the product algebra. Then, for every outside configuration τ, with $\rho = \tau_{m+1}\cdots\tau_{M-1}$, there are matrices E_j such that

$$A^j_b(A^j_\rho A^j_a(\mu_j^{M+1}X_j)) = A^j_b E_j A^j_a$$

for every block j and all boundary letters a, b.

Proof. ✓ Proof The local constraint of ψ at the cyclic interval beginning at M gives

$$\sum_j R_{M,\tau}\left(\Gamma^{A_j}_{M+1}(\mu_j^{M+1}X_j)\right) \in \bigvee_j G_{m+2}(A^j)$$

by Lemma 13.259. Applying Lemma 13.268 gives the displayed equations. $\square$

Lemma 13.271 (Last crossing-window component membership from a chain vector). Lean ✓ Stmt *Let $B = \bigoplus_j \mu_j A^j$, and assume $\mu_j \neq 0$ for every block j. Suppose*

$$\psi = \Gamma^B_{M+1}\left(\bigoplus_j X_j\right), \qquad \psi \in \mathcal{G}_{M+1,m+2}(B).$$

If the length-m simultaneous word tuples span the product algebra, then for every outside configuration τ and every block j,

$$R_{M,\tau}\left(\Gamma^{A_j}_{M+1}(\mu_j^{M+1} X_j)\right) \in G_{m+2}(A^j).$$

Proof. ✓ Proof Lemma 13.259 applied to the window beginning at the last site gives

$$\sum_j R_{M,\tau}\left(\Gamma^{A_j}_{M+1}(\mu_j^{M+1} X_j)\right) \in \bigvee_j G_{m+2}(A^j).$$

Lemma 13.269 then applies to each block component. □

Lemma 13.272 (Boundary-crossing intervals from a right boundary-matrix identity). Lean ✓ Stmt *Assume $0 < N$, $L \leq N$, and let the cyclic interval beginning at i cross the boundary cut, so $N < i + L$. Put $a = i + L - N$. Suppose that there is a matrix Y such that for every word β of length a,*

$$\mu_j^N X_j A^j_\beta = A^j_\beta Y.$$

Then

$$R_{i,\tau}\left(\Gamma^{A_j}_N(\mu_j^N X_j)\right) \in G_L(A_j).$$

Proof. ✓ Proof Let

$$E := Y A^j_{\tau_a} \cdots A^j_{\tau_{i-1}}.$$

Then, for every word β on the segment $0, \dots, a-1$,

$$\mu_j^N X_j A^j_\beta A^j_{\tau_a} \cdots A^j_{\tau_{i-1}} = A^j_\beta E.$$

Lemma 13.265 gives the stated local constraint. □

Lemma 13.273 (Boundary-crossing cyclic intervals suffice). Lean ✓ Stmt *Assume $0 < N$ and $L \leq N$. Fix block-diagonal boundary conditions X_j. Suppose that, for every block j, every cyclic interval beginning at i with $N < i + L$, and every configuration τ,*

$$R_{i,\tau}\left(\Gamma^{A_j}_N(\mu_j^N X_j)\right) \in G_L(A_j).$$

Then, for every block j,

$$\Gamma^{A_j}_N(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof By definition, membership in $\mathcal{G}_{N,L}(A_j)$ is the collection of all cyclic length-L local constraints. For an interval beginning at i, either $i + L \leq N$ or $N < i + L$. In the first case use Lemma 13.260. In the second case use the displayed hypothesis. □

Lemma 13.274 (Right boundary-matrix identities give single-block periodic-boundary constraints). Lean ✓ Stmt *Assume $0 < N$ and $L \le N$. Fix block-diagonal boundary conditions X_j. Suppose that, for every block j, every boundary-crossing interval beginning at i, and every configuration τ, there is a matrix $Y_{j,i,\tau}$ such that, for every word β of length $a = i + L - N$,*

$$\mu_j^N X_j A_\beta^j = A_\beta^j Y_{j,i,\tau}.$$

Then, for every block j,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof Lemma 13.272 turns the displayed boundary identity into the local constraint for each boundary-crossing interval. Lemma 13.273 then yields

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j),$$

handling both boundary-crossing intervals $N < i + L$ and non-crossing intervals $i + L \le N$. □

Lemma 13.275 (Spanning matrix identities give single-block constraints). Lean ✓ Stmt *Assume $0 < N$, $L \le N$, and that for every block j, the length-$N - L$ word products of A_j span the full matrix algebra. Suppose that, for every block j, every boundary-crossing interval beginning at i, and every outside word ρ of length $N - L$, there is a matrix $E_{j,i,\rho}$ such that for every word β before the cut, of length $i + L - N$,*

$$\mu_j^N X_j A_\beta^j A_\rho^j = A_\beta^j E_{j,i,\rho}.$$

Then, for every block j,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof For fixed j, i, apply Theorem 13.77 to the family

$$Z_\beta = \mu_j^N X_j A_\beta^j, \qquad F_\beta = A_\beta^j.$$

Since the length-$N - L$ outside-word products span the full matrix algebra, there is a single matrix $Y_{j,i}$ such that, for every β,

$$\mu_j^N X_j A_\beta^j = A_\beta^j Y_{j,i}.$$

Lemma 13.274 then gives the periodic-boundary constraints for each block. □

Lemma 13.276 (Matrix identities under block injectivity give single-block constraints). Lean ✓ Stmt *Assume $0 < N$, $L \le N$, and $L + L_0 \le N$. Suppose each block A_j is L_0-block-injective and satisfies the normalization equation*

$$\sum_a A_a^j A_a^{j\dagger} = I.$$

Suppose also that the boundary-crossing matrix identities of Lemma 13.275 hold for every outside word ρ of length $N - L$. Then, for every block j,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof Since $L + L_0 \le N$, the complementary length satisfies $L_0 \le N - L$. Theorem 8.79 therefore gives, for every block j,

$$S_{N-L}(A_j) = M_{D_j}(\mathbb{C}).$$

Apply Lemma 13.275. □

Lemma 13.277 (Normalization for words). Lean ✓ Stmt *If*

$$\sum_a A_a A_a^\dagger = I,$$

then, for every length M,

$$\sum_\rho A_\rho A_\rho^\dagger = I,$$

where ρ ranges over all words of length M.

Proof. ✓ Proof The case $M = 0$ is the identity matrix. For the induction step, write a word of length $M+1$ as $a\rho$. Since $A_{a\rho} = A_a A_\rho$,

$$\sum_{a,\rho} A_a A_\rho A_\rho^\dagger A_a^\dagger = \sum_a A_a \left(\sum_\rho A_\rho A_\rho^\dagger \right) A_a^\dagger = \sum_a A_a A_a^\dagger = I.$$

□

Lemma 13.278 (Compatibility hypotheses under block injectivity give single-block constraints). Lean ✓ Stmt *Assume $0 < N$, $L \leq N$, and $L + L_0 \leq N$. Suppose each block A_j is L_0-block-injective and satisfies*

$$\sum_a A_a^j A_a^{j\dagger} = I.$$

Fix block-diagonal boundary conditions X_j. For every boundary-crossing interval beginning at i, suppose there are matrices $C_{i,\rho}^j$, indexed by outside words ρ of length $N - L$, such that for every word β before the cut, of length $i + L - N$,

$$A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j.$$

This is the formal word-coordinate counterpart of the source C^j, D^j comparison, with $D_\beta^j = (\mu_j^N X_j) A_\beta^j$. Then, for every block j,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof The normalization equation gives, by induction on word length,

$$\sum_\rho A_\rho^j A_\rho^{j\dagger} = I$$

for words ρ of length $N - L$. Applying Lemma 13.235 with $X = \mu_j^N X_j$ gives, for every outside word ρ, a matrix $E_{j,i,\rho}$ such that for every word β before the cut,

$$((\mu_j^N X_j) A_\beta^j) A_\rho^j = A_\beta^j E_{j,i,\rho}.$$

Lemma 13.276 then gives the periodic-boundary constraint for each block. □

Lemma 13.279 (Boundary representation and crossing constraints give periodic components). Lean ✓ Stmt *Assume $0 < N$, $L \leq N$, and $L + L_0 \leq N$. Suppose each block A_j is L_0-block-injective and satisfies*

$$\sum_a A_a^j A_a^{j\dagger} = I.$$

Let

$$\psi \in \mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right), \qquad \psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right).$$

If every boundary-crossing interval has the corresponding tail-word spanning property, then there are block-diagonal boundary conditions X_j with the displayed representation of ψ and, for every j,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof For a boundary-crossing interval, the local constraint on ψ and the boundary representation give a vector in $\bigvee_j G_L(A_j)$ by Lemma 13.259. The crossing comparison theorem supplies matrices $C^j_{i,\rho}$ satisfying

$$A^j_\beta C^j_{i,\rho} = ((\mu_j^N X_j)A^j_\beta)A^j_\rho.$$

Lemma 13.278 then gives the single-block periodic constraints. □

Lemma 13.280 (Trace-decomposition comparisons give single-block constraints)**.** Lean ✓ Stmt *Assume $0 < N$, $L \leq N$, and $L + L_0 \leq N$. Suppose each block A_j is L_0-block-injective and satisfies*

$$\sum_a A^j_a A^{j\dagger}_a = I.$$

Assume also that the simultaneous tuples $w \mapsto (A^1_w, \dots, A^g_w)$ at the middle-word length m span the full product algebra $\prod_j M_{D_j}(\mathbb{C})$. Fix complex numbers μ_j and block-diagonal boundary conditions X_j. For every boundary-crossing interval beginning at i, suppose there are matrices $C^j_{i,\rho}$, indexed by outside words ρ of length $N - L$, such that for every word β before the cut, outside word ρ, and middle word w,

$$\sum_j \operatorname{tr}\left(A^j_\beta C^j_{i,\rho} A^j_w\right) = \sum_j \operatorname{tr}\left(((\mu_j^N X_j)A^j_\beta)A^j_\rho A^j_w\right).$$

Then, for every block j,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof Applying Lemma 13.234 with the same scalars μ_j and boundary matrices X_j, for every block j and every boundary-crossing interval i, this gives a matrix E^j_i such that, for every word β before the cut,

$$(\mu_j^N X_j)A^j_\beta = A^j_\beta E^j_i.$$

Hence for every outside word ρ, taking $E_{j,i,\rho} = E^j_i A^j_\rho$ gives

$$((\mu_j^N X_j)A^j_\beta)A^j_\rho = A^j_\beta E_{j,i,\rho}.$$

These are precisely the block-diagonal boundary-condition equations of Lemma 13.276, which gives the periodic-boundary constraint for each block. □

Lemma 13.281 (Trace-decomposition comparisons give periodic-boundary single-block states)**.** Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.257, assume in addition that $L+L_0 \leq N$. Assume also that, for some middle-word length m, the simultaneous tuples $w \mapsto (A^1_w, \dots, A^g_w)$,* *where w ranges over words of length m in $\{0, \dots, d-1\}$, span the full product algebra $\prod_j M_{D_j}(\mathbb{C})$. Let*

$$\psi \in \mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right).$$

Suppose that whenever

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right),$$

there are matrices $C^j_{i,\rho}$ such that, for every boundary-crossing interval i, every outside word ρ of length $N-L$, every word β before the cut, and every word w of length m,

$$\sum_j \operatorname{tr}\left(A^j_\beta C^j_{i,\rho} A^j_w\right) = \sum_j \operatorname{tr}\left(((\mu_j^N X_j)A^j_\beta)A^j_\rho A^j_w\right).$$

Then there are block-diagonal boundary conditions X_j with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right)$$

and, for every j,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof By Lemma 13.257 there are matrices X_j with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right).$$

For this representation, the assumed trace-decomposition comparison gives matrices $C^j_{i,\rho}$; here the source matrix D^j_β has already been specialized to $(\mu_j^N X_j)A^j_\beta$. Lemma 13.280 then gives, for every j,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

□

Lemma 13.282 (BNT product span gives trace-decomposition periodic components)**.** Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.257, assume in addition that $L + L_0 \le N$ and*

$$L \ge (L_0+1) + (g-1)((L_0+1) + ((L_0+1) + (L_0+1))) + 1.$$

Put

$$m = (L_0+1) + (g-1)((L_0+1) + ((L_0+1) + (L_0+1))).$$

Let

$$\psi \in \mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right).$$

Suppose that whenever

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right),$$

there are matrices $C^j_{i,\rho}$ such that, for every boundary-crossing interval i, every outside word ρ of length $N-L$, every word β before the cut, and every word w of length m,

$$\sum_j \operatorname{tr}\left(A^j_\beta C^j_{i,\rho} A^j_w\right) = \sum_j \operatorname{tr}\left(((\mu_j^N X_j) A^j_\beta) A^j_\rho A^j_w\right).$$

Then there are block-diagonal boundary conditions X_j with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right)$$

and, for every j,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof Lemma 13.246 gives

$$\operatorname{span}\{(A^1_w, \dots, A^g_w) : |w| = m\} = \prod_j M_{D_j}(\mathbb{C}).$$

Applying Lemma 13.281 with this middle-word span gives the stated block-diagonal boundary conditions and the periodic-boundary constraint in each block. ☐

Lemma 13.283 (C^j boundary comparison gives periodic single-block states)**.** Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.257, assume in addition that $L + L_0 \le N$. Let*

$$\psi \in \mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right).$$

Suppose that whenever

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right),$$

there are matrices $C^j_{i,\rho}$ such that, for every boundary-crossing interval i, every outside word ρ of length $N-L$, and every word β before the cut,

$$A^j_\beta C^j_{i,\rho} = ((\mu_j^N X_j) A^j_\beta) A^j_\rho.$$

Then there are block-diagonal boundary conditions X_j with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right)$$

and, for every j,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof By Lemma 13.257 there are matrices X_j with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right).$$

For these boundary conditions, the assumed C^j boundary comparison supplies the matrices $C^j_{i,\rho}$. Lemma 13.278 gives, for every j,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

☐

Lemma 13.284 (Matrix identities give periodic single-block states). Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.257, assume in addition that $L + L_0 \le N$. Let*

$$\psi \in \mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right).$$

Suppose that whenever

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right),$$

the corresponding boundary conditions satisfy the block-diagonal boundary-condition equations: for every boundary-crossing interval i, every word β before the cut, and every outside word ρ of length $N - L$,

$$\mu_j^N X_j A_\beta^j A_\rho^j = A_\beta^j E_{j,i,\rho}.$$

Then there are block-diagonal boundary conditions X_j with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right)$$

and, for every j,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof By Lemma 13.257 there are matrices X_j with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right).$$

These X_j satisfy the assumed identities, so Lemma 13.276 gives, for every j,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

□

13.14.2 Periodic-boundary ground space for a block-diagonal tensor

The periodic ground-space theorem states that, for sufficiently long periodic systems, the periodic-boundary ground space of a parent Hamiltonian is generated by a basis of normal tensors of the initial tensor A [CPGSV21, Theorem IV.16]. The lemmas in this subsection are conditional reductions toward that statement. Their hypotheses explicitly include a boundary inclusion, a block-diagonal boundary representation, or periodic-boundary membership for the single-block states; in the source theorem these conditions are obtained from the boundary-condition comparison in the periodic ring. The source proof [PGVWC07] writes this comparison with $C_{i_1}^j$, $D_{i_{m+1}}^j$, and the normalized matrix E^j; the symbols β and ρ below are formal word variables introduced after opening the periodic cut, not additional terminology from the paper. In these coordinates the source comparison becomes

$$A_\beta^j C_{i,\rho}^j = (\mu_j^N X_j A_\beta^j) A_\rho^j,$$

which is the formal word-coordinate counterpart of $A_{i_{m+1}}^j C_{i_1}^j = D_{i_{m+1}}^j A_{i_1}^j$ with $D_\beta^j = (\mu_j^N X_j) A_\beta^j$.

Lemma 13.285 (The block-diagonal periodic-boundary space lies between two block sums). Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.254, one has*

$$\bigvee_{j=1}^{g} \mathcal{G}_{N,L}(A_j) \subseteq \mathcal{G}_{N,L}\left(\bigoplus_{j=1}^{g} \mu_j A_j\right) \subseteq \bigvee_{j=1}^{g} G_N(A_j),$$

and the sum defining the right-hand side is internal.

Proof. ✓ Proof The first inclusion is Lemma 13.208. The second inclusion and internal directness are Lemma 13.255. ☐

Lemma 13.286 (Boundary inclusion gives the block-diagonal periodic-boundary equality). Lean ✓ Stmt *Let*

$$B = \bigoplus_{j=1}^{g} \mu_j A_j, \qquad \mu_j \neq 0 \quad (1 \leq j \leq g).$$

If the periodic-boundary comparison with block-diagonal boundary conditions gives

$$\mathcal{G}_{N,L}(B) \subseteq \bigvee_{j=1}^{g} \mathcal{G}_{N,L}(A_j),$$

then

$$\mathcal{G}_{N,L}(B) = \bigvee_{j=1}^{g} \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof The reverse inclusion is Lemma 13.208; combine the two inclusions. ☐

Lemma 13.287 (Componentwise periodic-boundary decomposition gives the reverse inclusion). Lean ✓ Stmt *Let*

$$B = \bigoplus_{j=1}^{g} \mu_j A_j.$$

Suppose that every vector $\psi \in \mathcal{G}_{N,L}(B)$ can be written as

$$\psi = \sum_{j=1}^{g} \psi_j, \qquad \psi_j \in \mathcal{G}_{N,L}(A_j).$$

Then

$$\mathcal{G}_{N,L}(B) \subseteq \bigvee_{j=1}^{g} \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof For $\psi \in \mathcal{G}_{N,L}(B)$, choose the displayed decomposition. Each summand lies in the corresponding subspace $\mathcal{G}_{N,L}(A_j)$, hence their finite sum lies in $\bigvee_j \mathcal{G}_{N,L}(A_j)$. ☐

Lemma 13.288 (Block-diagonal boundary vectors in the periodic-boundary block sum). Lean ✓ Stmt *Let $B = \bigoplus_{j=1}^{g} \mu_j A_j$. Fix block-diagonal boundary conditions X_j. Suppose that, for every j,*

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Then

$$\Gamma_N^{B}\left(\bigoplus_{j=1}^{g} X_j\right) \in \bigvee_{j=1}^{g} \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof The block-diagonal boundary formula gives

$$\Gamma_N^B\left(\bigoplus_{j=1}^{g} X_j\right) = \sum_{j=1}^{g} \Gamma_N^{A_j}(\mu_j^N X_j).$$

Each summand belongs to the corresponding block periodic-boundary space by assumption, so the finite sum lies in the linear span of these spaces. □

Lemma 13.289 (Block-diagonal boundary representation gives the reverse inclusion)**.** Lean ✓ Stmt *Let $B = \bigoplus_{j=1}^{g} \mu_j A_j$. Suppose every $\psi \in \mathcal{G}_{N,L}(B)$ admits block-diagonal boundary conditions X_j such that*

$$\psi = \Gamma_N^B\left(\bigoplus_{j=1}^{g} X_j\right),$$

and, for every j,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Then

$$\mathcal{G}_{N,L}(B) \subseteq \bigvee_{j=1}^{g} \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof Apply Lemma 13.288 to the block-diagonal boundary representation of each $\psi \in \mathcal{G}_{N,L}(B)$. □

Lemma 13.290 (Block-diagonal boundary representation reduces to periodic-boundary equality)**.** Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.254, suppose every vector $\psi \in \mathcal{G}_{N,L}(B)$ has block-diagonal boundary conditions X_j such that*

$$\psi = \Gamma_N^B\left(\bigoplus_{j=1}^{g} X_j\right),$$

and, for every j,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Then

$$\mathcal{G}_{N,L}(B) = \bigvee_{j=1}^{g} \mathcal{G}_{N,L}(A_j),$$

and the sum defining $\bigvee_j G_N(A_j)$ is internal.

Proof. ✓ Proof The block-diagonal boundary representation gives

$$\mathcal{G}_{N,L}(B) \subseteq \bigvee_{j=1}^{g} \mathcal{G}_{N,L}(A_j)$$

by Lemma 13.289. The equality follows from Lemma 13.286; the internality statement is Lemma 13.255. □

Lemma 13.291 (Boundary-condition identities reduce to periodic-boundary equality). Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.290, assume in addition that $L + L_0 \le N$. Assume further that, for every*

$$\psi \in \mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right)$$

and every block-diagonal boundary representation

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right),$$

the block-diagonal boundary-condition equations hold: for every block j, boundary-crossing interval i, word β before the cut, and outside word ρ of length $N - L$, there is a matrix $E_{j,i,\rho}$ such that

$$\mu_j^N X_j A_\beta^j A_\rho^j = A_\beta^j E_{j,i,\rho}.$$

Then

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the N-site ground spaces $G_N(A_j) = \mathcal{G}_{N,N}(A_j)$ have internal sum.

Proof. ✓ Proof For each ψ, Lemma 13.284 gives matrices X_j with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right)$$

and, for every j, the membership

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$$

holds. Lemma 13.290 then gives

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the internality of the sum of the $G_N(A_j)$. □

Lemma 13.292 (C^j, D^j boundary-condition comparisons reduce to periodic-boundary equality). Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.290, assume in addition that $L + L_0 \le N$. Assume further that, for every*

$$\psi \in \mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right)$$

and every block-diagonal boundary representation

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right),$$

there are matrices $C^j_{i,\rho}$ such that, for every boundary-crossing interval i, every outside word ρ of length $N-L$, and every word β before the cut,

$$A^j_\beta C^j_{i,\rho} = ((\mu_j^N X_j)A^j_\beta)A^j_\rho.$$

Then

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the N-site ground spaces $G_N(A_j) = \mathcal{G}_{N,N}(A_j)$ have internal sum.

Proof. ✓ Proof For each ψ, Lemma 13.283 gives matrices X_j with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right)$$

and, for every j,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Lemma 13.290 then gives

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the internality of the sum of the $G_N(A_j)$. □

Lemma 13.293 (C^j, D^j comparisons give the periodic block ground-space equality)**.** Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.292, one has*

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \bigvee_j \mathcal{G}_{N,L}(A_j).$$

This records only the ground-space equality conclusion obtained from the C^j, D^j boundary-condition comparison in [PGVWC07, Theorem 12, proof lines 1446–1456]; the independence of the N-site spaces is a separate conclusion of Lemma 13.292. The statement is still in the length-L_0 injectivity range displayed in that lemma, not the shorter range in [PGVWC07, Theorem 12].

Proof. ✓ Proof The claim is the equality part of Lemma 13.292. □

Lemma 13.294 (C^j, D^j comparisons at boundary-crossing intervals reduce to periodic-boundary equality)**.** Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.290, assume in addition that $L + L_0 \le N$. Assume also that, for every boundary-crossing interval beginning at i, the simultaneous products*

$$w \longmapsto (A^1_w, \dots, A^g_w), \qquad |w| = N - i,$$

span the product algebra $\prod_j M_{D_j}(\mathbb{C})$. Then

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the N-site ground spaces $G_N(A_j) = \mathcal{G}_{N,N}(A_j)$ have internal sum. The present formal statement is the equality-level consequence of the boundary-condition comparison in [PGVWC07, Theorem 12, proof lines 1436–1451] in the length-L_0 injectivity range

$$L \geq (L_0 + 1) + 3(g - 1)(L_0 + 1) + 1.$$

The source theorem has the shorter bound

$$L \geq 3(g - 1)(L_0 + 1) + 1.$$

Proof. ✓ Proof For each ψ, Lemma 13.290 gives block-diagonal boundary conditions X_j. For a boundary-crossing interval, Lemma 13.259 puts the sum of the restricted block components in $\bigvee_j G_L(A_j)$. The stated product-span hypothesis and Theorem 13.263 give the matrices $C^j_{i,\rho}$ satisfying

$$A^j_\beta C^j_{i,\rho} = ((\mu_j^N X_j)A^j_\beta)A^j_\rho.$$

Lemma 13.278 then gives

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$$

for every j. Applying Lemma 13.290 again gives the displayed equality and internality. □

Lemma 13.295 (Boundary-crossing comparisons give the periodic block ground-space equality)**.** Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.294, one has*

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \bigvee_j \mathcal{G}_{N,L}(A_j).$$

This records only the ground-space equality conclusion of [PGVWC07, Theorem 12]; the independence of the N-site spaces is a separate conclusion of Lemma 13.294. The statement is still in the length-L_0 injectivity range displayed in that lemma, not the shorter range in [PGVWC07, Theorem 12].

Proof. ✓ Proof The claim is the equality part of Lemma 13.294. □

Lemma 13.296 (Trace-decomposition comparisons reduce to periodic-boundary equality)**.** Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.290, assume in addition that $L + L_0 \leq N$. Assume also that, for some middle-word length m, the simultaneous tuples $w \mapsto (A^1_w, \dots, A^g_w)$, where w ranges over words of length m in $\{0, \dots, d-1\}$, span the full product algebra $\prod_j M_{D_j}(\mathbb{C})$. Assume further that, for every*

$$\psi \in \mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right)$$

and every block-diagonal boundary representation

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right),$$

there are matrices $C^j_{i,\rho}$ such that, for every boundary-crossing interval i, every outside word ρ of length $N - L$, every word β before the cut, and every word w of length m,

$$\sum_j \mathrm{tr}\left(A^j_\beta C^j_{i,\rho} A^j_w\right) = \sum_j \mathrm{tr}\left(((\mu_j^N X_j)A^j_\beta)A^j_\rho A^j_w\right).$$

Then

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the N-site ground spaces $G_N(A_j) = \mathcal{G}_{N,N}(A_j)$ have internal sum.

Proof. ✓ Proof For each ψ, Lemma 13.281 gives matrices X_j with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right)$$

and, for every j,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Lemma 13.290 then gives

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the internality of the sum of the $G_N(A_j)$. □

Lemma 13.297 (BNT product span gives trace-decomposition ground-space equality)**.** Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.290, assume in addition that $L + L_0 \leq N$ and*

$$L \geq (L_0 + 1) + (g - 1)((L_0 + 1) + ((L_0 + 1) + (L_0 + 1))) + 1.$$

Put

$$m = (L_0 + 1) + (g - 1)((L_0 + 1) + ((L_0 + 1) + (L_0 + 1))).$$

Assume that, for every

$$\psi \in \mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right)$$

and every block-diagonal boundary representation

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right),$$

there are matrices $C^j_{i,\rho}$ such that, for every boundary-crossing interval i, every outside word ρ of length $N - L$, every word β before the cut, and every word w of length m,

$$\sum_j \operatorname{tr}\left(A^j_\beta C^j_{i,\rho} A^j_w\right) = \sum_j \operatorname{tr}\left(((\mu_j^N X_j) A^j_\beta) A^j_\rho A^j_w\right).$$

Then

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the N-site ground spaces $G_N(A_j) = \mathcal{G}_{N,N}(A_j)$ have internal sum.

Proof. ✓ Proof Lemma 13.246 gives

$$\operatorname{span}\{(A^1_w, \dots, A^g_w) : |w| = m\} = \prod_j M_{D_j}(\mathbb{C}).$$

Applying Lemma 13.296 with this middle-word span gives the equality of periodic-boundary ground spaces and the internality of the sum of the N-site spaces. □

Lemma 13.298 (Boundary decomposition reduces to periodic-boundary equality). Lean ✓ Stmt
Let $A_1, \dots, A_g$ be normalized representatives of a basis of normal tensors: each representative is irreducible, the family is left-canonical, the self-overlaps are normalized, and no two distinct same-dimensional representatives are gauge-phase equivalent, and let

$$B = \bigoplus_{j=1}^{g} \mu_j A_j, \qquad \mu_j \neq 0.$$

Assume each block is injective at length L_0, assume $\sum_a A_a^j A_a^{j\dagger} = I$ for every j, and suppose

$$L \geq (L_0 + 1) + (g-1)3(L_0+1) + 1, \qquad N \geq L.$$

If every vector $\psi \in \mathcal{G}_{N,L}(B)$ can be written as

$$\psi = \sum_{j=1}^{g} \psi_j, \qquad \psi_j \in \mathcal{G}_{N,L}(A_j),$$

then

$$\mathcal{G}_{N,L}(B) = \bigvee_{j=1}^{g} \mathcal{G}_{N,L}(A_j),$$

and the sum $\bigvee_j G_N(A_j)$ is internal.

Proof. ✓ Proof Lemma 13.287 gives

$$\mathcal{G}_{N,L}(B) \subseteq \bigvee_{j=1}^{g} \mathcal{G}_{N,L}(A_j).$$

The blockwise inclusion gives the reverse containment, hence equality. The internality of $\bigvee_j G_N(A_j)$ is the directness conclusion of Lemma 13.255. □

13.15 Periodic-boundary ground space generated by a basis of normal tensors

For a block-diagonal tensor $\bigoplus_j \mu_j A_j$ with $\mu_j \neq 0$, this section relates three ground-space statements. Each block A_j contributes the local space $G_m(A_j)$ on an open length-m segment, and we write $S_m := \sum_{j=1}^{g} G_m(A_j)$ for their linear sum; internal directness is proved only under the BNT separation hypotheses and the stated length ranges. On the N-site ring the parent Hamiltonian has the periodic-boundary ground space $\mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$. The results below establish the open-segment recursion $\mathbb{C}^d \otimes S_m \cap S_m \otimes \mathbb{C}^d = S_{m+1}$ and, conditional on the block-diagonal boundary-condition comparison in the periodic ring, reduce the periodic-boundary ground space to the span of the vectors $V^{(N)}(A_j)$ associated to the chosen basis of normal tensors:

$$\operatorname{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

Theorem 13.299 (Block-diagonal local recursion and boundary-condition comparison). Not ready
Let $A_1, \dots, A_g$ be the blocks in a block-injective canonical form, and assume that their MPV families are pairwise different:

$$j \neq k \quad \Longrightarrow \quad \exists n \geq 1,\ V^{(n)}(A_j) \neq V^{(n)}(A_k).$$

Let L_0 be a common injectivity length. Assume the multi-block range used in [PGVWC07, Theorem 12]:

$$g \geq 2, \qquad L_0 \geq 1, \qquad L \geq 3(g-1)(L_0+1)+1.$$

Set

$$S_m := \bigoplus_{j=1}^{g} G_m(A_j).$$

The open-segment recursion in [PGVWC07, Theorem 12] says that, for every $m \geq L$,

$$\mathbb{C}^d \otimes S_m \cap S_m \otimes \mathbb{C}^d = S_{m+1}.$$

Let $H_{S_L,N}$ denote the periodic parent Hamiltonian whose local ground space is S_L. The periodic-boundary conclusion of the same theorem is

$$\ker H_{S_L,N} = \operatorname{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}$$

for every $N \geq L + L_0$.

Proof. Not ready The source proof takes a vector ψ in the left-hand side and writes its two boundary descriptions as

$$\psi = \sum_{j=1}^{g} \alpha_j = \sum_{j=1}^{g} \beta_j, \qquad \alpha_j \in \mathbb{C}^d \otimes G_m(A_j), \qquad \beta_j \in G_m(A_j) \otimes \mathbb{C}^d.$$

In the source notation, the two components have boundary matrices $C^j_{i_1}$ and $D^j_{i_{m+1}}$: the first family leaves the initial physical index open, and the second family leaves the final physical index open.

$$\alpha_j = \sum_{i_1,\dots,i_{m+1}} \operatorname{tr}\Big(A^j_{i_{m+1}} C^j_{i_1} A^j_{i_2} \cdots A^j_{i_m}\Big) |i_1 \cdots i_{m+1}\rangle,$$
$$\beta_j = \sum_{i_1,\dots,i_{m+1}} \operatorname{tr}\Big(D^j_{i_{m+1}} A^j_{i_1} A^j_{i_2} \cdots A^j_{i_m}\Big) |i_1 \cdots i_{m+1}\rangle.$$

The direct-sum separation lemma used in [PGVWC07, Theorem 12], together with injectivity of each block, gives the blockwise matrix identity

$$A^j_{i_{m+1}} C^j_{i_1} = D^j_{i_{m+1}} A^j_{i_1}$$

for every j, i_1, and i_{m+1}. For fixed j, set

$$E^j := \sum_a C^j_a A^{j\,\dagger}_a.$$

The printed formula in [PGVWC07, Theorem 12, proof lines 1449–1451] omits the adjoint on A^j_a; the adjointed form is forced by the displayed normalization below. In the normalization used in the source proof, $\sum_a A^j_a A^{j\,\dagger}_a = \mathbb{1}$, so Lemma 13.232 gives

$$D^j_b = \sum_a D^j_b A^j_a A^{j\,\dagger}_a = \sum_a A^j_b C^j_a A^{j\,\dagger}_a = A^j_b E^j.$$

Hence

$$A^j_b C^j_a = A^j_b E^j A^j_a$$

for every a, b, and cyclicity of the trace gives

$$\alpha_j = \sum_{i_1,\dots,i_{m+1}} \operatorname{tr}\Big(A^j_{i_1} \cdots A^j_{i_m} A^j_{i_{m+1}} E^j\Big) \, |i_1 \cdots i_{m+1}\rangle = \Gamma^{A_j}_{m+1}(E^j) \in G_{m+1}(A_j).$$

Since $\psi = \sum_j \alpha_j$, this gives

$$\psi \in \bigoplus_j G_{m+1}(A_j).$$

For one block the preceding implication is

$$\mathbb{C}^d \otimes G_m(A_j) \cap G_m(A_j) \otimes \mathbb{C}^d \subseteq G_{m+1}(A_j).$$

The reverse inclusion is blockwise:

$$G_{m+1}(A_j) \subseteq \mathbb{C}^d \otimes G_m(A_j) \cap G_m(A_j) \otimes \mathbb{C}^d.$$

Thus

$$\mathbb{C}^d \otimes G_m(A_j) \cap G_m(A_j) \otimes \mathbb{C}^d = G_{m+1}(A_j).$$

With

$$S_m := \bigoplus_{j=1}^{g} G_m(A_j),$$

the preceding calculation is the open-segment recursion

$$\mathbb{C}^d \otimes S_m \cap S_m \otimes \mathbb{C}^d = S_{m+1}$$

for every $m \geq L$. For $B = \bigoplus_j \mu_j A_j$ with every $\mu_j \neq 0$, Lemma 13.252 gives

$$G_L(B) = \bigvee_j G_L(A_j).$$

In the range where the sums $\bigvee_j G_m(A_j)$ are direct, this identifies $G_L(B)$ with the subspace denoted in the source by $S = \bigoplus_j \mathcal{G}^{A_j}_L$. The passage from the open-segment recursion to the N-site ring is the separate boundary-condition comparison in the periodic ring in [PGVWC07, Theorem 12]. The source periodic-boundary conclusion is

$$\ker H_{S_L,N} = \operatorname{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

$\square$

Lemma 13.300 (Conditional reduction to block-diagonal boundary conditions)**.** Not ready *Let $B^i = \bigoplus_j \mu_j A^i_j$, and assume $1 < L$, $N \geq L+1$, and $0 < m$. Assume that every virtual sector projection P_j lies in the pulled-back span of the length-m words B^ω, and assume that every vector $\psi \in \mathcal{G}_{N,L}\big(\bigoplus_j \mu_j A_j\big)$ has a boundary matrix X with*

$$\psi = \Gamma^B_N(X), \qquad X B^\omega = B^\omega X$$

for the boundary-crossing words ω. Then

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) \subseteq \sum_j \mathcal{G}_{N,L}(A_j),$$

with equality after the forward inclusion. This is a conditional form of the block-diagonal boundary-condition step in [CPGSV21, lines 2126–2128].

Proof. Not ready The commutant reduction makes the pulled-back boundary matrix block diagonal. Restricting the same word-commutation identities to a diagonal block and using the injectivity of that block shows, by the scalar-commutant lemma, that each diagonal block is scalar. Write Γ_N^B for the length-N boundary-to-state map of the tensor B. For a block-diagonal boundary matrix,

$$\Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right) = \sum_j \mu_j^N \Gamma_N^{A_j}(X_j).$$

If $X_j = \lambda_j I$, this is a finite sum of scalar multiples of the vectors $V^{(N)}(A_j)$. The reverse inclusion and equality statements are reformulations of this calculation, combined with the forward inclusion and the blockwise uniqueness theorem. □

Lemma 13.301 (Block-diagonal boundary reduction from blockwise word span)**.** Not ready *Let $A^i = \bigoplus_j \mu_j A_j^i$ be in block-injective canonical form, and assume $1 < L$, $N \geq L+1$, and $0 < m$. If*

$$\operatorname{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = m\} = \prod_j M_{D_j}(\mathbb{C})$$

and every vector in $\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right)$ comes with a boundary matrix representation commuting with all length-m direct-sum words, then

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) \subseteq \sum_j \mathcal{G}_{N,L}(A_j).$$

Combined with the forward inclusion, this gives the corresponding conditional equality. In the parent-ground-space formulation, the same hypotheses imply containment in $\operatorname{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}$.

Proof. Not ready The blockwise word-span hypothesis gives

$$P_j \in \operatorname{span}\left\{ \left(\bigoplus_r \mu_r A_r\right)^\omega : |\omega| = m \right\}.$$

If X commutes with all displayed word products, then $XP_j = P_j X$ for every j, hence $X = \bigoplus_j X_j$. The direct-sum boundary formula

$$\Gamma_N^{\oplus_j \mu_j A_j}\left(\bigoplus_j X_j\right) = \sum_j \mu_j^N \Gamma_N^{A_j}(X_j)$$

gives the reverse inclusion, and the blockwise uniqueness theorem gives the stated span containment. □

Lemma 13.302 (Block-diagonal boundary reduction from block-separating equations)**.** Not ready *Let $A^i = \bigoplus_j \mu_j A_j^i$ be in block-injective canonical form, with $A_1, \dots, A_g$ a basis of normal tensors. Assume $1 < L$ and $N \geq L+1$, and suppose that for each block k there are coefficients $c_\omega^{(k)}$ such that*

$$\sum_{|\omega|=S} c_\omega^{(k)} A_j^\omega = \begin{cases} I, & j = k, \\ 0, & j \neq k. \end{cases}$$

If every vector in $\mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$ has a boundary matrix representation commuting with all length-$(1+S)$ direct-sum words, then

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) \subseteq \sum_j \mathcal{G}_{N,L}(A_j).$$

Together with the forward inclusion, this gives the corresponding conditional equality. In the parent-ground-space formulation, the same hypotheses imply containment in $\mathrm{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}$. *This is the block-injective ground-space argument of [CPGSV21, lines 2120–2128], where the re-growing step is restricted to block-diagonal boundary conditions.*

Proof. Not ready Finite-length injectivity gives a block length at which each block has the required physical–virtual correspondence. In the current length-one specialization, Lemma 2.30 supplies the corresponding one-block injectivity. The displayed separation equations imply

$$\mathrm{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = 1 + S\} = \prod_j M_{D_j}(\mathbb{C}).$$

This is Lemma 13.225 with $L = 1$. Therefore Lemma 13.301 applies with $m = 1 + S$. Its conclusion is the displayed containment, and the equality follows after adding the forward inclusion. □

Theorem 13.303 (Periodic-boundary containment in the BNT span). Not ready *If $A^i = \bigoplus_j \mu_j A_j^i$ is in block-injective canonical form, where $A_1, \dots, A_g$ is a basis of normal tensors, assume $g \geq 2$ and $\mu_j \neq 0$ for every j. Let $L_0 \geq 1$ be a common injectivity length for the blocks, and suppose*

$$L_0 < L, \qquad L \geq 3(g-1)(L_0+1)+1, \qquad N \geq L + L_0,$$

and assume the block-diagonal boundary-condition comparison in the periodic ring,

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) \subseteq \bigvee_{j=1}^{g} \mathcal{G}_{N,L}(A_j).$$

then

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) \subseteq \mathrm{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

Proof. Not ready The length hypotheses are the conservative range used in [PGVWC07, Theorem 12]. The shorter range stated in [CPGSV21, lines 2126–2128] requires the strengthened block-diagonal re-growing step. These hypotheses do not by themselves give the displayed boundary-condition comparison in the periodic ring. That comparison follows only after the block-diagonal boundary conditions have been compared across the windows crossing the chosen periodic-boundary cut. Since $g \geq 2$ and $L_0 \geq 1$,

$$L \geq 3(g-1)(L_0+1)+1 \geq 3L_0 + 4 > L_0, \qquad N \geq L + L_0 \geq L + 1,$$

Because $A_1, \dots, A_g$ is a basis of normal tensors, the block MPV families are non-redundant:

$$j \neq k \quad \Longrightarrow \quad \exists n \geq 1,\ V^{(n)}(A_j) \neq V^{(n)}(A_k).$$

By the assumed block-diagonal boundary-condition comparison,

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) \subseteq \bigvee_j \mathcal{G}_{N,L}(A_j).$$

Theorem 13.124 gives

$$\mathcal{G}_{N,L}(A_j) = \operatorname{span}\{V^{(N)}(A_j)\}$$

for every j. Substituting these one-block identities into the join gives the displayed containment. $\square$

Theorem 13.304 (Periodic-boundary equality with the BNT span). Not ready *Under the hypotheses of Theorem 13.303, the corresponding periodic-boundary ground space is exactly the span of the BNT component vectors:*

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \operatorname{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

Proof. Not ready We prove the two inclusions separately. The inclusion

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) \subseteq \operatorname{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}$$

is exactly Theorem 13.303. For the reverse inclusion, the nonzero-weight hypothesis and Lemma 13.206 show that each vector $V^{(N)}(A_j)$ lies in $\mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$. Since this ground space is linear, it contains every linear combination of such vectors, and hence contains $\operatorname{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}$. Thus

$$\operatorname{span}\{V^{(N)}(A_j) : j = 1, \dots, g\} \subseteq \mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right),$$

and the two spaces are equal. $\square$

Remark. A separate kernel-identification statement for these direct-sum block families would convert the periodic-boundary ground-space statement into the corresponding statement for $\ker(H_N)$.

Chapter 14

Exponential Decay of Correlations

This chapter derives the transfer-matrix expression for thermodynamic-limit correlators of a normal MPS and the resulting exponential decay estimate. The key mechanism is spectral: after removing the leading eigenvalue contribution, the connected part is governed by subleading eigenvalues of the transfer map.

When the MPS tensor generates a parent Hamiltonian (Chapter 13), the correlator quantifies spatial correlations in the ground state. The exponential decay rate is set by the spectral properties of the transfer map, i.e. by the modulus of its subleading eigenvalues.

The spectral analysis of the transfer map and the mixed transfer operator, including the eigenvalue bound $\rho(F_{AB}) \le 1$, the strict transfer-operator gap $\rho(F_{AB}) < 1$ for non-equivalent blocks, and the resulting overlap decay, is developed in Chapter 7. This chapter focuses on the *correlator* side: one-point and two-point expectations, their spectral decomposition, and quantitative bounds on decay rates.

14.1 Connected correlators from the transfer map

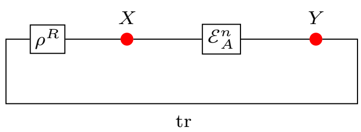
Definition 14.1 (One-point expectation)**.** Lean ✓ Stmt Fix an MPS tensor and a right fixed point of its transfer map. Write the tensor as A and the fixed point as ρ^R. Under the hypotheses of Theorem 6.11, that fixed point exists and is unique up to scaling; we normalize it so that $\mathrm{tr}(\rho^R) = 1$. For a local observable $X \in M_D(\mathbb{C})$ define

$$\langle X_0 \rangle := \mathrm{tr}(X\rho^R).$$

Definition 14.2 (Two-point expectation at distance n)**.** Lean ✓ Stmt For observables $X, Y \in M_D(\mathbb{C})$ and $n \ge 0$,

$$\langle X_0 Y_n \rangle := \mathrm{tr}\big(Y\, \mathcal{E}_A^n(X\rho^R)\big)\,.$$

Equivalently: insert X at site 0, propagate by $\mathcal{E}_A^n$, then insert Y and take the trace.



Definition 14.3 (Connected correlator). Lean ✓ Stmt The connected two-point function is

$$C(X,Y;n) := \langle X_0 Y_n \rangle - \langle X_0 \rangle \langle Y_0 \rangle.$$

14.2 Spectral expansion and exponential decay

Theorem 14.4 (Sum of exponentials (spectral expansion interface)). *Lean* *✓ Stmt* *If coefficients $c_j \in \mathbb{C}$ and eigenvalues $\lambda_j \in \mathbb{C}$ ($j = 1, \ldots, D^2 - 1$) satisfy, for all $n \geq 0$,*

$$C(X,Y;n) = \sum_{j=1}^{D^2-1} c_j \, \lambda_j^n,$$

then the connected correlator equals the stated sum of exponentials. The source [CPGSV21, Section II.B.3] states that the connected correlator is a sum of D^2-1 pure exponentials $C(X,Y;n) = \sum_{j\geq 2}^{D^2} c_{XY}(j)\lambda_j^n$, which always exists for a normal MPS via the transfer-map spectral decomposition.

Proof. ✓ Proof Given the expansion hypothesis, the equality is immediate. The source [CPGSV21, Section II.B.3] derives this expansion from the spectral decomposition of $\mathcal{E}_A^n$ restricted to the complement of the fixed-point eigenspace: the leading eigenvalue $\lambda_1 = 1$ contributes $\langle X_0 \rangle \langle Y_0 \rangle$, which is subtracted in the connected correlator, leaving the sum over subleading eigenvalues. □

Theorem 14.5 (Exponential decay bound (interface)). *Lean* *✓ Stmt* *If a constant $C_{XY} \in \mathbb{R}$ and $\lambda_2 \in \mathbb{C}$ satisfy, for all $n \geq 0$,*

$$|C(X,Y;n)| \leq C_{XY} \, |\lambda_2|^n,$$

then the connected correlator satisfies that exponential decay bound. The source [CPGSV21, Section II.B.3] obtains this by combining the sum-of-exponentials expansion with the triangle inequality, and Sec. 4 provides the transfer-map gap condition $|\lambda_j| < 1$ for the subleading eigenvalues.

Proof. ✓ Proof Given the bound hypothesis, the estimate is immediate. The source [CPGSV21, Section II.B.3] obtains this bound by applying the triangle inequality to the sum-of-exponentials expansion and using the transfer-map gap condition $|\lambda_j| \leq |\lambda_2| < 1$ for all subleading eigenvalues. □

Remark 14.6 (Transfer-map spectral hypothesis). The hypothesis "$|\lambda_2| < 1$" in Theorem 14.5 is a transfer-map gap condition: all non-leading eigenvalues of $\mathcal{E}_A$ lie strictly inside the unit disk. It is distinct from the Hamiltonian spectral gap of Chapter 13. The complementary transfer-map gap $\rho(\mathcal{E}_A - P) < 1$ is established for primitive channels (Theorem 4.57, Chapter 4). The mixed-transfer gap needed for block separation is supplied for irreducible trace-preserving tensors (Theorem 7.28), and quantitative single-tensor transfer-map bounds for injective tensors are recorded in Theorem 14.11. Thus the analytical input needed to make Theorems 14.4 and 14.5 unconditional is already present; what remains is extracting the spectral expansion coefficients c_j, λ_j from the eigendecomposition of $\mathcal{E}_A$ (Issue #1447).

Definition 14.7 (Correlation length). Lean ✓ Stmt The correlation length associated with a subleading eigenvalue λ_2 of the transfer map ($|\lambda_2| < 1$) is

$$\xi := -\frac{1}{\log |\lambda_2|}.$$

Lemma 14.8 (Correlation length is positive). *Lean* *✓ Stmt* *If $|\lambda_2| < 1$ and $\lambda_2 \neq 0$, then $\xi > 0$.*

Proof. ✓ Proof Since $0 < |\lambda_2| < 1$, we have $\log |\lambda_2| < 0$, so $\xi = -1/\log |\lambda_2| > 0$. □

14.3 Quantitative transfer-map gap bounds

The transfer-map gap theorems in Chapter 4 establish $\rho(\mathcal{E}_A - P) < 1$ for primitive channels qualitatively. This section gives the *quantitative* refinements with explicit constants and rates for the single-tensor transfer map $\mathcal{E}_A$.

Theorem 14.9 (Exponential convergence of injective primitive channels). Lean ✓ Stmt *Fix an injective, trace-preserving MPS tensor A with positive definite fixed point ρ. Let*

$$P(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)}\rho$$

be the corresponding fixed-point projection. Then there exist $C > 0$ and $0 < \delta \leq 1$ such that for all $n \geq 0$ and $X \in M_D(\mathbb{C})$,

$$\|\mathcal{E}_A^n(X) - P(X)\| \leq C\,(1-\delta)^n\,\|X\|.$$

The transfer-map gap δ exists by primitivity (Theorem 4.57, the complementary transfer-map gap).

Proof. ✓ Proof Injectivity implies primitivity of the transfer map. The complementary spectral radius $\rho(\mathcal{E}_A - P) < 1$ then follows from the primitive transfer-map gap theorem for the complementary map. The Gelfand formula yields a geometric bound $\|(\mathcal{E}_A - P)^n\| \leq C\,(1-\delta)^n$ for suitable constants, giving the claimed estimate. □

Theorem 14.10 (Correlation length bound). Lean ✓ Stmt *For an injective trace-preserving MPS tensor, there exist $C > 0$ and $\xi > 0$ such that for all $n \geq 0$ and all traceless $X \in M_D(\mathbb{C})$ (i.e. $\mathrm{tr}(X) = 0$),*

$$\|\mathcal{E}_A^n(X)\| \leq C\,e^{-n/\xi}\|X\|.$$

Traceless matrices lie in $\ker P$, where P is the fixed-point projection. Since $\mathcal{E}_A - P$ has spectral radius strictly less than 1, the Gelfand formula gives exponential decay at rate $\xi = -1/\log\rho(\mathcal{E}_A - P)$.

Proof. ✓ Proof For traceless X we have $P(X) = 0$, so $\mathcal{E}_A^n(X) = (\mathcal{E}_A - P)^n(X)$. The exponential convergence theorem gives a geometric bound $C\,(1-\delta)^n\,\|X\|$. Rewriting $(1-\delta)^n = e^{-n/\xi}$ with $\xi = -1/\log(1-\delta)$ yields the claimed estimate. □

Theorem 14.11 (Transfer-map gap from injectivity). Lean ✓ Stmt *For an injective trace-preserving MPS tensor, there exists $\delta > 0$ such that every eigenvalue $\mu \neq 1$ of the transfer map satisfies $|\mu| \leq 1 - \delta$.*

Proof. ✓ Proof Injectivity implies eventual full Kraus rank. This yields primitivity, hence a complementary transfer-map gap. Since the transfer map has only finitely many eigenvalues, one may choose a uniform $\delta > 0$ separating 1 from the remaining spectrum. □

14.4 Relation to parent Hamiltonians

Remark 14.12. When the MPS tensor A generates a parent Hamiltonian $H_N(A, L)$ (Definition 13.17) whose ground state is the matrix product vector (Lemma 13.21), the exponential decay bound (Theorem 14.5) shows that ground-state correlations decay with a finite correlation length ξ.

14.5 Renormalization fixed points and zero correlation length

Definition 14.13 (Renormalization fixed point). Lean ✓ Stmt An MPS tensor A is a *renormalization fixed point* (RFP) when its transfer map is idempotent, $\mathcal{E}_A \circ \mathcal{E}_A = \mathcal{E}_A$. See [CPGSV16, Definition 3.2].

Theorem 14.14 (Kraus-isometry criterion for RFP). Lean ✓ Stmt *Suppose there exists an isometry $V : \mathbb{C}^d \to \mathbb{C}^{d^2}$, written as coefficients $V_{(i_1,i_2),j}$, such that*

$$A^{i_1} A^{i_2} = \sum_{j=0}^{d-1} V_{(i_1,i_2),j} A^j \tag{14.1}$$

for all $i_1, i_2 \in \{0, \dots, d-1\}$. Then A is a renormalization fixed point. This is the backward direction of [CPGSV16, Theorem 3.1].

Proof. ✓ Proof Expanding $\mathcal{E}_A^2$ gives a double Kraus sum indexed by pairs (i_1, i_2). Substituting the decomposition (14.1) and using the isometry relation $V^\dagger V = 1$ collapses the double sum to the single Kraus family $\{A^j\}$, hence $\mathcal{E}_A^2 = \mathcal{E}_A$. □

Theorem 14.15 (Renormalization-fixed-point characterization). Lean ✓ Stmt *An MPS tensor A is a renormalization fixed point if and only if there exists an isometry $V : \mathbb{C}^d \to \mathbb{C}^{d^2}$, written as coefficients $V_{(i_1,i_2),j}$ with $V^\dagger V = 1$, such that*

$$A^{i_1} A^{i_2} = \sum_{j=0}^{d-1} V_{(i_1,i_2),j} A^j \tag{14.2}$$

for all $i_1, i_2 \in \{0, \dots, d-1\}$. This is [CPGSV16, Theorem 3.1].

Proof. ✓ Proof The backward implication is Theorem 14.14. For the forward implication, expanding $\mathcal{E}_A^2 = \mathcal{E}_A$ as a Kraus sum gives, for every X,

$$\sum_{i_1,i_2} (A^{i_1} A^{i_2})\, X\, (A^{i_1} A^{i_2})^\dagger = \sum_j A^j\, X\, (A^j)^\dagger.$$

Theorem 16.26 then supplies the isometry V relating the two Kraus families. □

Definition 14.16 (Single-block local orthogonality convention). Lean ✓ Stmt For the single-block convention used here, local orthogonality means

$$\mathcal{E}_A^2 = \mathcal{E}_A.$$

The source local-orthogonality condition for a BNT family is instead the collection of mixed-sector equations for distinct components,

$$\mathcal{E}_{j,j'} = 0.$$

Thus this definition records the one-block idempotence convention.

Definition 14.17 (BNT local orthogonality). Lean ✓ Stmt Let $(A_j)_j$ be the BNT components of a tensor in canonical form. The family is *locally orthogonal* when, for every pair of distinct components,

$$\mathcal{E}_{j,j'} = \sum_i A_j^i \otimes \overline{A_{j'}^i} = 0.$$

This is the mixed-sector condition of [CPGSV16, Definition 3.5].

Definition 14.18 (BNT zero correlation length). Lean ✓ Stmt Let $(A_j)_j$ be a basis of normal tensors for A. The BNT zero-correlation-length condition is that

$$\mathrm{CID}(A)$$

and, for every pair of distinct BNT components,

$$\mathcal{E}_{j,j'} = 0.$$

This is [CPGSV16, Definition 3.6].

Theorem 14.19 (RFP normal tensor is injective). Lean ✓ Stmt *If A is a nonzero normal RFP tensor, then A is injective.*

Proof. ✓ Proof The RFP condition $\mathcal{E}_A^2 = \mathcal{E}_A$ implies that the cumulative span $\mathrm{span}\{A^w : |w| \leq n\}$ stabilises after one step. Since A is normal (hence has an injective blocked form), the one-step span already equals $M_D(\mathbb{C})$, so A is injective. □

Theorem 14.20 (Left-canonical normal RFP tensor is injective). Lean ✓ Stmt *Assume $D > 0$. Let A be a normal renormalization fixed point such that*

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

Then A is injective. This is the left-canonical injectivity step in [CPGSV16, Appendix B].

Proof. ✓ Proof The normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ forces at least one matrix A^i to be nonzero, since otherwise the left-hand side would vanish. Hence A is a nonzero normal renormalization fixed point, and Theorem 14.19 applies. □

Theorem 14.21 (Unitary diagonal fixed point for a left-canonical normal RFP). Lean ✓ Stmt *Assume $D > 0$. Let A be a normal renormalization fixed point such that*

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

Then there exist a unitary U and a diagonal positive definite matrix Λ such that, for $B^i := U^\dagger A^i U$,

$$\sum_i (B^i)^\dagger B^i = \mathbb{1} \qquad \text{and} \qquad \mathcal{E}_B(\Lambda) = \Lambda.$$

This is the diagonal fixed-point reduction in [CPGSV16, Appendix B].

Proof. ✓ Proof By Theorem 14.20, the tensor A is injective. Therefore its transfer map is irreducible, and hence A is irreducible as a tensor. Applying Theorem 9.20 gives a unitary conjugate B for which the trace-preserving normalization remains valid and a diagonal positive definite matrix Λ with $\mathcal{E}_B(\Lambda) = \Lambda$. □

Theorem 14.22 (Rank-one classification for RFP transfer maps). Lean ✓ Stmt *Let A be an injective left-canonical RFP tensor with positive-definite fixed point ρ of the transfer map $\mathcal{E}_A$. Then $\mathcal{E}_A = P_\rho$, where $P_\rho(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)}\rho$ is the rank-one fixed-point projection.*

Proof. ✓ Proof The exponential convergence bound gives $\|\mathcal{E}_A^n(X) - P_\rho(X)\| \leq C(1-\delta)^n \|X\|$. Idempotence of $\mathcal{E}_A$ implies $\mathcal{E}_A^{n+1} = \mathcal{E}_A$ for all $n \geq 0$, so $\|\mathcal{E}_A(X) - P_\rho(X)\| \leq C(1-\delta)^{n+1}\|X\|$ for all n. Taking $n \to \infty$ gives $\mathcal{E}_A = P_\rho$. □

Theorem 14.23 (Full structural form for RFP tensors). Lean ✓ Stmt *A normal tensor A in canonical form II that is a renormalization fixed point admits a decomposition $A^i = X\Lambda U^i X^{-1}$, where Λ is diagonal positive, $\sum_i (U^i)^\dagger U^i = I$, and*

$$\sum_i \overline{(U^i)_{\alpha,\beta}}\,(U^i)_{\alpha',\beta'} = D^{-1}\delta_{\alpha,\alpha'}\delta_{\beta,\beta'}.$$

Proof. ✓ Proof Combine the diagonal fixed-point reduction with the rank-one classification of injective left-canonical RFP tensors. One then writes the resulting rank-one projector in a normalized matrix-unit Kraus form, applies Kraus freedom to identify the pair-index coefficients, and transports the decomposition back through the conjugation. The matrix-unit normalization contributes the factor D^{-1} in the pair-index equation. □

Theorem 14.24 (Per-block isometry canonical form). Lean ✓ Stmt *When each block A_k of a multi-block tensor is a normal, left-canonical renormalization fixed point, that block admits the isometry decomposition $A_k^i = X_k \Lambda_k U_k^i X_k^{-1}$, with X_k invertible, Λ_k diagonal positive, and $U_k = (U_k^i)$ satisfies $\sum_i (U_k^i)^\dagger U_k^i = I$ and the pair-index orthonormality condition*

$$\sum_i \overline{(U_k^i)_{\alpha,\beta}}\,(U_k^i)_{\alpha',\beta'} = D_k^{-1}\delta_{\alpha,\alpha'}\delta_{\beta,\beta'}.$$

This is the blockwise form of the isometry canonical form (Theorem 14.23); see [CPGSV16, Section 3]. The source additionally imposes the normalization $\operatorname{tr}(\Lambda_k) = 1$; the statement here gives positive Λ_k without it. That normalization is genuine rather than a conjugation gauge, since rescaling $\Lambda_k \mapsto \Lambda_k/\operatorname{tr}(\Lambda_k)$ factors out as an overall scalar on A_k^i (conjugation by X_k preserves the scale); the statement is thus the un-normalized isometry form. Scope restriction (source isometry). *Corollary III.cor3 also invokes the joint isometry condition*

$$\sum_i (U_k^i)_{\alpha,\beta}\,\overline{(U_\ell^i)_{\alpha',\beta'}} = \delta_{k,\ell}\delta_{\alpha,\alpha'}\delta_{\beta,\beta'}.$$

The theorem above records the contracted per-block condition $\sum_i (U_k^i)^\dagger U_k^i = I$ and the diagonal pair-index equation with right-hand side $D_k^{-1}\delta_{\alpha,\alpha'}\delta_{\beta,\beta'}$. It does not impose the trace-normalization $\operatorname{tr}(\Lambda_k) = 1$, and it does not include the cross-block equations for $k \neq \ell$. This restriction is recorded in `docs/paper-gaps/cpsv16_rfp_isometry_scope.tex`.

Proof. ✓ Proof Apply Theorem 14.23 to each block. □

Theorem 14.25 (Canonical-form blocks are injective). Lean ✓ Stmt *For a family of tensors in canonical form, every block is injective. This is a direct projection of the injectivity field of the canonical-form predicate; see [CPGSV16, Section 2.3].*

Proof. ✓ Proof Immediate from the `block_injective` field of `IsCanonicalForm`. □

Theorem 14.26 (RG flow convergence for canonical-form tensors). Lean ✓ Stmt *For a tensor in canonical form with blocks (A_k) and weights (μ_k), the iterated blocking $\mathcal{E}_{A_k}^{2^n}$ converges pointwise to an idempotent linear map $\mathcal{E}_\infty$ as $n \to \infty$. That is, there exists $\mathcal{E}_\infty$ with $\mathcal{E}_\infty^2 = \mathcal{E}_\infty$ such that $\mathcal{E}_{A_k}^{2^n}(\rho) \to \mathcal{E}_\infty(\rho)$ for all density matrices ρ. This is [CPGSV16, Appendix B].*

Proof. ✓ Proof Each block is injective and left-canonical, so its transfer map is a primitive channel with a unique positive-definite fixed point ρ_0. The exponential convergence bound $\|\mathcal{E}^n(X) - P(X)\| \le C(1-\delta)^n\|X\|$ (Theorem 14.9) then gives pointwise convergence $\mathcal{E}^n(X) \to P(X)$, and composing with the subsequence $2^n \to \infty$ yields the result. The witness $\mathcal{E}_\infty = P$ is the rank-one fixed-point projection, which is idempotent. □

Remark 14.27. When the transfer map is idempotent ($\mathcal{E}^2 = \mathcal{E}$), connected correlations become separation-independent for all separations $n \geq 1$. Informally, this corresponds to zero correlation length ($\xi = 0$). The full spectral equivalence (idempotence iff vanishing subleading spectrum) is deferred here.

Definition 14.28 (Correlations independent of distance)**.** Lean ✓ Stmt For a positive definite right fixed point $\rho^R > 0$, an MPS tensor has *correlations independent of distance* (CID) when for all local observables X and Y and all separations $n, m \geq 1$,

$$C(X, Y; n) = C(X, Y; m).$$

Definition 14.29 (Zero correlation length)**.** Lean ✓ Stmt In the single-block convention, an MPS tensor has *zero correlation length* when

$$\mathcal{E}_A^2 = \mathcal{E}_A \qquad \text{and} \qquad \mathrm{CID}(A)$$

both hold. The source definition is the BNT condition in Definition 14.18.

Theorem 14.30 (CID implies RFP)**.** Lean ✓ Stmt *If an MPS tensor admits a positive definite right fixed point of its transfer map and has correlations independent of distance, then its transfer map is idempotent. This is the reverse direction of [CPGSV16, Theorem 3.8].*

Proof. ✓ Proof Trace nondegeneracy and invertibility of the fixed point reduce idempotence to equality of the connected correlator at separations $n = 2$ and $n = 1$. □

Theorem 14.31 (Zero correlation length iff idempotent transfer)**.** Lean ✓ Stmt *Under the single-block convention above,*

$$(\mathcal{E}_A^2 = \mathcal{E}_A \text{ and } \mathrm{CID}(A)) \quad \Longleftrightarrow \quad \mathcal{E}_A^2 = \mathcal{E}_A.$$

This is the idempotence/CID part of [CPGSV16, Theorem 3.8]. The full source theorem for canonical-form tensors also uses Definition 14.18.

Proof. ✓ Proof The forward direction extracts the idempotence hypothesis. The reverse uses $\mathcal{E}^n = \mathcal{E}$ for $n \geq 1$ to show the connected correlator is separation-independent. □

Theorem 14.32 (Single-block RFP iff ZCL inside a canonical form)**.** Lean ✓ Stmt *Let* $A^i = \bigoplus_k \mu_k A_k^i$ *be a tensor in canonical form. For each* k*, the tensor* A_k *is a renormalization fixed point if and only if* A_k *has zero correlation length.*

This is the single-block consequence of [CPGSV16, Theorem 3.10]. The ZCL condition in the source theorem is the BNT-family condition: CID together with the local orthogonality equations

$$\mathbb{E}_{j,j'} = 0 \qquad (j \neq j'),$$

as in Definition 14.18.

Proof. ✓ Proof Apply Theorem 14.31 to the chosen canonical-form block. □

Chapter 15

Concrete Examples

This chapter collects concrete MPS tensors from [CPGSV21, Section III]. Each example is specified by its physical dimension d, bond dimension D, and explicit matrix family $\{A^i\}$. Key structural properties (injectivity, RFP status, symmetry) are stated for each tensor.

15.1 The GHZ state

The Greenberger–Horne–Zeilinger (GHZ) state is the prototypical non-injective, renormalization-fixed-point MPS.

Definition 15.1 (GHZ tensor). Lean ✓ Stmt Set $d = D = 2$. The GHZ tensor is the family $\{A^0, A^1\} \subset M_2(\mathbb{C})$ defined by

$$A^0 = |0\rangle\langle 0| = \mathrm{diag}(1,0), \qquad A^1 = |1\rangle\langle 1| = \mathrm{diag}(0,1).$$

For a system of N sites the resulting MPV is the GHZ state

$$|\mathrm{GHZ}_N\rangle = |0\rangle^{\otimes N} + |1\rangle^{\otimes N}.$$

Definition 15.2 (Two-site computational vector). Lean ✓ Stmt For $a, b \in \{0,1\}$, the vector

$$|ab\rangle \in (\{0,1\}^2 \to \mathbb{C})$$

is the indicator of the configuration

$$(\sigma_1, \sigma_2) = (a, b).$$

Definition 15.3 (Constant product vector). Lean ✓ Stmt For $a \in \{0,1\}$ and $N \geq 0$,

$$|a\rangle^{\otimes N}(\sigma_1, \dots, \sigma_N) = \begin{cases} 1, & (\sigma_1, \dots, \sigma_N) = (a, \dots, a), \\ 0, & \text{otherwise.} \end{cases}$$

Theorem 15.4 (GHZ transfer map is idempotent). Lean ✓ Stmt *The transfer map of the GHZ tensor satisfies $\mathcal{E}_A^2 = \mathcal{E}_A$.*

Proof. ✓ Proof The transfer map acts by $\mathcal{E}_A(X)_{ij} = \delta_{ij} X_{ij}$ (extraction of the diagonal). This operation is idempotent. □

Theorem 15.5 (GHZ tensor is an RFP). Lean ✓ Stmt *The GHZ tensor is a renormalization fixed point.*

Proof. ✓ Proof Immediate from Theorem 15.4. □

Theorem 15.6 (GHZ tensor is not injective). Lean ✓ Stmt *The GHZ tensor is not injective:* $\operatorname{span}_{\mathbb{C}}\{A^0, A^1\}$ *is the two-dimensional space of diagonal matrices in* $M_2(\mathbb{C})$*, which does not equal* $M_2(\mathbb{C})$*.*

Proof. ✓ Proof Every element of $\operatorname{span}_{\mathbb{C}}\{A^0, A^1\}$ has zero off-diagonal entries, so the matrix with all entries equal to 1 is not in the span. □

Theorem 15.7 ($\mathbb{Z}_2$ on-site symmetry of the GHZ tensor). Lean ✓ Stmt *Let* $\mathbb{Z}_2 = \{1, \sigma_x\}$ *act on* $\mathbb{C}^2$ *by the Pauli-X matrix* σ_x*. The GHZ tensor is on-site symmetric under this action: for each* $g \in \mathbb{Z}_2$*, the twisted tensor* $\widetilde{A}_g$ *defines the same MPV family as* A*, i.e.* $\mathcal{V}(A) = \mathcal{V}(\widetilde{A}_g)$*.*

Proof. ✓ Proof The identity element is trivial. For the generator $g = \sigma_x$, the twisted tensor satisfies $\widetilde{A}_g^i = A^{1-i}$, i.e. the two matrices are swapped. Conjugation by σ_x implements the same swap, giving gauge equivalence and hence the same MPV family. □

Definition 15.8 (Virtual $\mathbb{Z}_2$ representation of the GHZ tensor). Lean ✓ Stmt The virtual $\mathbb{Z}_2$ representation on the bond space is the homomorphism $\mathbb{Z}_2 \to M_2(\mathbb{C})$ sending the identity to I and the generator to $\sigma_z = \operatorname{diag}(1, -1)$.

Theorem 15.9 (Virtual $\mathbb{Z}_2$ symmetry of the GHZ tensor). Lean ✓ Stmt *The matrix* σ_z *commutes with every GHZ matrix* A^i*. A non-scalar matrix commuting with all* A^i *is the hallmark of a non-injective (*$\mathbb{Z}_2$*-injective) tensor: the on-site symmetry of Theorem 15.7 swaps the two one-dimensional injective blocks* $A^0 \leftrightarrow A^1$*, while* σ_z *acts on each block by a phase.*

Proof. ✓ Proof Both A^0 and A^1 are diagonal, so each commutes with the diagonal matrix σ_z. □

Theorem 15.10 (GHZ order parameter is a transfer-map fixed point). Lean ✓ Stmt *The order-parameter operator* σ_z *is a fixed point of the GHZ transfer map,* $\mathcal{E}_A(\sigma_z) = \sigma_z$*.*

Proof. ✓ Proof Since $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ and both A^i are diagonal, the claim is the entrywise computation $\mathcal{E}_A(\sigma_z) = \sigma_z$. □

Remark 15.11. A primitive (injective) tensor has a one-dimensional transfer-map fixed-point space, spanned by a positive matrix that equals the identity only in a normalized gauge. That the GHZ transfer map fixes the additional non-scalar operator σ_z is the transfer-matrix signature of the extra non-decaying mode behind its long-range ferromagnetic correlations.

Theorem 15.12 (GHZ zero correlation length). Lean ✓ Stmt *The GHZ tensor has zero correlation length: its idempotent transfer map (Theorem 15.5) gives correlations independent of the separation. This contrasts with the AKLT state (Theorem 15.35), whose subleading eigenvalue* $-\frac{1}{3}$ *produces a finite correlation length* $1/\log 3$*.*

Proof. ✓ Proof Immediate from the idempotence of the GHZ transfer map and the equivalence of zero correlation length with transfer-map idempotence (Theorem 14.31). □

15.2 The AKLT state

The Affleck–Kennedy–Lieb–Tasaki (AKLT) state is the canonical example of a normal (block-injective) MPS with non-trivial symmetry-protected topological (SPT) order.

Definition 15.13 (AKLT tensor). Lean ✓ Stmt Set $d = 3$ (spin-1) and $D = 2$. The AKLT tensor is the family $\{A^0, A^1, A^2\} \subset M_2(\mathbb{C})$ defined via the Pauli matrices by

$$A^0 = \frac{1}{\sqrt{3}}\sigma_z, \qquad A^1 = \frac{\sqrt{2}}{\sqrt{3}}|0\rangle\langle 1|, \qquad A^2 = -\frac{\sqrt{2}}{\sqrt{3}}|1\rangle\langle 0|,$$

where $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$, so equivalently $A^1 = \frac{1}{\sqrt{3}}\sigma_+$ and $A^2 = -\frac{1}{\sqrt{3}}\sigma_-$ for $\sigma_+ = \sqrt{2}|0\rangle\langle 1|$ and $\sigma_- = \sqrt{2}|1\rangle\langle 0|$. Equivalently, the matrices are the Clebsch–Gordan coefficients coupling spin-1 and spin-$\frac{1}{2}$ to spin-$\frac{1}{2}$.

Theorem 15.14 (AKLT tensor is not injective). Lean ✓ Stmt *The AKLT tensor is not injective at a single site:* $\mathrm{span}_{\mathbb{C}}\{A^0, A^1, A^2\}$ *is the three-dimensional space of traceless matrices* $\mathfrak{sl}(2, \mathbb{C})$*, which is a proper subspace of* $M_2(\mathbb{C})$*.*

Theorem 15.15 (AKLT tensor is normal). Lean ✓ Stmt *The transfer map of the AKLT tensor has a unique eigenvalue of maximal modulus (a simple spectral-radius eigenvalue); in particular the AKLT tensor is normal. Concretely, the length-*2 *blocked tensor is injective.*

Theorem 15.16 ($\mathbb{Z}_2 \times \mathbb{Z}_2$ on-site symmetry of the AKLT tensor). Lean ✓ Stmt *The AKLT tensor is on-site symmetric under the* $\mathbb{Z}_2 \times \mathbb{Z}_2$ *subgroup of* $\mathrm{SO}(3)$ *generated by two commuting* π*-rotations about orthogonal axes, acting on the spin-*1 *physical index.*

Theorem 15.17 (Anticommuting virtual gauges of the AKLT symmetry). Lean ✓ Stmt *The two virtual matrices that implement the symmetry of Theorem 15.16, namely* $i\sigma_y$ *and* σ_z*, anti-commute.*

Definition 15.18 (AKLT factor system). Lean ✓ Stmt The AKLT factor system on $\mathbb{Z}_2 \times \mathbb{Z}_2$ is $\omega(g, h) = (-1)^{(g_1+g_2)h_1}$ (the value -1 when $g_1 + g_2 = 1$ and $h_1 = 1$, and 1 otherwise). It is the factor system of the explicit projective representation with virtual gauges $i\sigma_y$ and σ_z; the diagonal term $g_1 h_1$ relative to the cluster case reflects $(i\sigma_y)^2 = -I$.

Lemma 15.19 (AKLT commutator phase). Lean ✓ Stmt *The commutator phase of the AKLT factor system on the two generators is* -1*.*

Theorem 15.20 (The AKLT factor system is a 2-cocycle). Lean ✓ Stmt *The AKLT factor system is a genuine* 2*-cocycle, being the factor system of an explicit projective representation, so it represents an element of* $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathrm{U}(1))$*.*

Theorem 15.21 (The AKLT $\mathbb{Z}_2\times\mathbb{Z}_2$ phase is non-trivial). Lean ✓ Stmt *The AKLT factor system (Theorem 15.20), built from the anticommuting gauges* $i\sigma_y$ *and* σ_z *(Theorem 15.17), is not a coboundary: its commutator phase on the two generators is* -1*. Its class in* $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathrm{U}(1)) \cong \mathbb{Z}_2$ *is therefore non-trivial, so the AKLT chain realizes a non-trivial SPT phase rather than a trivial product state.*

Remark 15.22. This $\mathbb{Z}_2 \times \mathbb{Z}_2$ is the smallest subgroup of $\mathrm{SO}(3)$ that still carries a non-trivial 2-cocycle. That minimality, and the full continuous classification over $\mathrm{SO}(3)$, are not formalized (Remark 15.32).

Definition 15.23 (Blocked AKLT tensor). Lean ✓ Stmt The length-2 blocked AKLT tensor groups two physical sites into one, with blocked physical dimension $9 = 3^2$ and the same bond dimension $D = 2$.

Definition 15.24 (Blocked $\mathbb{Z}_2 \times \mathbb{Z}_2$ on-site action). Lean ✓ Stmt The length-2 blocked on-site action of $\mathbb{Z}_2 \times \mathbb{Z}_2$ on the 9-dimensional blocked physical space is the Kronecker square

$$g \longmapsto P_g \otimes P_g$$

of the single-site spin-1 π-rotation representation P_g that implements the symmetry of Theorem 15.16. Identifying the blocked physical index with the pair of single-site indices, the entry of $P_g \otimes P_g$ at $(i_1 i_2, j_1 j_2)$ is the product $(P_g)_{i_1 j_1}(P_g)_{i_2 j_2}$ over the two grouped sites. This is the on-site action that makes the blocked twist agree with blocking the single-site twist.

Theorem 15.25 (The blocked AKLT tensor is injective). Lean ✓ Stmt *The length-2 blocked AKLT tensor is injective: its nine matrices span $M_2(\mathbb{C})$.*

Proof. ✓ Proof This is the blocked-tensor form of the normality of the single-site tensor (Theorem 15.15): the single-site tensor is 2-block injective, which is exactly injectivity of its length-2 blocking. □

Theorem 15.26 ($\mathbb{Z}_2 \times \mathbb{Z}_2$ on-site symmetry of the blocked AKLT tensor). Lean ✓ Stmt *The length-2 blocked AKLT tensor is on-site symmetric under $\mathbb{Z}_2 \times \mathbb{Z}_2$, with each group element acting by the Kronecker square $P_g \otimes P_g$ of the corresponding single-site spin-1 π-rotation P_g (Definition 15.24).*

Proof. ✓ Proof Writing A for the single-site tensor and $A^{[2]}$ for its length-2 blocking, twisting the blocked tensor by $P_g \otimes P_g$ equals blocking the twisted single-site tensor,

$$\left(A^{[2]}\right)^{P_g \otimes P_g} = \left(A^{P_g}\right)^{[2]},$$

where the superscript denotes the twist. Since the single-site tensor is symmetric, A^{P_g} has the same matrix product vector family as A (Theorem 15.16), and blocking preserves the matrix product vector family, so $\left(A^{[2]}\right)^{P_g \otimes P_g}$ has the same matrix product vector family as $A^{[2]}$. □

Theorem 15.27 (The single-site $\mathbb{Z}_2 \times \mathbb{Z}_2$ action is unitary). Lean ✓ Stmt *Each single-site spin-1 π-rotation matrix P_g of the $\mathbb{Z}_2 \times \mathbb{Z}_2$ action is unitary: $P_g P_g^\dagger = \mathbb{1}$.*

Theorem 15.28 (The AKLT state has string order under its $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry). Lean ✓ Stmt *For every element g of $\mathbb{Z}_2 \times \mathbb{Z}_2$, the length-2 blocked AKLT tensor has string order in the sense of Definition 12.40 for the twist by g, with the maximally mixed boundary state $\Lambda = \frac{1}{2}\mathbb{1}$ as its stationary boundary state.*

Proof. ✓ Proof The blocked tensor is injective (Theorem 15.25) and on-site symmetric under $\mathbb{Z}_2 \times \mathbb{Z}_2$ (Theorem 15.26), and each group element acts by a real symmetric involutive matrix, hence unitarily. The maximally mixed state $\Lambda = \frac{1}{2}\mathbb{1}$ is positive definite and has trace 1. The single-site AKLT matrices satisfy $\sum_i A^i (A^i)^\dagger = \mathbb{1}$ and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. These identities propagate to the length-2 blocked tensor $A^{[2]}$ whose matrices are the products $A^{i_1} A^{i_2}$: the blocked transfer map is unital,

$$\mathcal{E}_{A^{[2]}}(\mathbb{1}) = \sum_{i_1, i_2} A^{i_1} A^{i_2} (A^{i_2})^\dagger (A^{i_1})^\dagger = \mathbb{1},$$

and $\Lambda = \frac{1}{2}\mathbb{1}$ is a fixed point of the adjoint transfer map,

$$\mathcal{E}^{\dagger}_{A^{[2]}}(\Lambda) = \sum_{i_1,i_2} (A^{i_2})^{\dagger}(A^{i_1})^{\dagger}\,\Lambda\, A^{i_1} A^{i_2} = \Lambda.$$

The first identity sums the right-canonical relation $\sum_i A^i (A^i)^{\dagger} = \mathbb{1}$ over the inner letter and then the outer; the second sums the left-canonical relation $\sum_i (A^i)^{\dagger} A^i = \mathbb{1}$ over the outer letter and then the inner, using $\Lambda = \frac{1}{2}\mathbb{1}$. These are exactly the hypotheses of Theorem 12.64, which then yields string order for every g. Like the cluster state, the AKLT state is an explicit witness of that criterion, sharing the same virtual-boundary caveat on the string order parameter. □

Definition 15.29 (Cartesian AKLT tensor). Lean ✓ Stmt The Cartesian presentation of the AKLT tensor has $d = 3$, $D = 2$, and $A^x = \sigma_x$, $A^y = \sigma_y$, $A^z = \sigma_z$, the three Pauli matrices. This is the rotationally natural form used in [CPGSV21] (around line 1159); it is the AKLT state in the Cartesian basis of the spin-1 site, the counterpart of the $|m\rangle$-basis form of Definition 15.13.

Theorem 15.30 (The spin-$\frac{1}{2}$ cover is surjective onto SO(3)). Lean ✓ Stmt *Every rotation $R \in \mathrm{SO}(3)$ is realized by conjugating the Pauli vector by some $U \in \mathrm{SU}(2)$: there is $U \in \mathrm{SU}(2)$ with $\left(\frac{1}{2}\operatorname{tr}(\sigma_i U \sigma_j U^{-1})\right)_{ij} = R$. Equivalently, the adjoint double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$ is surjective. The proof factors $R = R_z(\alpha)\, R_x(\beta)\, R_z(\gamma)$ by its ZXZ Euler decomposition and maps each coordinate rotation to an explicit SU(2) generator: $\operatorname{diag}(e^{-i\theta/2}, e^{i\theta/2}) \mapsto R_z(\theta)$ and $\left(\begin{smallmatrix} \cos\frac{\beta}{2} & -i\sin\frac{\beta}{2} \\ -i\sin\frac{\beta}{2} & \cos\frac{\beta}{2} \end{smallmatrix}\right) \mapsto R_x(\beta)$.*

Theorem 15.31 (SO(3) on-site symmetry of the AKLT tensor). Lean ✓ Stmt *The Cartesian AKLT tensor is on-site symmetric under the spin-1 (defining) representation of SO(3) on the physical index.*

Remark 15.32. The bond index carries the projective spin-$\frac{1}{2}$ representation: the gauge for a rotation is either of its two SU(2) preimages, so the assignment is a representation only up to sign and factors through the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$. Its class is the non-trivial element of $H^2(\mathrm{SO}(3), \mathrm{U}(1)) \cong \mathbb{Z}_2$, witnessing the SPT phase.

15.2.1 Correlation length

The AKLT state is the canonical example of a finite correlation length. Its transfer map has leading eigenvalue 1 and a single subleading eigenvalue $-\frac{1}{3}$ of multiplicity three; the three Pauli matrices are the subleading eigenvectors and the identity is the leading one. The subleading modulus $\frac{1}{3}$ gives the correlation length $\xi = 1/\log 3$, and connected correlations decay as $(\frac{1}{3})^n$.

Theorem 15.33 (Pauli eigenvectors of the AKLT transfer map). Lean Lean Lean ✓ Stmt *Each of the three Pauli matrices σ_x, σ_y, σ_z is an eigenvector of the AKLT transfer map with eigenvalue $-\frac{1}{3}$:*

$$\mathcal{E}_A(\sigma_k) = -\tfrac{1}{3}\,\sigma_k, \qquad k \in \{x, y, z\}.$$

Proof. ✓ Proof Each identity is an entrywise computation from the AKLT matrices, using $(\sqrt{3})^2 = 3$ and $(\sqrt{2})^2 = 2$ to clear the prefactors. □

Theorem 15.34 (Subleading eigenvalue of the AKLT transfer map). Lean Lean Lean Lean ✓ Stmt *The value 1 is the leading eigenvalue of the AKLT transfer map, with the identity $\mathbb{1}$ as eigenvector, and $-\frac{1}{3}$ is a subleading eigenvalue, with σ_z as eigenvector. The subleading modulus is therefore $\frac{1}{3}$.*

Theorem 15.35 (AKLT correlation length). Lean Lean ✓ Stmt *The AKLT correlation length, computed from the subleading eigenvalue $\lambda_2 = -\frac{1}{3}$, equals*

$$\xi = -\frac{1}{\log|\lambda_2|} = \frac{1}{\log 3}.$$

It is finite and positive.

Proof. ✓ Proof The subleading modulus is $|-\frac{1}{3}| = \frac{1}{3}$, so rewriting the correlation length at $\lambda_2 = -\frac{1}{3}$ gives $\xi = -1/\log\frac{1}{3} = 1/\log 3$. Since $0 < \frac{1}{3} < 1$, positivity is the specialization of Lemma 14.8 to $\lambda_2 = -\frac{1}{3}$. □

15.3 The Majumdar–Ghosh state

The Majumdar–Ghosh state is the exact dimer ground state of the spin-$\frac{1}{2}$ chain $H = \sum S_i \cdot S_{i+1} + \frac{1}{2}\sum S_i \cdot S_{i+2}$. On an even periodic chain it is the symmetric superposition of the two nearest-neighbour singlet coverings $(1,2)(3,4)\cdots$ and $(2,3)(4,5)\cdots(N,1)$, the one-dimensional analogue of a resonating valence-bond state.

Definition 15.36 (Majumdar–Ghosh tensor). Lean ✓ Stmt Set $d = 2$ (spin-$\frac{1}{2}$) and $D = 3$. The Majumdar–Ghosh tensor is the family $\{A^0, A^1\} \subset M_3(\mathbb{C})$ defined by

$$A^0 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ \frac{1}{\sqrt{2}} & 0 & 0 \end{pmatrix}, \qquad A^1 = \begin{pmatrix} 0 & 0 & 1 \\ -\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The three bond levels are the two spin-$\frac{1}{2}$ values of an open singlet partner together with the closed inter-dimer link; the relative sign $-\frac{1}{\sqrt{2}}$ is the antisymmetry of the singlet. The equality between the matrix product vectors of these matrices and the even-ring dimer superposition, with normalization and cyclic convention fixed, is a separate verification, not part of this definition.

Theorem 15.37 (Majumdar–Ghosh tensor is in left-canonical form). Lean ✓ Stmt *The Majumdar–Ghosh tensor is isometric: $(A^0)^\dagger A^0 + (A^1)^\dagger A^1 = \mathbb{1}$.*

Proof. ✓ Proof Direct multiplication of the two displayed matrices gives $(A^0)^\dagger A^0 = \mathrm{diag}(\frac{1}{2}, 1, 0)$ and $(A^1)^\dagger A^1 = \mathrm{diag}(\frac{1}{2}, 0, 1)$. □

Theorem 15.38 (Transfer map of the Majumdar–Ghosh tensor). Lean ✓ Stmt *The transfer map of the Majumdar–Ghosh tensor sends the identity to the diagonal matrix $\mathcal{E}_A(\mathbb{1}) = \mathrm{diag}(2, \frac{1}{2}, \frac{1}{2})$.*

Proof. ✓ Proof Substitute the two matrices in $\mathcal{E}_A(\mathbb{1}) = A^0(A^0)^\dagger + A^1(A^1)^\dagger$ and multiply. □

Theorem 15.39 (Majumdar–Ghosh tensor is not injective). Lean ✓ Stmt *The Majumdar–Ghosh tensor is not injective: every element of $\mathrm{span}_{\mathbb{C}}\{A^0, A^1\}$ has vanishing $(2,2)$ entry, so the span is a proper subspace of $M_3(\mathbb{C})$.*

Proof. ✓ Proof The matrix unit E_{22} has nonzero $(2,2)$ entry, while every linear combination of A^0 and A^1 has zero $(2,2)$ entry. □

Theorem 15.40 (Majumdar–Ghosh tensor is not normal). Lean ✓ Stmt *The Majumdar–Ghosh tensor is not normal: no blocking length makes the products of that length span the full matrix algebra $M_3(\mathbb{C})$. This non-normality is the algebraic signature expected from the two-periodic dimer-covering structure.*

Proof. ✓ Proof The bond carries a two-periodic grading with complementary support sets $\{(0,1),(0,2),(1,0),(2,0)\}$ and $\{(0,0),(1,1),(1,2),(2,1),(2,2)\}$ that are exchanged by left multiplication with a generator. Let E_{ij} denote the (i,j) matrix unit. Then, for every blocking length N,

$$N \text{ odd} \Rightarrow (\forall w,\ |w| = N \Rightarrow (A^w)_{22} = 0) \Rightarrow E_{22} \notin \operatorname{span}_{\mathbb{C}}\{A^w : |w| = N\},$$
$$N \text{ even} \Rightarrow (\forall w,\ |w| = N \Rightarrow (A^w)_{10} = 0) \Rightarrow E_{10} \notin \operatorname{span}_{\mathbb{C}}\{A^w : |w| = N\}.$$

Hence no fixed-length word span is all of $M_3(\mathbb{C})$. □

Remark 15.41. The review writes the tensor in the valence-bond shorthand $A = [|0\rangle((02| + (20|) + |1\rangle((12| + (21|)] \otimes Y$ with the singlet $Y = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. Read literally as $(ab| = |a\rangle\langle b|$ on a separate three-level factor tensored with the two-level singlet space, that shorthand would describe a six-dimensional bond. Its two-site contractions do not have the singlet-pair support pattern: same-letter traces can be nonzero while opposite-letter traces vanish. The shorthand is therefore treated here as a schematic of the valence-bond construction: contracting the singlet Y leaves a matrix product operator on a three-dimensional bond, with the antisymmetry of Y supplying the relative sign. The contracted tensor is the $D = 3$ representative in Definition 15.36. The source-notation gap and the missing matrix-product-vector identification are recorded in `docs/paper-gaps/rmp_majumdar_ghosh_tensor_gap.tex`.

15.4 The W state

The W state on N sites is the single-excitation symmetric state

$$|W_N\rangle = |10\cdots 0\rangle + |010\cdots 0\rangle + \cdots + |0\cdots 01\rangle,$$

the equal-weight superposition of all configurations with exactly one excitation. Unlike the preceding examples it has no translation-invariant representation with bond dimension bounded independently of N; it is presented with open boundary conditions, where the matrix product is closed by a left and a right boundary vector rather than by a trace.

Definition 15.42 (Open-boundary contraction)**.** Lean ✓ Stmt For a tensor A with bond dimension D, a left boundary covector $v_L \in \mathbb{C}^D$, a right boundary vector $v_R \in \mathbb{C}^D$, and a word $w = (i_1, \dots, i_k)$, the open-boundary contraction is the bilinear pairing

$$(v_L|\, A^{i_1} A^{i_2} \cdots A^{i_k}\, |v_R) = v_L^{\top} \left(A^{i_1} \cdots A^{i_k}\right) v_R.$$

Definition 15.43 (Open-boundary state)**.** Lean ✓ Stmt The open-boundary state on N sites assigns to each configuration $\sigma = (\sigma_1, \dots, \sigma_N)$ the amplitude $(v_L|\, A^{\sigma_1} \cdots A^{\sigma_N}\, |v_R)$.

Definition 15.44 (W tensor and boundary vectors)**.** Lean Lean Lean ✓ Stmt Set $d = D = 2$. The W tensor is the site-independent family

$$A^0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad A^1 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

with left boundary covector $(l| = (0|$ and right boundary vector $|r) = |1)$. Here A^1 is the single raising operator: it sends $|1) \mapsto |0)$ and annihilates $|0)$, and squares to zero.

Definition 15.45 (W-state indicator)**.** Lean ✓ Stmt For $N \geq 0$, the W-state amplitude is the indicator of the single-excitation configurations:

$$|W_N\rangle(\sigma_1, \dots, \sigma_N) = \begin{cases} 1, & \text{exactly one } \sigma_j = 1, \\ 0, & \text{otherwise.} \end{cases}$$

Theorem 15.46 (The open-boundary contraction is the W state). Lean ✓ Stmt *For every N, the open-boundary state of the W tensor with boundary vectors $(0|$ and $|1)$ equals the W state:*

$$|W_N\rangle = \sum_\sigma (0| A^{\sigma_1} \cdots A^{\sigma_N} |1)\, |\sigma\rangle.$$

Proof. ✓ Proof Because A^0 is the identity and A^1 is the raising operator with $A^1A^1 = 0$, the ordered product $A^{\sigma_1} \cdots A^{\sigma_N}$ depends only on the number of excitations of σ: it is the identity with no excitation, the raising operator with one excitation, and zero with two or more. Reading off the $(0, 1)$ matrix element against the boundary vectors therefore gives 1 on the single-excitation configurations and 0 elsewhere, which is the W-state amplitude. □

Theorem 15.47 (Three-site single excitations). Lean Lean Lean ✓ Stmt *On three sites, each of the three single-excitation configurations $|100\rangle$, $|010\rangle$, $|001\rangle$ receives W-state amplitude 1, while the vacuum $|000\rangle$ and the doubly-excited $|110\rangle$ receive amplitude 0.*

Proof. ✓ Proof Direct instances of Theorem 15.46: the excitation count is one in the first three cases and zero or two in the others. □

Remark 15.48 (No fixed-bond translation-invariant family). The open-boundary representation above is not translation invariant: the chain is closed by boundary vectors, not by a trace. The review observes that there is no translation-invariant representation whose bond dimension is bounded independently of N: any translation-invariant matrix product state equal to $|W_N\rangle$ must have bond dimension growing with N, satisfying a bound of the form $D^3 \log D = \Omega(N)$. The review obtains this by combining the matrix product state representation theory of [PGVWC07] with the quantitative quantum Wielandt inequality. That asymptotic lower bound is not formalized here; it is recorded as a deferred result in `docs/paper-gaps/rmp_w_state_ti_bound.tex`.

15.5 The cluster state

The cluster state is a universal resource for measurement-based quantum computation and provides another example of a non-trivial SPT phase.

Definition 15.49 (Cluster-state tensor). Lean ✓ Stmt Set $d = D = 2$. The cluster-state tensor is the family $\{A^0, A^1\} \subset M_2(\mathbb{C})$ defined by

$$A^0 = |+\rangle\langle 0| = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \qquad A^1 = |-\rangle\langle 1| = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix},$$

where $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. The review [CPGSV21] writes the reflected convention $A^0 = |0\rangle\langle +|$, $A^1 = |1\rangle\langle -|$; the transpose taken here is the same state read in the opposite direction and carries the same SPT order.

Theorem 15.50 (Cluster-state tensor is not injective). Lean ✓ Stmt *The cluster-state tensor is not injective:* $\mathrm{span}_{\mathbb{C}}\{A^0, A^1\} = \mathrm{span}_{\mathbb{C}}\{|+\rangle\langle 0|, |-\rangle\langle 1|\}$ *has dimension* 2*, not* $\dim M_2(\mathbb{C}) = 4$.

Proof. ✓ Proof Both A^0 and A^1 are rank-one matrices with linearly independent column spaces, so they are linearly independent. Their span is two-dimensional, which is strictly less than $\dim M_2(\mathbb{C}) = 4$. □

Theorem 15.51 (Cluster-state tensor is normal). Lean ✓ Stmt *Although the cluster-state tensor is not injective at a single site, its length-2 blocking is injective: the four length-2 products span $M_2(\mathbb{C})$. In particular the cluster-state tensor is normal.*

Remark 15.52. Although the cluster-state tensor is not injective at length 1, the length-2 blocked tensor is injective. Using $\langle 0|\pm\rangle = \frac{1}{\sqrt{2}}$, $\langle 1|+\rangle = \frac{1}{\sqrt{2}}$, and $\langle 1|-\rangle = -\frac{1}{\sqrt{2}}$, the four products $A^{ij} = A^i A^j$ are

$$A^{00} = \tfrac{1}{\sqrt{2}}|+\rangle\langle 0|, \quad A^{01} = \tfrac{1}{\sqrt{2}}|+\rangle\langle 1|, \quad A^{10} = \tfrac{1}{\sqrt{2}}|-\rangle\langle 0|, \quad A^{11} = -\tfrac{1}{\sqrt{2}}|-\rangle\langle 1|. \tag{15.1}$$

Since $\{|+\rangle, |-\rangle\}$ and $\{\langle 0|, \langle 1|\}$ are bases, these four rank-one matrices are linearly independent and span $M_2(\mathbb{C})$.

Theorem 15.53 ($\mathbb{Z}_2 \times \mathbb{Z}_2$ on-site symmetry of the blocked cluster-state tensor). Lean ✓ Stmt *The length-2 blocked cluster-state tensor is on-site symmetric under $\mathbb{Z}_2 \times \mathbb{Z}_2$. The two generators act on the 4-dimensional blocked physical space $(\mathbb{C}^2)^{\otimes 2}$ by $\sigma_x \otimes I$ and $I \otimes \sigma_x$, which commute and each square to the identity, defining a linear representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$.*

Theorem 15.54 (Anticommuting virtual gauges of the cluster-state symmetry). Lean ✓ Stmt *The virtual matrices implementing the two generators of Theorem 15.53, namely $V_1 = \sigma_z$ and $V_2 = \sigma_x$, anticommute ($V_1 V_2 = -V_2 V_1$).*

Definition 15.55 (Cluster-state factor system). Lean ✓ Stmt The cluster-state factor system on $\mathbb{Z}_2 \times \mathbb{Z}_2$ is $\omega(g, h) = (-1)^{g_2 h_1}$ (the value -1 when $g_2 = 1$ and $h_1 = 1$, and 1 otherwise). It is the factor system of the explicit projective representation with virtual gauges σ_z and σ_x, both squaring to the identity, so there is no diagonal term.

Lemma 15.56 (Cluster-state commutator phase). Lean ✓ Stmt *The commutator phase of the cluster-state factor system on the two generators is -1.*

Theorem 15.57 (The cluster-state factor system is a 2-cocycle). Lean ✓ Stmt *The cluster-state factor system is a genuine 2-cocycle, being the factor system of an explicit projective representation, so it represents an element of $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathrm{U}(1))$.*

Theorem 15.58 (The cluster-state $\mathbb{Z}_2 \times \mathbb{Z}_2$ phase is non-trivial). Lean ✓ Stmt *The cluster-state factor system (Theorem 15.57), built from the anticommuting gauges σ_z and σ_x (Theorem 15.54), is not a coboundary: its commutator phase on the two generators is -1. Its class in $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathrm{U}(1)) \cong \mathbb{Z}_2$ is therefore non-trivial, so the cluster state realizes a non-trivial SPT phase.*

Theorem 15.59 (The cluster state has string order under its $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry). Lean ✓ Stmt *For every element g of $\mathbb{Z}_2 \times \mathbb{Z}_2$, the length-2 blocked cluster-state tensor has string order in the sense of Definition 12.40 for the twist by g, with the maximally mixed boundary state $\Lambda = \frac{1}{2}\mathbb{1}$ as its stationary boundary state.*

Proof. ✓ Proof The blocked tensor is injective (Remark 15.52) and on-site symmetric under $\mathbb{Z}_2 \times \mathbb{Z}_2$ (Theorem 15.53), and each group element acts by a real symmetric involutive permutation matrix, hence unitarily. The maximally mixed state $\Lambda = \frac{1}{2}\mathbb{1}$ is positive definite and has trace 1. The four blocked matrices in (15.1) satisfy $\sum_i A^i (A^i)^\dagger = \mathbb{1}$ and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, so the transfer map is unital and Λ is a fixed point of the adjoint transfer map. These are exactly the hypotheses of Theorem 12.64, which then yields string order for every g. The cluster state is thus the first explicit witness of that criterion. □

Theorem 15.60 (Blocked cluster transfer map is idempotent). Lean Lean ✓ Stmt *The length-2 blocked cluster transfer map satisfies $\mathcal{E}_A^2 = \mathcal{E}_A$. In closed form it sends every X to the scalar $(X_{00} + X_{11})/2$ times the identity, so its image is one-dimensional.*

Proof. ✓ Proof The single-site cluster transfer map is not idempotent, but the four blocked matrices give the closed form $\mathcal{E}_A(X) = \frac{1}{2}(X_{00} + X_{11})\,\mathbb{1}$, which is manifestly idempotent. □

Theorem 15.61 (Cluster zero correlation length). Lean ✓ Stmt *The length-2 blocked cluster tensor has zero correlation length: its idempotent transfer map (Theorem 15.60) gives correlations independent of the separation. Like the GHZ state, and unlike the AKLT state, the cluster state is a renormalization fixed point with zero correlation length.*

Proof. ✓ Proof Immediate from the idempotence of the blocked cluster transfer map and the equivalence of zero correlation length with transfer-map idempotence (Theorem 14.31). □

15.6 Parent Hamiltonian statements for the examples

The parent-Hamiltonian construction of [CPGSV21, lines 1988–2000] is recorded here in the form

$$\mathcal{G}_L(A) = \left\{ \sum_{i_1,\dots,i_L} \operatorname{tr}(A^{i_1} \cdots A^{i_L} X)|i_1 \cdots i_L\rangle : X \in M_D(\mathbb{C}) \right\}.$$

Thus

$$\ker h_L = \mathcal{G}_L(A), \qquad H_N^{(L)} = \sum_i (h_L)_i.$$

We write

$$\mathrm{GS}(H) = \ker(H - E_0(H)I)$$

for the ground eigenspace.

Theorem 15.62 (GHZ two-site parent space). Lean ✓ Stmt *For the GHZ tensor A of Definition 15.1,*

$$\mathcal{G}_2(A) = \operatorname{span}_{\mathbb{C}}\{|00\rangle, |11\rangle\}.$$

Proof. ✓ Proof By definition,

$$\Gamma_2(X)(\sigma_1, \sigma_2) = \operatorname{tr}(A^{\sigma_1} A^{\sigma_2} X).$$

For the GHZ matrices $A^0 = \operatorname{diag}(1,0)$ and $A^1 = \operatorname{diag}(0,1)$, whenever $\sigma_1 \neq \sigma_2$,

$$A^{\sigma_1} A^{\sigma_2} = 0.$$

Hence

$$\Gamma_2(X) = X_{00}|00\rangle + X_{11}|11\rangle,$$

and the image of Γ_2 is $\operatorname{span}_{\mathbb{C}}\{|00\rangle, |11\rangle\}$. □

Theorem 15.63 (Constant GHZ product vectors satisfy the cyclic constraints). Lean ✓ Stmt *For $N \geq 2$ and $a \in \{0,1\}$,*

$$|a\rangle^{\otimes N} \in \mathcal{G}_{N,2}(A_{\mathrm{GHZ}}).$$

Proof. ✓ Proof For a cyclic nearest-neighbour window and outside assignment τ, set

$$R_{i,\tau} := |a\rangle^{\otimes N}\big|_{[i,i+1],\tau}.$$

For $b, c \in \{0,1\}$ with $b \neq c$, the completed configuration is not constant, hence

$$R_{i,\tau}(b,c) = 0.$$

Therefore

$$R_{i,\tau} \in \{v : v(0,1) = v(1,0) = 0\} = \operatorname{span}_{\mathbb{C}}\{|00\rangle, |11\rangle\} = \mathcal{G}_2(A_{\mathrm{GHZ}}),$$

so $|a\rangle^{\otimes N} \in \mathcal{G}_{N,2}(A_{\mathrm{GHZ}})$. □

Theorem 15.64 (GHZ periodic nearest-neighbour ground space). Lean ✓ Stmt *For $N \geq 2$, the common nearest-neighbour cyclic ground space of the GHZ tensor satisfies*

$$\mathcal{G}_{N,2}(A_{\mathrm{GHZ}}) = \mathrm{span}_{\mathbb{C}}\{|0\rangle^{\otimes N}, |1\rangle^{\otimes N}\}.$$

This is the cyclic-window form of the two-fold degeneracy in [CPGSV21, line 2205].

Proof. ✓ Proof Let $\psi \in \mathcal{G}_{N,2}(A_{\mathrm{GHZ}})$. For every cyclic nearest-neighbour window and outside assignment τ,

$$\psi|_{[i,i+1],\tau} \in \mathcal{G}_2(A_{\mathrm{GHZ}}) = \mathrm{span}_{\mathbb{C}}\{|00\rangle, |11\rangle\}.$$

Taking the outside assignment from the word σ gives

$$\psi|_{[i,i+1],\sigma}(\sigma_i, \sigma_{i+1}) = \psi(\sigma).$$

Hence a mixed nearest-neighbour word has zero coefficient:

$$\sigma_i \neq \sigma_{i+1} \quad \Longrightarrow \quad \psi(\sigma) = 0.$$

A nonconstant binary cyclic word has a mixed edge,

$$\sigma \notin \{(0, \dots, 0), (1, \dots, 1)\} \quad \Longrightarrow \quad \exists i,\ \sigma_i \neq \sigma_{i+1}.$$

Therefore

$$\psi = \psi(0, \dots, 0)|0\rangle^{\otimes N} + \psi(1, \dots, 1)|1\rangle^{\otimes N}.$$

Conversely, Theorem 15.63 gives

$$|0\rangle^{\otimes N},\ |1\rangle^{\otimes N} \in \mathcal{G}_{N,2}(A_{\mathrm{GHZ}}),$$

and hence their span lies in $\mathcal{G}_{N,2}(A_{\mathrm{GHZ}})$. □

Remark 15.65 (GHZ parent interaction). For the GHZ tensor of Definition 15.1,

$$\mathcal{G}_2(A) = \mathrm{span}\{|00\rangle, |11\rangle\}, \qquad h_{\mathrm{GHZ}} = |01\rangle\langle 01| + |10\rangle\langle 10| = I - |00\rangle\langle 00| - |11\rangle\langle 11|.$$

The periodic nearest-neighbour parent Hamiltonian therefore satisfies

$$H_{\mathrm{GHZ},N} = \sum_i (h_{\mathrm{GHZ}})_i, \qquad \ker H_{\mathrm{GHZ},N} = \mathrm{span}\{|0\rangle^{\otimes N}, |1\rangle^{\otimes N}\},$$

using the cyclic-window equality

$$\mathcal{G}_{N,2}(A_{\mathrm{GHZ}}) = \mathrm{span}\{|0\rangle^{\otimes N}, |1\rangle^{\otimes N}\}$$

from Theorem 15.64. This is the ferromagnetic Ising parent Hamiltonian [CPGSV21, lines 1194–1196 and line 2205]; see also [FGSW+15, Section 3] for the displayed two-site space and interaction.

Remark 15.66 (Cluster-state stabilizer convention). The one-dimensional cluster tensor in Definition 15.49 is the reflected convention of the tensor displayed in [CPGSV21, lines 2364–2369]. The sourced stabilizer is

$$K_i^{ZXZ} = \sigma_i^z \sigma_{i+1}^x \sigma_{i+2}^z.$$

The corresponding positive parent interaction is

$$h_i^{ZXZ} = \frac{1}{2}(I - K_i^{ZXZ}), \qquad H_{\mathrm{cl},N}^{ZXZ} = \sum_i h_i^{ZXZ}, \qquad \dim\ker H_{\mathrm{cl},N}^{ZXZ} = 1$$

[PGVWC07, local TeX lines 374–387]. Thus an XZX formula should be used for this tensor only after establishing an equivalence such as

$$U H_{\mathrm{cl},N}^{ZXZ} U^\dagger = H_{\mathrm{cl},N}^{XZX} \quad \text{or} \quad T H_{\mathrm{cl},N}^{ZXZ} T^{-1} = H_{\mathrm{cl},N}^{XZX}.$$

Remark 15.67 (AKLT parent Hamiltonian and edge modes). For the AKLT tensor of Definition 15.13, with $\mathcal{G}_{[a,b]}$ denoting the local ground space on sites $a, \dots, b$,

$$L_0 = 2, \qquad \mathcal{G}_{[1,2]} \cap \mathcal{G}_{[2,3]} = \mathcal{G}_{[1,3]}$$

[CPGSV21, line 2095]. The displayed local spin-chain Hamiltonian is

$$H_{\mathrm{AKLT},N} = \sum_i \left(\vec{S}_i \cdot \vec{S}_{i+1} + \frac{1}{3}(\vec{S}_i \cdot \vec{S}_{i+1})^2 \right),$$

up to an additive constant [PGVWC07, local TeX lines 340–349]; [SPGC11, lines 2126–2130]. The spin-2 projector form is related to the polynomial Hamiltonian by the spin-addition identity

$$P_{i,i+1}^{(2)} = \frac{1}{2} \left(\vec{S}_i \cdot \vec{S}_{i+1} + \frac{1}{3}(\vec{S}_i \cdot \vec{S}_{i+1})^2 + \frac{2}{3} I \right)$$

for two spin-1 particles. For open boundary conditions,

$$\dim \mathrm{GS}(H_{\mathrm{AKLT},N}^{\mathrm{open}}) = 4, \qquad \dim(\mathrm{GS}(H_{\mathrm{AKLT},N}^{\mathrm{open}}) \cap \mathcal{H}_{S=0}) = 1$$

[CPGSV21, lines 1169–1174].

Chapter 16

Channel Representations and Normal Forms

This chapter collects the representation material for finite-dimensional quantum channels from [Wol12, Chapter 2]: the Choi–Jamiołkowski isomorphism, Kraus representation theorems, Stinespring and Naimark dilations, ordered completely positive maps, Radon–Nikodym and open-system representations, trace-pairing expansions, SVD and Lorentz normal forms, and the channel determinant. These results are important for the later channel theory, but they are not prerequisites for the matrix-product-state Fundamental Theorem.

16.1 Representations

The following definitions support the Choi–Jamiolkowski isomorphism (Theorem 16.6) and the representation theorems of [Wol12, Chapter 2].

Definition 16.1 (Partial traces). Lean Lean ✓ Stmt Let $X \in M_{d \cdot d'}(\mathbb{C})$ be a bipartite matrix indexed by

$$(\{0, \ldots, d-1\} \times \{0, \ldots, d'-1\})^2.$$

The *left partial trace* $\mathrm{tr}_A(X)$ and *right partial trace* $\mathrm{tr}_B(X)$ are the $d' \times d'$ and $d \times d$ matrices defined by

$$(\mathrm{tr}_A(X))_{ij} = \sum_k X_{(k,i),(k,j)}, \qquad (\mathrm{tr}_B(X))_{ij} = \sum_k X_{(i,k),(j,k)}.$$

Definition 16.2 (Maximally entangled state). Lean ✓ Stmt The *maximally entangled state* on $\mathbb{C}^d \otimes \mathbb{C}^d$ is the rank-one projector

$$|\Omega\rangle\langle\Omega|, \quad |\Omega\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |j, j\rangle.$$

Concretely, $(|\Omega\rangle\langle\Omega|)_{(i_1,i_2),(j_1,j_2)} = \frac{1}{d}\delta_{i_1 j_1}\delta_{i_2 j_2}$ and $\mathrm{tr}(|\Omega\rangle\langle\Omega|) = 1$.

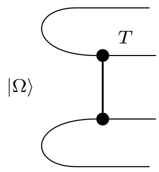
Definition 16.3 (Tensor product of a map with identity). Lean ✓ Stmt Given a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, the *tensor extension* $T \otimes \mathrm{id}$ acts on bipartite matrices $X \in M_{D \times D}(\mathbb{C})$ by applying T to each "slice":

$$(T \otimes \mathrm{id})(X)_{(i_1,i_2),(j_1,j_2)} = (T(X^{(i_2,j_2)}))_{i_1 j_1},$$

where $X^{(i_2,j_2)}_{ab} = X_{(a,i_2),(b,j_2)}$ is the bipartite slice.

16.2 Choi–Jamiołkowski isomorphism

The Choi–Jamiołkowski isomorphism bends the input legs of the channel T around the maximally entangled pair $|\Omega\rangle$, presenting the map as the matrix $\tau = (T \otimes \mathrm{id})|\Omega\rangle\langle\Omega|$ on the doubled space:



Definition 16.4 (Choi matrix). Lean ✓ Stmt The *Choi matrix* of a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is

$$\tau = (T \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|) \in M_{D\times D}(\mathbb{C}).$$

Concretely, $\tau_{(i_1,i_2),(j_1,j_2)} = \frac{1}{D}(T(E_{i_2 j_2}))_{i_1 j_1}$ where $E_{i_2 j_2}$ is the matrix unit. This is the Choi–Jamiolkowski convention of [Wol12, Proposition 2.1].

Theorem 16.5 (Choi matrix of the identity map). Lean ✓ Stmt *The identity channel has Choi matrix* $|\Omega\rangle\langle\Omega|$.

Proof. ✓ Proof Apply the identity map in the definition of the Choi matrix. □

Theorem 16.6 (Complete positivity iff the Choi matrix is positive semidefinite). Lean ✓ Stmt *A linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is completely positive (in the Kraus sense) if and only if its Choi matrix $\tau \geq 0$.*

Proof. ✓ Proof ($\Rightarrow$) If $T(X) = \sum_i K_i X K_i^\dagger$, then $\tau = \sum_i |v_i\rangle\langle v_i|$ where $(v_i)_{(a,b)} = c\, K_i(a,b)$ and $c = 1/\sqrt{D}$. This is a sum of rank-one PSD matrices.

($\Leftarrow$) If $\tau \geq 0$, write $\tau = \sum_m |v_m\rangle\langle v_m|$ by spectral decomposition, define $K_m(a,b) = v_m(a,b)/c$, and recover $T(X) = \sum_m K_m X K_m^\dagger$ by linearity on matrix units. □

Theorem 16.7 (Trace preservation implies the partial-trace condition). Lean ✓ Stmt *If T is trace-preserving, then* $\mathrm{tr}_A(\tau) = \frac{1}{D}\mathbb{1}_D$.

Proof. ✓ Proof $(\mathrm{tr}_A(\tau))_{ij} = \mathrm{tr}(T(\Omega^{(ij)})) = \mathrm{tr}(\Omega^{(ij)})$ by trace preservation, where $\Omega^{(ij)} = \frac{1}{D}E_{ij}$ is the (i,j)-slice of $|\Omega\rangle\langle\Omega|$. Taking the trace gives $\frac{1}{D}\delta_{ij}$. □

Theorem 16.8 (Hermiticity preservation and the Choi matrix). Lean ✓ Stmt *The Choi matrix τ is Hermitian if and only if $T(B^\dagger) = T(B)^\dagger$ for all $B \in M_D(\mathbb{C})$.*

Proof. ✓ Proof The Choi matrix entries satisfy $\tau_{(a,i),(b,j)} = \frac{1}{D}(T(E_{ij}))_{ab}$. Hermiticity of τ says $\overline{\tau_{(b,j),(a,i)}} = \tau_{(a,i),(b,j)}$, which unravels to $T(E_{ji})_{ba} = \overline{T(E_{ij})_{ab}}$, i.e., $T(E_{ij}^\dagger) = T(E_{ij})^\dagger$. By linearity this extends to all B. □

Theorem 16.9 (Trace of the Choi matrix of a trace-preserving map). Lean ✓ Stmt *If T is trace-preserving, then* $\mathrm{tr}(\tau) = 1$.

Proof. ✓ Proof $\mathrm{tr}(\tau) = \mathrm{tr}(\mathrm{tr}_A(\tau)) = \mathrm{tr}(\frac{1}{D}\mathbb{1}_D) = 1$. □

16.3 Representation corollaries for channel decompositions

The next three corollaries follow from the Choi/Kraus/Stinespring representations already developed above. They correspond to [Wol12, Propositions 2.2–2.4].

Theorem 16.10 (Polarization of sesquilinear sandwiches). Lean ✓ Stmt *For any $A, B, X \in M_D(\mathbb{C})$,*

$$4\,AXB^\dagger = (A+B)X(A+B)^\dagger - (A-B)X(A-B)^\dagger + i\,(A+iB)X(A+iB)^\dagger - i\,(A-iB)X(A-iB)^\dagger.$$

Each of the four summands on the right is a single-Kraus CP map, so every sesquilinear sandwich decomposes into a signed complex linear combination of CP maps.

Proof. ✓ Proof Working entrywise, reduces to the scalar polarization identity $4\,\alpha\,\overline{\delta} = \sum_{k=0}^{3} i^k(\alpha + i^k\beta)\overline{(\gamma + i^k\delta)}$ in $\mathbb{C}$, which holds after substituting $i^2 = -1$. □

Theorem 16.11 (CP decomposition of sandwich sums). Lean ✓ Stmt *For any finite Kraus-like families $\{A_i\}, \{B_i\}$, the map $T(X) = \sum_i A_i X B_i^\dagger$ satisfies $4\,T = T_1 - T_2 + i\,T_3 - i\,T_4$ with each T_k a CP map given explicitly by a single-side Kraus sum of the combined families.*

Proof. ✓ Proof Apply the polarization identity summand-by-summand. □

Theorem 16.12 (No information without disturbance). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be linear. If $T(|v\rangle\langle v|) = |v\rangle\langle v|$ for every $v \in \mathbb{C}^D$, then $T = \mathrm{id}$. In particular, a quantum channel preserving every pure state equals the identity.*

Proof. ✓ Proof Rank-one outer products $|u\rangle\langle v|$ polarize into rank-one self-outer-products:

$$4\,|u\rangle\langle v| = \sum_{k=0}^{3} i^k |u + i^k v\rangle\langle u + i^k v|,$$

so T preserves every $|u\rangle\langle v|$. Since the matrix units $E_{ij} = |e_i\rangle\langle e_j|$ span $M_D(\mathbb{C})$, T equals the identity on a spanning set, hence everywhere. □

Definition 16.13 (Pure-state ensemble density). Lean ✓ Stmt Given a finite family $\{\psi_i\}_{i\in\iota}$ of (unnormalized) vectors in $\mathbb{C}^D$, its *pure-ensemble density* is $\rho = \sum_i |\psi_i\rangle\langle\psi_i|$. Weights $p_i \geq 0$ with $\sum p_i = 1$ can be absorbed by replacing $\psi_i \mapsto \sqrt{p_i}\psi_i$.

Theorem 16.14 (Isometric mixing preserves the density). Lean ✓ Stmt *If two pure-state ensembles $\{\psi_i\}_{i\in\iota_1}$ and $\{\phi_j\}_{j\in\iota_2}$ are related by an isometric mixing matrix $V \in \mathbb{C}^{\iota_1\times\iota_2}$ with $V^\dagger V = \mathbb{1}$ and $\psi_i = \sum_j V_{ij}\phi_j$, then they induce the same pure-ensemble density operator. This is the sufficient direction of the Hughston–Jozsa–Wootters theorem; the converse is Theorem 16.15.*

Proof. ✓ Proof Expanding and applying the orthogonality relation $\sum_i V_{ij}\overline{V_{ij'}} = \delta_{jj'}$ from $V^\dagger V = \mathbb{1}$:

$$\sum_i |\psi_i\rangle\langle\psi_i| = \sum_{j,j'} \left(\sum_i V_{ij}\overline{V_{ij'}}\right) |\phi_j\rangle\langle\phi_{j'}| = \sum_j |\phi_j\rangle\langle\phi_j|.$$

□

Theorem 16.15 (Hughston–Jozsa–Wootters converse). Lean ✓ Stmt *If two pure-state ensembles $\{\psi_i\}_{i\in\iota_1}$ and $\{\phi_j\}_{j\in\iota_2}$ induce the same pure-ensemble density operator and $|\iota_2| \leq |\iota_1|$, then there exists a tall isometric mixing matrix $V \in \mathbb{C}^{\iota_1\times\iota_2}$ with $V^\dagger V = \mathbb{1}$ and $\psi_i = \sum_j V_{ij}\phi_j$. The cardinality hypothesis is what makes V a tall isometry; the symmetric case $|\iota_1| \leq |\iota_2|$ follows by swapping the roles of the ensembles.*

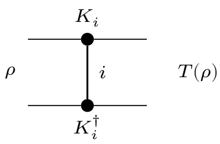
Proof. ✓ Proof Embed each vector ψ_i, ϕ_j as the 0-th column of a $D \times D$ matrix with zeros elsewhere; call these K_i, $L_j \in M_D(\mathbb{C})$. A direct entry-wise computation shows that for any $X \in M_D(\mathbb{C})$ both $\sum_i K_i X K_i^\dagger$ and $\sum_j L_j X L_j^\dagger$ collapse to $X_{00} \cdot \rho$, so the density equality $\rho_\psi = \rho_\phi$ forces the two Kraus families to define the same CP map. Theorem 16.26 then supplies an isometry V with $V^\dagger V = \mathbb{1}$ and $K_i = \sum_j V_{ij} L_j$; reading the equation off at column 0 recovers the vector relation $\psi_i = \sum_j V_{ij} \phi_j$. □

Theorem 16.16 (Hughston–Jozsa–Wootters equivalence). Lean ✓ Stmt *Under the cardinality hypothesis $|\iota_2| \le |\iota_1|$, two pure-state ensembles induce the same density operator iff they are related by a tall isometric mixing matrix.*

Proof. ✓ Proof Combine the two directions above. □

16.4 Kraus representation theorem

A Kraus representation writes the channel as $T(\rho) = \sum_i K_i \rho K_i^\dagger$. In tensor-network form the Kraus index i is summed between the K_i layer and the $K_i^\dagger$ layer:



Theorem 16.17 (Trace preservation implies Kraus normalization). Lean ✓ Stmt *If $T(X) = \sum_i K_i X K_i^\dagger$ is trace-preserving, then $\sum_i K_i^\dagger K_i = \mathbb{1}$.*

Proof. ✓ Proof For any N, $\operatorname{tr}((\sum_i K_i^\dagger K_i)\, N) = \sum_i \operatorname{tr}(K_i^\dagger K_i N) = \sum_i \operatorname{tr}(K_i N K_i^\dagger) = \operatorname{tr}(T(N)) = \operatorname{tr}(N)$. Non-degeneracy of the trace pairing forces $\sum_i K_i^\dagger K_i = \mathbb{1}$. □

Theorem 16.18 (Kraus normalization implies trace preservation). Lean ✓ Stmt *If $\sum_i K_i^\dagger K_i = \mathbb{1}$, then $T(X) = \sum_i K_i X K_i^\dagger$ is trace-preserving.*

Proof. ✓ Proof $\operatorname{tr}(T(X)) = \sum_i \operatorname{tr}(K_i X K_i^\dagger) = \sum_i \operatorname{tr}(K_i^\dagger K_i X) = \operatorname{tr}((\sum_i K_i^\dagger K_i)\, X) = \operatorname{tr}(X)$. □

Theorem 16.19 (Unitality implies Kraus unital normalization). Lean ✓ Stmt *If $T(X) = \sum_i K_i X K_i^\dagger$ satisfies $T(\mathbb{1}) = \mathbb{1}$, then*

$$\sum_i K_i K_i^\dagger = \mathbb{1}.$$

Proof. ✓ Proof Evaluate the Kraus formula at the identity matrix. □

Theorem 16.20 (Unitary freedom in Kraus operators). Lean ✓ Stmt *Let U be a unitary $r \times r$ matrix ($U^\dagger U = \mathbb{1}$) and suppose $K_j = \sum_\ell U_{j\ell} \tilde{K}_\ell$. Then $\{K_j\}$ and $\{\tilde{K}_\ell\}$ define the same Kraus map: for every $X \in M_D(\mathbb{C})$,*

$$\sum_j K_j X K_j^\dagger = \sum_\ell \tilde{K}_\ell X \tilde{K}_\ell^\dagger.$$

Proof. ✓ Proof Substitute the combination formula, expand the triple sum over j, ℓ, ℓ', swap summation order, and use the entry-wise form of $U^\dagger U = \mathbb{1}$ to collapse to $\ell = \ell'$ diagonal terms. □

Theorem 16.21 (Isometric mixing preserves the Kraus map). Lean ✓ Stmt *Let W be an isometry between two Kraus index spaces, so $W^\dagger W = \mathbb{1}$. If*

$$K_j = \sum_\ell W_{j\ell}\widetilde{K}_\ell,$$

then the Kraus families $\{K_j\}$ and $\{\widetilde{K}_\ell\}$ define the same completely positive map.

Proof. ✓ Proof Expand the Kraus sums, interchange the finite summations, and use $W^\dagger W = \mathbb{1}$ to collapse the coefficient matrix. □

Theorem 16.22 (The transition matrix of orthonormal Kraus families is unitary). Lean ✓ Stmt *Let $\{K_j\}_{j=0}^{r-1}$ and $\{K'_j\}_{j=0}^{r-1}$ be two Hilbert–Schmidt orthonormal Kraus families, and suppose*

$$K_j = \sum_{\ell=0}^{r-1} U_{j\ell}\, K'_\ell.$$

Then the transition matrix U is unitary: $U^\dagger U = \mathbb{1}_r$.

Proof. ✓ Proof Expand $\operatorname{tr}(K_j^\dagger K_i)$ using the change of basis. Hilbert–Schmidt orthonormality of both Kraus families identifies the Gram matrix with the identity, yielding the matrix identity $UU^\dagger = \mathbb{1}_r$, hence also $U^\dagger U = \mathbb{1}_r$. □

Theorem 16.23 (Dual map equality from primal map equality). Lean ✓ Stmt *If two Kraus families satisfy $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all X, then $\sum_\alpha B_\alpha^\dagger Y B_\alpha = \sum_j A_j^\dagger Y A_j$ for all Y.*

Proof. ✓ Proof Use the trace pairing: for all X, $\operatorname{tr}(X^\dagger(\sum B_\alpha^\dagger Y B_\alpha)) = \operatorname{tr}((\sum B_\alpha X^\dagger B_\alpha^\dagger)Y) = \operatorname{tr}((\sum A_j X^\dagger A_j^\dagger)Y) = \operatorname{tr}(X^\dagger(\sum A_j^\dagger Y A_j))$. Nondegeneracy of the trace pairing gives the result. □

Theorem 16.24 (Equal Stinespring Gramians). Lean ✓ Stmt *If two Kraus families define the same CPM, then $\sum_\alpha B_\alpha^\dagger B_\alpha = \sum_j A_j^\dagger A_j$.*

Proof. ✓ Proof Specialise Theorem 16.23 at $Y = \mathbb{1}$. □

Theorem 16.25 (Rectangular Kraus freedom). Lean ✓ Stmt *If $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all X and $r_2 \le r_1$ (where $\{B_\alpha\}_{\alpha=0}^{r_1-1}$ is the larger family and $\{A_j\}_{j=0}^{r_2-1}$ the smaller), then there exists a rectangular isometry V ($r_1 \times r_2$, $V^\dagger V = \mathbb{1}_{r_2}$) such that $B_\alpha = \sum_j V_{\alpha j} A_j$.*

Proof. ✓ Proof Pad the smaller family to size r_1, establish Gram matrix equality $M_B^\dagger M_B = M_{A'}^\dagger M_{A'}$ from dual map equality, construct a partial isometry via inner product preservation on the range, and extend to a full isometry V with $V^\dagger V = \mathbb{1}_{r_2}$. □

Theorem 16.26 (Rectangular Kraus freedom with general finite index types). Lean ✓ Stmt *Variant of Theorem 16.25 with general finite index sets ι_1, ι_2 in place of standard index sets of cardinalities r_1 and r_2.*

Proof. ✓ Proof Reindex by choosing enumerations of those finite index sets and apply Theorem 16.25. □

Theorem 16.27 (Isometric freedom of Kraus representations). Lean ✓ Stmt *Let $\{B_\alpha\}_{\alpha\in\iota_1}$ and $\{A_j\}_{j\in\iota_2}$ be finite Kraus families with $|\iota_2| \le |\iota_1|$. Then the following are equivalent:*

1. $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ *for all* $X \in M_D(\mathbb{C})$.

2. *There exists a matrix* $V = (V_{\alpha j})$ *with* $V^\dagger V = \mathbb{1}$ *such that* $B_\alpha = \sum_j V_{\alpha j} A_j$ *for every* α.

Proof. ✓ Proof The implication (1) ⇒ (2) is Theorem 16.26. Conversely, expand $\sum_\alpha B_\alpha X B_\alpha^\dagger$ using $B_\alpha = \sum_j V_{\alpha j} A_j$, interchange the finite sums, and use $V^\dagger V = \mathbb{1}$ to collapse the coefficient matrix to the diagonal. □

Theorem 16.28 (Unitary freedom of Kraus representations)**.** *Lean* *✓ Stmt* *Let* $\{B_\alpha\}_{\alpha \in \iota}$ *and* $\{A_j\}_{j \in \iota}$ *be two Kraus families with the same finite index set. Then the following are equivalent:*

1. $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ *for all* $X \in M_D(\mathbb{C})$.

2. *There exists a unitary matrix* $U = (U_{\alpha j})$ *such that* $B_\alpha = \sum_j U_{\alpha j} A_j$ *for every* α.

Proof. ✓ Proof Apply Theorem 16.27 with $|\iota| = |\iota|$. In the square case an isometry matrix is unitary, and the converse is immediate. □

Remark 16.29 (Choi rank and minimal Kraus cardinality). Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt If a completely positive map admits a Kraus representation with exactly r operators, then its Choi matrix is a sum of r rank-one outer products. Consequently,

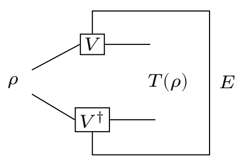
$$\mathrm{rank}(\tau_E) \le r.$$

Conversely, diagonalizing the positive semidefinite Choi matrix and keeping only the nonzero eigenvalue summands produces a Kraus family with exactly $\mathrm{rank}(\tau_E)$ operators. Hence the Choi rank is precisely the minimal Kraus cardinality.

Proof. ✓ Proof For the forward implication, expand the Choi matrix using the Kraus formula and write $\tau_E = \sum_j v_j v_j^\dagger$ with one vector v_j for each Kraus operator. The column space of this sum lies in the span of the r vectors $\{v_j\}$, so the rank is at most r. For the converse, use the spectral decomposition of the positive semidefinite Choi matrix, discard the zero-eigenvalue terms, and rescale the remaining eigenvectors into Kraus operators. □

16.5 Stinespring dilation

The Stinespring dilation realizes the channel through an isometry V into a larger space, $T(\rho) = \mathrm{tr}_E[V \rho V^\dagger]$; tracing out the environment E returns the channel:



Definition 16.30 (Stinespring isometry)**.** Lean ✓ Stmt Given Kraus operators $\{K_j\}_{j=0}^{r-1}$ with $K_j \in M_D(\mathbb{C})$, the *Stinespring isometry* $V : \mathbb{C}^D \to \mathbb{C}^D \otimes \mathbb{C}^r$ is the $(D \cdot r) \times D$ matrix

$$V_{(i,j),k} = (K_j)_{ik}.$$

This implements $V = \sum_j K_j \otimes |j\rangle$.

Lemma 16.31 (Entry formula for the Stinespring isometry). Lean ✓ Stmt *For indices $i, k \in \{0, \dots, D-1\}$ and $j \in \{0, \dots, r-1\}$,*

$$V_{(i,j),k} = (K_j)_{ik}.$$

Theorem 16.32 (Stinespring Gram identity). Lean ✓ Stmt *The Stinespring isometry satisfies*

$$V^\dagger V = \sum_{j=0}^{r-1} K_j^\dagger K_j.$$

Theorem 16.33 (Stinespring dual representation). Lean ✓ Stmt *For Kraus operators $\{K_j\}$ and $A \in M_D(\mathbb{C})$,*

$$V^\dagger \left(A \otimes \mathbb{1}_r\right) V = \sum_j K_j^\dagger A\, K_j = T^*(A).$$

This is the Stinespring representation of the dual *(Heisenberg picture) map T^*.*

Proof. ✓ Proof Compute $(V^\dagger(A \otimes \mathbb{1}_r)V)_{ab}$ by summing over the product index (i, j); the δ_{jl} from $\mathbb{1}_r$ collapses the sum over l, giving $\sum_j K_j^\dagger A K_j$. ☐

Theorem 16.34 (The Stinespring isometry condition iff trace preservation). Lean ✓ Stmt $V^\dagger V = \mathbb{1}_D$ *if and only if* $\sum_j K_j^\dagger K_j = \mathbb{1}_D$*. In particular, V is an isometry precisely when the Kraus map is trace-preserving.*

Proof. ✓ Proof $(V^\dagger V)_{ab} = \sum_{(i,j)} \overline{V_{(i,j),a}}\, V_{(i,j),b} = \sum_j \sum_i \overline{(K_j)_{ia}}\, (K_j)_{ib} = (\sum_j K_j^\dagger K_j)_{ab}$. ☐

Theorem 16.35 (Stinespring Schrödinger representation). Lean ✓ Stmt *The Kraus map $T(\rho) = \sum_j K_j \rho K_j^\dagger$ equals the partial trace over the dilation space:*

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V \rho V^\dagger)_{(i,k),(j,k)} = \mathrm{tr}_r(V \rho V^\dagger).$$

This is the Schrödinger-picture Stinespring representation.

Proof. ✓ Proof Expand $(V \rho V^\dagger)_{(i,k),(j,k)}$ and sum over k to recover $\sum_k K_k \rho K_k^\dagger$. ☐

Theorem 16.36 (Existence of a Stinespring dilation). Lean ✓ Stmt *Every completely positive map admits an ancilla dimension r, Kraus operators $\{K_j\}_{j=0}^{r-1}$, and the concrete representation $\pi(A) = A \otimes \mathbb{1}_r$ such that*

$$E(A) = V^\dagger \pi(A) V.$$

Proof. ✓ Proof Choose a Kraus representation of E and apply the explicit Heisenberg-picture Stinespring formula. ☐

Theorem 16.37 (CPTP maps admit isometric Stinespring dilations). Lean ✓ Stmt *Every quantum channel admits a Stinespring dilation whose matrix V is an isometry, equivalently $V^\dagger V = \mathbb{1}$.*

Proof. ✓ Proof Choose Kraus operators for the channel and use trace preservation to obtain the Stinespring isometry condition $V^\dagger V = \mathbb{1}$. ☐

16.6 Ordered CP maps, Radon–Nikodym, and open-system representation

The next three results form the structural part of [Wol12, Theorems 2.3–2.5]: the ordered CP-map theorem, the Radon–Nikodym theorem for completely positive maps, and the open-system representation. They refine the Stinespring dilation by tracking how two CP maps related by domination or decomposition sit inside a common dilation space.

Definition 16.38 (CP partial order). Lean ✓ Stmt For linear maps $S, T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, write $T \le S$ (in the CP order) when $S - T$ is completely positive.

Definition 16.39 (Block-top contraction). Lean ✓ Stmt For natural numbers r, s, let $C = C_{r,s} \in \mathbb{C}^{r\times(r+s)}$ be the rectangular $r \times (r+s)$ matrix whose rows are the first r rows of the identity on $\mathbb{C}^{r+s}$: $C_{ij} = 1$ if $j = i < r$ and 0 otherwise.

Lemma 16.40 (C is a co-isometry). Lean ✓ Stmt *The block-top matrix satisfies $CC^\dagger = \mathbb{1}_r$.*

Proof. ✓ Proof Direct entrywise computation: $(CC^\dagger)_{ii'} = \sum_j C_{ij}\overline{C_{i'j}}$ collapses to $\delta_{ii'}$. □

Lemma 16.41 (C is a contraction). Lean ✓ Stmt *The block-top matrix satisfies $C^\dagger C \le \mathbb{1}_{r+s}$.*

Proof. ✓ Proof The matrix $C^\dagger C$ is diagonal with a 1 on the first r coordinates and a 0 on the last s, so $\mathbb{1} - C^\dagger C$ is positive semidefinite. □

Lemma 16.42 (Intertwining of Stinespring isometries). Lean ✓ Stmt *Let $K : \{0, \dots, r-1\} \to M_D(\mathbb{C})$ and $L : \{0, \dots, s-1\} \to M_D(\mathbb{C})$ be two Kraus families. Write $K +\!\!+ L$ for their concatenation as a family indexed by $\{0, \dots, r+s-1\}$. Then*

$$V_K = (\mathbb{1}_D \otimes C_{r,s})V_{K+\!\!+L}.$$

Proof. ✓ Proof Entrywise expansion of both sides: the Kronecker product with $C_{r,s}$ picks out the first r coordinates of the dilation space, matching the action of $V_{K+\!\!+L}$ on those coordinates, which is precisely V_K. □

Theorem 16.43 (Ordered CP maps via Stinespring contractions). Lean ✓ Stmt *Let $T_1, T_2 : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be CP maps with $T_1 \le T_2$. Then there exist an ancilla dimension m, Heisenberg-form Stinespring matrices V_1, V_2 realizing T_1, T_2 via $T_i(A) = V_i^\dagger(A \otimes \mathbb{1})V_i$, and a contraction $\widetilde{C}$ on the dilation space with $\widetilde{C}^\dagger\widetilde{C} \le \mathbb{1}$ such that*

$$V_1 = (\mathbb{1}_D \otimes \widetilde{C})\, V_2.$$

This is [Wol12, Theorem 2.3].

Proof. ✓ Proof Choose a Heisenberg-form Kraus family K_1 of T_1 with Stinespring matrix $V_1 = V_{K_1}$, and Kraus operators L of the CP map $T_2 - T_1$ in Schrödinger orientation. Let $L^\dagger$ denote the conjugate-transposed family. Set $m = r_1 + s$ and let $K_2 = K_1 +\!\!+ L^\dagger$. Then $T_2(A) = T_1(A) + (T_2 - T_1)(A) = \sum_i (K_1^i)^\dagger A K_1^i + \sum_j L_j A L_j^\dagger$, and the latter equals $\sum_i (K_1^i)^\dagger A K_1^i + \sum_j (L_j^\dagger)^\dagger A L_j^\dagger$ which is the Heisenberg form for K_2, yielding $T_2(A) = V_{K_2}^\dagger(A \otimes \mathbb{1})V_{K_2}$. Taking $\widetilde{C} = C_{r_1,s}$ the intertwining $V_{K_1} = (\mathbb{1}_D \otimes \widetilde{C})V_{K_2}$ is Lemma 16.42, and $\widetilde{C}^\dagger\widetilde{C} \le \mathbb{1}$ is Lemma 16.41. □

Definition 16.44 (Block-diagonal projectors on the dilation space). Lean Lean ✓ Stmt For r, s, let $P_{\mathrm{top}} = C_{r,s}^\dagger C_{r,s}$ and $P_{\mathrm{bot}} = \mathbb{1} - P_{\mathrm{top}}$. Both are PSD (for P_{bot}, by Lemma 16.41) and $P_{\mathrm{top}} + P_{\mathrm{bot}} = \mathbb{1}$.

Theorem 16.45 (Radon–Nikodym theorem for finite quantum instruments)**.** Lean ✓ Stmt *Let $\{T_i\}_{i\in I}$ be a nonempty finite family of completely positive maps, let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ satisfy $\sum_{i\in I} T_i = T$, and suppose that T has a supplied Stinespring representation*

$$T(A) = V^\dagger(A \otimes \mathbb{1}_r)V.$$

Then there are positive semidefinite operators $P_i \in M_r(\mathbb{C})$ with

$$\sum_{i\in I} P_i = \mathbb{1}_r.$$

For every $i \in I$ and $A \in M_D(\mathbb{C})$,

$$T_i(A) = V^\dagger(A \otimes P_i)V$$

This is [Wol12, Theorem 2.4], with the nonempty-family condition made explicit; see `docs/paper-gaps/wolf_radon_nikodym_nonempty_family.tex`.

Proof. ✓ Proof Choose Kraus families for the component maps T_i and group all their Kraus operators into one finite family. Add zero Kraus operators to one component so that this grouped family has at least as many rows as the supplied Stinespring dilation. Write the grouped Kraus operators as B_α, with a label map $\ell(\alpha) \in I$, and write the Kraus operators obtained from the blocks of V as A_j. Rectangular Kraus freedom gives a matrix W with

$$W^\dagger W = \mathbb{1}_r, \qquad B_\alpha = \sum_{j=0}^{r-1} W_{\alpha j} A_j.$$

For the fibre over i, set

$$(C_i)_{\alpha j} = \begin{cases} \overline{W_{\alpha j}}, & \ell(\alpha) = i, \\ 0, & \ell(\alpha) \neq i, \end{cases} \qquad P_i = C_i^\dagger C_i.$$

Then $P_i \geq 0$ and $\sum_i P_i = \mathbb{1}_r$ follows from $W^\dagger W = \mathbb{1}_r$. Finally,

$$V^\dagger(A \otimes P_i)V = \sum_{\ell(\alpha)=i} B_\alpha A B_\alpha^\dagger = T_i(A),$$

which proves the claim. ☐

Theorem 16.46 (Binary Radon–Nikodym theorem for CP maps)**.** Lean ✓ Stmt *Let $T_1, T_2 : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be completely positive. There exist an ancilla dimension m, a Kraus family K with Stinespring matrix $V = V_K$, and positive semidefinite operators $P_1, P_2 : \mathbb{C}^m \to \mathbb{C}^m$ with $P_1 + P_2 = \mathbb{1}_m$ such that, for $i = 1, 2$ and every $A \in M_D(\mathbb{C})$,*

$$T_i(A) = V^\dagger(A \otimes P_i)V.$$

Remark 16.47 (Relation with Wolf's Radon–Nikodym theorem)*.* Theorem 16.45 is the source-faithful finite-family statement relative to a supplied Stinespring dilation. Theorem 16.46 is the earlier binary block-diagonal corollary in which the common dilation is constructed from the Kraus families of T_1 and T_2.

Proof. ✓ Proof Take Heisenberg-form Kraus families K_1 for T_1 and Schrödinger-form L for T_2, and form $K = K_1 +\!\!+ L^\dagger$ on $\mathbb{C}^{r_1+s}$. Choose $P_1 = P_{\text{top}}$, $P_2 = P_{\text{bot}}$ as in Definition 16.44. The Kronecker identity $A \otimes (C^\dagger C) = (\mathbb{1} \otimes C)^\dagger (A \otimes \mathbb{1})(\mathbb{1} \otimes C)$ rewrites $V^\dagger (A \otimes P_{\text{top}})V$ as $((\mathbb{1} \otimes C)V)^\dagger (A \otimes \mathbb{1})((\mathbb{1} \otimes C)V)$, which equals $V_{K_1}^\dagger (A \otimes \mathbb{1}) V_{K_1} = T_1(A)$ by the intertwining Lemma 16.42. For the complementary block, $A \otimes P_{\text{bot}} = A \otimes \mathbb{1} - A \otimes P_{\text{top}}$ and sandwiching by V yields $(T_1 + T_2)(A) - T_1(A) = T_2(A)$. □

Theorem 16.48 (Open-system representation of quantum channels). Lean ✓ Stmt *Every quantum channel $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ admits a Stinespring isometry V on an ancilla space $\mathbb{C}^r$ such that*

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V \rho V^\dagger)_{(i,k),(j,k)} = \operatorname{tr}_r(V \rho V^\dagger).$$

Proof. ✓ Proof Apply the existence of an isometric Stinespring dilation (Theorem 16.37) and the Schrödinger-picture identity (Theorem 16.35). □

Theorem 16.49 (Open-system representation, partial-trace form). Lean ✓ Stmt *Every quantum channel T admits an isometric Stinespring dilation V such that $T(\rho) = \operatorname{tr}_E(V \rho V^\dagger)$.*

Proof. ✓ Proof Rewrite the componentwise identity of Theorem 16.48 as a partial trace over the ancilla factor. □

Remark 16.50 (Unitary form)*.* The fully unitary open-system form $T(\rho) = \operatorname{tr}_E(U(\rho \otimes |\phi\rangle\langle\phi|)U^\dagger)$ of [Wol12, Theorem 2.5] follows from the isometric form by extending the Stinespring isometry V to a unitary U on $\mathbb{C}^D \otimes \mathbb{C}^r$; this uses orthonormal basis extension and is left to a later theorem.

16.7 POVMs and Naimark dilation

Definition 16.51 (Positive operator-valued measure). Lean ✓ Stmt A *positive operator-valued measure* with n outcomes on $\mathbb{C}^D$ is a family $\{E_i\}_{i=0}^{n-1}$ of positive semidefinite operators on $\mathbb{C}^D$ satisfying the resolution of identity

$$\sum_{i=0}^{n-1} E_i = \mathbb{1}_D.$$

Definition 16.52 (Naimark Kraus square roots). Lean Lean ✓ Stmt For a POVM $\{E_i\}$, each effect $E_i \geq 0$ admits a square-root factorisation $E_i = M_i^\dagger M_i$ with $M_i \in M_D(\mathbb{C})$. The operators M_i are the *Naimark Kraus square roots.*

Theorem 16.53 (Naimark square roots satisfy the Kraus normalization). Lean ✓ Stmt *For a POVM $\{E_i\}$ with square roots $E_i = M_i^\dagger M_i$, one has*

$$\sum_i M_i^\dagger M_i = \mathbb{1}.$$

Proof. ✓ Proof Sum the identities $E_i = M_i^\dagger M_i$ and use the defining resolution of the identity for the POVM. □

Definition 16.54 (Naimark isometry). Lean ✓ Stmt The *Naimark isometry* $V : \mathbb{C}^D \to \mathbb{C}^D \otimes \mathbb{C}^n$ of a POVM is the Stinespring-type construction

$$V = \sum_{i=0}^{n-1} M_i \otimes |i\rangle, \qquad V_{(a,i),\,k} = (M_i)_{a,k}.$$

Definition 16.55 (Naimark projective measurement). Lean Lean ✓ Stmt The *Naimark projectors* on $\mathbb{C}^D \otimes \mathbb{C}^n$ are

$$P_i = \mathbb{1}_D \otimes |i\rangle\langle i|, \qquad (P_i)_{(a,b),(c,d)} = \delta_{a,c}\delta_{b,i}\delta_{d,i}.$$

Theorem 16.56 (Naimark isometry condition). Lean ✓ Stmt *The Naimark isometry satisfies $V^\dagger V = \mathbb{1}_D$.*

Proof. ✓ Proof The Stinespring Gram identity gives $V^\dagger V = \sum_i M_i^\dagger M_i = \sum_i E_i = \mathbb{1}_D$ by the resolution of identity. $\square$

Theorem 16.57 (Naimark projectors form a projective measurement). Lean Lean Lean Lean ✓ Stmt *The Naimark projectors satisfy $P_i^2 = P_i$, $P_i^\dagger = P_i$, $P_iP_j = 0$ for $i \neq j$, and $\sum_i P_i = \mathbb{1}_{\mathbb{C}^D \otimes \mathbb{C}^n}$.*

Proof. ✓ Proof Each identity follows from direct entrywise computation using the delta-function form of $(P_i)_{(a,b),(c,d)}$. $\square$

Theorem 16.58 (Naimark dilation). Lean ✓ Stmt *Every POVM $\{E_i\}_{i=0}^{n-1}$ arises as a projective measurement on a dilation: for the isometry V and projectors P_i above,*

$$E_i = V^\dagger P_i V.$$

Proof. ✓ Proof Computing entrywise, $(V^\dagger P_i V)_{k,\ell} = \sum_a \overline{(M_i)_{a,k}}\,(M_i)_{a,\ell} = (M_i^\dagger M_i)_{k,\ell} = (E_i)_{k,\ell}$. $\square$

Theorem 16.59 (Existential Naimark dilation). Lean ✓ Stmt *For every POVM $\{E_i\}$ on $\mathbb{C}^D$ there exist a dilation dimension r, an isometry $V : \mathbb{C}^D \to \mathbb{C}^D \otimes \mathbb{C}^r$, and a projective measurement $\{P_i\}$ on the dilation satisfying $E_i = V^\dagger P_i V$.*

Proof. ✓ Proof Take $r = n$ and the explicit witnesses $V = \sum_i M_i \otimes |i\rangle$ and $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$. $\square$

Definition 16.60 (Naimark dilation as a structure). Lean ✓ Stmt A *Naimark dilation* of a POVM $\{E_i\}$ consists of an isometry V into a larger Hilbert space together with a projective measurement $\{P_i\}$ there such that $E_i = V^\dagger P_i V$ for all i.

Theorem 16.61 (Canonical dilation satisfies the Naimark dilation axioms). Lean ✓ Stmt *The explicit isometry $V = \sum_i M_i \otimes |i\rangle$ and projectors $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$ satisfy the defining axioms of a Naimark dilation.*

Proof. ✓ Proof Combine the previously proved identities $V^\dagger V = \mathbb{1}_D$, $P_i^2 = P_i$, $P_i^\dagger = P_i$, $P_iP_j = 0$ for $i \neq j$, $\sum_i P_i = \mathbb{1}$, and $V^\dagger P_i V = E_i$. $\square$

Theorem 16.62 (Concrete uniqueness for canonical projectors). Lean ✓ Stmt *Let $V : \mathbb{C}^D \to \mathbb{C}^D \otimes \mathbb{C}^n$ satisfy $V^\dagger P_i V = E_i$ for the canonical projectors $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$. Then there exists an isometry W on the dilated space such that $V = WV_0$, where V_0 is the canonical Naimark isometry of $\{E_i\}$.*

Proof. ✓ Proof For each outcome i, the i-th block of V has Gram matrix E_i. Comparing with the canonical square root M_i gives $V_i^\dagger V_i = M_i^\dagger M_i$, hence $V_i = U_i M_i$ for a unitary U_i on $\mathbb{C}^D$. Assemble the U_i block-diagonally to obtain an isometry $W = \bigoplus_i U_i$ on $\mathbb{C}^D \otimes \mathbb{C}^n$, and then $V = W \sum_i M_i \otimes |i\rangle = WV_0$. $\square$

Definition 16.63 (POVM from a PSD resolution of identity on a dilation). Lean ✓ Stmt Given an isometry $V : \mathbb{C}^D \to \mathbb{C}^{d'}$ with $V^\dagger V = \mathbb{1}_D$ and a family $\{P_i\}_{i=0}^{n-1}$ of positive semidefinite operators on $\mathbb{C}^{d'}$ summing to the identity, the pulled-back operators

$$E_i := V^\dagger P_i V$$

form a POVM. In particular, any projective measurement on the dilation (a special case of a PSD resolution of identity) pulls back to a POVM.

Definition 16.64 (Quantum instrument). Lean ✓ Stmt A *quantum instrument* with n outcomes is a family $\{\Phi_i\}_{i=0}^{n-1}$ of completely positive maps on $M_D(\mathbb{C})$ whose sum $\sum_i \Phi_i$ is trace-preserving.

Definition 16.65 (Instrument operations). Lean Lean Lean Lean ✓ Stmt Associated to an instrument are the total channel $\sum_i \Phi_i$, the unnormalized update $\rho \mapsto \Phi_i(\rho)$ for each outcome i, the outcome probability $p_i(\rho) = \operatorname{tr}(\Phi_i(\rho))$, and the normalized posterior state $\Phi_i(\rho)/p_i(\rho)$ whenever $p_i(\rho) \neq 0$.

Theorem 16.66 (Instrument total map is a channel). *Lean* *✓ Stmt* *The total map $\sum_i \Phi_i$ of an instrument is a quantum channel.*

Proof. ✓ Proof Complete positivity of the sum follows from closure of CP maps under finite addition; trace preservation is built into the definition. □

Theorem 16.67 (Conservation of probability for instruments). *Lean* *✓ Stmt* *For every state $\rho \in M_D(\mathbb{C})$, the outcome probabilities $p_i(\rho) := \operatorname{tr}(\Phi_i(\rho))$ sum to $\operatorname{tr}(\rho)$.*

Proof. ✓ Proof Linearity of trace and trace preservation of $\sum_i \Phi_i$ give $\sum_i \operatorname{tr}(\Phi_i(\rho)) = \operatorname{tr}((\sum_i \Phi_i)(\rho)) = \operatorname{tr}(\rho)$. □

Theorem 16.68 (Non-negativity of instrument probabilities). *Lean* *✓ Stmt* *If $\rho \in M_D(\mathbb{C})$ is positive semidefinite, then each outcome probability $p_i(\rho) = \operatorname{tr}(\Phi_i(\rho))$ is a non-negative real number.*

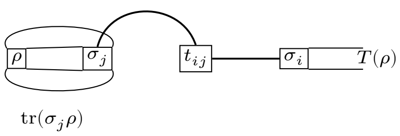
Proof. ✓ Proof Each Φ_i is completely positive, hence positive, so $\Phi_i(\rho)$ is positive semidefinite and its trace is a non-negative real. □

16.8 Trace-pairing expansion in transfer-matrix form

The trace-pairing expansion separates a linear map into the scalar input pairing, the transfer coefficients, and the output basis element:

$$T(\rho) = \sum_i \sum_j t_{ij} \operatorname{tr}(\sigma_j \rho)\, \sigma_i.$$

The diagram records the same factorization by closing the input matrix with σ_j, weighting the resulting scalar by t_{ij}, and producing the output matrix through σ_i:



Definition 16.69 (Trace-self-dual basis). Lean ✓ Stmt A basis $\{\sigma_i\}_i$ of $M_D(\mathbb{C})$ is *trace-self-dual* when its coordinate functionals are given by trace pairing:

$$X = \sum_i \operatorname{tr}(\sigma_i X)\sigma_i.$$

Definition 16.70 (Trace-pairing coefficients). Lean ✓ Stmt For a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ and a trace-self-dual basis $\{\sigma_i\}$, define coefficients

$$t_{ij} := \operatorname{tr}(\sigma_i T(\sigma_j)).$$

Theorem 16.71 (Trace-pairing expansion of a linear map). *Lean* *✓ Stmt* *If $\{\sigma_i\}$ is trace-self-dual, then every linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ admits the expansion*

$$T(\rho) = \sum_i \sum_j t_{ij} \operatorname{tr}(\sigma_j \rho)\sigma_i, \qquad t_{ij} = \operatorname{tr}(\sigma_i T(\sigma_j)).$$

Proof. ✓ Proof Expand ρ and each $T(\sigma_j)$ in the chosen basis and substitute. The trace-self-dual identity replaces coordinates by traces, giving the stated double-sum formula. □

16.9 SVD normal form (existence)

[Wol12, Section 2.3] discusses two families of normal forms for the transfer-matrix representation of a quantum channel: the *singular value decomposition* (SVD) and, for qubit channels, the *Lorentz normal form* obtained by $\mathrm{SL}(2,\mathbb{C})$ filtering operations. The SVD normal form is proved below. The Lorentz normal form theorem (Theorem 16.83) requires the compactness and minimisation argument for $\mathrm{SL}(2,\mathbb{C})$ filterings in [Wol12, Propositions 2.9 and 2.11]; see Section 16.10.

Theorem 16.72 (SVD for positive semidefinite matrices). *Lean* *✓ Stmt* *Every positive semidefinite matrix $M \in M_D(\mathbb{C})$ admits a decomposition*

$$M = U \operatorname{diag}(\sigma) U^\dagger,$$

where $U \in \mathcal{U}(D)$ is unitary and $\sigma : \{0, \dots, D-1\} \to \mathbb{R}_{\geq 0}$ has non-negative entries.

Proof. ✓ Proof The spectral theorem for Hermitian matrices provides an orthonormal eigenbasis with real eigenvalues. Positive semidefiniteness forces those eigenvalues to be non-negative, and the decomposition is written in SVD form with U the eigenvector unitary and σ the eigenvalue sequence. □

Theorem 16.73 (SVD existence for invertible matrices). *Lean* *✓ Stmt* *Every invertible complex square matrix $M \in M_D(\mathbb{C})$ admits a singular value decomposition*

$$M = U \operatorname{diag}(\sigma) V^\dagger,$$

with $U, V \in \mathcal{U}(D)$ unitary and $\sigma_i > 0$ for all i.

Proof. ✓ Proof Apply the spectral theorem to the positive definite matrix $M^\dagger M$ to obtain $M^\dagger M = V \operatorname{diag}(\lambda) V^\dagger$ with $\lambda_i > 0$. Set $\sigma_i := \sqrt{\lambda_i}$ and $\Sigma := \operatorname{diag}(\sigma)$; since Σ is a real positive diagonal, it is self-adjoint and invertible. Define $U := MV\Sigma^{-1}$. Then $U^\dagger U = \Sigma^{-1} V^\dagger (M^\dagger M) V \Sigma^{-1} = \Sigma^{-1} \operatorname{diag}(\lambda) \Sigma^{-1} = \mathbb{1}$, so U is unitary, and $U\Sigma V^\dagger = MV(\Sigma^{-1}\Sigma)V^\dagger = MVV^\dagger = M$. □

Theorem 16.74 (SVD representation of a transfer matrix)**.** Lean ✓ Stmt *Every invertible transfer matrix $\widehat{T}$ of a linear super-operator $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ admits a singular value decomposition $\widehat{T} = U \operatorname{diag}(\sigma) V^\dagger$ with U, V unitary on $\mathbb{C}^D \otimes \mathbb{C}^D$ and $\sigma_i > 0$.*

Proof. ✓ Proof Direct specialisation of Theorem 16.73 to the index type $\{0, \dots, D-1\} \times \{0, \dots, D-1\}$ used by the transfer-matrix representation. □

Remark 16.75 (Sorted / unique singular values). Theorem 16.73 produces the singular values as an unordered family, without the usual convention that σ_i are sorted in non-increasing order or the statement that they are uniquely determined by M. Downstream applications in [Wol12, Section 2.3], such as trace-norm identities, polar decomposition, and the Lorentz normal form, will typically require the sorted / unique variant; that refinement is future work.

16.10 Lorentz normal form

This section states the existence theorems from [Wol12, Section 2.3].

Definition 16.76 (SL-filtering operation)**.** An *SL-filtering* for $D \times D$ matrices is a completely positive map of the form $\Phi(X) = SXS^\dagger$ where $S \in M_D(\mathbb{C})$ satisfies $\det S = 1$. Such maps are invertible and have Kraus rank 1.

Definition 16.77 (Doubly-stochastic map)**.** A linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *doubly-stochastic* if $T(\mathbb{1}) \propto \mathbb{1}$ and the reduced density matrix $\operatorname{tr}_1[\tau]$ of its Choi matrix $\tau = (T \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|)$ is proportional to the identity. This is the normal form in [Wol12, Proposition 2.8].

Lemma 16.78 (Infimum attainment for SL-filterings)**.** *For a positive-definite Choi matrix τ, the infimum of $\operatorname{tr}[(S_2 \otimes_k S_1)\tau(S_2 \otimes_k S_1)^\dagger]$ over $S_1, S_2 \in M_D(\mathbb{C})$ with $\det S_1 = \det S_2 = 1$ is attained.* ***Not yet proved****—requires compactness of bounded subsets of $\mathrm{SL}(D, \mathbb{C})$ and the extreme-value theorem.*

Theorem 16.79 (Generic normal form for CP maps)**.** *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a completely positive map whose Choi matrix is positive-definite (equivalently, T has full Kraus rank). Then there exist SL-filterings Φ_1, Φ_2 such that $\Phi_2 \circ T \circ \Phi_1$ is doubly-stochastic.* ***Not yet proved****—proof blocked on Lemma 16.78.*

Definition 16.80 (Diagonal Lorentz normal form)**.** For a completely positive, trace-preserving qubit channel $T' : M_2(\mathbb{C}) \to M_2(\mathbb{C})$, write $\widehat{T'}$ for its matrix in the normalized Pauli basis $\{\sigma_0/\sqrt{2}, \sigma_1/\sqrt{2}, \sigma_2/\sqrt{2}, \sigma_3/\sqrt{2}\}$. The channel is in *diagonal Lorentz normal form* when it is unital and every off-diagonal entry of $\widehat{T'}$ is zero.

Definition 16.81 (Non-diagonal Lorentz normal form)**.** For a completely positive, trace-preserving qubit channel $T' : M_2(\mathbb{C}) \to M_2(\mathbb{C})$, write $\widehat{T'}$ in the normalized Pauli basis. The channel is in *non-diagonal Lorentz normal form* when, for some $x \in [0, 1]$,

$$\widehat{T'} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & x/\sqrt{3} & 0 & 0 \\ 0 & 0 & x/\sqrt{3} & 0 \\ 2/3 & 0 & 0 & 1/3 \end{pmatrix}.$$

Trace preservation supplies the first row of the displayed matrix.

Definition 16.82 (Singular Lorentz normal form)**.** For a completely positive, trace-preserving qubit channel $T' : M_2(\mathbb{C}) \to M_2(\mathbb{C})$, write $\widehat{T'}$ in the normalized Pauli basis. The channel is in *singular Lorentz normal form* when

$$\widehat{T'} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

equivalently, every input state is mapped to $(1+\sigma_3)/2$. Trace preservation supplies the first row of the displayed matrix.

Theorem 16.83 (Lorentz normal form for qubit channels)**.** *For every qubit channel $T : M_2(\mathbb{C}) \to M_2(\mathbb{C})$, there exist $\mathrm{SL}(2,\mathbb{C})$-filterings Φ_1, Φ_2 such that the filtered channel $T' = \Phi_2 \circ T \circ \Phi_1$ is in one of the three Lorentz normal forms: diagonal, non-diagonal, or singular.* ***Not yet proved****—proof blocked on Lemma 16.78 and the Lorentz group classification of $\mathrm{SL}(2,\mathbb{C})$ orbits.*

16.11 Determinant of a quantum channel

Definition 16.84 (Channel determinant)**.** Lean ✓ Stmt For a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, the *channel determinant* $\det T$ is the determinant of the matrix of T with respect to the standard matrix-unit basis of $M_D(\mathbb{C})$. This is the quantity studied in [Wol12, Section 6.1].

Remark 16.85 (Channel determinant as an operator determinant). Lean ✓ Stmt The channel determinant agrees with the ordinary determinant of the linear endomorphism $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$.

Definition 16.86 (Unitary channel)**.** Lean Lean ✓ Stmt For a unitary matrix $U \in \mathcal{U}(D)$, the *unitary channel* is $T(\rho) = U\rho U^\dagger$. It is automatically a quantum channel.

Theorem 16.87 (Unitary channels have determinant one)**.** *Lean* *Lean* *✓ Stmt* *If $T(\rho) = U\rho U^\dagger$ is a unitary channel, then $\det T = 1$ and hence $|\det T| = 1$.*

Proof. ✓ Proof Vectorization identifies T with a Kronecker product $\overline{U} \otimes U$, whose determinant is $\overline{\det U}^D (\det U)^D = 1$. ☐

Theorem 16.88 (Determinant bound for positive trace-preserving maps)**.** *Lean* *Lean* *✓ Stmt* *For any positive trace-preserving map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, $|\det T| \le 1$. This is [Wol12, Theorem 6.1(1)].*

Proof. ✓ Proof Every eigenvalue μ of T satisfies $|\mu| \le 1$ (positivity and trace preservation force spectral radius ≤ 1). Since $\det T$ is the product of the eigenvalues (counted with algebraic multiplicity), the bound $|\det T| = \prod_i |\mu_i| \le 1$ follows by induction. ☐

Theorem 16.89 (Determinant one iff the channel is unitary)**.** *Lean* *✓ Stmt* *For a CPTP map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$,*

$$|\det T| = 1 \iff \exists U \in \mathcal{U}(D), \quad T(\rho) = U\rho U^\dagger.$$

This is [Wol12, Theorem 6.1(2)].

Proof. ✓ Proof Reverse direction is Theorem 16.87. Forward direction: $|\det T| = 1$ together with $|\mu| \leq 1$ for every eigenvalue μ (from trace preservation and positivity) forces every eigenvalue to satisfy $|\mu| = 1$. Hence T is bijective. A bijective CPTP map is a $*$-automorphism: the Heisenberg dual $T^*(Y) = \sum_i K_i^\dagger Y K_i$ is unital (since T is TP), so the Kadison–Schwarz inequality $T^*(Y^\dagger Y) \geq T^*(Y)^\dagger T^*(Y)$ becomes an equality for every Y. Hence T^*, and therefore T, preserves products. By the Skolem–Noether theorem, every $*$-automorphism of $M_D(\mathbb{C})$ has the form $T(\rho) = U\rho U^\dagger$ for some unitary U. Kraus freedom then gives $K_i = c_i U$ with $\sum_i |c_i|^2 = 1$. □

Chapter 17

Quantum Dynamical Semigroups

This chapter develops the theory of one-parameter semigroups of quantum channels and characterizes their generators via the GKSL (Gorini–Kossakowski–Sudarshan–Lindblad) theorem, following [Wol12, Chapter 7].

Section 17.1 establishes definitions and the basic exponential form (Proposition 17.14). Section 17.2 gives the Duhamel formula and perturbation bound. Section 17.3 develops the generator theory and proves the GKSL theorem.

17.1 Dynamical semigroups and the exponential form

Definition 17.1 (Dynamical semigroup). Lean ✓ Stmt A family of linear maps $T : \mathbb{R} \to (M_D(\mathbb{C}) \to_\ell M_D(\mathbb{C}))$ is a *dynamical semigroup* if

1. (semigroup law) $T_{t+s} = T_t \circ T_s$ for all $t, s \geq 0$; and
2. (initial condition) $T_0 = \mathrm{id}$.

This is [Wol12, Equation (7.1)].

Definition 17.2 (Continuous dynamical semigroup). Lean ✓ Stmt A dynamical semigroup T is *norm-continuous* if the map $t \mapsto T_t$ is continuous in the operator norm topology. In finite dimension this coincides with strong continuity.

Definition 17.3 (Exponential semigroup). Lean Lean ✓ Stmt Given a generator $L \in \mathrm{End}_{\mathbb{C}}(M_D(\mathbb{C}))$, the *exponential semigroup* is

$$T_t := e^{tL} = \sum_{k=0}^{\infty} \frac{(tL)^k}{k!}.$$

Theorem 17.4 (Semigroup law for the exponential). Lean Lean ✓ Stmt *For all $t, s \in \mathbb{R}$, $e^{(t+s)L} = e^{tL} \cdot e^{sL}$.*

Proof. ✓ Proof Since $(t \cdot L)$ commutes with $(s \cdot L)$, this follows from the exponential addition formula for commuting elements in a Banach algebra. □

Theorem 17.5 (Exponential semigroup at time 0). Lean ✓ Stmt *The continuous-linear-map exponential satisfies $e^{0L} = \mathbb{1}$.*

Proof. ✓ Proof The exponential of the zero endomorphism is the identity. □

Theorem 17.6 (Continuity of the exponential semigroup). Lean Lean ✓ Stmt *The map $t \mapsto e^{tL}$ is continuous in the operator norm topology.*

Proof. ✓ Proof The exponential is analytic (convergence radius $= \infty$), hence continuous. □

Theorem 17.7 (Derivative of the exponential semigroup). Lean ✓ Stmt *For all $t \in \mathbb{R}$,*

$$\frac{d}{dt}e^{tL} = e^{tL} \cdot L.$$

This is [Wol12, Equation (7.2)].

Proof. ✓ Proof Chain rule applied to the composition $\mathbb{R} \xrightarrow{t \mapsto t \cdot L} \mathrm{End}(M_D(\mathbb{C})) \xrightarrow{\exp} \mathrm{End}(M_D(\mathbb{C}))$. □

Theorem 17.8 (Derivative at the origin). Lean ✓ Stmt *One has*

$$\left.\frac{d}{dt}e^{tL}\right|_{t=0} = L.$$

Proof. ✓ Proof This is Theorem 17.7 at $t = 0$ together with Theorem 17.5. □

Theorem 17.9 (Generator uniqueness). Lean ✓ Stmt *If $e^{tL} = e^{tL'}$ for all $t \geq 0$, then $L = L'$.*

Proof. ✓ Proof Both semigroups agree on $[0, \infty)$, so their derivatives within $[0, \infty)$ at $t = 0$ agree. Since $[0, \infty)$ has unique differentials at 0, $L = L'$. □

Theorem 17.10 (CLM and linear-map forms coincide). Lean ✓ Stmt *The linear-map exponential semigroup and the continuous-linear-map exponential semigroup agree under the canonical identification of endomorphisms with continuous linear endomorphisms.*

Proof. ✓ Proof This is a direct unpacking of the definition of the linear-map exponential semigroup. □

Theorem 17.11 (Pointwise derivative formula). Lean ✓ Stmt *For every $X \in M_D(\mathbb{C})$ and every $t \in \mathbb{R}$,*

$$\frac{d}{dt}(e^{tL}(X)) = e^{tL}(L(X)).$$

Proof. ✓ Proof Evaluate the derivative of the continuous-linear-map semigroup at X and transport the statement through Theorem 17.10. □

Theorem 17.12 (Linear-map exponential semigroup at time 0). Lean ✓ Stmt *The linear-map exponential semigroup satisfies $e^{0L} = \mathbb{1}$.*

Proof. ✓ Proof Transport Theorem 17.5 through Theorem 17.10. □

Theorem 17.13 (Exponential semigroup as a dynamical semigroup). Lean ✓ Stmt *The family $t \mapsto e^{tL}$ satisfies the semigroup law and the initial condition, hence is a dynamical semigroup.*

Proof. ✓ Proof Transport the continuous-linear-map semigroup law from Theorem 17.4 through Theorem 17.10; the initial condition is Theorem 17.12. □

Theorem 17.14 (Every continuous semigroup is exponential). Lean ✓ Stmt *Every norm-continuous dynamical semigroup T on $M_D(\mathbb{C})$ is of the form*

$$T_t = e^{tL}$$

for some generator $L \in \mathrm{End}_{\mathbb{C}}(M_D(\mathbb{C}))$ and every $t \geq 0$. This is [Wol12, Proposition 7.1].

Proof. ✓ Proof Since $T_0 = \mathbb{1}$ and T is continuous, the integral $M_\varepsilon = \int_0^\varepsilon T_s\, ds$ is invertible for small $\varepsilon > 0$. Using the semigroup property, $T_t = M_\varepsilon^{-1}(M_{t+\varepsilon} - M_t)$ is differentiable for $t \geq 0$. ODE uniqueness then gives $T_t = e^{tL}$ for all $t \geq 0$. □

17.2 Perturbation theory

Theorem 17.15 (Duhamel formula). Lean ✓ Stmt *Let $\Delta = L' - L$. For $t \geq 0$,*

$$e^{tL'} - e^{tL} = \int_0^t e^{(t-s)L} \Delta\, e^{sL'}\, ds.$$

This is [Wol12, Lemma 7.1].

Proof. ✓ Proof Define $f(s) = e^{(t-s)L} e^{sL'}$. Then $f'(s) = e^{(t-s)L} \Delta\, e^{sL'}$ (Theorem 17.7), and $e^{tL'} - e^{tL} = f(t) - f(0) = \int_0^t f'(s)\, ds$. □

Theorem 17.16 (Perturbation bound). Lean ✓ Stmt *For $t \geq 0$,*

$$\|e^{tL'} - e^{tL}\| \leq t\, \|\Delta\| \cdot \sup_{s\in[0,t]} \|e^{sL}\| \cdot \sup_{s\in[0,t]} \|e^{sL'}\|,$$

where $\Delta = L' - L$. This is [Wol12, Corollary 7.1].

Proof. ✓ Proof Apply the triangle inequality and submultiplicativity of the operator norm to the Duhamel integral (Theorem 17.15). □

Corollary 17.17 (Perturbation bound under unit-norm control). Lean ✓ Stmt *For $t \geq 0$, if $\|e^{sL}\| \leq 1$ and $\|e^{sL'}\| \leq 1$ for every $s \in [0, t]$, then*

$$\|e^{tL'} - e^{tL}\| \leq t\, \|L' - L\|.$$

Proof. ✓ Proof Apply Theorem 17.16 and bound both suprema by 1. □

Lemma 17.18 (Dyson-series summability from factorial majorant). Lean ✓ Stmt *Assume a factorial majorant of Dyson–Phillips iterates:*

$$\|D_n(t)\| \leq M\, \frac{(t\, \|L' - L\|\, M)^n}{n!}.$$

Then the Dyson series is summable in operator norm.

Proof. ✓ Proof This is the Weierstrass M-test step, using the scalar convergence of $\sum_n a^n/n!$ and lifting via norm-bounded summability. □

Theorem 17.19 (Factorial norm bound for Dyson iterates). Lean ✓ Stmt *For $s \in [0, t]$ and $M = \sup_{u\in[0,t]} \|e^{uL}\|$,*

$$\|\widetilde{T}_s^{(n)}\| \leq M\, \frac{(s\, \|\Delta\|\, M)^n}{n!}.$$

Proof. ✓ Proof Induction on n. The base case uses the semigroup supremum bound. The inductive step bounds the integrand pointwise by $M\|\Delta\| \cdot M(u\|\Delta\|M)^n/n!$ and integrates $\int_0^s u^n\,du = s^{n+1}/(n+1)$, recovering the factorial. □

Corollary 17.20 (Dyson series summability). Lean ✓ Stmt *The Dyson–Phillips series* $\sum_n \widetilde{T}_t^{(n)}$ *converges in operator norm for every* $t \geq 0$.

Proof. ✓ Proof Compose the factorial norm bound (Theorem 17.19) with the Weierstrass M-test step (Lemma 17.18). □

Corollary 17.21 (Factorial norm bound at the final time). Lean ✓ Stmt *For* $t \geq 0$*, setting* $s = t$ *in Theorem 17.19 gives*

$$\|\widetilde{T}_t^{(n)}\| \leq M\,\frac{(t\,\|\Delta\|\,M)^n}{n!},$$

where $M = \sup_{u\in[0,t]} \|e^{uL}\|$.

Proof. ✓ Proof This is the special case $s = t$ of Theorem 17.19. □

Theorem 17.22 (Continuity of Dyson iterates). Lean ✓ Stmt *Each Dyson iterate* $t \mapsto \widetilde{T}_t^{(n)}$ *is continuous in the time parameter.*

Proof. ✓ Proof By induction on n. The base case is continuity of the matrix exponential. For the inductive step, the parametric integral $t \mapsto \int_0^t e^{(t-s)L}\Delta\,\widetilde{T}_s^{(n)}\,ds$ is continuous by the continuous-parameter primitive theorem; the pasting lemma extends this to all of $\mathbb{R}$ since $\widetilde{T}_t^{(n+1)} = 0$ for $t \leq 0$. □

Theorem 17.23 (Factorial bound on the Dyson remainder). Lean ✓ Stmt *For* $s \in [0,t]$*,* $M = \sup_{u\in[0,t]} \|e^{uL}\|$*, and* $M' = \sup_{u\in[0,t]} \|e^{uL'}\|$*:*

$$\|T'_s - \textstyle\sum_{n<N} \widetilde{T}_s^{(n)}\| \leq M'\,\frac{(s\,\|\Delta\|\,M)^N}{N!}.$$

Proof. ✓ Proof Induction on N. The base case is the semigroup supremum bound. The inductive step uses the telescoping remainder identity $R_{N+1}(s) = \int_0^s e^{(s-u)L}\Delta\,R_N(u)\,du$ (derived from the Duhamel formula), bounds the integrand pointwise, and integrates to recover the factorial. □

Theorem 17.24 (Dyson series identity [Wol12, Equation (7.13)]). Lean ✓ Stmt *For* $t \geq 0$*, the Dyson–Phillips series equals the perturbed semigroup:*

$$\sum_{n=0}^{\infty} \widetilde{T}_t^{(n)} = e^{tL'}.$$

Proof. ✓ Proof The partial-sum remainder $\|e^{tL'} - \sum_{n<N} \widetilde{T}_t^{(n)}\| \leq M'\,(t\|\Delta\|M)^N/N!$ tends to zero as $N \to \infty$, since $c^N/N! \to 0$ for any constant c. Together with the summability of the series (Corollary 17.20), this identifies the sum via uniqueness of limits. □

Corollary 17.25 (Tsum form of the Dyson series). Lean ✓ Stmt $\sum_{n=0}^{\infty} \widetilde{T}_t^{(n)} = e^{tL'}$ *(tsum form).*

Proof. ✓ Proof Immediate from Theorem 17.24. □

17.3 GKSL/Lindblad generators

This section characterizes generators of semigroups of completely positive and trace-preserving maps. The central result is the GKSL theorem (Theorem 17.54), which shows that such generators have the standard Lindblad form.

17.3.1 Generator decomposition and conditional complete positivity

Definition 17.26 (Generator decomposition). Lean ✓ Stmt A *generator decomposition* consists of a completely positive map $\phi : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ and a matrix $\kappa \in M_d(\mathbb{C})$, defining the linear map

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger.$$

This is [Wol12, Equation (7.14)].

Definition 17.27 (Conditional complete positivity). Lean ✓ Stmt A linear map $L : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ is *conditionally completely positive* (CCP) if it admits a generator decomposition, i.e. there exist a CP map ϕ and a matrix κ such that $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$. This is condition 1 of [Wol12, Proposition 7.2].

Definition 17.28 (Trace-annihilating map). Lean ✓ Stmt A linear map $L : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ is *trace-annihilating* if $\mathrm{tr}(L(\rho)) = 0$ for all $\rho \in M_d(\mathbb{C})$. This is the infinitesimal version of trace preservation.

Definition 17.29 (Trace constraint). Lean ✓ Stmt A generator decomposition (ϕ, κ) satisfies the *trace constraint* if $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, where ϕ^* is the Hilbert–Schmidt adjoint. This is the condition in [Wol12, Equation (7.20)].

Theorem 17.30 (Trace constraint implies trace-annihilating). Lean ✓ Stmt *If (ϕ, κ) satisfies $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, then $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ is trace-annihilating.*

Proof. ✓ Proof Let $\phi(\rho) = \sum_i K_i\rho K_i^\dagger$. By the cyclic property of trace, $\mathrm{tr}(L(\rho)) = \mathrm{tr}((\sum_i K_i^\dagger K_i)\rho) - \mathrm{tr}(\kappa\rho) - \mathrm{tr}(\kappa^\dagger\rho) = \mathrm{tr}((\kappa + \kappa^\dagger)\rho) - \mathrm{tr}(\kappa\rho) - \mathrm{tr}(\kappa^\dagger\rho) = 0$. □

17.3.2 The Lindblad form

Definition 17.31 (Lindblad form). Lean ✓ Stmt A *Lindblad form* consists of a Hermitian matrix $H = H^\dagger$ (the Hamiltonian) and a family of matrices $\{L_j\}_{j=0}^{r-1}$ (the Lindblad operators), defining the linear map

$$L(\rho) = i[\rho, H] + \sum_j \Big(L_j\rho L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\}_+ \Big),$$

where $[A, B] = AB - BA$ and $\{A, B\}_+ = AB + BA$. This is [Wol12, Equation (7.21)].

Theorem 17.32 (Lindblad form is trace-annihilating). Lean ✓ Stmt *Every Lindblad form defines a trace-annihilating linear map.*

Proof. ✓ Proof The Hamiltonian part satisfies $\mathrm{tr}(i[\rho, H]) = 0$ by the cyclic property. Each dissipator term satisfies $\mathrm{tr}(L_j\rho L_j^\dagger) = \mathrm{tr}(L_j^\dagger L_j\rho) = \mathrm{tr}(\rho L_j^\dagger L_j)$, so the three terms sum to zero. □

Theorem 17.33 (Lindblad form equals generator decomposition). Lean ✓ Stmt *A Lindblad form with Hamiltonian H and operators $\{L_j\}$ defines the same linear map as the generator decomposition (ϕ, κ) with $\phi(\rho) = \sum_j L_j\rho L_j^\dagger$ and $\kappa = iH + \frac{1}{2}\sum_j L_j^\dagger L_j$. This is [Wol12, Equation (7.24)].*

Proof. ✓ Proof Compute $\kappa^\dagger = -iH + \frac{1}{2}\sum_j L_j^\dagger L_j$ using $H^\dagger = H$ and $(A^\dagger A)^\dagger = A^\dagger A$. Setting $S = \sum_j L_j^\dagger L_j$, expand $\phi(\rho) - \kappa\rho - \rho\kappa^\dagger = \sum_j L_j \rho L_j^\dagger - iH\rho - \frac{1}{2}S\rho + i\rho H - \frac{1}{2}\rho S$, which is $i[\rho, H] + \sum_j (L_j \rho L_j^\dagger - \frac{1}{2}L_j^\dagger L_j \rho - \frac{1}{2}\rho L_j^\dagger L_j)$. □

Theorem 17.34 (Lindblad form is CCP)**.** Lean ✓ Stmt *Every Lindblad form defines a conditionally completely positive map.*

Proof. ✓ Proof By Theorem 17.33, the Lindblad form equals a generator decomposition with ϕ CP. □

Definition 17.35 (Commutator form of Lindblad equation)**.** Lean ✓ Stmt The *commutator form* of the Lindblad equation writes the generator as

$$L(\rho) = i[\rho, H] + \frac{1}{2}\sum_j ([L_j, \rho L_j^\dagger] + [L_j \rho, L_j^\dagger]),$$

where $[A, B] = AB - BA$. This is [Wol12, Equation (7.22)].

Theorem 17.36 (Commutator form equals standard Lindblad form)**.** Lean ✓ Stmt *The commutator form (Definition 17.35) defines the same linear map as the standard Lindblad form (Definition 17.31).*

Proof. ✓ Proof Expanding the commutator brackets $[L_j, \rho L_j^\dagger]$ and $[L_j \rho, L_j^\dagger]$ yields the standard dissipator terms. □

17.3.3 Characterization of conditional complete positivity

Theorem 17.37 (Conditional complete positivity implies projected Choi positivity)**.** Lean ✓ Stmt *If L is CCP and $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$, then*

$$P((L \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|))P \geq 0.$$

This is [Wol12, Proposition 7.2].

Proof. ✓ Proof Write $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$. The left- and right-multiplication terms are annihilated by P, so the projected Choi matrix reduces to the projected Choi matrix of ϕ, which is positive. □

Theorem 17.38 (Projected Choi positivity implies conditional complete positivity)**.** Lean ✓ Stmt *If L is Hermiticity-preserving and $P((L \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|))P \geq 0$ where $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$, then L is CCP. This is [Wol12, Proposition 7.2].*

Proof. ✓ Proof Decompose the Hermitian Choi matrix as $\tau_L = Q - |\psi\rangle\langle\Omega| - |\Omega\rangle\langle\psi|$ with $Q \geq 0$ supported on $|\Omega\rangle^\perp$. The Choi–Jamiolkowski isomorphism applied to Q gives the CP map ϕ, and $(\kappa \otimes \mathbb{1})|\Omega\rangle = |\psi\rangle$ defines κ. □

17.3.4 Completely positive semigroups and conditionally completely positive generators

Theorem 17.39 (CP semigroups have CCP generators)**.** Lean ✓ Stmt *If e^{tL} is CP for all $t \geq 0$, then L is CCP. This is [Wol12, Proposition 7.3].*

Proof. ✓ Proof From $(e^{tL} \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|) \geq 0$, expand at infinitesimal t and project onto P to get $P(L \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|)P \geq 0$. Then Theorem 17.38 gives CCP. □

Theorem 17.40 (CCP generators yield CP semigroups). Lean ✓ Stmt *If L is CCP, then e^{tL} is CP for all $t \geq 0$. This is [Wol12, Proposition 7.3].*

Proof. ✓ Proof Approximate e^{tL} by the completely positive Euler steps

$$\rho \mapsto (\mathbb{1} - h\kappa)\rho(\mathbb{1} - h\kappa)^\dagger + h\phi(\rho), \qquad h = \frac{t}{n+1}.$$

Each step is CP, hence so are its powers. These powers converge in operator norm to e^{tL}, and the cone of CP maps is closed. □

Theorem 17.41 (CP semigroups iff CCP generators). Lean ✓ Stmt *e^{tL} is CP for all $t \geq 0$ if and only if L is CCP.*

Proof. ✓ Proof Combine Theorems 17.39 and 17.40. □

17.3.5 Freedom in generator representation

Theorem 17.42 (Traceless Kraus operators exist). Lean ✓ Stmt *Assume $d > 0$. Given any Kraus operators $\{K_j\}$, there exist $c_j \in \mathbb{C}$ such that $K_j' = K_j + c_j\mathbb{1}$ is traceless. This is part of [Wol12, Proposition 7.4].*

Proof. ✓ Proof Set $c_j = -\mathrm{tr}(K_j)/d$. □

Theorem 17.43 (Shift invariance of generators). Lean ✓ Stmt *If $K_j' = K_j + c_j\mathbb{1}$ and κ' is adjusted per [Wol12, Equation (7.19)], then (ϕ', κ') and (ϕ, κ) define the same generator.*

Proof. ✓ Proof Direct algebraic expansion. □

17.3.6 Uniqueness of the traceless Lindblad form

Definition 17.44 (Traceless Kraus operators). Lean ✓ Stmt A Lindblad form has *traceless Kraus operators* if $\mathrm{tr}(L_j) = 0$ for every j.

Theorem 17.45 (Uniqueness of the completely positive part). Lean ✓ Stmt *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then their CP parts agree: $\phi = \phi'$. This is part of [Wol12, Proposition 7.4(2)].*

Proof. ✓ Proof Compare projected Choi matrices and use Choi injectivity. □

Theorem 17.46 (Uniqueness of the drift term modulo phase). Lean ✓ Stmt *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then their drift matrices differ by an imaginary scalar: $\kappa' = \kappa + i\lambda\mathbb{1}$ for some $\lambda \in \mathbb{R}$. This is part of [Wol12, Proposition 7.4(2)].*

Proof. ✓ Proof After identifying $\phi = \phi'$ (Theorem 17.45), the generator equality forces $\Delta\kappa \cdot \rho + \rho \cdot (\Delta\kappa)^\dagger = 0$ for all ρ. Setting $\rho = \mathbb{1}$ gives $\Delta\kappa + (\Delta\kappa)^\dagger = 0$ (skew-Hermiticity), which converts the equation to $[\Delta\kappa, \rho] = 0$ for all ρ. The scalar commutant lemma gives $\Delta\kappa = c \cdot \mathbb{1}$, and skew-Hermiticity forces $c = i\lambda$ for some $\lambda \in \mathbb{R}$. □

Theorem 17.47 (Combined uniqueness of the traceless Lindblad form). Lean ✓ Stmt *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then $\phi = \phi'$ and $\kappa' = \kappa + i\lambda\mathbb{1}$ for some $\lambda \in \mathbb{R}$.*

Proof. ✓ Proof Combine Theorems 17.45 and 17.46. □

17.3.7 Trace-annihilation and trace preservation

Theorem 17.48 (Trace-annihilating generators yield trace-preserving semigroups). Lean ✓ Stmt
If L is trace-annihilating, then e^{tL} is trace-preserving for every $t \in \mathbb{R}$.

Proof. ✓ Proof For fixed ρ, the function $f(t) = \operatorname{tr}(e^{tL}(\rho))$ has derivative $f'(t) = \operatorname{tr}(L(e^{tL}(\rho))) = 0$, so f is constant and $\operatorname{tr}(e^{tL}(\rho)) = f(t) = f(0) = \operatorname{tr}(\rho)$. ☐

Theorem 17.49 (Trace-preserving semigroups have trace-annihilating generators). Lean ✓ Stmt
If e^{tL} is trace-preserving for every $t \geq 0$, then L is trace-annihilating.

Proof. ✓ Proof Differentiate the identity $\operatorname{tr}(e^{tL}(\rho)) = \operatorname{tr}(\rho)$ at $t = 0$ from the right. ☐

17.3.8 The GKSL/Lindblad theorem

Definition 17.50 (GKSL generator). Lean ✓ Stmt A linear map $L : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ is a *GKSL generator* if e^{tL} is a quantum channel (CPTP) for all $t \geq 0$.

Theorem 17.51 (A trace-constrained generator decomposition gives a GKSL generator). Lean ✓ Stmt *If $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ with ϕ CP and $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, then L is a GKSL generator. This is [Wol12, Theorem 7.1, Equation (7.20)].*

Proof. ✓ Proof CP follows from Theorem 17.40. The trace constraint implies trace-annihilation by Theorem 17.30, so Theorem 17.48 gives trace preservation. ☐

Theorem 17.52 (GKSL generators admit a trace-constrained generator decomposition). Lean ✓ Stmt *If L is a GKSL generator, then there exist a CP map ϕ and a matrix κ such that*

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$$

and $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$.

Proof. ✓ Proof Apply Theorem 17.39 to obtain a CCP decomposition. Then apply Theorem 17.49 to the trace-preserving part of the semigroup and rewrite the resulting trace-annihilation condition as $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$. ☐

Theorem 17.53 (GKSL iff conditional complete positivity and trace annihilation). Lean ✓ Stmt
L is a GKSL generator if and only if L is CCP and trace-annihilating.

Proof. ✓ Proof Forward: apply Theorem 17.39 to the CP part and Theorem 17.49 to the TP part. Reverse: combine Theorem 17.40 with Theorem 17.48. ☐

Theorem 17.54 (GKSL iff Lindblad form). Lean ✓ Stmt *L is a GKSL generator if and only if*

$$L(\rho) = i[\rho, H] + \sum_j \left(L_j \rho L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\}_+ \right)$$

with $H = H^\dagger$. This is [Wol12, Theorem 7.1].

Proof. ✓ Proof Forward: apply Theorem 17.52 to write $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ with trace constraint. Then choose traceless Kraus operators by Theorem 17.42 and rewrite κ as $iH + \frac{1}{2}\phi^*(\mathbb{1})$; Theorem 17.33 gives the Lindblad formula. Reverse: Theorems 17.34 and 17.32 show that every Lindblad form satisfies the right-hand side of Theorem 17.53. ☐

17.3.9 Kossakowski matrix form

Definition 17.55 (Kossakowski form). Lean ✓ Stmt A *Kossakowski form* consists of $H = H^\dagger$, a finite family of matrices $\{F_k\}_{k=0}^{n-1}$, and a positive semidefinite matrix $C \in M_n(\mathbb{C})$, defining

$$L(\rho) = i[\rho, H] + \tfrac{1}{2} \sum_{k,l} C_{lk}([F_k, \rho F_l^\dagger] + [F_k \rho, F_l^\dagger]).$$

In Wolf's formula one takes $\{F_k\}$ to be a basis of traceless matrices; here we keep only the algebraic data needed for the conversion to Lindblad form.

Theorem 17.56 (Kossakowski form iff Lindblad form). Lean ✓ Stmt *A linear map admits a Kossakowski form iff it admits a Lindblad form.*

Proof. ✓ Proof Forward: diagonalize $C = B^\dagger B$ and set $L_j = \sum_k B_{jk} F_k$. Reverse: keep the same Hamiltonian and family $F_k = L_k$, and take $C = \mathbb{1}$. □

17.4 Primitivity and irreducibility of QDS

17.4.1 Auxiliary spectral and semigroup lemmas

Definition 17.57 (Quantum dynamical semigroup). Lean ✓ Stmt A quantum dynamical semigroup is a norm-continuous dynamical semigroup whose time slices are quantum channels for every $t \geq 0$.

Theorem 17.58 (Semigroup power identity). Lean ✓ Stmt *For $t \geq 0$ and $n \in \mathbb{N}$, one has $T_{nt} = (T_t)^n$.*

Theorem 17.59 (Eigenvector propagation under powers). Lean ✓ Stmt *If $E(v) = \mu v$, then $E^n(v) = \mu^n v$ for all $n \in \mathbb{N}$.*

Lemma 17.60 (Density matrices are nonzero). Lean ✓ Stmt *Every density matrix is nonzero.*

Theorem 17.61 (Compression invariance is stable under powers). Lean ✓ Stmt *If a corner compression is invariant under E, it remains invariant under every power E^n.*

Theorem 17.62 (Generator eigenvectors along the exponential semigroup). Lean ✓ Stmt *If $L(X) = \mu X$, then $e^{tL}(X) = e^{t\mu} X$ for all $t \in \mathbb{R}$.*

Theorem 17.63 (Spectral mapping for semigroup generators). Lean ✓ Stmt *If μ is in the spectrum of L, then $e^{t\mu}$ is in the spectrum of e^{tL}.*

Theorem 17.64 (Root-of-unity rigidity for phases). Lean ✓ Stmt *If $e^{it\theta}$ is a root of unity for every $t > 0$, then $\theta = 0$.*

Theorem 17.65 (Peripheral generator eigenvalues are purely imaginary). Lean ✓ Stmt *Peripheral spectral points of the generator have vanishing real part.*

Theorem 17.66 (Peripheral cardinality bound). Lean ✓ Stmt *The number of peripheral eigenvalues is bounded by* $\dim(M_D(\mathbb{C}))$.

Theorem 17.67 (Bounded order under peripheral power-closure). Lean ✓ Stmt *If all powers of a peripheral eigenvalue remain peripheral, then some positive power not exceeding* $\dim(M_D(\mathbb{C}))$ *equals* 1.

Theorem 17.68 (Peripheral powers stay peripheral under irreducibility). Lean ✓ Stmt *For an irreducible channel with faithful fixed point, every power of a peripheral eigenvalue is again peripheral.*

Theorem 17.69 (Continuous-linear-map power evaluation). Lean ✓ Stmt *Evaluating powers of a linear map agrees with powers in its continuous linear realization.*

Theorem 17.70 (Spectral-radius criterion from eigenvalue bounds). Lean ✓ Stmt *If all eigenvalues satisfy $|\lambda| < 1$, then the spectral radius is < 1.*

Theorem 17.71 (Primitive powers annihilate trace-zero matrices). Lean ✓ Stmt *If E is an irreducible primitive channel with faithful fixed density σ, then $E^n(X) \to 0$ for every matrix X with $\mathrm{tr}(X) = 0$.*

Proof. ✓ Proof Decompose E^n into the fixed-point projection and its complementary part. On the complement, every eigenvalue has modulus < 1, so the complementary powers tend to 0; the trace-zero hypothesis removes the fixed-point component. □

Theorem 17.72 (Fixed points of an irreducible channel are trace multiples). Lean ✓ Stmt *If E is an irreducible channel with faithful fixed density σ, then every nonzero fixed point V of E satisfies $V = \mathrm{tr}(V)\sigma$.*

Proof. ✓ Proof Subtract the trace multiple $\mathrm{tr}(V)\sigma$ and apply the vanishing theorem for trace-zero fixed points of an irreducible channel. □

Corollary 17.73 (Nonzero fixed points have nonzero trace). Lean ✓ Stmt *Every nonzero fixed point of an irreducible channel has nonzero trace.*

Proof. ✓ Proof Otherwise the preceding theorem would force the fixed point to vanish. □

Theorem 17.74 (Peripheral eigenvectors become periodic fixed points). Lean ✓ Stmt *For every positive time t, a quantum dynamical semigroup irreducible at every positive time has the following property: every peripheral eigenvalue μ of T_t admits a nonzero eigenvector V and a positive integer p such that $T_t(V) = \mu V$ and $T_{pt}(V) = V$.*

Proof. ✓ Proof Peripheral powers remain peripheral for an irreducible channel with faithful fixed point. The bounded-order lemma then gives $\mu^p = 1$, and the semigroup power identity turns the eigenvector equation for T_t into a fixed-point equation for T_{pt}. □

Theorem 17.75 (Peripheral eigenvectors with nonzero trace). Lean ✓ Stmt *For every positive time t, under the same hypotheses every peripheral eigenvalue of T_t has a nonzero eigenvector with nonzero trace.*

Proof. ✓ Proof Apply Theorem 17.74 and then use the previous corollary at the irreducible time pt. □

Theorem 17.76 (Trace-preserving eigenvectors force eigenvalue 1). Lean ✓ Stmt *If E is trace-preserving, $E(V) = \mu V$, and $\mathrm{tr}(V) \neq 0$, then $\mu = 1$.*

Proof. ✓ Proof Take traces in $E(V) = \mu V$ and use trace preservation. □

Theorem 17.77 (Peripheral collapse under irreducibility at all times). Lean ✓ Stmt *If a quantum dynamical semigroup is irreducible at every positive time, then every peripheral eigenvalue of every positive-time slice equals 1.*

Proof. ✓ Proof Combine Theorem 17.75 with Theorem 17.76. □

Theorem 17.78 (Primitivity from irreducibility at all times). Lean ✓ Stmt *If a quantum dynamical semigroup is irreducible at every positive time, then every positive-time slice is primitive.*

Proof. ✓ Proof The unique faithful fixed density gives the one-dimensional peripheral spectrum criterion, and Theorem 17.77 collapses the peripheral set to $\{1\}$. □

Remark 17.79 (Semigroup irreducible/primitive equivalence). ✓ Stmt The semigroup analog of the irreducible/primitive equivalence starts from one irreducible time slice, constructs a primitive smaller time step, and lifts the conclusion to the full semigroup (Theorems 17.95 and 17.96).

Remark 17.80 (Reduction to a smaller time step for semigroups). ✓ Stmt Starting from one irreducible time t_0, we construct a positive time step $u = t_0/N$ with $T_{t_0} = (T_u)^N$, prove peripheral eigenvalue collapse at time u, and deduce primitivity of T_u.

Theorem 17.81 (The stationary density is fixed for all times). Lean ✓ Stmt *If $t_0 > 0$, T_{t_0} is irreducible, and σ is its unique fixed density matrix, then $T_u(\sigma) = \sigma$ for every $u \geq 0$.*

Proof. ✓ Proof By semigroup commutativity, $T_{t_0}(T_u(\sigma)) = T_u(T_{t_0}(\sigma)) = T_u(\sigma)$, and uniqueness of the fixed density at time t_0 identifies $T_u(\sigma)$ with σ. □

Theorem 17.82 (Fixed points at the irreducible time are trace multiples). Lean ✓ Stmt *If T_{t_0} is irreducible with faithful fixed density σ, then every fixed point V of T_{t_0} satisfies $V = \operatorname{tr}(V)\sigma$.*

Proof. ✓ Proof This is the fixed-point trace-scalar theorem for the irreducible channel T_{t_0}. □

Definition 17.83 (Residual slice index). Lean ✓ Stmt For $u \geq 0$ and $s > 0$, the residual slice index is

$$m_n = \left\lfloor \frac{nu}{s} \right\rfloor.$$

Definition 17.84 (Residual slice time). Lean ✓ Stmt For $u \geq 0$ and $s > 0$, the residual slice time is the remainder

$$r_n = s \operatorname{fract}\left(\frac{nu}{s}\right).$$

Theorem 17.85 (Residual slice decomposition). Lean Lean ✓ Stmt *For $u \geq 0$ and $s > 0$, one has $r_n \in [0, s]$ and*

$$nu = m_n s + r_n.$$

Proof. ✓ Proof This is the floor-plus-fraction decomposition of nu/s, multiplied by s. □

Theorem 17.86 (A residual slice can vanish). Lean ✓ Stmt *Suppose $u \geq 0$ and $s > 0$, and that $T_s(\delta) = \delta$ and $T_{nu}(\delta) \to 0$ as $n \to \infty$. Then there exists $a \in [0, s]$ such that $T_a(\delta) = 0$.*

Proof. ✓ Proof The residual slice times lie in the compact interval $[0, s]$, so a subsequence converges to some $a \in [0, s]$. The residual decomposition rewrites $T_{nu}(\delta)$ in terms of $T_{r_n}(\delta)$, and continuity of the semigroup identifies the limit with $T_a(\delta)$. □

Theorem 17.87 (Reduced-time eigenvector has nonzero trace). Lean ✓ Stmt *Under the reduced-time hypotheses $T_{t_0} = (T_u)^{(\dim M_D(\mathbb{C}))!}$, if μ is a peripheral eigenvalue of T_u, then T_u admits a μ-eigenvector with nonzero trace.*

Theorem 17.88 (Peripheral collapse at the reduced time step). Lean ✓ Stmt *Under the same hypotheses, every peripheral eigenvalue of T_u equals 1.*

Theorem 17.89 (Primitivity at the reduced time step). Lean ✓ Stmt *Under the same hypotheses, T_u is primitive.*

Theorem 17.90 (Irreducibility descends to a reduced time step). Lean ✓ Stmt *If $T_{t_0} = (T_u)^N$ and T_{t_0} is irreducible, then T_u is irreducible.*

Proof. ✓ Proof Any invariant compression for T_u remains invariant under powers, hence also for T_{t_0}; irreducibility of T_{t_0} rules this out. □

Theorem 17.91 (Existence of an irreducible reduced time step). Lean ✓ Stmt *If $t_0 > 0$, T_{t_0} is irreducible, and σ is fixed for all times, then there exists a positive time step u such that*

$$T_{t_0} = (T_u)^{(\dim M_D(\mathbb{C}))!},$$

the slice T_u is a channel, T_u is irreducible, and $T_u(\sigma) = \sigma$.

Proof. ✓ Proof Take $u = t_0/N$ with $N = (\dim M_D(\mathbb{C}))!$. The semigroup power identity gives the factorization of T_{t_0}, channelity comes from the semigroup hypotheses, and Theorem 17.90 gives irreducibility of T_u. □

Theorem 17.92 (Fixed points of the primitive reduced time step are trace-scalars). Lean ✓ Stmt *If T_u is primitive and σ is the common fixed density, then every fixed point of any positive-time slice T_s has the form $\tau = \operatorname{tr}(\tau)\sigma$.*

Theorem 17.93 (Existence of a primitive reduced time step). Lean ✓ Stmt *If T_{t_0} is irreducible for some $t_0 > 0$, then there exists a positive reduced time step u such that T_u is primitive.*

Theorem 17.94 (Irreducibility propagates to all positive times). Lean ✓ Stmt *If T_{t_0} is irreducible for some $t_0 > 0$, then every positive-time slice T_s is irreducible.*

Proof. ✓ Proof Let σ be the unique faithful fixed density of T_{t_0}. By Theorem 17.81, it is fixed for all times, and Theorem 17.82 makes the fixed-point space of T_{t_0} one-dimensional. Choose a primitive reduced time step by Theorem 17.93; then the trace-scalar description of fixed points from Theorem 17.92 gives the uniqueness criterion for irreducibility at every positive time. □

Theorem 17.95 (An irreducible semigroup is primitive). Lean ✓ Stmt *Let $T_t = e^{tL}$ be a quantum dynamical semigroup. If T_{t_0} is irreducible for some $t_0 > 0$, then T_t is primitive (and irreducible) for every $t > 0$. This is [Wol12, Proposition 7.5].*

Proof. ✓ Proof First apply Theorem 17.94 to propagate irreducibility from the single time t_0 to every positive time. Then apply Theorem 17.78 to conclude primitivity of every positive-time slice. □

Theorem 17.96 (Irreducibility iff primitivity for a quantum dynamical semigroup). Lean ✓ Stmt *For a quantum dynamical semigroup $T_t = e^{tL}$:*

$$(\exists\, t_0 > 0,\ T_{t_0}\ \textit{irreducible}) \iff (\forall\, t > 0,\ T_t\ \textit{primitive and irreducible}).$$

This is [Wol12, Proposition 7.5].

Proof. ✓ Proof Forward: apply Theorem 17.95. Reverse: take $t_0 = 1$ (or any positive time); the right-hand side already includes irreducibility of T_t for every $t > 0$. □

17.5 Kernel of the adjoint Liouvillian

Definition 17.97 (Faithful stationary state). Lean ✓ Stmt A generator L has a faithful stationary state if there exists a positive definite density matrix ρ_0 with $L(\rho_0) = 0$.

Definition 17.98 (Adjoint dissipator). Lean ✓ Stmt For a Lindblad operator L_j, the adjoint dissipator on observables is

$$\mathcal{D}^*_{L_j}(A) = L_j^\dagger A L_j - \tfrac{1}{2} L_j^\dagger L_j A - \tfrac{1}{2} A L_j^\dagger L_j.$$

Definition 17.99 (Adjoint generator). Lean ✓ Stmt The adjoint generator of a Lindblad form is

$$L^*(A) = i[H, A] + \sum_j \mathcal{D}^*_{L_j}(A).$$

Theorem 17.100 (Trace-pairing characterization of the adjoint generator). Lean ✓ Stmt *For every density matrix ρ and every observable A,*

$$\operatorname{tr}(\rho\, L^*(A)) = \operatorname{tr}(L(\rho)\, A).$$

Proof. ✓ Proof Expand the Schrödinger and Heisenberg formulas and apply cyclicity of the trace term by term. □

Definition 17.101 (Commutant of the Lindblad data). Lean ✓ Stmt The commutant of a Lindblad form is the set of matrices commuting with H, with every Lindblad operator L_j, and with every adjoint $L_j^\dagger$.

Definition 17.102 (Adjoint kernel). Lean ✓ Stmt The adjoint kernel is the set of observables A satisfying $L^*(A) = 0$.

Theorem 17.103 (Commutant lies in the adjoint kernel). Lean Lean ✓ Stmt *If an observable commutes with H, with every L_j, and with every $L_j^\dagger$, then it lies in* $\ker(L^*)$.

Proof. ✓ Proof The Hamiltonian commutator term vanishes, and each adjoint dissipator $\mathcal{D}^*_{L_j}(A)$ is zero under the commutation hypotheses. □

Theorem 17.104 (Adjoint kernel lies in the commutant). Lean ✓ Stmt *Let L be a GKSL generator in Lindblad form with Hamiltonian H and Lindblad operators $\{L_j\}$. Define the adjoint generator L^* by $\operatorname{tr}(\rho\, L^*(A)) = \operatorname{tr}(L(\rho)\, A)$. If L has a faithful stationary state $\rho_0 > 0$ with $L(\rho_0) = 0$, then for any A:*

$$L^*(A) = 0 \implies [A, H] = [A, L_j] = [A, L_j^\dagger] = 0 \ \forall j.$$

That is, $\ker(L^*) \subseteq \{H, L_j, L_j^\dagger\}'$. *This is [Wol12, Theorem 7.2].*

Proof. ✓ Proof From $L^*(A) = 0$ and $L^*(A^\dagger) = 0$, compute $L^*(A^\dagger A) = \sum_j [A, L_j]^\dagger [A, L_j]$. Faithfulness of ρ_0 and $\operatorname{tr}(\rho_0 L^*(A^\dagger A)) = 0$ force $[A, L_j] = 0$ for all j. Similarly $[A, L_j^\dagger] = 0$, and then $L^*(A) = 0$ directly gives $[A, H] = 0$. □

Theorem 17.105 (Adjoint kernel equals the commutant). Lean ✓ Stmt *Under the same hypotheses as Theorem 17.104,*

$$\ker(L^*) = \{H, L_j, L_j^\dagger\}',$$

i.e. the adjoint kernel equals the commutant of $\{H, L_j, L_j^\dagger\}$. This is [Wol12, Theorem 7.2].

Proof. ✓ Proof The inclusion $\supseteq$ is the easy direction (commutant elements are in the kernel by a direct calculation). The reverse $\subseteq$ is Theorem 17.104. □

17.6 Reducibility of quantum dynamical semigroups

The four equivalent reducibility conditions of [Wol12, Proposition 7.6] are as follows.

Definition 17.106 (Nontrivial projection). Lean ✓ Stmt A projection is nontrivial if it is orthogonal and neither 0 nor $\mathbb{1}$.

Definition 17.107 (Rank-deficient fixed density). Lean ✓ Stmt Condition (1): there exists a density matrix ρ_0 with nontrivial kernel such that $e^{tL}(\rho_0) = \rho_0$ for all $t \geq 0$.

Definition 17.108 (Rank-deficient kernel element). Lean ✓ Stmt Condition (2): there exists a density matrix ρ_0 with nontrivial kernel satisfying $L(\rho_0) = 0$.

Definition 17.109 (Invariant compression). Lean ✓ Stmt Condition (3): there exists a nontrivial orthogonal projector P such that $e^{tL}(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$ for all $t \geq 0$.

Definition 17.110 (Block-upper-triangular Lindblad form). Lean ✓ Stmt Condition (4): there exists a nontrivial projector P and a Lindblad form $(H, \{L_j\})$ for L such that $(1-P)L_jP = 0$ and $(1-P)\kappa P = 0$ for all j, where $\kappa = iH + \frac{1}{2}\sum_j L_j^\dagger L_j$.

Definition 17.111 (Generator-preserved compression). Lean ✓ Stmt For a projection P, the generator preserves the compression $PM_d(\mathbb{C})P$ if

$$P\,L(PXP)\,P = L(PXP)$$

for every $X \in M_d(\mathbb{C})$.

Definition 17.112 (Reducible quantum dynamical semigroup). Lean ✓ Stmt A quantum dynamical semigroup is reducible if it has a nontrivial invariant compression.

Theorem 17.113 (Semigroup invariance implies generator invariance). Lean ✓ Stmt *If the semigroup preserves $PM_d(\mathbb{C})P$ for all non-negative times, then the generator preserves the same compression.*

Proof. ✓ Proof Differentiate the identity $Pe^{tL}(PXP)P = e^{tL}(PXP)$ at $t = 0$. □

Theorem 17.114 (Generator invariance implies semigroup invariance). Lean ✓ Stmt *If the generator preserves $PM_d(\mathbb{C})P$, then so does the exponential semigroup e^{tL} for every $t \geq 0$.*

Proof. ✓ Proof Apply the compression map termwise to the exponential series. Every iterate of L preserves the compression, so the whole series does as well. □

Theorem 17.115 (Reducibility as a generator-level criterion). Lean ✓ Stmt *A quantum dynamical semigroup is reducible if and only if its generator preserves some nontrivial compression.*

Proof. ✓ Proof Combine Theorems 17.113 and 17.114. □

Theorem 17.116 (Rank-deficient fixed densities and kernel elements). Lean ✓ Stmt *A rank-deficient density matrix is a fixed point of e^{tL} for all $t \geq 0$ if and only if it lies in* $\ker L$*. This is [Wol12, Proposition 7.6, (1)⇔(2)].*

Proof. ✓ Proof The equivalence $L(\rho) = 0 \iff e^{tL}(\rho) = \rho$ for all $t \geq 0$ follows from the generator–semigroup correspondence: differentiate at $t = 0$ for the forward direction, and exponentiate for the reverse. □

Theorem 17.117 (A rank-deficient fixed density yields an invariant compression). Lean ✓ Stmt *If L is a GKSL generator and there exists a rank-deficient fixed density matrix, then the semigroup preserves a nontrivial compression. This is [Wol12, Proposition 7.6, (1)⇒(3)].*

Proof. ✓ Proof Let ρ_0 be a rank-deficient fixed density. Let P be the support projection of ρ_0. Since ρ_0 is rank-deficient, $P \neq \mathbb{1}$; since $\rho_0 \neq 0$, $P \neq 0$. Channel positivity and trace preservation force e^{tL} to preserve $PM_D(\mathbb{C})P$. □

Theorem 17.118 (Invariant compression yields block-upper-triangular Lindblad data). Lean ✓ Stmt *If a GKSL generator preserves a nontrivial compression, then its Lindblad data can be chosen block-upper-triangular with respect to that compression.*

Proof. ✓ Proof First pass from semigroup invariance to generator invariance by Theorem 17.113. The generator-level vanishing condition forces $(1 - P)L_jP = 0$ and $(1 - P)\kappa P = 0$ through the sum-of-squares identity. □

Theorem 17.119 (Block-upper-triangular Lindblad data yield invariant compression). Lean ✓ Stmt *If the Lindblad data are block-upper-triangular with respect to a nontrivial projection, then the associated semigroup preserves that compression.*

Proof. ✓ Proof The block-upper-triangular relations imply generator-level compression invariance; then apply Theorem 17.114. □

Theorem 17.120 (Invariant compression and triangular Lindblad data). Lean Lean ✓ Stmt *For a GKSL generator L, the semigroup preserves a nontrivial compression if and only if the Lindblad data can be chosen block-upper-triangular. This is [Wol12, Proposition 7.6, (3)⇔(4)].*

Proof. ✓ Proof Combine Theorems 17.119 and 17.118. □

Theorem 17.121 (Triangular Lindblad data give deficient kernels). Lean ✓ Stmt *If the Lindblad data of a GKSL generator are block-upper-triangular with respect to some nontrivial projector P, then there exists a density matrix ρ_0 with nontrivial kernel satisfying $L(\rho_0) = 0$. This is [Wol12, Proposition 7.6, (4)⇒(2)].*

Proof. ✓ Proof The block-upper-triangular structure implies L preserves the compression $PM_d(\mathbb{C})P$. Therefore the semigroup e^{tL} also preserves $PM_d(\mathbb{C})P$ for all $t \geq 0$. By the Brouwer fixed-point theorem, $e^{(1/m)L}$ has a fixed density matrix ρ_m supported in $PM_d(\mathbb{C})P$ for each m. Passing to a subsequential limit gives a density matrix ρ_0 with $P\rho_0P = \rho_0$ and $L(\rho_0) = 0$. Since $P \neq \mathbb{1}$, the matrix ρ_0 has nontrivial kernel. □

Theorem 17.122 (Equivalent formulations of reducibility). Lean ✓ Stmt *For a GKSL generator $L : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, the four conditions*

1. *rank-deficient fixed density,*
2. *rank-deficient kernel element,*
3. *invariant compression, and*
4. *block-upper-triangular Lindblad form*

are equivalent. This is [Wol12, Proposition 7.6].

Proof. ✓ Proof The cycle (1)↔(2), (1)→(3), (3)↔(4), and (4)→(2) closes the equivalence chain. □

17.6.1 Sufficient conditions for non-reducibility

Definition 17.123 (Lindblad span). Lean ✓ Stmt The $\mathbb{C}$-linear span of the Lindblad operators $\{L_j\}$.

Definition 17.124 (Hermitian-closed Lindblad span). Lean ✓ Stmt The span of the Lindblad operators is closed under Hermitian conjugation, i.e. it forms a $*$-subspace of $M_d(\mathbb{C})$.

Definition 17.125 (Trivial Lindblad commutant). Lean ✓ Stmt The commutant of the Lindblad family $\{L_j\}'$ equals $\mathbb{C} \cdot \mathbb{1}$, i.e. the Lindblad operators act irreducibly on $M_d(\mathbb{C})$.

Definition 17.126 (Kossakowski rank). Lean ✓ Stmt The minimal number of Lindblad operators across all GKSL representations of L, which equals the rank of the Kossakowski matrix in the orthonormal-basis representation.

Theorem 17.127 (κ block vanishing from generator compression). Lean ✓ Stmt *Let $(H, \{L_j\})$ be a Lindblad form with $\kappa = iH + \frac{1}{2}\sum_j L_j^\dagger L_j$, and let P be an orthogonal projection such that the generator preserves the compression $PM_d(\mathbb{C})P$. If $(1-P)L_jP = 0$ for every j, then $(1-P)\kappa P = 0$.*

Proof. ✓ Proof From $L = \varphi - \kappa(\cdot) - (\cdot)\kappa^\dagger$ and $(1-P)L(P) = 0$: the $(1-P)\varphi(P)$ term vanishes because each $(1-P)L_jP = 0$, and the $(1-P)(P\kappa^\dagger)$ term vanishes because $(1-P)P = 0$. What remains is $(1-P)\kappa P = 0$. ☐

Theorem 17.128 (Full algebra generation forbids block-upper-triangular form). Lean ✓ Stmt *Let $(H, \{L_j\})$ be a Lindblad form. If the algebra generated by $\{L_j\}$ and κ is the entire matrix algebra $M_d(\mathbb{C})$, then the Lindblad data do not admit a block-upper-triangular decomposition.*

Proof. ✓ Proof Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection P. By Theorem 17.127, $(1-P)\kappa P = 0$. Induction on elements of the subalgebra shows $(1-P)AP = 0$ for every generator. Since the generated algebra is all of $M_d(\mathbb{C})$, every matrix satisfies $(1-P)AP = 0$, forcing $P = 0$ or $P = \mathbb{1}$ — a contradiction. ☐

Theorem 17.129 (Hermitian span with trivial commutant forbids triangular form). Lean ✓ Stmt *If the Lindblad span is closed under Hermitian conjugation and its commutant is $\mathbb{C} \cdot \mathbb{1}$, then the Lindblad data do not admit a block-upper-triangular decomposition.*

Proof. ✓ Proof Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection P. The block vanishing $(1-P)L_jP = 0$ and Hermitian closure give $PL_j(1-P) = 0$ as well, so $[P, L_j] = 0$ for all j. But then P lies in the commutant of the Lindblad span, contradicting triviality. ☐

Theorem 17.130 (Large Kossakowski rank forbids block-upper-triangular form). Lean ✓ Stmt *If the Kossakowski rank satisfies*

$$\mathrm{rank}(C) + d \geq d^2 + 1,$$

then no block-upper-triangular Lindblad form exists.

Proof. ✓ Proof A block-upper-triangular traceless Lindblad family lies in a proper subspace whose dimension is at most $d^2 - d$. The formal rank minimization argument then contradicts the large-rank bound. ☐

Theorem 17.131 (No block-upper-triangular Lindblad form implies non-reducibility). Lean ✓ Stmt *If a GKSL generator admits no block-upper-triangular Lindblad form, then the associated quantum dynamical semigroup is not reducible.*

Proof. ✓ Proof A reducible semigroup would have an invariant compression, and Theorem 17.118 would then produce a block-upper-triangular Lindblad form. □

Theorem 17.132 (Full algebra generation implies non-reducibility). Lean ✓ Stmt *If the algebra generated by* $\{L_j\}$ *and* κ *is the entire matrix algebra* $M_d(\mathbb{C})$*, then the QDS is not reducible. This is [Wol12, Corollary 7.2(1)].*

Proof. ✓ Proof By Theorem 17.128, no block-upper-triangular decomposition exists. Then apply Theorem 17.131. □

Theorem 17.133 (Hermitian span and trivial commutant imply non-reducibility). Lean ✓ Stmt *If the Lindblad span is Hermitian-closed and its commutant is trivial, then the QDS is not reducible. This is [Wol12, Corollary 7.2(2)].*

Proof. ✓ Proof By Theorem 17.129, no block-upper-triangular decomposition exists. Then apply Theorem 17.131. □

Theorem 17.134 (Large Kossakowski rank implies non-reducibility). Lean ✓ Stmt *If* $\operatorname{rank}(C) > d^2 - d$*, then the QDS is not reducible. This is [Wol12, Corollary 7.2(3)].*

Proof. ✓ Proof The rank hypothesis rules out block-upper-triangular Lindblad data by Theorem 17.130. Then apply Theorem 17.131. □

Chapter 18

Operator Convexity and Jensen Inequalities

This chapter collects the operator convexity/concavity results, trace inequalities, the operator Jensen inequality for positive maps, and the Lieb concavity theorem. These are standard results in matrix analysis ([Bha97, Chapter V]; [Wol12, Theorems 5.12 and 5.13]; Hansen–Pedersen's Jensen inequality [HP82]; and Lieb's concavity theorem [Lie73]). They are provided here as cited inputs, since the corresponding matrix-analysis proofs for real powers $x \mapsto x^p$ on the Loewner order and for the matrix logarithm are not developed elsewhere in this document.

18.1 Diagonal Jensen inequality

Lemma 18.1 (Diagonal Jensen inequality). Lean ✓ Stmt *Let $f : \mathbb{R} \to \mathbb{R}$ be convex on $[0, \infty)$, let A be a positive semidefinite matrix on a finite-dimensional space indexed by n, and let $v \in \mathbb{C}^n$ satisfy $\langle v, v\rangle = 1$. Then*

$$f(\Re\langle v, Av\rangle) \le \Re\langle v, f(A)\, v\rangle,$$

where $f(A)$ is defined by the Hermitian continuous functional calculus.

Proof. ✓ Proof Write $A = U \operatorname{diag}(\mu)\, U^*$ by the spectral theorem and set $w = U^* v$. Since U is unitary, $p_i := |w_i|^2$ satisfies $\sum_i p_i = \|v\|^2 = 1$, so (p_i) is a probability distribution on n. The eigenvalues μ_i lie in $[0, \infty)$ because A is positive semidefinite. A direct computation yields $\Re\langle v, Av\rangle = \sum_i p_i \mu_i$ and $\Re\langle v, f(A)v\rangle = \sum_i p_i f(\mu_i)$, so the scalar Jensen inequality applied to the weights (p_i) and points (μ_i) gives the conclusion. See [Bha97, Chapter V]. □

18.2 Trace concavity and convexity of matrix powers

Theorem 18.2 (Trace concavity of real powers). Lean Lean ✓ Stmt *For $p \in [0, 1]$, PSD matrices A_1, A_2, and $t \in [0, 1]$:*

$$t\,\Re\operatorname{tr}(A_1^p) + (1-t)\Re\operatorname{tr}(A_2^p) \le \Re\operatorname{tr}((tA_1 + (1-t)A_2)^p).$$

Follows from operator concavity of $x \mapsto x^p$ composed with trace monotonicity on the Loewner order.

Proof. ✓ Proof Let $A = tA_1 + (1-t)A_2$ and let $\{\psi_j\}$ with eigenvalues $\mu_j \ge 0$ be an orthonormal eigenbasis of A. A direct computation gives $\mathfrak{R}\,\mathrm{tr}(A^p) = \sum_j \mu_j^p$ and $\mu_j = t\,a_j + (1-t)\,b_j$ where $a_j = \mathfrak{R}\langle\psi_j, A_1\psi_j\rangle$, $b_j = \mathfrak{R}\langle\psi_j, A_2\psi_j\rangle$. Scalar concavity of $x \mapsto x^p$ on $[0,\infty)$ yields $\mu_j^p \ge t\,a_j^p + (1-t)\,b_j^p$, and the diagonal Jensen inequality (Lemma 18.1, concave form) gives $a_j^p \le \mathfrak{R}\langle\psi_j, A_1^p\psi_j\rangle$ (and similarly for b_j). Summing over j and using $\sum_j \mathfrak{R}\langle\psi_j, M\psi_j\rangle = \mathfrak{R}\,\mathrm{tr}\,M$ yields the conclusion. See [Bha97, Chapter V]. ☐

Theorem 18.3 (Trace convexity of real powers). Lean Lean ✓ Stmt *For $p \in [1,2]$, PSD matrices A_1, A_2, and $t \in [0,1]$:*

$$\mathfrak{R}\,\mathrm{tr}((tA_1 + (1-t)A_2)^p) \le t\,\mathfrak{R}\,\mathrm{tr}(A_1^p) + (1-t)\mathfrak{R}\,\mathrm{tr}(A_2^p).$$

Follows from operator convexity of $x \mapsto x^p$ composed with trace monotonicity on the Loewner order.

Proof. ✓ Proof Identical eigenbasis reduction as in Theorem 18.2, with scalar convexity (rather than concavity) of $x \mapsto x^p$ on $[0,\infty)$ for $p \ge 1$ and the convex form of the diagonal Jensen inequality. ☐

18.3 Operator Jensen inequality for positive maps

Definition 18.4 (Finite-POVM dilation construction). Lean Lean ✓ Stmt For a finite family of matrices $\{C_i\}_{i\in\iota}$ and a defect block S, one forms an isometric dilation whose rows are the adjoints of the matrices C_i together with the adjoint of S. One also forms the scalar block-diagonal matrix whose ι-blocks carry prescribed weights and whose defect block carries a single scalar.

Lemma 18.5 (Compression auxiliary for Jensen). Lean ✓ Stmt *Let Y be positive definite and let W be an isometry. Then*

$$(W^*YW)^{-1} \le W^*Y^{-1}W. \tag{18.1}$$

Proof. ✓ Proof Apply the Schur-complement criterion to the block matrix $\begin{pmatrix} Y^{-1} & W \\ W^* & W^*YW \end{pmatrix}$. The lower-right Schur complement is zero, hence positive semidefinite, and compressing the resulting upper-left Schur complement by W gives the inequality (18.1). ☐

Lemma 18.6 (Finite-POVM compression identities). Lean Lean Lean Lean Lean Lean ✓ Stmt *If the defect block satisfies*

$$SS^* = \mathbb{1} - \sum_{i\in\iota} C_iC_i^*, \tag{18.2}$$

then the dilation is an isometry, compressing the weighted scalar block-diagonal matrix gives

$$\sum_{i\in\iota} w_i\,C_iC_i^* + t\,SS^*,$$

the defect relation rewrites as

$$\sum_{i\in\iota} C_iC_i^* + SS^* = \mathbb{1},$$

and strict positivity of all weights together with $t > 0$ implies positive definiteness of the scalar block-diagonal matrix. If all weights are nonzero and the defect scalar is nonzero, the inverse diagonal has reciprocal weights, and compressing this inverse gives

$$\sum_{i\in\iota} w_i^{-1}\,C_iC_i^* + t^{-1}\,SS^*.$$

Proof. ✓ Proof Expand the block matrix multiplication entrywise. The defect identity (18.2) gives the isometry relation after summing the ι-blocks and the defect block, and the weighted compression identity follows from the same expansion with the scalar diagonal inserted. Positive definiteness uses strict positivity of every diagonal entry, while the inverse formula uses the nonvanishing of every diagonal entry. □

Lemma 18.7 (Finite-POVM resolvent inequality). Lean ✓ Stmt *Let $w_i \geq 0$ and $t > 0$. If the defect block satisfies*

$$SS^* = \mathbb{1} - \sum_{i\in\iota} C_i C_i^*,$$

then

$$\left(\sum_{i\in\iota} w_i\, C_i C_i^* + t\,\mathbb{1}\right)^{-1} \leq \sum_{i\in\iota} (w_i + t)^{-1}\, C_i C_i^* + t^{-1}\, SS^*. \tag{18.3}$$

Proof. ✓ Proof Apply the compression inequality to the positive definite scalar block-diagonal matrix with entries $w_i + t$ and defect entry t. The compression identities rewrite the compressed matrix as $\sum_i w_i C_i C_i^* + t\mathbb{1}$ and its inverse compression as the upper bound (18.3). □

Theorem 18.8 (Jensen for concave real powers). Lean Not ready *For a positive subunital map T, a positive semidefinite matrix A, and $p \in [0,1]$:*

$$T(A^p) \leq (TA)^p.$$

This is [Wol12, Theorem 5.13] for concave $f(x) = x^p$.

Theorem 18.9 (Map form of Jensen for concave real powers). Lean ✓ Stmt *For a positive subunital map T, a positive semidefinite matrix A, and $p \in [0,1]$, the inequality $T(A^p) \leq (TA)^p$ holds.*

Proof. ✓ Proof Direct application of the axiom. □

Theorem 18.10 (Jensen for convex real powers). Lean Not ready *For a positive subunital map T, a positive semidefinite matrix A, and $p \in [1,2]$:*

$$(TA)^p \leq T(A^p).$$

This is [Wol12, Theorem 5.11] for the subunital operator-convex case $f(x) = x^p$.

Theorem 18.11 (Map form of Jensen for convex real powers). Lean ✓ Stmt *For a positive subunital map T, a positive semidefinite matrix A, and $p \in [1,2]$, the inequality $(TA)^p \leq T(A^p)$ holds.*

Proof. ✓ Proof Direct application of the axiom. □

Theorem 18.12 (Jensen for concave log). Lean Not ready *For a positive* unital *map T and positive-definite A:*

$$T(\log A) \leq \log(TA).$$

This is [Wol12, Theorem 5.13] for $f = \log$. Requires unitality ($T(\mathbb{1}) = \mathbb{1}$), not merely subunitality.

Theorem 18.13 (Map form of Jensen for the concave logarithm). Lean ✓ Stmt *For a positive unital map T and a positive-definite matrix A, the inequality $T(\log A) \leq \log(TA)$ holds.*

Proof. ✓ Proof Direct application of the axiom. □

18.4 Lieb concavity theorem

Theorem 18.14 (Lieb concavity). Lean Not ready *For $s \in [0,1]$, any matrix K, and positive-definite matrices A_1, A_2, B_1, B_2, the map $(A, B) \mapsto \operatorname{tr}(K^\dagger A^s K B^{1-s})$ is jointly concave:*

$$t\,\Re\operatorname{tr}(K^\dagger A_1^s K B_1^{1-s})+(1-t)\Re\operatorname{tr}(K^\dagger A_2^s K B_2^{1-s}) \leq \Re\operatorname{tr}(K^\dagger(tA_1+(1-t)A_2)^s K(tB_1+(1-t)B_2)^{1-s}). \tag{18.4}$$

See [Lie73, And79].

Theorem 18.15 (Map form of Lieb concavity). Lean ✓ Stmt *For $s \in [0,1]$, any matrix K, and positive-definite matrices A_1, A_2, B_1, B_2, the function $(A, B) \mapsto \Re\operatorname{tr}(K^\dagger A^s K B^{1-s})$ is jointly concave, satisfying the inequality (18.4) of Theorem 18.14.*

Proof. ✓ Proof Direct application of the axiom. □

Corollary 18.16 (Lieb concavity, $K = \mathbb{1}$ case). Lean ✓ Stmt *For $s \in [0,1]$, $t \in [0,1]$, and positive-definite matrices A_1, A_2, B_1, B_2, the map $(A, B) \mapsto \Re\operatorname{tr}(A^s B^{1-s})$ is jointly concave:*

$$t\,\Re\operatorname{tr}(A_1^s B_1^{1-s}) + (1-t)\Re\operatorname{tr}(A_2^s B_2^{1-s}) \leq \Re\operatorname{tr}((tA_1 + (1-t)A_2)^s(tB_1 + (1-t)B_2)^{1-s}).$$

Obtained from Theorem 18.15 by taking $K = \mathbb{1}$.

Proof. ✓ Proof Specialize Theorem 18.15 at $K = \mathbb{1}$ and simplify $\mathbb{1}^\dagger = \mathbb{1}$ together with $\mathbb{1} \cdot X = X \cdot \mathbb{1} = X$. □

Corollary 18.17 (Concavity of Lieb's map in the first argument). Lean ✓ Stmt *For $s \in [0,1]$, $t \in [0,1]$, matrix K, and positive-definite matrices A_1, A_2, B, the map $A \mapsto \Re\operatorname{tr}(K^\dagger A^s K B^{1-s})$ is concave:*

$$t\,\Re\operatorname{tr}(K^\dagger A_1^s K B^{1-s}) + (1-t)\Re\operatorname{tr}(K^\dagger A_2^s K B^{1-s}) \leq \Re\operatorname{tr}(K^\dagger(tA_1 + (1-t)A_2)^s K B^{1-s}).$$

Proof. ✓ Proof Specialize Theorem 18.15 at $B_1 = B_2 = B$, so that the convex combination collapses: $tB + (1-t)B = B$. □

Corollary 18.18 (Concavity of Lieb's map in the second argument). Lean ✓ Stmt *For $s \in [0,1]$, $t \in [0,1]$, matrix K, and positive-definite matrices A, B_1, B_2, the map $B \mapsto \Re\operatorname{tr}(K^\dagger A^s K B^{1-s})$ is concave:*

$$t\,\Re\operatorname{tr}(K^\dagger A^s K B_1^{1-s}) + (1-t)\Re\operatorname{tr}(K^\dagger A^s K B_2^{1-s}) \leq \Re\operatorname{tr}(K^\dagger A^s K(tB_1 + (1-t)B_2)^{1-s}).$$

Proof. ✓ Proof Specialize Theorem 18.15 at $A_1 = A_2 = A$, so that the convex combination collapses: $tA + (1-t)A = A$. □

Chapter 19

Quantum Entropy

This chapter develops the von Neumann entropy and its basic properties for finite-dimensional quantum systems.

19.1 Von Neumann entropy

Definition 19.1 (Von Neumann entropy). Lean ✓ Stmt For a Hermitian matrix $\rho \in M_D(\mathbb{C})$ with eigenvalues $\lambda_0, \ldots, \lambda_{D-1}$, the *von Neumann entropy* is

$$S(\rho) = -\sum_{i=0}^{D-1} \lambda_i \log \lambda_i,$$

where $0 \log 0 := 0$.

Theorem 19.2 (Entropy is non-negative). Lean ✓ Stmt *For any density matrix ρ, $S(\rho) \geq 0$.*

Proof. ✓ Proof Each eigenvalue λ_i of a density matrix satisfies $0 \leq \lambda_i \leq 1$, and $-x \log x \geq 0$ on $[0, 1]$. □

Lemma 19.3 (Eigenvalue sum). Lean ✓ Stmt *The eigenvalues of a density matrix sum to 1.*

Proof. ✓ Proof Follows from $\mathrm{tr}(\rho) = \sum_i \lambda_i = 1$. □

Theorem 19.4 (Eigenvalue bound). Lean ✓ Stmt *Each eigenvalue of a density matrix lies in $[0, 1]$.*

Proof. ✓ Proof Non-negativity comes from positive semidefiniteness. The upper bound follows because the eigenvalues are non-negative and sum to 1. □

Theorem 19.5 (Entropy upper bound). Lean ✓ Stmt *For a density matrix $\rho \in M_D(\mathbb{C})$ with $D \geq 1$,*

$$S(\rho) \leq \log D.$$

Proof. ✓ Proof By Jensen's inequality applied to the concave function $-x \log x$, the entropy is maximized when all eigenvalues equal $1/D$. □

Theorem 19.6 (Entropy bounded by log of rank). Lean ✓ Stmt *For a density matrix ρ of rank r,*

$$S(\rho) \le \log r.$$

This refines the dimension bound: only the nonzero eigenvalues contribute to the entropy, and there are exactly r of them.

Proof. ✓ Proof The entropy is the $-x \log x$ sum over all eigenvalues; the $D - r$ zero eigenvalues contribute nothing. Jensen's inequality applied to $-x \log x$ over the r nonzero eigenvalues, with uniform weights $1/r$, gives $S(\rho) \le \log r$, the maximum attained when each nonzero eigenvalue equals $1/r$. The rank equals the number of nonzero eigenvalues of the Hermitian matrix ρ. □

Lemma 19.7 (Entropy from characteristic polynomial roots). Lean ✓ Stmt *The von Neumann entropy of a Hermitian matrix is the $-x \log x$ sum over the real parts of the roots of its characteristic polynomial χ_ρ:*

$$S(\rho) = \sum_{\lambda \in \mathrm{roots}(\chi_\rho)} (-\operatorname{Re}(\lambda) \log \operatorname{Re}(\lambda)).$$

Proof. ✓ Proof The eigenvalues of a Hermitian matrix are exactly the roots of its characteristic polynomial, counted with multiplicity, so the eigenvalue sum defining $S(\rho)$ equals the displayed sum over roots. □

Theorem 19.8 (Entropy is invariant under reindexing). Lean ✓ Stmt *Let ρ be a Hermitian matrix indexed by a finite set J, and let $e : I \to J$ be a bijection from a finite set I. The reindexed matrix on I with entries $\rho_{e(i)\,e(j)}$ has the same von Neumann entropy as ρ.*

Proof. ✓ Proof By Lemma 19.7 the entropy depends only on the characteristic polynomial. Reindexing conjugates ρ by a permutation matrix, so

$$\chi_{(\rho_{e(i)\,e(j)})} = \chi_\rho,$$

and the entropies coincide. □

Lemma 19.9 (Cyclic invariance of the charpoly-root entropy sum). Lean ✓ Stmt *Let $A \in M_{m\times n}(\mathbb{C})$ and $B \in M_{n\times m}(\mathbb{C})$. Then the charpoly-root entropy sum is invariant under the cyclic swap $AB \mapsto BA$:*

$$\sum_{\lambda \in \mathrm{roots}(\chi_{AB})} (-\operatorname{Re}(\lambda) \log \operatorname{Re}(\lambda)) = \sum_{\mu \in \mathrm{roots}(\chi_{BA})} (-\operatorname{Re}(\mu) \log \operatorname{Re}(\mu)),$$

with roots counted with algebraic multiplicity.

Proof. ✓ Proof The rectangular characteristic-polynomial identity gives

$$X^n \chi_{AB} = X^m \chi_{BA}.$$

Thus the two characteristic polynomials have the same nonzero roots, with multiplicity; the additional zero roots contribute nothing because $0 \log 0 = 0$. □

Theorem 19.10 (Entropy of AB equals entropy of BA). Lean ✓ Stmt *For matrices $A \in M_{m\times n}(\mathbb{C})$ and $B \in M_{n\times m}(\mathbb{C})$ such that AB and BA are Hermitian,*

$$S(AB) = S(BA).$$

Proof. ✓ Proof By Lemma 19.7, the entropy of a Hermitian matrix is its charpoly-root entropy sum. Lemma 19.9 identifies these two sums for AB and BA. □

19.2 Tripartite partial traces

Definition 19.11 (Partial trace over A). Lean ✓ Stmt For a tripartite matrix ρ_{ABC} on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B} \otimes \mathbb{C}^{d_C}$, the partial trace over A is

$$(\operatorname{tr}_A \rho_{ABC})_{(b_1,c_1)(b_2,c_2)} = \sum_{a=0}^{d_A-1} (\rho_{ABC})_{(a,b_1,c_1)(a,b_2,c_2)}.$$

Definition 19.12 (Partial trace over C). Lean ✓ Stmt The partial trace over C:

$$(\operatorname{tr}_C \rho_{ABC})_{(a_1,b_1)(a_2,b_2)} = \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a_1,b_1,c)(a_2,b_2,c)}.$$

Definition 19.13 (Partial trace over AC). Lean ✓ Stmt The partial trace over A and C:

$$(\operatorname{tr}_{AC} \rho_{ABC})_{b_1 b_2} = \sum_{a=0}^{d_A-1} \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a,b_1,c)(a,b_2,c)}.$$

Lemma 19.14 (Partial trace preserves Hermiticity). *Lean* *✓ Stmt* *If ρ_{ABC} is Hermitian, then $\operatorname{tr}_A(\rho_{ABC})$, $\operatorname{tr}_C(\rho_{ABC})$, and $\operatorname{tr}_{AC}(\rho_{ABC})$ are all Hermitian. The same holds for bipartite partial traces $\operatorname{tr}_A(\rho_{AB})$ and $\operatorname{tr}_B(\rho_{AB})$.*

Proof. ✓ Proof Follows from $\overline{\rho_{ji}} = \rho_{ij}$ applied entry-wise inside the summation defining each partial trace. □

19.3 Strong subadditivity

Theorem 19.15 (Strong subadditivity). *Lean* *Not ready* *For a tripartite density matrix ρ_{ABC},*

$$S(\rho_{ABC}) + S(\rho_B) \le S(\rho_{AB}) + S(\rho_{BC}).$$

Definition 19.16 (SSA equality). Lean ✓ Stmt A tripartite density matrix ρ_{ABC} satisfies *SSA equality* if

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC}).$$

Definition 19.17 (Quantum Markov decomposition). Lean ✓ Stmt A *Hayashi Markov decomposition* of a tripartite state ρ_{ABC} consists of a finite direct-sum decomposition

$$H_B \cong \bigoplus_j H_{B_j^L} \otimes H_{B_j^R},$$

together with a unitary change of basis on B, a probability vector $(p_j)_j$, and density matrices $\rho_{AB_j^L}$ and $\rho_{B_j^R C}$ such that, in the adapted basis, the state becomes

$$\bigoplus_j p_j\, \rho_{AB_j^L} \otimes \rho_{B_j^R C}.$$

The terminology follows Hayashi's presentation of quantum Markov structure [Hay06]; the block decomposition used by the MPDO argument is the structure theorem of [HJPW04].

Theorem 19.18 (Hayashi SSA-equality characterization). Lean Not ready *For a tripartite density matrix ρ_{ABC},*

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC})$$

holds if and only if ρ_{ABC} admits a quantum Markov decomposition on the middle subsystem B.

This equality criterion is cited here as an input for the later MPDO arguments. It is the equality criterion needed by Appendix C of arXiv:1606.00608. The equality conditions are reviewed in [Hay06, Rus02]; the structural block-decomposition formulation used for MPDOs appears in [HJPW04].

19.4 Mutual information

Definition 19.19 (Mutual information). Lean ✓ Stmt The *quantum mutual information* of a bipartite state ρ_{AB} is

$$I(A{:}B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}).$$

Theorem 19.20 (Mutual information is non-negative). Lean ✓ Stmt *For any bipartite density matrix ρ_{AB}, $I(A{:}B) \geq 0$.*

Proof. ✓ Proof Apply strong subadditivity (Theorem 19.15) with trivial B (one-dimensional middle system). The SSA inequality $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$ reduces to subadditivity $S(\rho_{AC}) \leq S(\rho_A) + S(\rho_C)$, which gives $I(A{:}C) \geq 0$. □

19.5 Entropy formulations

This section states the entropy formulations used later in the development: von Neumann entropy, strong subadditivity, quantum Markov decomposition, and mutual information. These statements are cited from the standard entropy literature and supply the entropy-theoretic input for the later MPDO arguments.

Definition 19.21 (Entropy formulation of von Neumann entropy). Lean ✓ Stmt This formulation has the same value $S(\rho) = -\sum_i \lambda_i \log \lambda_i$ as in Definition 19.1.

Theorem 19.22 (Entropy formulation of strong subadditivity). Lean ✓ Stmt *For any tripartite density matrix ρ_{ABC} on $A \otimes B \otimes C$,*

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

This formulation introduces no new axiom: it is the same strong-subadditivity statement as Theorem 19.15. The underlying axiom's proof in [LR73] is deferred to the umbrella entropy issue.

Definition 19.23 (Entropy formulation of quantum Markov decomposition). Lean ✓ Stmt This is the entropy formulation of the Hayashi Markov decomposition from Definition 19.17.

Theorem 19.24 (SSA equality iff quantum Markov decomposition). Lean ✓ Stmt *For any tripartite density matrix ρ_{ABC}, equality in strong subadditivity holds if and only if ρ_{ABC} admits a quantum Markov decomposition on the middle subsystem B.*

This formulation introduces no new axiom: it is the same equality criterion as Theorem 19.18.

Definition 19.25 (Entropy formulation of mutual information). Lean ✓ Stmt This formulation has the same value $I(A{:}B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$ as in Definition 19.19.

Theorem 19.26 (Subadditivity from trivial-middle SSA). Lean ✓ Stmt *For a tripartite density matrix ρ_{ABC} with* $\dim B = 1$,

$$S(\rho_{ABC}) \le S(\rho_{AB}) + S(\rho_{BC}).$$

Proof. ✓ Proof Apply Theorem 19.22 with one-dimensional middle subsystem. The reduced middle state has trace 1 by Theorem 19.32, hence its entropy vanishes by Theorem 19.33. ☐

19.6 Mutual information: monotonicity and area-law bound

The two inequalities below are the downstream MPDO-facing consequences of the entropy inequalities in this chapter. The monotonicity inequality is the strong-subadditivity content underlying the MPDO mutual-information monotonicity $I_L \le I_{L+1}$ (arXiv:1606.00608, Proposition C.1); the elementary area-law bound is the single-site entropy bound underlying the MPDO area-law bound $I_L \le 4 \log D$ (arXiv:1606.00608, cited from the Wolf area-law bound). Both inequalities follow directly from strong subadditivity (taken as an axiom) and the single-system $S(\rho) \le \log D$ bound; neither introduces a new axiom.

Theorem 19.27 (Entropy nonnegativity for finite index sets). Lean ✓ Stmt *For any PSD Hermitian matrix ρ with $\mathrm{tr}(\rho) = 1$ on an arbitrary finite index set, $S(\rho) \ge 0$. This is the analog of Theorem 19.2 for arbitrary finite index sets: it does not require the index set to be $\{0, \dots, D-1\}$, so it applies to bipartite density matrices on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$.*

Proof. ✓ Proof The eigenvalues of a PSD matrix are non-negative, and eigenvalues of a trace-1 Hermitian matrix with non-negative eigenvalues are bounded above by 1 (each single eigenvalue is at most the total sum). Apply $x \log x \le 0$ on $[0, 1]$ pointwise and sum. ☐

Theorem 19.28 (Mutual information is monotone under enlargement). Lean ✓ Stmt *For any tripartite density matrix ρ_{ABC} on $A \otimes B \otimes C$,*

$$I(A{:}B)_{\mathrm{tr}_C \rho_{ABC}} \le I(A{:}BC)_{\rho_{ABC}},$$

where the left-hand side is the bipartite mutual information of the reduced state $\rho_{AB} = \mathrm{tr}_C(\rho_{ABC})$ and the right-hand side is evaluated by expanding $I(A{:}BC) = S(\rho_A) + S(\rho_{BC}) - S(\rho_{ABC})$.

Proof. ✓ Proof Expanding both sides in entropy form and cancelling the common $S(\rho_A)$ term, the inequality reduces to strong subadditivity $S(\rho_{ABC}) + S(\rho_B) \le S(\rho_{AB}) + S(\rho_{BC})$. The bipartite B-reduced state of ρ_{AB} matches the tripartite B-reduced state $\mathrm{tr}_{AC}(\rho_{ABC})$, ensuring the $S(\rho_B)$ term is the one supplied by Theorem 19.22. ☐

Theorem 19.29 (Elementary area-law bound on mutual information). Lean ✓ Stmt *For a bipartite density matrix ρ_{AB} on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ with $d_A, d_B \ge 1$ whose single-system reduced states are obtained by partial trace,*

$$I(A{:}B) \le \log d_A + \log d_B.$$

Proof. ✓ Proof The reduced states are density matrices because partial trace preserves positivity and trace. Combine $S(\rho_A) \le \log d_A$ and $S(\rho_B) \le \log d_B$ (Theorem 19.5) with $S(\rho_{AB}) \ge 0$ (Theorem 19.27). ☐

19.7 Trivial-factor corollaries

The partial traces and von Neumann entropy are introduced in Chapter 19. This supplement collects elementary trace-preservation identities and the dimension-1 entropy bound that follow directly from those definitions. They are listed as separate results because each proof requires only unfolding definitions and re-indexing finite sums.

Theorem 19.30 (Partial trace over A preserves the full trace). Lean ✓ Stmt *For any tripartite matrix* ρ_{ABC}*,*

$$\operatorname{tr}(\rho_{ABC}) = \operatorname{tr}(\operatorname{tr}_A(\rho_{ABC})).$$

Proof. ✓ Proof Unfolding the definitions,

$$\operatorname{tr}(\operatorname{tr}_A(\rho_{ABC})) = \sum_{b,c} \sum_{a} (\rho_{ABC})_{(a,b,c)(a,b,c)} = \operatorname{tr}(\rho_{ABC}).$$

□

Theorem 19.31 (Partial trace over C preserves the full trace). Lean ✓ Stmt *For any tripartite matrix* ρ_{ABC}*,*

$$\operatorname{tr}(\rho_{ABC}) = \operatorname{tr}(\operatorname{tr}_C(\rho_{ABC})).$$

Proof. ✓ Proof Unfolding the definitions,

$$\operatorname{tr}(\operatorname{tr}_C(\rho_{ABC})) = \sum_{a,b} \sum_{c} (\rho_{ABC})_{(a,b,c)(a,b,c)} = \operatorname{tr}(\rho_{ABC}).$$

□

Theorem 19.32 (Partial trace over AC preserves the full trace). Lean ✓ Stmt *For any tripartite matrix* ρ_{ABC}*,*

$$\operatorname{tr}(\rho_{ABC}) = \operatorname{tr}(\operatorname{tr}_{AC}(\rho_{ABC})).$$

Proof. ✓ Proof Unfolding the definitions and interchanging the outer sums,

$$\operatorname{tr}(\operatorname{tr}_{AC}(\rho_{ABC})) = \sum_{b} \sum_{a,c} (\rho_{ABC})_{(a,b,c)(a,b,c)} = \operatorname{tr}(\rho_{ABC}).$$

□

Theorem 19.33 (Entropy vanishes in dimension 1). Lean ✓ Stmt *If* $\rho \in M_1(\mathbb{C})$ *is Hermitian and* $\operatorname{tr}(\rho) = 1$*, then*

$$S(\rho) = 0.$$

Proof. ✓ Proof The unique eigenvalue equals the trace, hence equals 1, and $-1 \cdot \log 1 = 0$:

$$\lambda_0 = \operatorname{tr}(\rho) = 1, \qquad S(\rho) = -\lambda_0 \log \lambda_0 = -1 \cdot \log 1 = 0.$$

□

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